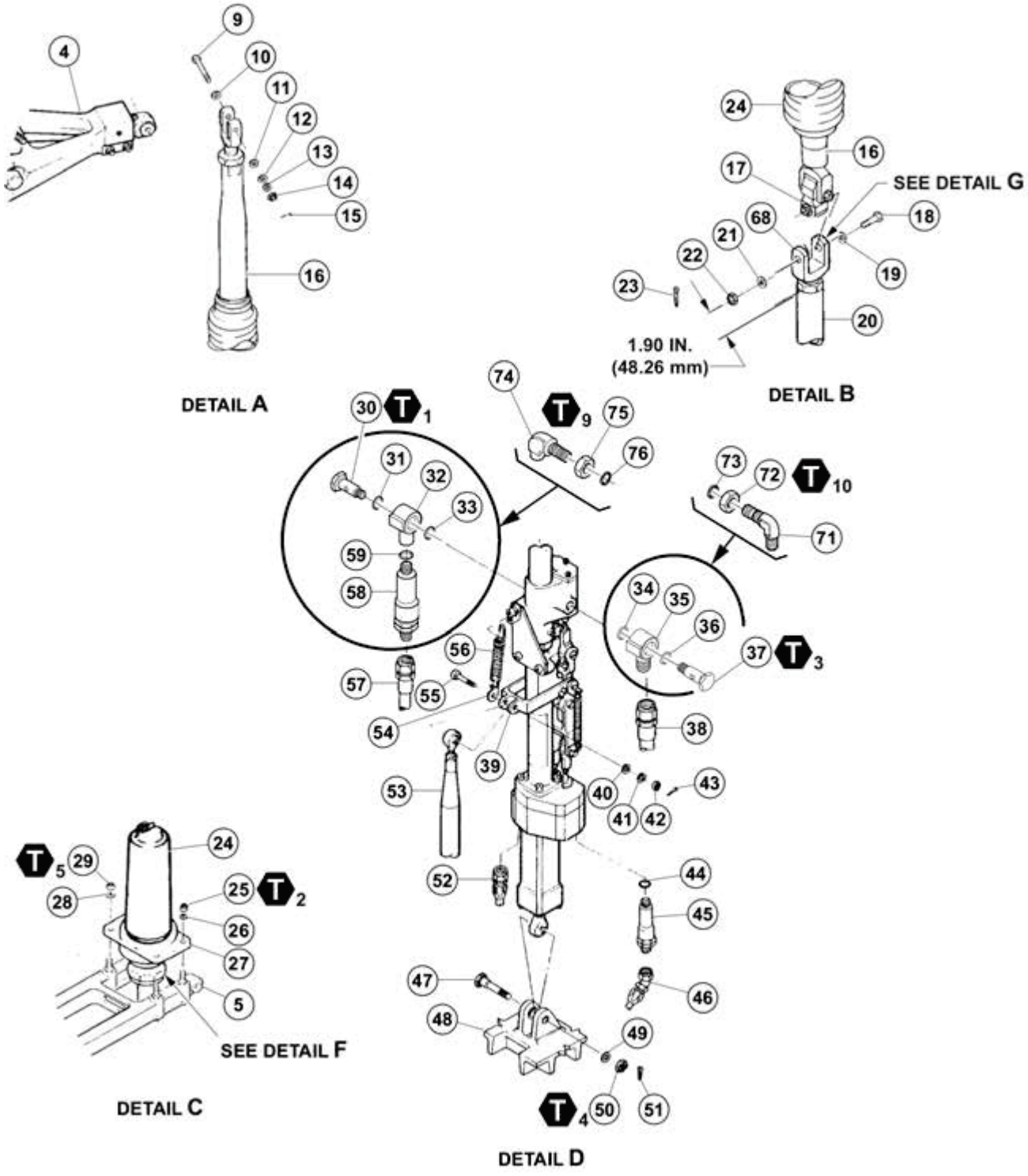
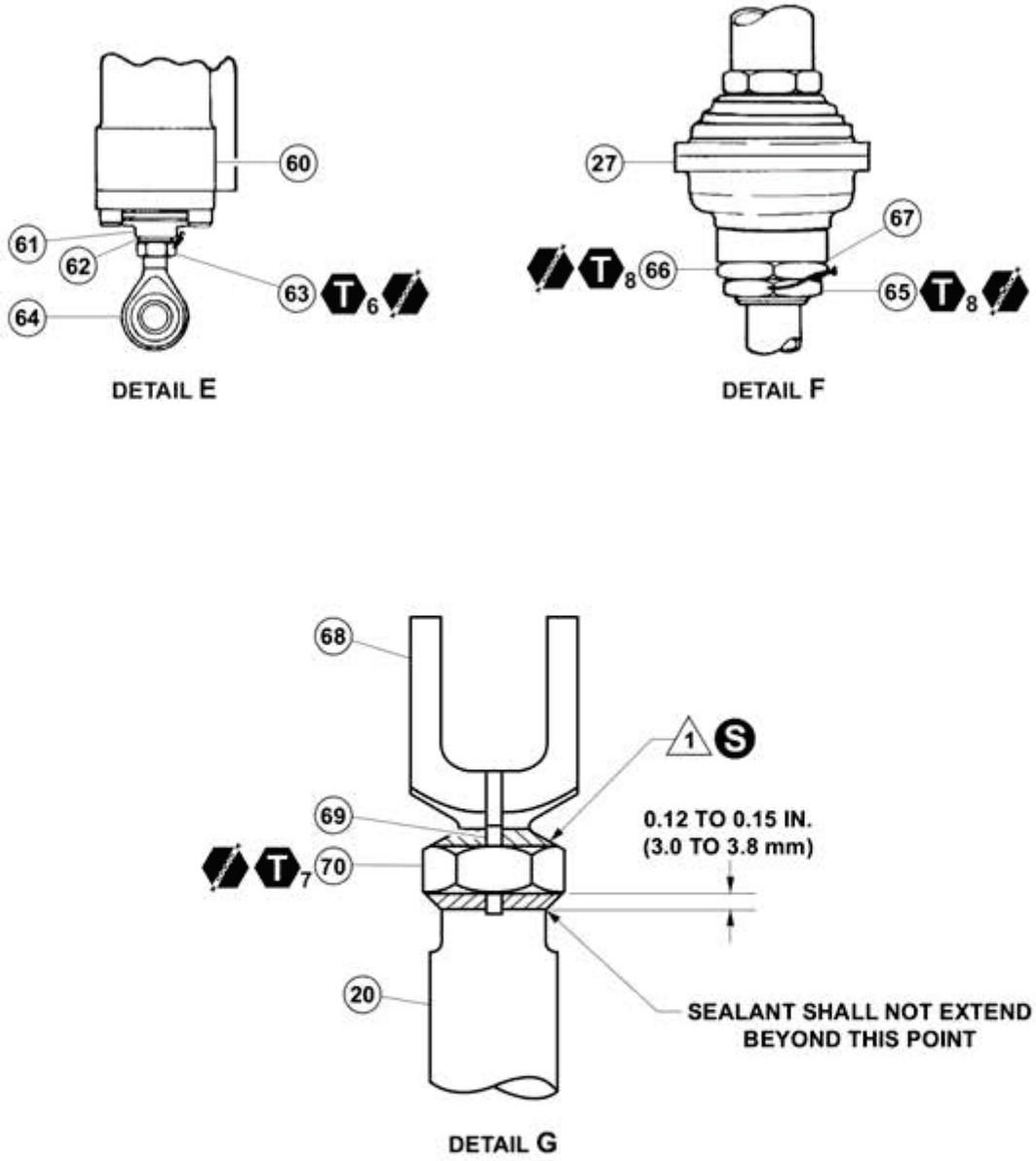


ICN-412-A-671100-B-97499-00008-A-001-01

Figure 1. Collective Flight Control Cylinder (Sheet 2 of 4)



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Figure 1. Collective Flight Control Cylinder (Sheet 3 of 4)

1. Pump
2. Reservoir
3. Swashplate horn
4. Collective lever
5. Support
6. Collective flight control cylinder
7. Cyclic flight control cylinder
8. Cyclic flight control cylinder
9. Bolt
10. Washer (AN960C516)
11. Washer (AN960C516)
12. Washer (AN960C516)
13. Washer (AN960C516L)
14. Nut
15. Cotter pin
16. Extension tube
17. Universal
18. Bolt
19. Washer (AN960C516)
20. Upper piston
21. Washer (AN960C516)
22. Nut
23. Cotter pin
24. Boot
25. Nut
26. Aluminum washer
27. Support bracket
28. Aluminum washer
29. Nut
30. Bolt
31. Preformed packing
32. Elbow
33. Preformed packing
34. Preformed packing
35. Elbow
36. Preformed packing
37. Bolt
38. Hose
39. Lever
40. Washer
41. Washer
42. Nut
43. Cotter pin
44. Preformed packing
45. Bypass check valve
46. Hose
47. Bolt
48. Support
49. Aluminum washer
50. Nut
51. Cotter pin
52. Hose
53. Tube
54. Spring clip
55. Bolt
56. Spring
57. Hose
58. Bypass check valve
59. Preformed packing
60. Actuator
61. Lower piston
62. Locking device
63. Nut
64. Rod end bearing
65. Nut
66. Bearing nut
67. Keyed washer
68. Clevis
69. Lock
70. Jam nut
71. Elbow
72. Nut
73. Packing
74. Elbow
75. Nut
76. Packing

LOCKWIRE (C-405)

S SEALANT (C-308)

T1 40 TO 65 IN-LBS (4.51 TO 7.34 Nm)

T2 75 TO 95 IN-LBS (8.47 TO 10.73 Nm)

T3 75 TO 125 IN-LBS (8.47 TO 14.12 Nm)

T4 95 TO 110 IN-LBS (10.73 TO 12.42 Nm)

T5 120 TO 160 IN-LBS (13.55 TO 18.07 Nm)

T6 200 TO 250 IN-LBS (22.59 TO 28.54 Nm)

T7 660 TO 780 IN-LBS (74.56 TO 88.12 Nm)

T8 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

T9 135 TO 145 IN-LBS (15.25 TO 16.38 Nm)

T10 215 TO 245 IN-LBS (24.30 TO 27.68 Nm)

NOTE

1. Apply sealant (C-308) around jam nut (70), lock (69), and exposed threads of clevis (68).

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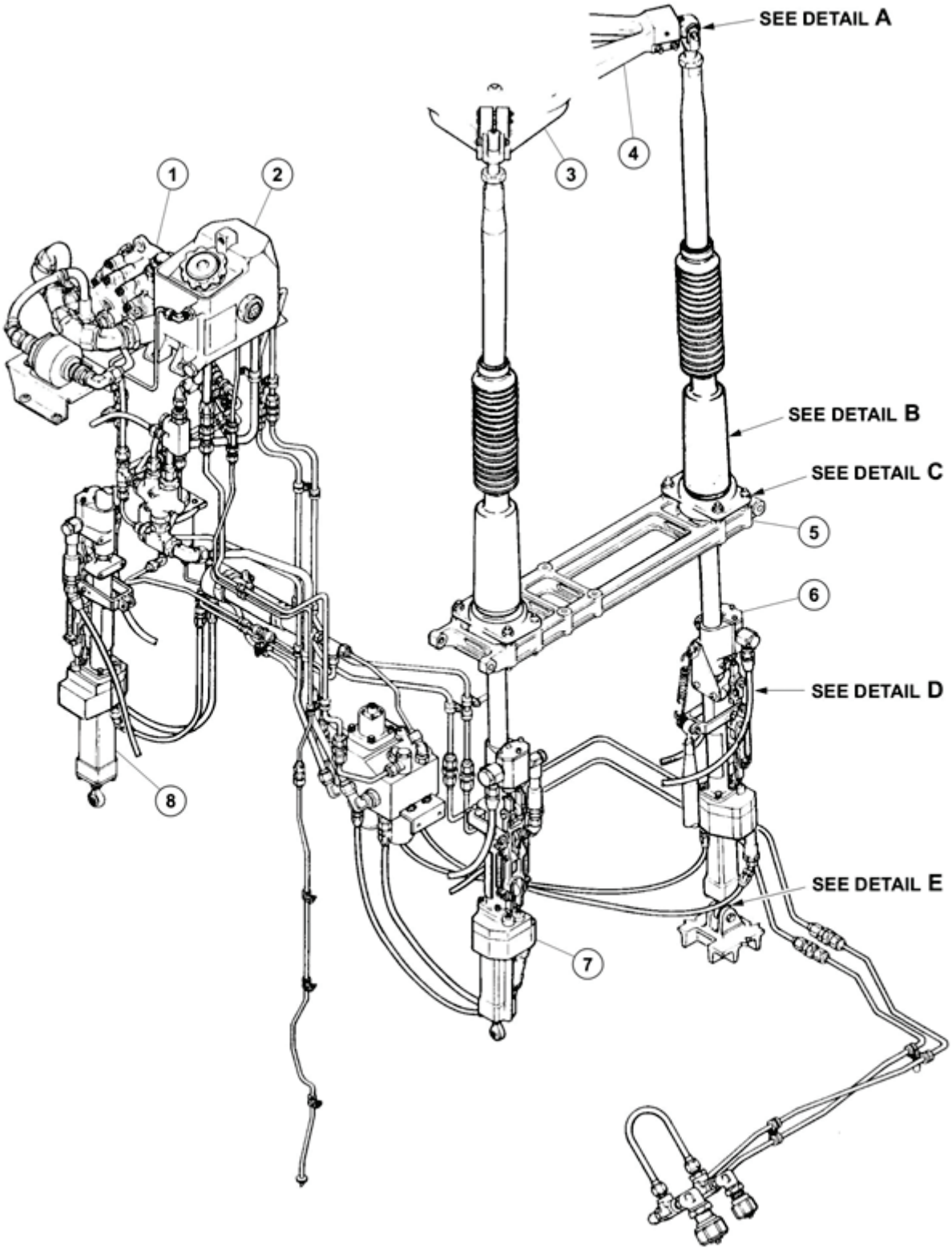
Figure 1. Collective Flight Control Cylinder (Sheet 4 of 4)

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Collective Flight Control Cylinder - Inspection

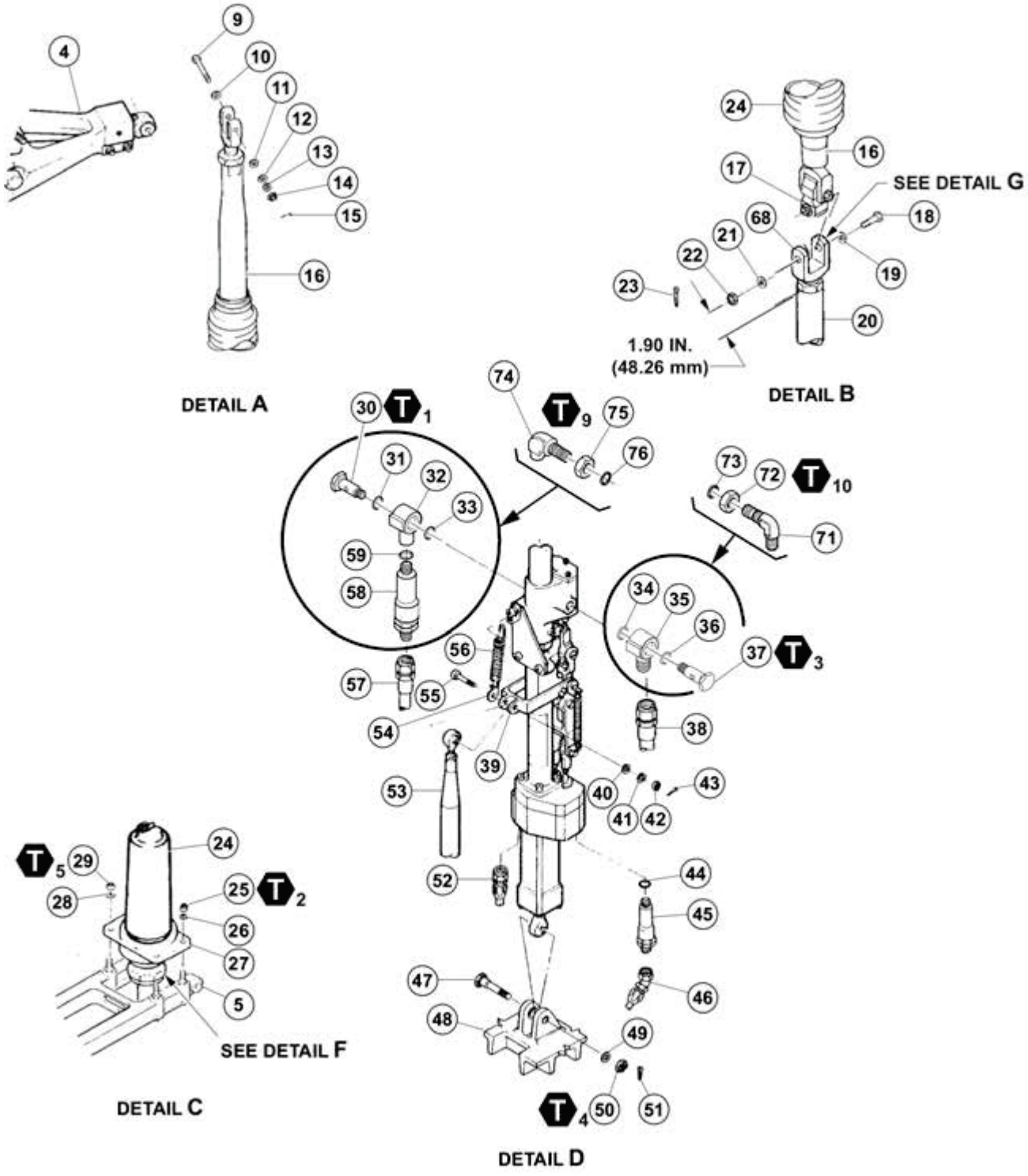
Procedure

1. Examine the collective flight control cylinder (6, Figure 1) for damage and signs of leakage.
2. Examine the exposed portion of pistons for wear and corrosion.
3. Examine the bearings for looseness (BHT-ALL-SPM, Standard Practices Manual).



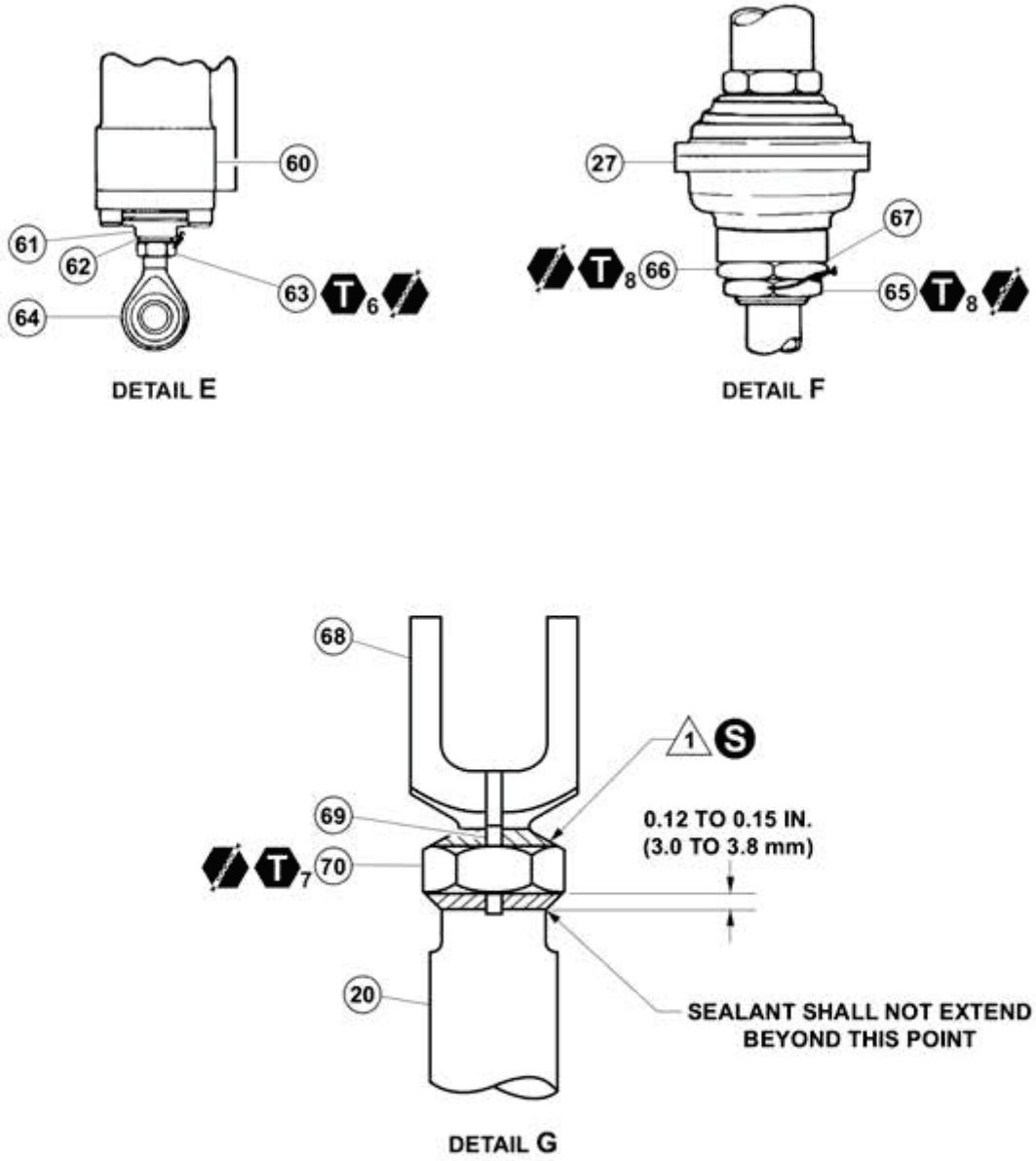
ICN-412-A-671100-B-97499-00007-A-001-01

Figure 1. Collective Flight Control Cylinder (Sheet 1 of 4)



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Figure 1. Collective Flight Control Cylinder (Sheet 2 of 4)



ICN-412-A-671100-B-97499-00009-A-001-01

Figure 1. Collective Flight Control Cylinder (Sheet 3 of 4)

1. Pump
2. Reservoir
3. Swashplate horn
4. Collective lever
5. Support
6. Collective flight control cylinder
7. Cyclic flight control cylinder
8. Cyclic flight control cylinder
9. Bolt
10. Washer (AN960C516)
11. Washer (AN960C516)
12. Washer (AN960C516)
13. Washer (AN960C516L)
14. Nut
15. Cotter pin
16. Extension tube
17. Universal
18. Bolt
19. Washer (AN960C516)
20. Upper piston
21. Washer (AN960C516)
22. Nut
23. Cotter pin
24. Boot
25. Nut
26. Aluminum washer
27. Support bracket
28. Aluminum washer
29. Nut
30. Bolt
31. Preformed packing
32. Elbow
33. Preformed packing
34. Preformed packing
35. Elbow
36. Preformed packing
37. Bolt
38. Hose
39. Lever
40. Washer
41. Washer
42. Nut
43. Cotter pin
44. Preformed packing
45. Bypass check valve
46. Hose
47. Bolt
48. Support
49. Aluminum washer
50. Nut
51. Cotter pin
52. Hose
53. Tube
54. Spring clip
55. Bolt
56. Spring
57. Hose
58. Bypass check valve
59. Preformed packing
60. Actuator
61. Lower piston
62. Locking device
63. Nut
64. Rod end bearing
65. Nut
66. Bearing nut
67. Keyed washer
68. Clevis
69. Lock
70. Jam nut
71. Elbow
72. Nut
73. Packing
74. Elbow
75. Nut
76. Packing

LOCKWIRE (C-405)

S SEALANT (C-308)

T1 40 TO 65 IN-LBS (4.51 TO 7.34 Nm)

T2 75 TO 95 IN-LBS (8.47 TO 10.73 Nm)

T3 75 TO 125 IN-LBS (8.47 TO 14.12 Nm)

T4 95 TO 110 IN-LBS (10.73 TO 12.42 Nm)

T5 120 TO 160 IN-LBS (13.55 TO 18.07 Nm)

T6 200 TO 250 IN-LBS (22.59 TO 28.54 Nm)

T7 660 TO 780 IN-LBS (74.56 TO 88.12 Nm)

T8 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

T9 135 TO 145 IN-LBS (15.25 TO 16.38 Nm)

T10 215 TO 245 IN-LBS (24.30 TO 27.68 Nm)

NOTE

1. Apply sealant (C-308) around jam nut (70), lock (69), and exposed threads of clevis (68).

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Figure 1. Collective Flight Control Cylinder (Sheet 4 of 4)

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Collective Flight Control Cylinder - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-002).
Hydraulic fluid	C-072	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-072).
Corrosion preventive compound	C-104	AR	
Corrosion preventive compound	C-127	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

1. Cut and remove the lockwire and loosen the nut (65 and 66, Figure 1) so the support bracket (27) may be turned easily (Detail F).

2. Coat threads of the piston rod clevis (68) with corrosion preventive compound (C-104). Install the clevis (68) with the jam nut (70) and the lock (69) in the upper piston (20) to dimension of 1.90 inches (48.26 mm) between end of the upper piston (20) and center of holes in the clevis (68).



AVOID GETTING SEALANT ON PISTON ROD SURFACE OR SCRAPER AND ROD SEAL DAMAGE MAY OCCUR.

3. Torque the jam nut (70) 660 to 780 inch-pounds (74.56 to 88.12 Nm) and safety with the lockwire (C-405). Cover the jam nut (70), lock (69), and threads of the clevis (68) with the sealant (C-308) (Detail G).

4. Put the universal (17), attached to the tube (16), into the clevis (68) on the upper piston (20) and install the bolt (18), washers (19 and 21), nut (22), and the cotter pin (23) (Detail B).

5. Raise the bottom of the boot (24) to expose the universal (17) and the clevis (68). Apply the corrosion preventive compound (C-104) on the universal (17) and the clevis (68) as necessary. Apply the corrosion preventive compound (C-104) on the mount bearing and in the housing cavity. Put the lower end of the boot in groove of the support bracket (27).

6. Guide the collective flight control cylinder (6) carefully into the helicopter. Put the support bracket (27) on the studs in the support (5) (Detail C). Install the aluminum washers (26) and the torque nuts (25) 75 to 95-inch pounds (8.47 to 10.73 Nm). Install the aluminum washer (28) and the torque nut (29) 120 to 160 inch-pounds (13.55 to 18.07 Nm).

7. Pull the lower piston (61) full down and push the actuator (60) full up (Detail E). If hole in the rod end bearing (64) does not align with hole in the support (48), adjust the rod end bearing (64).



INCORRECT TORQUE TECHNIQUE CAN DESTROY TAB ON THE LOCKING DEVICE (62). THE ROD END BEARING (64) MUST BE HELD TIGHT TO PREVENT THE DAMAGE TO THE LOCKING DEVICE (62).

8. Remove the lockwire and loosen the nut (63) and turn the rod end bearing (64) to align with the bolt (47) hole.

9. Torque the nut (63) 200 to 250 inch-pounds (22.59 to 28.24 Nm) and safety to the locking device (62) with the lockwire (C-405).

10. Examine the locking device (62) to make sure that the tab is in groove of the rod end bearing (64) and that it was not damaged when the nut (63) was tightened. Replace the locking device (62) if the tab is twisted or damaged.

11. Align the collective flight control cylinder (6) into the support (48) as follows:

11.1. Move the lower piston rod (61) up and clear of the support (48). The Lower piston rod (61) must be able to be moved freely down, up, and then down, and the bolt (47) replaced without any pressure, forcing, binding, or bending of the piston rod (61) in the fore, aft, left, or right direction. One person must guide the extension tube (16) to prevent damage when conducting this step and subsequent steps.

11.2. If the cylinder (6) meets requirements of the Step 11.1., it is considered in alignment. If the cylinder (6) is aligned, accomplish the Step 11.8. through Step 11.9.. If the cylinder (6) is misaligned, accomplish the Step 11.3. through Step 11.9..

11.3. Loosen the nut (65) and the bearing nut (66) so the cylinder (6) is loose.

11.4. Move the rod end bearing (64) up approximately 0.5 inch (12.7 mm) above the support (48). Turn the lower end of the cylinder (6) in a 3.0 inch (76.2 mm) circle.

11.5. Put the rod end bearing (64) directly above the lugs of the support (48) and move the piston (61) down and temporarily install the bolt (47).

11.6. Tighten the bearing nut (66) by hand, approximately 70 inch-pounds (7.91 Nm).

11.7. Remove the bolt (47) and move the rod end bearing (64) up and out of the support (48), then move down. The rod end bearing (64) must be free of any side loading. Do the Step 11.3. through Step 11.7. again until free of side loading.

11.8. Install the bolt (47). Torque the bearing nut (66) 900 to 1100 inch-pounds (101.68 to 124.28 Nm). Torque the jam nut (65) 900 to 1100 inch-pounds (101.68 to 124.28 Nm). Safety the nuts (65 and 66) to the keyed washer (67) with the lockwire (C-405).

11.9. If the cylinder (6) still meets requirements of the Step 11.1., install the aluminum washer (49) and the nut (50) and torque 95 to 110 inch-pounds (10.73 to 12.42 Nm). Install the cotter pin (51).

NOTE

If the collective flight control cylinder (6) still cannot be installed without the preload, it is permitted to add the one washer AN960C416L (maximum) on two aft studs of the support (5) between the support bracket (27) and the support (5). Or, it is permitted to add the one washer AN960C416L (maximum) and the one washer AN960C516L (maximum) on each of the two forward studs between the support bracket (27) and the support (5) .

Use the washers (12 and 13) as required to get the proper engagement of the cotter pin (15) and the nut (14).

12. Put the extension tube (16) on the collective lever (4). Install the bolt (9), washer (10, 11, 12, and 13), and the nut (14). Install the cotter pin (15) (Detail A).

13. After the install and torque of the cylinder assembly and before the install of the boot (24), fill the mount bearing housing cavity as required with the corrosion preventive compound (C-104) or corrosion preventive compound (C-127). Attach the boot (24) on the universal (17) and mount the bearing housing, as shown in the Figure 2.

14. Attach the tube (53, Figure 1) to the lever (39) with the bolt (55), spring clip (54), washers (40 and 41), nut (42), and the cotter pin (43). Connect the spring (56) to the spring clip (54) (Detail D).

NOTE

SN: 33001-33213 or 36001-36300:

Refer to Step 15. through Step 17. for the installation of new preformed packings.

15. Lubricate the new preformed packings (31, 33, 34, and 36) with the hydraulic fluid (C-002) or hydraulic fluid (C-072). Install the preformed packings (31, 33, 34, and 36), elbows (32 and 35), and the bolts (30 and 37).

16. Torque the bolt (30) 40 to 65 inch-pounds (4.51 to 7.34 Nm).

17. Torque the bolt (37) 75 to 125 inch-pounds (8.47 to 14.12 Nm).

NOTE

SN: 36301-Subsequent:

Refer to Step 18. through Step 20. for the installation of new preformed packings.

Applicable: SN: 36301-Subsequent:

18. Lubricate the new preformed packings (73 and 76) with the hydraulic fluid (C-002) or hydraulic fluid (C-072). Install the preformed packings (73 and 76), elbows (71 and 74), and the nuts (72 and 75).

19. Torque the nut (72) 215 to 245 inch-pounds (24.30 to 27.68 Nm).

20. Torque the nut (75) 135 to 145 inch-pounds (15.25 to 16.38 Nm).

21. Lubricate the new preformed packings (44 and 59) with the hydraulic fluid (C-002) or hydraulic fluid (C-072). Install the preformed packings (44 and 59) and the bypass check valves (45 and 58), if removed. Connect the hoses (38, 46, 52, and 57). Observe the hydraulic systems No. 1 and No. 2 labels.

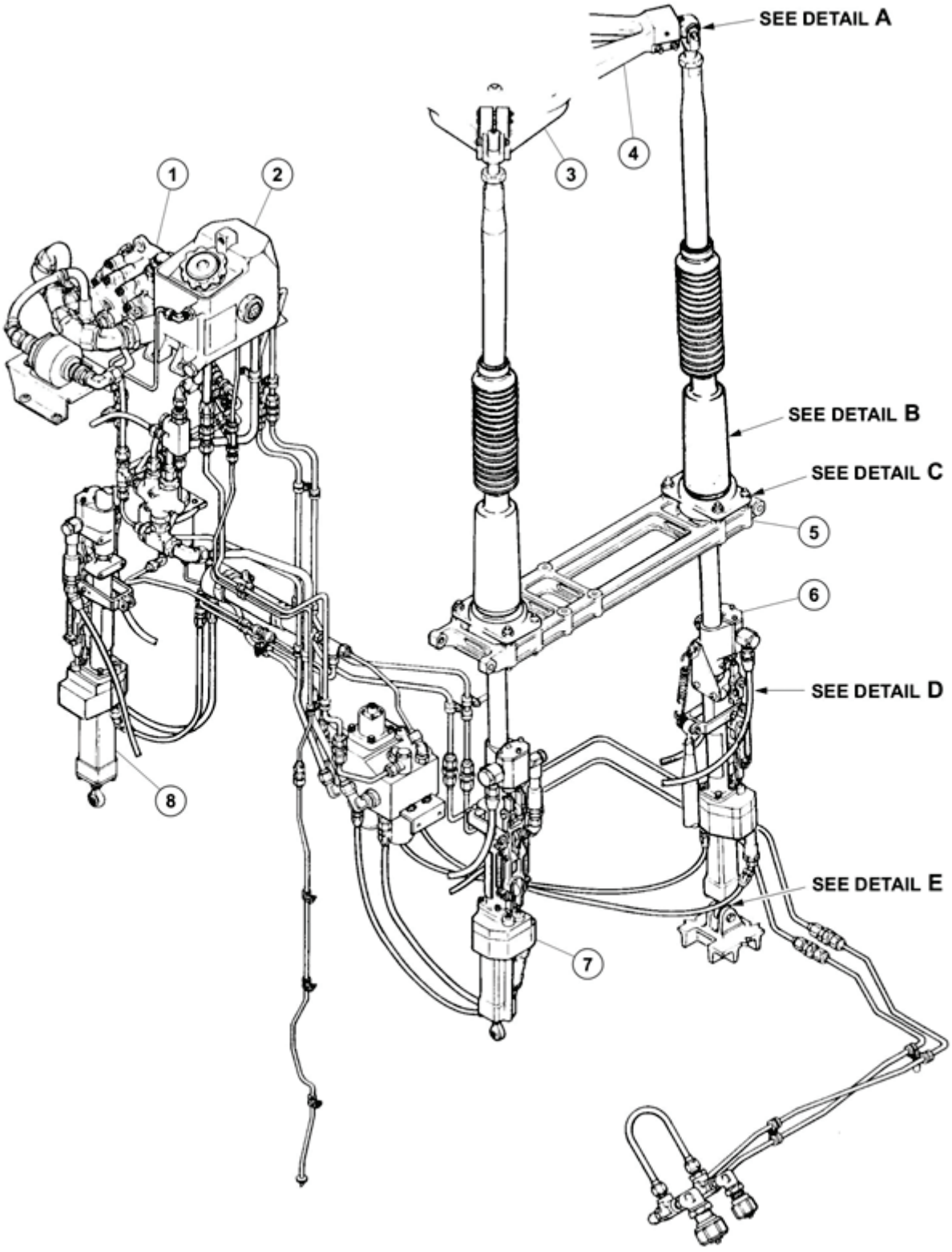
22. Bleed the hydraulic system (412-A-29-00-00-00A-009A-A / 00610).

23. Energize the hydraulic system and operate the collective actuator through 10 complete cycles. Visually examine the system for leaks.

24. Make sure that the collective rigging is within the limits (412-A-67-01-00-01A-276A-A / 01323).

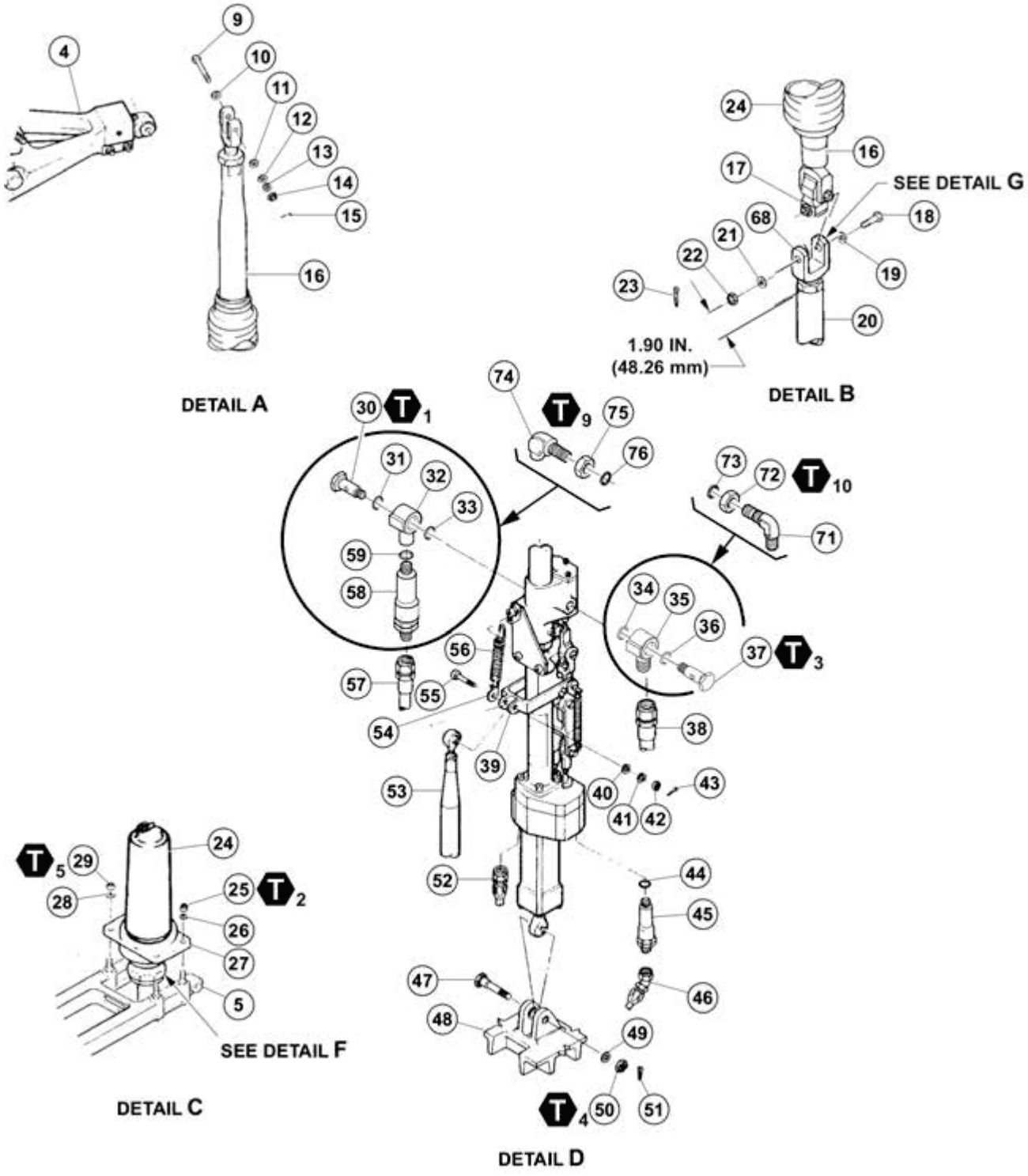
25. Install the pylon access panel removed (412-A-52-40-00-02A-720A-A / 00840).

26. Install the aft pylon fairing assembly (412-A-71-10-00-01A-720A-A / 01607).



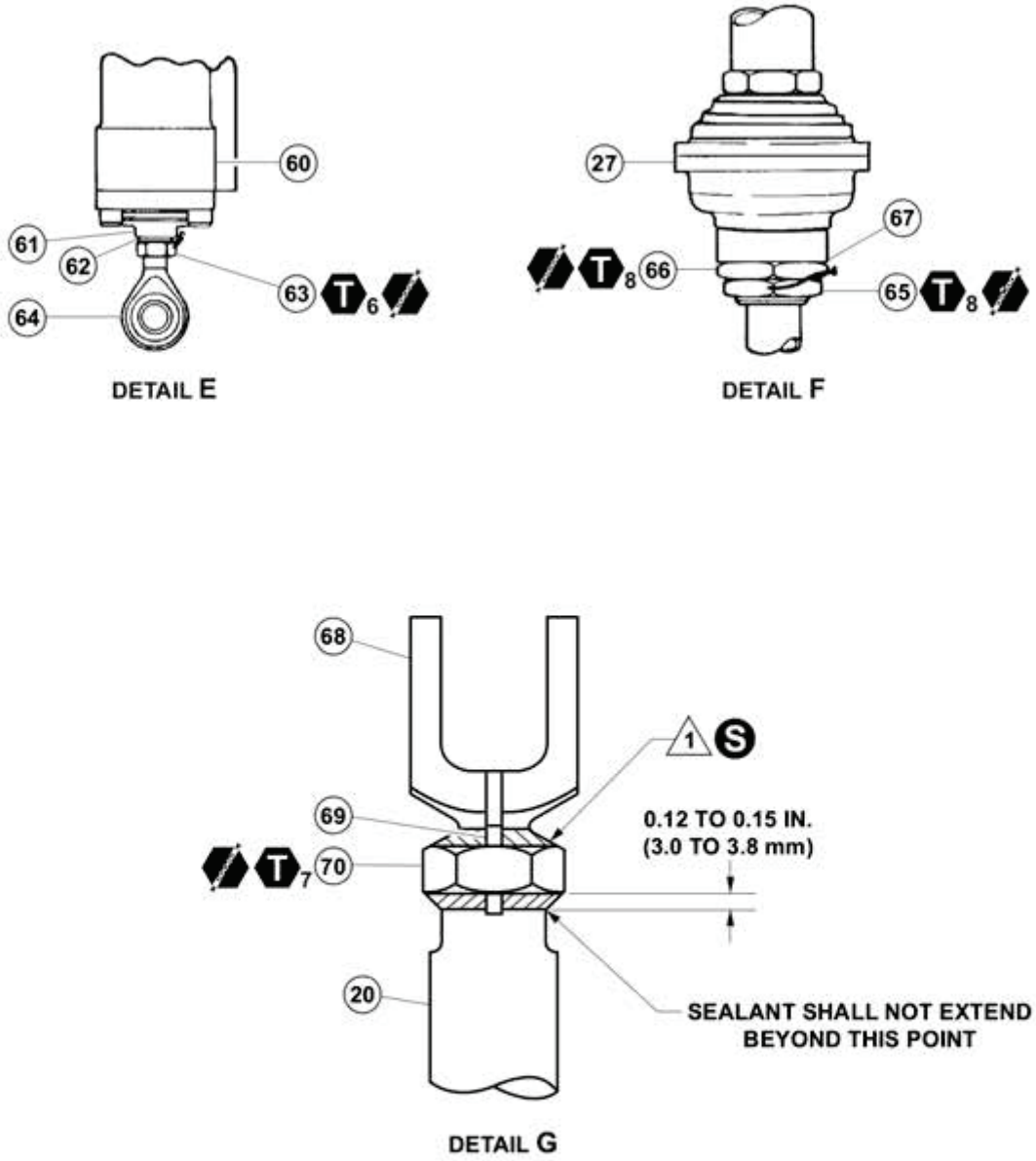
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Figure 1. Collective Flight Control Cylinder (Sheet 1 of 4)



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Figure 1. Collective Flight Control Cylinder (Sheet 2 of 4)



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Figure 1. Collective Flight Control Cylinder (Sheet 3 of 4)

1. Pump
2. Reservoir
3. Swashplate horn
4. Collective lever
5. Support
6. Collective flight control cylinder
7. Cyclic flight control cylinder
8. Cyclic flight control cylinder
9. Bolt
10. Washer (AN960C516)
11. Washer (AN960C516)
12. Washer (AN960C516)
13. Washer (AN960C516L)
14. Nut
15. Cotter pin
16. Extension tube
17. Universal
18. Bolt
19. Washer (AN960C516)
20. Upper piston
21. Washer (AN960C516)
22. Nut
23. Cotter pin
24. Boot
25. Nut
26. Aluminum washer
27. Support bracket
28. Aluminum washer
29. Nut
30. Bolt
31. Preformed packing
32. Elbow
33. Preformed packing
34. Preformed packing
35. Elbow
36. Preformed packing
37. Bolt
38. Hose
39. Lever
40. Washer
41. Washer
42. Nut
43. Cotter pin
44. Preformed packing
45. Bypass check valve
46. Hose
47. Bolt
48. Support
49. Aluminum washer
50. Nut
51. Cotter pin
52. Hose
53. Tube
54. Spring clip
55. Bolt
56. Spring
57. Hose
58. Bypass check valve
59. Preformed packing
60. Actuator
61. Lower piston
62. Locking device
63. Nut
64. Rod end bearing
65. Nut
66. Bearing nut
67. Keyed washer
68. Clevis
69. Lock
70. Jam nut
71. Elbow
72. Nut
73. Packing
74. Elbow
75. Nut
76. Packing

LOCKWIRE (C-405)

S SEALANT (C-308)

T1 40 TO 65 IN-LBS (4.51 TO 7.34 Nm)

T2 75 TO 95 IN-LBS (8.47 TO 10.73 Nm)

T3 75 TO 125 IN-LBS (8.47 TO 14.12 Nm)

T4 95 TO 110 IN-LBS (10.73 TO 12.42 Nm)

T5 120 TO 160 IN-LBS (13.55 TO 18.07 Nm)

T6 200 TO 250 IN-LBS (22.59 TO 28.54 Nm)

T7 660 TO 780 IN-LBS (74.56 TO 88.12 Nm)

T8 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

T9 135 TO 145 IN-LBS (15.25 TO 16.38 Nm)

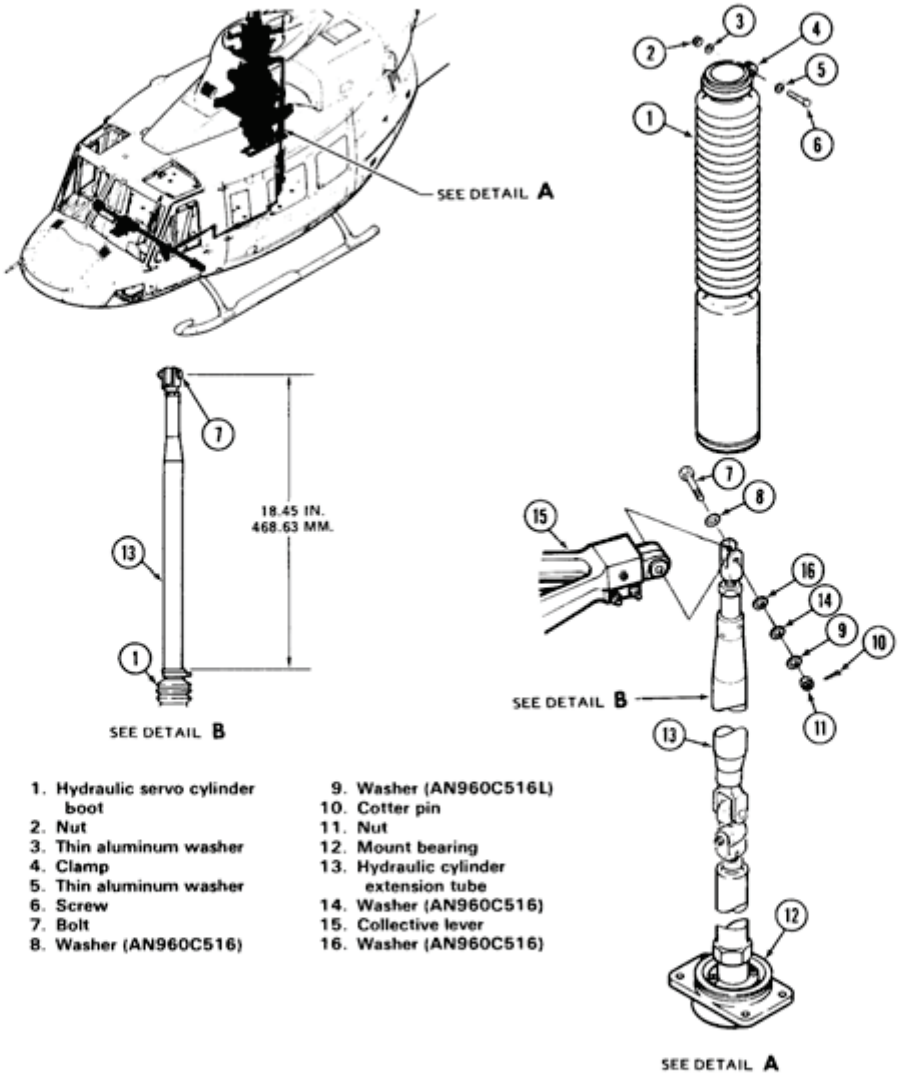
T10 215 TO 245 IN-LBS (24.30 TO 27.68 Nm)

NOTE

1. Apply sealant (C-308) around jam nut (70), lock (69), and exposed threads of clevis (68).

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Figure 1. Collective Flight Control Cylinder (Sheet 4 of 4)



1. Hydraulic servo cylinder boot
2. Nut
3. Thin aluminum washer
4. Clamp
5. Thin aluminum washer
6. Screw
7. Bolt
8. Washer (AN960C516)
9. Washer (AN960C516L)
10. Cotter pin
11. Nut
12. Mount bearing
13. Hydraulic cylinder extension tube
14. Washer (AN960C516)
15. Collective lever
16. Washer (AN960C516)

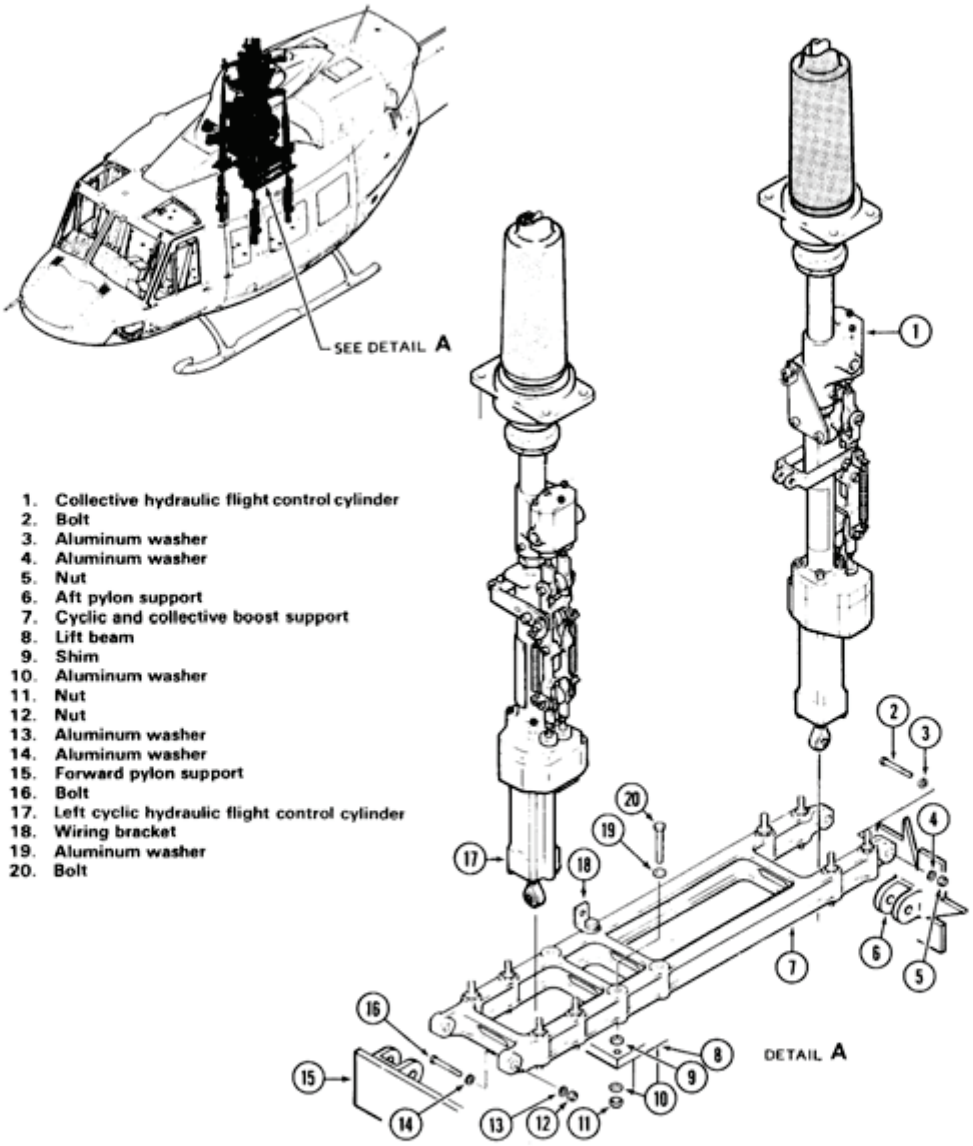
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Figure 2. Hydraulic Servo Cylinder Boot (Sheet 1 of 1)

Left Cyclic and Collective Boost Support - Removal

Procedure

1. Remove transmission from helicopter (412-A-63-21-02-00A-520A-A / 01013).
2. Remove collective hydraulic flight control cylinder (1, Figure 1) (412-A-67-11-00-06A-520A-A / 01346).
3. Remove left cyclic hydraulic flight control cylinder (17) (412-A-67-12-00-06A-520A-A / 01420).
4. Remove four nuts (11), one wiring bracket (18), four aluminum washers (19), four aluminum washers (10), four bolts (20). Remove peel shims (9).
5. Remove two nuts (5), two aluminum washers (3), two aluminum washers (4), and two bolts (2) that release the cyclic and collective boost support (7) from aft pylon support (6).
6. Remove two nuts (12), two aluminum washers (13), two aluminum washers (14), and two bolts (16) that release the cyclic and collective boost support (7) from forward pylon support (15). Remove cyclic and collective boost support (7).



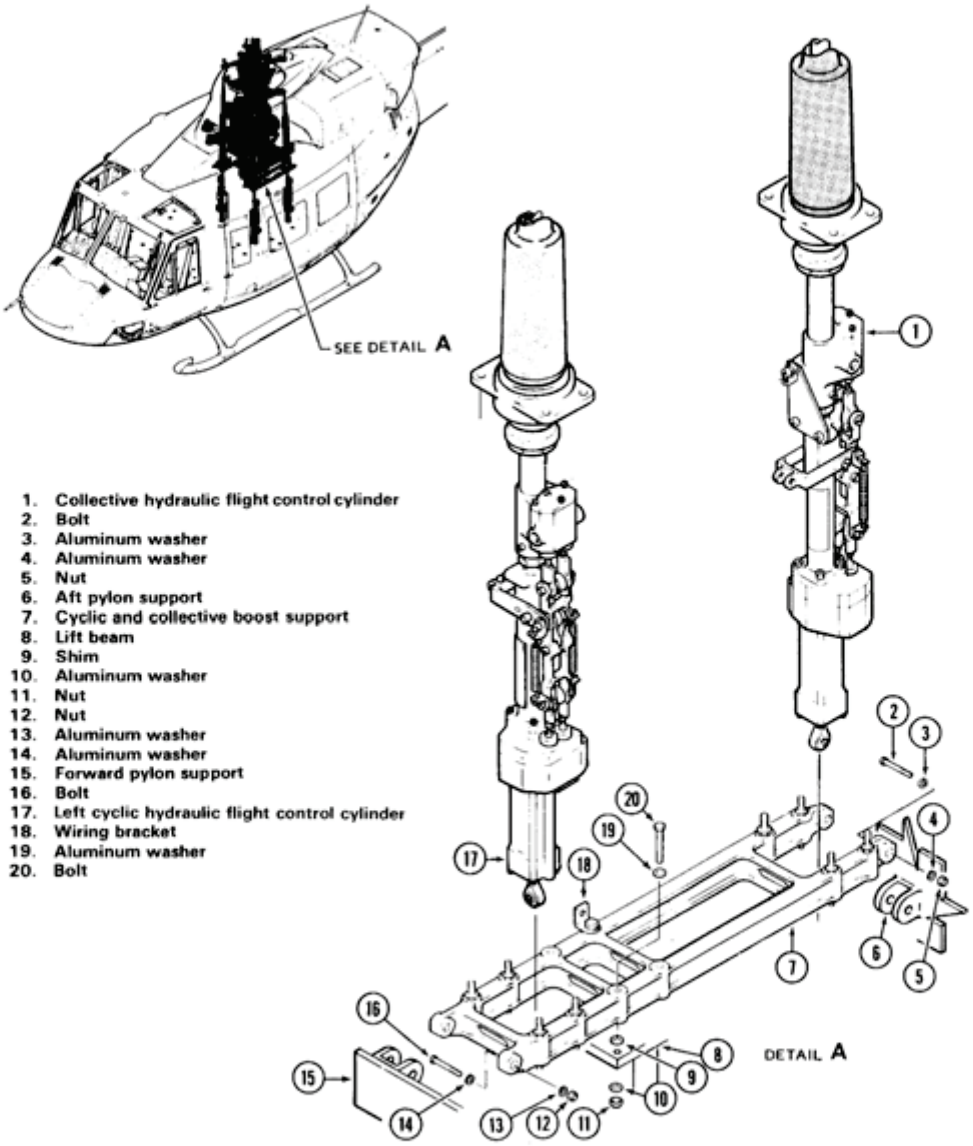
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Figure 1. Left Cyclic and Collective Boost Support (Sheet 1 of 1)

Left Cyclic and Collective Boost Support - Inspection

Procedure

1. Examine cyclic and collective boost support (7, Figure 1) (BHT-412-CR&O).
2. Examine bushings for correct installation in support.



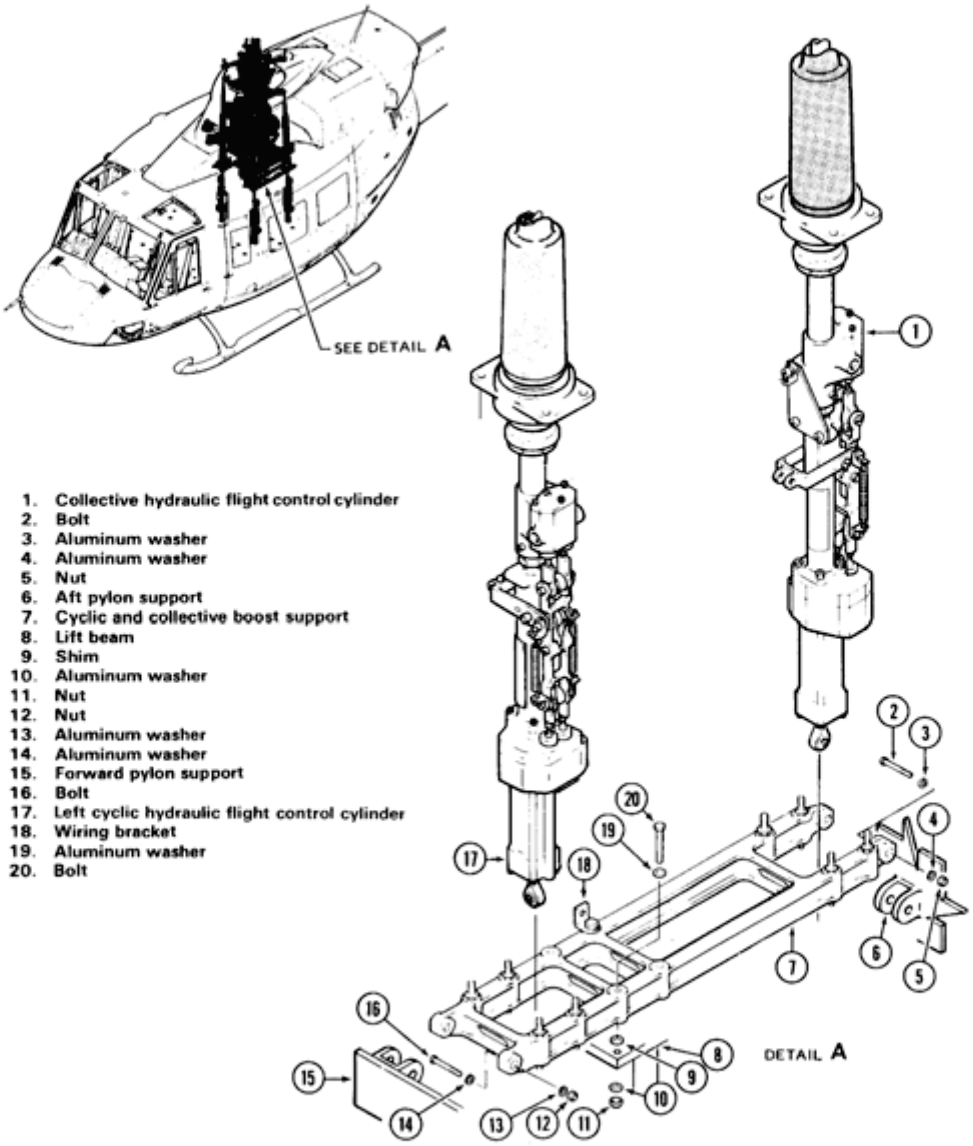
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Figure 1. Left Cyclic and Collective Boost Support (Sheet 1 of 1)

Left Cyclic and Collective Boost Support - Installation

Procedure

1. Put cyclic and collective boost support (7, Figure 1) into position in helicopter.
2. Install bolts (2), aluminum washers (3), aluminum washers (4), and nuts (5).
3. Install bolts (16), aluminum washers (14), aluminum washers (13), and nuts (12).
4. Measure gap between cyclic and collective boost support (7) and lift beam (8). Peel shims (9) to get a maximum gap of 0.004 inch (0.101 mm). Install bolts (20), wiring bracket (18), aluminum washers (19), aluminum washers (10), and nuts (11).
5. Install left cyclic hydraulic flight control cylinder (17) (412-A-67-12-00-06A-720A-A / 01422).
6. Install collective hydraulic flight control cylinder (1) (412-A-67-11-00-06A-720A-A / 01348).
7. Install transmission (412-A-63-21-02-00A-720B-A / 01016).



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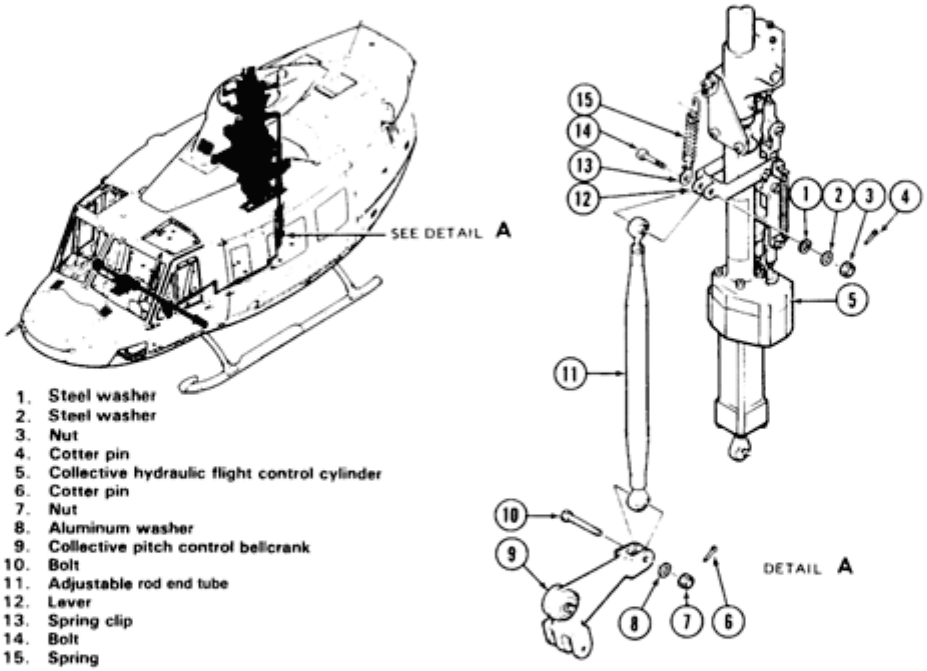
Figure 1. Left Cyclic and Collective Boost Support (Sheet 1 of 1)

Adjustable Rod End Tube - Removal

Procedure

1. Remove the cotter pin (6, Figure 1), nut (7), aluminium washer (8), and bolt (10) to remove the adjustable rod end tube (11) from the collective pitch control bellcrank (9).

2. Disconnect spring (15) from spring clip (13). Remove cotter pin (4), nut (3), steel washers (2 and 1), and bolt (14), and disconnect adjustable rod end tube (11). Keep the bolt (14), spring clip (13), and spring (15).



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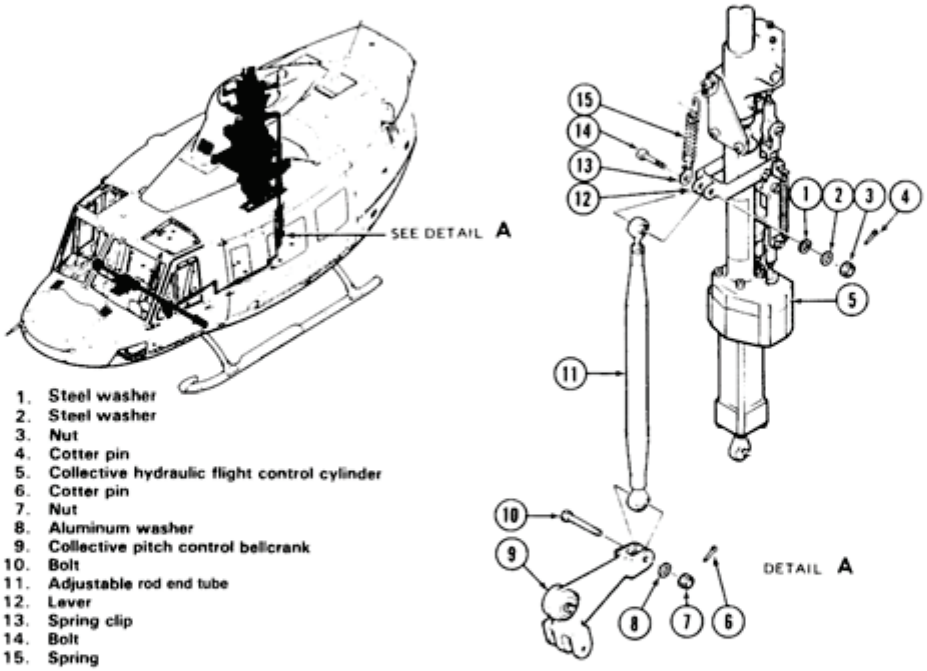
Figure 1. Adjustable Rod End Tube (Sheet 1 of 1)

Adjustable Rod End Tube - Inspection

Procedure

1. Examine adjustable rod end tube (11, Figure 1) (BHT-412-CR&O).

2. Examine rod end bearings in adjustable rod end tube (11) for correct installation and wear (BHT-ALL-SPM, Standard Practices Manual).



ICN-412-A-671100-B-97499-00011-A-001-01

Figure 1. Adjustable Rod End Tube (Sheet 1 of 1)

Adjustable Rod End Tube - Installation

Procedure

1. Place spring clip (13, Figure 1) and adjustable rod end tube (11) into position on lever (12) and secure with bolt (14), washer (1 and 2), nut (3), and cotter pin (4).

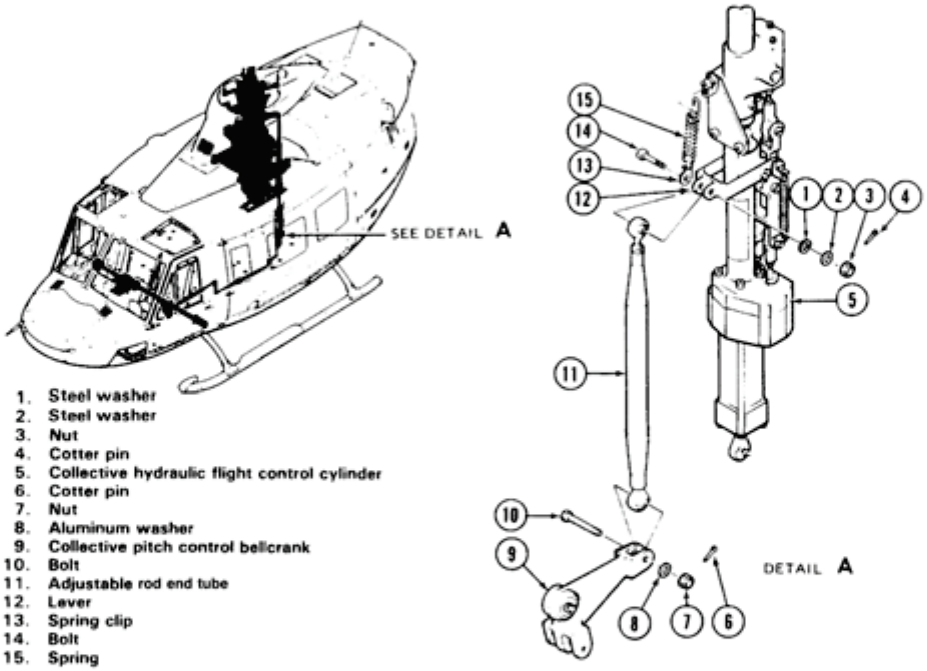
2. Place adjustable rod end tube (11) into position in bellcrank (9) and secure with bolt (10), washer (8), nut (7), and cotter pin (6).

NOTE

Exposed thread at upper rod end bearing shall not exceed 1.0 in. (25.4 mm).

3. Move collective controls through full travel and check for binding and/or obstruction of control movement.

4. Verify collective control rigging within limits (412-A-67-01-00-01A-276A-A / 01323).



ICN-412-A-671100-B-97499-00011-A-001-01

Figure 1. Adjustable Rod End Tube (Sheet 1 of 1)

Pitch Control Bellcrank - Removal

Procedure

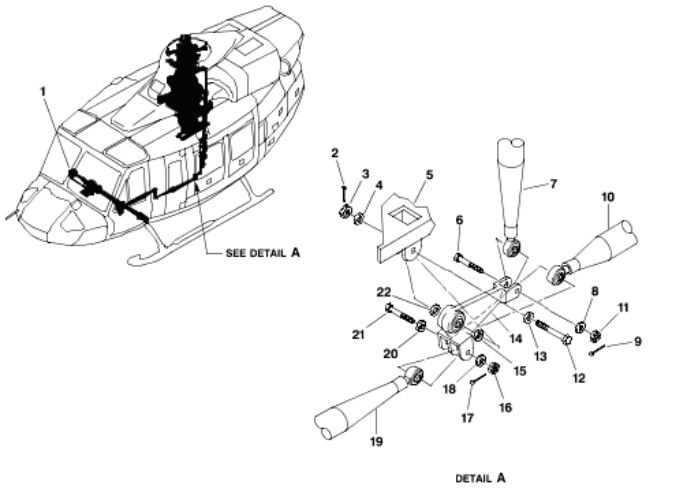
NOTE

Access to bellcrank is provided through large access in bottom of helicopter.

1. Remove adjustable rod end tube (7, Figure 1) from collective pitch control bellcrank (14) by removing cotter pin (9), nut (11), aluminum washer (8) and bolt (6).

2. Remove droop compensator control tube (10) and adjustable control tube (19) from bellcrank (14) by removing cotter pin (17), nut (16), aluminum washers (18 and 20) and bolt (21).

3. Remove bellcrank (14) from support (5) by removing cotter pin (2), nut (3), aluminum washers (4 and 13) and bolt (12). Remove bellcrank (14) and washers (22 and 15).



1. Pilot collective stick
2. Cotter pin
3. Nut
4. Aluminum washer
5. Support
6. Bolt
7. Adjustable rod end tube
8. Aluminum washer
9. Cotter pin
10. Droop compensator control tube
11. Nut
12. Bolt
13. Aluminum washer
14. Collective pitch control bellcrank
15. Aluminum washer
16. Nut
17. Cotter pin
18. Aluminum washer
19. Adjustable collective control tube
20. Aluminum washer
21. Bolt
22. Aluminum washer

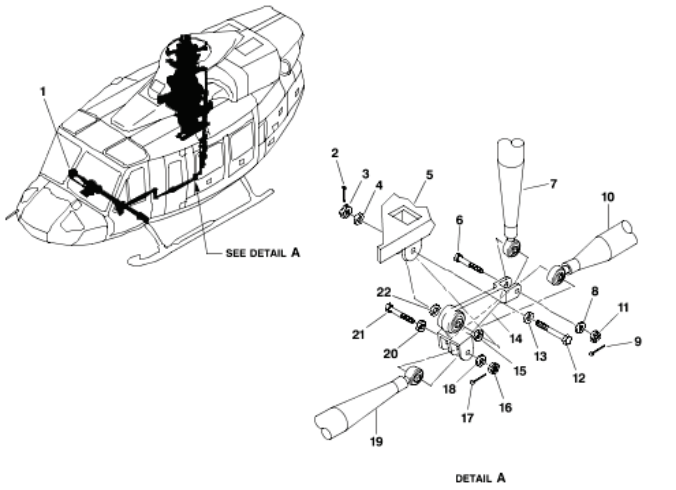
ICN-412-A-671100-B-97499-00012-A-001-01

Figure 1. Collective Control Bellcrank (Sheet 1 of 1)

Pitch Control Bellcrank - Inspection

Procedure

1. Inspect bellcrank (14, Figure 1) for mechanical and corrosion damage (BHT-412-CR&O).



1. Pilot collective stick
2. Cotter pin
3. Nut
4. Aluminum washer
5. Support
6. Bolt
7. Adjustable rod end tube
8. Aluminum washer
9. Cotter pin
10. Droop compensator control tube
11. Nut
12. Bolt
13. Aluminum washer
14. Collective pitch control bellcrank
15. Aluminum washer
16. Nut
17. Cotter pin
18. Aluminum washer
19. Adjustable collective control tube
20. Aluminum washer
21. Bolt
22. Aluminum washer

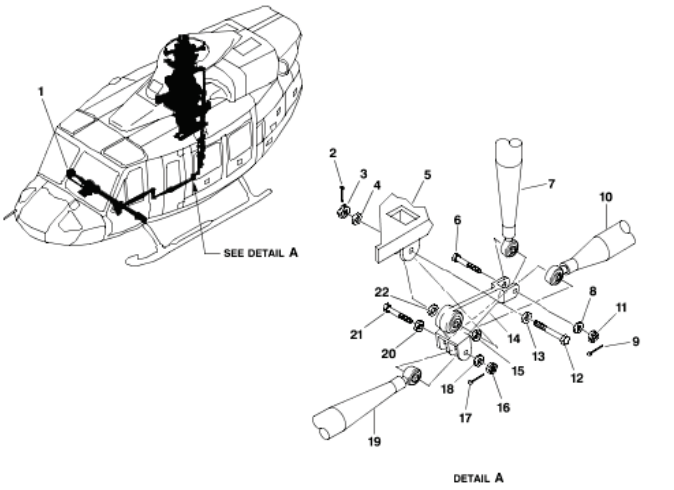
ICN-412-A-671100-B-97499-00012-A-001-01

Figure 1. Collective Control Bellcrank (Sheet 1 of 1)

Pitch Control Bellcrank - Installation

Procedure

1. Place bellcrank (14, Figure 1) in position in support (5). Use washer (15 and 22) to a maximum of four to obtain proper fit of bellcrank (14) in support (5). Install bolt (12), washer (13 and 4), nut (3) and cotter pin (2).
2. Place tube (19) and tube (10) into bellcrank (14). Install bolt (21), washers (20 and 18), nut (16), and cotter pin (17).
3. Place tube (7) into bellcrank (14). Install bolt (6), washer (8), nut (11), and cotter pin (9).
4. Move pilot collective stick (1) through full travel and check for binding and/or obstruction of control movement.
5. Verify collective rigging within limits (412-A-67-01-00-01A-276A-A / 01323).



1. Pilot collective stick
2. Cotter pin
3. Nut
4. Aluminum washer
5. Support
6. Bolt
7. Adjustable rod end tube
8. Aluminum washer
9. Cotter pin
10. Droop compensator control tube
11. Nut
12. Bolt
13. Aluminum washer
14. Collective pitch control bellcrank
15. Aluminum washer
16. Nut
17. Cotter pin
18. Aluminum washer
19. Adjustable collective control tube
20. Aluminum washer
21. Bolt
22. Aluminum washer

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Figure 1. Collective Control Bellcrank (Sheet 1 of 1)

Collective Force Gradient - Removal

Procedure

NOTE

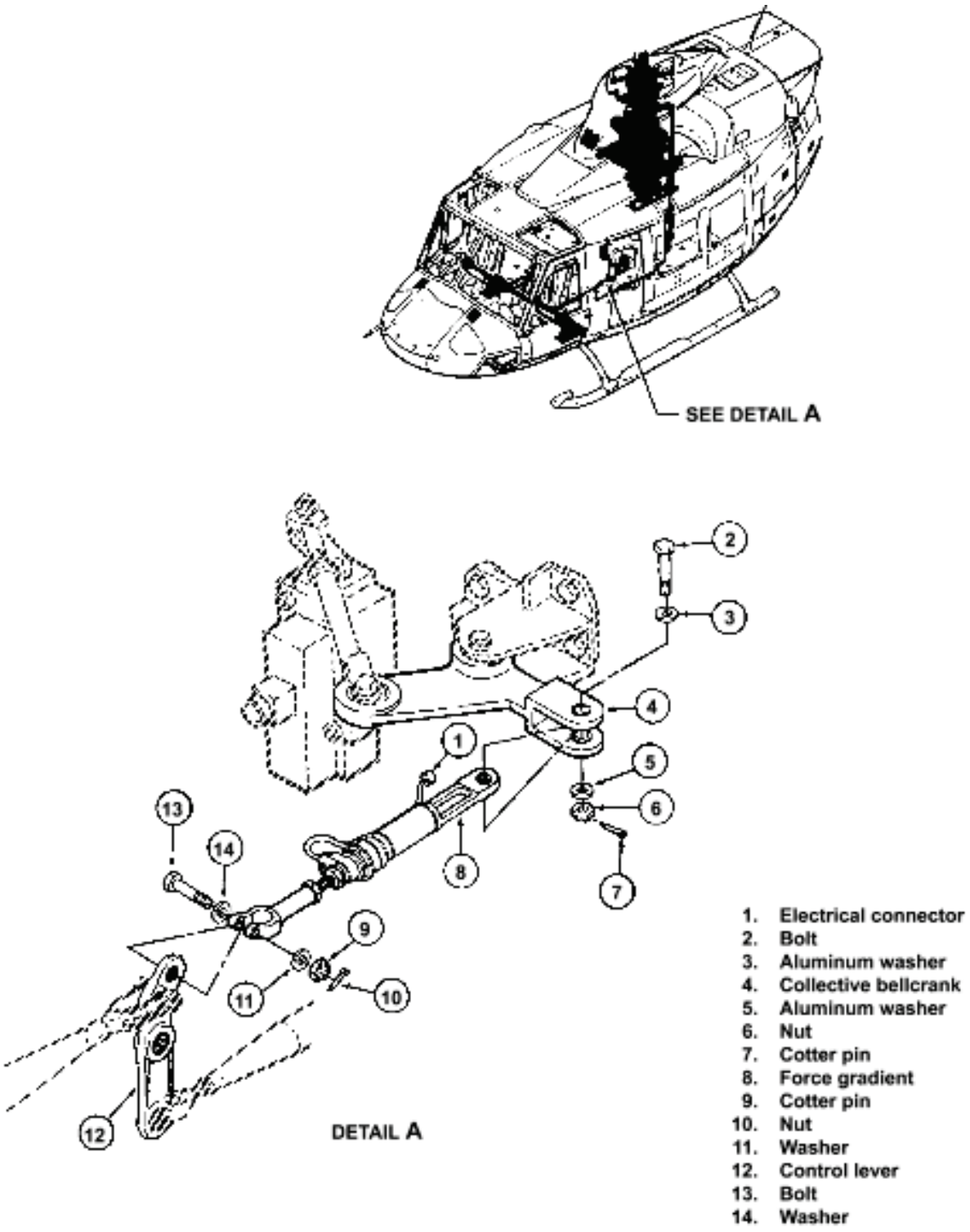
Applicable to helicopters with 4-axis Flight Director kit installed.

1. Remove access panel from cabin floor forward of pylon.

2. Disconnect electrical connector (1, Figure 1).

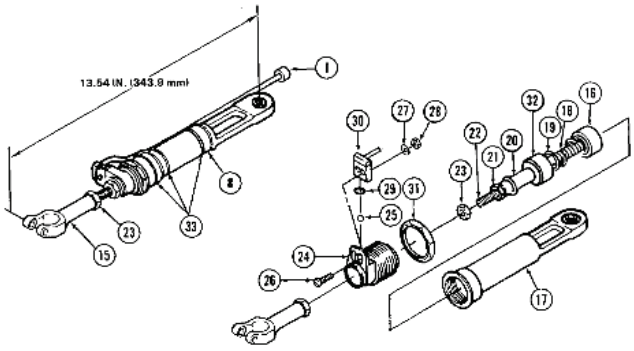
3. Remove force gradient (8) from collective bellcrank (4) by removing cotter pin (7), nut (6), washers (5 and 3), and bolt (2).

4. Remove force gradient (8) from control lever (12) by removing cotter pin (10), nut (9), washers (11 and 14), and bolt (13).



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Figure 1. Collective Force Gradient (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 2)



15. Clevis	22. Shaft	28. Nut
16. Guide	23. Jam nut	29. Packing
17. Cylinder	24. Cap	30. Switch and cable assembly
18. Spring	25. Ball	31. Jam nut
19. Spring	26. Screw	32. Spring assembly
20. Spacer	27. Washer	33. Nylon braid
21. Jam nut		

ICN-412-A-671100-B-97499-00014-A-001-01

Figure 1. Collective Force Gradient (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 2 of 2)

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Collective Force Gradient - Inspection

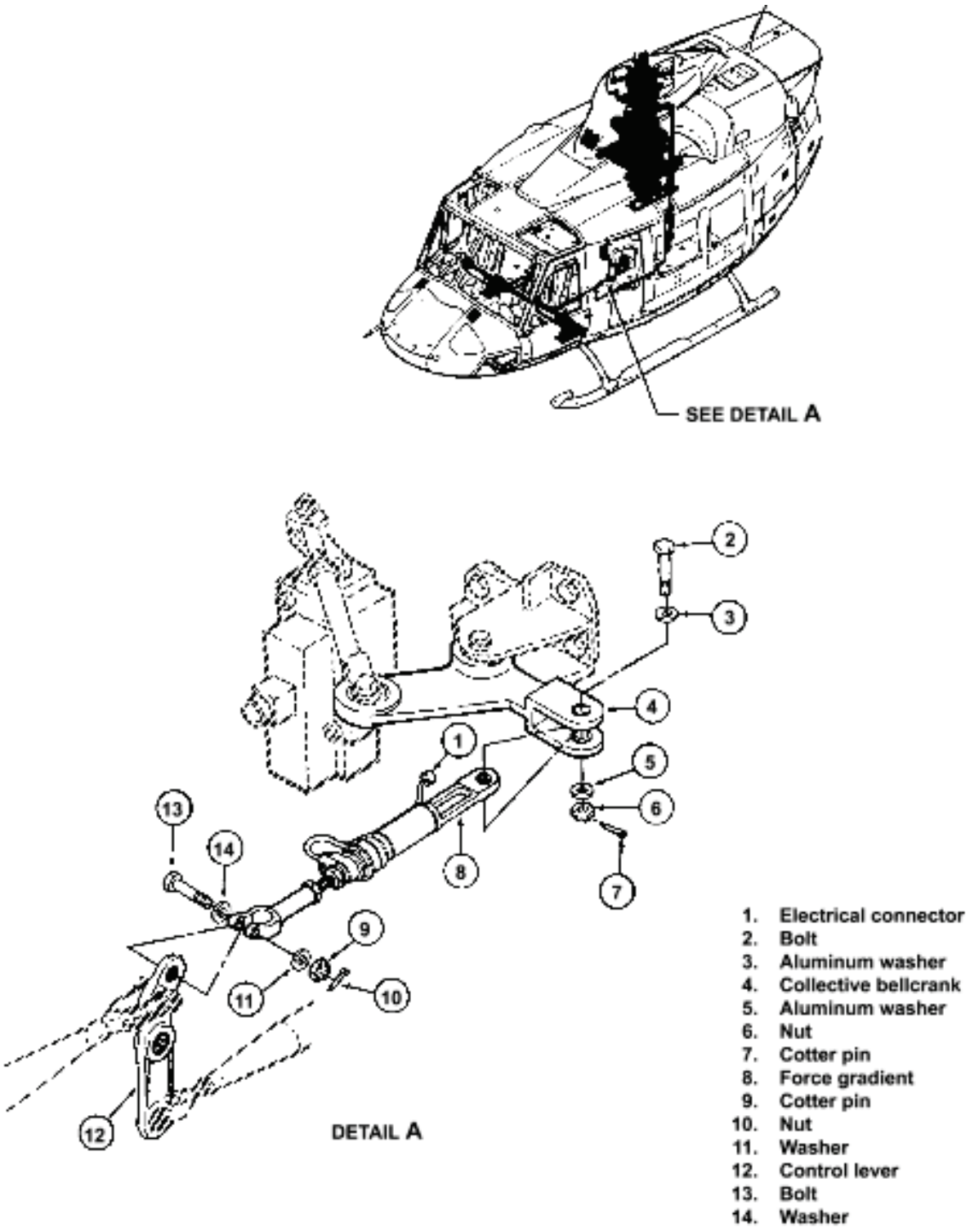
Procedure

NOTE

Applicable to helicopters with 4-axis Flight Director kit installed.

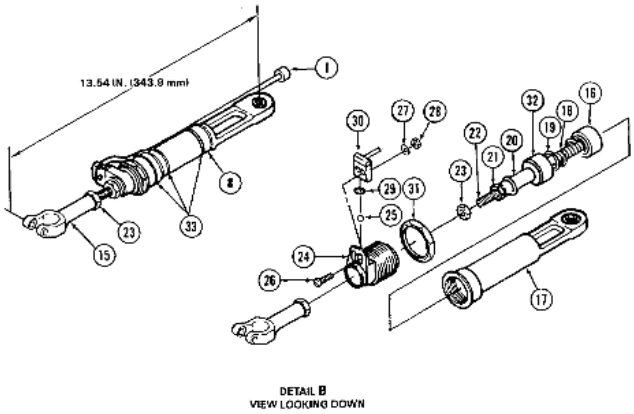
1. Inspect force gradient (8, Figure 1) for corrosion and mechanical damage (BHT-412-CR&O). Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is not acceptable.

2. Inspect bearing for secure installation in housing and for wear (BHT-ALL-SPM, Standard Practices Manual).



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Figure 1. Collective Force Gradient (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 2)



15. Clevis	22. Shaft	28. Nut
16. Guide	23. Jam nut	29. Packing
17. Cylinder	24. Cap	30. Switch and cable assembly
18. Spring	25. Ball	31. Jam nut
19. Spring	26. Screw	32. Spring assembly
20. Spacer	27. Washer	33. Nylon braid
21. Jam nut		

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Figure 1. Collective Force Gradient (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 2 of 2)

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Collective Force Gradient - Repair

Procedure

NOTE

Applicable to helicopters with 4-axis Flight Director kit installed.

1. Repair of force gradient (8) consists of replacement of unserviceable components and adjustment of spring preload in accordance with 412-A-67-11-00-10A-720A-A / 01361, step 1.

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Collective Force Gradient - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure



ENSURE THAT FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO INSTALLATION ON ROTARY ACTUATOR.

NOTE

The collective force gradient (8, Figure 1) is similar in appearance to the force gradients (8) used in the cyclic controls but has a different part number. Ensure that correct part number force gradient (8) is installed for the collective controls system.

NOTE

Applicable to helicopters with 4-axis Flight Director kit installed.

1. If not previously accomplished, adjust spring preload on force gradient (8) to 40 to 41 pounds (177.9 to 182.4 N) as follows:

1.1. Loosen jam nut (23) and remove clevis (15) from shaft (22).

1.2. Cut nylon braids (33) holding switch and cable assembly (30) to force gradient (8).

1.3. Remove nut (28), washer (27), screw (26), packing (29), and ball (25) from cap (24), and switch and cable assembly (30).

1.4. Cut lockwire and loosen jam nut (31), remove cap (24) from cylinder (17).

1.5. Remove spring assembly (32) from cylinder (17).

1.6. Loosen jam nut (21) while holding shaft (22) at wrench flat. Adjust spring preload to 40 to 41 pounds (177.9 to 182.4 N) by turning spacer (20), (using wrench flat) clockwise against spring assembly (32). Tighten jam nut (21) against spacer (20).

1.7. Position spring assembly (32) in cylinder (17) and install cap (24) until all end play is eliminated. Torque jam nut (31) to 100 pounds (11.298 Nm) and lockwire (C-405) to prevent turning in either direction. After assembly verify spring preload of 40 to 41 pounds (177.9 to 182.4 N).

1.8. Install ball (25), packing (29), and switch and cable assembly (30) on cap (24).

1.9. Install screws (26), washers (27), nuts (28) in cap (24) and switch and cable assembly (30). Do not tighten screws at this time.

1.10. Install clevis (15) on shaft (22). Adjust length of force gradient (8) as shown in . Tighten jam nut (23) to clevis (15).

1.11. Push and hold switch and cable assembly (30), and ball (25) against cap (24). Hold this position to initially adjust switch and cable assembly (30).

1.12. Adjust switch to actuate within ±0.050 inch (1.27 mm) linear shaft travel of spring assembly (32) and secure screws (26).

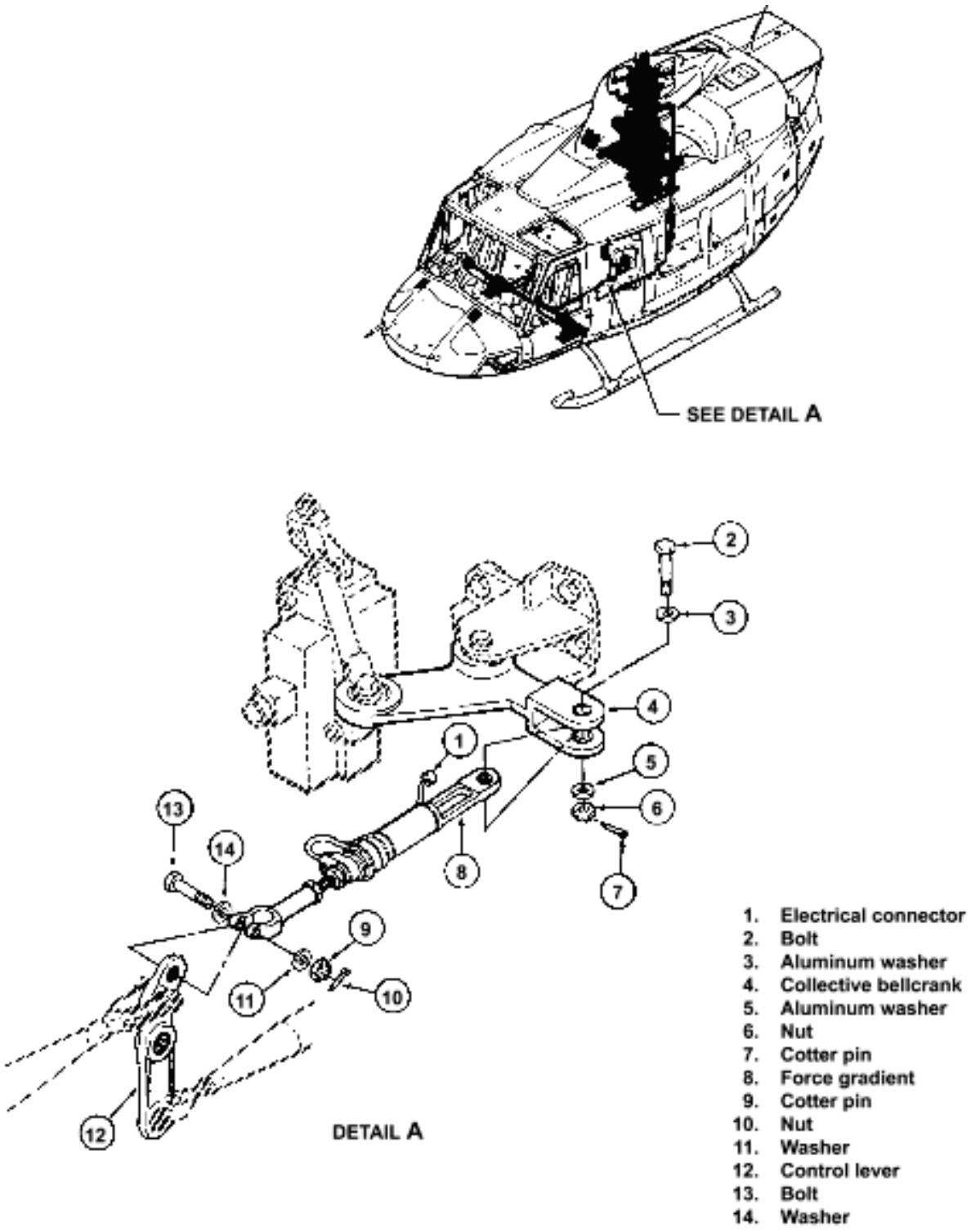
1.13. Install nylon braid (33), holding switch, and cable assembly (30) to force gradient (8).

2. Place force gradient (8) into position on bellcrank (3). Install bolt (2), steel washers (3 and 5), nut (6), and cotter pin (7).

3. Place force gradient (8) into position on control lever (12). Install bolt (13), steel washers (14 and 11), nut (10), and cotter pin (9).

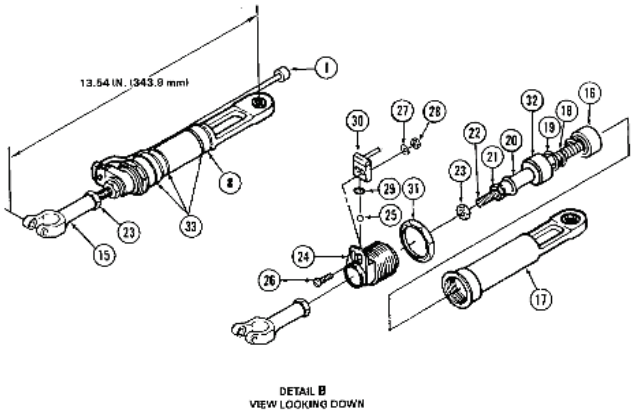
4. Connect electrical connector (1).

5. Install access panel in cabin floor forward of pylon.



ICN-412-A-671100-B-97499-00013-A-001-01

Figure 1. Collective Force Gradient (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 2)



15. Clevis
16. Guide
17. Cylinder
18. Spring
19. Spring
20. Spacer
21. Jam nut
22. Shaft
23. Jam nut
24. Cap
25. Ball
26. Screw
27. Washer
28. Nut
29. Packing
30. Switch and cable assembly
31. Jam nut
32. Spring assembly
33. Nylon braid

ICN-412-A-671100-B-97499-00014-A-001-01

Figure 1. Collective Force Gradient (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 2 of 2)

Collective Bellcrank - Removal

Procedure

NOTE

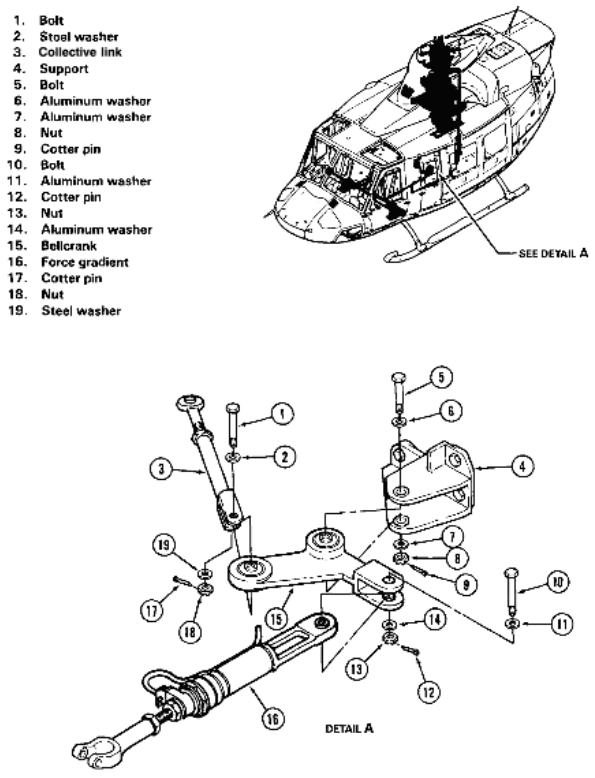
Applicable to helicopters with 4-axis AFCS/Flight Director Kit installed.

1. Remove access panel from cabin floor forward of pylon.

2. Remove collective link (3, Figure 1) from collective bellcrank (15) by removing cotter pin (17), nut (18), washers (2 and 19), and bolt (1).

3. Remove force gradient (16) from bellcrank (15) by removing cotter pin (12), nuts (13), washers (11 and 14), and bolt (10).

4. Remove bellcrank (15) from support (4) by removing cotter pin (9), nut (8), washers (6 and 7), and bolt (5).



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Figure 1. Collective Bellcrank (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 1)

Collective Bellcrank - Inspection

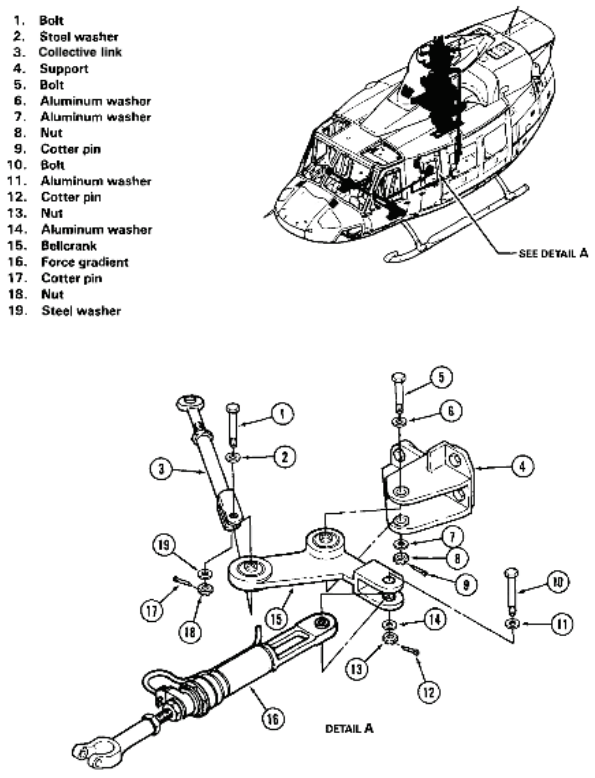
Procedure

NOTE

Applicable to helicopters with 4-axis AFCS/Flight Director Kit installed.

1. Inspect bellcrank (15, Figure 1) for mechanical and corrosion damage in accordance with BHT-412-CR&O.

2. Inspect bearings in bellcrank (15) for secure installation in bellcrank and for wear (BHT-ALL-SPM, Standard Practices Manual).



ICN-412-A-671100-B-97499-00015-A-001-01

Figure 1. Collective Bellcrank (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 1)

Collective Bellcrank - Installation

Procedure

NOTE

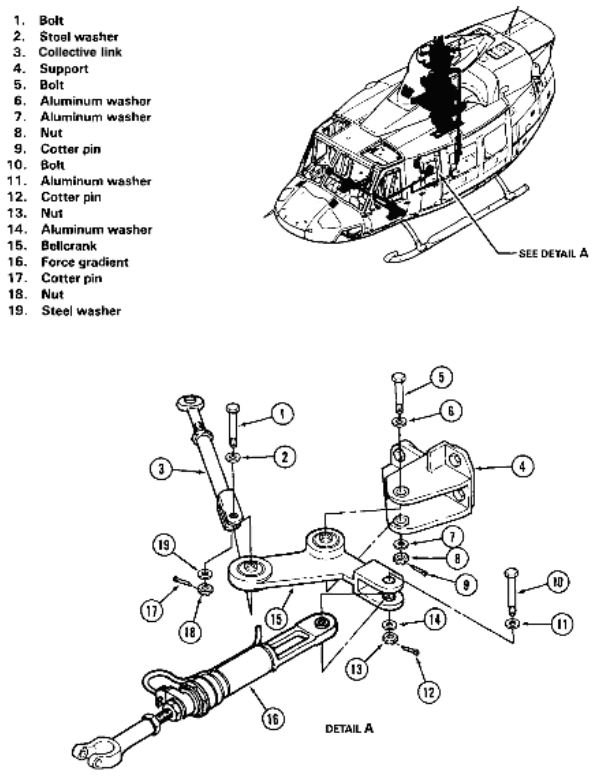
Applicable to helicopters with 4-axis AFCS/Flight Director Kit installed.

1. Place bellcrank (15, Figure 1) into position on support (4). Install bolt (5), aluminum washers (6 and 7), nut (8), and cotter pin (9).

2. Position force gradient (16) on bellcrank (15) and install bolt (10), aluminum washers (11 and 14), nut (13), and cotter pin (12).

3. Position collective link (3) on bellcrank (15) and install bolt (1), steel washers (2 and 19), nut (18), and cotter pin (17).

4. Install access panel in cabin floor forward of pylon.



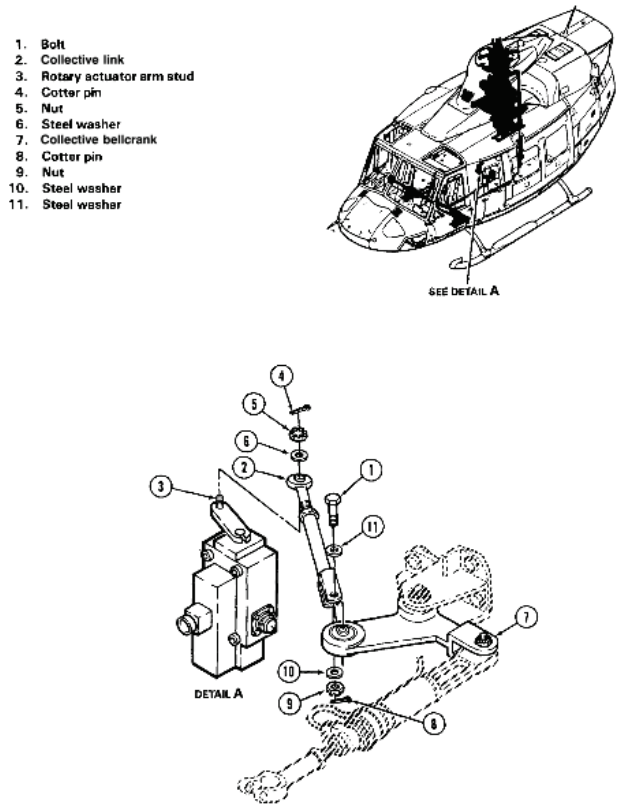
ICN-412-A-671100-B-97499-00015-A-001-01

Figure 1. Collective Bellcrank (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 1)

Collective Link - Removal

Procedure

1. Remove access panel from cabin floor forward of pylon.
2. Remove collective link (2, Figure 1) from collective bellcrank (7) by removing cotter pin (8), nut (9), washers (10 and 11), and bolt (1).
3. Remove collective link (2) from rotary actuator arm stud (3) by removing cotter pin (4), nut (5), and washer (6).



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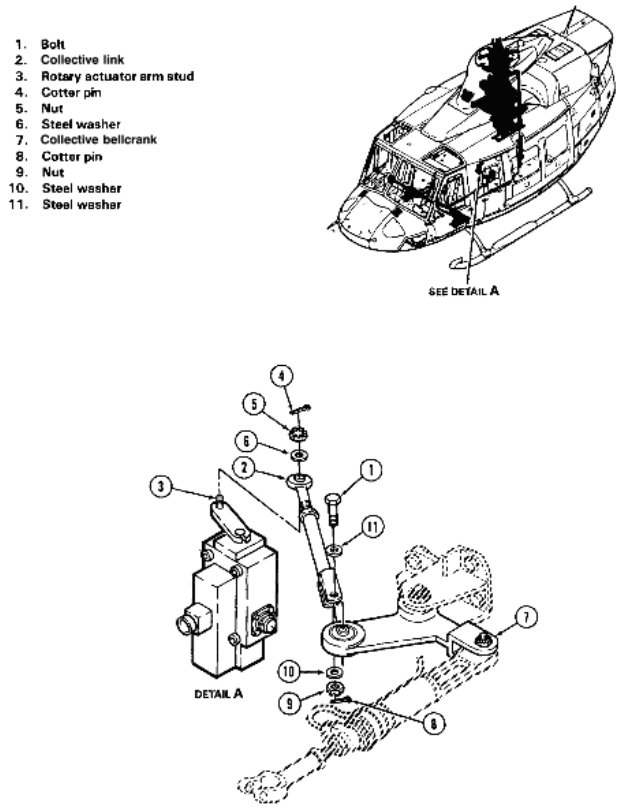
Figure 1. Collective Link (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 1)

Collective Link - Inspection

Procedure

1. Inspect link (2, Figure 1) for mechanical and corrosion damage in accordance with BHT-412-CR&O.

2. Inspect bearing in link (2) for security and for wear (BHT-ALL-SPM, Standard Practices Manual).



ICN-412-A-671100-B-97499-00016-A-001-01

Figure 1. Collective Link (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 1)

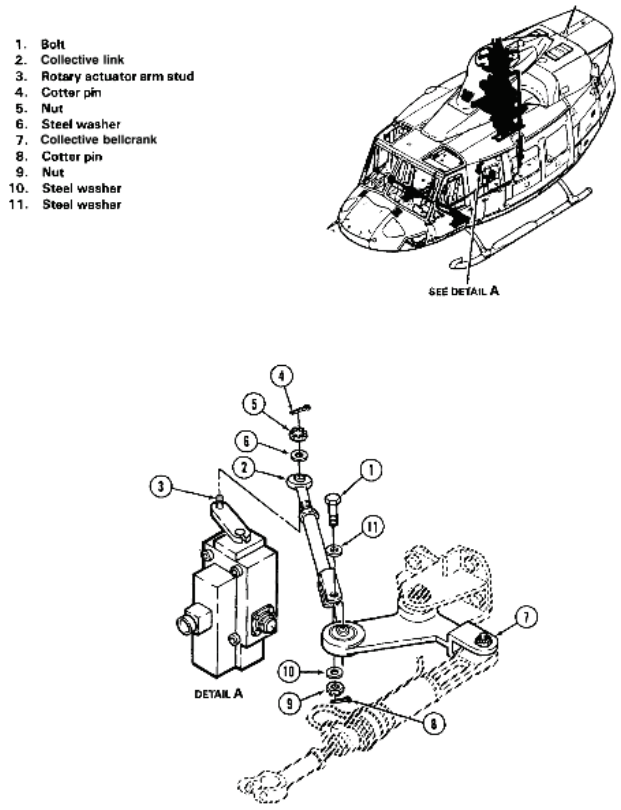
Collective Link - Installation

Procedure

1. Place link (2, Figure 1) into position on bellcrank (7). Install bolt (1), steel washers (10 and 11), nut (9), and cotter pin (8).

2. Position link (2) on rotary actuator arm stud (3). Install steel washer (6), nut (5), and cotter pin (4).

3. Install access panel in cabin floor forward of pylon.



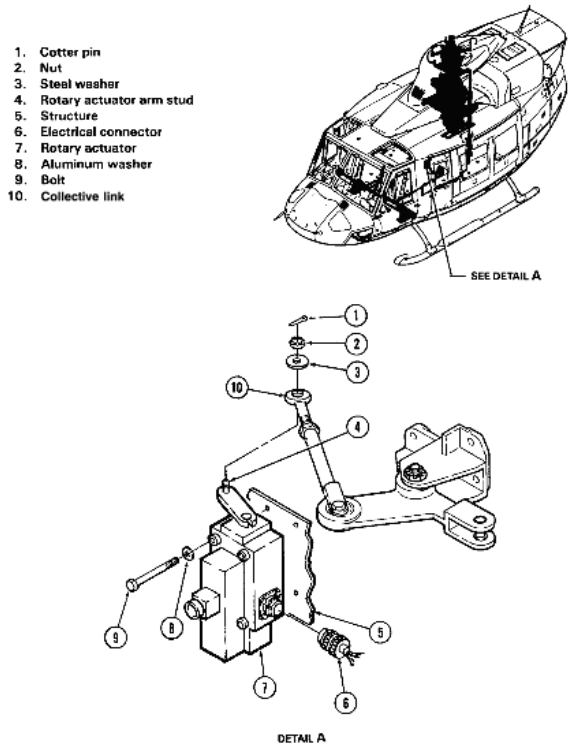
ICN-412-A-671100-B-97499-00016-A-001-01

Figure 1. Collective Link (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 1)

Collective Rotary Actuator - Removal

Procedure

1. Remove access panel from cabin floor forward of pylon.
2. Disconnect electrical connector (6, Figure 1) from rotary actuator (7).
3. Remove collective link (10) from rotary actuator arm stud (4) by removing cotter pin (1), nut (2), and steel washer (3).
4. Remove rotary actuator (7) from structure (5) by removing bolts (9) and washers (8).



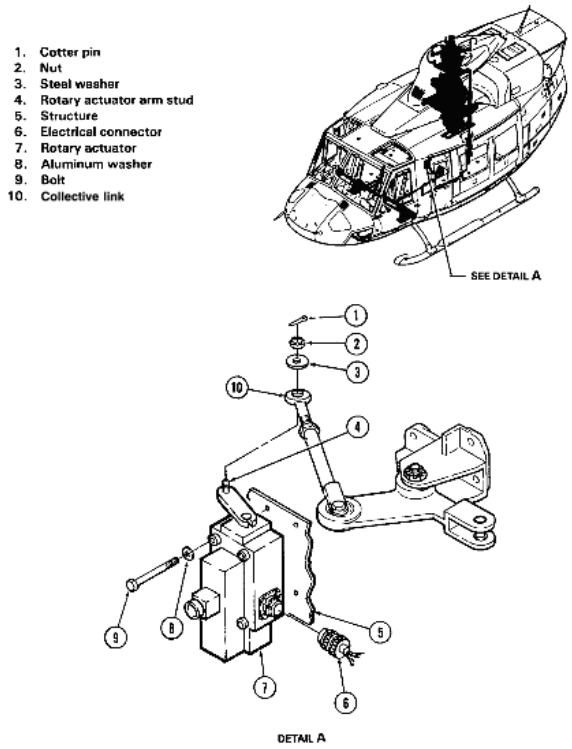
ICN-412-A-671100-B-97499-00017-A-001-01

Figure 1. Collective Rotary Actuator (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 1)

Collective Rotary Actuator - Inspection

Procedure

1. Inspect rotary actuator (7, Figure 1) for mechanical and corrosion damage and wear (BHT-ALL-SPM, Standard Practices Manual).
2. Inspect rotary actuator arm stud (4) for damaged threads. No damage that can be detected visually is acceptable.
3. Inspect electrical receptacle for bent pins and damaged threads. Superficial damage is acceptable without repair.



ICN-412-A-671100-B-97499-00017-A-001-01

Figure 1. Collective Rotary Actuator (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 1)

Collective Rotary Actuator - Repair

Procedure

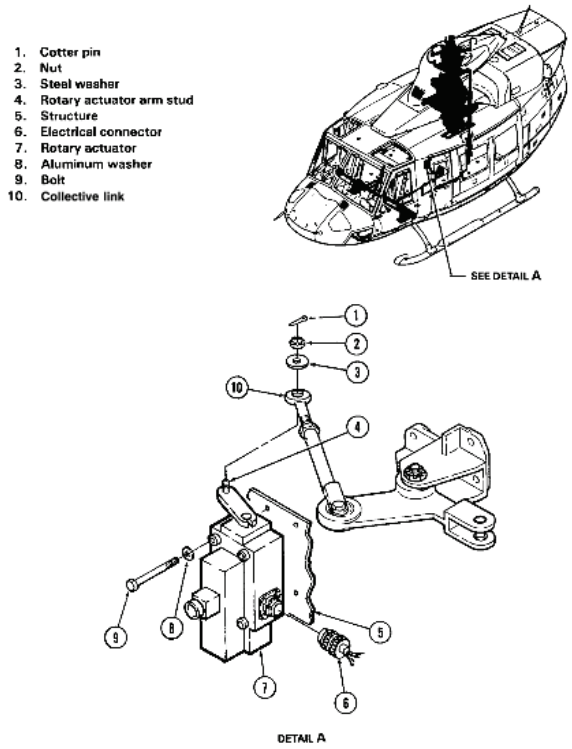
1. Refer to vendor manual for rotary actuator repair instructions.

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Collective Rotary Actuator - Installation

Procedure

1. Place rotary actuator (7, Figure 1) into position on structure (5). Install four aluminum washers (8), and four bolts (9).
2. Position link (10) on rotary actuator arm stud (4) and install steel washer (3), nut (2), and cotter pin (1).
3. Connect electrical connector (6) to rotary actuator (7).
4. Install access panel in cabin floor forward of pylon.



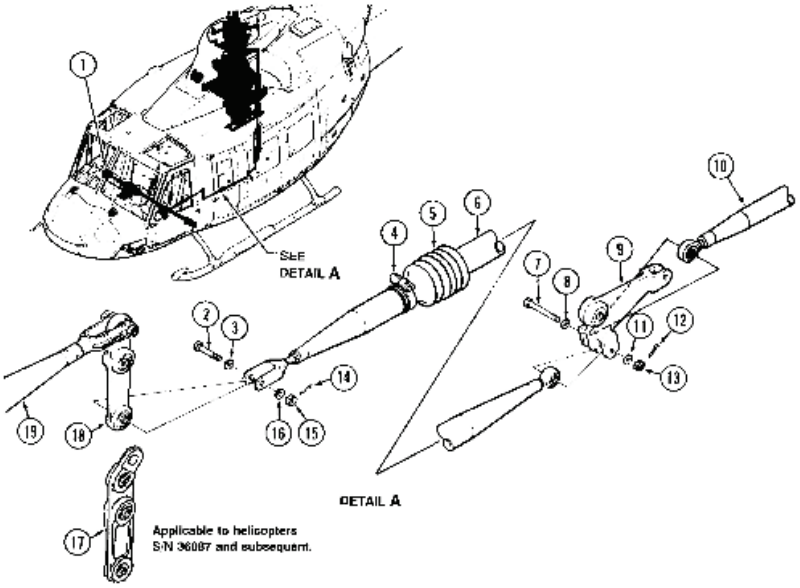
ICN-412-A-671100-B-97499-00017-A-001-01

Figure 1. Collective Rotary Actuator (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 1)

Adjustable Collective Control Tube - Removal

Procedure

1. Remove access panel in aft cabin floor just forward of pylon (412-A-52-40-00-02A-520A-A / 00837).
2. Remove adjustable collective control tube (6, Figure 1) from control lever (17 or 18) by removing cotter pin (14), nut (15), washers (16 and 3) and bolt (2).
3. Remove adjustable collective control tube (6) and control tube, droop compensator (10) from bellcrank (9) by removing cotter pin (12), nut (13), washers (11 and 8) and bolt (7).
4. Loosen clamp (4) and slide forward off of boot (5).
5. Slide adjustable collective control tube (6) forward out of helicopter. Place control tube, droop compensator (10) on airframe structure.



1. Pilot collective stick
2. Bolt
3. Washer (AN960C416)
4. Clamp
5. Boot
6. Adjustable collective control tube
7. Bolt
8. Aluminum washer
9. Bellcrank
10. Control tube, droop compensator
11. Aluminum washer
12. Cotter pin
13. Nut
14. Cotter pin
15. Nut
16. Washer (AN960C416)
17. Control lever
18. Control lever
19. Flight control tube

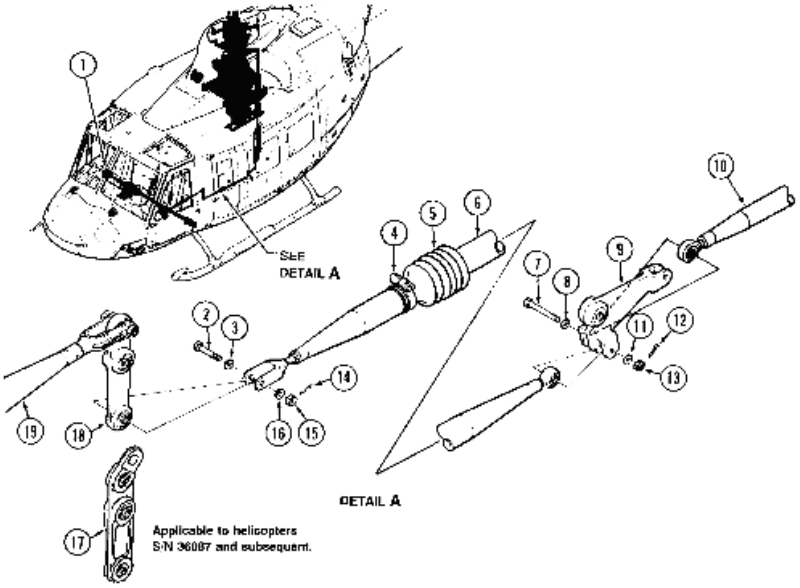
ICN-412-A-671100-B-97499-00018-A-001-01

Figure 1. Adjustable Collective Control Tube (Sheet 1 of 1)

Adjustable Collective Control Tube - Inspection

Procedure

1. Inspect adjustable collective control tube (6, Figure 1) (BHT-412-CR&O).
2. Inspect rod end bearing for secure installation in tube and for wear (BHT-ALL-SPM, Standard Practices Manual).



1. Pilot collective stick
2. Bolt
3. Washer (AN960C416)
4. Clamp
5. Boot
6. Adjustable collective control tube
7. Bolt
8. Aluminum washer
9. Bellcrank
10. Control tube, droop compensator
11. Aluminum washer
12. Cotter pin
13. Nut
14. Cotter pin
15. Nut
16. Washer (AN960C416)
17. Control lever
18. Control lever
19. Flight control tube

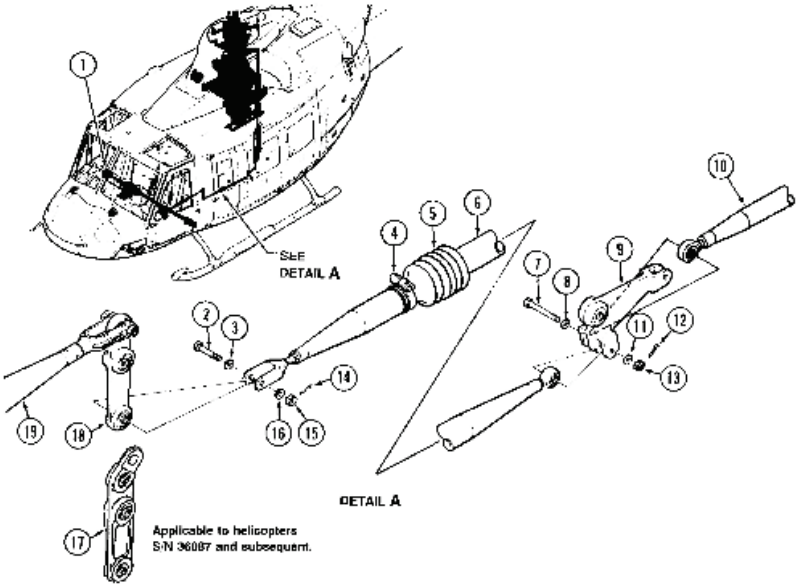
ICN-412-A-671100-B-97499-00018-A-001-01

Figure 1. Adjustable Collective Control Tube (Sheet 1 of 1)

Adjustable Collective Control Tube - Installation

Procedure

1. Slide adjustable collective control tube (6, Figure 1), with rod end bearing aft, into boot (5).
2. Position adjustable collective control tube (6) on control lever (17 or 18). Install bolt (2), washer (3), washer (16), nut (15) and cotter pin (14).
3. Position adjustable collective control tube (6) and control tube, droop compensator (10) on bellcrank (9). Install bolt (7), washer (8), washer (11), nut (13) and cotter pin (12).
4. Position forward end of boot (5) on tube (6) 7.9 in. (200.7 mm) forward of bulkhead with collective control stick (1) full down and tighten clamp (4).
5. Move pilot collective stick (1) through full travel and check for interference and binding.
6. Verify collective rigging within limits (412-A-67-01-00-01A-276A-A / 01323).
7. Install access panel in the aft cabin floor just forward of pylon (412-A-52-40-00-02A-720A-A / 00840).



1. Pilot collective stick
2. Bolt
3. Washer (AN960C416)
4. Clamp
5. Boot
6. Adjustable collective control tube
7. Bolt
8. Aluminum washer
9. Bellcrank
10. Control tube, droop compensator
11. Aluminum washer
12. Cotter pin
13. Nut
14. Cotter pin
15. Nut
16. Washer (AN960C416)
17. Control lever
18. Control lever
19. Flight control tube

ICN-412-A-671100-B-97499-00018-A-001-01

Figure 1. Adjustable Collective Control Tube (Sheet 1 of 1)

Control Lever and Stop Assembly - Removal

Procedure

1. Remove access panel from cabin floor forward of pylon (412-A-52-40-00-02A-520A-A / 00837).

NOTE

SN: 36069-Subsequent:

Collective stop assembly (19, Figure 1) is applicable.

NOTE

SN: 36087-Subsequent:

Control lever (11) is applicable.

2. Remove flight control tube (1, Figure 1) from control lever (11 or 12) by removing cotter pin (5), nut (4), washer (3) and bolt (2).

3. Remove adjustable collective control tube (10) from control lever (11 or 12) by removing cotter pin (9), nut (8), washer (7) and bolt (6).

NOTE

The following step is only applicable to helicopters with 4-axis AFCS/Flight Director kit installed.

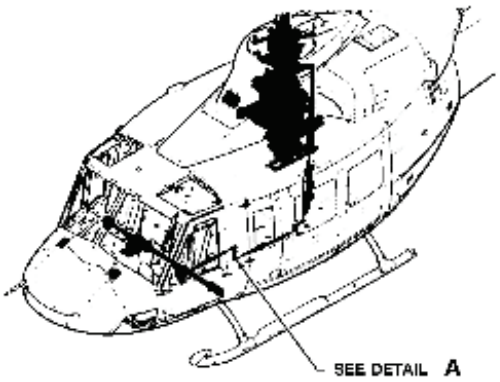
4. Remove cotter pin (10, Figure 2), nut (9), washers (11 and 14), and bolt (13).

5. Remove cotter pin (16, Figure 1), nut (15), washers (14), and bolt (13). Remove control lever (11 or 12).

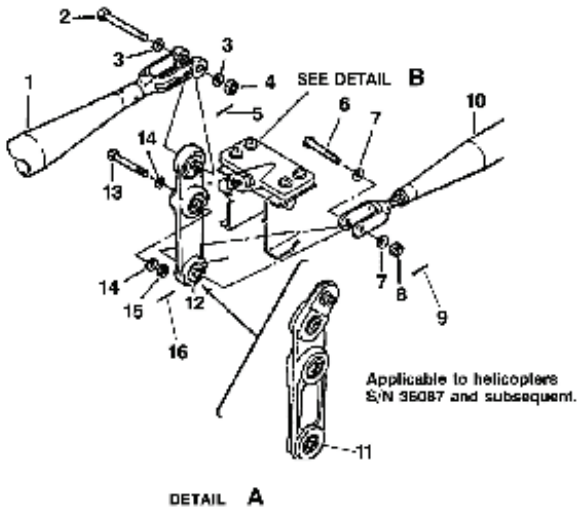
6. Loosen nut (18) and remove bolt (17) from stop assembly (19). Remove nut (18) from bolt (17).

7. Remove screws (20) and washers (21) from stop assembly (19).

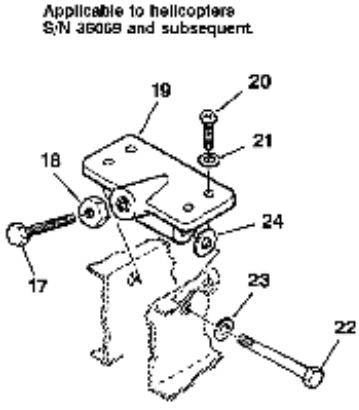
8. Remove bolt (22), washer (23), and shim (24). Remove stop assembly (19).



1. Flight control tube
2. Bolt
3. Aluminum washer
4. Nut
5. Cotter pin
6. Bolt
7. Washer
8. Nut
9. Cotter pin
10. Adjustable collective control tube
11. Control lever
12. Control lever
13. Bolt
14. Aluminum washer
15. Nut
16. Cotter pin
17. Bolt
18. Jam nut
19. Stop assembly
20. Screw
21. Washer
22. Bolt
23. Washer
24. Shim



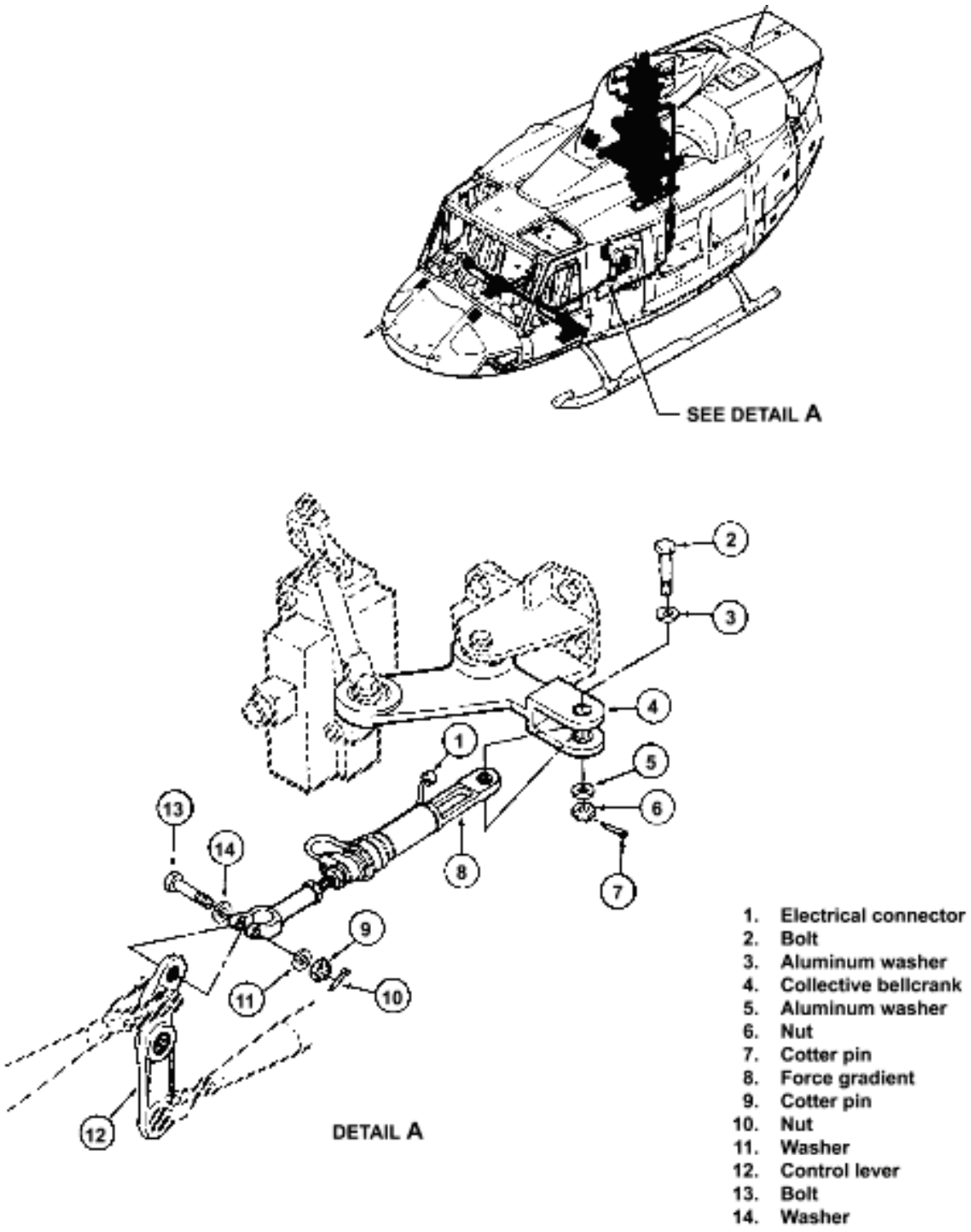
DETAIL A



DETAIL B

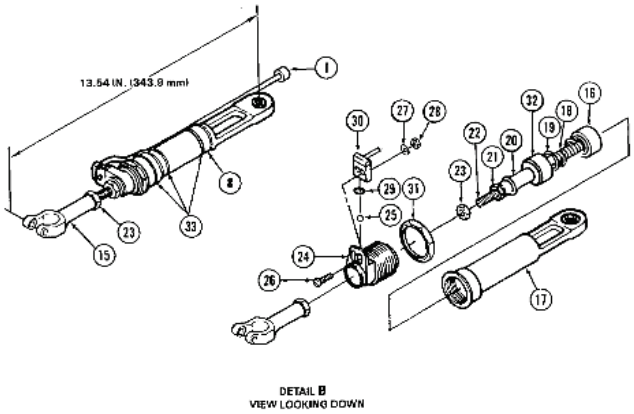
ICN-412-A-671100-B-97499-00019-A-001-01

Figure 1. Collective Control Lever and Stop Assembly (Sheet 1 of 1)



ICN-412-A-671100-B-97499-00013-A-001-01

Figure 2. Collective Force Gradient (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 2)



15. Clevis
16. Guide
17. Cylinder
18. Spring
19. Spring
20. Spacer
21. Jam nut
22. Shaft
23. Jam nut
24. Cap
25. Ball
26. Screw
27. Washer
28. Nut
29. Packing
30. Switch and cable assembly
31. Jam nut
32. Spring assembly
33. Nylon braid

ICN-412-A-671100-B-97499-00014-A-001-01

Figure 2. Collective Force Gradient (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 2 of 2)

Control Lever and Stop Assembly - Inspection

Procedure

NOTE

SN: 36069-Subsequent:

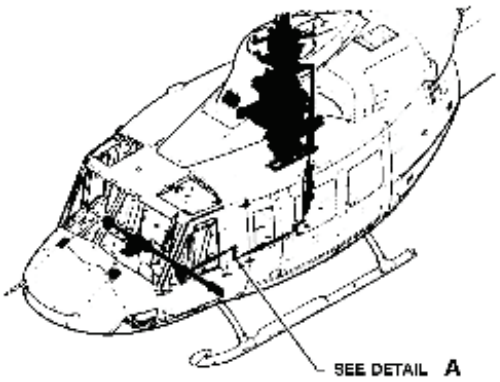
Collective stop assembly (19, Figure 1) is applicable.

NOTE

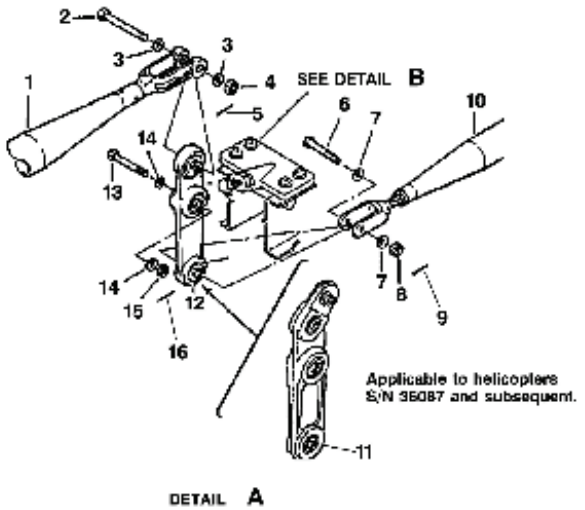
SN: 36087-Subsequent:

Control lever (11) is applicable.

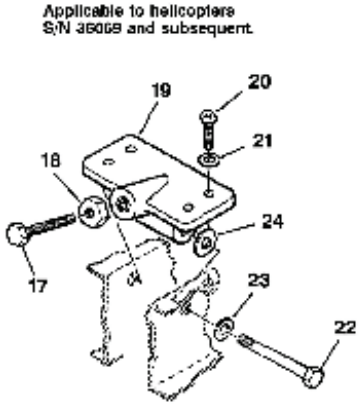
1. Inspect control lever (11 or 12, Figure 1) for mechanical and corrosion damage (BHT-412-CR&O).
2. Inspect bearings in control lever (11 or 12) for secure installation in lever and for wear (BHT-ALL-SPM, Standard Practices Manual).
3. Inspect stop assembly (19) and its attaching hardware for wear, damage, or corrosion.



1. Flight control tube
2. Bolt
3. Aluminum washer
4. Nut
5. Cotter pin
6. Bolt
7. Washer
8. Nut
9. Cotter pin
10. Adjustable collective control tube
11. Control lever
12. Control lever
13. Bolt
14. Aluminum washer
15. Nut
16. Cotter pin
17. Bolt
18. Jam nut
19. Stop assembly
20. Screw
21. Washer
22. Bolt
23. Washer
24. Shim



DETAIL A



DETAIL B

ICN-412-A-671100-B-97499-00019-A-001-01

Figure 1. Collective Control Lever and Stop Assembly (Sheet 1 of 1)

Control Lever and Stop Assembly - Installation

Procedure

NOTE

SN: 36069-Subsequent:

Collective stop assembly (19, Figure 1) is applicable.

NOTE

SN: 36087-Subsequent:

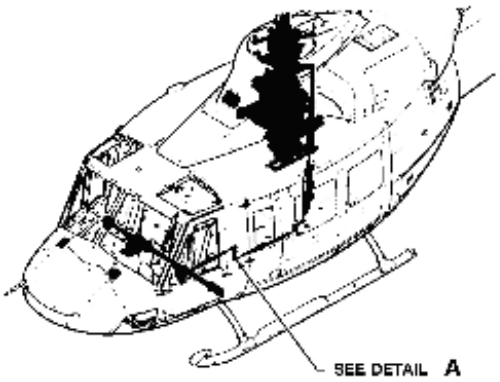
Control lever (11) is applicable.

1. Install stop assembly (19, Figure 1) and attach with shim (24), washers (23), and bolt (22).
2. Install washers (21) and screw (20) in the top of the stop assembly (19).
3. Place nut (18) onto bolt (17) and install in the stop assembly (19).
4. Adjust stop assembly (19) to the proper limits (412-A-67-01-00-01A-276A-A / 01323).
5. Place control lever (11 or 12) into position in structure with long end down. Install bolt (13), washers (14), nut (15), and cotter pin (16).
6. Position adjustable collective control tube (10) and control lever (11 or 12) and install bolt (6), washers (7), nut (8), and cotter pin (9).
7. Position flight control tube (1) on control lever (11 or 12) and install bolt (2), washers (3), nut (4) and cotter pin (5).

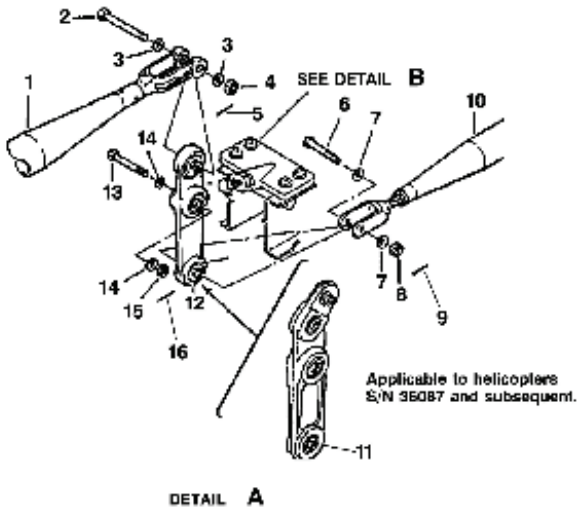
NOTE

The following step is only applicable to helicopters with 4-axis AFCS/Flight Director kit installed.

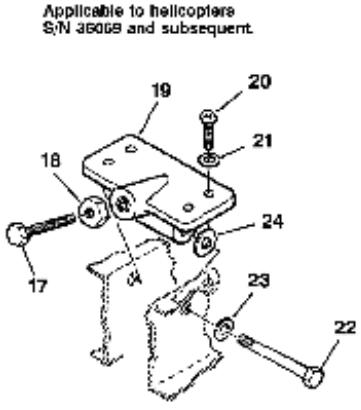
8. Position collective force gradient (8, Figure 2) on control lever (12) and install bolt (13), washers (14 and 11), nut (9), and cotter pin (10).
9. Move collective controls through full travel and check for binding and/or obstruction of controls.
10. Verify collective rigging is within limits (412-A-67-01-00-01A-276A-A / 01323).
11. Install access panel in cabin floor forward of pylon (412-A-52-40-00-02A-720A-A / 00840).



1. Flight control tube
2. Bolt
3. Aluminum washer
4. Nut
5. Cotter pin
6. Bolt
7. Washer
8. Nut
9. Cotter pin
10. Adjustable collective control tube
11. Control lever
12. Control lever
13. Bolt
14. Aluminum washer
15. Nut
16. Cotter pin
17. Bolt
18. Jam nut
19. Stop assembly
20. Screw
21. Washer
22. Bolt
23. Washer
24. Shim



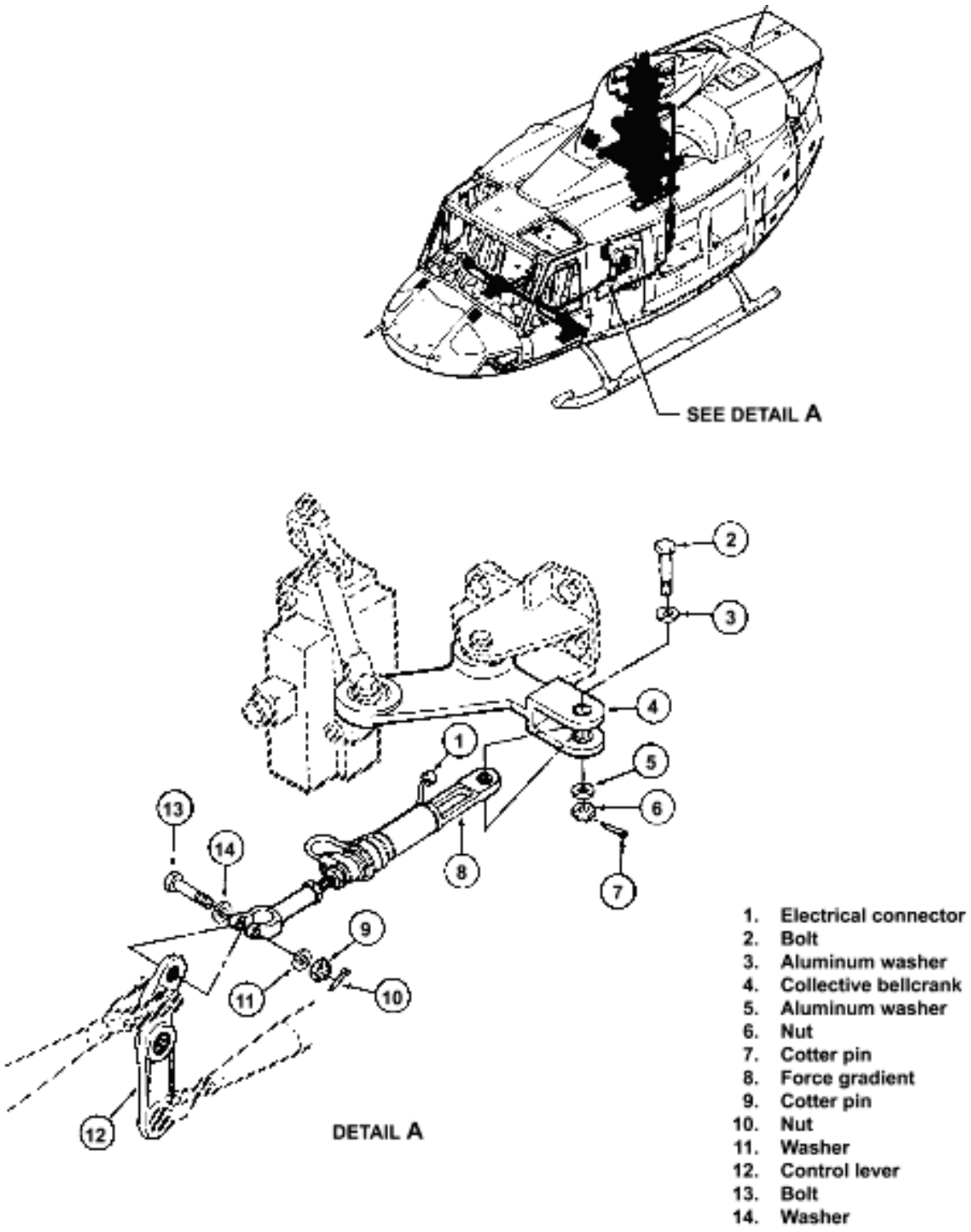
DETAIL A



DETAIL B

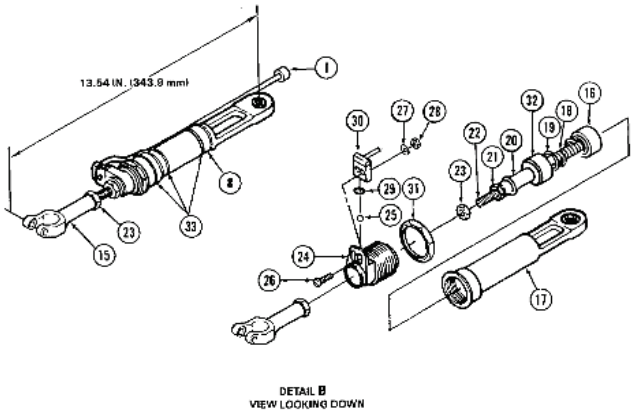
ICN-412-A-671100-B-97499-00019-A-001-01

Figure 1. Collective Control Lever and Stop Assembly (Sheet 1 of 1)



ICN-412-A-671100-B-97499-00013-A-001-01

Figure 2. Collective Force Gradient (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 1 of 2)



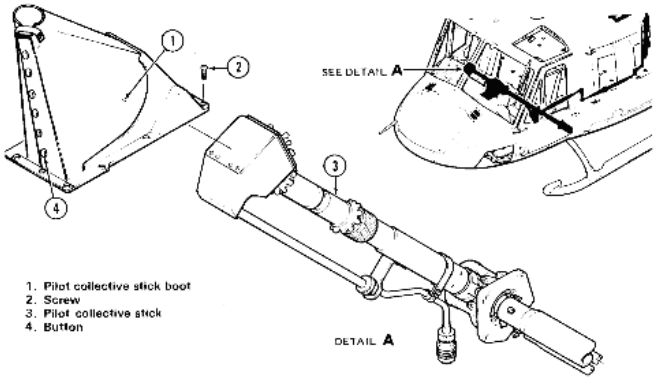
15. Clevis
16. Guide
17. Cylinder
18. Spring
19. Spring
20. Spacer
21. Jam nut
22. Shaft
23. Jam nut
24. Cap
25. Ball
26. Screw
27. Washer
28. Nut
29. Packing
30. Switch and cable assembly
31. Jam nut
32. Spring assembly
33. Nylon braid

ICN-412-A-671100-B-97499-00014-A-001-01

Figure 2. Collective Force Gradient (Applicable to Helicopters with 4-Axis AFCS/Flight Director Kit Installed) (Sheet 2 of 2)

Pilot Collective Boot - Description

The pilot collective stick boot (1, Figure 1) is mounted around the base of the pilot collective stick to prevent foreign objects and dirt from fouling controls. The boot is made of synthetic fabric with a metal base to secure the boot to the floor. Snaps provide a tight fit to the stick. A similar boot is installed on the copilot stick.



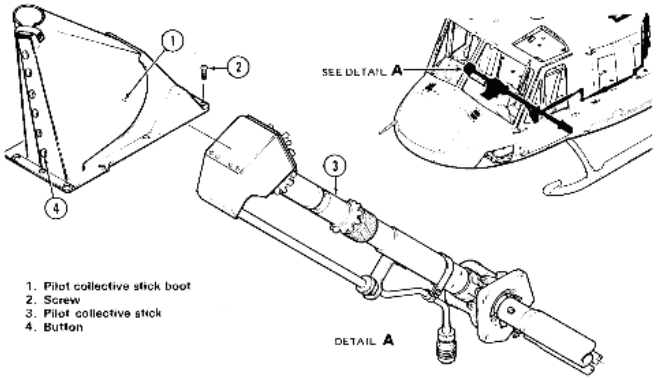
ICN-412-A-671100-B-97499-00020-A-001-01

Figure 1. Pilot Collective Stick Boot (Sheet 1 of 1)

Pilot Collective Boot - Removal

Procedure

1. Unsnap the buttons (4, Figure 1).
2. Remove the screws (2).
3. Remove the pilot collective stick boot (1).



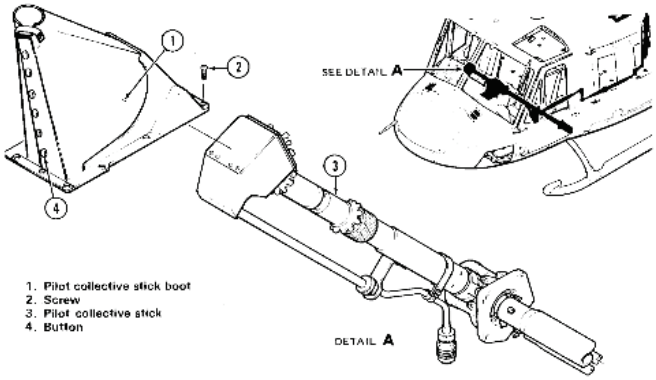
ICN-412-A-671100-B-97499-00020-A-001-01

Figure 1. Pilot Collective Stick Boot (Sheet 1 of 1)

Pilot Collective Boot - Inspection

Procedure

1. Inspect the pilot collective stick boot (1, Figure 1) for tears and deterioration.
2. Inspect for defective buttons (4).



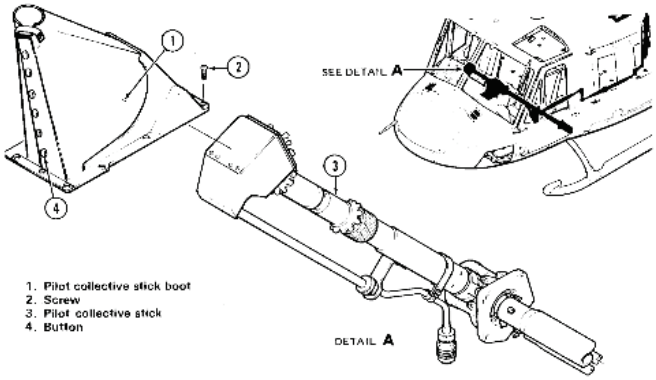
ICN-412-A-671100-B-97499-00020-A-001-01

Figure 1. Pilot Collective Stick Boot (Sheet 1 of 1)

Pilot Collective Boot - Installation

Procedure

1. Slide the pilot collective stick boot (1, Figure 1) over the pilot collective stick (3).
2. Install the screws (2) to secure the boot (1) to the floor.
3. Fasten the buttons (4).

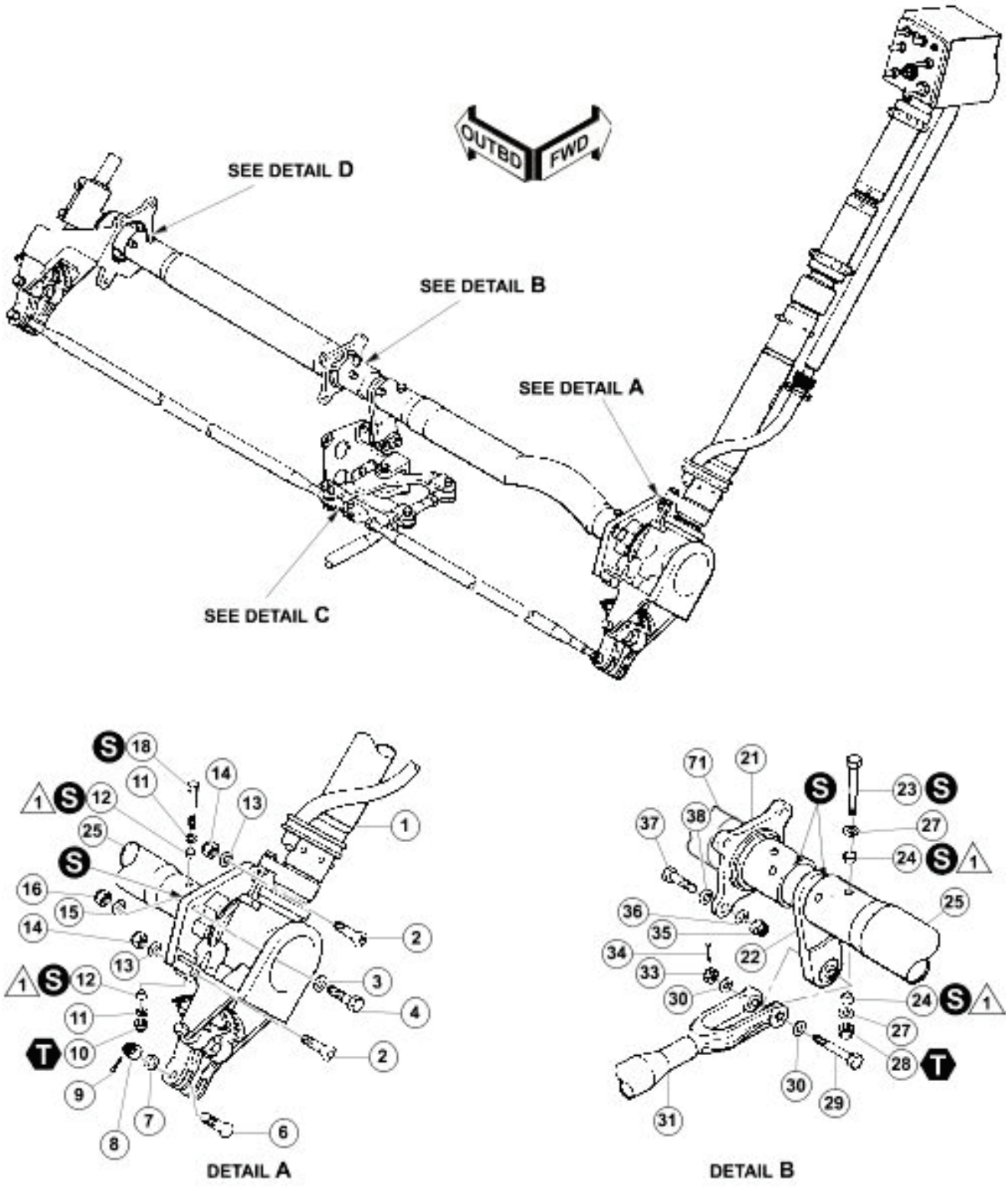


ICN-412-A-671100-B-97499-00020-A-001-01

Figure 1. Pilot Collective Stick Boot (Sheet 1 of 1)

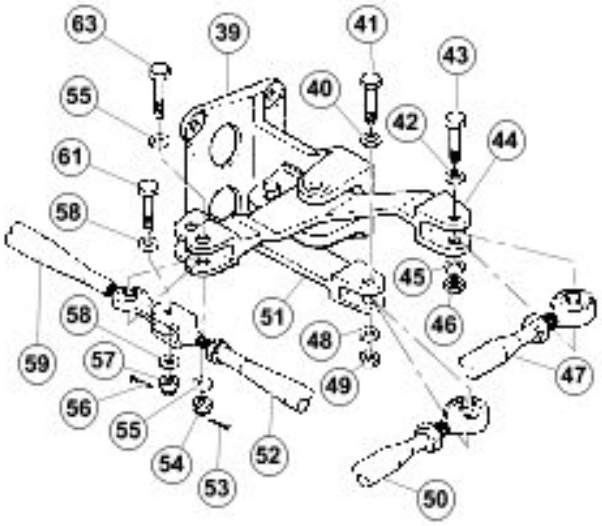
Collective Jackshaft - Description

The collective jackshaft (Figure 1) installation consists of the jackshafts, pilot collective stick, throttle interconnect tubes, bellcranks, and supports. A downlock is mounted on the floor below the stick. An elbow and support are provided to allow a copilot collective stick to be installed.

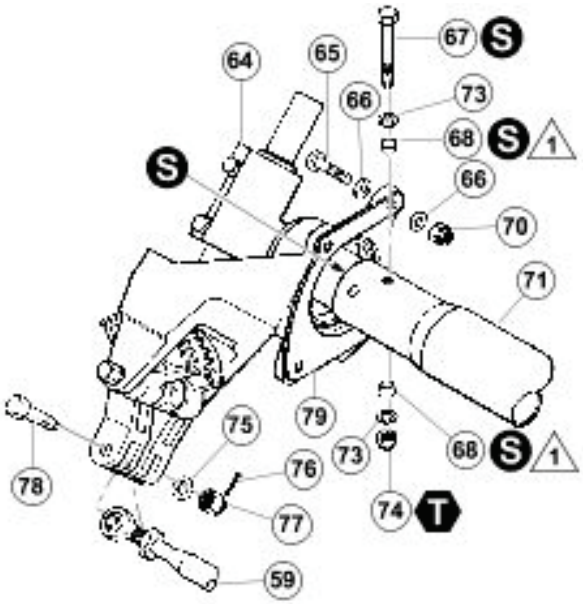


ICN-412-A-671100-B-97499-00021-A-001-01

Figure 1. Collective Jackshaft (Sheet 1 of 2)



DETAIL C



DETAIL D

1. Stick assembly
2. Screw
3. Washer (NAS1197-416)
4. Bolt
5. Deleted
6. Bolt
7. Thin aluminum washer
8. Nut
9. Cotter pin
10. Nut
11. Washer
12. Tapered bushing
13. Aluminum washer
14. Nut
15. Aluminum washer
16. Nut
17. Deleted
18. Bolt
19. Deleted
20. Deleted
21. Support
22. Lever
23. Bolt
24. Tapered bushing
25. Tube
26. Deleted
27. Washer
28. Nut
29. Bolt
30. Aluminum washer
31. Flight control tube
32. Deleted
33. Nut
34. Cotter pin
35. Nut
36. Aluminum washer
37. Bolt
38. Washer (NAS1197-416)
39. Support
40. Washer
41. Bolt
42. Washer
43. Bolt
44. Bellcrank
45. Washer
46. Nut
47. Tube
48. Washer
49. Nut
50. Tube
51. Bellcrank
52. Tube
53. Cotter pin
54. Nut
55. Aluminum washer
56. Cotter pin
57. Nut
58. Aluminum washer
59. Tube
60. Deleted
61. Bolt
62. Deleted
63. Bolt
64. Elbow and support
65. Bolt
66. Aluminum washer
67. Bolt
68. Tapered bushing
69. Deleted
70. Nut
71. Tube
72. Deleted
73. Washer
74. Nut
75. Thin aluminum washer
76. Cotter pin
77. Nut
78. Bolt
79. Shim

S SEALANT (C-251)

T 50 TO 70 IN-LBS (5.65 TO 7.90 Nm)

NOTE

1 For helicopters S/N 36487 and subsequent, if protrusion of tapered bushing (P/N 20-037-4-7) is less than 0.030 inch (0.76 mm), replace tapered bushing with P/N 20-037-4-8.

ICN-412-A-671100-B-97499-00022-A-001-01

Figure 1. Collective Jackshaft (Sheet 2 of 2)

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Collective Jackshaft - Removal

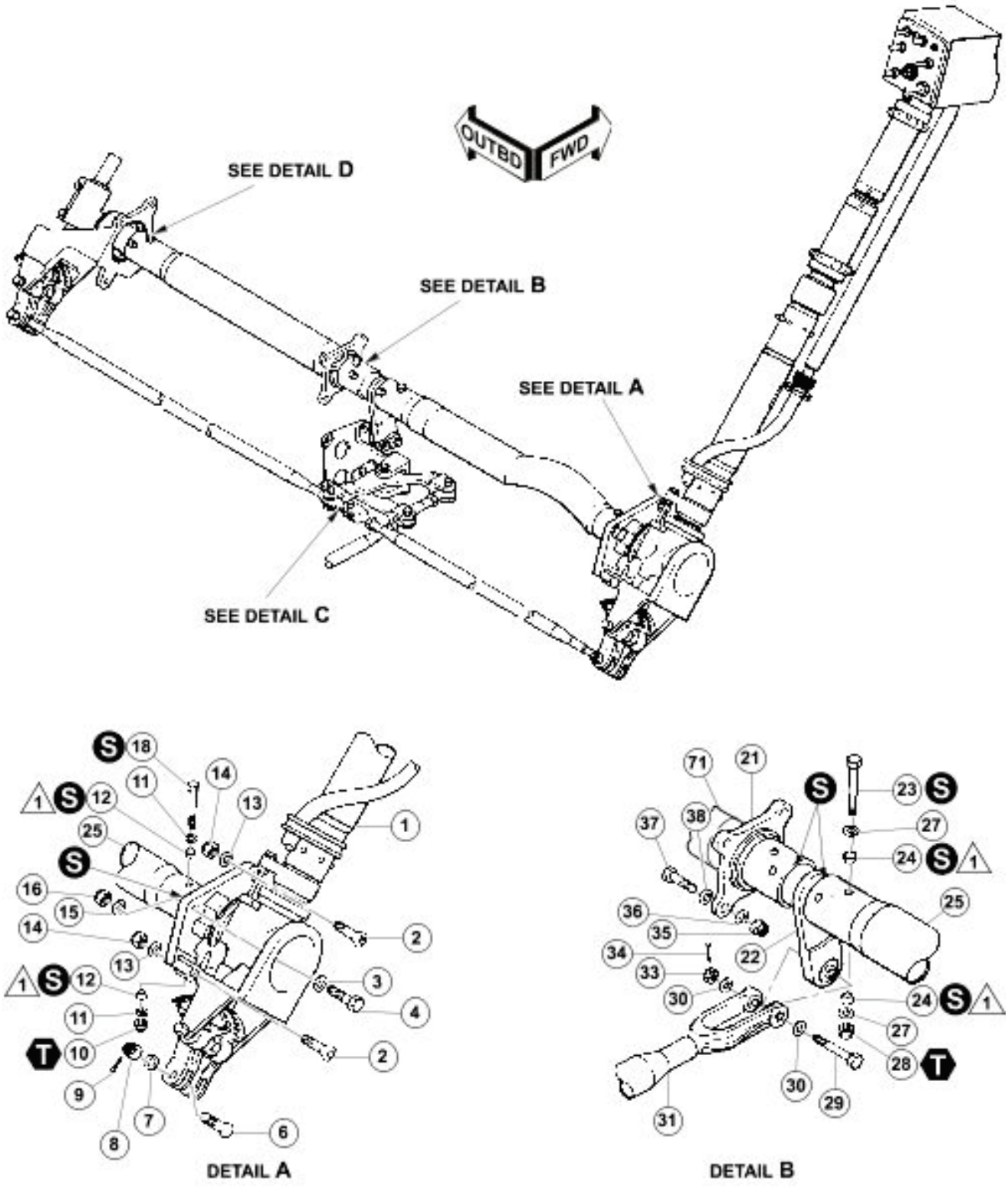
Procedure

1. Remove the pilot collective stick boot (412-A-67-11-00-16A-520A-A / 01379).
2. Remove the following access panels (412-A-52-40-00-02A-520A-A / 00837):
   - Cabin floor aft of pedestal
   - Cabin floor under left crew seat
   - Cabin floor left side of left crew seat
3. Remove the access panel from the lower left side of the helicopter just below the aft corner of the left crew door (412-A-52-40-00-02A-520A-A / 00837).
4. Disconnect the collective stick electrical connector.
5. Disconnect the tubes (52, Figure 1) from the stick assembly (1) by removing the cotter pin (9), nut (8), washers (7), and bolt (6).
6. Remove the bolts (18), tapered bushings (12), washers (11), and nuts (10). Index the bushings for installation in the original location.
7. Remove the nuts (14), washers (13), and screws (2) from the stick assembly (1) and remove the nuts (16), washers (3 and 15), and bolts (4).

NOTE

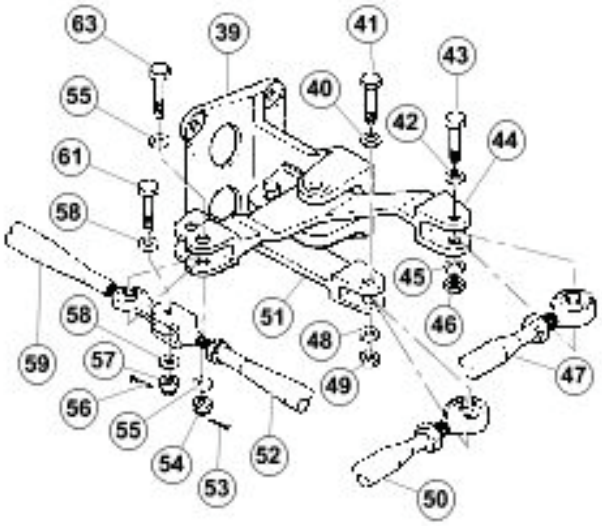
Make sure the bearing in the support (21) is not damaged in Step 8.

8. Tap the stick assembly (1) with a soft mallet to slide the stick out of the tube (25). Remove the stick assembly.
9. Remove the cotter pin (34), nut (33), aluminum washers (30), and bolt (29). Disconnect the flight control tube (31).
10. Remove the bolts (23), tapered bushings (24), washers (27), and nuts (28). Index the bushings for installation in the original location.
11. Slide the tube (25) to the right to disengage the tube from the lever (22). Remove the tube.
12. Remove the lever (22) from the support (21).
13. To remove the tubes (52), remove the cotter pin (56), nut (57), aluminum washers (58 and 60), and bolt (61). Discard the cotter pin.
14. To remove the tube (50) from the bellcrank (51), remove the nut (49), washers (48 and 40), and bolt (41).
15. To remove the tube (59), remove the cotter pin (53), nut (54), aluminum washers (55), and bolts (63). Discard the cotter pin.
16. To remove the tube (47) from the bellcrank (44), remove the nut (46), washers (45 and 42), and bolt (43).
17. Remove the cotter pins (76), nuts (77), thin aluminum washers (75), and bolts (78) from the tubes (59). Discard the cotter pins.
18. Remove the bolts (67), tapered bushings (68), washers (73), and nuts (74). Index the bushings for installation in the original location.
19. Remove the nuts (70), aluminum washers (66), and bolts (65).
20. Disengage the elbow and the support (64) from the tube (71) and remove from the helicopter.
21. Remove the tube (71).
22. Remove the nuts (35), washers (36 and 38), and bolts (37). Remove the support (21).

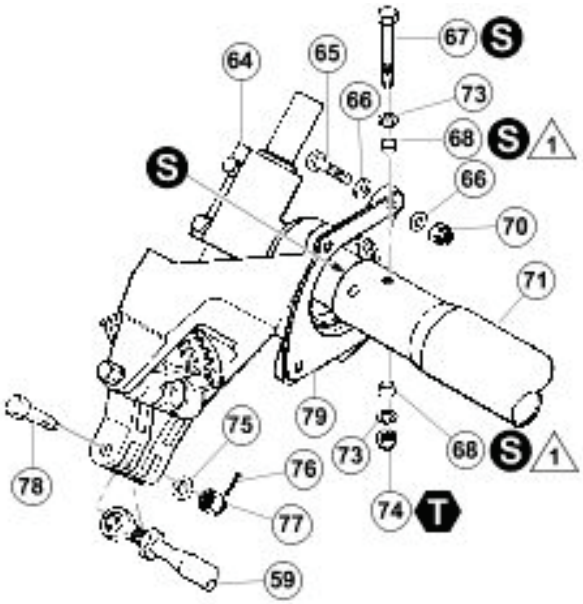


ICN-412-A-671100-B-97499-00021-A-001-01

Figure 1. Collective Jackshaft (Sheet 1 of 2)



DETAIL C



DETAIL D

1. Stick assembly
2. Screw
3. Washer (NAS1197-416)
4. Bolt
5. Deleted
6. Bolt
7. Thin aluminum washer
8. Nut
9. Cotter pin
10. Nut
11. Washer
12. Tapered bushing
13. Aluminum washer
14. Nut
15. Aluminum washer
16. Nut
17. Deleted
18. Bolt
19. Deleted
20. Deleted
21. Support
22. Lever
23. Bolt
24. Tapered bushing
25. Tube
26. Deleted
27. Washer
28. Nut
29. Bolt
30. Aluminum washer
31. Flight control tube
32. Deleted
33. Nut
34. Cotter pin
35. Nut
36. Aluminum washer
37. Bolt
38. Washer (NAS1197-416)
39. Support
40. Washer
41. Bolt
42. Washer
43. Bolt
44. Bellcrank
45. Washer
46. Nut
47. Tube
48. Washer
49. Nut
50. Tube
51. Bellcrank
52. Tube
53. Cotter pin
54. Nut
55. Aluminum washer
56. Cotter pin
57. Nut
58. Aluminum washer
59. Tube
60. Deleted
61. Bolt
62. Deleted
63. Bolt
64. Elbow and support
65. Bolt
66. Aluminum washer
67. Bolt
68. Tapered bushing
69. Deleted
70. Nut
71. Tube
72. Deleted
73. Washer
74. Nut
75. Thin aluminum washer
76. Cotter pin
77. Nut
78. Bolt
79. Shim

S SEALANT (C-251)

T 50 TO 70 IN-LBS (5.65 TO 7.90 Nm)

NOTE

1 For helicopters S/N 36487 and subsequent, if protrusion of tapered bushing (P/N 20-037-4-7) is less than 0.030 inch (0.76 mm), replace tapered bushing with P/N 20-037-4-8.

ICN-412-A-671100-B-97499-00022-A-001-01

Figure 1. Collective Jackshaft (Sheet 2 of 2)

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Collective Jackshaft - Inspection

Procedure

1. Inspect the stick assembly (1, Figure 1) as follows:

1.1. Inspect the stick assembly for mechanical and corrosion damage. Superficial mechanical and corrosion damage is acceptable if polished out and the repaired area refinished to the original condition.

1.2. Inspect wiring for secure installation and damaged insulation.

1.3. Inspect the switches for secure installation.

1.4. Inspect the throttle controls for smooth operation. If necessary, adjust the throttle control gears (412-A-67-01-00-01A-276A-A / 01323).

2. Inspect the support (21) as follows:

2.1. Inspect the support for mechanical and corrosion damage (BHT-412-CR&O).

2.2. Inspect the bearing for security and wear (BHT-ALL-SPM, Standard Practices Manual).

3. Inspect the tubes (25, 52, 59, and 71) for mechanical and corrosion damage (BHT-412-CR&O).

4. Inspect the elbow and support (64) as follows:

4.1. Inspect the elbow and support (64) for mechanical and corrosion damage (BHT-412-CR&O).

4.2. Inspect the power control bellcranks in the base of the elbow and support (64) for freedom of movement.

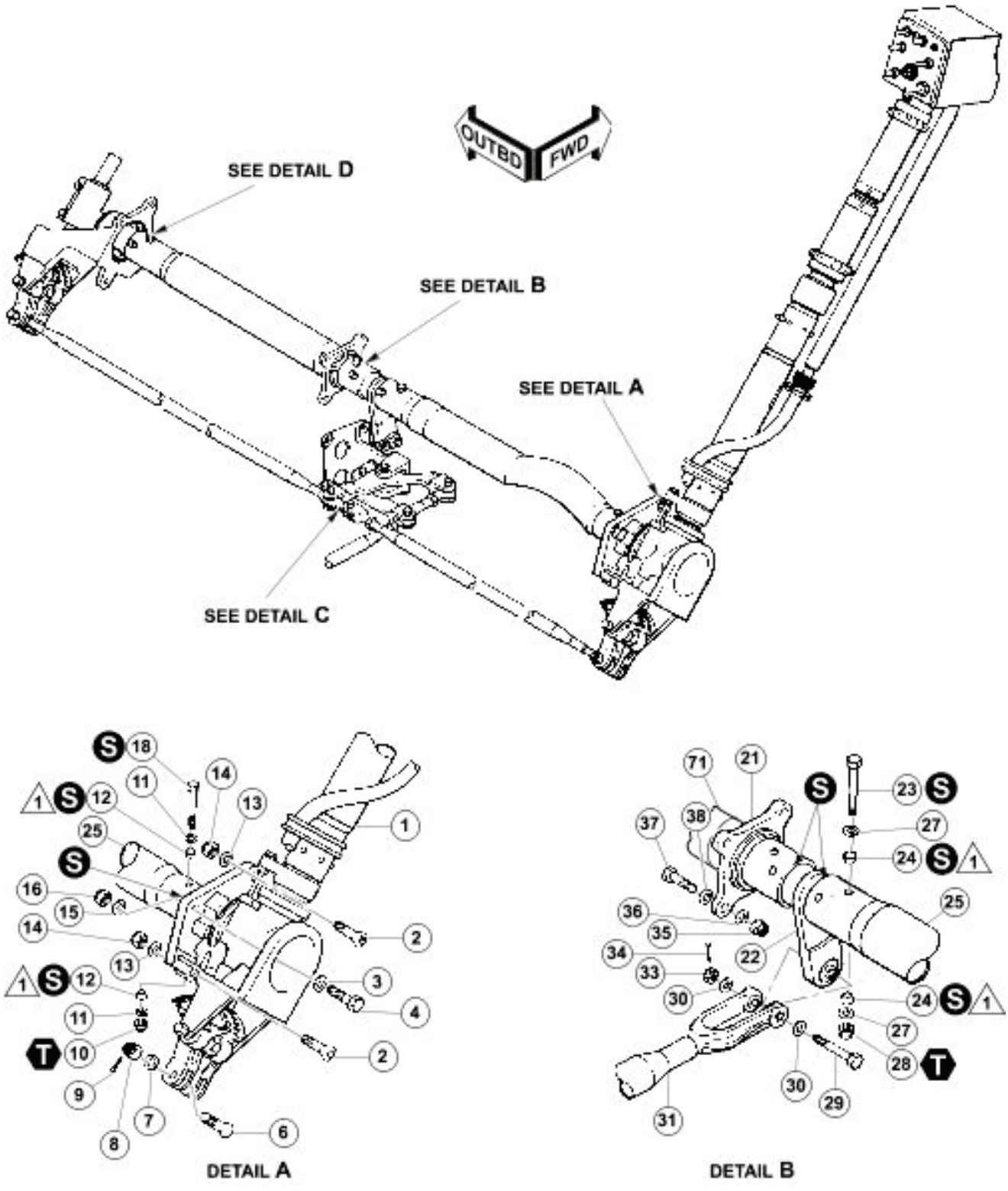
4.3. Inspect the bearings for security and wear (BHT-ALL-SPM, Standard Practices Manual).

5. Inspect the installation of the tapered bushings (12, 24, and 68) as follows:

5.1. Verify that the protrusion of the tapered bushings (12, 24, and 68) is less than 0.030 inch (0.76 mm), as shown in Figure 2. If less than 0.030 inch (0.76 mm), replace the part as required (412-A-67-11-00-17A-685A-A / 01385).

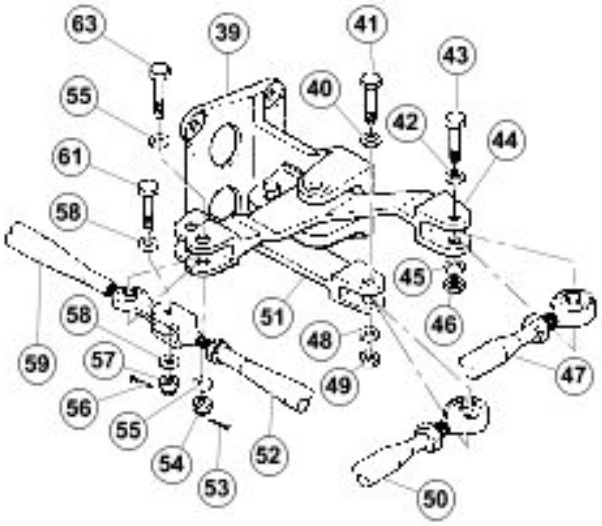
Applicable: SN: 36487-Subsequent:

5.2. If the protrusion of the tapered bushings (12, 24, and 68) (P/N 20-037-4-7) is less than 0.030 inch (0.76 mm), replace with the tapered bushing (P/N 20-037-4-8).

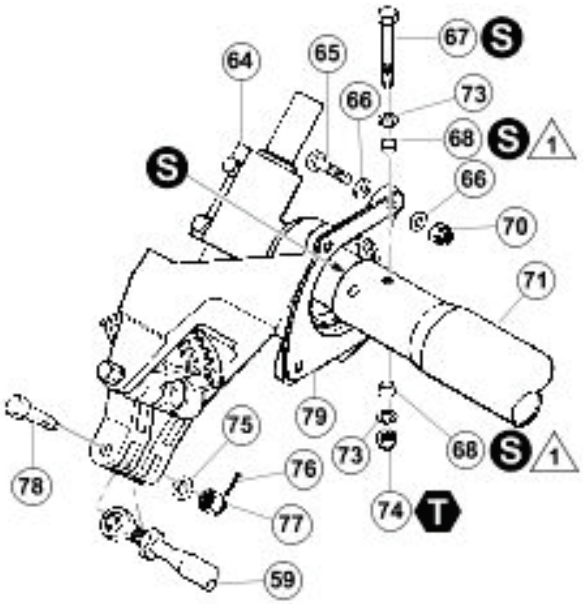


ICN-412-A-671100-B-97499-00021-A-001-01

Figure 1. Collective Jackshaft (Sheet 1 of 2)



DETAIL C



DETAIL D

1. Stick assembly
2. Screw
3. Washer (NAS1197-416)
4. Bolt
5. Deleted
6. Bolt
7. Thin aluminum washer
8. Nut
9. Cotter pin
10. Nut
11. Washer
12. Tapered bushing
13. Aluminum washer
14. Nut
15. Aluminum washer
16. Nut
17. Deleted
18. Bolt
19. Deleted
20. Deleted
21. Support
22. Lever
23. Bolt
24. Tapered bushing
25. Tube
26. Deleted
27. Washer
28. Nut
29. Bolt
30. Aluminum washer
31. Flight control tube
32. Deleted
33. Nut
34. Cotter pin
35. Nut
36. Aluminum washer
37. Bolt
38. Washer (NAS1197-416)
39. Support
40. Washer
41. Bolt
42. Washer
43. Bolt
44. Bellcrank
45. Washer
46. Nut
47. Tube
48. Washer
49. Nut
50. Tube
51. Bellcrank
52. Tube
53. Cotter pin
54. Nut
55. Aluminum washer
56. Cotter pin
57. Nut
58. Aluminum washer
59. Tube
60. Deleted
61. Bolt
62. Deleted
63. Bolt
64. Elbow and support
65. Bolt
66. Aluminum washer
67. Bolt
68. Tapered bushing
69. Deleted
70. Nut
71. Tube
72. Deleted
73. Washer
74. Nut
75. Thin aluminum washer
76. Cotter pin
77. Nut
78. Bolt
79. Shim

S SEALANT (C-251)

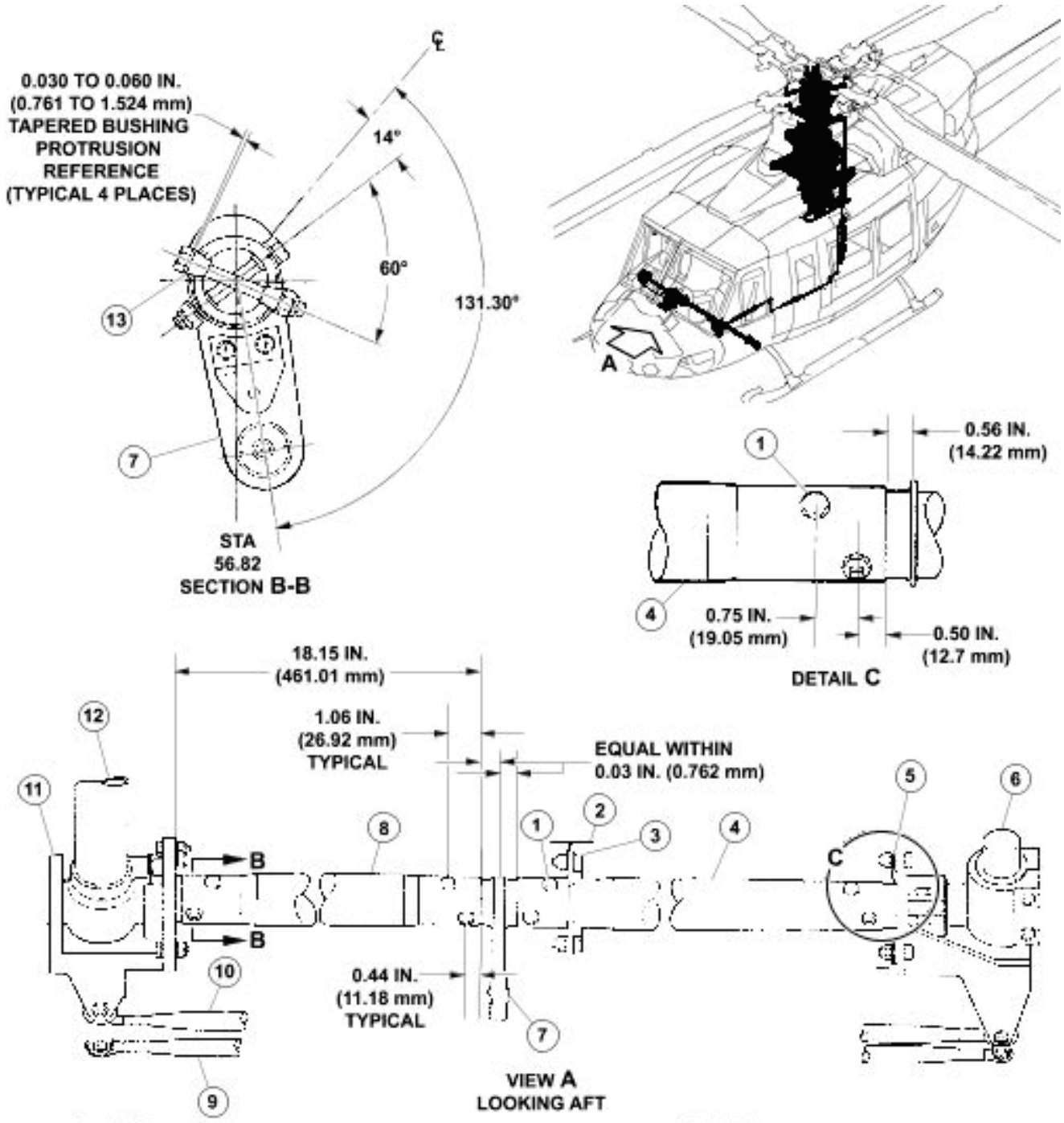
T 50 TO 70 IN-LBS (5.65 TO 7.90 Nm)

NOTE

1 For helicopters S/N 36487 and subsequent, if protrusion of tapered bushing (P/N 20-037-4-7) is less than 0.030 inch (0.76 mm), replace tapered bushing with P/N 20-037-4-8.

ICN-412-A-671100-B-97499-00022-A-001-01

Figure 1. Collective Jackshaft (Sheet 2 of 2)



ICN-412-A-671100-B-97499-00024-A-001-01

Figure 2. Collective Jackshaft Assembly (Sheet 1 of 1)

Collective Jackshaft - Repair

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Reamer	T27872-2	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Epoxy polyamide primer	C-204	AR	

Procedure

NOTE

If the stick assembly (1, Figure 1), tube (25), lever (22), tube (71), or elbow and support (64) is being replaced, the new component must be drilled and reamed to pick up holes existing in original components. If holes have been drilled and reamed, proceed to 412-A-67-11-00-17A-720A-A / 01386.

1. Replace the collective jackshaft as follows:

1.1. Assemble the tubes (25 and 71, Figure 1), lever (22), elbow and support (64), and stick assembly (1) to the dimensions and angles shown in Figure 2.

1.2. Clamp the collective jackshaft in a suitable holding fixture.

1.3. Hold the dimensions, as shown in Figure 2.

1.4. Center the punch and drill new holes through the tubes (25 and 71, Figure 1), lever (22), elbow and support (64), and stick assembly (1) with an F drill (0.257 inch). If the existing holes are not visible, use a No. 20 drill (0.161 inch) initially to permit the hole to be centered with the existing holes.

1.5. Insert the reamer (T27872-2) into the hole so the pilot projects through the part being reamed. Ream one side for the installation of the tapered bushings so the bushing will extend 0.030 to 0.060 inch (0.76 to 1.52 mm) (Figure 2, Section B-B).

1.6. Insert the reamer (T27872-2) into the opposite hole. Insert a tapered bushing into the reamed hole to act as a pilot for the reamer. Ream to depth for the installation of the tapered bushings as described in Step 1.5..

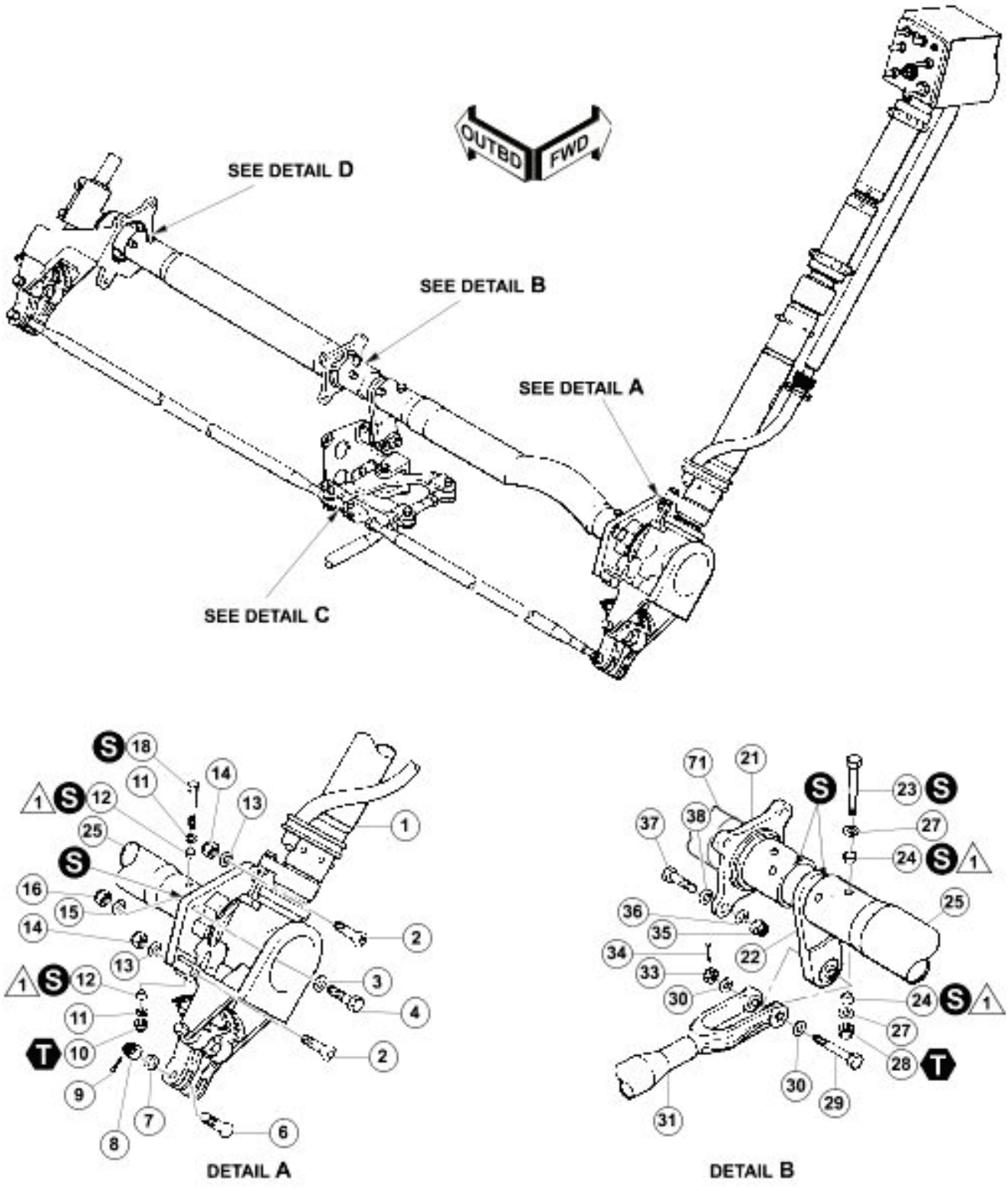
1.7. Repeat Step 1.4. through Step 1.6. for the remaining holes.

1.8. Disassemble and remove the burrs and metal particles.

1.9. Repeat Step 1.4. through Step 1.8. for the remaining holes.

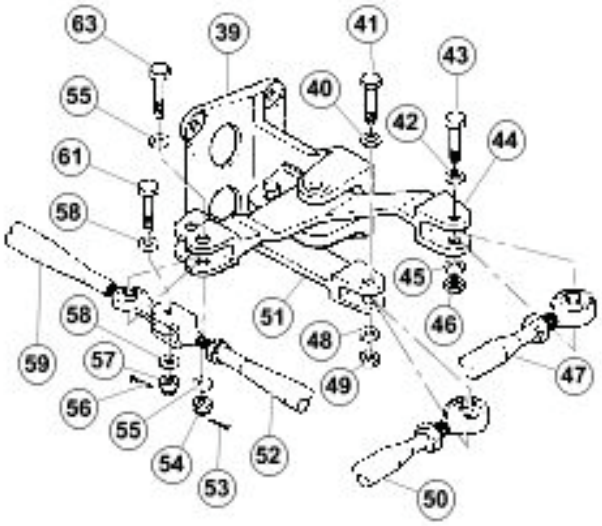
1.10. Remove the collective jackshaft from the holding fixture and disassemble. Remove all burrs and metal particles.

1.11. Apply epoxy polyamide primer (C-204) to all bare metal areas.

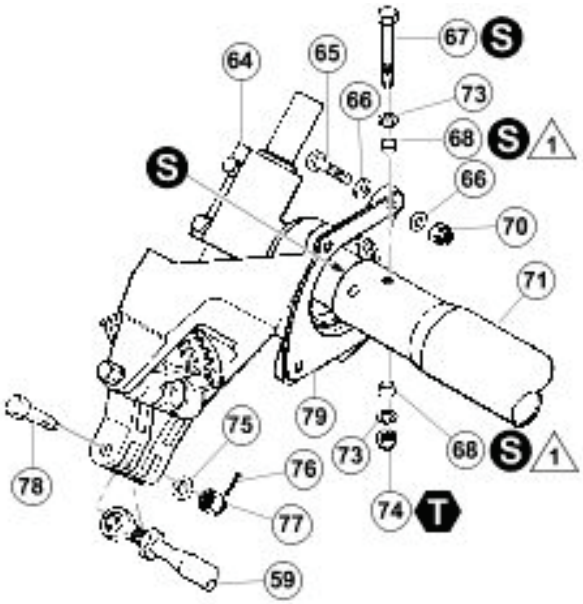


ICN-412-A-671100-B-97499-00021-A-001-01

Figure 1. Collective Jackshaft (Sheet 1 of 2)



DETAIL C



DETAIL D

1. Stick assembly
2. Screw
3. Washer (NAS1197-416)
4. Bolt
5. Deleted
6. Bolt
7. Thin aluminum washer
8. Nut
9. Cotter pin
10. Nut
11. Washer
12. Tapered bushing
13. Aluminum washer
14. Nut
15. Aluminum washer
16. Nut
17. Deleted
18. Bolt
19. Deleted
20. Deleted
21. Support
22. Lever
23. Bolt
24. Tapered bushing
25. Tube
26. Deleted
27. Washer
28. Nut
29. Bolt
30. Aluminum washer
31. Flight control tube
32. Deleted
33. Nut
34. Cotter pin
35. Nut
36. Aluminum washer
37. Bolt
38. Washer (NAS1197-416)
39. Support
40. Washer
41. Bolt
42. Washer
43. Bolt
44. Bellcrank
45. Washer
46. Nut
47. Tube
48. Washer
49. Nut
50. Tube
51. Bellcrank
52. Tube
53. Cotter pin
54. Nut
55. Aluminum washer
56. Cotter pin
57. Nut
58. Aluminum washer
59. Tube
60. Deleted
61. Bolt
62. Deleted
63. Bolt
64. Elbow and support
65. Bolt
66. Aluminum washer
67. Bolt
68. Tapered bushing
69. Deleted
70. Nut
71. Tube
72. Deleted
73. Washer
74. Nut
75. Thin aluminum washer
76. Cotter pin
77. Nut
78. Bolt
79. Shim

S SEALANT (C-251)

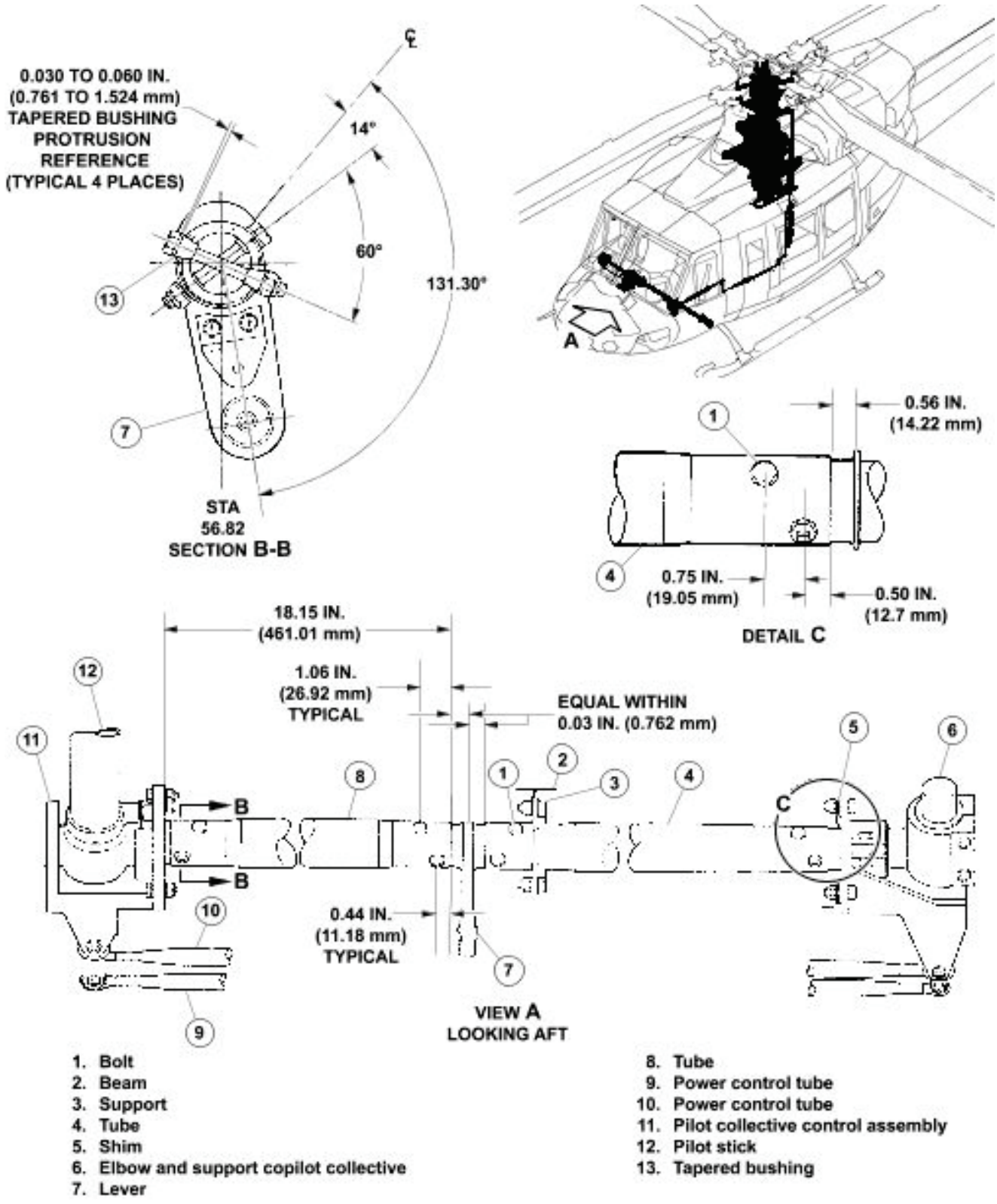
T 50 TO 70 IN-LBS (5.65 TO 7.90 Nm)

NOTE

1 For helicopters S/N 36487 and subsequent, if protrusion of tapered bushing (P/N 20-037-4-7) is less than 0.030 inch (0.76 mm), replace tapered bushing with P/N 20-037-4-8.

ICN-412-A-671100-B-97499-00022-A-001-01

Figure 1. Collective Jackshaft (Sheet 2 of 2)



ICN-412-A-671100-B-97499-00024-A-001-01

Figure 2. Collective Jackshaft Assembly (Sheet 1 of 1)

Collective Jackshaft - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-251	AR	
Barrier tape	C-456	AR	

Procedure

1. Apply the barrier tape (C-456) between the faying surfaces and put the support (21, Figure 1) in position in the helicopter and install the bolts (37), washers (38), aluminum washers (36), and nuts (35). Tighten the bolts (37).

2. Put the tube (71), lever (22), and tube (25) in position in the helicopter and fit together.

3. Put the tubes (52 and 59) in position in the helicopter.

NOTE

Adjust the shim (79) to minimize lateral deflection of the shafts during installation.

4. Apply the barrier tape (C-456) between the faying surfaces of the structure and the shims (79), then position the elbow and support (64) in the helicopter.

5. Engage the tube (71) on the elbow and support (64).

6. Install the bolts (65), aluminum washer (66), and the nuts (70).

7. Make sure that a shim 212-001-168-1 (79) is installed between the structure and the elbow and support (64).

8. Apply the barrier tape (C-456) between the faying surfaces of the structure and the stick assembly (1), then place the stick assembly in the helicopter.

9. Engage the tube (25) onto the stick assembly (1).

10. Install the screws (2), aluminum washers (13), and the nuts (14)

11. Install the bolts (4), washers (3), aluminum washers (15), and nuts (16). Tighten the nuts (14 and 16).

NOTE

After torquing, the tapered bushings (12) must protrude within the 0.030 to 0.060 inch (0.761 to 1.525 mm) limits shown in the Figure 2.

12. Connect the tube (71, Figure 1) to the elbow and support (64) as follows:

12.1. Align the holes in the tube (71) with the holes in the elbow and support (64).

12.2. At two locations, wet install the bolt (67) and the tapered bushings (68) with the sealant (C-251) and the washers (73) and the nut (74).

12.3. At two locations, torque the nut (74) 50 to 70 inch-pounds (5.65 to 7.90 Nm).

12.4. Cover the bolt (67) ends with the sealant (C-251).

12.5. Apply a bead of the sealant (C-251) to cover the edges of the tube (71) and the elbow and support (64).

13. Connect the tubes (25 and 71) to the lever (22) as follows:

13.1. Align the holes in the tubes (25 and 71) with the holes in the lever (22).

13.2. At four locations, wet install the bolts (23) and the tapered bushings (24) with the sealant (C-251) and the washers (27) and the nut (74).

13.3. At four locations, torque the nut (74) 50 to 70 inch-pounds (5.65 to 7.90 Nm).

13.4. Cover the bolt (23) ends with the sealant (C-251).

13.5. Apply a bead of the sealant (C-251) to cover the edges of the tubes (25 and 71) and the lever (22).

14. Connect the tube (25) to the stick assembly (1) as follows:

14.1. Align the holes in the tube (25) with the holes in the stick assembly (1).

14.2. At two locations, wet install the bolt (18) and the tapered bushings (12) with the sealant (C-251) and the washers (1) and the nut (10).

14.3. At two locations, torque the nut (10) 50 to 70 inch-pounds (5.65 to 7.90 Nm).

14.4. Cover the bolt (18) ends with the sealant (C-251).

14.5. Apply a bead of the sealant (C-251) to cover the edges of the tube (25) and the stick assembly (1).

15. Connect the tubes (52) and the stick assembly (1) with the bolts (6), thin aluminum washers (7), and nuts (8). Tighten the nuts (8). Safety the nuts (8) with the new cotter pins (9).

16. Put the tubes (59) in position and install the bolts (78), thin aluminum washers (75), and nuts (77). Tighten the nuts and safety with new cotter pins (76).

17. Secure the inboard end of the tubes (59), with the bolts (63), aluminum washer (55), and nuts (54). Tighten the nuts and safety with new cotter pins (53).

18. Secure the tubes (52) to the tubes (59) with the bolts (61), washers (60), aluminum washers (58), and nuts (57). Tighten the nuts and safety with new cotter pins (56).

19. Install the tube (47) on the bellcrank (44) with the bolt (43), washers (42 and 45), and nut (46). Tighten the nut (46).

20. Install the tube (50) on the bellcrank (51) with the bolt (41), washers (40 and 48), and nut (49). Tighten the nut (46).

21. Connect the flight control tube (31) to the lever (22) with the bolt (29), washer (30), and nut (33). Tighten the nut (33) and safety with a new cotter pin (34).

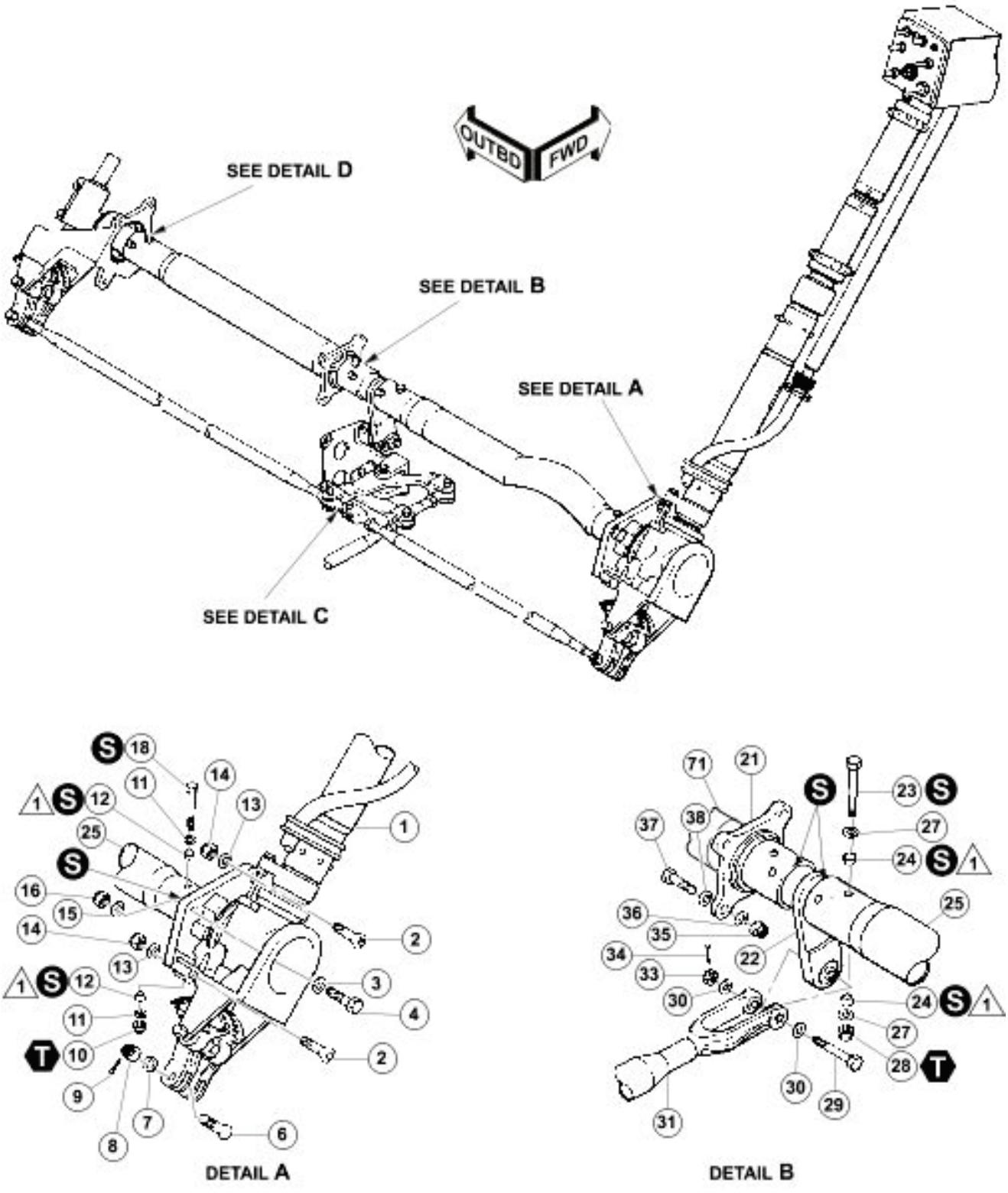
22. Connect the stick assembly (1) electrical connector.

23. Rig the collective flight controls (412-A-67-01-00-01A-276A-A / 01323).

24. Rig the power controls (throttle) (412-A-76-10-00-00A-276A-A / 01725).

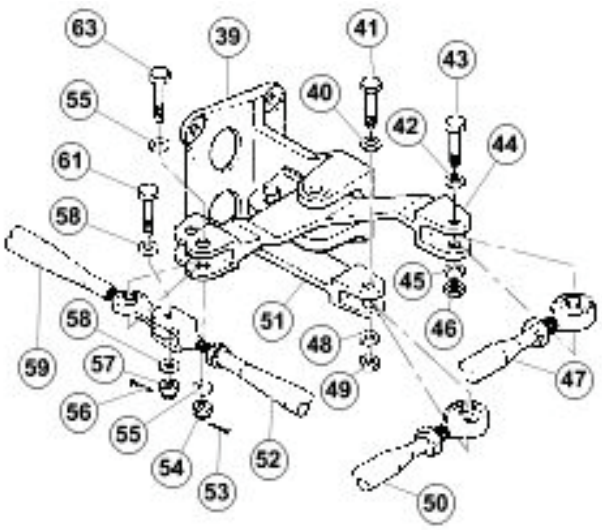
25. Install the pilot collective stick boot (412-A-67-11-00-16A-720A-A / 01381).

26. Install the following access panels (412-A-52-40-00-02A-720A-A / 00840):
- Cabin floor aft of pedestal
- Cabin floor under left crew seat
- Cabin floor left side of left crew seat.

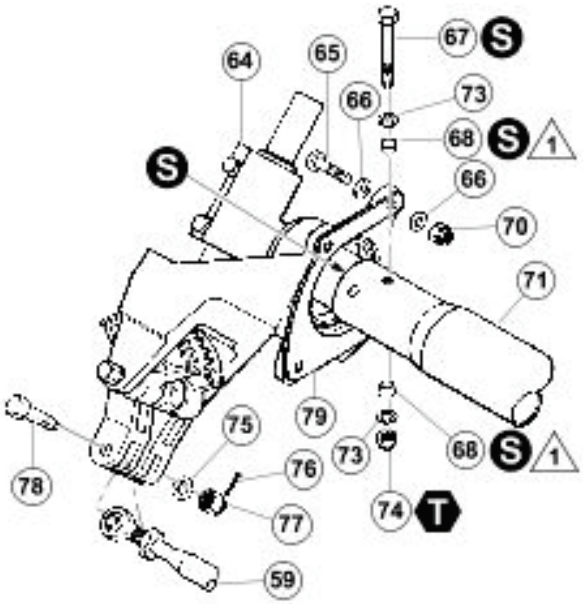


ICN-412-A-671100-B-97499-00021-A-001-01

Figure 1. Collective Jackshaft (Sheet 1 of 2)



DETAIL C



DETAIL D

1. Stick assembly
2. Screw
3. Washer (NAS1197-416)
4. Bolt
5. Deleted
6. Bolt
7. Thin aluminum washer
8. Nut
9. Cotter pin
10. Nut
11. Washer
12. Tapered bushing
13. Aluminum washer
14. Nut
15. Aluminum washer
16. Nut
17. Deleted
18. Bolt
19. Deleted
20. Deleted
21. Support
22. Lever
23. Bolt
24. Tapered bushing
25. Tube
26. Deleted
27. Washer
28. Nut
29. Bolt
30. Aluminum washer
31. Flight control tube
32. Deleted
33. Nut
34. Cotter pin
35. Nut
36. Aluminum washer
37. Bolt
38. Washer (NAS1197-416)
39. Support
40. Washer
41. Bolt
42. Washer
43. Bolt
44. Bellcrank
45. Washer
46. Nut
47. Tube
48. Washer
49. Nut
50. Tube
51. Bellcrank
52. Tube
53. Cotter pin
54. Nut
55. Aluminum washer
56. Cotter pin
57. Nut
58. Aluminum washer
59. Tube
60. Deleted
61. Bolt
62. Deleted
63. Bolt
64. Elbow and support
65. Bolt
66. Aluminum washer
67. Bolt
68. Tapered bushing
69. Deleted
70. Nut
71. Tube
72. Deleted
73. Washer
74. Nut
75. Thin aluminum washer
76. Cotter pin
77. Nut
78. Bolt
79. Shim

S SEALANT (C-251)

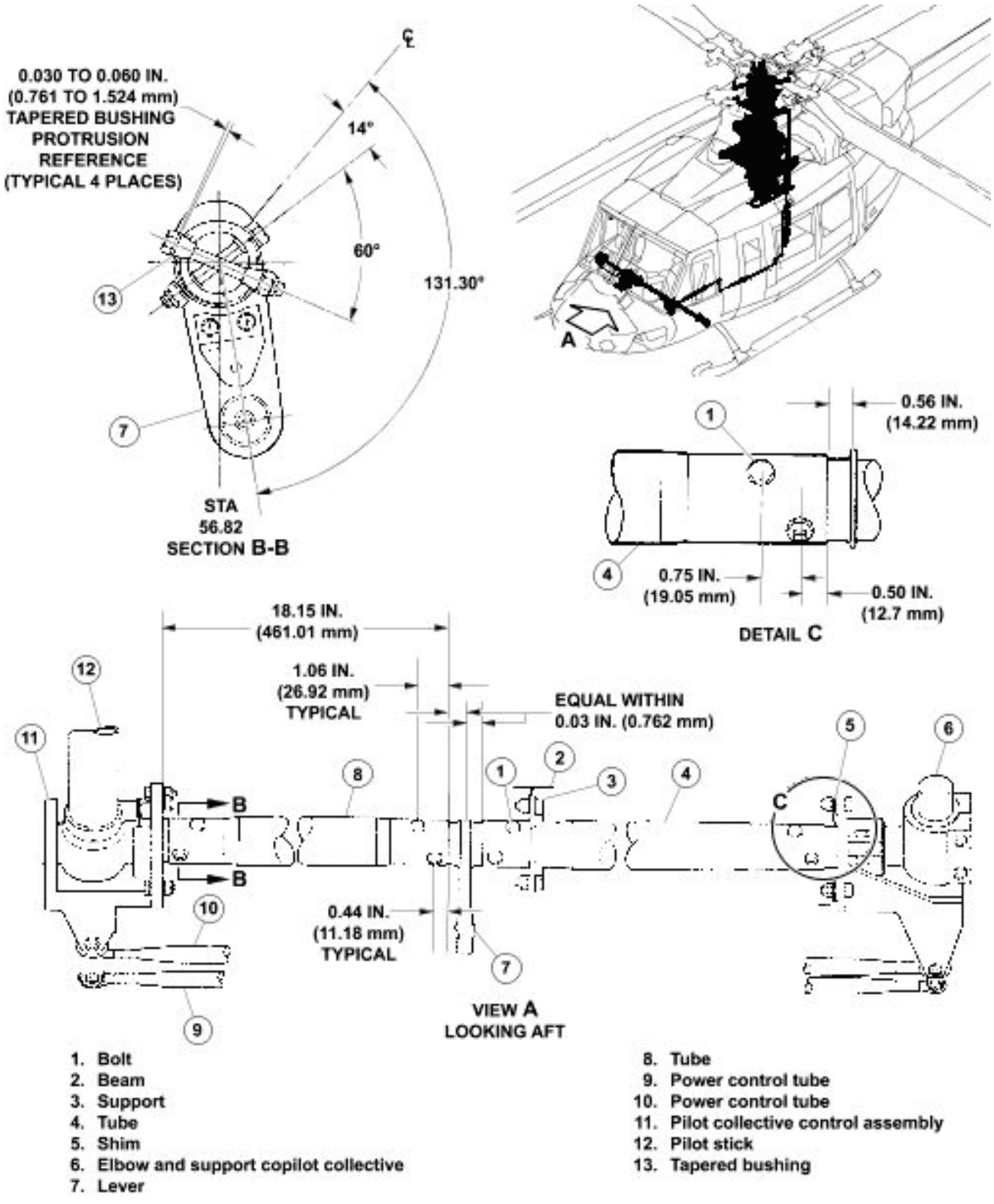
T 50 TO 70 IN-LBS (5.65 TO 7.90 Nm)

NOTE

1 For helicopters S/N 36487 and subsequent, if protrusion of tapered bushing (P/N 20-037-4-7) is less than 0.030 inch (0.76 mm), replace tapered bushing with P/N 20-037-4-8.

ICN-412-A-671100-B-97499-00022-A-001-01

Figure 1. Collective Jackshaft (Sheet 2 of 2)



ICN-412-A-671100-B-97499-00024-A-001-01

Figure 2. Collective Jackshaft Assembly (Sheet 1 of 1)

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Collective Transducers - Removal

Procedure

NOTE

SN: 33001-36036:

Helicopters use a single control motion transducer (2, Figure 1) 412-074-101-101. The transducer has one element and is installed with the Flight Director Kit (412-706-111-103).

NOTE

SN: 36037-36086:

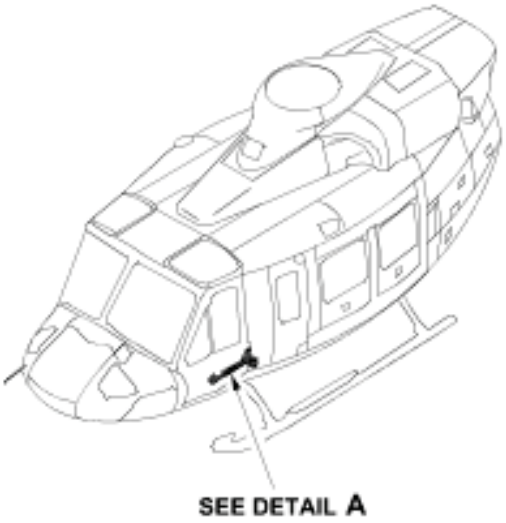
Helicopters use one control motion transducer (3) 412-074-101-103. The transducer has two elements and is installed as part of the basic ship. Transducer element one is used by the Flight Director Kit (412-706-111-107), if installed. Transducer element two is used by the Flight Data Recorder (FDR) , if installed.

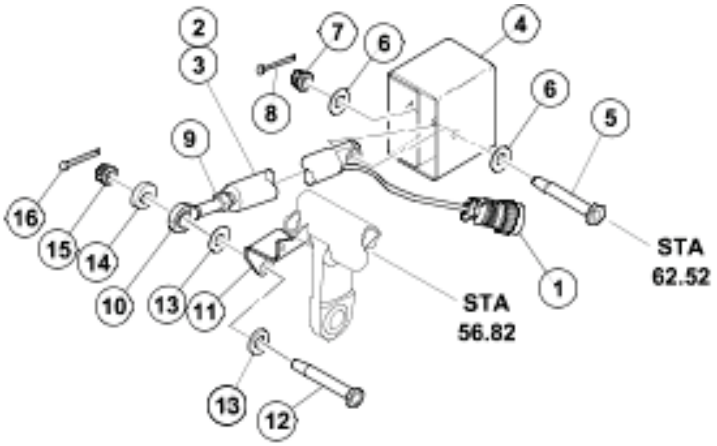
1. Remove the access panel from the cabin floor forward of the pylon.

2. Disconnect the electrical connector (1).

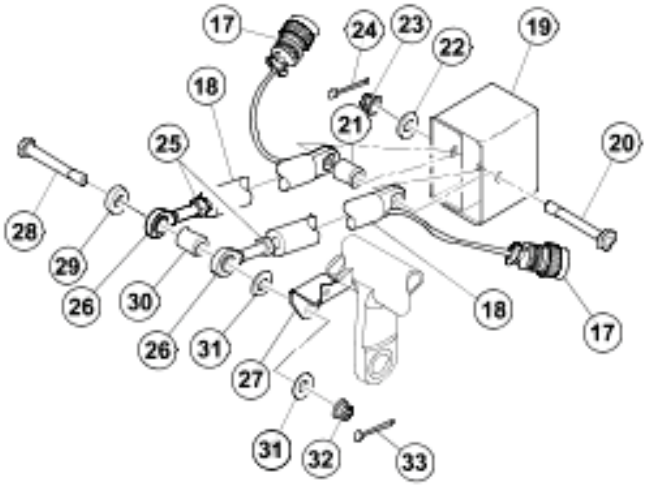
3. To disconnect the control motion transducer (2 or 3) from the support (4), remove the cotter pin (8), nut (7), washers (6), and bolt (5).

4. To disconnect the transducer rod end (10) from the bracket (11), remove the cotter pin (16), nut (15), spacer (14), washers (13), and bolt (12). Remove the transducer (2 or 3).

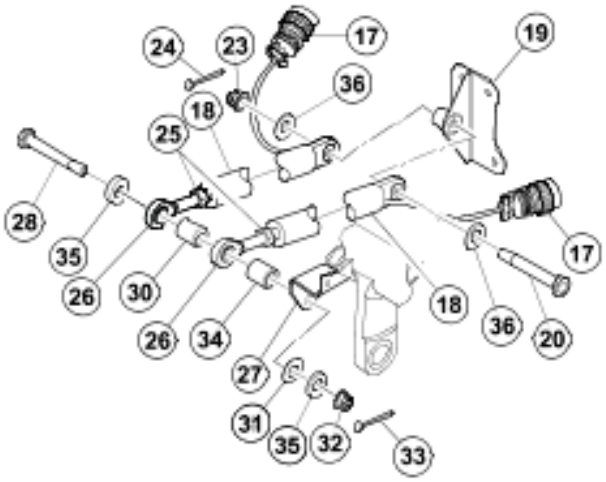




HELICOPTERS S/N 33001 THROUGH 36086



HELICOPTERS S/N 36087 THROUGH 36693



HELICOPTERS S/N 36694 AND SUBSEQUENT

DETAIL A

1. Connector
2. Control motion transducer
3. Control motion transducer
4. Support
5. Bolt
6. Washer
7. Nut
8. Cotter pin
9. Jam nut
10. Rod end
11. Bracket
12. Bolt
13. Washer
14. Spacer
15. Nut
16. Cotter pin
17. Connector
18. Control motion transducer
19. Support
20. Bolt
21. Spacer
22. Spacer
23. Nut
24. Cotter pin
25. Jam nut
26. Rod end
27. Bracket
28. Bolt
29. Spacer
30. Spacer
31. Washer
32. Nut
33. Cotter pin
34. Spacer
35. Washer
36. Washer

ICN-412-A-671100-B-97499-00023-A-001-01

Figure 1. Collective Transducers (Sheet 1 of 1)

Collective Transducers - Removal

Procedure

NOTE

SN: 36087-36693:

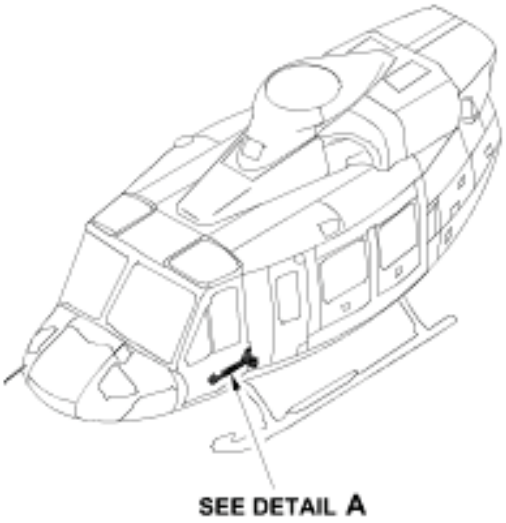
Helicopters use two control motion transducers (18, Figure 1) 412-074-101-103. Each transducer has two elements. One element on each transducer is used as an input to the autopilot and the second element of one transducer is used for the Flight Data Recorder (FDR) , if installed.

1. Remove the access panel from the cabin floor forward of the pylon.

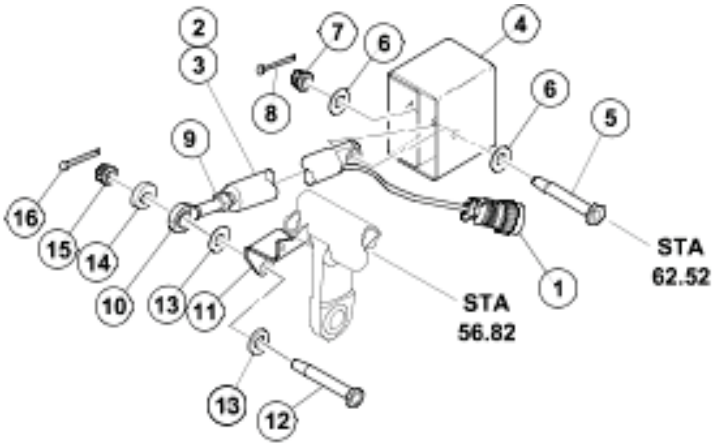
2. Disconnect the electrical connectors (17).

3. To detach the transducers (18) from the support (19), remove the cotter pin (24), nut (23), spacer (21), washer (22), and bolt (20).

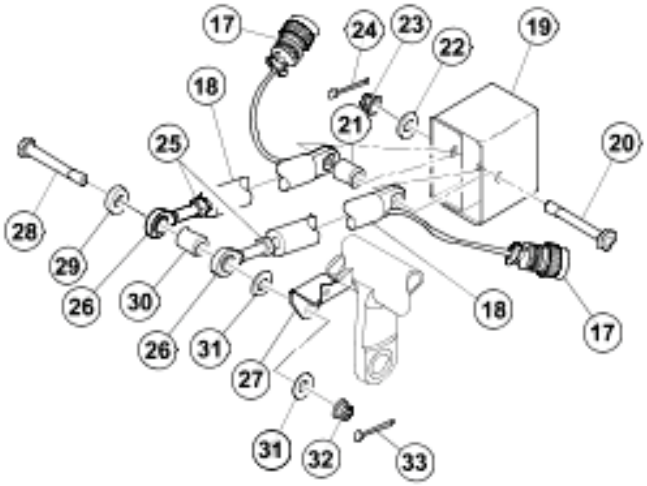
4. To detach the transducer rod ends (26) from the bracket (27), remove the cotter pin (33), nut (32), washers (31), spacers (29 and 30), and bolt (28). Remove the transducer (18).



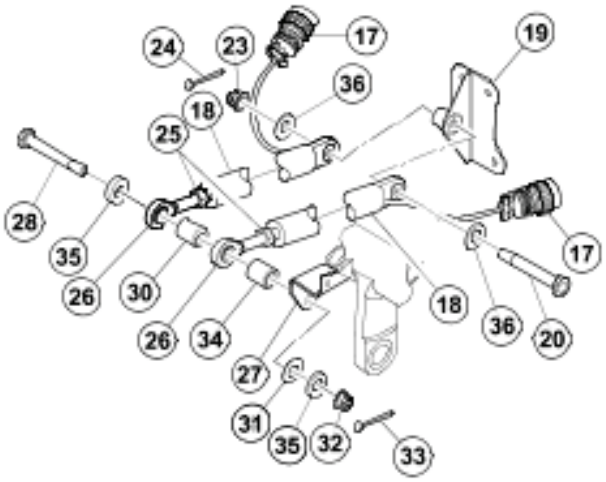
1. Connector
2. Control motion transducer
3. Control motion transducer
4. Support
5. Bolt
6. Washer
7. Nut
8. Cotter pin
9. Jam nut
10. Rod end
11. Bracket
12. Bolt
13. Washer
14. Spacer
15. Nut
16. Cotter pin
17. Connector
18. Control motion transducer
19. Support
20. Bolt
21. Spacer
22. Spacer
23. Nut
24. Cotter pin
25. Jam nut
26. Rod end
27. Bracket
28. Bolt
29. Spacer
30. Spacer
31. Washer
32. Nut
33. Cotter pin
34. Spacer
35. Washer
36. Washer



HELICOPTERS S/N 33001 THROUGH 36086



HELICOPTERS S/N 36087 THROUGH 36693



HELICOPTERS S/N 36694 AND SUBSEQUENT

DETAIL A

ICN-412-A-671100-B-97499-00023-A-001-01

Figure 1. Collective Transducers (Sheet 1 of 1)

Collective Transducers - Removal

Procedure

NOTE

SN: 36694-Subsequent:

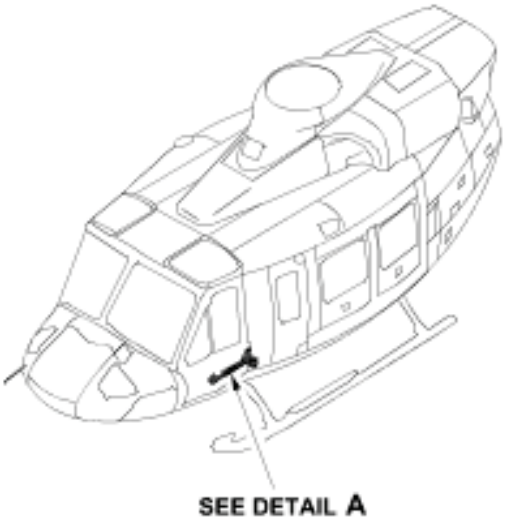
Helicopters use two control motion transducers (18, Figure 1) 412-074-101-103. Each transducer has two elements. One element on each transducer is used as an input to the autopilot and the second element of one transducer is used for the Flight Data Recorder (FDR) , if installed.

1. Remove the access panel from the cabin floor forward of the pylon.

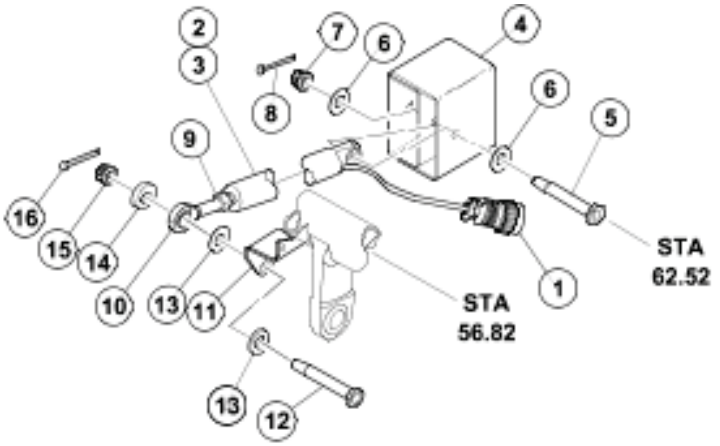
2. Disconnect the electrical connectors (17).

3. To detach the transducers (18) from the support (19), remove the cotter pin (24), nut (23), washers (35), and bolt (20).

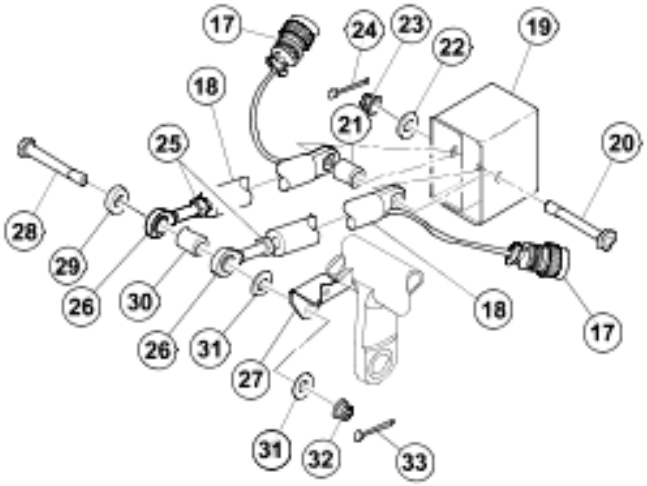
4. To detach the transducer rod ends (26) from the bracket (27), remove the cotter pin (33), nut (32), washers (31 and 36), spacers (30 and 34), and bolt (28). Remove the transducers (18).



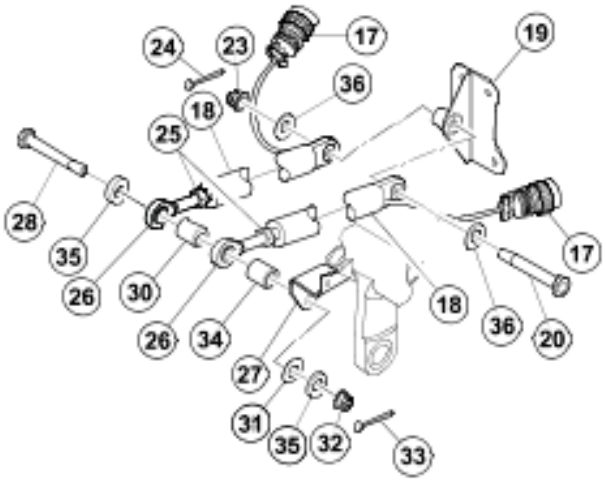
1. Connector
2. Control motion transducer
3. Control motion transducer
4. Support
5. Bolt
6. Washer
7. Nut
8. Cotter pin
9. Jam nut
10. Rod end
11. Bracket
12. Bolt
13. Washer
14. Spacer
15. Nut
16. Cotter pin
17. Connector
18. Control motion transducer
19. Support
20. Bolt
21. Spacer
22. Spacer
23. Nut
24. Cotter pin
25. Jam nut
26. Rod end
27. Bracket
28. Bolt
29. Spacer
30. Spacer
31. Washer
32. Nut
33. Cotter pin
34. Spacer
35. Washer
36. Washer



HELICOPTERS S/N 33001 THROUGH 36086



HELICOPTERS S/N 36087 THROUGH 36693



HELICOPTERS S/N 36694 AND SUBSEQUENT

DETAIL A

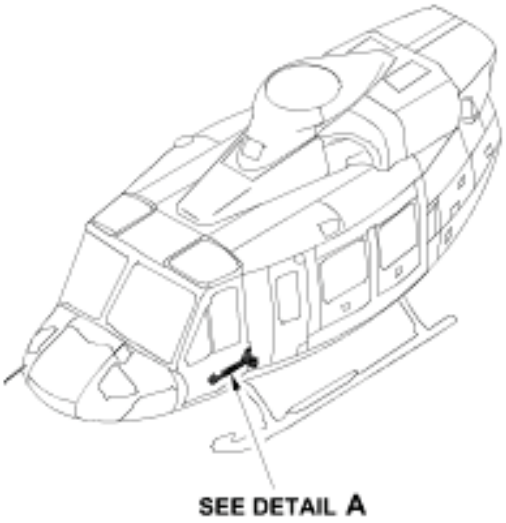
ICN-412-A-671100-B-97499-00023-A-001-01

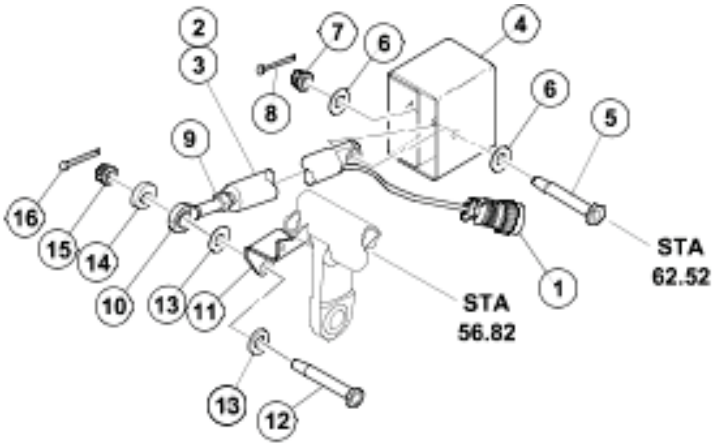
Figure 1. Collective Transducers (Sheet 1 of 1)

Collective Transducers - Inspection

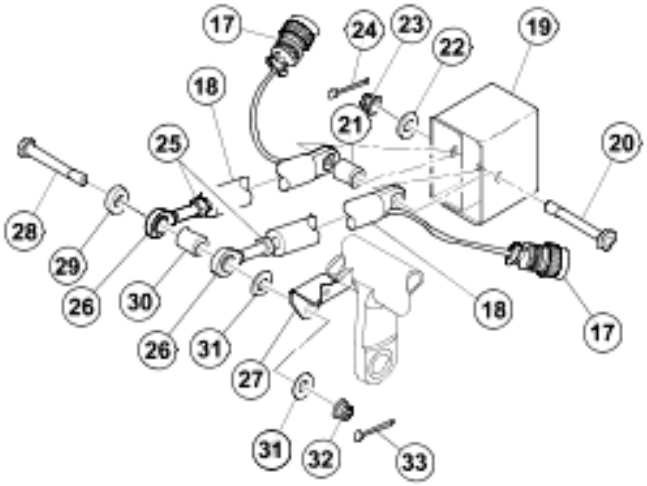
Procedure

1. Inspect the control motion transducers (2, 3, or 18, Figure 1) for mechanical and corrosion damage.
2. Inspect the bearings in the control motion transducers (2, 3, or 18) for secure installation and wear (BHT-ALL-SPM, Standard Practices Manual).
3. Inspect the electrical connectors (1 or 17) for bent or corroded pins and damaged threads. Superficial damage is acceptable without repair.

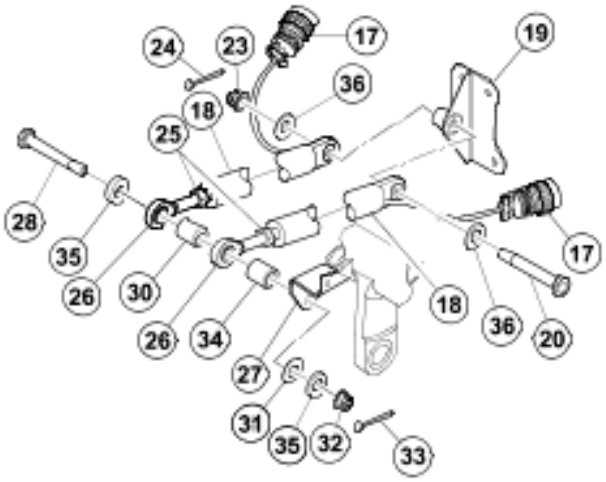




HELICOPTERS S/N 33001 THROUGH 36086



HELICOPTERS S/N 36087 THROUGH 36693



HELICOPTERS S/N 36694 AND SUBSEQUENT

DETAIL A

1. Connector
2. Control motion transducer
3. Control motion transducer
4. Support
5. Bolt
6. Washer
7. Nut
8. Cotter pin
9. Jam nut
10. Rod end
11. Bracket
12. Bolt
13. Washer
14. Spacer
15. Nut
16. Cotter pin
17. Connector
18. Control motion transducer
19. Support
20. Bolt
21. Spacer
22. Spacer
23. Nut
24. Cotter pin
25. Jam nut
26. Rod end
27. Bracket
28. Bolt
29. Spacer
30. Spacer
31. Washer
32. Nut
33. Cotter pin
34. Spacer
35. Washer
36. Washer

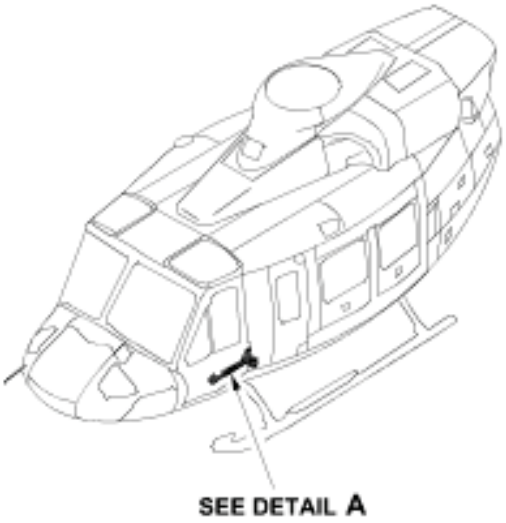
ICN-412-A-671100-B-97499-00023-A-001-01

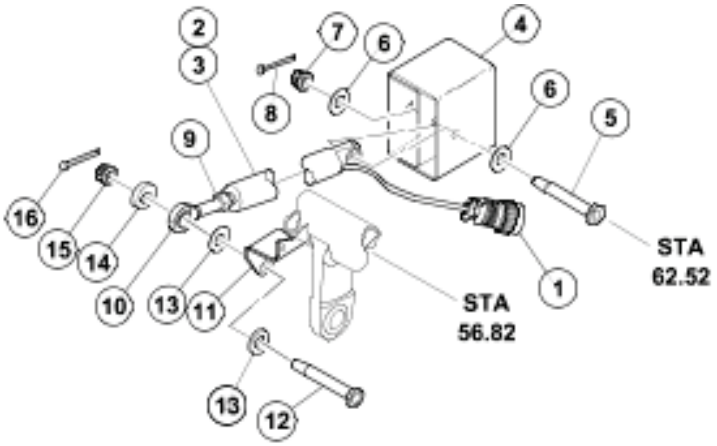
Figure 1. Collective Transducers (Sheet 1 of 1)

Collective Transducers - Repair

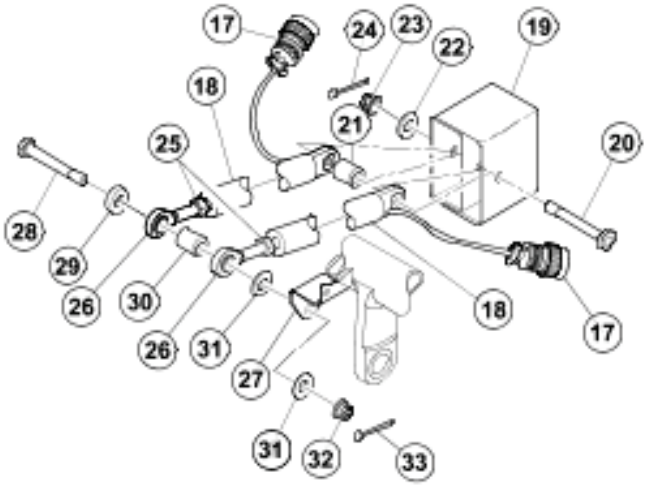
Procedure

1. The control motion transducers (2, 3, or 18, Figure 1) are not repairable and must be replaced when defective.

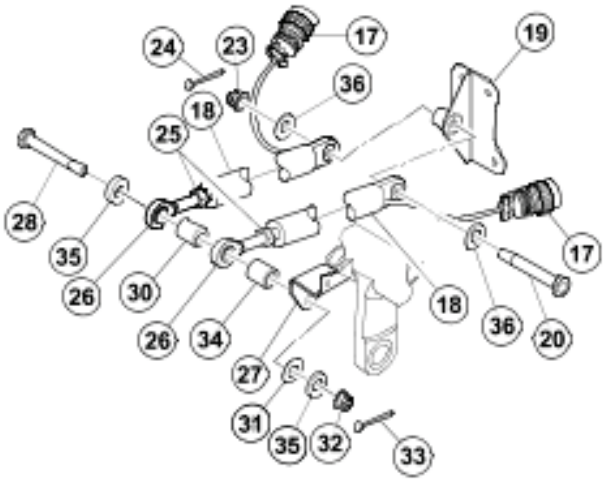




HELICOPTERS S/N 33001 THROUGH 36086



HELICOPTERS S/N 36087 THROUGH 36693



HELICOPTERS S/N 36694 AND SUBSEQUENT

DETAIL A

1. Connector
2. Control motion transducer
3. Control motion transducer
4. Support
5. Bolt
6. Washer
7. Nut
8. Cotter pin
9. Jam nut
10. Rod end
11. Bracket
12. Bolt
13. Washer
14. Spacer
15. Nut
16. Cotter pin
17. Connector
18. Control motion transducer
19. Support
20. Bolt
21. Spacer
22. Spacer
23. Nut
24. Cotter pin
25. Jam nut
26. Rod end
27. Bracket
28. Bolt
29. Spacer
30. Spacer
31. Washer
32. Nut
33. Cotter pin
34. Spacer
35. Washer
36. Washer

ICN-412-A-671100-B-97499-00023-A-001-01

Figure 1. Collective Transducers (Sheet 1 of 1)

Collective Transducers - Installation

Procedure

NOTE

SN: 33001-36036:

Helicopters use a single control motion transducer (2, Figure 1) 412-074-101-101. The transducer has one element and is installed with the Flight Director kit (412-706-111-103).

NOTE

SN: 36037-36086:

Helicopters use one control motion transducer (3) 412-074-101-103. The transducer has two elements and is installed as part of the basic ship. Transducer element one is used by the Flight Director kit (412-706-111-107), if installed. Transducer element two is used by the Flight Data Recorder (FDR) , if installed.

1. Position the control motion transducer (2 or 3) in the support (4) and install the bolt (5), washers (6), nut (7), and cotter pin (8).

2. Position the collective stick against the up and down stop to check the control motion transducer (2 or 3) for bottoming out in either direction, as if it were installed. Make sure that the control motion transducer can travel beyond the up and down stop of the collective stick. If adjustments are needed, loosen the jam nut (9) and turn the rod end (10).

3. Attach the control motion transducer (2 or 3) to the bracket (11) with the bolt (12), washers (13), spacer (14), nut (15), and cotter pin (16).

NOTE

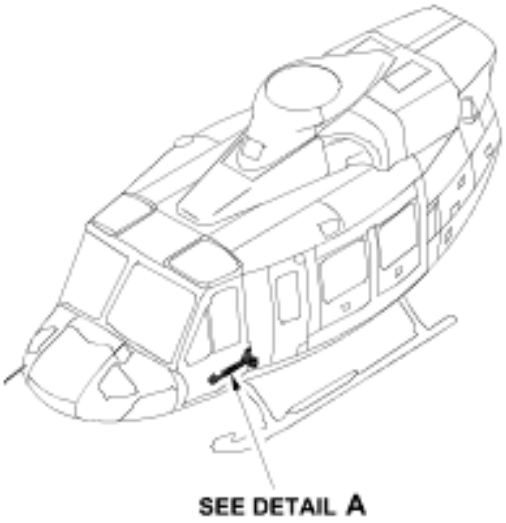
SN: 36037-36086:

The following step is only applicable to helicopters with transducer 412-074-101-103 installed.

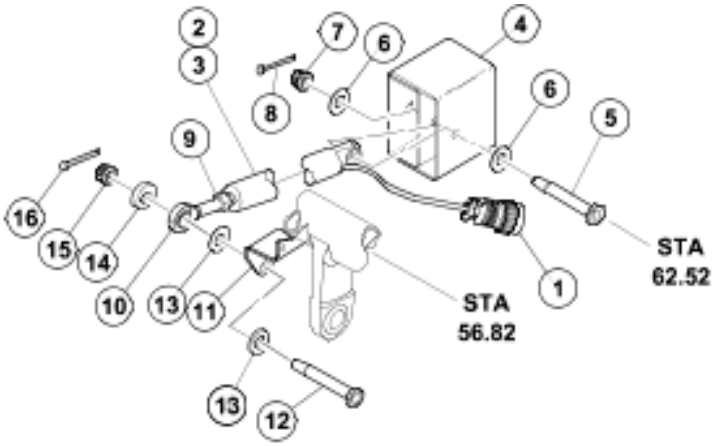
4. Perform the collective transducer adjustment (412-A-67-11-00-18A-271A-A / 01395).

5. Connect the electrical connectors (1).

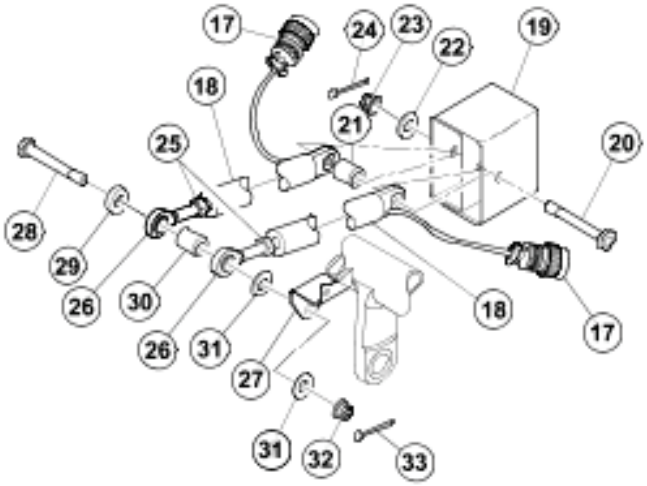
6. Install the access panel in the cabin floor, forward of the pylon.



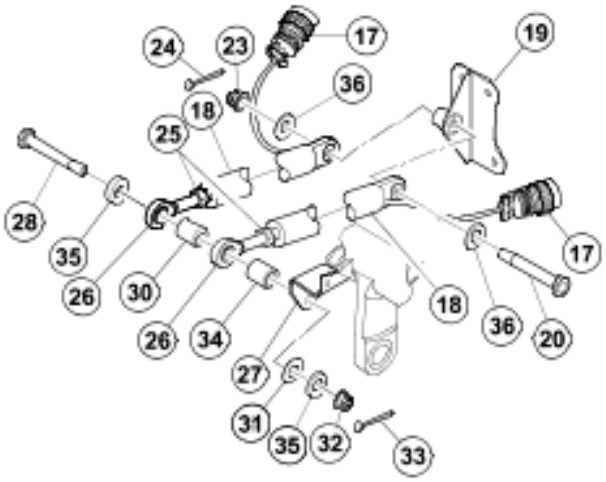
1. Connector
2. Control motion transducer
3. Control motion transducer
4. Support
5. Bolt
6. Washer
7. Nut
8. Cotter pin
9. Jam nut
10. Rod end
11. Bracket
12. Bolt
13. Washer
14. Spacer
15. Nut
16. Cotter pin
17. Connector
18. Control motion transducer
19. Support
20. Bolt
21. Spacer
22. Spacer
23. Nut
24. Cotter pin
25. Jam nut
26. Rod end
27. Bracket
28. Bolt
29. Spacer
30. Spacer
31. Washer
32. Nut
33. Cotter pin
34. Spacer
35. Washer
36. Washer



HELICOPTERS S/N 33001 THROUGH 36086



HELICOPTERS S/N 36087 THROUGH 36693



HELICOPTERS S/N 36694 AND SUBSEQUENT

DETAIL A

ICN-412-A-671100-B-97499-00023-A-001-01

Figure 1. Collective Transducers (Sheet 1 of 1)

Collective Transducers - Installation

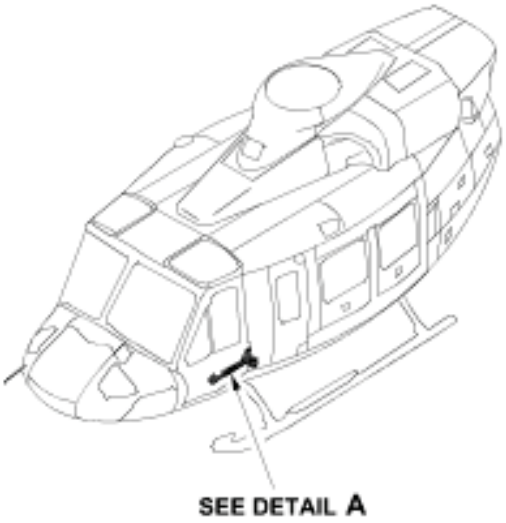
Procedure

NOTE

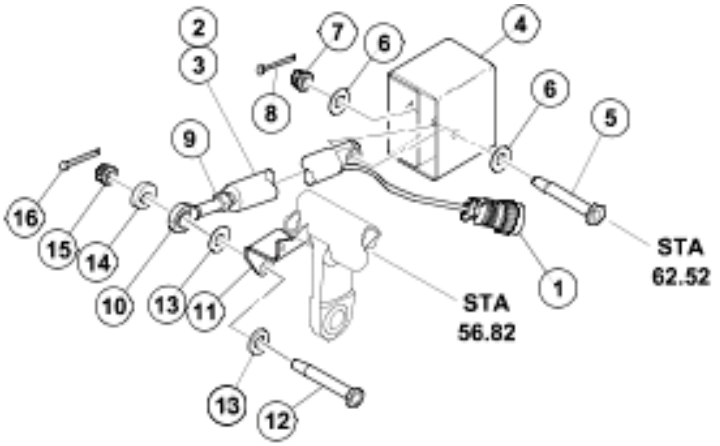
SN: 36087-36693:

Helicopters use two control motion transducers (18, Figure 1) 412-074-101-103. Each transducer has two elements. One element on each transducer is used as an input to the autopilot and the second element of one transducer is used for the Flight Data Recorder (FDR) , if installed.

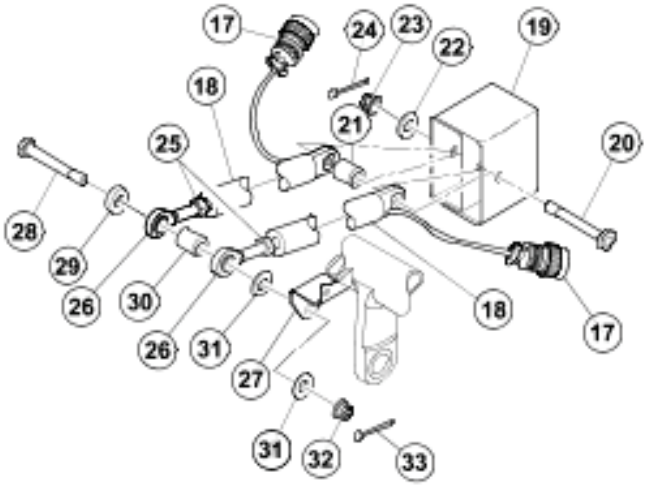
1. Position the control motion transducers (18) in the support (19) and install the bolt (20), spacer (21), washer (22), nut (23), and cotter pin (24).
2. Position the collective stick against the up and down stop to check the control motion transducers (18) for bottoming out in either direction, as if they were installed. Make sure that the transducer can travel beyond the up and down stop of the collective stick. If adjustments are needed, loosen the jam nut (25) and turn the rod ends (26).
3. To attach the transducer rod ends (26) to the bracket (27), install the bolt (28), spacers (29 and 30), washers (31), nut (32), and cotter pin (33).
4. Perform the collective transducer adjustment (412-A-67-11-00-18A-271A-A / 01395).
5. Connect the electrical connectors (17).
6. Install the access panel in the cabin floor, forward of the pylon.



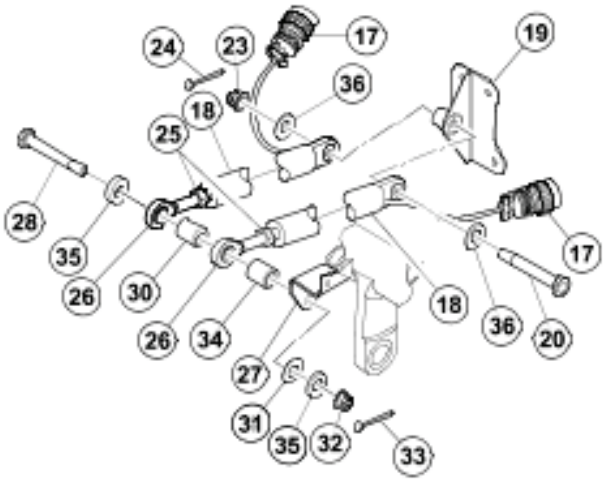
1. Connector
2. Control motion transducer
3. Control motion transducer
4. Support
5. Bolt
6. Washer
7. Nut
8. Cotter pin
9. Jam nut
10. Rod end
11. Bracket
12. Bolt
13. Washer
14. Spacer
15. Nut
16. Cotter pin
17. Connector
18. Control motion transducer
19. Support
20. Bolt
21. Spacer
22. Spacer
23. Nut
24. Cotter pin
25. Jam nut
26. Rod end
27. Bracket
28. Bolt
29. Spacer
30. Spacer
31. Washer
32. Nut
33. Cotter pin
34. Spacer
35. Washer
36. Washer



HELICOPTERS S/N 33001 THROUGH 36086



HELICOPTERS S/N 36087 THROUGH 36693



HELICOPTERS S/N 36694 AND SUBSEQUENT

DETAIL A

ICN-412-A-671100-B-97499-00023-A-001-01

Figure 1. Collective Transducers (Sheet 1 of 1)

Collective Transducers - Installation

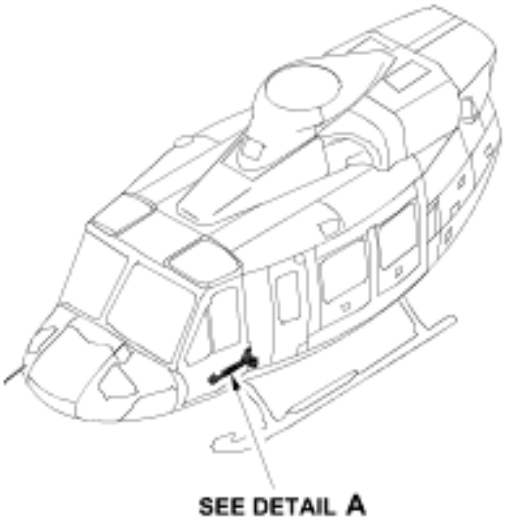
Procedure

NOTE

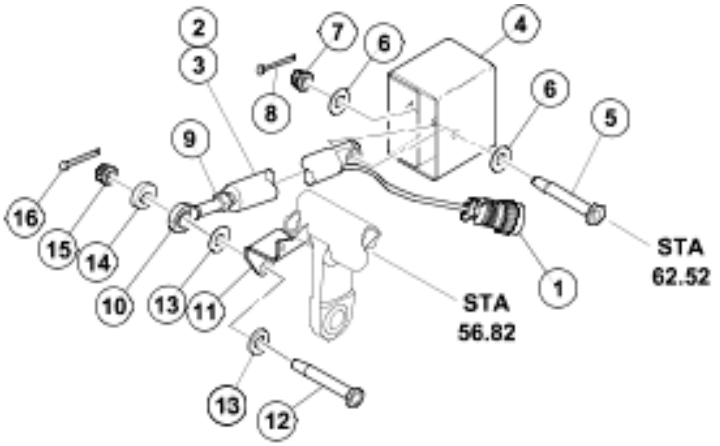
SN: 36694-Subsequent:

Helicopters use two control motion transducers (18, Figure 1) 412-074-101-103. Each transducer has two elements. One element on each transducer is used as an input to the autopilot and the second element of one transducer is used for the Flight Data Recorder (FDR) , if installed.

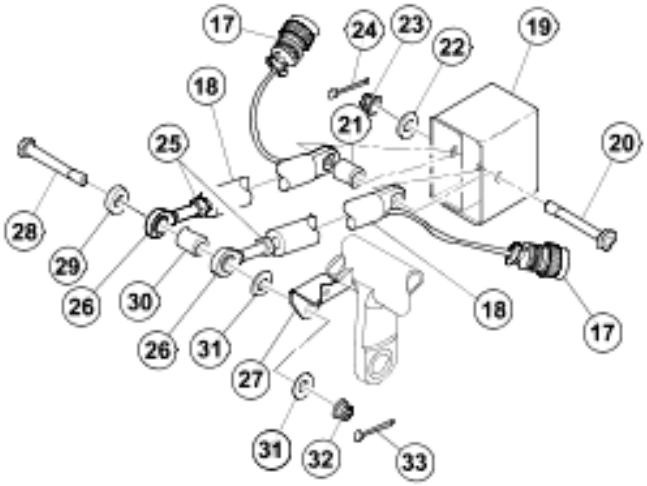
1. Position the control motion transducers (18) in the support (19) and install the bolt (20), washers (35), nut (23), and cotter pin (24).
2. Position the collective stick against the up and down stop to check the control motion transducers (18) for bottoming out in either direction, as if they were installed. Make sure that the transducer can travel beyond the up and down stop of the collective stick. If adjustments are needed, loosen the jam nut (25) and turn the rod ends (26).
3. To attach the transducer rod ends (26) to the bracket (27), install the bolt (28), spacers (30 and 34), washers (31 and 36), nut (32), and cotter pin (33).
4. Perform the collective transducer adjustment (412-A-67-11-00-18A-271A-A / 01395).
5. Connect the electrical connectors (17).
6. Install the access panel in the cabin floor forward of the pylon.



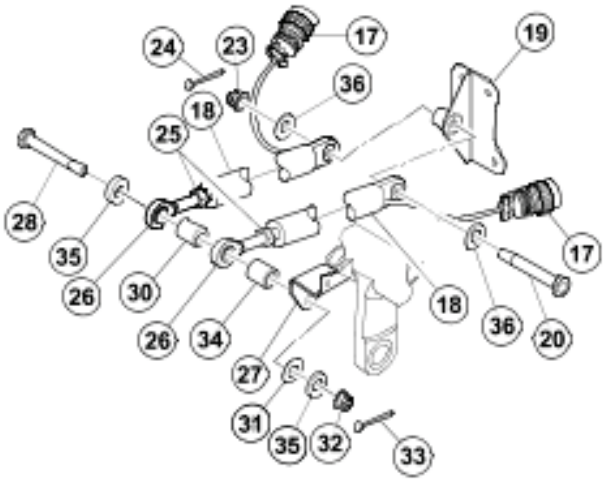
1. Connector
2. Control motion transducer
3. Control motion transducer
4. Support
5. Bolt
6. Washer
7. Nut
8. Cotter pin
9. Jam nut
10. Rod end
11. Bracket
12. Bolt
13. Washer
14. Spacer
15. Nut
16. Cotter pin
17. Connector
18. Control motion transducer
19. Support
20. Bolt
21. Spacer
22. Spacer
23. Nut
24. Cotter pin
25. Jam nut
26. Rod end
27. Bracket
28. Bolt
29. Spacer
30. Spacer
31. Washer
32. Nut
33. Cotter pin
34. Spacer
35. Washer
36. Washer



HELICOPTERS S/N 33001 THROUGH 36086



HELICOPTERS S/N 36087 THROUGH 36693



HELICOPTERS S/N 36694 AND SUBSEQUENT

DETAIL A

ICN-412-A-671100-B-97499-00023-A-001-01

Figure 1. Collective Transducers (Sheet 1 of 1)

Collective Transducers - Adjustment

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Calibration Tool (Pot Calibrator)	412-274-001-101	1	

Procedure

NOTE

SN: 36037-Subsequent:

Electrical adjustment is required for the dual element control motion transducers.

1. Do the collective control motion transducer (3 and/or 18, Figure 1, Detail A) adjustment as follows:

NOTE

Dual element control motion transducers 412-074-101-103 are aligned and positioned using the calibration tool (pot calibrator) (412-274-001-101). This calibration tool is used to electrically align the control motion transducers installed on the helicopter to be used as control position sensors for the flight data recorder. This tool will ensure the control motion transducers are aligned within ±0.5% of the prescribed position.

1.1. Connect the hydraulic power cart and turn on.

1.2. Position the pilot collective stick against the down stop.

1.3. Use the calibration tool (pot calibrator) (412-274-001-101) (4, Figure 2) (pot calibrator) to turn the calibration power switch (5) to the OFF position. Connect the calibration box connector (3) to one of the collective control motion transducers (1) electrical connector (2). Turn the selector knob (6) to the collective (CLTV) position.

1.4. Turn the calibration power switch (5) to the ON position.

NOTE

Any time the LO BAT lamp comes on, replace the batteries (use 9 VDC alkaline batteries). Low battery is set to illuminate at 12 ±1 volt.

1.5. If a RED lamp comes on, the control motion transducer (1) shaft shall be adjusted in the direction indicated. An extend indication means extend the transducer shaft by screwing it into the rod end bearing (10 or 26, Figure 1) (in clockwise direction when looking at rod end). A retract indication means retract the transducer shaft by unscrewing it from the rod end bearing (in counterclockwise direction when looking at the rod end). Adjust the transducer shaft until the GREEN IN RANGE lamp comes on.

1.6. Tighten the jam nut (9 or 25) on the transducer shaft and verify that the GREEN IN RANGE lamp is on and the RED lamps are off.

1.7. Move the calibration power switch (5, Figure 2) to the OFF position. Disconnect the calibration box connector (3) from the transducer electrical connector (2).

1.8. Connect the transducer electrical connector (2) to the electrical connector on the bulkhead.

1.9. Do the calibration adjustment on the remaining control motion transducer (1). Refer to Step 1.1. through Step 1.8..

2. An alternate method for control motion transducer (3 and/or 18, Figure 1, Detail A) adjustment is as follows:

2.1. Connect the hydraulic power cart and turn on.

2.2. Make sure the control motion transducer (3 or 18) is installed on the structure only.

2.3. Loosen the jam nut (9 or 25) on the transducer so the shaft can rotate freely.

2.4. Measure and record the resistance (T) between pins D and F of the transducer electrical connector. Resistance should be 5000 ±10% ohms.

2.5. Multiply the resistance (T) x 11.3% = __. Record value of T.

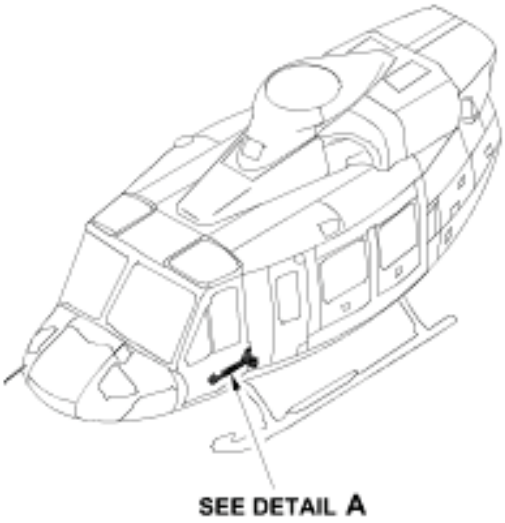
2.6. Measure and record the resistance (R) between pins E and F of the transducer electrical connector. Adjust the transducer shaft until resistance (R) equals the value of Step 2.5. ±25 ohms.

2.7. Tighten the jam nut on the transducer shaft and verify that the resistance is still within limits.

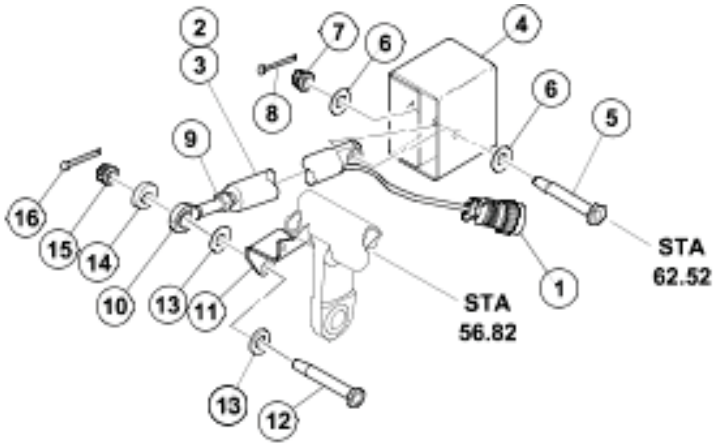
2.8. Connect the transducer electrical connector to the bulkhead.

2.9. Do the calibration adjustment on the remaining control motion transducer (1). Refer to Step 2.1. through Step 2.8..

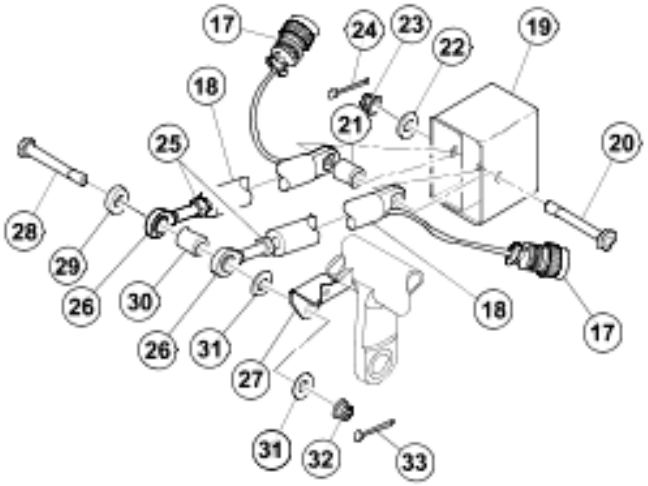
2.10. Turn off the hydraulic power cart and disconnect from the helicopter.



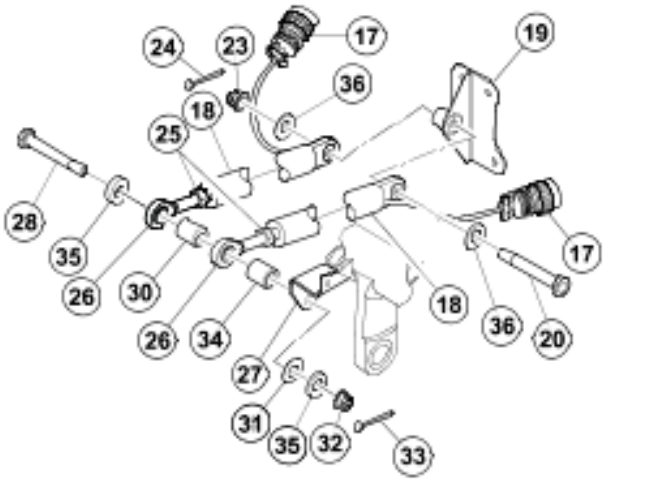
1. Connector
2. Control motion transducer
3. Control motion transducer
4. Support
5. Bolt
6. Washer
7. Nut
8. Cotter pin
9. Jam nut
10. Rod end
11. Bracket
12. Bolt
13. Washer
14. Spacer
15. Nut
16. Cotter pin
17. Connector
18. Control motion transducer
19. Support
20. Bolt
21. Spacer
22. Spacer
23. Nut
24. Cotter pin
25. Jam nut
26. Rod end
27. Bracket
28. Bolt
29. Spacer
30. Spacer
31. Washer
32. Nut
33. Cotter pin
34. Spacer
35. Washer
36. Washer



HELICOPTERS S/N 33001 THROUGH 36086



HELICOPTERS S/N 36087 THROUGH 36693

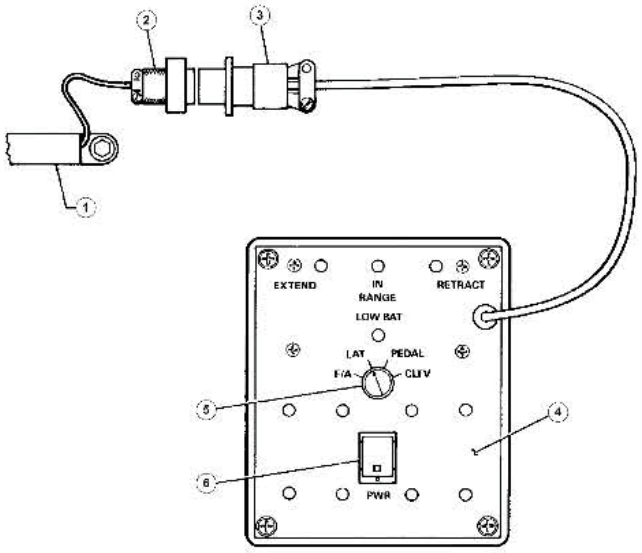


HELICOPTERS S/N 36694 AND SUBSEQUENT

DETAIL A

ICN-412-A-671100-B-97499-00023-A-001-01

Figure 1. Collective Transducers (Sheet 1 of 1)



1. Control motion transducer (Typical)
2. Electrical connector
3. Calibration box connector
4. Calibration tool (412-274-001-101)
5. Calibration power switch
6. Selector knob

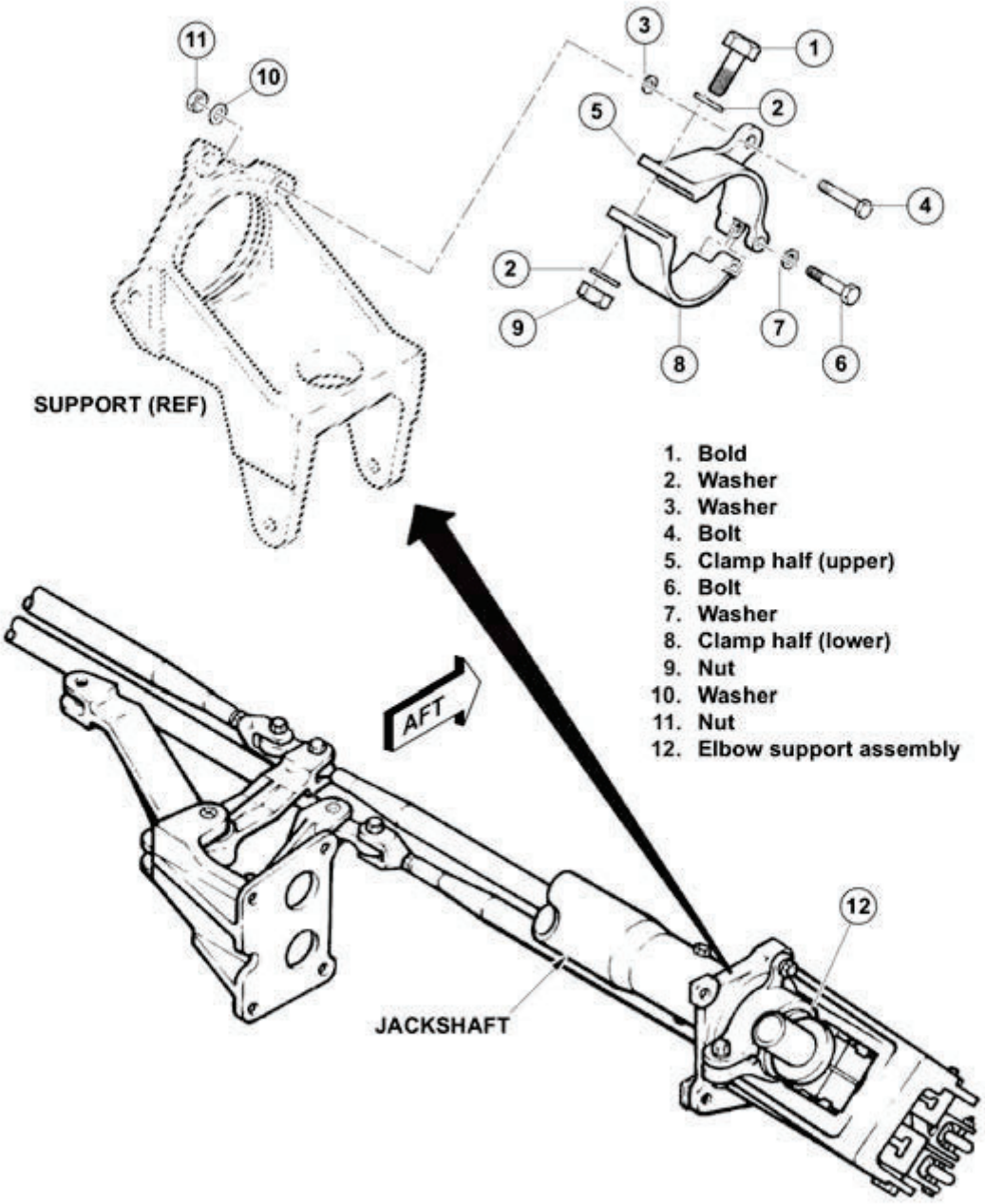
ICN-412-A-671100-B-97499-00025-A-001-01

Figure 2. Control Motion Transducer Adjustment (Sheet 1 of 1)

Friction Clamp - Removal

Procedure

1. Remove the copilot seat and access the panel below the seat (412-A-52-40-00-02A-520A-A / 00837).
2. Remove the bolt (4, Figure 1), washers (3 and 10), and nut (11) that attach the clamp halves (5 and 8) to the elbow support assembly (12).
3. Remove the bolt (1), washers (2), and nut (9).
4. Remove the bolt (6) and the washer (7) and remove the two halves of the clamp assembly from the elbow support assembly (12).



ICN-412-A-671100-B-97499-00026-A-001-01

Figure 1. Collective Friction Clamp Assembly (Sheet 1 of 1)

Friction Clamp - Inspection

Procedure

1. Inspect the clamp assembly for cracks or deformation. No cracks are allowed. Any deformation that prevents proper operation is cause for rejection.

2. Inspect the friction lining for excessive wear. If wear prevents proper operation, replace the friction liner.

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Friction Clamp - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
MEK	C-309	AR	
Adhesive	C-363	AR	
Abrasive pad	C-407	AR	
Abrasive cloth or paper	C-423	AR	
Anti-fretting buffer	C-465	AR	

Procedure

1. Remove all the old liner material and adhesive from the clamp halves (5 and 8, Figure 1) with 180 grit abrasive cloth or paper (C-423).

2. Rinse the clamp halves in clean water and dry with clean cloths or a heat gun.

3. Cut the anti-fretting buffer (C-465) material to approximately 1.0 x 3.0 inches (25.4 x 76.2 mm) size to match the inside areas of the clamp halves.

4. Lightly abrade the Dacron surface (pink side) of the liner material with 180 grit abrasive cloth or paper (C-423).

NOTE

The dark (Teflon) side of liner material is friction side.

5. Mix the adhesive (C-363) according to the manufacturers instructions.

NOTE

Pot life of adhesive not to exceed 30 minutes at 75°F (24°C).

6. Apply adhesive to the inner surface of both clamp halves and to the Dacron (pink) surface of the liner material. Install the liner in the clamp halves.

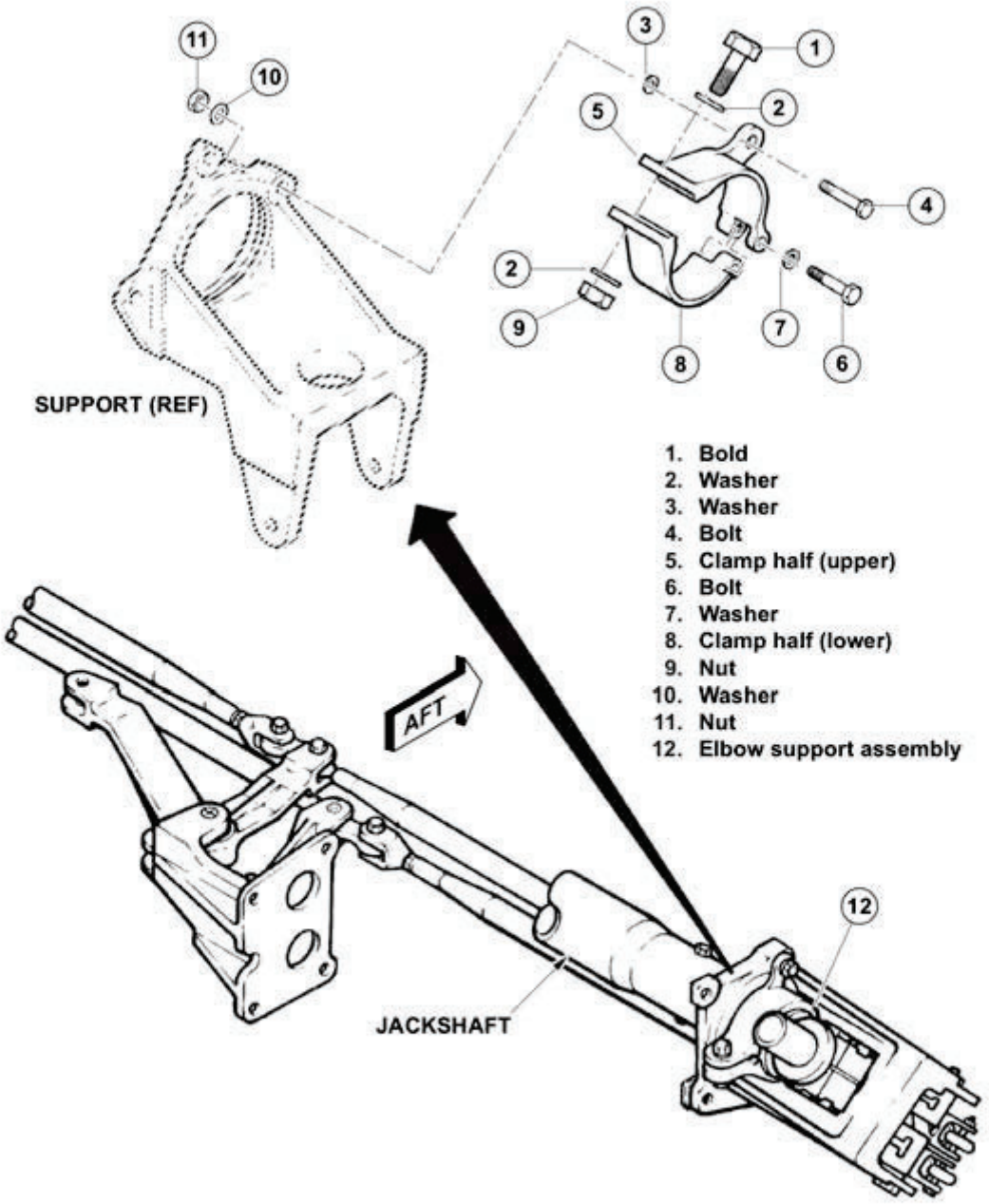
7. Install the clamp halves around the cylindrical workaid 2.03 to 2.09 inches (51.56 to 53.09 mm) diameter and install the bolt (6) and the washer (7).

8. Install the bolt (1), washers (2) and nut (9) through the tongues of the clamp halves (5 and 8) and tighten the nut to 1 inch-pound (0.11 Nm) greater than the nut tare torque.

9. Remove the adhesive squeeze-out. Allow the adhesive to cure 24 hours at 75°F (24°C) or heat the assembly to 175 to 190°F (80 to 88°C) for 1 hour.

10. After completely cured, remove the clamp assembly from the workaid and scuff the friction surface of the liner with 180 grit abrasive cloth or paper (C-423), followed by an abrasive pad (C-407), until a uniform finish is attained.

11. Wash the clamp assembly with mild soap and water followed by a rinse of MEK (C-309) to remove all grit and foreign material.



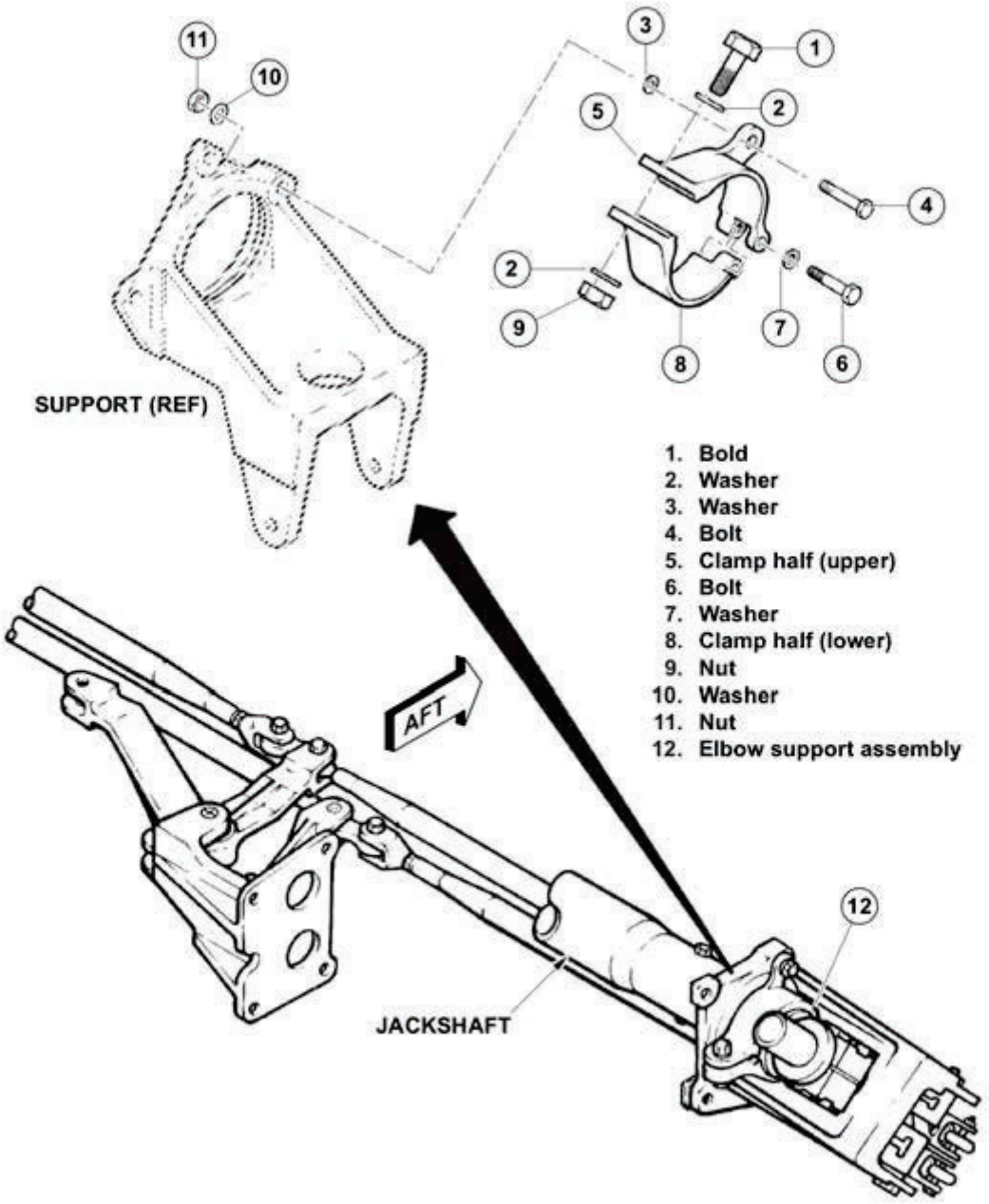
ICN-412-A-671100-B-97499-00026-A-001-01

Figure 1. Collective Friction Clamp Assembly (Sheet 1 of 1)

Friction Clamp - Installation

Procedure

1. Place the clamp halves around the elbow support assembly (12, Figure 1).
2. Install the washer (7) and the bolt (6).
3. Install the bolt (1), washers (2), and nut (9). Tighten the nut only enough to hold the clamp halves in contact with the elbow.
4. Align the hole in the tang on the upper clamp half (5) with the hole in the support and install the bolt (4) with the washer (3), between the clamp and support, and the nut (11) with the washer (10) under the nut. Tighten the nut.
5. Adjust the minimum collective friction (412-A-67-01-00-01A-276A-A / 01323).
6. Install the floor access panel and the seat (412-A-52-40-00-02A-720A-A / 00840).



ICN-412-A-671100-B-97499-00026-A-001-01

Figure 1. Collective Friction Clamp Assembly (Sheet 1 of 1)

Cyclic Controls - Description

The cyclic controls (Figure 1) consists of controls and components from pilot cyclic stick (22) up to and including extension tubes (1 and 34).

The AFCS cyclic control system incorporates two series actuators, one connected in roll system and one connected in pitch system. A control tube and idler arm are attached to series actuator in pitch system.

Four transducers, two in roll system and two in pitch system, are used to convert mechanical movement to electrical signals for injecting pilot control commands to AFCS.

Two transducers are attached to right end of cyclic jackshaft providing a fore and aft cyclic signal. Two transducers are attached to a forward cyclic bellcrank providing a lateral cyclic signal.

A force gradient assembly connected to an electromechanical rotary actuator is also installed in each cyclic system. Cyclic roll system force gradient attaches to a bellcrank forward of jackshaft. The cyclic pitch system force gradient attaches directly to left end of cyclic jackshaft.

A support assembly and idler are used to connect pitch series actuator and a pitch control tube, which is attached to mixing lever assembly.

Two proximity switches and two cyclic centering (CYC CTR) caution lights are installed by kit 412-704-115-101 for aid in centering swashplate. The lights are located in pilot and copilot instrument panel. The two proximity switches are located underneath floor near mixing lever assembly.

NOTE

Cyclic control AFCS system components forward of mixing lever assembly (28) are identified as "lateral cyclic" or "fore and aft cyclic" as applicable, i.e., "lateral cyclic force gradient".

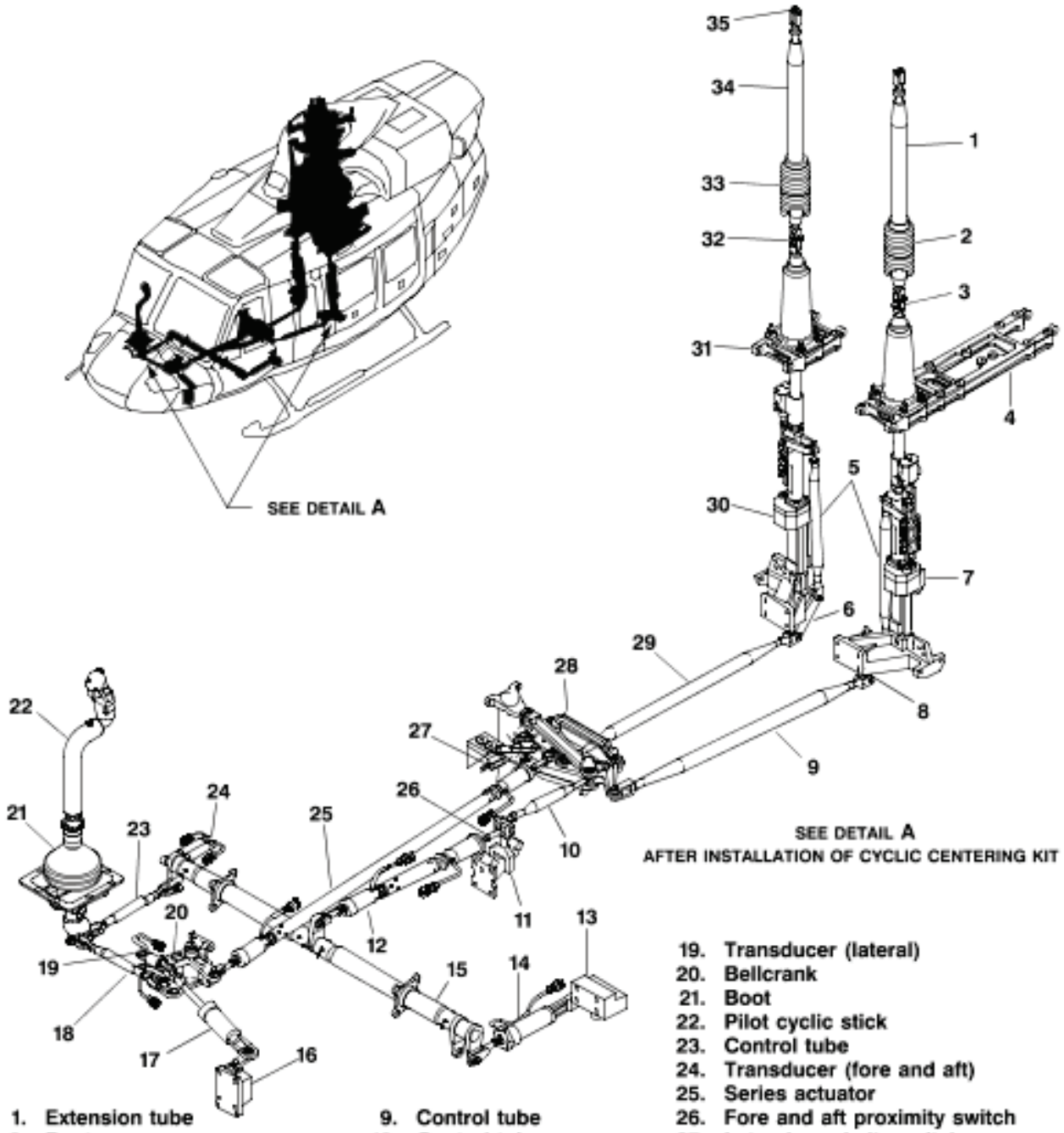
Cyclic control system components aft of mixing lever assembly (28) are identified as "left cyclic" or "right cyclic" as applicable, i.e., "left cyclic flight control cylinder".

The fore and aft control linkage and lateral control linkage are separate from cyclic stick to mixing lever (28). From mixing levers to swashplate effects of control linkages are combined and cannot be considered separately as to effect.

Actuation of pilot cyclic stick (22) causes hydraulic power assisted cyclic pitch change of main rotor blades.

The forward series actuators in each cyclic system are connected to Helipilot (HP-1) system. Aft series actuators in each cyclic system are connected to Helipilot (HP-2) system.

Refer to 412-A-98-00-00-10B-051A-A / 02570, for AFCS system wiring diagrams.



1. Extension tube
2. Boot
3. Universal
4. Left cyclic and collective boost support
5. Control tube
6. Bellcrank
7. Left cyclic flight control cylinder
8. Bellcrank
9. Control tube
10. Control tube
11. Idler
12. Series actuator
13. Rotary actuator
14. Force gradient
15. Cyclic jackshaft
16. Rotary actuator
17. Force gradient
18. Control tube
19. Transducer (lateral)
20. Bellcrank
21. Boot
22. Pilot cyclic stick
23. Control tube
24. Transducer (fore and aft)
25. Series actuator
26. Fore and aft proximity switch
27. Lateral proximity switch
28. Mixing lever assembly
29. Control tube
30. Right cyclic flight control cylinder
31. Right cyclic boost support
32. Universal
33. Boot
34. Extension tube
35. Bolt

ICN-412-A-671200-B-97499-00001-A-001-01

Figure 1. Cyclic Controls (Sheet 1 of 1)

Hydraulic Cylinder Extension Tube - Removal

Procedure

NOTE

The removal procedure for hydraulic cylinder extension tube (20, Figure 1) in left cyclic system is given. The removal procedure for corresponding hydraulic cylinder extension tube (23) in right cyclic system is the same.

1. Remove forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).

Applicable: SN: 33001-33100:

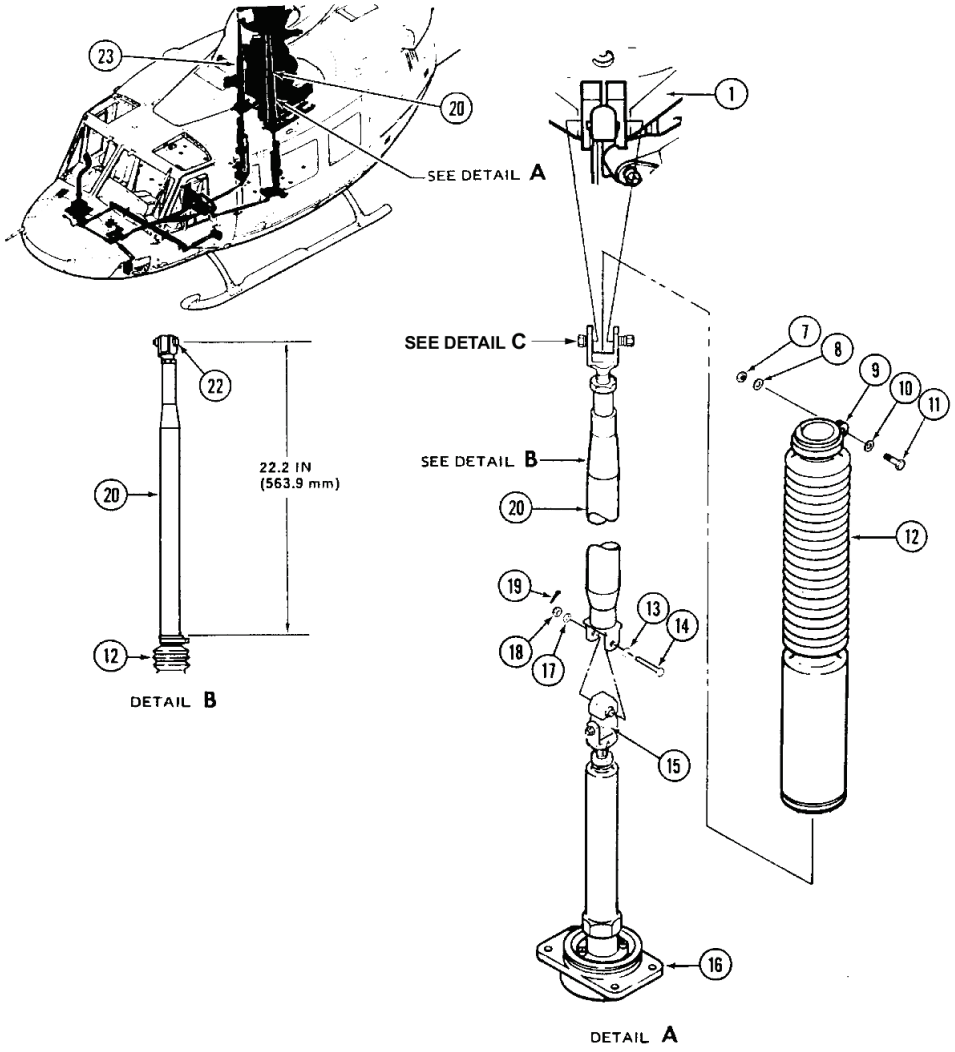
2. *Disconnect hydraulic cylinder extension tube (20) from swashplate (1) by removing cotter pin (6), nut (5), washers (4, 3, 2 and 21) and bolt (22).*

Applicable: SN: 33101-33213 or 36001-36999 or 37002-37999 or 38001-38999 or 39101-39999:

3. *Disconnect hydraulic cylinder extension tube (20) from swashplate (1) by removing cotter pin (6), nut (5), washers (4, 3 and 21) and bolt (22).*

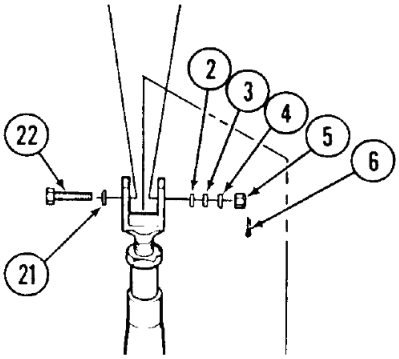
4. Remove nut (7), thin aluminum washers (8 and 10) and screw (11). Loosen bottom of boot (12) from lip on mount (16) and remove boot and clamp (9) from helicopter.

5. Remove cotter pin (19), nut (18), washers (13 and 17) and bolt (14). Separate hydraulic cylinder extension tube (20) from hydraulic cylinder extension universal (15).

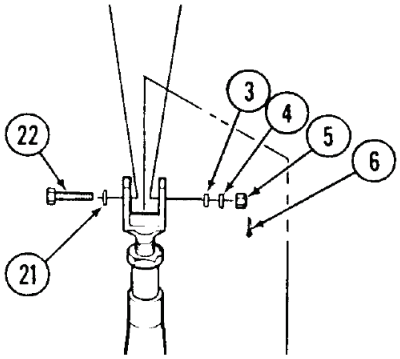


ICN-412-A-671200-B-97499-00002-A-002-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 1 of 2)



S/N 33001 through 33100



S/N 33101 through 33213, 36001 through 36999, 37002 through 37999, 38001 through 38999 and 39101 through 39999

DETAIL C

1. Swashplate
2. Washer (AN960C516)
3. Washer (AN960C516)
4. Washer (AN960C516L)
5. Nut
6. Cotter pin
7. Nut
8. Thin aluminum washer
9. Clamp
10. Thin aluminum washer
11. Screw
12. Boot
13. Washer (AN960C516)
14. Bolt
15. Hydraulic cylinder extension universal
16. Mount
17. Washer (AN960C516)
18. Nut
19. Cotter pin
20. Hydraulic cylinder extension tube
21. Washer (AN960C516)
22. Bolt
23. Hydraulic cylinder extension tube

ICN-412-A-671200-B-97499-02935-A-001-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 2 of 2)

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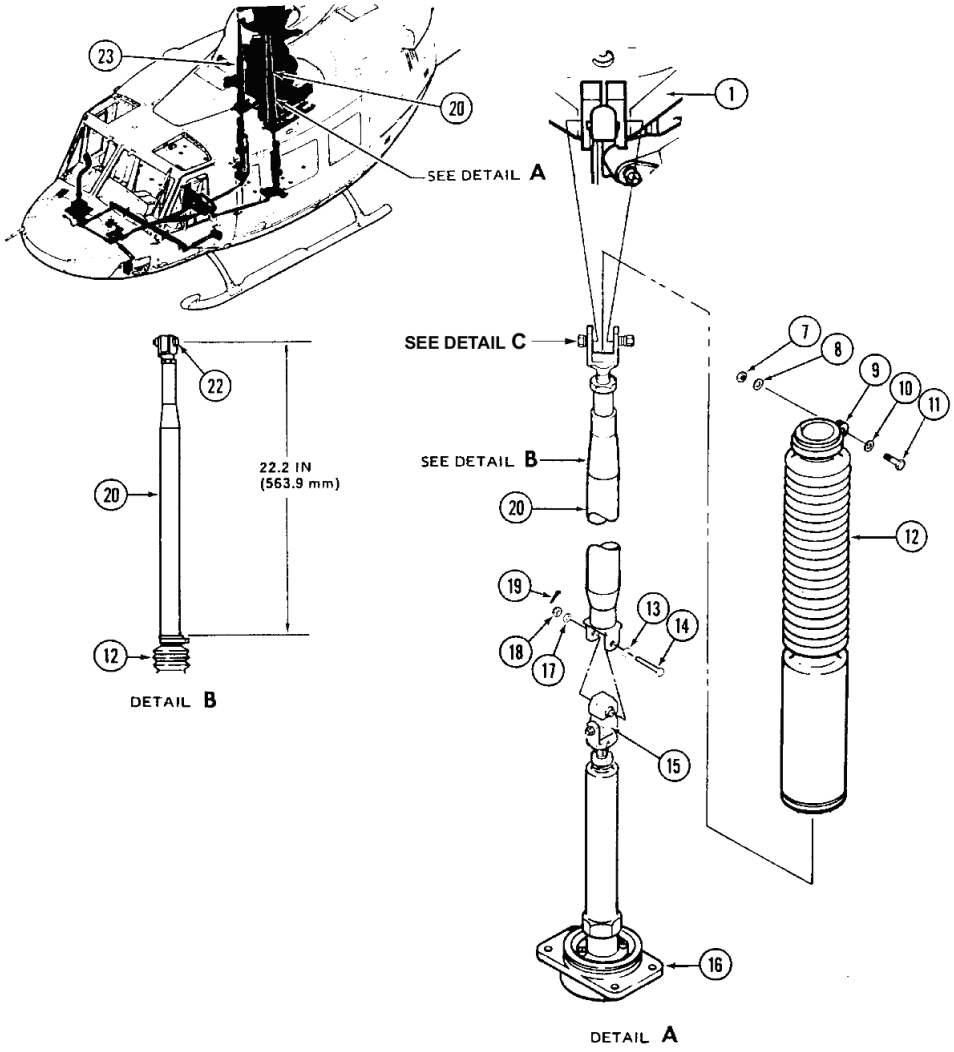
Hydraulic Cylinder Extension Tube - Inspection

Procedure

1.

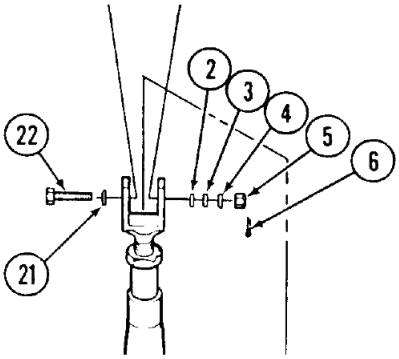
NOTE

The inspection procedure for hydraulic cylinder extension tube (20 and 23, Figure 1) in left cyclic system is contained in BHT-412-CR&O.

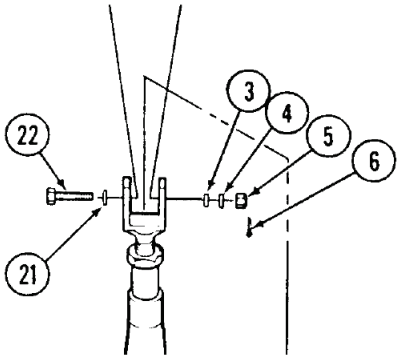


ICN-412-A-671200-B-97499-00002-A-002-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 1 of 2)



S/N 33001 through 33100



S/N 33101 through 33213, 36001 through 36999, 37002 through 37999, 38001 through 38999 and 39101 through 39999

DETAIL C

1. Swashplate
2. Washer (AN960C516)
3. Washer (AN960C516)
4. Washer (AN960C516L)
5. Nut
6. Cotter pin
7. Nut
8. Thin aluminum washer
9. Clamp
10. Thin aluminum washer
11. Screw
12. Boot
13. Washer (AN960C516)
14. Bolt
15. Hydraulic cylinder extension universal
16. Mount
17. Washer (AN960C516)
18. Nut
19. Cotter pin
20. Hydraulic cylinder extension tube
21. Washer (AN960C516)
22. Bolt
23. Hydraulic cylinder extension tube

ICN-412-A-671200-B-97499-02935-A-001-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 2 of 2)

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Hydraulic Cylinder Extension Tube - Installation

Procedure

NOTE

The installation procedure for hydraulic cylinder extension tube (20, Figure 1) in left cyclic system is given. The installation procedure for hydraulic cylinder extension tube (23) in right cyclic system is the same.

1. *Put hydraulic cylinder extension tube (20) in its position with the lower end on the hydraulic cylinder extension universal (15). Install bolt (14), washers (13 and 17), nut (18), and cotter pin (19).*

2. *Put boot (12) and clamp (9) on the top of tube (20) and move down to attach the bottom of the boot (12) into the lip on mount (16).*

Applicable: SN: 33001-33100:

3. *Put the top of tube (20) on swashplate (1) and install bolt (22), washers (21, 2, 3, and 4) and nut (5). Safety the nut (5) with cotter pin (6).*

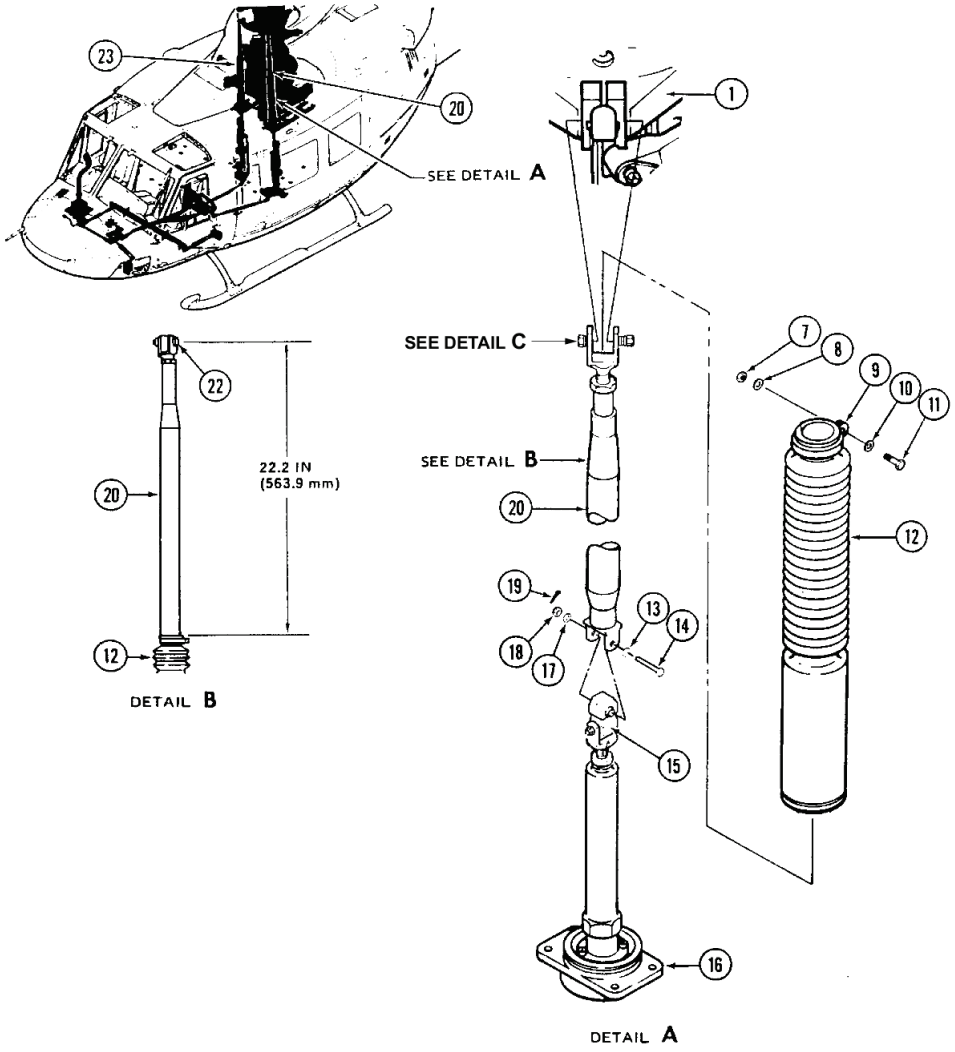
Applicable: SN: 33101-33213 or 36001-36999 or 37002-37999 or 38001-38999 or 39101-39999:

4. *Put the top of tube (20) on swashplate (1) and install bolt (22), washers (21, 3, and 4) and nut (5). Safety the nut (5) with cotter pin (6).*

5. *Put the top of boot (12)* 22.2 inches (563.9 mm) *in its position below the centerline of bolt (22, detail B). Install clamp (9), screw (11), washers (10 and 8), and nut (7).*

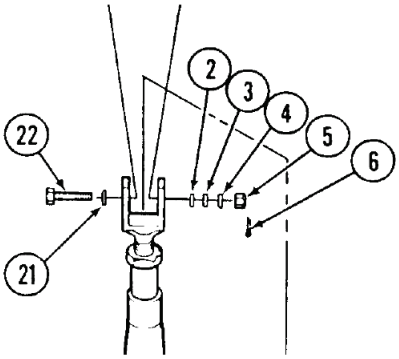
6. Install forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).

7. *Make sure that the cyclic rigging is within limits (*412-A-67-01-00-03A-276A-A / 01324 *or* 412-A-67-01-00-04A-276A-A / 01325*).*

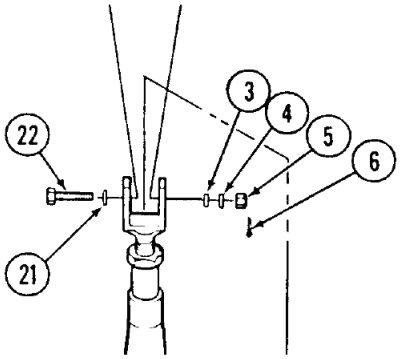


ICN-412-A-671200-B-97499-00002-A-002-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 1 of 2)



S/N 33001 through 33100



S/N 33101 through 33213, 36001 through 36999, 37002 through 37999, 38001 through 38999 and 39101 through 39999

DETAIL C

1. Swashplate
2. Washer (AN960C516)
3. Washer (AN960C516)
4. Washer (AN960C516L)
5. Nut
6. Cotter pin
7. Nut
8. Thin aluminum washer
9. Clamp
10. Thin aluminum washer
11. Screw
12. Boot
13. Washer (AN960C516)
14. Bolt
15. Hydraulic cylinder extension universal
16. Mount
17. Washer (AN960C516)
18. Nut
19. Cotter pin
20. Hydraulic cylinder extension tube
21. Washer (AN960C516)
22. Bolt
23. Hydraulic cylinder extension tube

ICN-412-A-671200-B-97499-02935-A-001-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 2 of 2)

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Hydraulic Servo Cylinder Boot - Removal

Procedure

NOTE

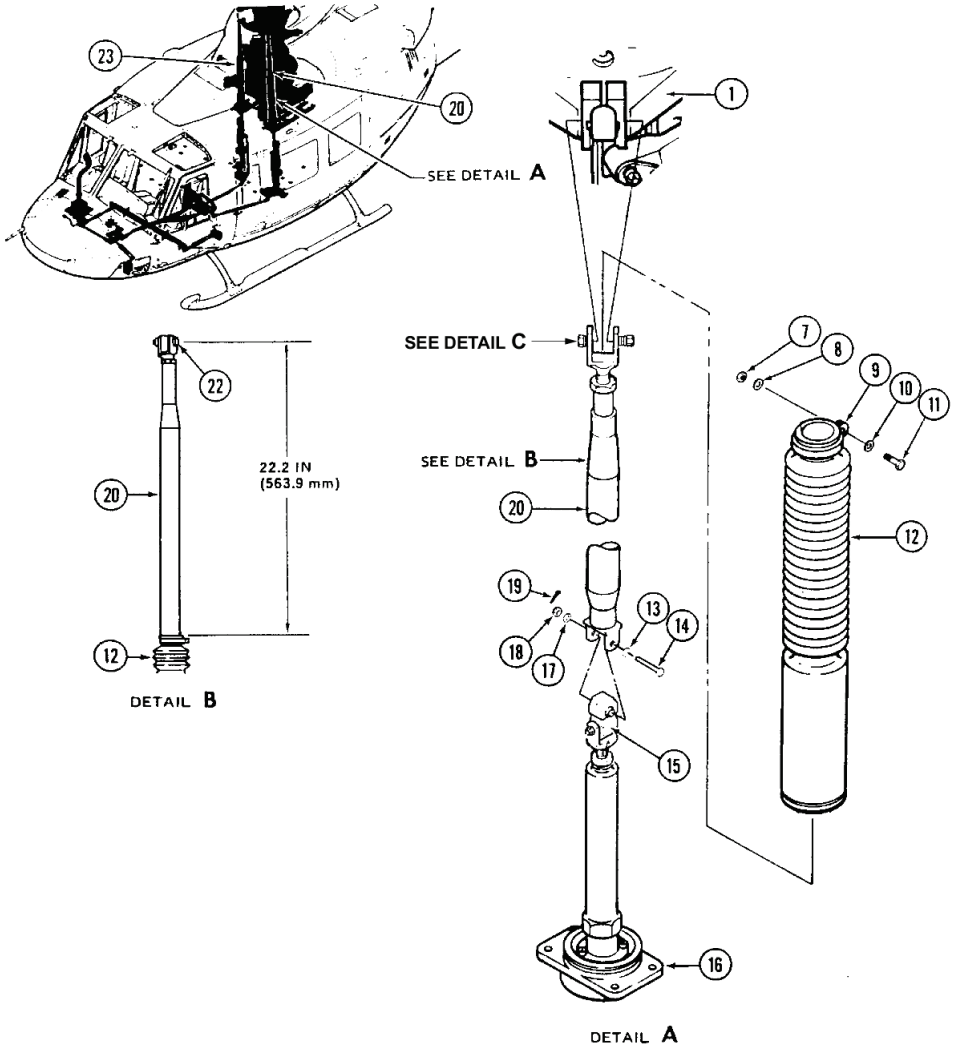
The maintenance procedures for boot (12, Figure 1) on left cyclic flight control cylinder are given. The procedures for boot on right cyclic flight control cylinder are the same.

1. Remove forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).

2. Disconnect hydraulic cylinder extension tube (20) from swashplate horn (1) by removing cotter pin (6), nut (5), washers (4, 3, 2 and 21) and bolt (22).

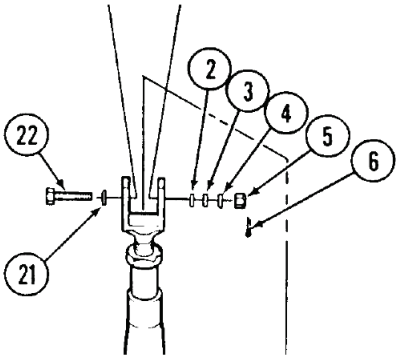
3. Remove nut (7), thin aluminum washers (8 and 10) and screw (11).

4. Loosen bottom of boot (12) from mount bearing (16) and slip boot off tube (20).

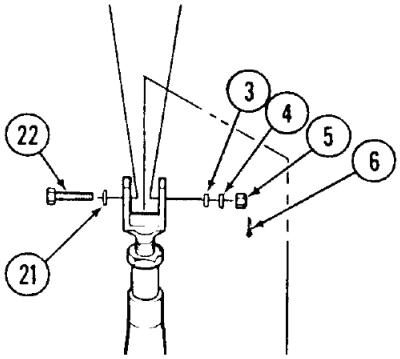


ICN-412-A-671200-B-97499-00002-A-002-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 1 of 2)



S/N 33001 through 33100



S/N 33101 through 33213, 36001 through 36999, 37002 through 37999, 38001 through 38999 and 39101 through 39999

DETAIL C

1. Swashplate
2. Washer (AN960C516)
3. Washer (AN960C516)
4. Washer (AN960C516L)
5. Nut
6. Cotter pin
7. Nut
8. Thin aluminum washer
9. Clamp
10. Thin aluminum washer
11. Screw
12. Boot
13. Washer (AN960C516)
14. Bolt
15. Hydraulic cylinder extension universal
16. Mount
17. Washer (AN960C516)
18. Nut
19. Cotter pin
20. Hydraulic cylinder extension tube
21. Washer (AN960C516)
22. Bolt
23. Hydraulic cylinder extension tube

ICN-412-A-671200-B-97499-02935-A-001-01

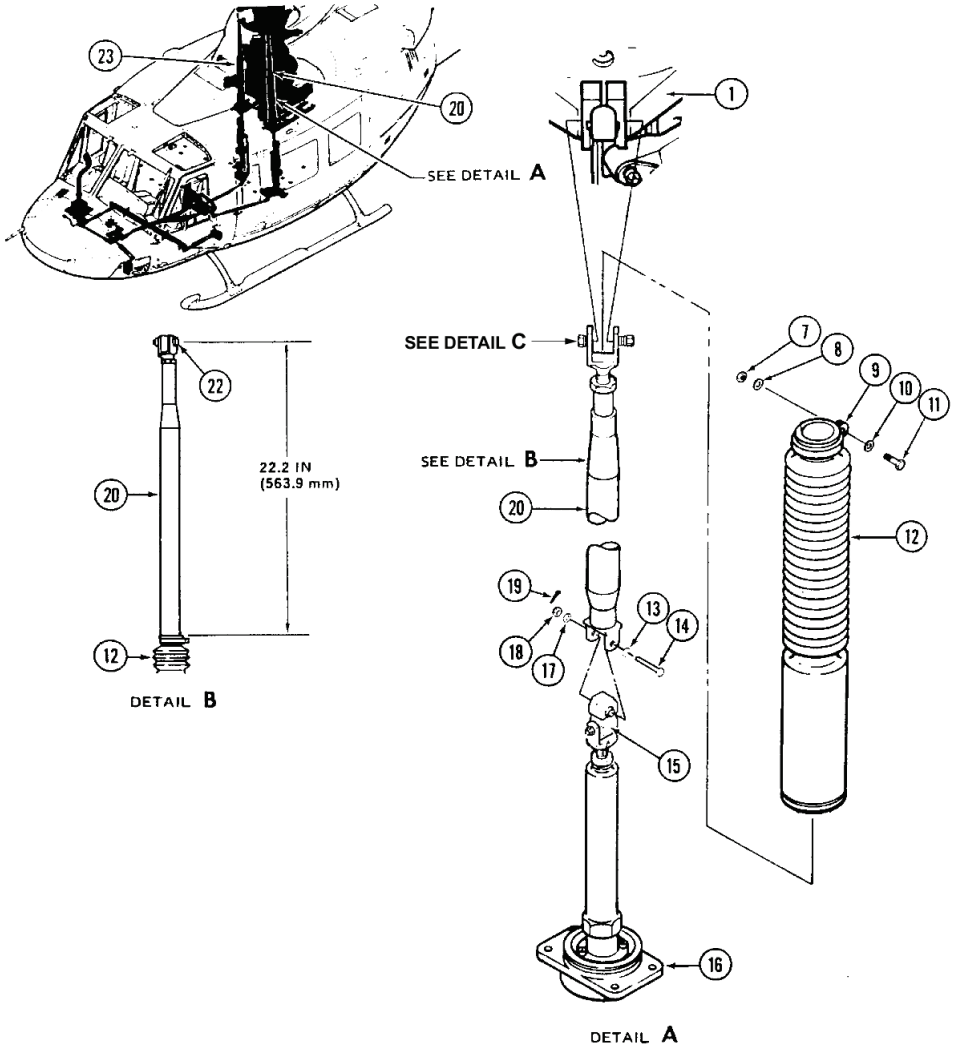
Figure 1. Hydraulic Cylinder Extension Tube (Sheet 2 of 2)

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Hydraulic Servo Cylinder Boot - Inspection

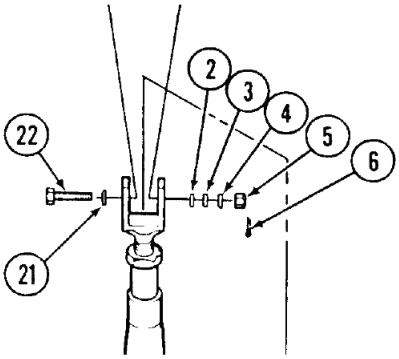
Procedure

1. Inspect boot (12, Figure 1) for tears and signs of deterioration. Torn or deteriorated boots are not acceptable and shall be replaced.

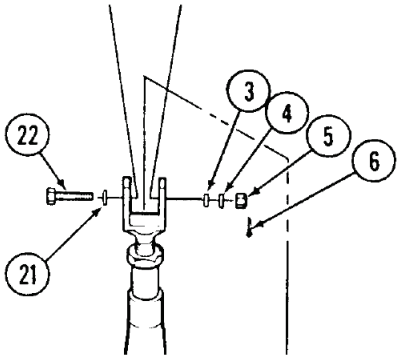


ICN-412-A-671200-B-97499-00002-A-002-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 1 of 2)



S/N 33001 through 33100



S/N 33101 through 33213, 36001 through 36999, 37002 through 37999, 38001 through 38999 and 39101 through 39999

DETAIL C

1. Swashplate
2. Washer (AN960C516)
3. Washer (AN960C516)
4. Washer (AN960C516L)
5. Nut
6. Cotter pin
7. Nut
8. Thin aluminum washer
9. Clamp
10. Thin aluminum washer
11. Screw
12. Boot
13. Washer (AN960C516)
14. Bolt
15. Hydraulic cylinder extension universal
16. Mount
17. Washer (AN960C516)
18. Nut
19. Cotter pin
20. Hydraulic cylinder extension tube
21. Washer (AN960C516)
22. Bolt
23. Hydraulic cylinder extension tube

ICN-412-A-671200-B-97499-02935-A-001-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 2 of 2)

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Hydraulic Servo Cylinder Boot - Installation

Procedure

NOTE

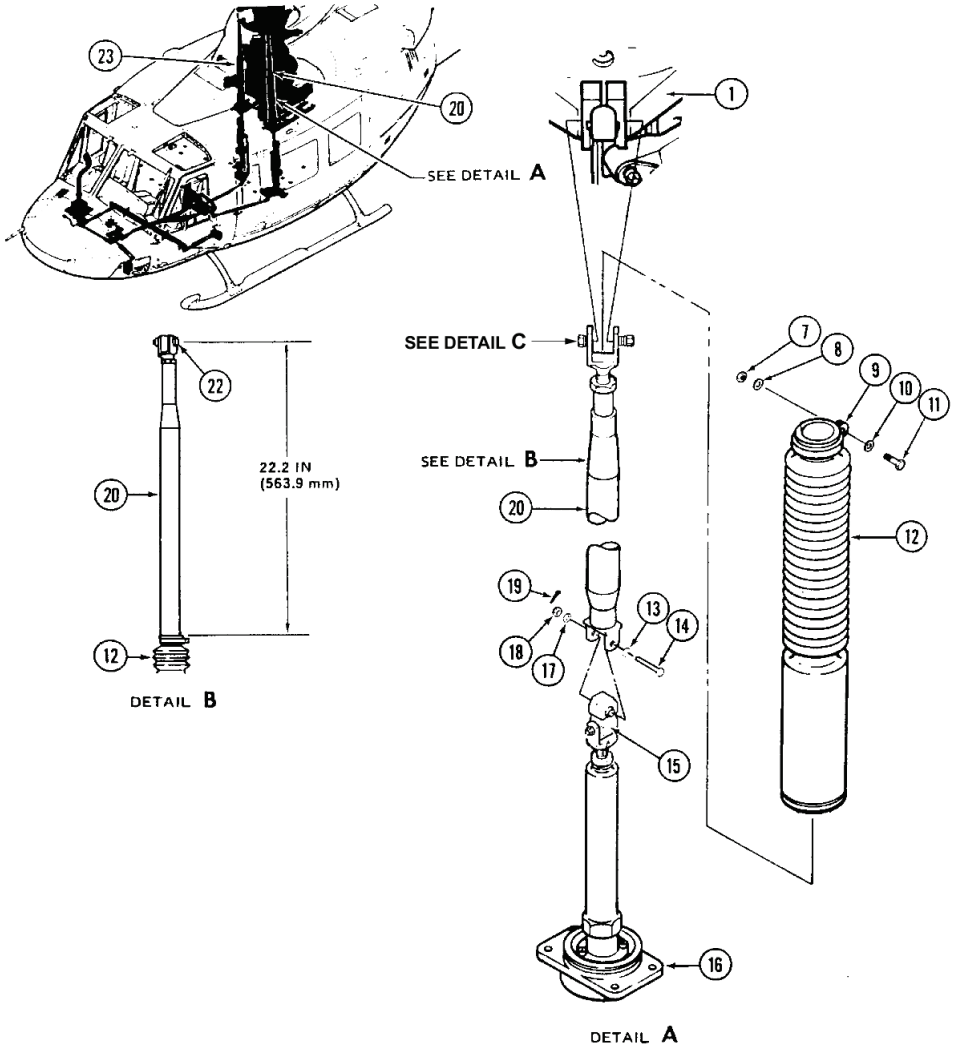
The maintenance procedures for boot (12, Figure 1) on left cyclic flight control cylinder are given. The procedures for boot on right cyclic flight control cylinder are the same.

1. Slide boot (12, Figure 1) over tube (20) and secure lower end of boot into lip on mount (16).

2. Position tube (20) on swashplate (1). Install bolt (22), washer (21), washers (2, 3, and 4), nut (5), and cotter pin (6).

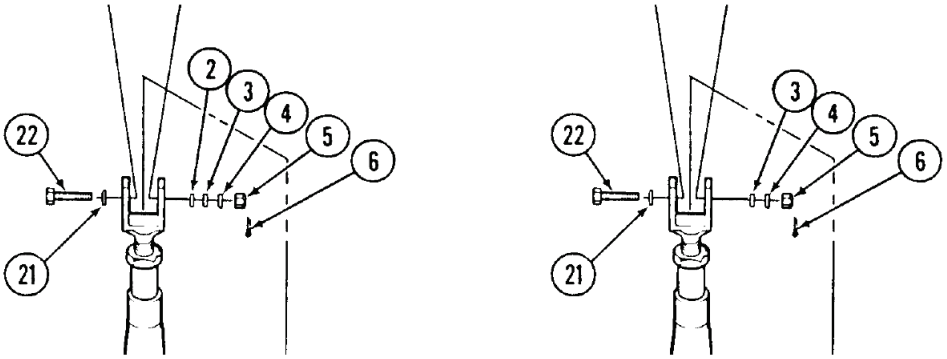
3. Position top of boot (12) 22.2 in. (563.9 mm) below centerline of bolt (22) (detail B). Install clamp (9), screw (11), washer (10), washer (8), and nut (7).

4. Install forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).



ICN-412-A-671200-B-97499-00002-A-002-01

Figure 1. Hydraulic Cylinder Extension Tube (Sheet 1 of 2)



S/N 33001 through 33100

S/N 33101 through 33213, 36001 through 36999, 37002 through 37999, 38001 through 38999 and 39101 through 39999

DETAIL C

1. Swashplate
2. Washer (AN960C516)
3. Washer (AN960C516)
4. Washer (AN960C516L)
5. Nut
6. Cotter pin
7. Nut
8. Thin aluminum washer
9. Clamp
10. Thin aluminum washer
11. Screw
12. Boot
13. Washer (AN960C516)
14. Bolt
15. Hydraulic cylinder extension universal
16. Mount
17. Washer (AN960C516)
18. Nut
19. Cotter pin
20. Hydraulic cylinder extension tube
21. Washer (AN960C516)
22. Bolt
23. Hydraulic cylinder extension tube

ICN-412-A-671200-B-97499-02935-A-001-01

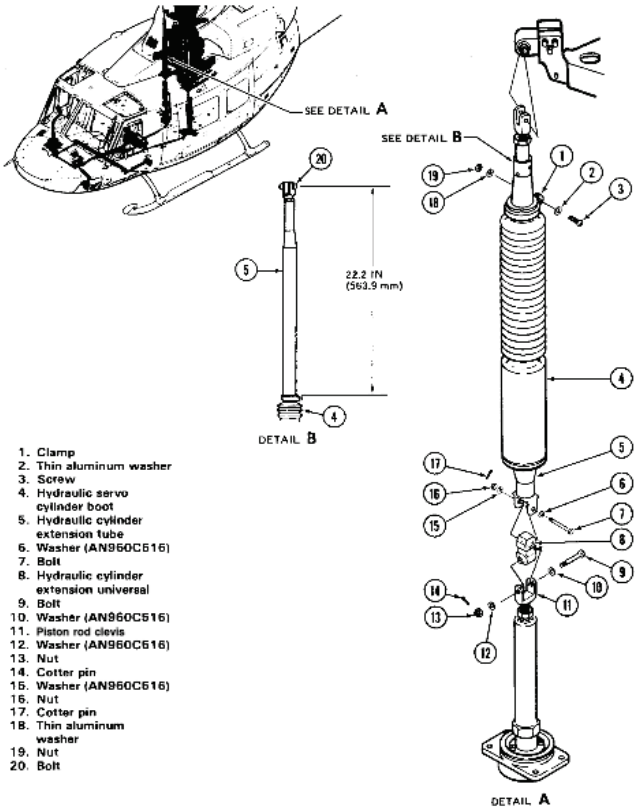
Figure 1. Hydraulic Cylinder Extension Tube (Sheet 2 of 2)

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Hydraulic Cylinder Extension Universal - Description

NOTE

The maintenance procedures for hydraulic cylinder extension universal (8, Figure 1) on right hydraulic cylinder extension tube (5) is given. Procedures for universal on left cylinder are the same.



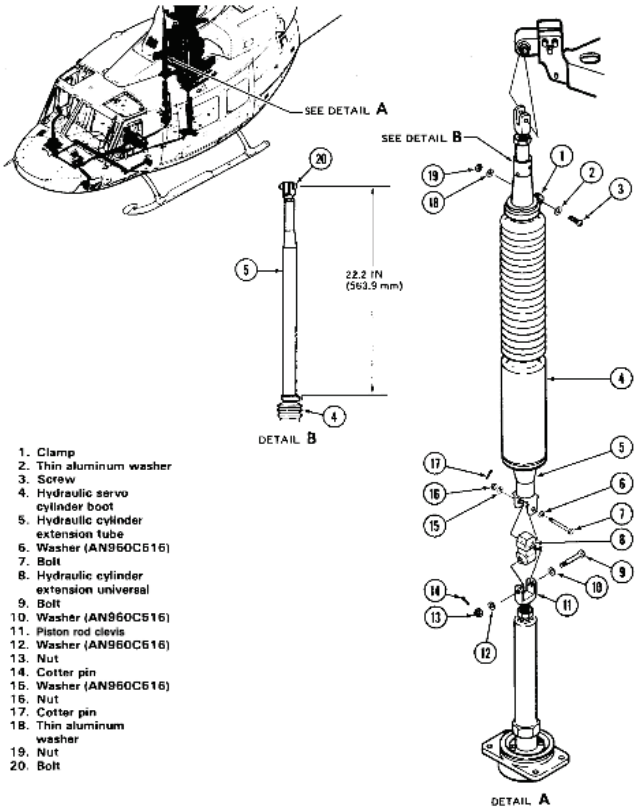
ICN-412-A-671200-B-97499-00003-A-001-01

Figure 1. Hydraulic Cylinder Extension Universal (Sheet 1 of 1)

Hydraulic Cylinder Extension Universal - Removal

Procedure

1. Remove forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).
2. Remove nut (19, Figure 1), thin aluminum washers (2 and 18) and screw (3). Slide boot (4) and clamp (1) upward to expose bolts (7 and 9).
3. Remove cotter pin (14), nut (13), washers (10 and 12) and bolt (9).
4. Remove cotter pin (17), nut (16), washers (6 and 15) and bolt (7). Remove hydraulic cylinder extension universal (8) from piston rod clevis (11).



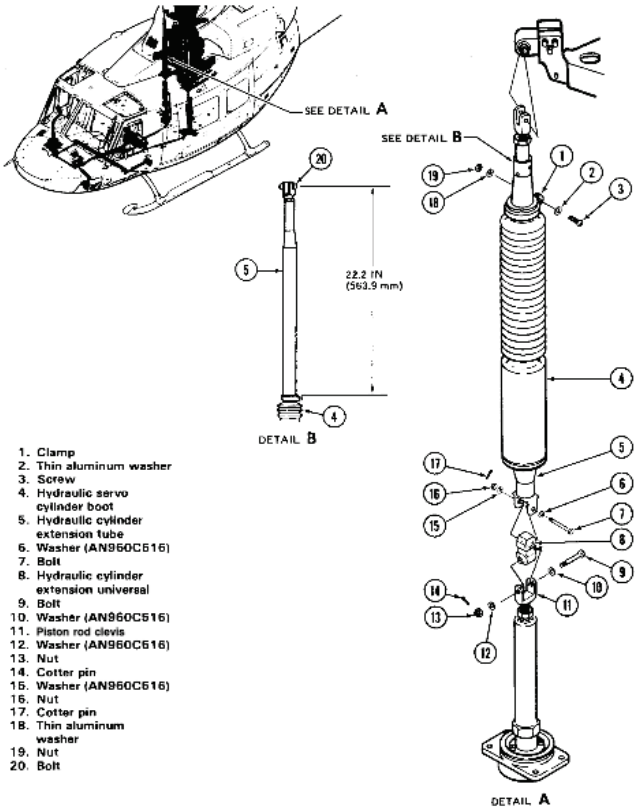
ICN-412-A-671200-B-97499-00003-A-001-01

Figure 1. Hydraulic Cylinder Extension Universal (Sheet 1 of 1)

Hydraulic Cylinder Extension Universal - Inspection

Procedure

1. Inspect hydraulic cylinder extension universal (8, Figure 1) for mechanical and corrosion damage (BHT-412-CR&O).



ICN-412-A-671200-B-97499-00003-A-001-01

Figure 1. Hydraulic Cylinder Extension Universal (Sheet 1 of 1)

Hydraulic Cylinder Extension Universal - Bearing Replacement

Procedure

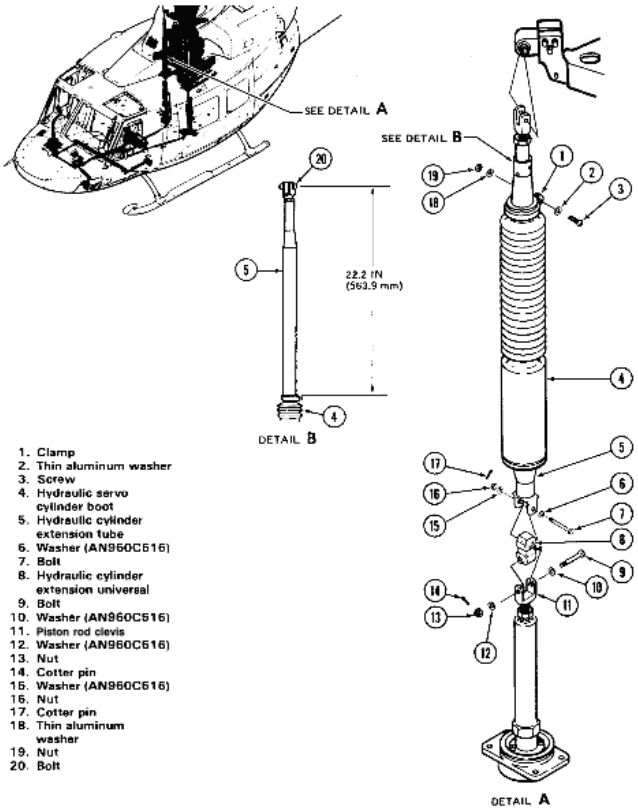
1. Refer to BHT-412-CR&O for bearing replacement procedure in universal assembly P/N 205-076-381, and in universal assembly P/N 412-076-620-101.

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Hydraulic Cylinder Extension Universal - Installation

Procedure

1. Place universal (8, Figure 1) in clevis of extension tube (5).
2. Install bolt (7), washers (6 and 15), nut (16), and cotter pin (17).
3. Place bottom of universal into piston rod clevis (11) and install bolt (9), washers (10 and 12), nut (13), and cotter pin (14).
4. Slide boot (4) and clamp (1) down 22.2 in. (563.9 mm) from centerline of top bolt (20). Install screw (3), washers (2 and 18), and nut (19).
5. Install forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).



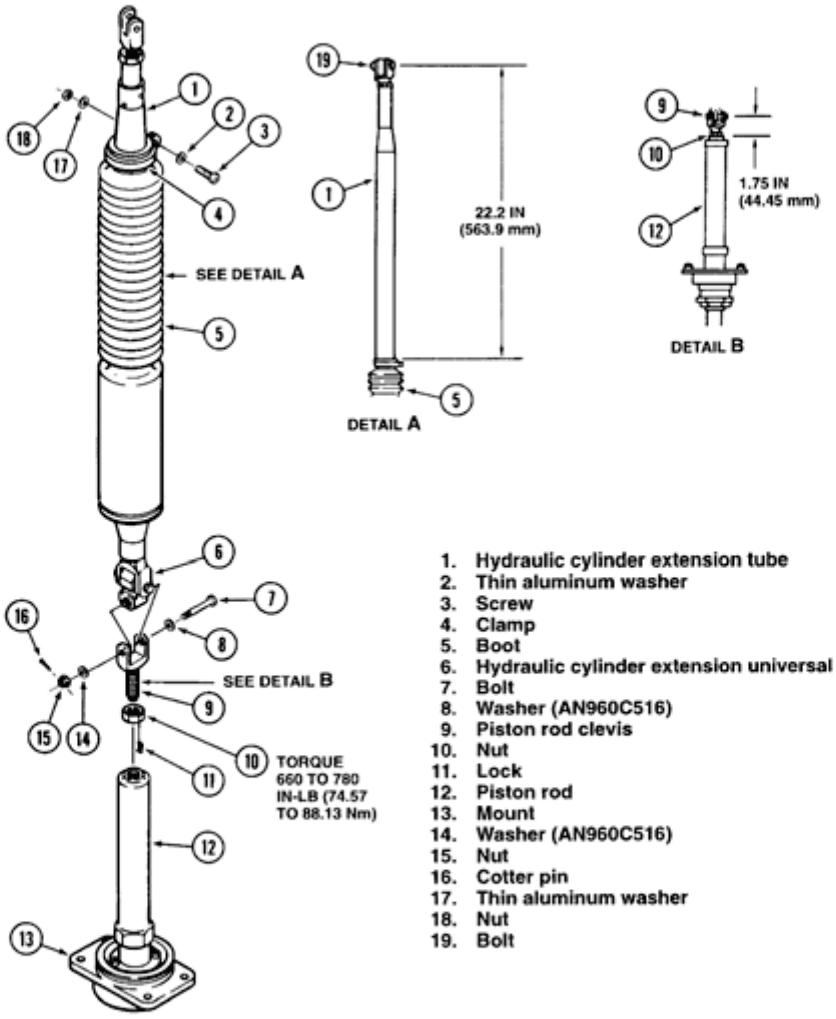
ICN-412-A-671200-B-97499-00003-A-001-01

Figure 1. Hydraulic Cylinder Extension Universal (Sheet 1 of 1)

Left Cyclic Piston Rod Clevis - Removal

Procedure

1. Remove forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).
2. Remove nut (18, Figure 1), thin aluminum washers (2 and 17) and screw (3).
3. Loosen bottom of boot (5) from lip on mount (13) and slide boot upward to expose universal assembly (6).
4. Remove cotter pin (16), nut (15), washers (8 and 14) and bolt (7).
5. Loosen nut (10). Remove piston rod clevis (9) from piston rod (12). Remove nut (10) and lock (11) from clevis (9).



ICN-412-A-671200-B-97499-00004-A-001-01

Figure 1. Left Cyclic Piston Rod Clevis (Sheet 1 of 1)

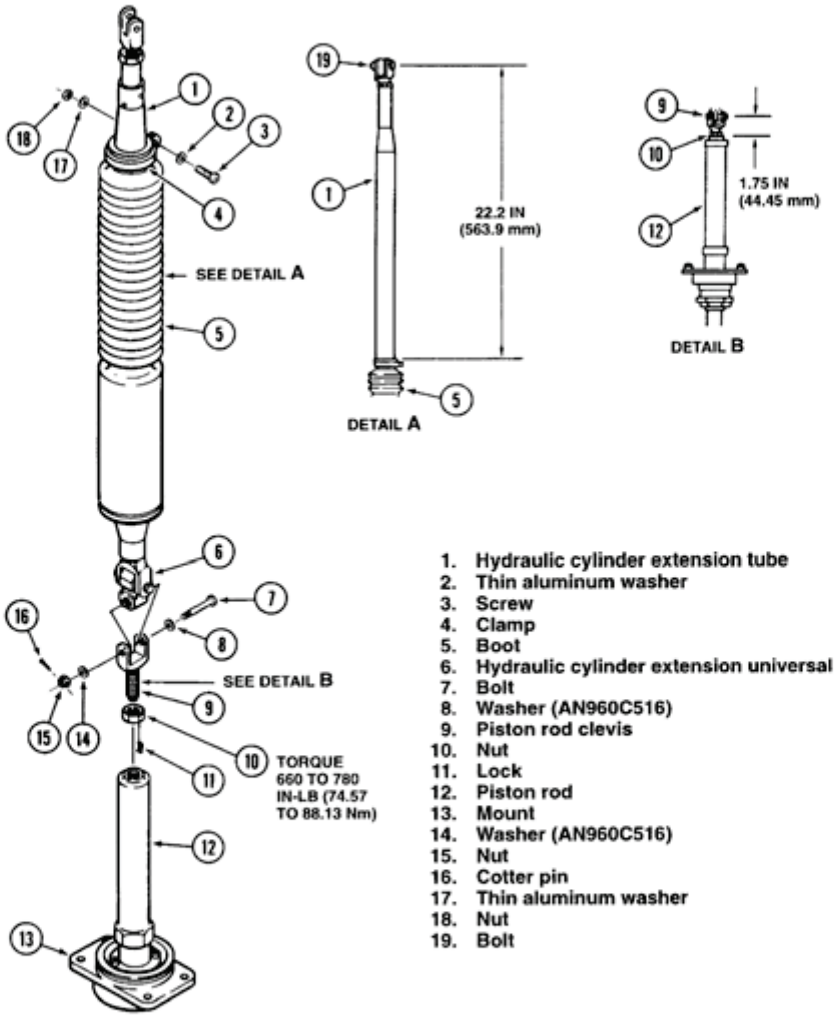
Left Cyclic Piston Rod Clevis - Inspection

Procedure

1. Inspect clevis (9, Figure 1) for mechanical and corrosion damage. Superficial mechanical and corrosion damage is acceptable if repaired (412-A-67-12-00-04A-685A-A / 01415).

2. Inspect bolt holes for elongation.

3. Inspect threads for damage. No visible thread damage is acceptable.



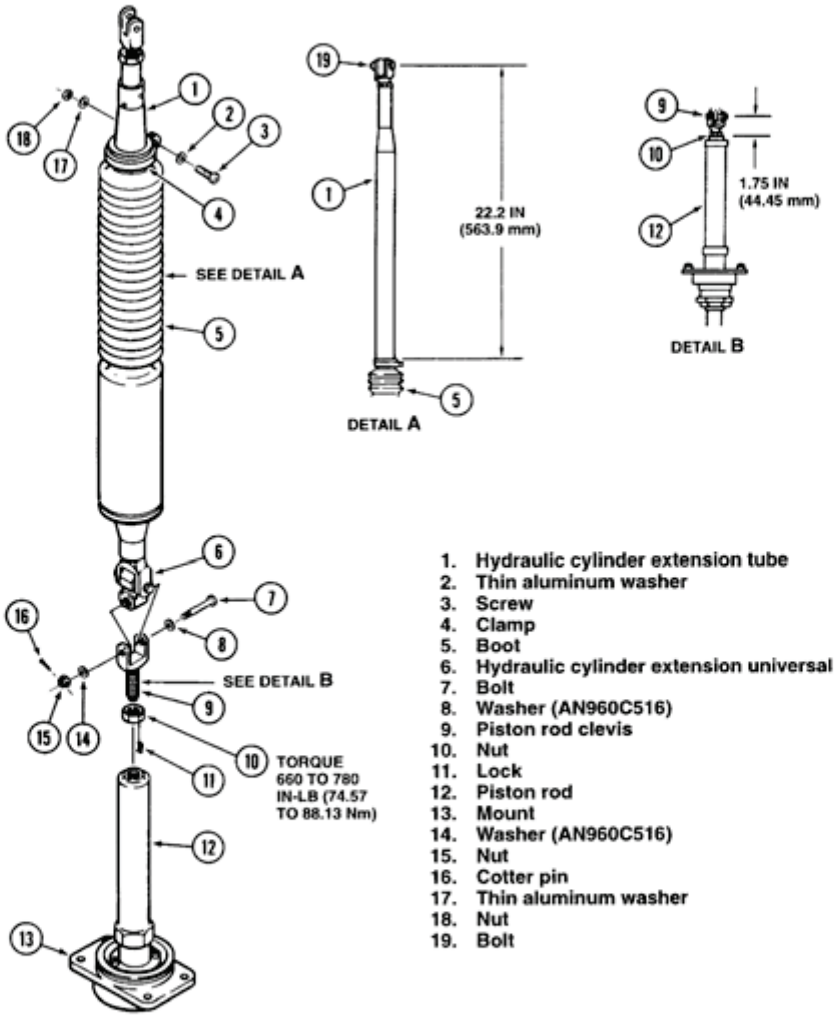
ICN-412-A-671200-B-97499-00004-A-001-01

Figure 1. Left Cyclic Piston Rod Clevis (Sheet 1 of 1)

Left Cyclic Piston Rod Clevis - Repair

Procedure

1. Repair piston rod clevis (9, Figure 1) (BHT-412-CR&O).



ICN-412-A-671200-B-97499-00004-A-001-01

Figure 1. Left Cyclic Piston Rod Clevis (Sheet 1 of 1)

Left Cyclic Piston Rod Clevis - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-104	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

1. Coat threads of clevis (9, Figure 1) with corrosion preventive compound (C-104).

2. Place lock (11) into nut (10). Install nut (10) and lock (11) on clevis (9).

3. Install clevis (9) into piston rod (12). Adjust clevis (9) to obtain 1.75 in. (44.45 mm) from clevis bolt hole centerline to top of piston rod (detail B).

4. Torque nut (10) 660 to 780 in.-lbs. (74.57 to 88.13 N-m). Secure nut with lockwire (C-405).



AVOID GETTING SEALANT ON PISTON ROD SURFACE OR SCRAPER AND ROD SEAL DAMAGE MAY OCCUR.

5. Coat nut (10), lock (11), lockwire and threads on clevis (9) with sealant (C-308). Refer to Figure 2, Sheet 3

6. Place universal (6, Figure 1) in clevis (9). Install bolt (7), washer (8), washer (14), nut (15), and cotter pin (16).

7. Slide boot (5) down over piston rod (12) and engage bottom of boot into lip on mount (13).

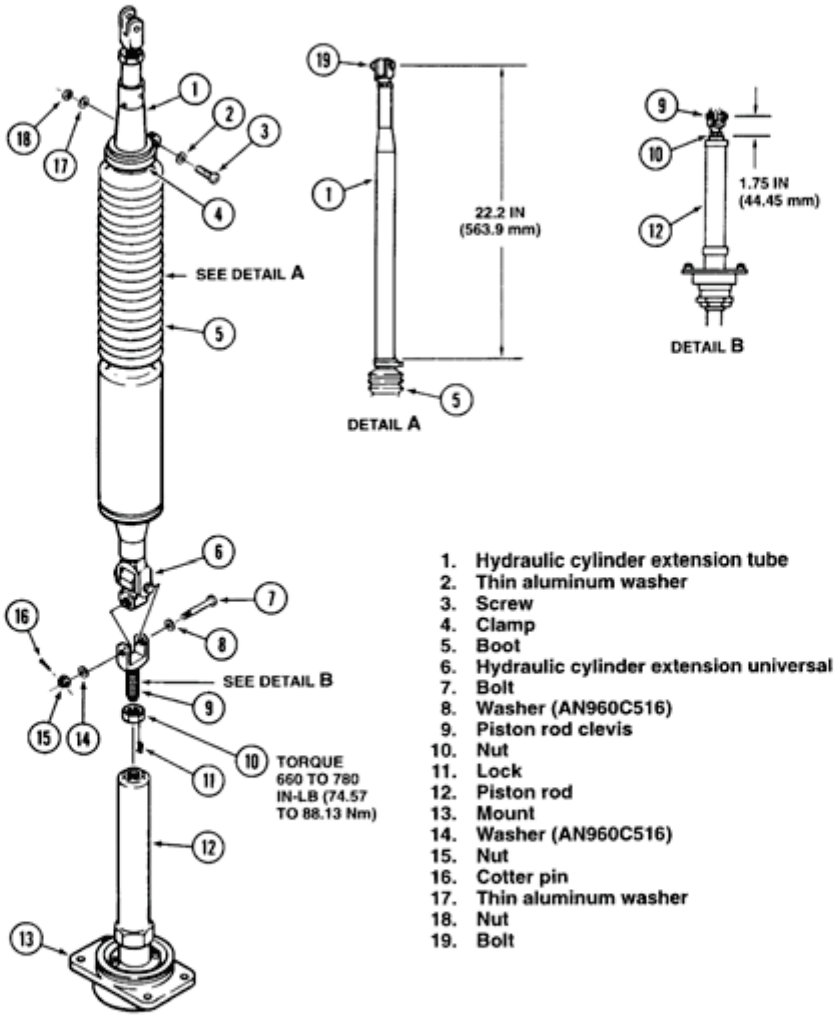
8. Position top of boot (5) 22.2 in. (563.9 mm) from centerline of bolt (19) (detail A).

9. Secure boot to hydraulic cylinder extension tube (1) with clamp (4) using screw (3), washer (2), washer (17) and nut (18).

10. Verify cyclic rigging is within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).

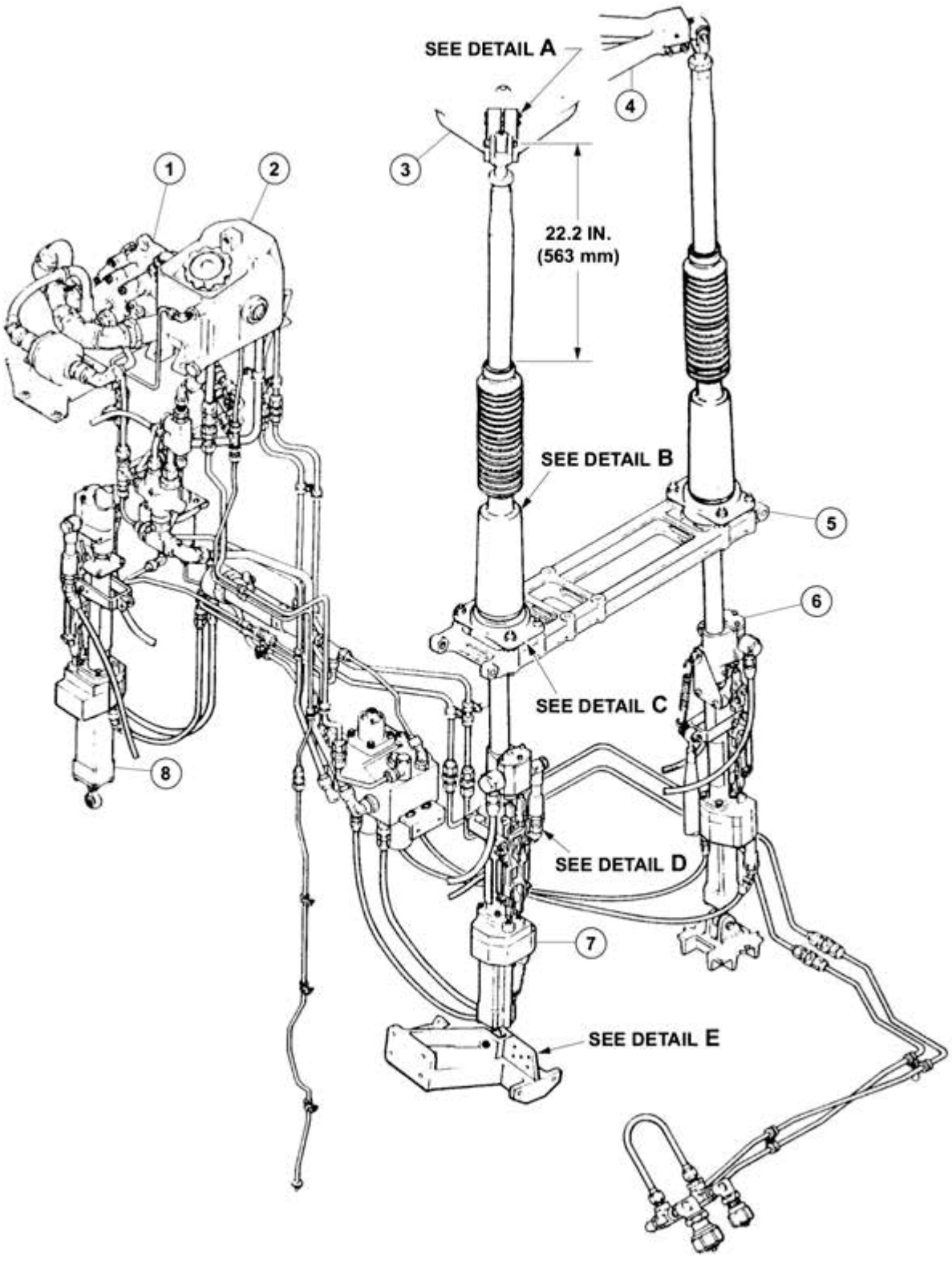
11. Move cyclic controls through full travel and check for interference and binding.

12. Install forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).



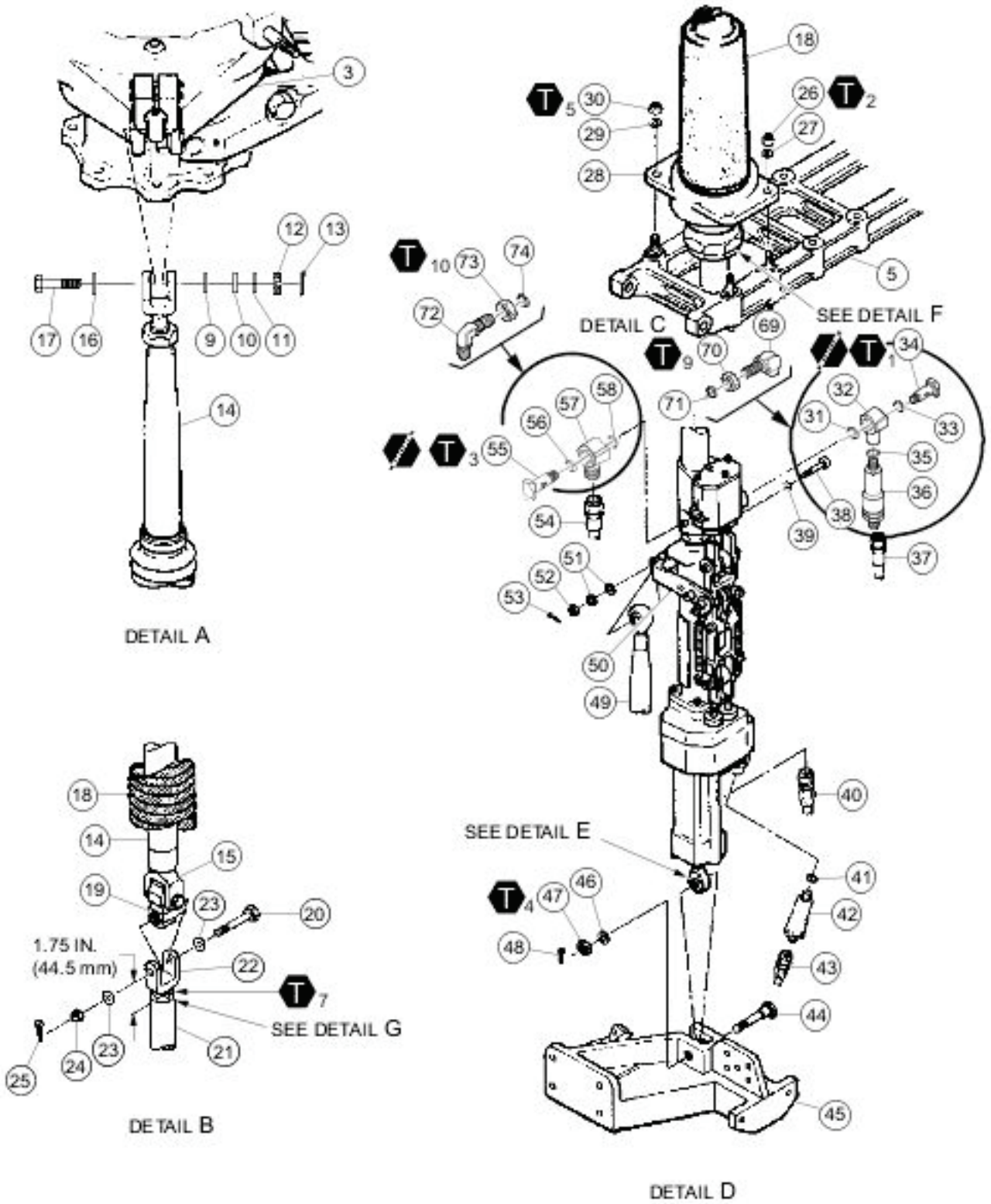
ICN-412-A-671200-B-97499-00004-A-001-01

Figure 1. Left Cyclic Piston Rod Clevis (Sheet 1 of 1)



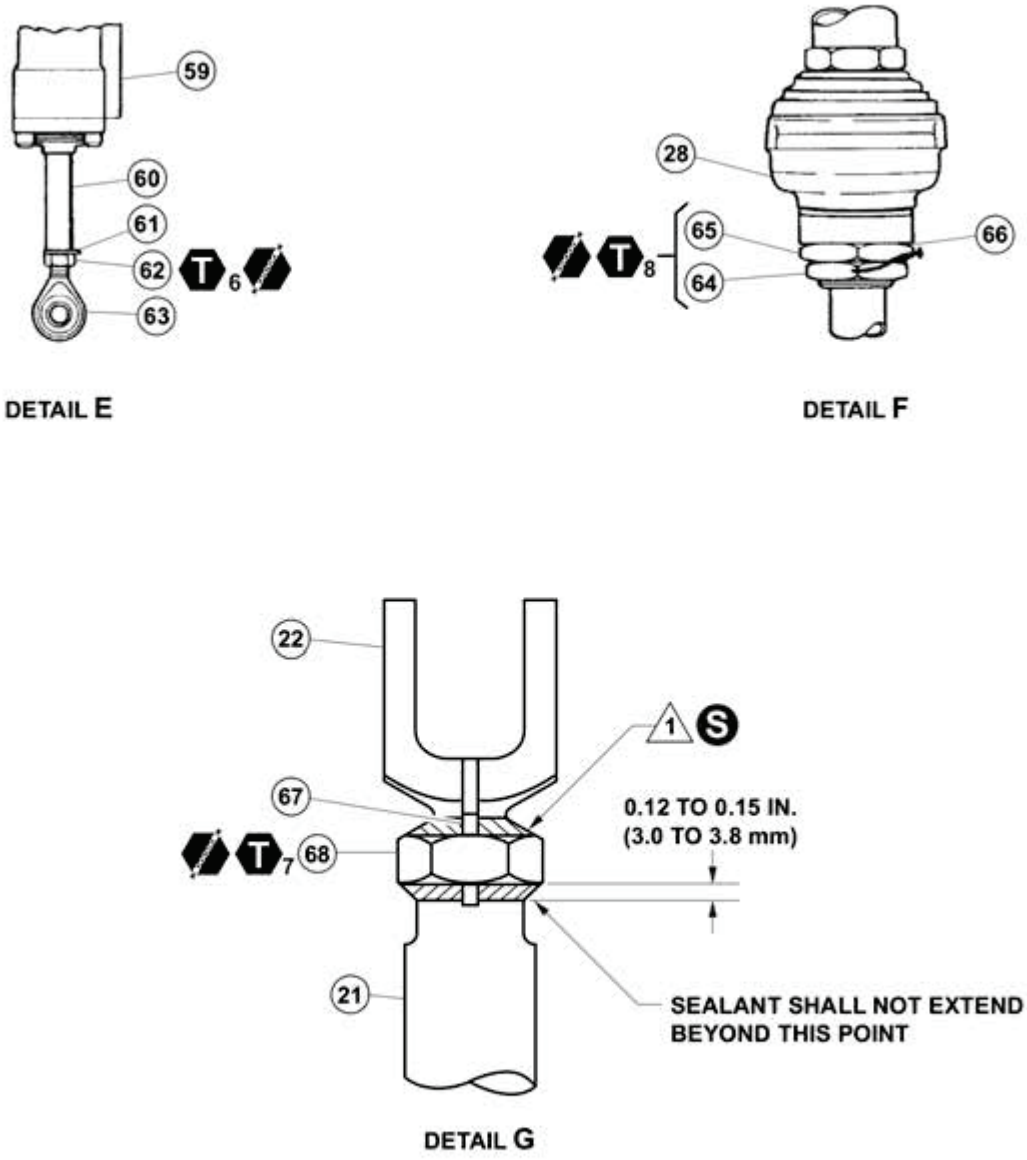
ICN-412-A-671200-B-97499-00006-A-001-01

Figure 2. Left Cyclic Flight Control Cylinder (Sheet 1 of 4)



ICN-412-A-671200-B-97499-00007-A-001-01

Figure 2. Left Cyclic Flight Control Cylinder (Sheet 2 of 4)



ICN-412-A-671200-B-97499-00008-A-001-01

Figure 2. Left Cyclic Flight Control Cylinder (Sheet 3 of 4)

1. Pump
2. Reservoir
3. Swashplate horn
4. Collective lever
5. Support
6. Collective flight control cylinder
7. Left cyclic flight control cylinder
8. Right cyclic flight control cylinder
9. Washer (AN960C516)
10. Washer (AN960C516)
11. Washer (AN960C516L)
12. Nut
13. Cotter pin
14. Extension tube
15. Clevis
16. Washer (AN960C516)
17. Bolt
18. Boot
19. Universal
20. Bolt
21. Upper piston
22. Piston rod clevis
23. Washer (AN960C516)
24. Nut
25. Cotter pin
26. Nut
27. Aluminum washer
28. Support bracket
29. Aluminum washer
30. Nut
31. Preformed packing
32. Elbow
33. Preformed packing
34. Bolt
35. Preformed packing
36. Bypass check valve
37. Hose
38. Bolt
39. Washer
40. Hose
41. Preformed packing
42. Bypass check valve
43. Hose
44. Bolt
45. Support
46. Aluminum washer
47. Nut
48. Cotter pin
49. Tube assembly
50. Lever
51. Washer
52. Nut
53. Cotter pin
54. Hose
55. Bolt
56. Preformed packing
57. Elbow
58. Preformed packing
59. Actuator
60. Lower piston
61. Locking device
62. Nut
63. Rod end bearing
64. Nut
65. Bearing nut
66. Keyed washer
67. Lock
68. Jam nut
69. Elbow
70. Nut
71. Packing
72. Elbow
73. Nut
74. Packing

LOCKWIRE (C-405)

SEALANT (C-308)

T1 40 TO 65 IN-LBS (4.51 TO 7.34 Nm)

T2 75 TO 95 IN-LBS (8.47 TO 10.73 Nm)

T3 75 TO 125 IN-LBS (8.47 TO 14.12 Nm)

T4 95 TO 110 IN-LBS (10.73 TO 12.42 Nm)

T5 120 TO 160 IN-LBS (13.55 TO 18.07 Nm)

T6 200 TO 250 IN-LBS (22.59 TO 28.54 Nm)

T7 660 TO 780 IN-LBS (74.56 TO 88.12 Nm)

T8 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

T9 135 TO 145 IN-LBS (15.25 TO 16.38 Nm)

T10 215 TO 245 IN-LBS (24.30 TO 27.68 Nm)

NOTE

1. Apply sealant (C-308) around jam nut (68), lock (67), and exposed threads of pistion rod clevis (22).

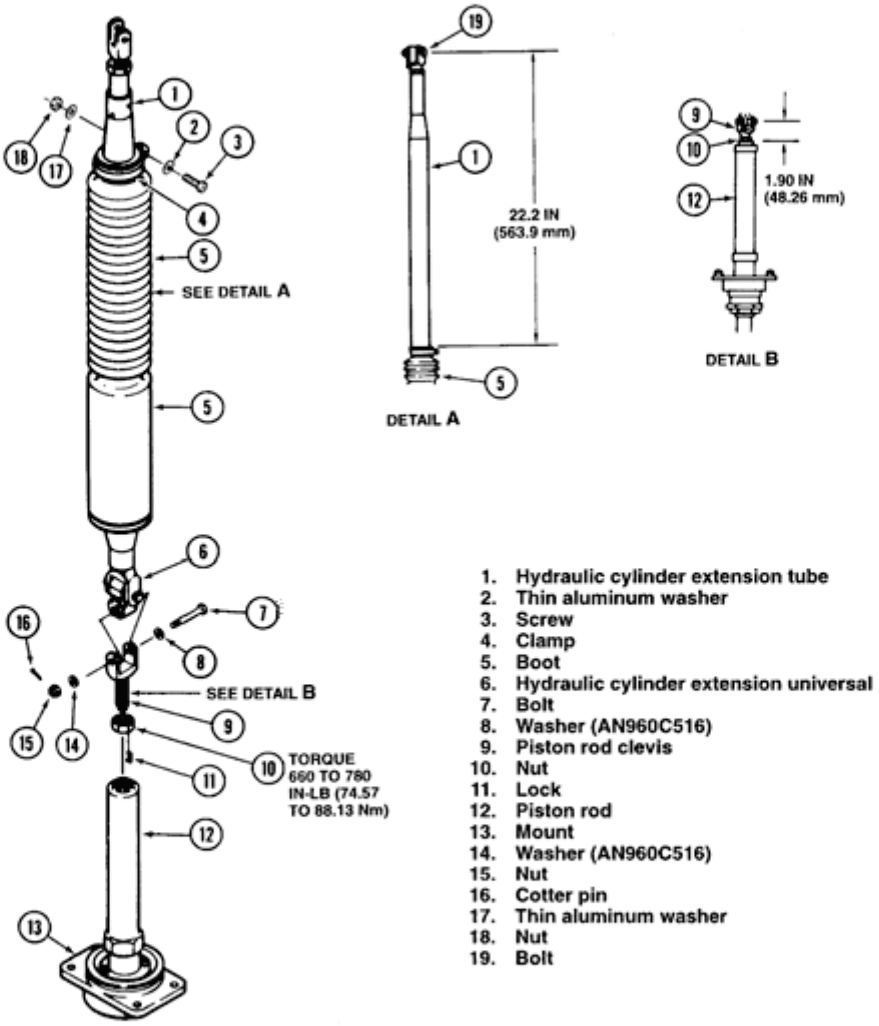
ICN-412-A-671000-B-97499-00937-A-001-01

Figure 2. Left Cyclic Flight Control Cylinder (Sheet 4 of 4)

Right Cyclic Piston Rod Clevis - Removal

Procedure

1. Remove forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).
2. Loosen bottom of boot (5, Figure 1) from lip in mount (13).
3. Remove nut (18), thin aluminum washers (2 and 17) and screw (3). Slide clamp (4) and boot (5) upward on hydraulic cylinder extension tube (1) to expose universal (6).
4. Remove cotter pin (16), nut (15), washers (8 and 14) and bolt (7).
5. Loosen nut (10). Remove piston rod clevis (9) from piston rod (12). Remove nut (10) and lock (11) from clevis (9).



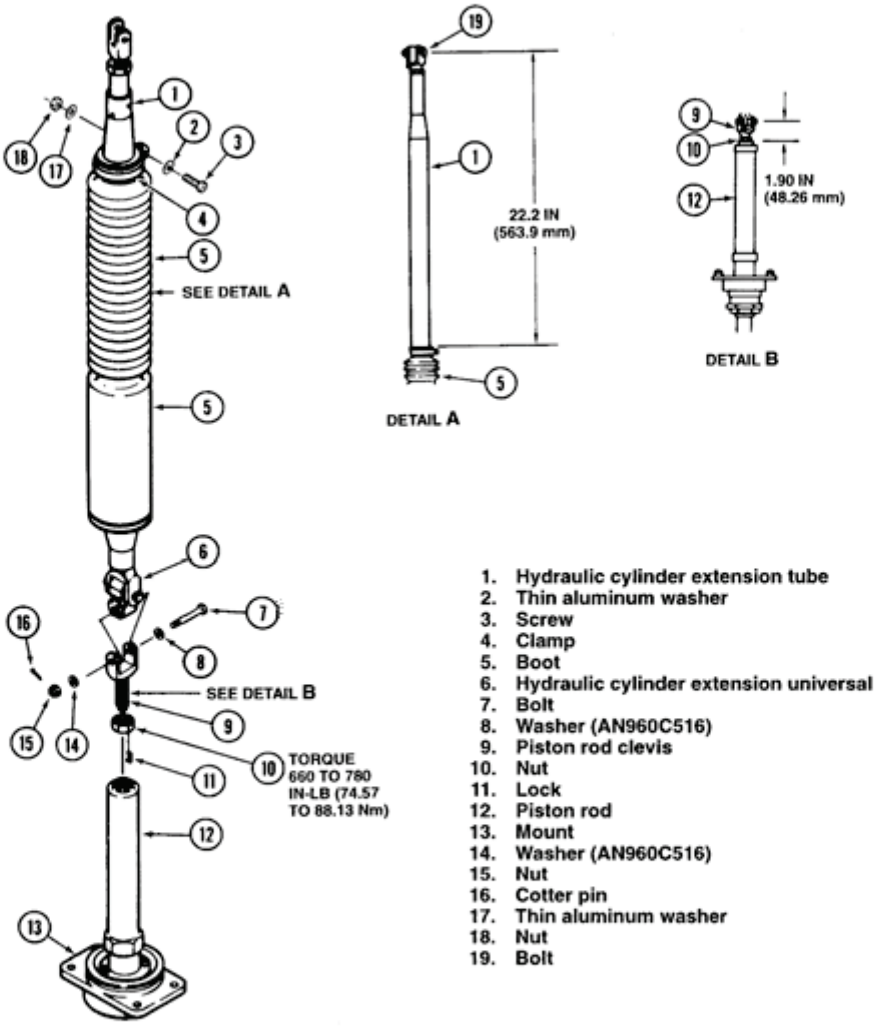
ICN-412-A-671200-B-97499-00005-A-001-01

Figure 1. Right Cyclic Piston Rod Clevis (Sheet 1 of 1)

Right Cyclic Piston Rod Clevis - Inspection

Procedure

1. Inspect piston rod clevis (9, Figure 1) for mechanical and corrosion damage (BHT-412-CR&O).
2. Inspect bolt holes for elongation.
3. Inspect threads for damage. No visible damage to threads is acceptable.



ICN-412-A-671200-B-97499-00005-A-001-01

Figure 1. Right Cyclic Piston Rod Clevis (Sheet 1 of 1)

Right Cyclic Piston Rod Clevis - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-104	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

1. Coat threads of clevis (9, Figure 1) with corrosion preventive compound (C-104). Place lock (11) into nut (10). Install nut (10) and lock (11) on clevis (9).

2. Adjust clevis (9) to obtain 1.90 in. (48.26 mm) between bolt hole centerline and top of piston rod (12) (detail B).

3. Torque nut (10) 660 to 780 in.-lbs. (74.57 to 88.13 N-m) and secure with lockwire (C-405).

4. Coat nut (10), lock (11), lockwire, and threads on clevis (9) with sealant (C-308). Refer to Figure 2, Sheet 3, Detail G.



DO NOT ALLOW THE SEALANT TO CONTACT THE PISTON ROD SURFACE OR SCRAPER AND ROD SEAL OR DAMAGE MAY OCCUR.

5. Put the universal (6, Figure 1) to the piston rod clevis (9). Install the bolt (7), washer (8), washer (14), nut (15), and cotter pin (16).

6. Move the boot (5) downward and put the bottom of the boot into lip in the mount (13).

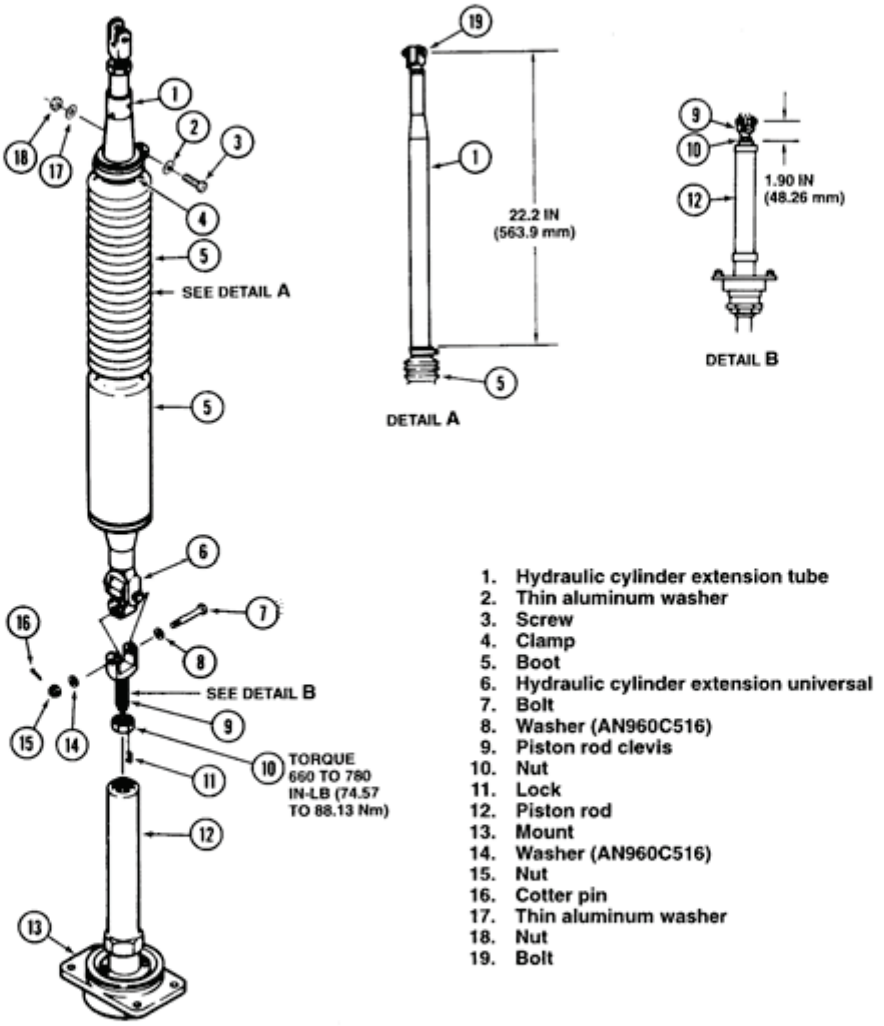
7. Put the top of boot (5) 22.2 inches (563.9 mm) below centerline bolt (19) (Detail A).

8. Attach the boot (5) to hydraulic cylinder extension tube (1) with the clamp (4), screw (3), washer (2), washer (17), and nut (18).

9. Make sure that the cyclic control rigging is within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).

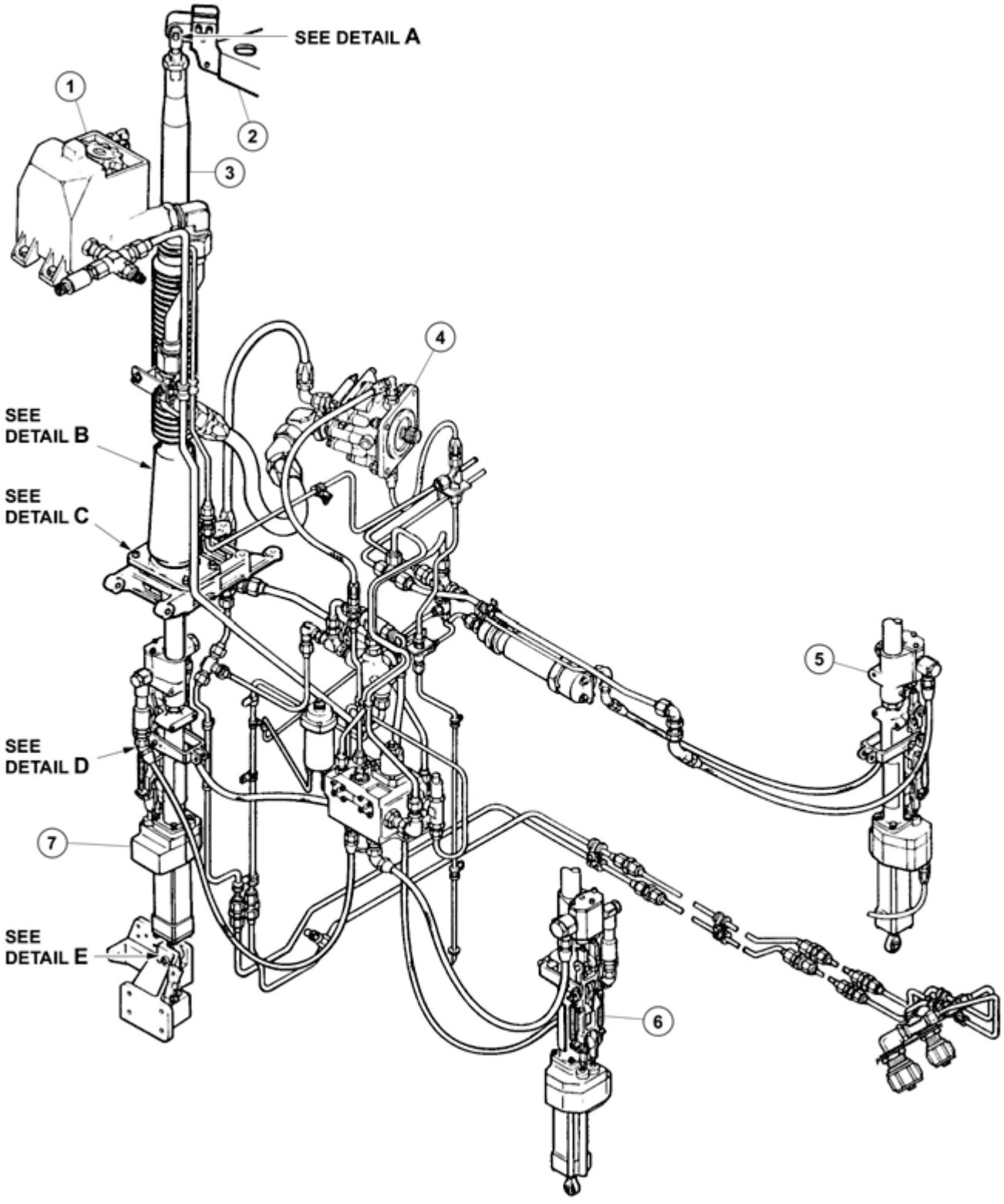
10. Move the cyclic controls through the full travel and check for the interference and binding.

11. Install the forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).



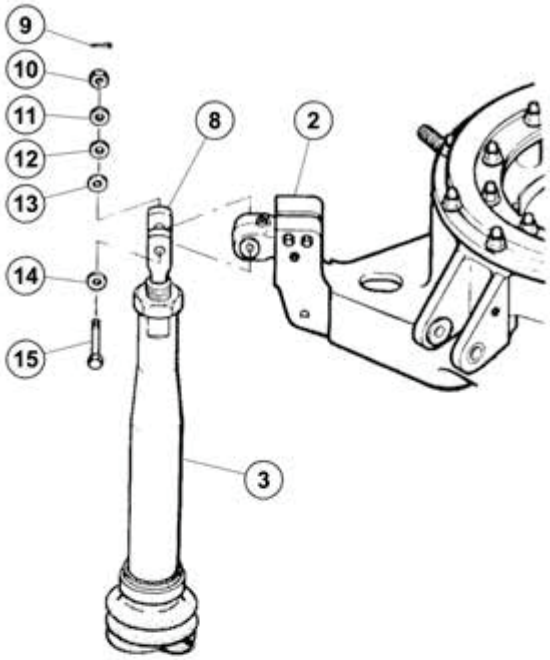
ICN-412-A-671200-B-97499-00005-A-001-01

Figure 1. Right Cyclic Piston Rod Clevis (Sheet 1 of 1)

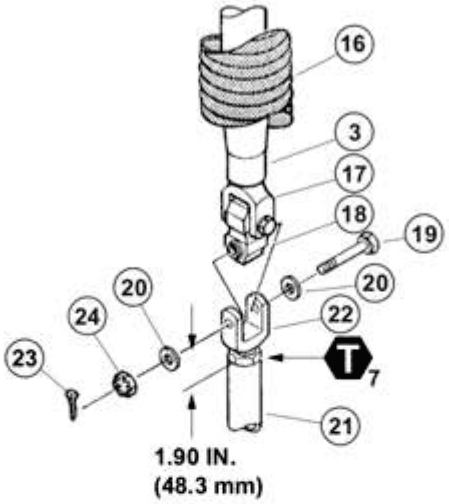


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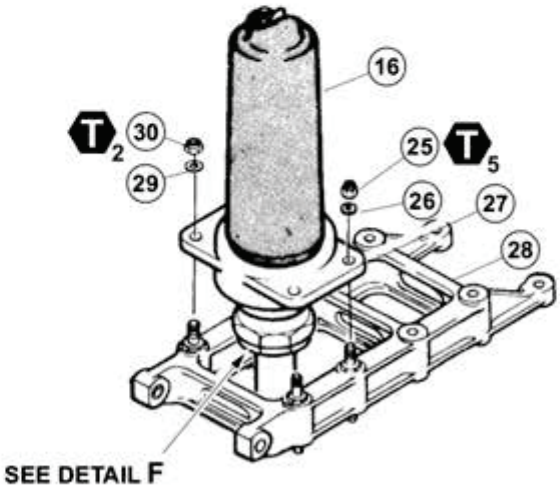
Figure 2. Right Cyclic Flight Control Cylinder (Sheet 1 of 4)



DETAIL A



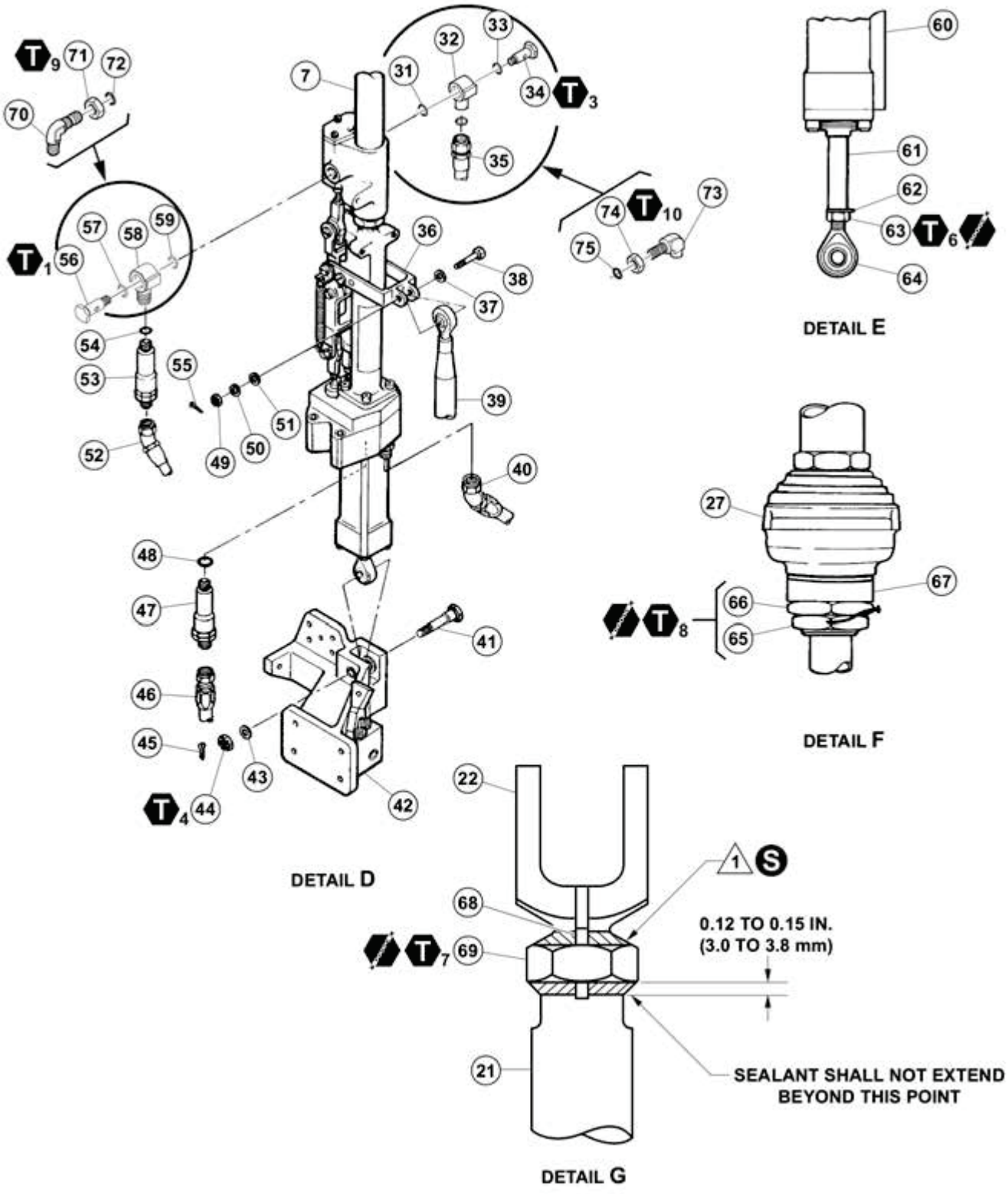
DETAIL B



DETAIL C

ICN-412-A-671200-B-97499-00010-A-001-01

Figure 2. Right Cyclic Flight Control Cylinder (Sheet 2 of 4)



ICN-412-A-671200-B-97499-00011-A-001-01

Figure 2. Right Cyclic Flight Control Cylinder (Sheet 3 of 4)

1. Reservoir
2. Swashplate horn
3. Extension tube
4. Pump
5. Collective flight control cylinder
6. Left cyclic flight control cylinder
7. Right cyclic flight control cylinder
8. Clevis
9. Cotter pin
10. Nut
11. Washer (AN960C516L)
12. Washer (AN960C516)
13. Washer (AN960C516)
14. Washer (AN960C516)
15. Bolt
16. Boot
17. Clevis
18. Universal
19. Bolt
20. Washer (AN960C516)
21. Upper piston
22. Clevis
23. Cotter pin
24. Nut
25. Nut
26. Aluminum washer
27. Support bracket
28. Support
29. Aluminum washer
30. Nut
31. Preformed packing
32. Elbow
33. Preformed packing
34. Bolt
35. Hose
36. Lever
37. Washer
38. Bolt
39. Tube assembly
40. Hose
41. Bolt
42. Support
43. Aluminum washer
44. Nut
45. Cotter pin
46. Hose
47. Bypass check valve
48. Preformed packing
49. Nut
50. Washer
51. Washer
52. Hose
53. Bypass check valve
54. Preformed packing
55. Cotter pin
56. Bolt
57. Preformed packing
58. Elbow
59. Preformed packing
60. Actuator
61. Lower piston
62. Locking device
63. Nut
64. Rod end bearing
65. Nut
66. Bearing nut
67. Keyed washer
68. Lock
69. Jam nut
70. Elbow
71. Nut
72. Packing
73. Elbow
74. Nut
75. Packing

LOCKWIRE (C-405)

S SEALANT (C-308)

T1 40 TO 65 IN-LBS (4.51 TO 7.34 Nm)

T2 75 TO 95 IN-LBS (8.47 TO 10.73 Nm)

T3 75 TO 125 IN-LBS (8.47 TO 14.12 Nm)

T4 95 TO 110 IN-LBS (10.73 TO 12.42 Nm)

T5 120 TO 160 IN-LBS (13.55 TO 18.07 Nm)

T6 200 TO 250 IN-LBS (22.59 TO 28.54 Nm)

T7 660 TO 780 IN-LBS (74.56 TO 88.12 Nm)

T8 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

T9 135 TO 145 IN-LBS (15.25 TO 16.38 Nm)

T10 215 TO 245 IN-LBS (24.30 TO 27.68 Nm)

NOTE

1. Apply sealant (C-308) around jam nut (69), lock (68), and exposed threads of clevis (22).

ICN-412-A-671200-B-97499-00012-A-001-01

Figure 2. Right Cyclic Flight Control Cylinder (Sheet 4 of 4)

Left Cyclic Flight Control Cylinder - Removal

Procedure

1. Remove the access panels that are around the pylon support structure in the aft cabin. Remove the forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).

2. Move the bottom of boot (18, Figure 1) to get the access. Remove the cotter pin (25), nut (24), washers (23), and bolt (20) (Detail B).

3. Remove the cotter pin (13), nut (12), washers (9, 10, 11, and 16), and the bolt (17) (Detail A). Remove the extension tube (14) with the attached universal (19).

Applicable: SN: 33001-33213 or 36001-36300:

4. Disconnect the hoses (37, 40, 43, and 54) (Detail D). Remove the bolts (34 and 55) and elbows (32 and 57) and discard the packings (31, 33, 56, and 58). Plug or cap the open ports to prevent contamination.

Applicable: SN: 36301-Subsequent:

5. Disconnect the hoses (37, 40, 43, and 54) (Detail D). Remove the elbows (69 and 72), nuts (70 and 73) and packings (71 and 74). Plug or cap the open ports to prevent contamination.

6. Remove the cotter pin (53), nut (52), two washers (51), bolt (38), and the washer (39).

7. Remove the nuts (26 and 30) and washers (27 and 29) (Figure 1 Detail C).

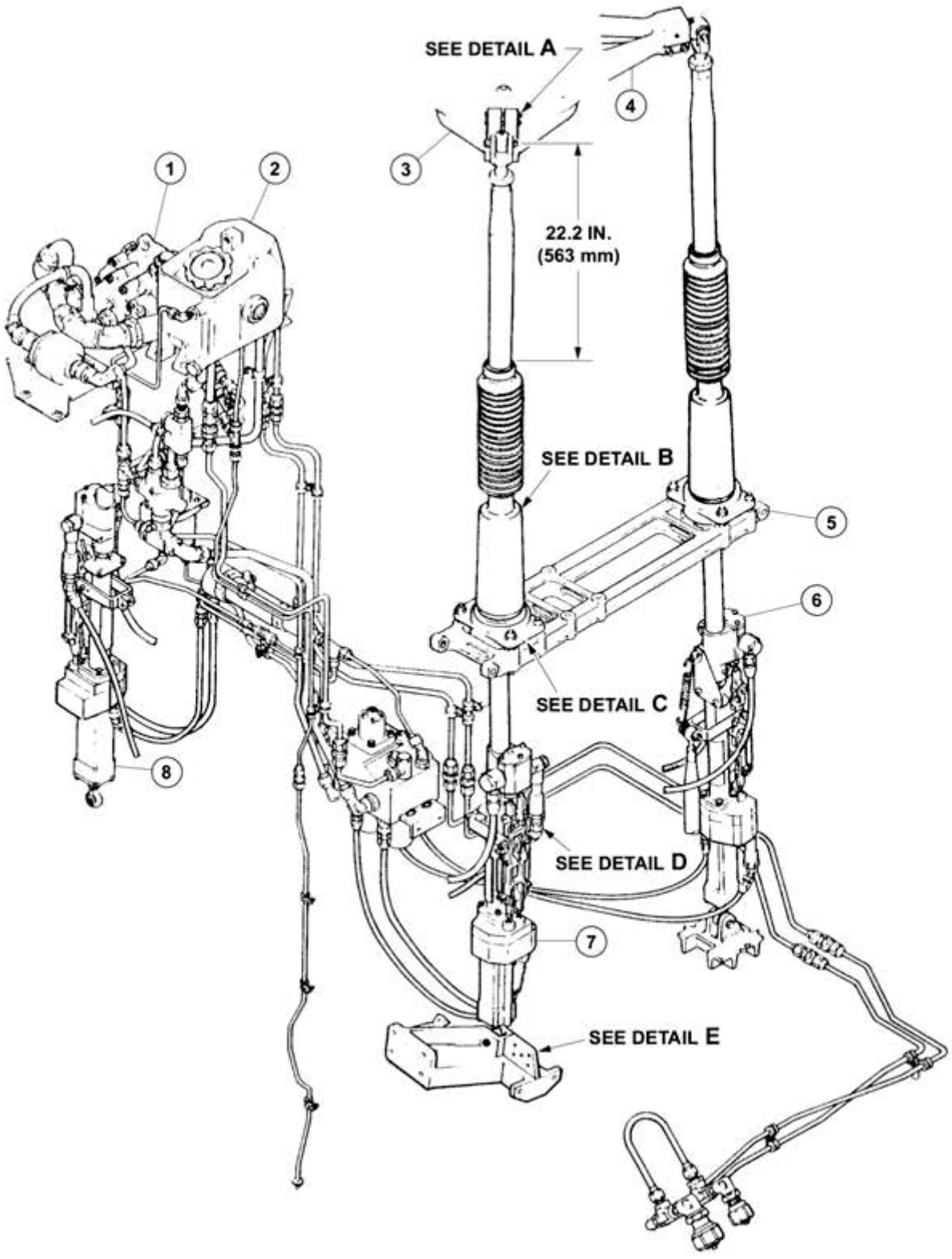
8. Remove the cotter pin (48), nut (47), washer (46), and the bolt (44).

NOTE

Turn the support bracket (28) to remove the cylinder assembly, if the transmission is installed. If the transmission has been removed, the cylinder assembly can be removed upward.

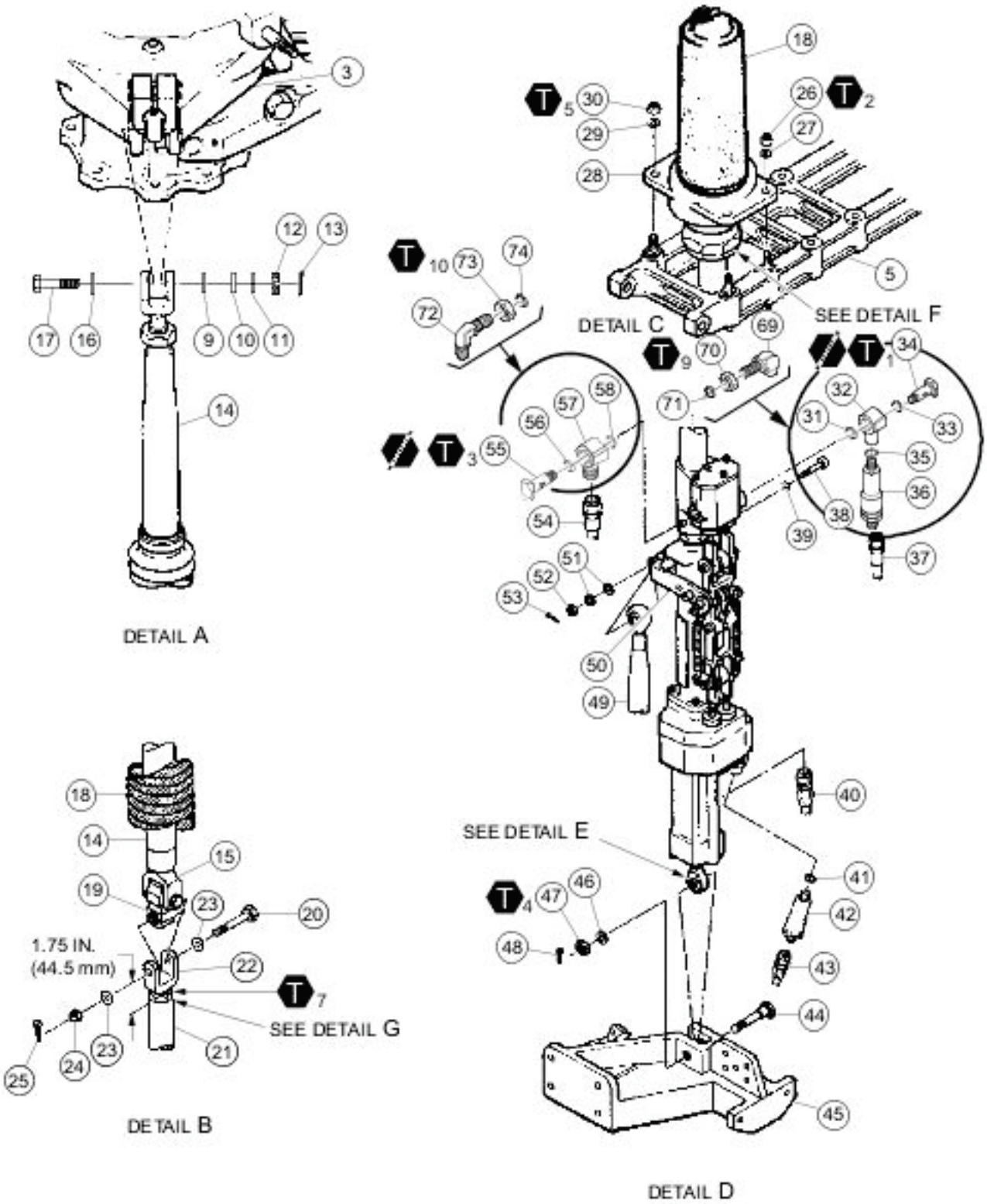
9. Carefully remove the left cyclic flight control cylinder (7) from the helicopter.

10. Loosen the jam nut (68) and remove the clevis (22) from the upper piston (21).



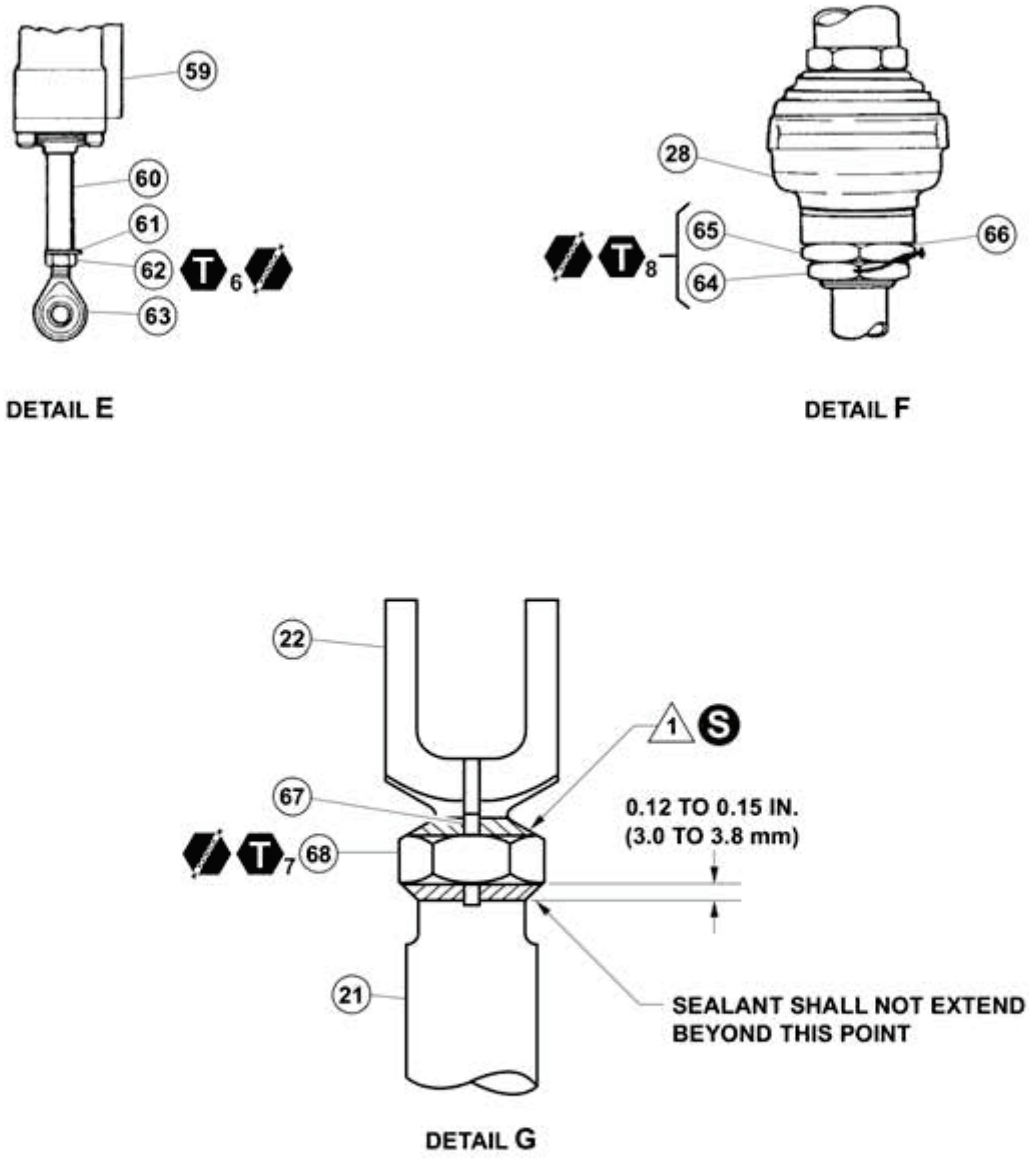
ICN-412-A-671200-B-97499-00006-A-001-01

Figure 1. Left Cyclic Flight Control Cylinder (Sheet 1 of 4)



ICN-412-A-671200-B-97499-00007-A-001-01

Figure 1. Left Cyclic Flight Control Cylinder (Sheet 2 of 4)



ICN-412-A-671200-B-97499-00008-A-001-01

Figure 1. Left Cyclic Flight Control Cylinder (Sheet 3 of 4)

1. Pump
2. Reservoir
3. Swashplate horn
4. Collective lever
5. Support
6. Collective flight control cylinder
7. Left cyclic flight control cylinder
8. Right cyclic flight control cylinder
9. Washer (AN960C516)
10. Washer (AN960C516)
11. Washer (AN960C516L)
12. Nut
13. Cotter pin
14. Extension tube
15. Clevis
16. Washer (AN960C516)
17. Bolt
18. Boot
19. Universal
20. Bolt
21. Upper piston
22. Piston rod clevis
23. Washer (AN960C516)
24. Nut
25. Cotter pin
26. Nut
27. Aluminum washer
28. Support bracket
29. Aluminum washer
30. Nut
31. Preformed packing
32. Elbow
33. Preformed packing
34. Bolt
35. Preformed packing
36. Bypass check valve
37. Hose
38. Bolt
39. Washer
40. Hose
41. Preformed packing
42. Bypass check valve
43. Hose
44. Bolt
45. Support
46. Aluminum washer
47. Nut
48. Cotter pin
49. Tube assembly
50. Lever
51. Washer
52. Nut
53. Cotter pin
54. Hose
55. Bolt
56. Preformed packing
57. Elbow
58. Preformed packing
59. Actuator
60. Lower piston
61. Locking device
62. Nut
63. Rod end bearing
64. Nut
65. Bearing nut
66. Keyed washer
67. Lock
68. Jam nut
69. Elbow
70. Nut
71. Packing
72. Elbow
73. Nut
74. Packing

LOCKWIRE (C-405)

SEALANT (C-308)

T1 40 TO 65 IN-LBS (4.51 TO 7.34 Nm)

T2 75 TO 95 IN-LBS (8.47 TO 10.73 Nm)

T3 75 TO 125 IN-LBS (8.47 TO 14.12 Nm)

T4 95 TO 110 IN-LBS (10.73 TO 12.42 Nm)

T5 120 TO 160 IN-LBS (13.55 TO 18.07 Nm)

T6 200 TO 250 IN-LBS (22.59 TO 28.54 Nm)

T7 660 TO 780 IN-LBS (74.56 TO 88.12 Nm)

T8 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

T9 135 TO 145 IN-LBS (15.25 TO 16.38 Nm)

T10 215 TO 245 IN-LBS (24.30 TO 27.68 Nm)

NOTE

1. Apply sealant (C-308) around jam nut (68), lock (67), and exposed threads of pistion rod clevis (22).

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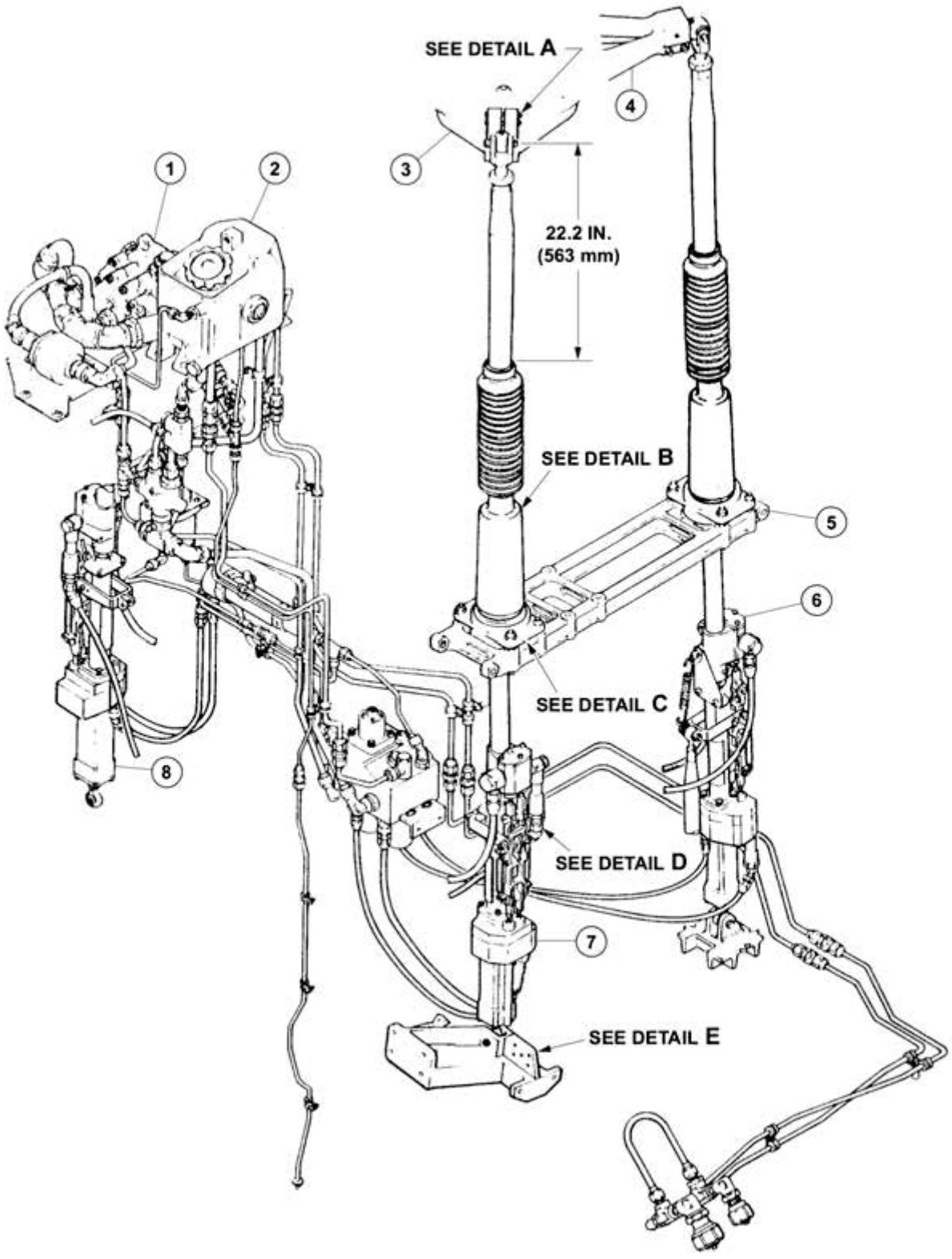
Figure 1. Left Cyclic Flight Control Cylinder (Sheet 4 of 4)

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Left Cyclic Flight Control Cylinder - Inspection

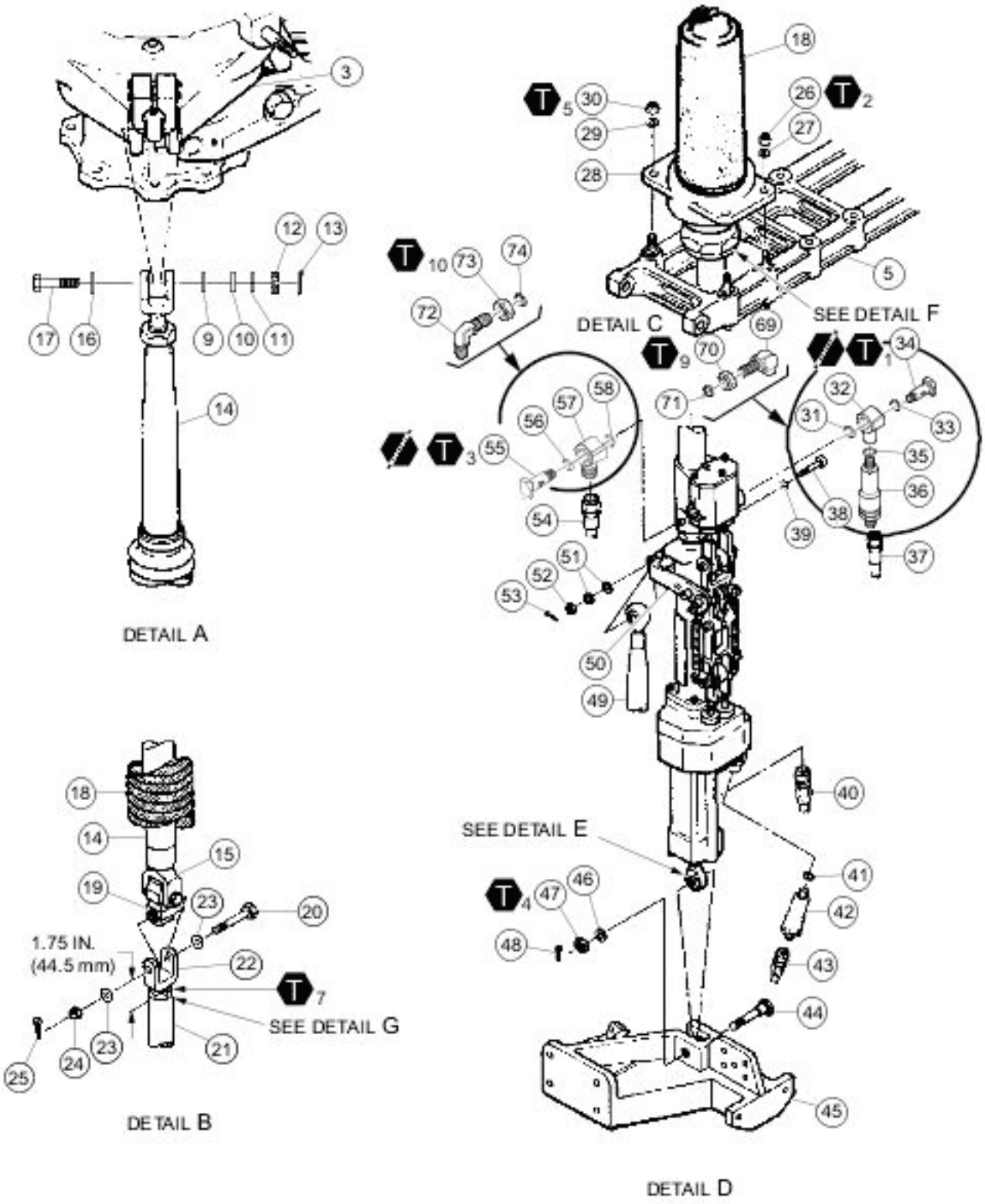
Procedure

1. Examine the cylinder (7, Figure 1) for damage and signs of leakage.
2. Examine the exposed areas of the piston for scratches and wear.
3. Examine the bearings for looseness (BHT-ALL-SPM, Standard Practices Manual).



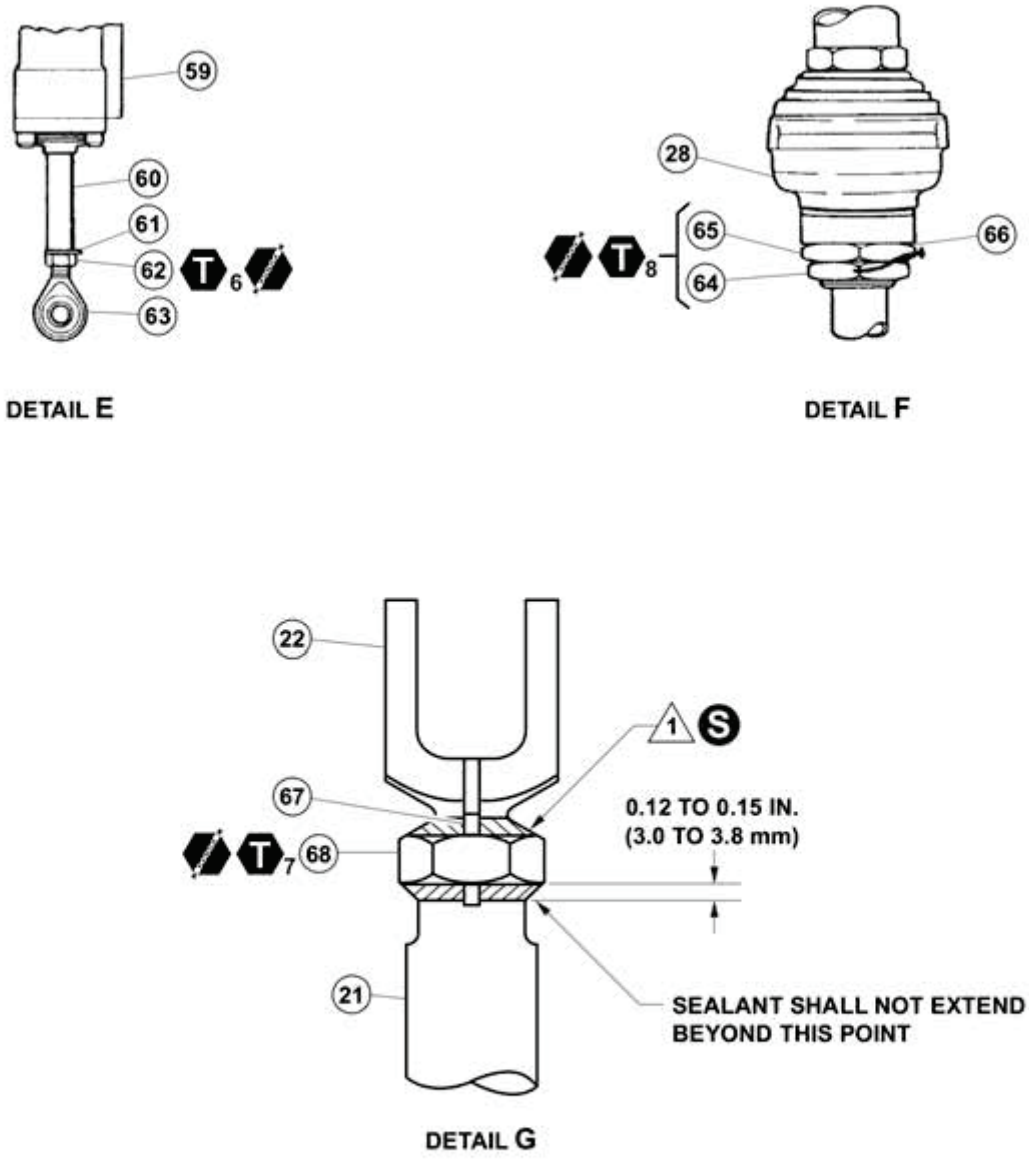
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Figure 1. Left Cyclic Flight Control Cylinder (Sheet 1 of 3)



ICN-412-A-671200-B-97499-00007-A-001-01

Figure 1. Left Cyclic Flight Control Cylinder (Sheet 2 of 3)



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Figure 1. Left Cyclic Flight Control Cylinder (Sheet 3 of 3)

Left Cyclic Flight Control Cylinder - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-002).
Hydraulic fluid	C-072	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-072).
Corrosion preventive compound	C-104	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

NOTE

When the transmission is installed, insert the lower end of the cylinder assembly through the passenger/cargo compartment (412-A-52-12-00-00A-042A-A / 00810) until the top surface clears the access hole in the forward island then lift the cylinder assembly through the cylinder support (5) until the support bracket (28, Figure 1) is past the attachment studs.

1. Cut and remove the lockwire, and loosen the nuts (65 and 66, Figure 1) so the support bracket (28) may be turned easily (Detail F).

2. Coat threads of the piston rod clevis (68) with corrosion preventive compound (C-104). Install the clevis (22) with the jam nut (68) and the lock (67) in the upper piston (21) to dimension of 1.75 inches (44.45 mm) between the end of the upper piston and center of the holes in the clevis.



AVOID GETTING SEALANT ON PISTON ROD SURFACE OR SCRAPER AND ROD SEAL DAMAGE MAY OCCUR.

3. Torque the jam nut (68) 660 to 780 inch-pounds (74.56 to 88.12 Nm) and safety with the lockwire (C-405). Cover the jam nut (68), lock (67), and threads of the clevis (22) with the sealant (C-308) (Detail G).

4. Place the universal (19), attached to the tube (14), into the clevis (22) on the upper piston (21) and install the bolt (20), washers (23), nut (24), and the cotter pin (25) (Detail B).

5. Lift the bottom of the boot (18) to expose the universal (19) and clevis (22). Apply the corrosion preventive compound (C-104) on the universal (19) and clevis (22) as required. Apply the corrosion preventive compound (C-104) on the mount bearing and in the housing cavity. Position the lower end of the boot (18) in groove of the support bracket (28).

6. Put the left cyclic flight control cylinder (7) carefully into the helicopter. Put the support bracket (28) over the studs in the support (5) (Detail C). Install the aluminum washers (27) and torque the nuts (26) 75 to 95 inch-pounds (8.47 to 10.73 Nm). Install the aluminum washer (29) and torque the nut (30) 120 to 160 inch-pounds (13.55 to 18.0 Nm).

7. Pull the lower piston (60) full down and push the actuator (59) full up (Detail E). If the hole in the rod end bearing (63) does not align with the hole in the support (45), adjust the rod end bearing (63).



INCORRECT TORQUE TECHNIQUE CAN DESTROY THE TAB ON THE LOCKING DEVICE (61). THE ROD END BEARING (63) MUST BE HELD SAFETY TO PREVENT DAMAGE TO THE LOCKING DEVICE (61).

7.1. Remove the lockwire and loosen the nut (62), and turn the rod end bearing (63) to align with the bolt (44) hole.

7.2. Torque the nut (62) to 200 to 250 inch-pounds (22.59 to 28.24 Nm) and safety to the locking device (61) with the lockwire (C-405).

7.3. Examine the locking device (61) to ensure the tab is in groove of the rod end bearing (63) and that it was not damaged when the nut (62) was tightened. Replace the locking device (61) if tab is twisted or damaged.

8. Align the left cyclic flight control cylinder (7) into the support (45) as follows:

8.1. Move the lower piston (60) rod up and clear of the support (45). The lower piston (60) rod must be able to be moved freely down, up, and then down, and the bolt (44) replaced without any pressure, forcing, binding, or bending of piston rod in the fore, aft, left, or right direction. One person must guide the extension tube (14) to prevent the damage when conducting this step and subsequent steps.

8.2. If the cylinder (7) meets the requirements of the Step 8.1., it is considered in alignment. If the cylinder (7) is aligned, accomplish the Step 8.8. through Step 8.9.. If the cylinder (7) is misaligned, accomplish the Step 8.3. through Step 8.9..

8.3. Loosen the nut (64) and bearing nut (65) so the cylinder (7) is loose.

8.4. Move the rod end bearing (63) up approximately 0.5 inch (12.7 mm) above the support (45). Turn the lower end of the cylinder (7) in a 3.0 inch (76.2 mm) circle.

8.5. Put the rod end bearing (63) directly on the lugs of the support (45) and move the piston (60) down and temporarily install the bolt (44).

8.6. Tighten the bearing nut (65) by hand (approximately 70 inch-pounds (7.90 Nm)).

8.7. Remove the bolt (44) and move the rod end bearing (64) up and out of the support (45), then move down. The rod end bearing (63) must be free of any side loading. Do Step 8.3. through Step 8.7. again until free of side loading.

NOTE

If the left cyclic flight control cylinder (7) still cannot be installed without the preload, it is permitted to add the one washer AN960C416L (maximum) on the two aft studs of the support (5) between the support bracket (28) and support (5) . Or, it is permitted to add the one washer AN960C416L (maximum) and the one washer AN960C516L (maximum) on each of the two forward studs between the support bracket (28) and support (5).

8.8. Install the bolt (44). Torque the bearing nut (65) 900 to 1100 inch-pounds (101.68 to 124.28 Nm). Torque the nut (64) 900 to 1100 inch-pounds (101.68 to 124.28 Nm). Safety the nuts (65 and 64) to the keyed washer (66) with the lockwire (C-405).

8.9. If the cylinder (7) still meets the requirements of the Step 8.1., install the aluminum washer (46) and nut (47) and torque the nut 95 to 110 inch-pounds (10.73 to 12.42 Nm). Install the cotter pin (48).

NOTE

Use the washers (10 and 12) as required to get the proper engagement of the cotter pin (13) and nut (12).

9. Put the extension tube (14) on the swashplate horn (3). Install the bolt (17), washers (16, 9, 10, and 11), and the nut (12). Install the cotter pin (13) (Detail A).

10. Put the tube assembly (49, Detail D) into the lever (50) and safety with the bolt (38), washer (39), washers (51), nut (52), and the cotter pin (53).

NOTE

SN: 33001-33213 or 36001-36300:

Refer to Step 11. through Step 13. for the installation of preformed packings.

11. Lubricate the new preformed packings (31, 33, 56, and 58) with the hydraulic fluid (C-002) or hydraulic fluid (C-072). Install the preformed packing (31 and 58), elbows (32 and 57), and the preformed packings (33 and 56) with the bolts (34 and 55).

12. Torque the bolt (34) 40 to 65 inch-pounds (4.52 to 7.34 Nm). Secure the bolt (34) with lockwire (C-405).

13. Torque the bolt (55) 75 to 125 inch-pounds (8.47 to 14.12 Nm). Secure the bolt (55) with lockwire (C-405).

NOTE

SN: 36301-Subsequent:

Refer to Step 14. through Step 16. for the installation of preformed packings.

14. Install the preformed packings (71 and 74), elbows (69 and 72), and nuts (70 and 73).

15. Torque the nut (70) 135 to 145 inch-pounds (15.25 to 16.38 Nm).

16. Torque the nut (73) 215 to 245 inch-pounds (24.30 to 27.68 Nm).

17. If the bypass check valves have been removed, lubricate the new preformed packings (35 and 41) with the hydraulic fluid (C-002) or hydraulic fluid (C-072). Install the preformed packings and the bypass check valves (36 and 42).

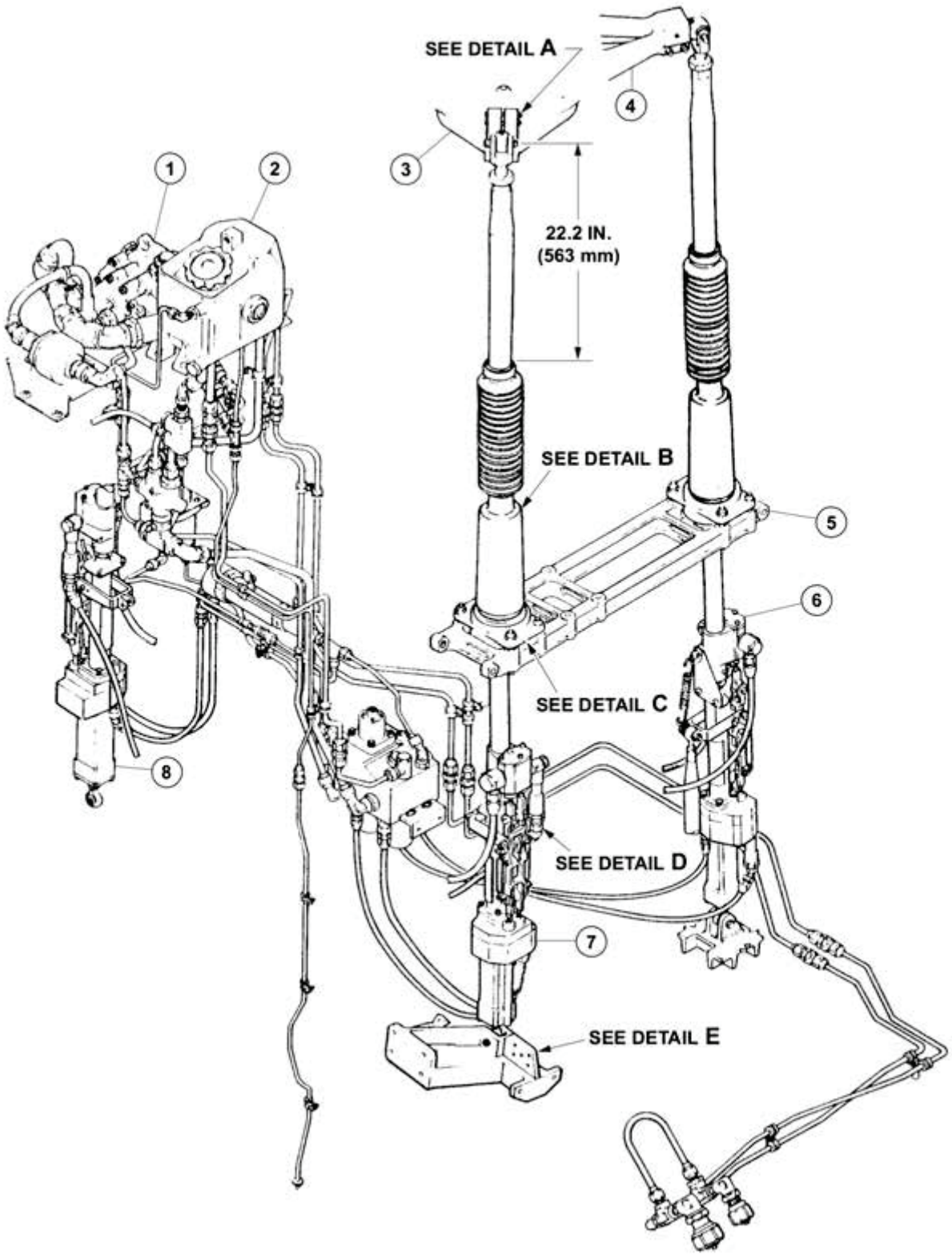
18. Connect the hoses (37, 40, 43, and 54) observe labels on the hydraulic systems No. 1 and No. 2.

19. Bleed and leak check the hydraulic system (412-A-29-00-00-00A-042A-A / 00612).

20. Energize the hydraulic system and operate the cyclic controls through 10 complete operations. Check for the leaks.

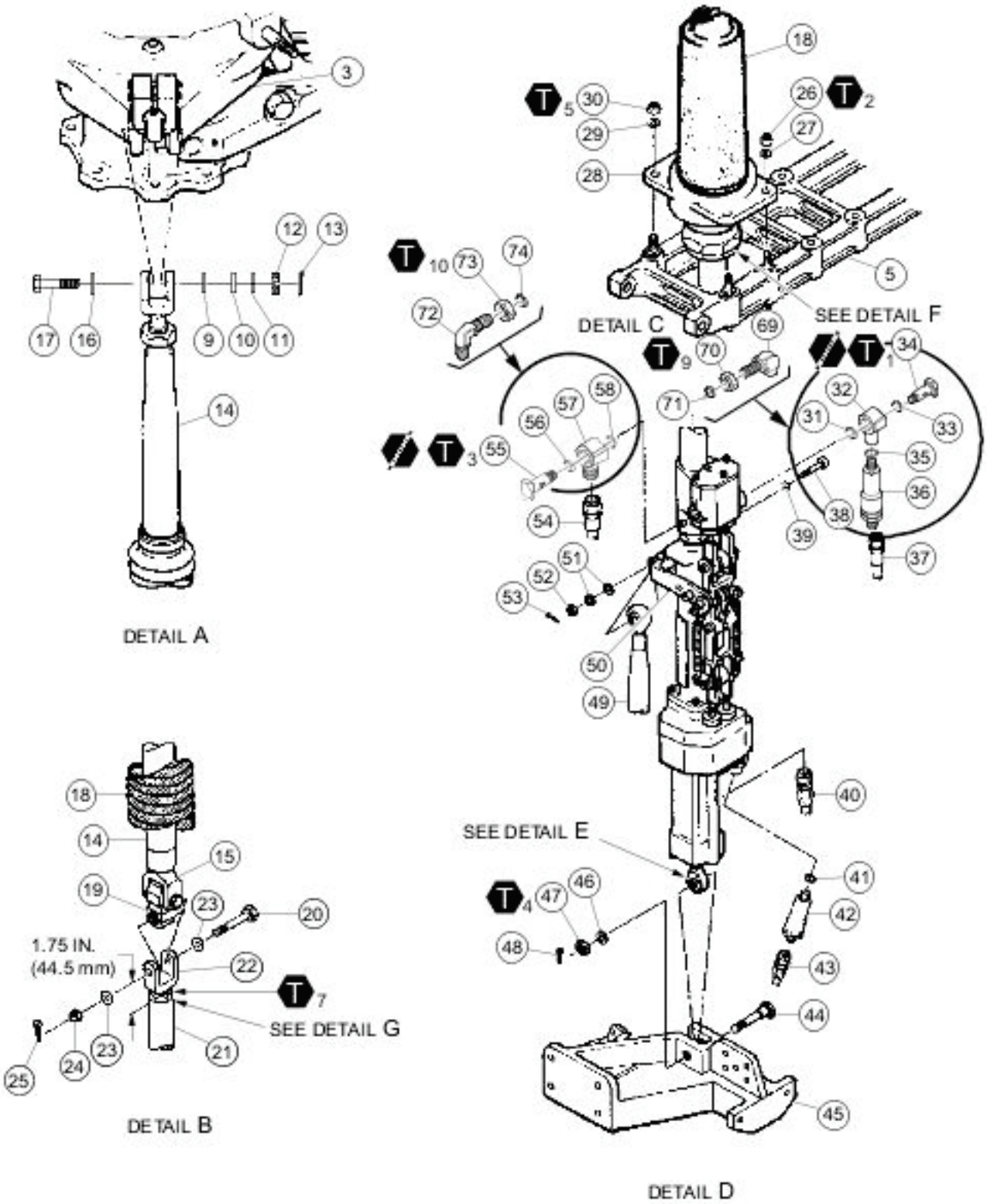
21. Make sure that the cyclic control rigging is within the limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).

22. Install the access panel on the pylon support. Install the forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).



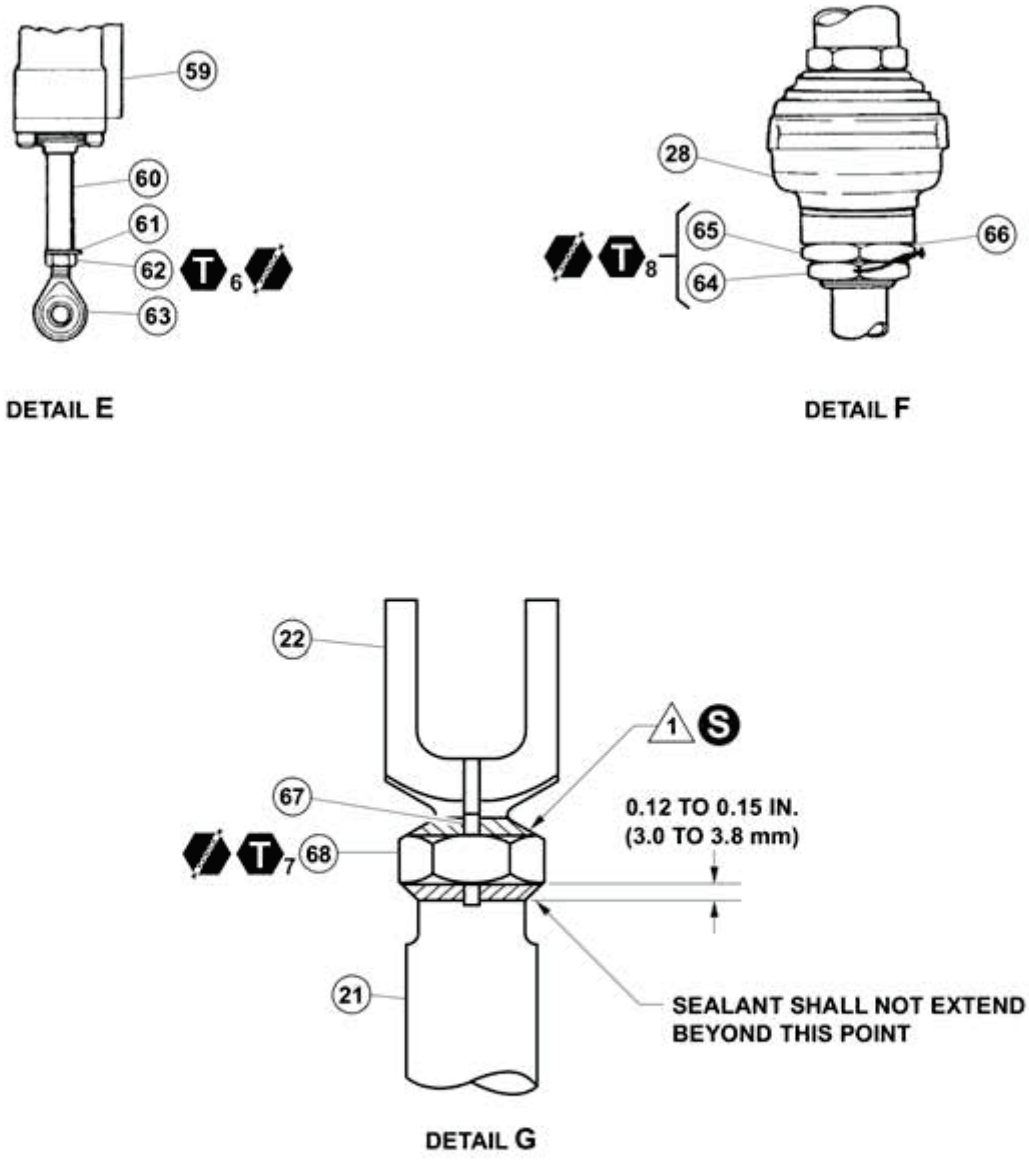
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Figure 1. Left Cyclic Flight Control Cylinder (Sheet 1 of 4)



ICN-412-A-671200-B-97499-00007-A-001-01

Figure 1. Left Cyclic Flight Control Cylinder (Sheet 2 of 4)



ICN-412-A-671200-B-97499-00008-A-001-01

Figure 1. Left Cyclic Flight Control Cylinder (Sheet 3 of 4)

1. Pump
2. Reservoir
3. Swashplate horn
4. Collective lever
5. Support
6. Collective flight control cylinder
7. Left cyclic flight control cylinder
8. Right cyclic flight control cylinder
9. Washer (AN960C516)
10. Washer (AN960C516)
11. Washer (AN960C516L)
12. Nut
13. Cotter pin
14. Extension tube
15. Clevis
16. Washer (AN960C516)
17. Bolt
18. Boot
19. Universal
20. Bolt
21. Upper piston
22. Piston rod clevis
23. Washer (AN960C516)
24. Nut
25. Cotter pin
26. Nut
27. Aluminum washer
28. Support bracket
29. Aluminum washer
30. Nut
31. Preformed packing
32. Elbow
33. Preformed packing
34. Bolt
35. Preformed packing
36. Bypass check valve
37. Hose
38. Bolt
39. Washer
40. Hose
41. Preformed packing
42. Bypass check valve
43. Hose
44. Bolt
45. Support
46. Aluminum washer
47. Nut
48. Cotter pin
49. Tube assembly
50. Lever
51. Washer
52. Nut
53. Cotter pin
54. Hose
55. Bolt
56. Preformed packing
57. Elbow
58. Preformed packing
59. Actuator
60. Lower piston
61. Locking device
62. Nut
63. Rod end bearing
64. Nut
65. Bearing nut
66. Keyed washer
67. Lock
68. Jam nut
69. Elbow
70. Nut
71. Packing
72. Elbow
73. Nut
74. Packing

LOCKWIRE (C-405)

SEALANT (C-308)

T1 40 TO 65 IN-LBS (4.51 TO 7.34 Nm)

T2 75 TO 95 IN-LBS (8.47 TO 10.73 Nm)

T3 75 TO 125 IN-LBS (8.47 TO 14.12 Nm)

T4 95 TO 110 IN-LBS (10.73 TO 12.42 Nm)

T5 120 TO 160 IN-LBS (13.55 TO 18.07 Nm)

T6 200 TO 250 IN-LBS (22.59 TO 28.54 Nm)

T7 660 TO 780 IN-LBS (74.56 TO 88.12 Nm)

T8 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

T9 135 TO 145 IN-LBS (15.25 TO 16.38 Nm)

T10 215 TO 245 IN-LBS (24.30 TO 27.68 Nm)

NOTE

1. Apply sealant (C-308) around jam nut (68), lock (67), and exposed threads of pistion rod clevis (22).

ICN-412-A-671000-B-97499-00937-A-001-01

Figure 1. Left Cyclic Flight Control Cylinder (Sheet 4 of 4)

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Right Cyclic Flight Control Cylinder - Removal

Procedure

1. Remove the access panels that are around the pylon support structure in the aft cabin. Remove the forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).

Applicable: SN: 33001-33213 or 36001-36300:

2. Disconnect the hoses (35, 40, 46, and 52, Figure 1, Detail D). Remove the bolts (34 and 56) and elbows (32 and 58). Plug or cap the open ports.

Applicable: SN: 36301-Subsequent:

3. Disconnect the hoses (35, 40, 46, and 52, Figure 1, Detail D). Remove the elbows (70 and 73), nuts (71 and 74) and packings (72 and 75). Plug or cap the open ports to prevent contamination.

4. Remove the cotter pin (55), nut (49), washers (37, 50, and 51), and bolt (38). Remove the tube assembly (39) from the lever (36).

5. Remove the cotter pin (45), nut (44), aluminum washer (43), and bolt (41) (Detail D).

6. Remove the nuts (25 and 30) and aluminum washers (26 and 29) (Detail C).

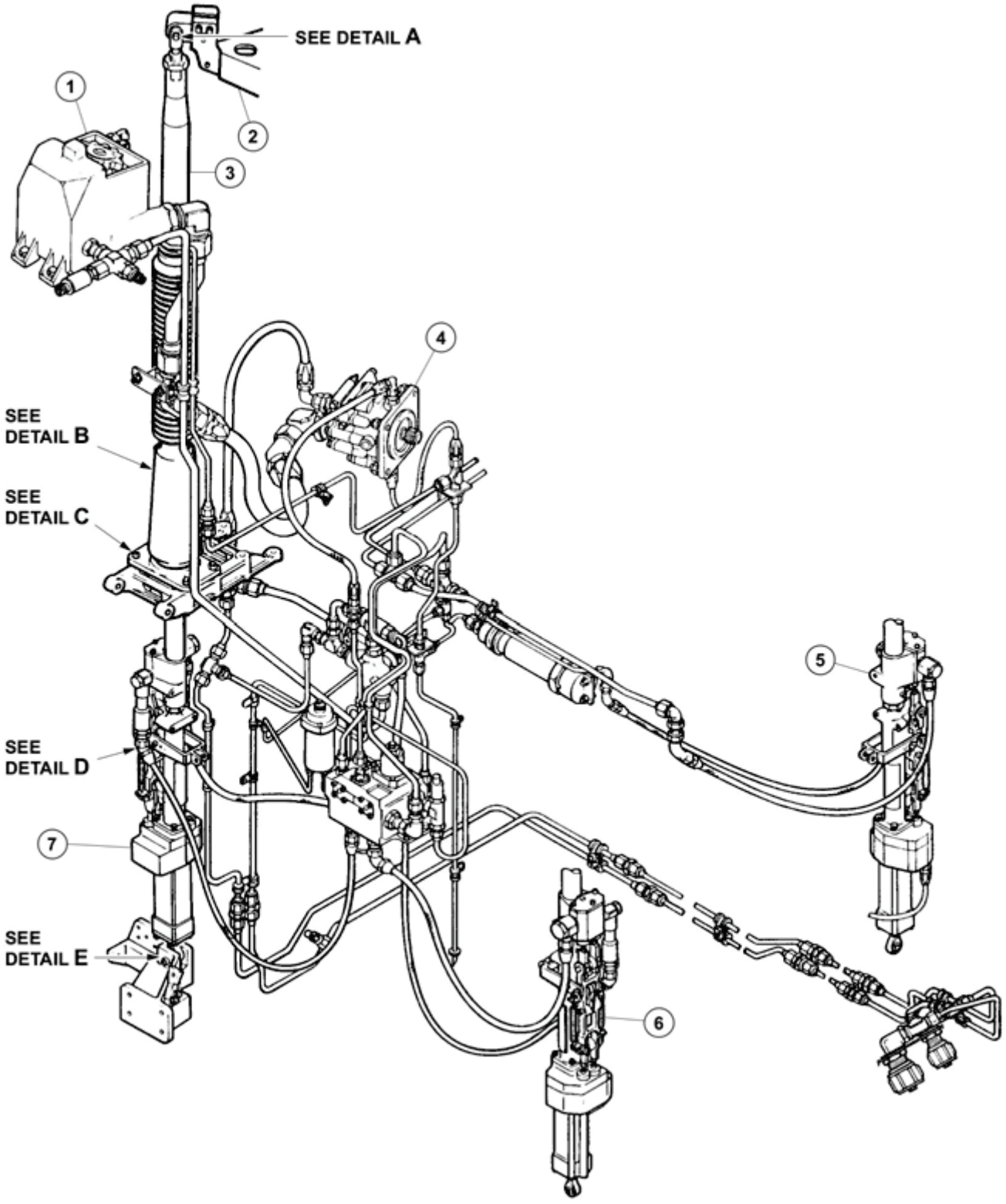
7. Remove the cotter pin (9, Detail A), nut (10), washers (11, 12, 13, and 14), and the bolt (15).

8. Carefully remove the right cyclic control cylinder (7) and extension tube (3) from the helicopter.

9. Move the boot (16) to give the access to the universal (18) (Detail B). Remove the cotter pin (23), nut (24), washers (20), and bolt (19).

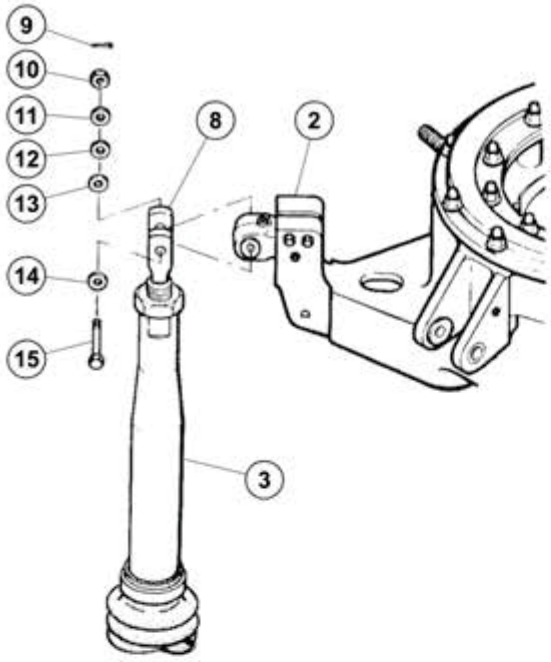
10. Remove the extension tube (3) with the attached universal (18) from the right cyclic flight control cylinder (7).

11. Loosen the jam nut (69) and remove the clevis (22) from the upper piston (21).

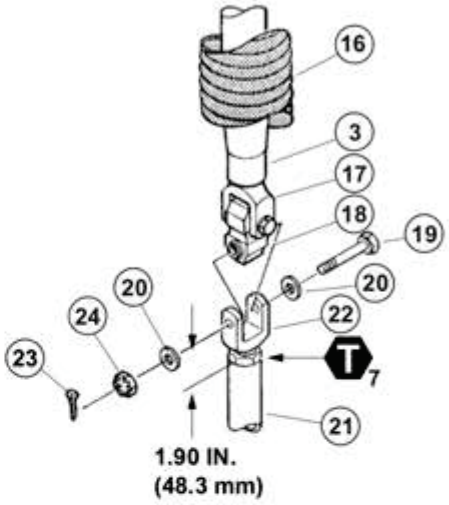


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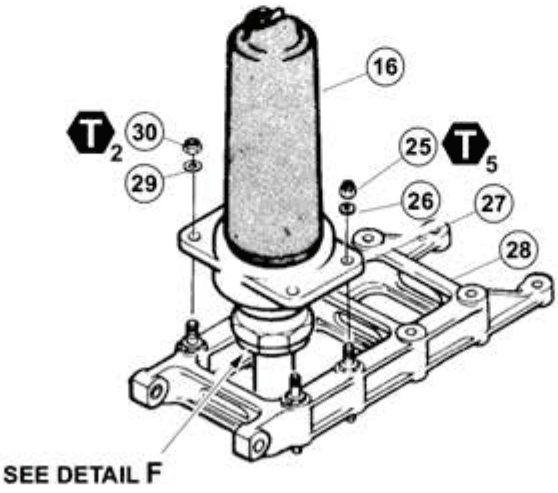
Figure 1. Right Cyclic Flight Control Cylinder (Sheet 1 of 4)



DETAIL A



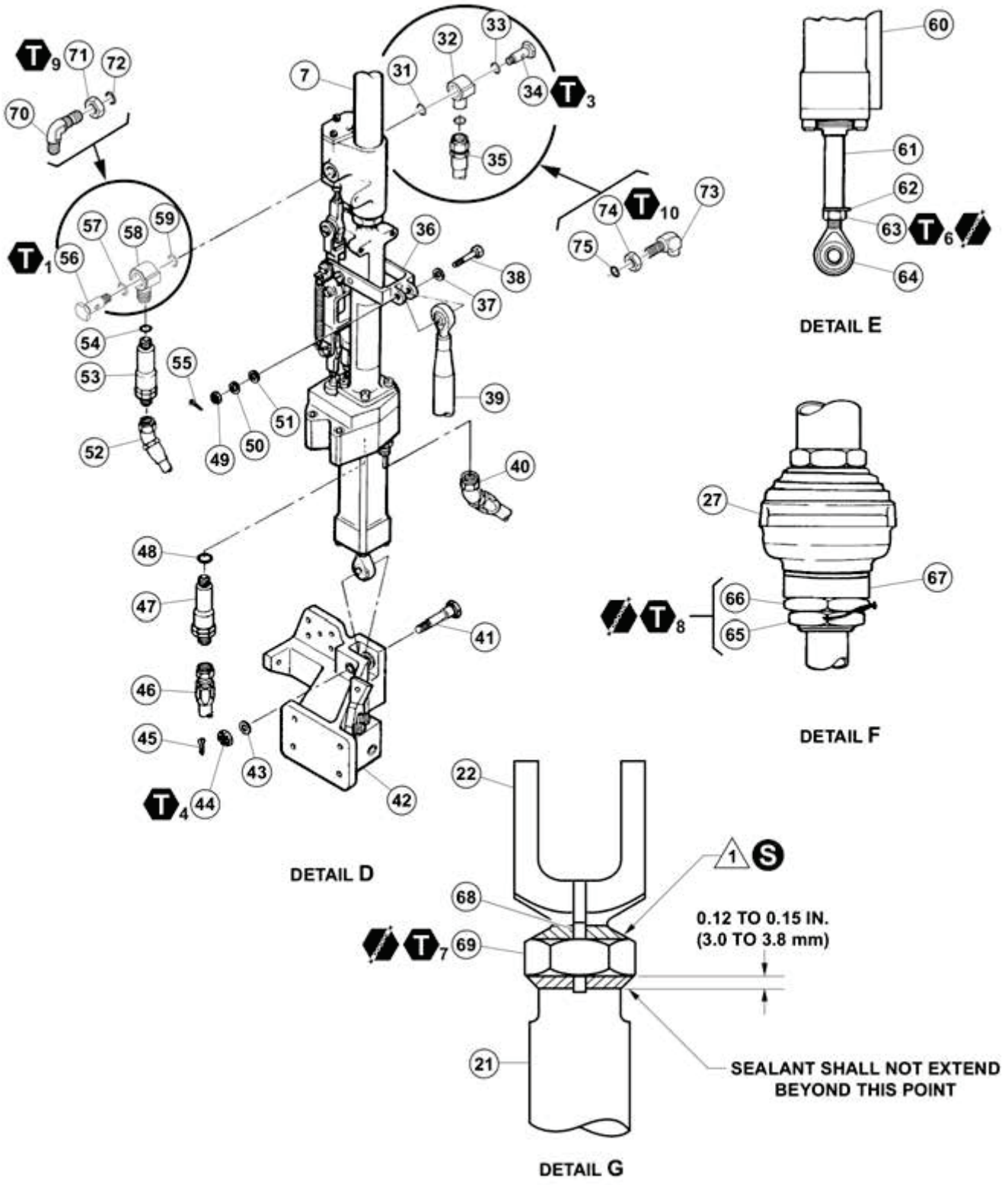
DETAIL B



DETAIL C

ICN-412-A-671200-B-97499-00010-A-001-01

Figure 1. Right Cyclic Flight Control Cylinder (Sheet 2 of 4)



ICN-412-A-671200-B-97499-00011-A-001-01

Figure 1. Right Cyclic Flight Control Cylinder (Sheet 3 of 4)

1. Reservoir
2. Swashplate horn
3. Extension tube
4. Pump
5. Collective flight control cylinder
6. Left cyclic flight control cylinder
7. Right cyclic flight control cylinder
8. Clevis
9. Cotter pin
10. Nut
11. Washer (AN960C516L)
12. Washer (AN960C516)
13. Washer (AN960C516)
14. Washer (AN960C516)
15. Bolt
16. Boot
17. Clevis
18. Universal
19. Bolt
20. Washer (AN960C516)
21. Upper piston
22. Clevis
23. Cotter pin
24. Nut
25. Nut
26. Aluminum washer
27. Support bracket
28. Support
29. Aluminum washer
30. Nut
31. Preformed packing
32. Elbow
33. Preformed packing
34. Bolt
35. Hose
36. Lever
37. Washer
38. Bolt
39. Tube assembly
40. Hose
41. Bolt
42. Support
43. Aluminum washer
44. Nut
45. Cotter pin
46. Hose
47. Bypass check valve
48. Preformed packing
49. Nut
50. Washer
51. Washer
52. Hose
53. Bypass check valve
54. Preformed packing
55. Cotter pin
56. Bolt
57. Preformed packing
58. Elbow
59. Preformed packing
60. Actuator
61. Lower piston
62. Locking device
63. Nut
64. Rod end bearing
65. Nut
66. Bearing nut
67. Keyed washer
68. Lock
69. Jam nut
70. Elbow
71. Nut
72. Packing
73. Elbow
74. Nut
75. Packing

LOCKWIRE (C-405)

S SEALANT (C-308)

T1 40 TO 65 IN-LBS (4.51 TO 7.34 Nm)

T2 75 TO 95 IN-LBS (8.47 TO 10.73 Nm)

T3 75 TO 125 IN-LBS (8.47 TO 14.12 Nm)

T4 95 TO 110 IN-LBS (10.73 TO 12.42 Nm)

T5 120 TO 160 IN-LBS (13.55 TO 18.07 Nm)

T6 200 TO 250 IN-LBS (22.59 TO 28.54 Nm)

T7 660 TO 780 IN-LBS (74.56 TO 88.12 Nm)

T8 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

T9 135 TO 145 IN-LBS (15.25 TO 16.38 Nm)

T10 215 TO 245 IN-LBS (24.30 TO 27.68 Nm)

NOTE

1. Apply sealant (C-308) around jam nut (69), lock (68), and exposed threads of clevis (22).

ICN-412-A-671200-B-97499-00012-A-001-01

Figure 1. Right Cyclic Flight Control Cylinder (Sheet 4 of 4)

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Right Cyclic Flight Control Cylinder - Inspection

Procedure

1. Examine the right cyclic flight control cylinder in same manner outlined for the left cylinder (412-A-67-12-00-06A-280A-A / 01421).

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Right Cyclic Flight Control Cylinder - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-002).
Hydraulic fluid	C-072	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-072).
Corrosion preventive compound	C-104	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

NOTE

When the transmission is installed, insert the lower end of cylinder assembly through the passenger/cargo compartment until the top surface clears the access hole in the forward island. Then, lift the cylinder assembly through the cylinder support (28) until the support bracket (27, Figure 1) is past the attachment studs.

1. Cut and remove the lockwire and loosen the nut (65 and 66, Figure 1) so that the support bracket (27) may be turned easily (Detail F).

2. Coat threads of the piston rod clevis (22) with corrosion preventive compound (C-104). Install the clevis (22) with the jam nut (69) and lock (68) in the upper piston (21) to dimension of 1.90 inches (48.3 mm) between the end of the piston (21) and center of the holes in the clevis (22).



AVOID GETTING SEALANT ON PISTON ROD SURFACE OR SCRAPER AND ROD SEAL DAMAGE MAY OCCUR.

3. Torque the jam nut (69) 660 to 780 inch-pounds (to 88.12 Nm) and safety with the lockwire (C-405). Cover the jam nut (69), lock (68), and threads of the clevis (22) with the sealant (C-308) (Detail G).

4. Put the universal (18), attached to the extension tube (3), into the clevis (22) on the upper piston (21) and install the bolt (19), washers (20), nut (24), and the cotter pin (23) (Detail B).

5. Lift the bottom of the boot (16) to show the universal (18) and clevis (22).Apply the corrosion preventive compound (C-104) on the universal (18) and clevis (22) as required. Apply the corrosion preventive compound (C-104) on the mount bearing and in the housing cavity. Put the lower end of the boot (16) in the groove of the support bracket (27).

6. Put the right cyclic flight control cylinder (7) carefully into the helicopter. Place the support bracket (27) on the studs in the support (28) (Detail C). Install the aluminum washers (29) and nuts (30). Torque the nuts 75 to 95 inch-pounds (8.47 to 10.73 Nm.

7. Install the aluminium washers (26) and nuts (25). Torque the nuts 120 to 160 inch-pounds (13.55 to 18.07 Nm).

8. Pull the lower piston (61) full down and push the actuator (60) full up (Detail E). If the hole in the rod end bearing (64) does not align with the hole in the support (42), adjust the rod end bearing (64) as follows:



INCORRECT TORQUE TECHNIQUE CAN DESTROY THE TAB ON THE LOCKING DEVICE (62). THE ROD END BEARING (64) MUST BE HELD SECURELY TO PREVENT THE DAMAGING OF THE LOCKING DEVICE (62).

8.1. Remove the lockwire and loosen the nut (63), and turn the rod end bearing (64) to align with the bolt (41) hole.

8.2. Torque the nut (63) to 200 to 250 inch-pounds (22.59 to 28.24 Nm) and safety to the locking device (62) with the lockwire (C-405).

8.3. Inspect the locking device (62) to make sure that the tab is in groove of the rod end bearing (64) and that it was not damaged when the nut (63) was tightened. Replace the locking device (62) if tab is twisted or damaged.

9. Align the right cyclic flight control cylinder (7) into the support (28) as follows:

9.1. Move the lower piston (61) rod up and clear of the support (28). The lower piston (61) rod must be able to be moved freely down, up, and then down, and the bolt (41) replaced without any pressure, forcing, binding, or bending of the piston rod in the fore, aft, left, or right direction. One person must guide the extension tube (3) to prevent the damage when conducting this step and subsequent steps.

9.2. If the cylinder (7) meets requirements of the Step 9.1., it is considered in the alignment. If the cylinder (7) is aligned, accomplish the Step 9.8. through Step 9.9.. If the cylinder (7) is misaligned, accomplish the Step 9.3. through Step 9.9..

9.3. Loosen the nut (65) and bearing nut (63) so the cylinder (7) is loose.

9.4. Move the rod end bearing (64) up approximately 0.5 inch (12.7 mm) above the support (42). Rotate the lower end of cylinder (7) in a 3.0 inch (76.2 mm) circle.

9.5. Put the rod end bearing (64) directly over the lugs of the support (42) and move the piston (61) down and temporarily install the bolt (41).

9.6. Tighten the bearing nut (66) by hand (approximately 70 inch-pounds (7.9 Nm)).

9.7. Remove the bolt (41) and move the rod end bearing (64) up and out of the support (42), then move down. The rod end bearing must be free of any side loading. Repeat the Step 9.3. through Step 9.7. until free of side loading.

NOTE

If the cylinder (7) still cannot be installed without the preload, it is permitted to add the one washer AN960C416L (maximum) on the two aft studs of the support (5) between the support bracket (27) and support (28). Or, it is permissible to add the one washer AN960C416L (maximum) and one washer AN960C516L (maximum) on each of the two forward studs between the support bracket (27) and support (28).

9.8. Install the bolt (41). Torque the bearing nut (66) 900 to 1100 inch-pounds (101.68 to 124 Nm). Torque the nut (65) 900 to 1100 inch-pounds (101.68 to 124 Nm). Safety the nuts (66 and 65) to the keyed washer (67) with the lockwire (C-405).

9.9. If the cylinder (7) still meets the requirements of the Step 9.1., install the aluminum washer (43) and nut (44) and torque 95 to 110 inch-pounds (10.73 to 12.42 Nm). Install the cotter pin (45).

NOTE

Use the washers (12 and 13) as required to get the proper engagement of the cotter pin (9) and nut (10).

10. Install the extension tube (3) on the swashplate horn (2). Install the bolt (15), washers (14, 13, 12, and 11), and the nut (10). Install the cotter pin (9) (Detail A).

11. Put the tube assembly (39) into the lever (36) and install the bolt (38), washer (37), washers (50 and 51), nut (49), and cotter pin (55) (Detail D).

NOTE

SN: 33001-33213 or 36001-36300:

Refer to Step 12. through Step 16. for the installation of preformed packing.

12. Lubricate the new preformed packings (31, 33, 57, and 59) with the hydraulic fluid (C-002) or hydraulic fluid (C-072).

13. Install the bolt (34), preformed packing (33), elbow (32), and preformed packing (31).

14. Torque the bolt (34) 75 to 125 inch-pounds (8.47 to 14.12 Nm).

15. Install the bolt (56), preformed packing (57), elbow (58), and the preformed packing (59).

16. Torque the bolt (56) 40 to 65 inch-pounds (4.52 to 7.34 Nm).

NOTE

SN: 36301-Subsequent:

Refer to Step 17. through Step 20. for the installation of preformed packing.

17. Lubricate the new preformed packings (72 and 75) with the hydraulic fluid (C-002) or hydraulic fluid (C-072).

18. Install the preformed packings (72 and 75), elbows (70 and 73), and nuts (71 and 74).

19. Torque the nut (71) 135 to 145 inch-pounds (15.25 to 16.38 Nm).

20. Torque the nut (74) 215 to 245 inch-pounds (24.30 to 27.68 Nm).

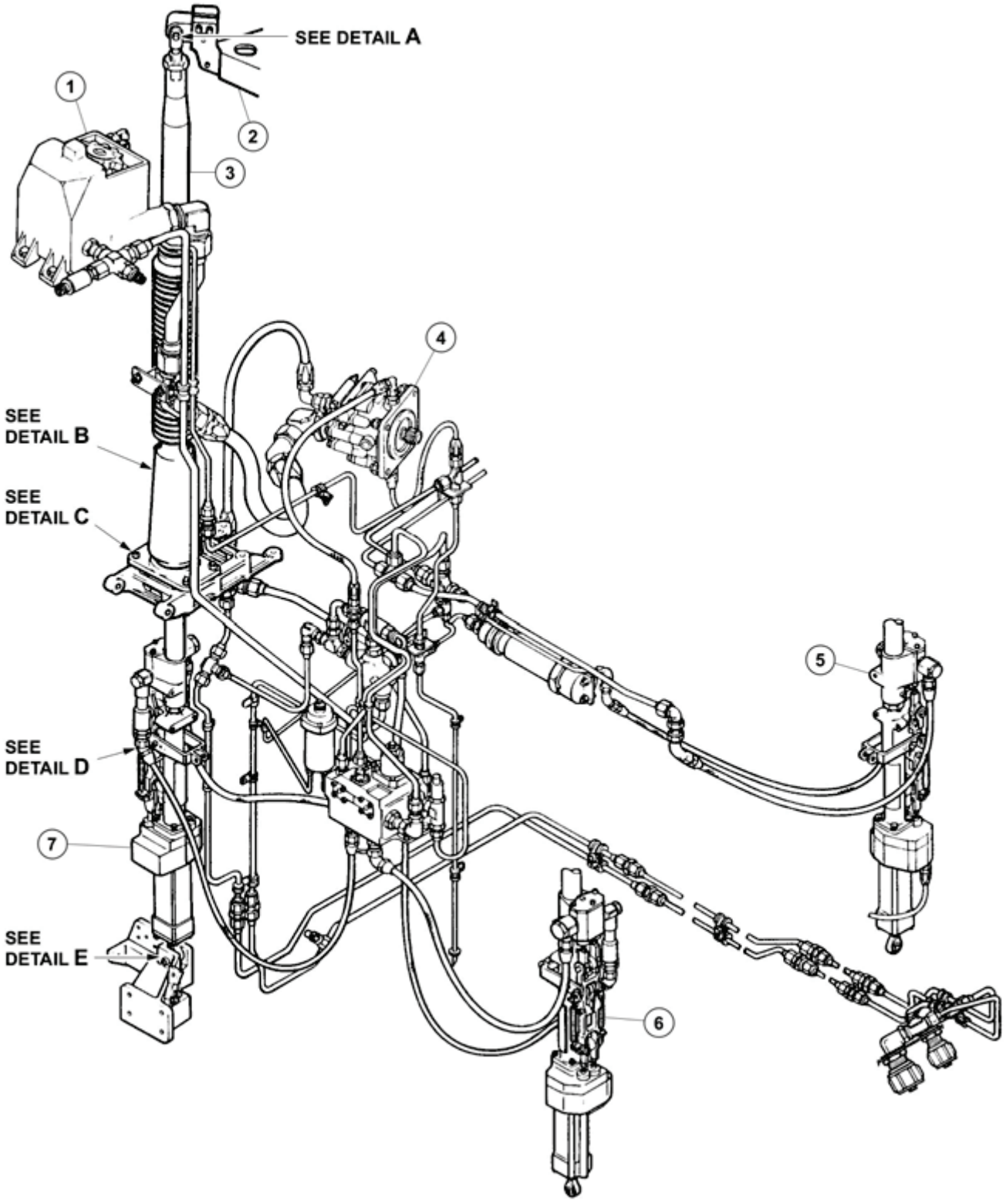
21. Lubricate the new preformed packings (48 and 54) with the hydraulic fluid (C-002) or hydraulic fluid (C-072). Install the preformed packings and bypass check valves (47 and 53), if removed (Detail D). Connect the hoses (35, 40, 46, and 52).

22. Bleed and inspect the hydraulic system (412-A-29-00-00-00A-042A-A / 00612).

23. Energize the hydraulic system and operate the cyclic controls through 10 complete operations. Make sure that there are no leaks.

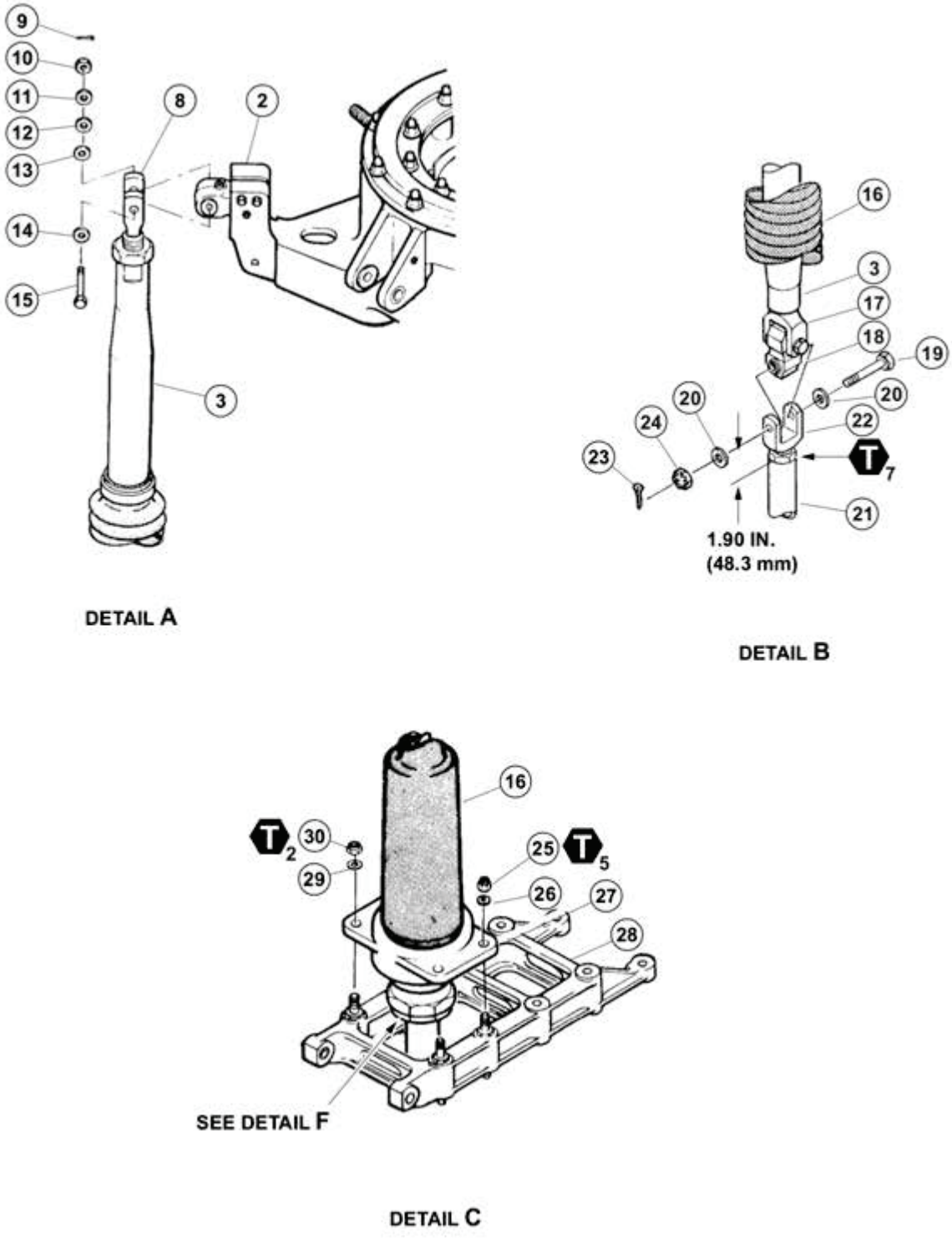
24. Make sure that the cyclic control rigging is within the limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).

25. Install the access panels on the pylon support. Install the forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).



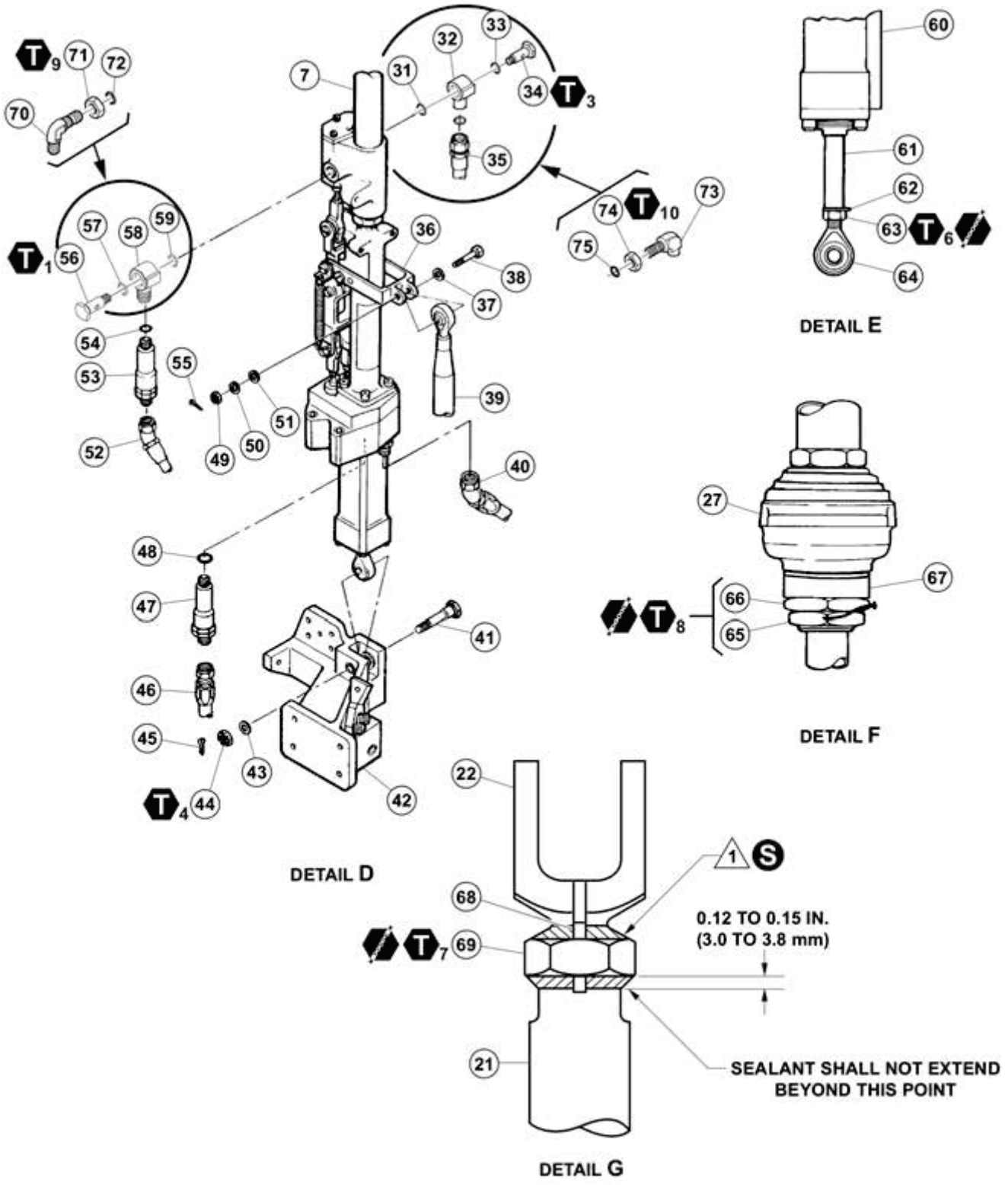
ICN-412-A-671200-B-97499-00009-A-001-01

Figure 1. Right Cyclic Flight Control Cylinder (Sheet 1 of 4)



ICN-412-A-671200-B-97499-00010-A-001-01

Figure 1. Right Cyclic Flight Control Cylinder (Sheet 2 of 4)



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Figure 1. Right Cyclic Flight Control Cylinder (Sheet 3 of 4)

1. Reservoir
2. Swashplate horn
3. Extension tube
4. Pump
5. Collective flight control cylinder
6. Left cyclic flight control cylinder
7. Right cyclic flight control cylinder
8. Clevis
9. Cotter pin
10. Nut
11. Washer (AN960C516L)
12. Washer (AN960C516)
13. Washer (AN960C516)
14. Washer (AN960C516)
15. Bolt
16. Boot
17. Clevis
18. Universal
19. Bolt
20. Washer (AN960C516)
21. Upper piston
22. Clevis
23. Cotter pin
24. Nut
25. Nut
26. Aluminum washer
27. Support bracket
28. Support
29. Aluminum washer
30. Nut
31. Preformed packing
32. Elbow
33. Preformed packing
34. Bolt
35. Hose
36. Lever
37. Washer
38. Bolt
39. Tube assembly
40. Hose
41. Bolt
42. Support
43. Aluminum washer
44. Nut
45. Cotter pin
46. Hose
47. Bypass check valve
48. Preformed packing
49. Nut
50. Washer
51. Washer
52. Hose
53. Bypass check valve
54. Preformed packing
55. Cotter pin
56. Bolt
57. Preformed packing
58. Elbow
59. Preformed packing
60. Actuator
61. Lower piston
62. Locking device
63. Nut
64. Rod end bearing
65. Nut
66. Bearing nut
67. Keyed washer
68. Lock
69. Jam nut
70. Elbow
71. Nut
72. Packing
73. Elbow
74. Nut
75. Packing

LOCKWIRE (C-405)

S SEALANT (C-308)

T1 40 TO 65 IN-LBS (4.51 TO 7.34 Nm)

T2 75 TO 95 IN-LBS (8.47 TO 10.73 Nm)

T3 75 TO 125 IN-LBS (8.47 TO 14.12 Nm)

T4 95 TO 110 IN-LBS (10.73 TO 12.42 Nm)

T5 120 TO 160 IN-LBS (13.55 TO 18.07 Nm)

T6 200 TO 250 IN-LBS (22.59 TO 28.54 Nm)

T7 660 TO 780 IN-LBS (74.56 TO 88.12 Nm)

T8 900 TO 1100 IN-LBS (101.68 TO 124.28 Nm)

T9 135 TO 145 IN-LBS (15.25 TO 16.38 Nm)

T10 215 TO 245 IN-LBS (24.30 TO 27.68 Nm)

NOTE

1. Apply sealant (C-308) around jam nut (69), lock (68), and exposed threads of clevis (22).

ICN-412-A-671200-B-97499-00012-A-001-01

Figure 1. Right Cyclic Flight Control Cylinder (Sheet 4 of 4)

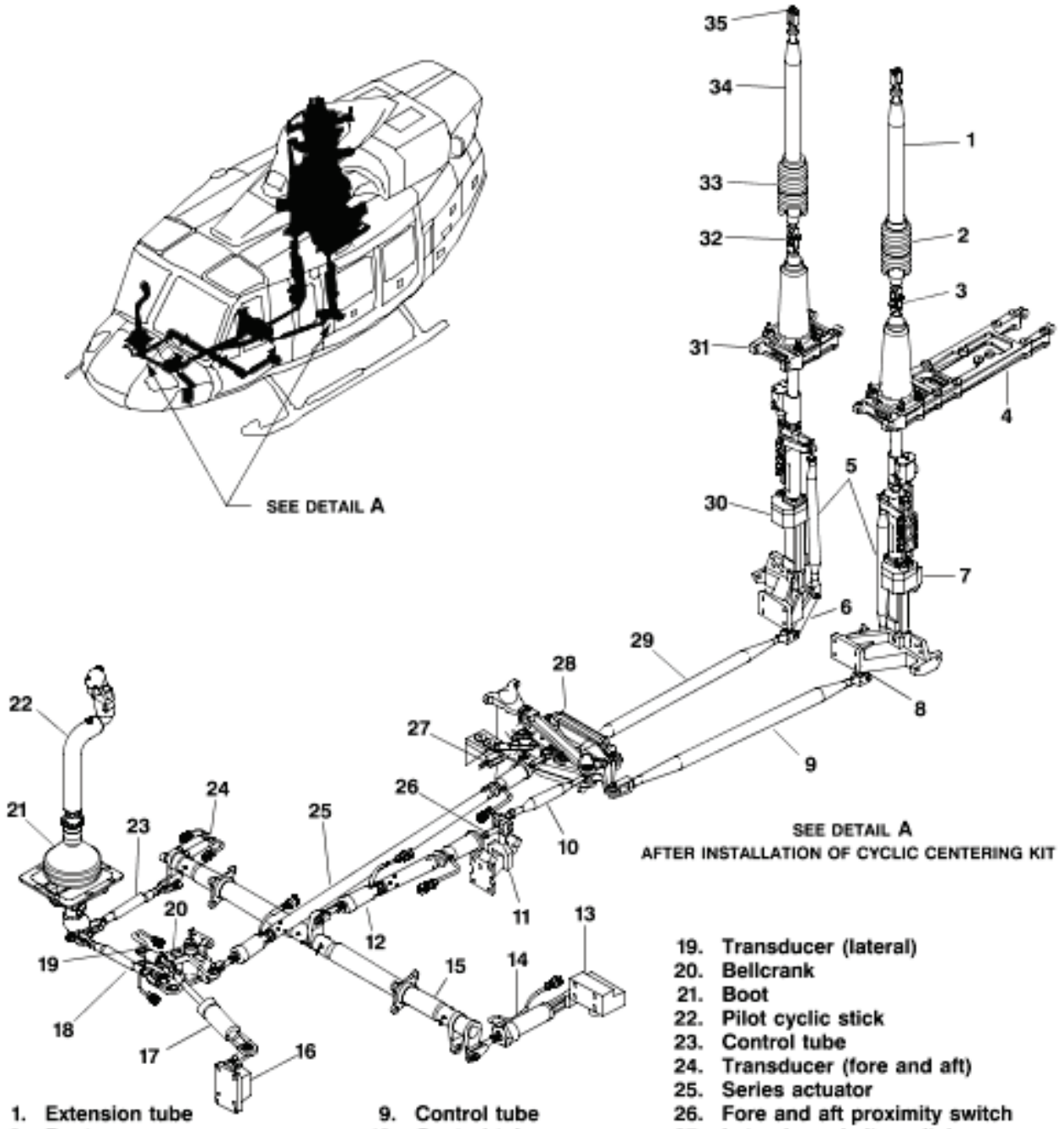
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Right Cyclic Boost Cylinder Support - Removal

Left cyclic and collective boost support (4, Figure 1) is for the left cyclic control and collective boost cylinders. Maintenance for the support is covered in the collective controls (412-A-67-11-00-07A-520A-A / 01349).

Procedure

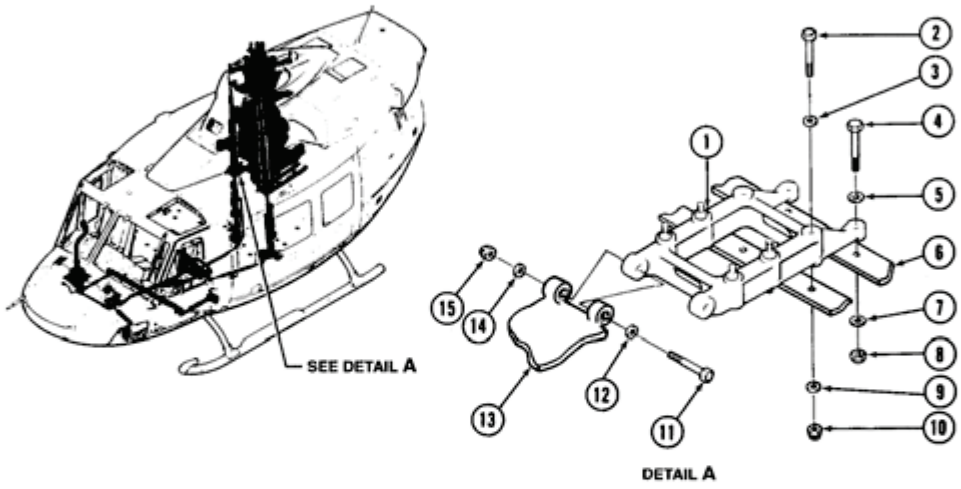
1. Remove the access panel on the right side and forward side of pylon structure inside the cabin (412-A-71-10-00-01A-520A-A / 01605).
2. Remove the right cyclic hydraulic flight control cylinder (412-A-67-12-00-07A-520A-A / 01423).
3. Remove the hydraulic system No. 1 hydraulic pump (412-A-29-10-00-01A-520A-A / 00620).
4. Remove the main rotor tachometer generator (412-A-63-21-02-06A-520A-A / 01165).
5. Remove the nut (10, Figure 2), two aluminum washers (3 and 9), and bolt (2).
6. Remove the nut (8), two aluminum washers (5 and 7), and bolt (4).
7. Remove the nut (15), two aluminum washers (12 and 14), and bolt (11). Remove the cyclic control boost cylinder support (1) from the lift beam (6).



1. Extension tube
2. Boot
3. Universal
4. Left cyclic and collective boost support
5. Control tube
6. Bellcrank
7. Left cyclic flight control cylinder
8. Bellcrank
9. Control tube
10. Control tube
11. Idler
12. Series actuator
13. Rotary actuator
14. Force gradient
15. Cyclic jackshaft
16. Rotary actuator
17. Force gradient
18. Control tube
19. Transducer (lateral)
20. Bellcrank
21. Boot
22. Pilot cyclic stick
23. Control tube
24. Transducer (fore and aft)
25. Series actuator
26. Fore and aft proximity switch
27. Lateral proximity switch
28. Mixing lever assembly
29. Control tube
30. Right cyclic flight control cylinder
31. Right cyclic boost support
32. Universal
33. Boot
34. Extension tube
35. Bolt

ICN-412-A-671200-B-97499-00001-A-001-01

Figure 1. Cyclic Controls (Sheet 1 of 1)



1. Cyclic control boost cylinder support
2. Bolt
3. Aluminum washer
4. Bolt
5. Aluminum washer
6. Lift beam
7. Aluminum washer
8. Nut
9. Aluminum washer
10. Nut
11. Bolt
12. Aluminum washer
13. Pylon structure
14. Aluminum washer
15. Nut

ICN-412-A-671200-B-97499-00013-A-001-01

Figure 2. Right Cyclic Control Boost Cylinder Support (Sheet 1 of 1)

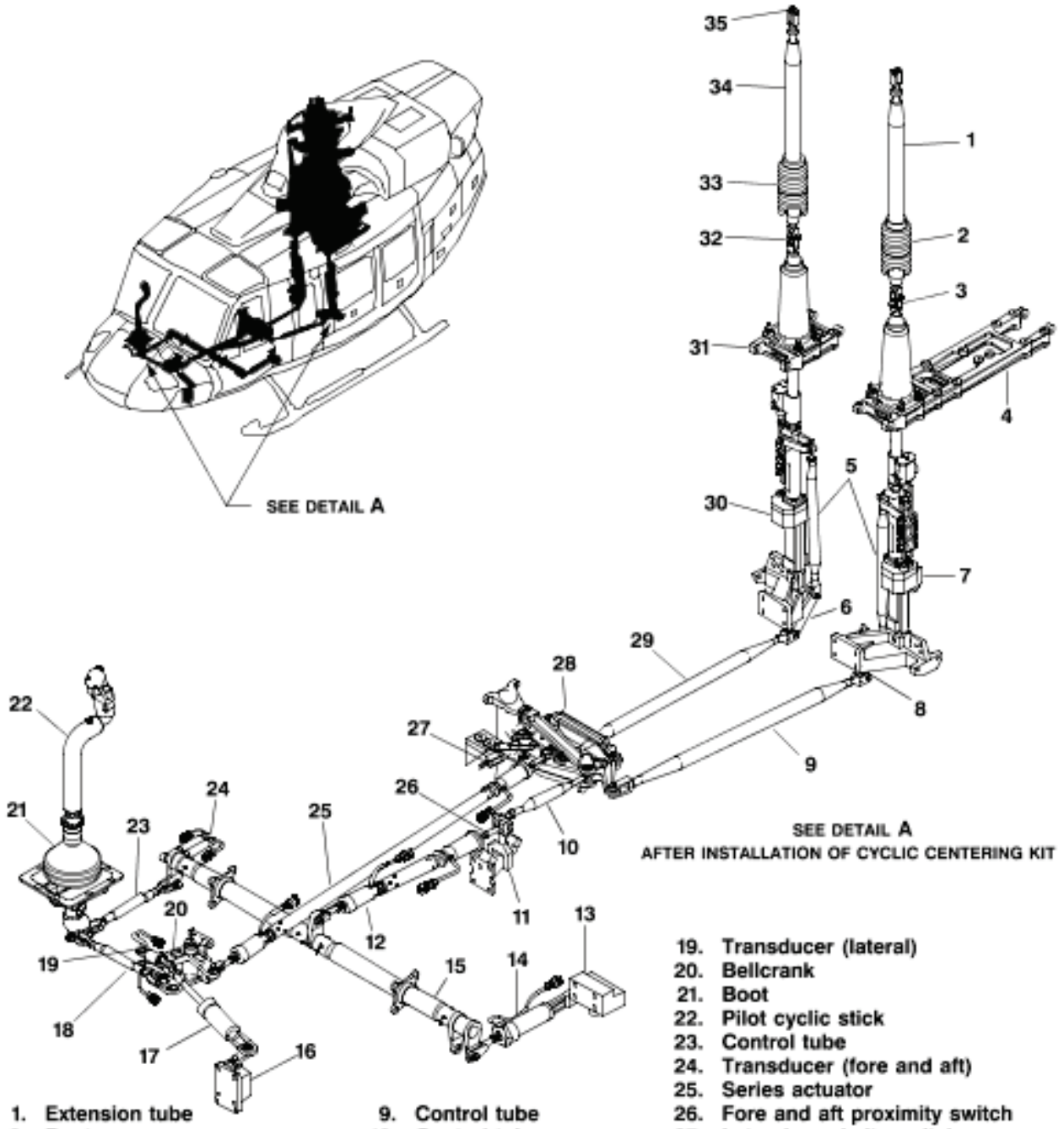
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Right Cyclic Boost Cylinder Support - Inspection

Left cyclic and collective boost support (4, Figure 1) is for the left cyclic control and collective boost cylinders. Maintenance for support is covered in collective controls 412-A-67-11-00-07A-280A-A / 01350).

Procedure

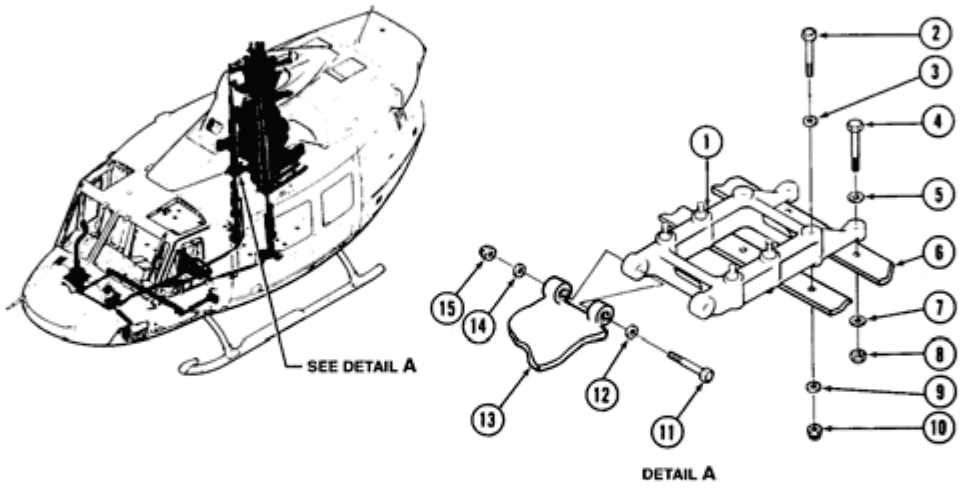
1. Inspect the cyclic control boost cylinder support (1, Figure 2) in accordance with BHT-412-CR&O.



1. Extension tube
2. Boot
3. Universal
4. Left cyclic and collective boost support
5. Control tube
6. Bellcrank
7. Left cyclic flight control cylinder
8. Bellcrank
9. Control tube
10. Control tube
11. Idler
12. Series actuator
13. Rotary actuator
14. Force gradient
15. Cyclic jackshaft
16. Rotary actuator
17. Force gradient
18. Control tube
19. Transducer (lateral)
20. Bellcrank
21. Boot
22. Pilot cyclic stick
23. Control tube
24. Transducer (fore and aft)
25. Series actuator
26. Fore and aft proximity switch
27. Lateral proximity switch
28. Mixing lever assembly
29. Control tube
30. Right cyclic flight control cylinder
31. Right cyclic boost support
32. Universal
33. Boot
34. Extension tube
35. Bolt

ICN-412-A-671200-B-97499-00001-A-001-01

Figure 1. Cyclic Controls (Sheet 1 of 1)



1. Cyclic control boost cylinder support
2. Bolt
3. Aluminum washer
4. Bolt
5. Aluminum washer
6. Lift beam
7. Aluminum washer
8. Nut
9. Aluminum washer
10. Nut
11. Bolt
12. Aluminum washer
13. Pylon structure
14. Aluminum washer
15. Nut

ICN-412-A-671200-B-97499-00013-A-001-01

Figure 2. Right Cyclic Control Boost Cylinder Support (Sheet 1 of 1)

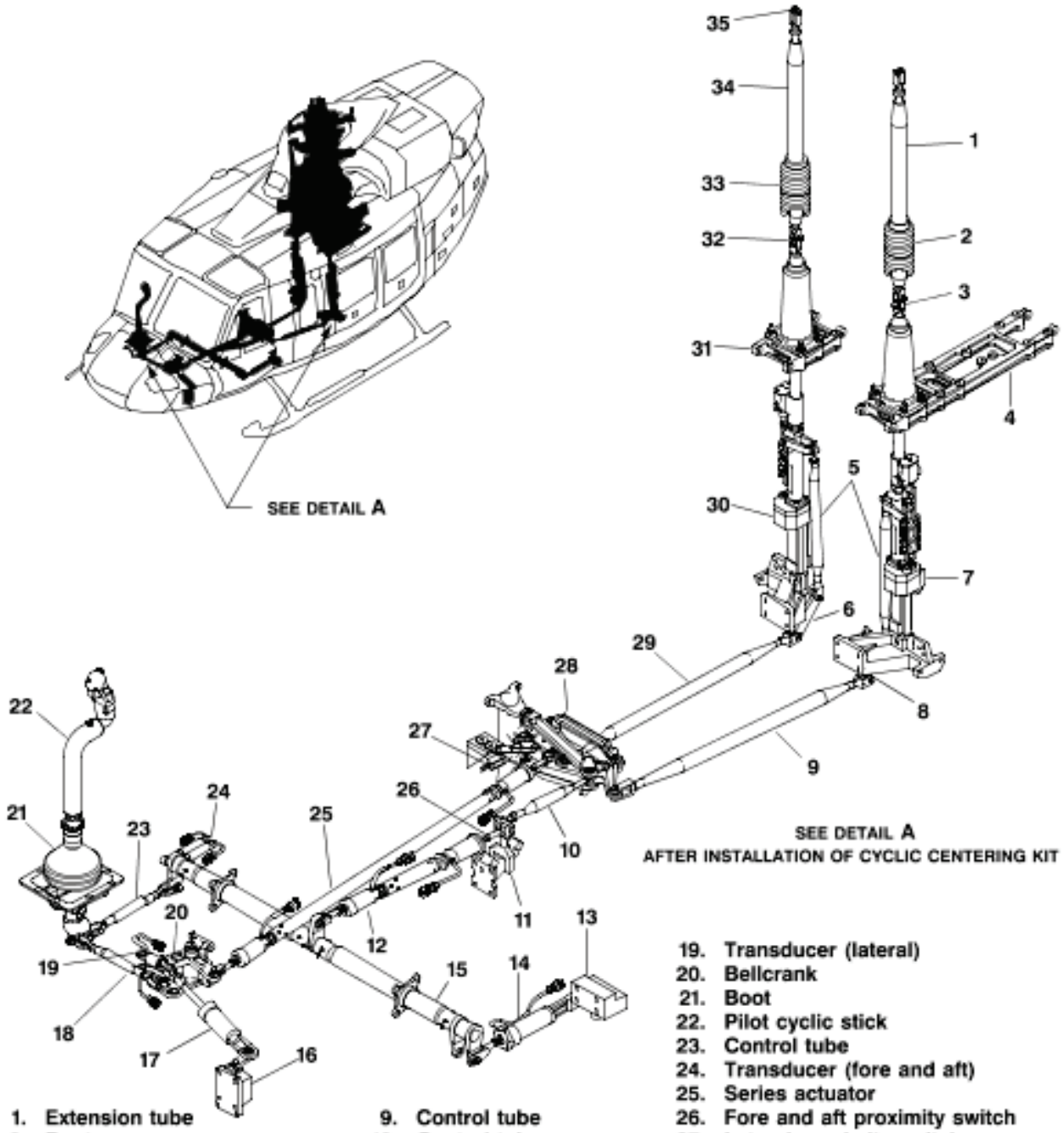
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Right Cyclic Boost Cylinder Support - Installation

Left cyclic and collective boost support (4, Figure 1) is for the left cyclic control and collective boost cylinders. Maintenance for support is covered in collective controls (412-A-67-11-00-07A-720A-A / 01351).

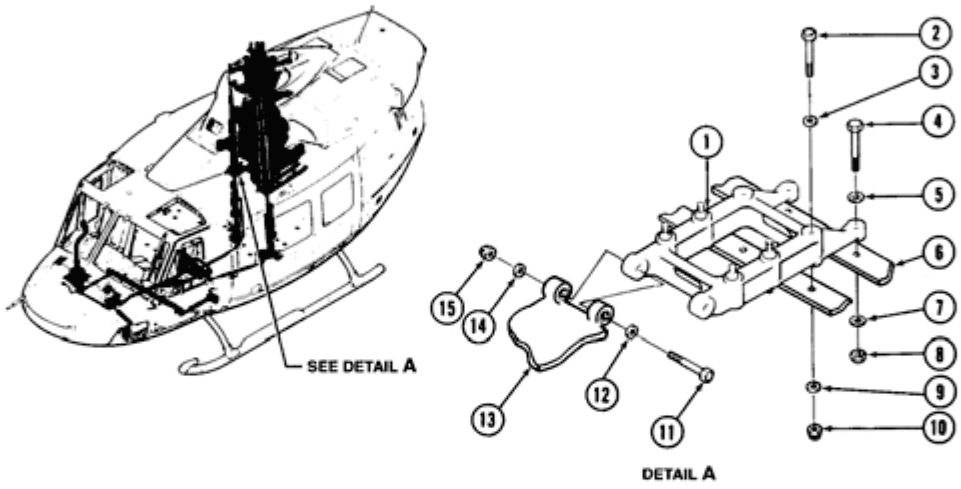
Procedure

1. Place the cyclic control boost cylinder support (1, Figure 2) into position in pylon structure (13) and lift beam (6).
2. Install the bolts (11), aluminum washers (12), aluminum washers (14), and nuts (15).
3. Install bolts (2), aluminum washers (3), aluminum washers (9), and nuts (10).
4. Install bolts (4), aluminum washers (5), aluminum washers (7), and nut (8).
5. Install main rotor tachometer generator (412-A-63-21-02-06A-720A-A / 01169).
6. Install No. 1 hydraulic system pump (412-A-29-10-00-01A-720A-A / 00623).
7. Install right cyclic flight control cylinder (412-A-67-12-00-07A-720A-A / 01425).
8. Move cyclic controls through full travel and check for interference and binding.
9. Make sure that the cyclic control rigging within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).
10. Install access panel on right side of pylon structure inside cabin (412-A-71-10-00-01A-720A-A / 01607).
11. Install access panel on forward side of pylon structure inside cabin (412-A-71-10-00-01A-720A-A / 01607).



ICN-412-A-671200-B-97499-00001-A-001-01

Figure 1. Cyclic Controls (Sheet 1 of 1)



1. Cyclic control boost cylinder support
2. Bolt
3. Aluminum washer
4. Bolt
5. Aluminum washer
6. Lift beam
7. Aluminum washer
8. Nut
9. Aluminum washer
10. Nut
11. Bolt
12. Aluminum washer
13. Pylon structure
14. Aluminum washer
15. Nut

ICN-412-A-671200-B-97499-00013-A-001-01

Figure 2. Right Cyclic Control Boost Cylinder Support (Sheet 1 of 1)

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Cyclic Control Tubes - Removal

Procedure

NOTE

The removal and installation procedures for control tube (8, Figure 1) are typical for left and right sides.

1. Remove access panel from center of aft cabin floor (412-A-52-40-00-02A-520A-A / 00837).

2. Remove cotter pin (1) and nut (2). Remove bolt (6), and washers (3) from lever (5).

3. Remove cotter pin (13) and nut (12). Remove bolt (10) and washers (11).

4. Remove control tube (8).

5. Remove cotter pin (18) and nut (19). Remove bolt (17) and washers (16).

6. Remove cotter pin (21) and nut (22). Remove bolt (24) and washers (23).

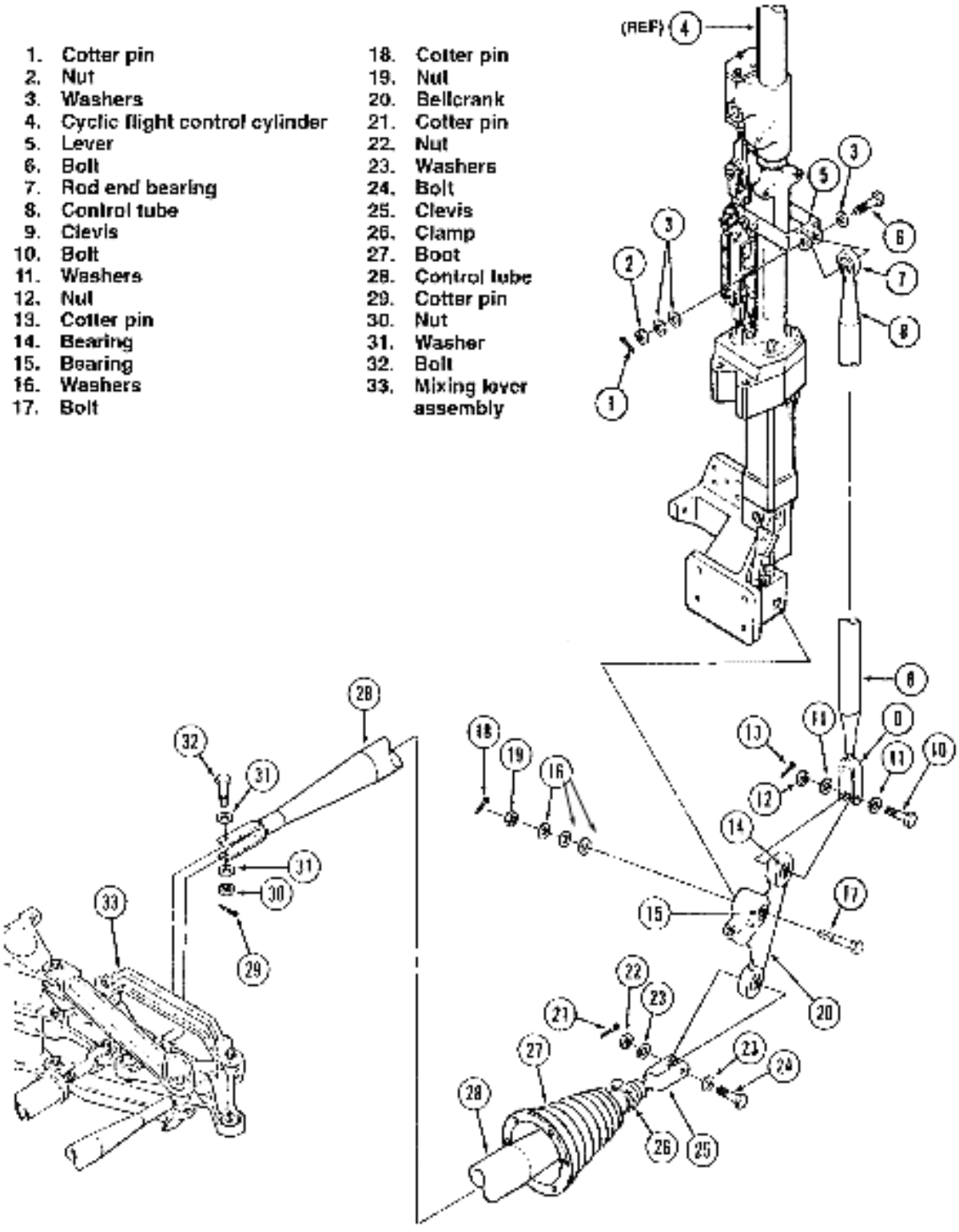
7. Remove bellcrank (20).

8. Remove cotter pin (29) and nut (30). Remove bolt (32) and washers (31).

9. Loosen clamp (26) and slide forward off of boot (27).

10. Slide control tube (28) forward out of helicopter.

1. Cotter pin
2. Nut
3. Washers
4. Cyclic flight control cylinder
5. Lever
6. Bolt
7. Rod end bearing
8. Control tube
9. Clevis
10. Bolt
11. Washers
12. Nut
13. Cotter pin
14. Bearing
15. Bearing
16. Washers
17. Bolt
18. Cotter pin
19. Nut
20. Bellcrank
21. Cotter pin
22. Nut
23. Washers
24. Bolt
25. Clevis
26. Clamp
27. Boot
28. Control tube
29. Cotter pin
30. Nut
31. Washer
32. Bolt
33. Mixing lever assembly



ICN-412-A-671200-B-97499-00014-A-001-01

Figure 1. Right Cyclic Control Tubes and Bellcranks (Sheet 1 of 1)

Cyclic Control Tubes - Inspection

Procedure

NOTE

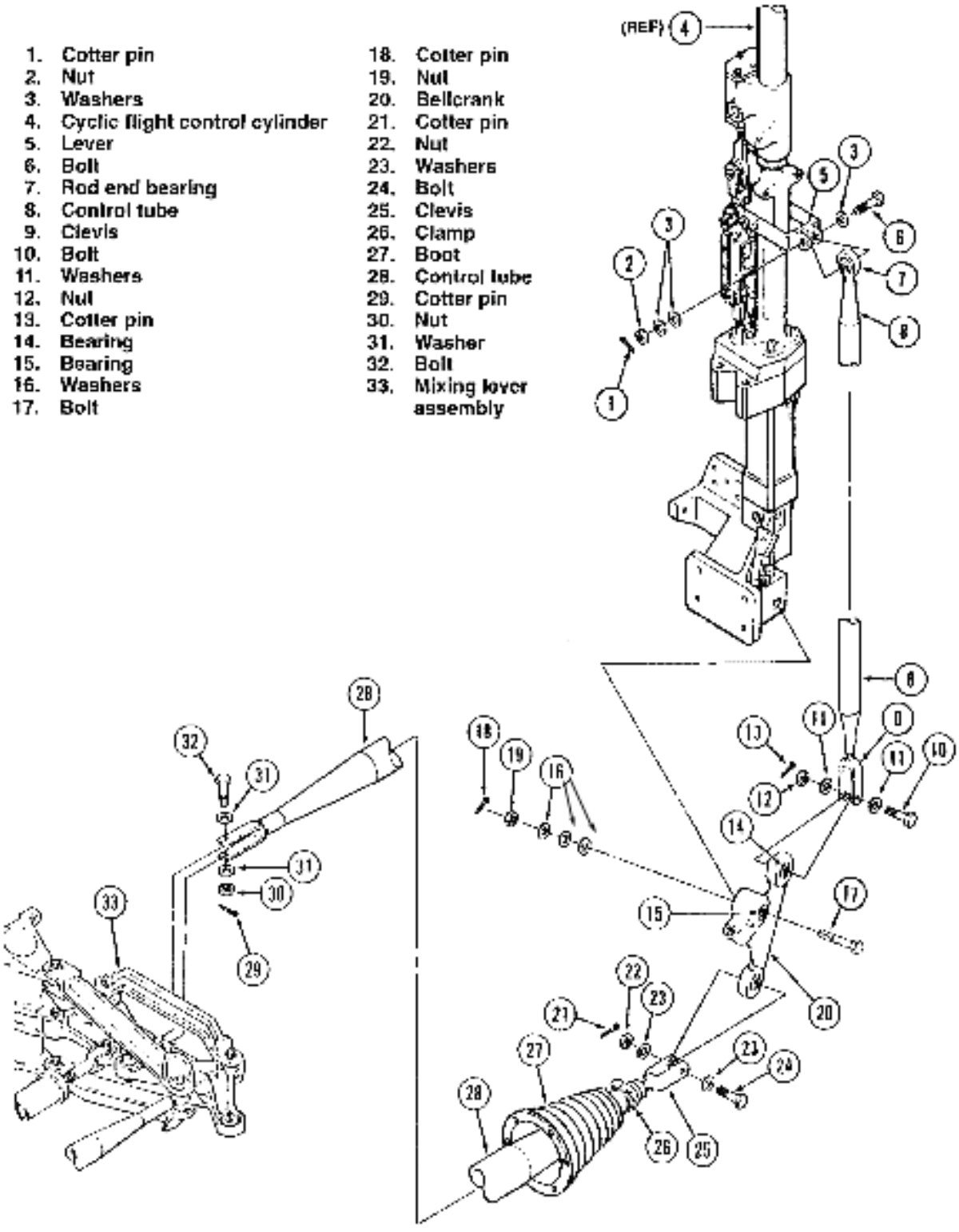
The inspection procedure for control tubes (8 and 28, Figure 1), clevises (9 and 25), and bellcrank (20) in left and right cyclic system are covered in the BHT-412-CR&O. Inserts and bearings in control tubes and bellcranks are covered in the BHT-ALL-SPM, Standard Practices Manual.

1. Inspect bolt holes for elongation.

2. Inspect threads for damage. No visible damage to threads is acceptable.

3. Inspect boot (27) for tears and signs of deterioration. Torn or deteriorated boots are not acceptable and shall be replaced.

1. Cotter pin
2. Nut
3. Washers
4. Cyclic flight control cylinder
5. Lever
6. Bolt
7. Rod end bearing
8. Control tube
9. Clevis
10. Bolt
11. Washers
12. Nut
13. Cotter pin
14. Bearing
15. Bearing
16. Washers
17. Bolt
18. Cotter pin
19. Nut
20. Bellcrank
21. Cotter pin
22. Nut
23. Washers
24. Bolt
25. Clevis
26. Clamp
27. Boot
28. Control tube
29. Cotter pin
30. Nut
31. Washer
32. Bolt
33. Mixing lever assembly



ICN-412-A-671200-B-97499-00014-A-001-01

Figure 1. Right Cyclic Control Tubes and Bellcranks (Sheet 1 of 1)

Cyclic Control Tubes - Repair

Procedure

1.

NOTE

Repair bellcranks and control tubes in accordance with the BHT-412-CR&O. Refer to the BHT-ALL-SPM, Standard Practices Manual for bearing replacement procedure.

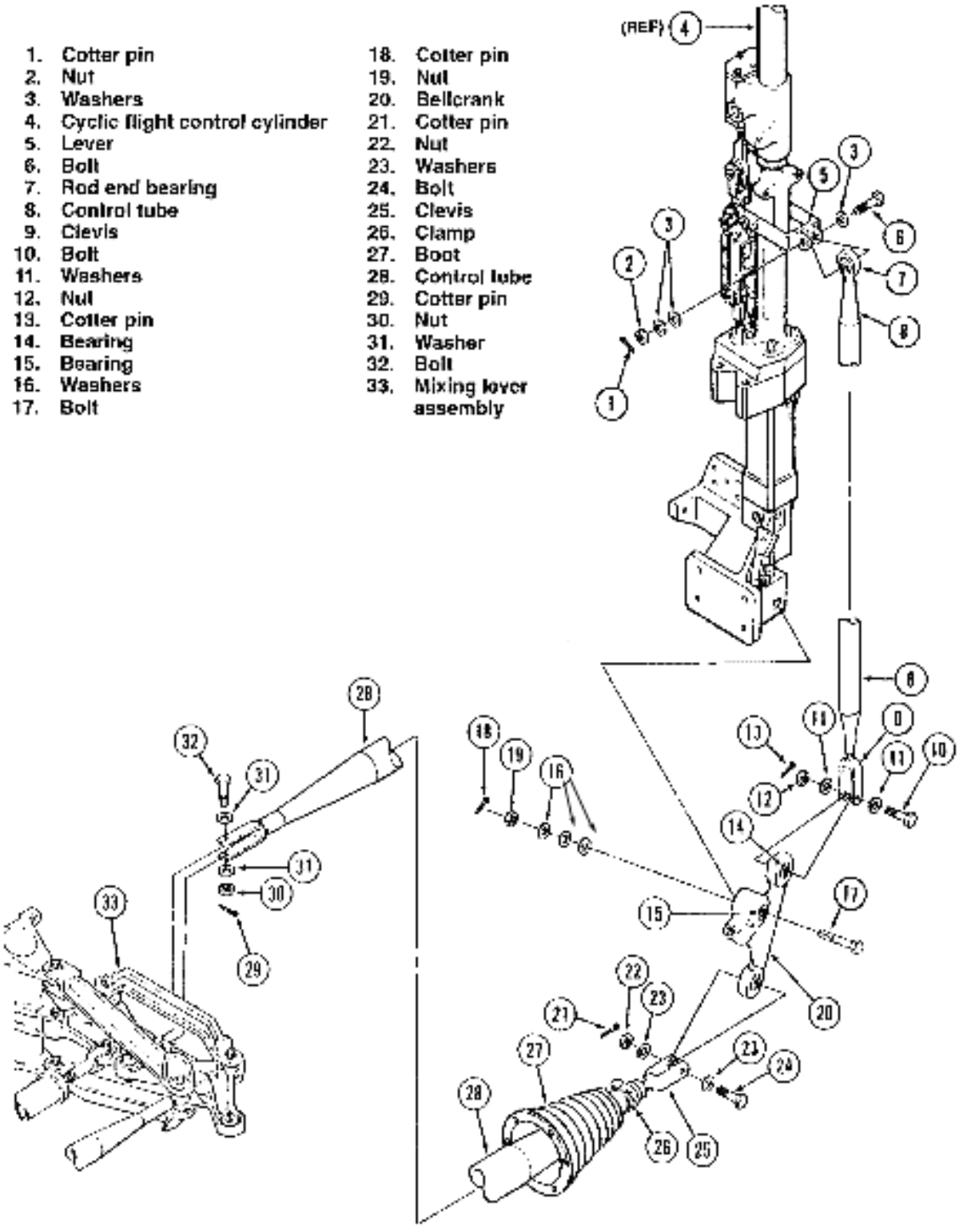
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Cyclic Control Tubes - Installation

Procedure

1. Slide control tube (28, Figure 1) aft through boot (27). Position control tube and install clamp (26) on boot and tighten. Do not tighten clamp at this time.
2. Install control tube (28) on mixing lever assembly (33) with bolt (32), two washers (31), nut (30), and cotter pin (29).
3. Install bellcrank (20) in clevis (25) of control tube (28) with bolt (24), two washers (23), nut (22), and cotter pin (21).
4. Install bellcrank (20) with bolt (17), three washers (16), nut (19), and cotter pin (18).
5. Install clevis (9) of control tube (8) on bellcrank (20) with bolt (10), two washers (11), nut (12), and cotter pin (13).
6. Install rod end bearing (7) of control tube (8) on lever (5) with bolt (6), washers (3), nut (2), and cotter pin (1).
7. With cyclic stick in neutral position, locate aft end of boot (27) on control tube (28) 6.95 inches (176.53 mm) aft of bulkhead attachment for the boot. Tighten the clamp (26).
8. Make sure that the cyclic control rigging is within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325) and that cyclic controls move freely through their entire travel with no binding or restriction.
9. Install access panel in the center aft cabin floor (412-A-52-40-00-02A-720A-A / 00840).

1. Cotter pin
2. Nut
3. Washers
4. Cyclic flight control cylinder
5. Lever
6. Bolt
7. Rod end bearing
8. Control tube
9. Clevis
10. Bolt
11. Washers
12. Nut
13. Cotter pin
14. Bearing
15. Bearing
16. Washers
17. Bolt
18. Cotter pin
19. Nut
20. Bellcrank
21. Cotter pin
22. Nut
23. Washers
24. Bolt
25. Clevis
26. Clamp
27. Boot
28. Control tube
29. Cotter pin
30. Nut
31. Washer
32. Bolt
33. Mixing lever assembly



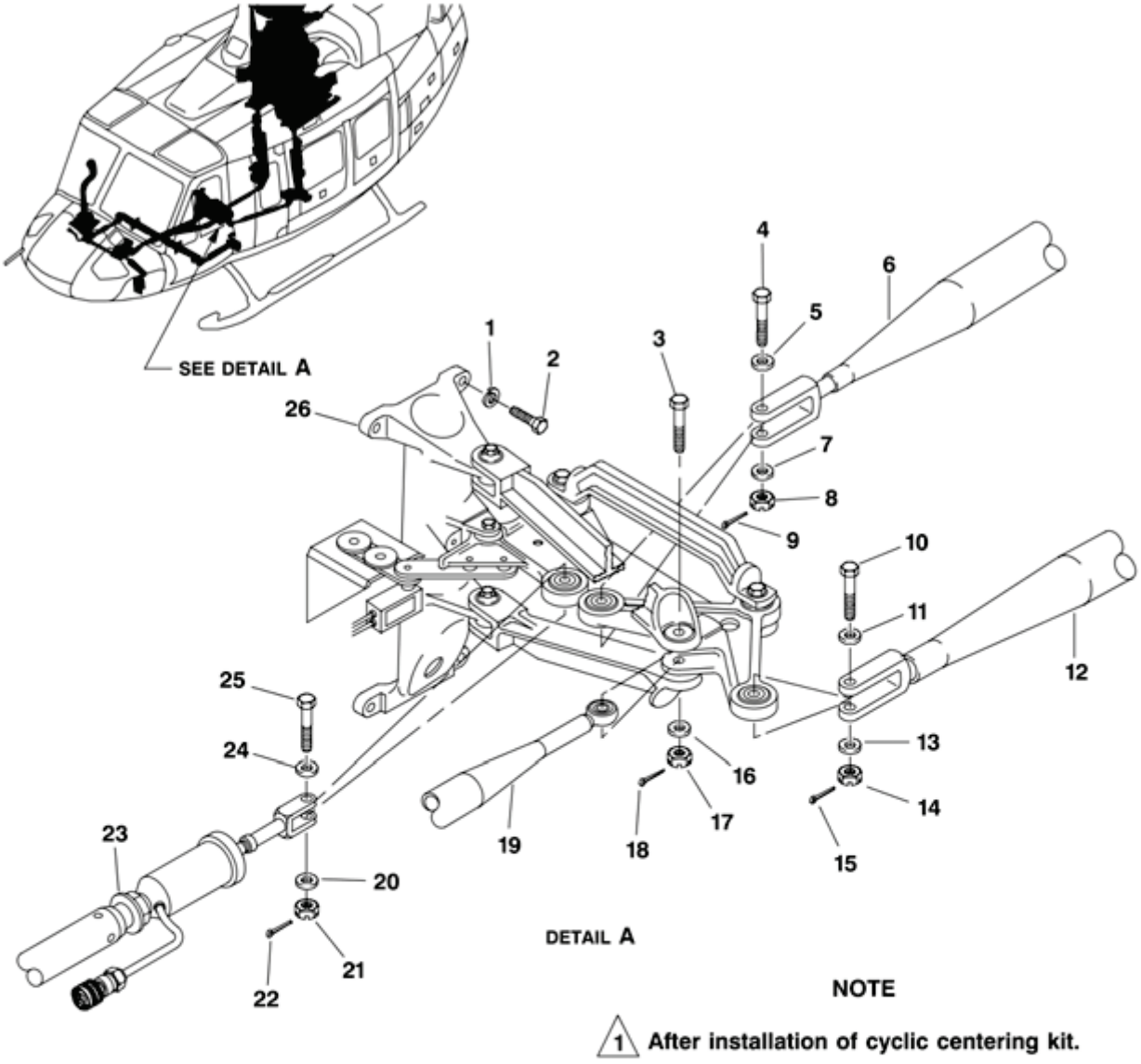
ICN-412-A-671200-B-97499-00014-A-001-01

Figure 1. Right Cyclic Control Tubes and Bellcranks (Sheet 1 of 1)

Mixing Lever Assembly - Removal

Procedure

1. Removal access panel in center of aft cabin floor (412-A-52-40-00-02A-520A-A / 00837).
2. Remove flight control tube (19, Figure 1) from mixing lever assembly (26) by removing cotter pin (18), nut (17), washer (16) and bolt (3).
3. Remove flight control tube (6) from mixing lever assembly (26) by removing cotter pin (9), nut (8), washers (7 and 5) and bolt (4).
4. Remove flight control tube (12) from mixing lever assembly (26) by removing cotter pin (15), nut (14), washers (13 and 11), and bolt (10).
5. Disconnect series actuator (23) by removing cotter pin (22), nut (21), aluminum washers (20 and 24), and bolt (25) from lever assembly (26).
6. Remove four bolts (2) and washers (1). Remove mixing lever assembly (26).



1. Washer (NAS1197-416)
2. Bolt
3. Bolt
4. Bolt
5. Washer (AN960C416)
6. Flight control tube
7. Washer (AN960C416)
8. Nut
9. Cotter pin
10. Bolt
11. Washer (AN960C416)
12. Flight control tube
13. Washer (AN960C416)
14. Nut
15. Cotter pin
16. Washer
17. Nut
18. Cotter pin
19. Flight control tube
20. Aluminum washer
21. Nut
22. Cotter pin
23. Series actuator
24. Aluminum washer
25. Bolt
26. Mixing lever assembly

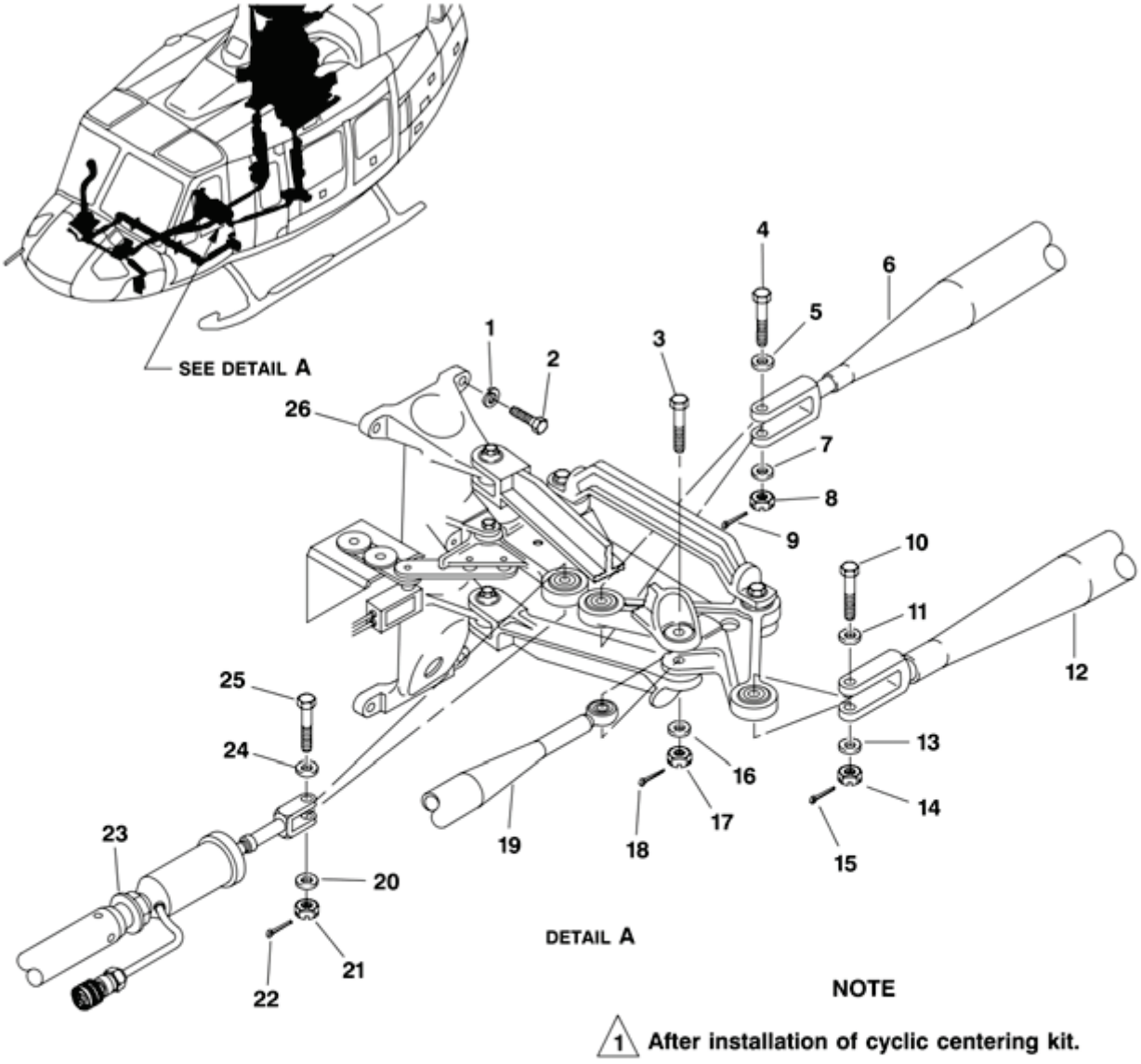
ICN-412-A-671200-B-97499-00015-A-001-01

Figure 1. Mixing Lever (Sheet 1 of 1)

Mixing Lever Assembly - Inspection

Procedure

1. Examine mixing lever assembly (26, Figure 1) for mechanical and corrosion damage per BHT-412-CR&O.



1. Washer (NAS1197-416)
2. Bolt
3. Bolt
4. Bolt
5. Washer (AN960C416)
6. Flight control tube
7. Washer (AN960C416)
8. Nut
9. Cotter pin
10. Bolt
11. Washer (AN960C416)
12. Flight control tube
13. Washer (AN960C416)
14. Nut
15. Cotter pin
16. Washer
17. Nut
18. Cotter pin
19. Flight control tube
20. Aluminum washer
21. Nut
22. Cotter pin
23. Series actuator
24. Aluminum washer
25. Bolt
26. Mixing lever assembly

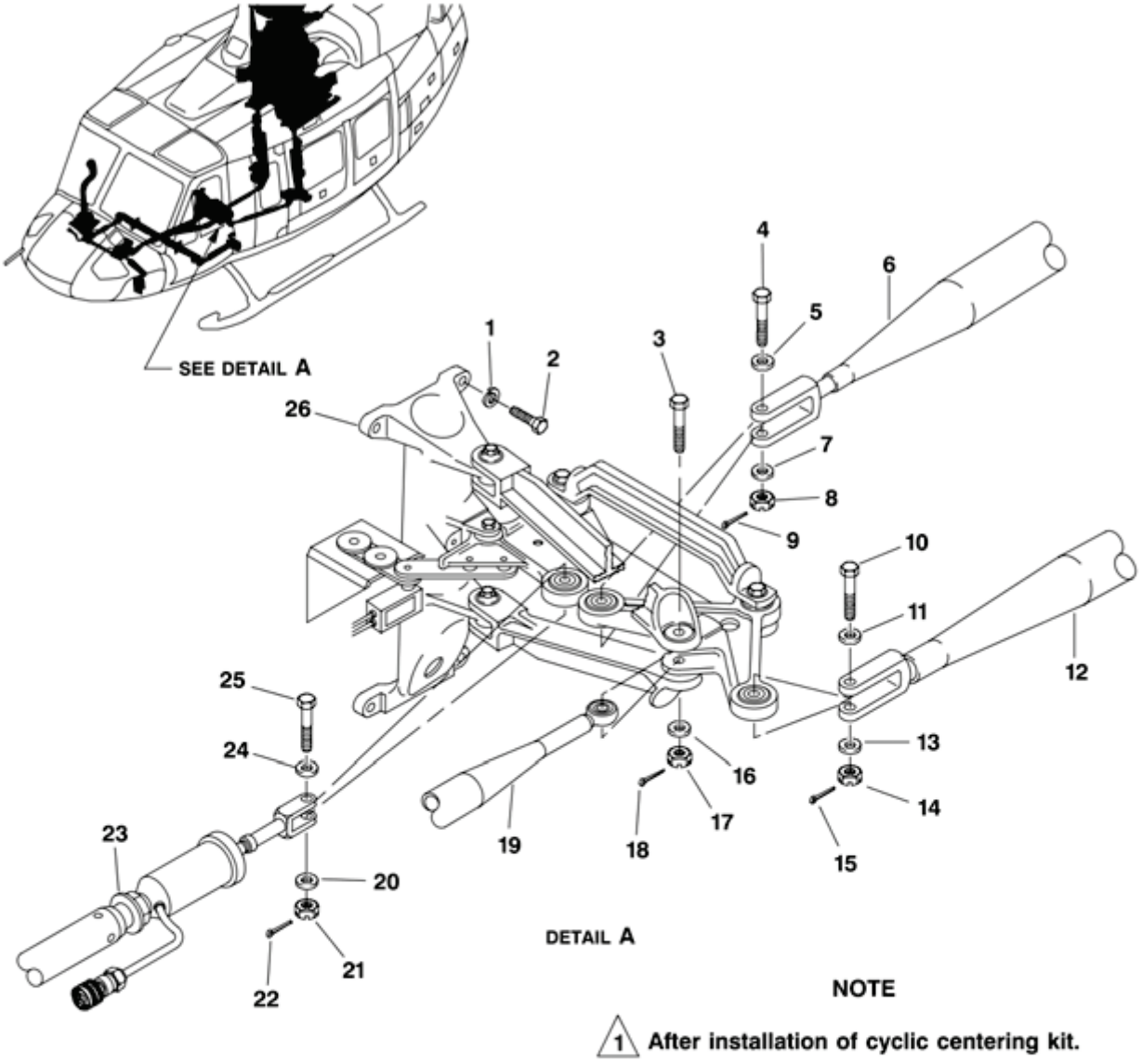
ICN-412-A-671200-B-97499-00015-A-001-01

Figure 1. Mixing Lever (Sheet 1 of 1)

Mixing Lever Assembly - Repair

Procedure

1. Repair mixing lever assembly (26, Figure 1) (BHT-412-CR&O).



1. Washer (NAS1197-416)
2. Bolt
3. Bolt
4. Bolt
5. Washer (AN960C416)
6. Flight control tube
7. Washer (AN960C416)
8. Nut
9. Cotter pin
10. Bolt
11. Washer (AN960C416)
12. Flight control tube
13. Washer (AN960C416)
14. Nut
15. Cotter pin
16. Washer
17. Nut
18. Cotter pin
19. Flight control tube
20. Aluminum washer
21. Nut
22. Cotter pin
23. Series actuator
24. Aluminum washer
25. Bolt
26. Mixing lever assembly

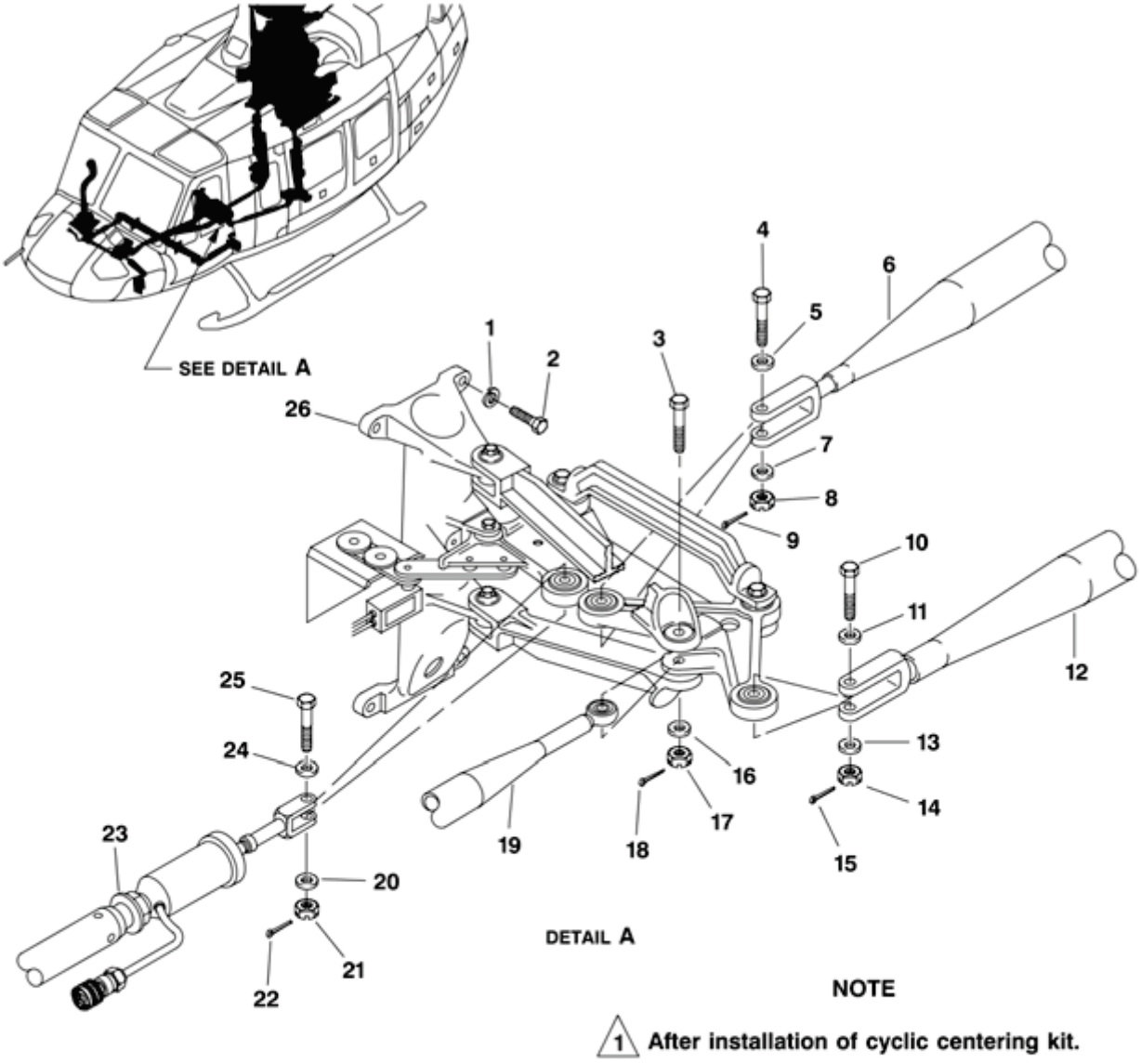
ICN-412-A-671200-B-97499-00015-A-001-01

Figure 1. Mixing Lever (Sheet 1 of 1)

Mixing Lever Assembly - Installation

Procedure

1. Put mixing lever assembly (26, Figure 1) in position in helicopter and install bolts (2) and washers (1).Place flight control tube (6) on lever assembly (26) and install bolt (4), washer (5 and 7), nut (8), and cotter pin (9).
2. Put flight control tube (12) on lever assembly (26) and install bolt (10), washers (11 and 13), nut (14), and cotter pin (15).
3. Put flight control tube (19) in lever assembly (26) and install bolt (3), washer (16), nut (17), and cotter pin (18).
4. Put series actuator (23) onto lever assembly (26) and install bolt (25), washers (24 and 20), nut (21), and cotter pin (22).
5. Move cyclic controls through full travel and check for obstructions and binding.
6. Make sure that the cyclic rigging within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).
7. Install access panel in center of aft cabin floor. (412-A-52-40-00-02A-720A-A / 00840).



1. Washer (NAS1197-416)
2. Bolt
3. Bolt
4. Bolt
5. Washer (AN960C416)
6. Flight control tube
7. Washer (AN960C416)
8. Nut
9. Cotter pin
10. Bolt
11. Washer (AN960C416)
12. Flight control tube
13. Washer (AN960C416)
14. Nut
15. Cotter pin
16. Washer
17. Nut
18. Cotter pin
19. Flight control tube
20. Aluminum washer
21. Nut
22. Cotter pin
23. Series actuator
24. Aluminum washer
25. Bolt
26. Mixing lever assembly

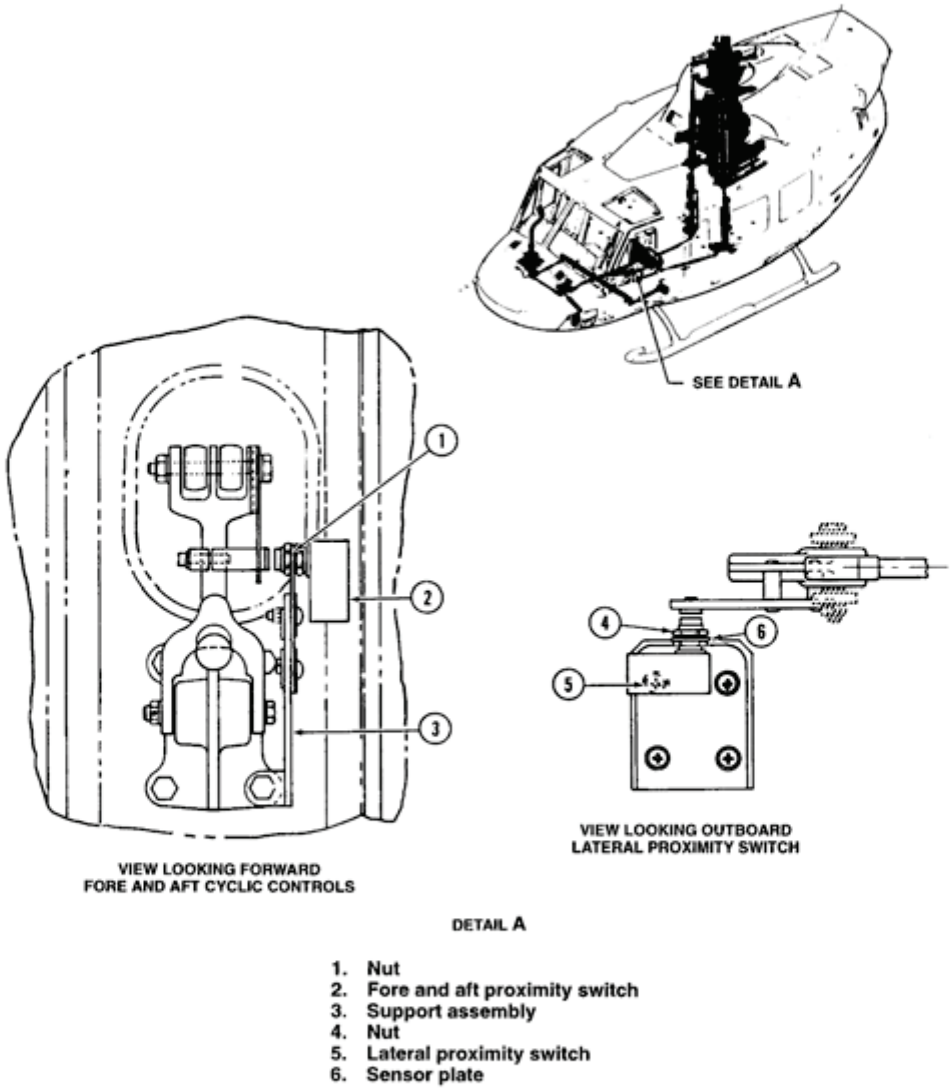
ICN-412-A-671200-B-97499-00015-A-001-01

Figure 1. Mixing Lever (Sheet 1 of 1)

Fore and Aft Proximity Switch - Removal

Procedure

1. Remove access panel in center of aft cabin floor (412-A-52-40-00-02A-520A-A / 00837).
2. Cut lockwire and remove nut (1, Figure 1) from fore and aft proximity switch (2).
3. Separate wires at splice to switch.
4. Remove fore and aft proximity switch (2) from support assembly (3).



1. Nut
2. Fore and aft proximity switch
3. Support assembly
4. Nut
5. Lateral proximity switch
6. Sensor plate

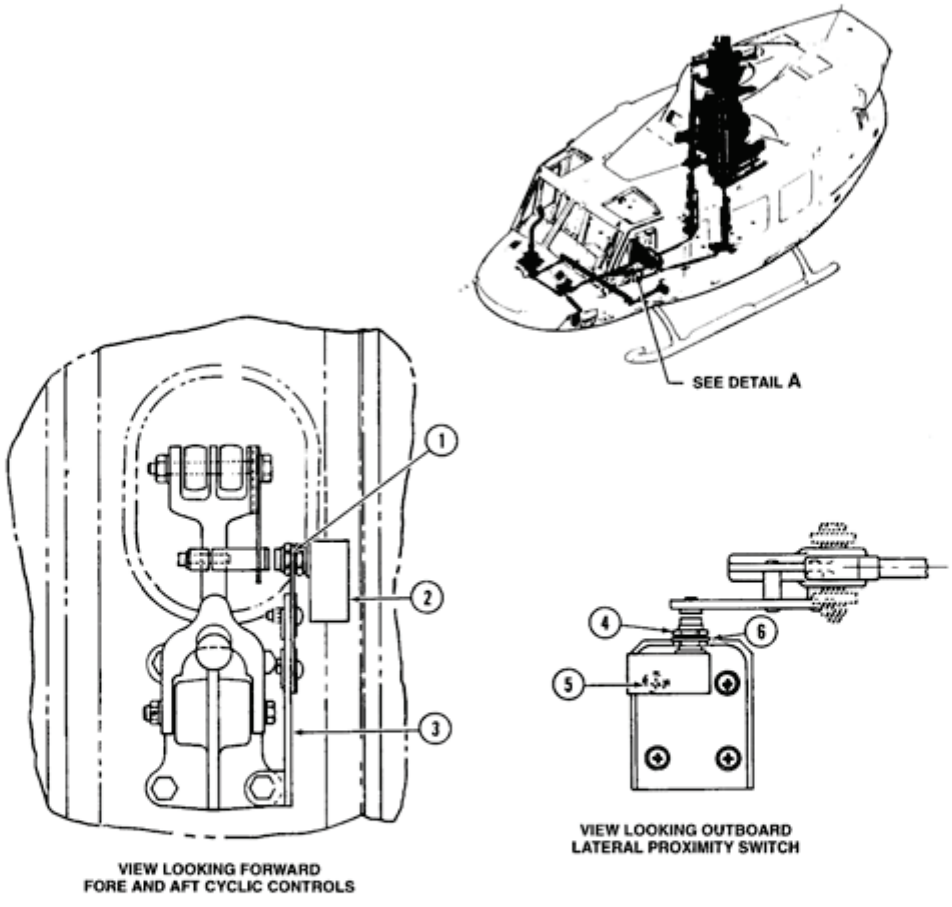
ICN-412-A-671200-B-97499-00016-A-001-01

Figure 1. Fore and Aft and Lateral Proximity Switches (Sheet 1 of 1)

Fore and Aft Proximity Switch - Inspection

Procedure

1. Examine fore and aft proximity switch (2, Figure 1) for cracks and security.
2. Examine wires for deterioration and looseness.



1. Nut
2. Fore and aft proximity switch
3. Support assembly
4. Nut
5. Lateral proximity switch
6. Sensor plate

ICN-412-A-671200-B-97499-00016-A-001-01

Figure 1. Fore and Aft and Lateral Proximity Switches (Sheet 1 of 1)

Fore and Aft Proximity Switch - Repair

Procedure

1. Tighten loose connections on wires.
2. Proximity switches are nonrepairable and shall be replaced if unserviceable.

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Fore and Aft Proximity Switch - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

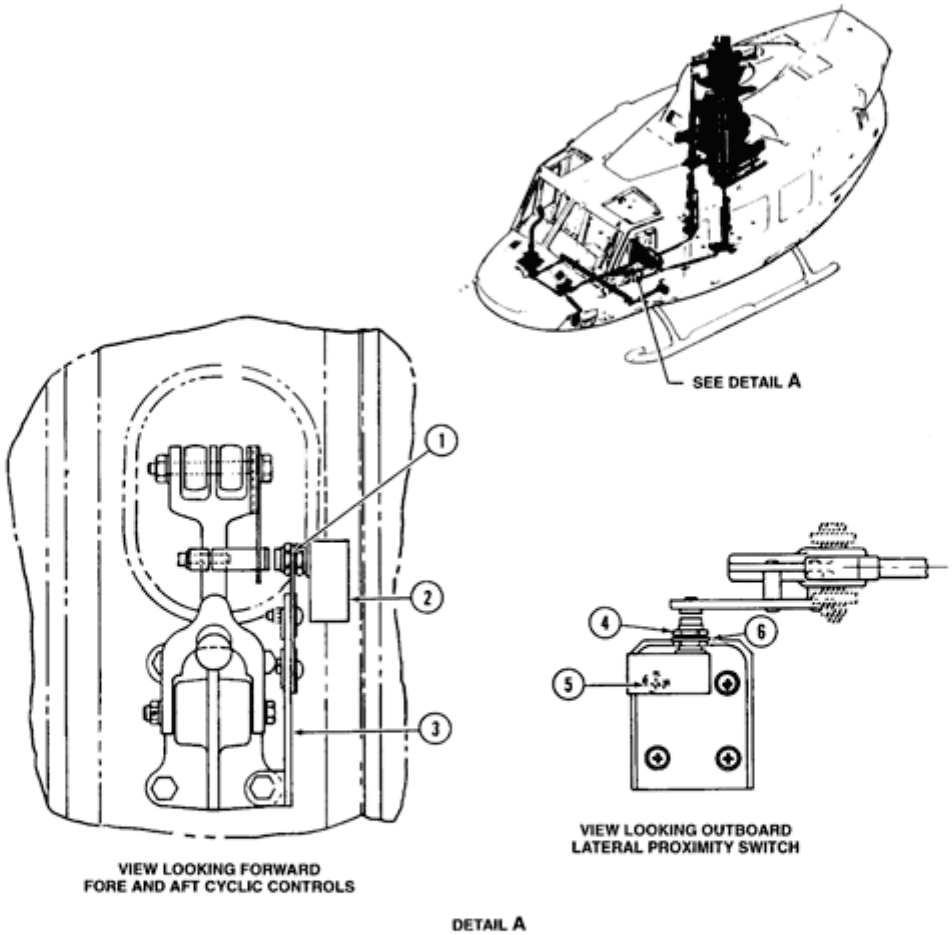
Procedure

1. Place fore and aft proximity switch (2, Figure 1) in support assembly (3) and loosely install nut (1). Do not tighten nut (1) at this time.

2. Connect wires at splice to switch.

3. Adjust switch as required (412-A-67-01-00-05A-276A-A / 01326). Tighten nut (1) and safety with lockwire (C-405).

4. Install access panel in center of cabin floor (412-A-52-40-00-02A-720A-A / 00840).



1. Nut
2. Fore and aft proximity switch
3. Support assembly
4. Nut
5. Lateral proximity switch
6. Sensor plate

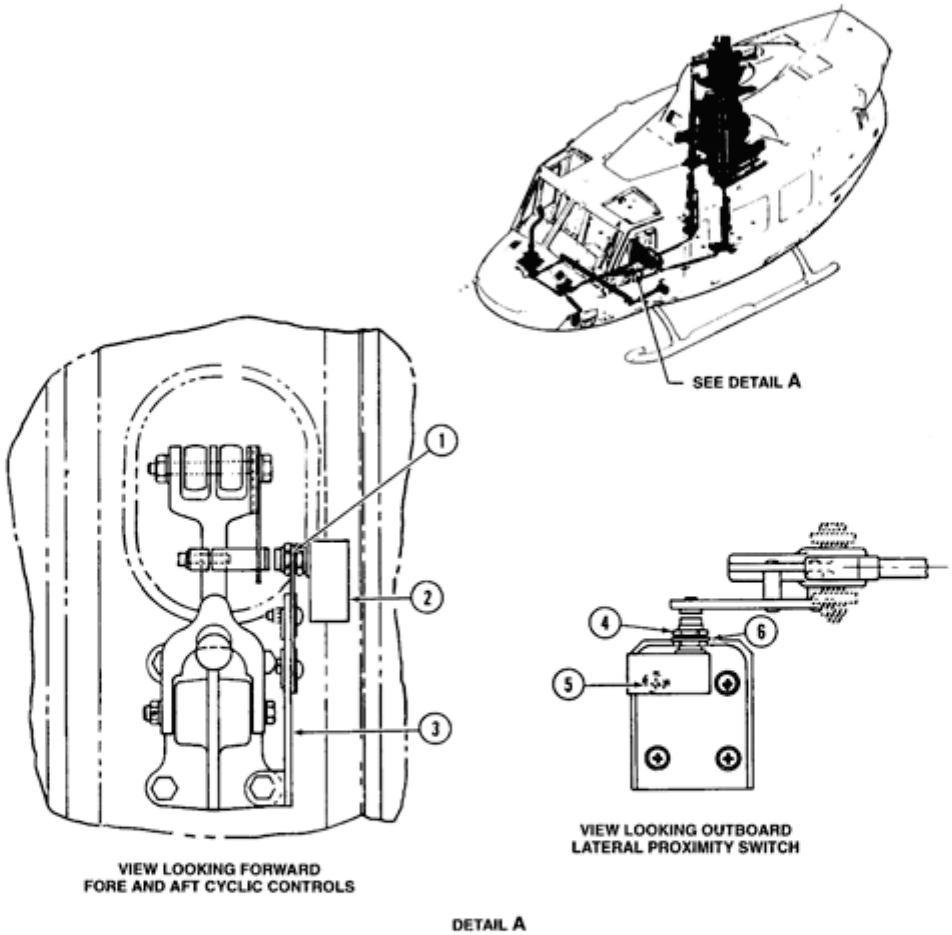
ICN-412-A-671200-B-97499-00016-A-001-01

Figure 1. Fore and Aft and Lateral Proximity Switches (Sheet 1 of 1)

Lateral Proximity Switch - Removal

Procedure

1. Remove access panel in center of aft cabin floor (412-A-52-40-00-02A-520A-A / 00837).
2. Cut lockwire and remove nut (4, Figure 1) from lateral proximity switch (5).
3. Separate wires at splice to switch.
4. Remove lateral proximity switch (5) from sensor plate (6).



1. Nut
2. Fore and aft proximity switch
3. Support assembly
4. Nut
5. Lateral proximity switch
6. Sensor plate

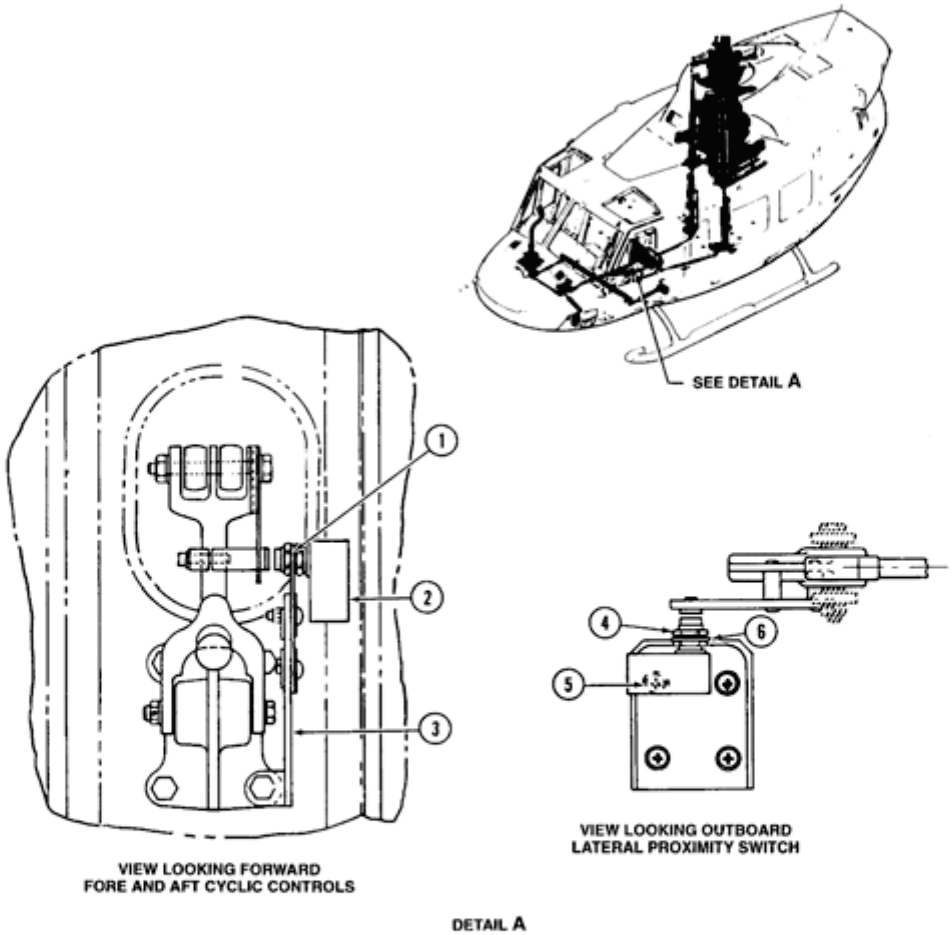
ICN-412-A-671200-B-97499-00016-A-001-01

Figure 1. Fore and Aft and Lateral Proximity Switches (Sheet 1 of 1)

Lateral Proximity Switch - Inspection

Procedure

1. Inspect lateral proximity switch (5, Figure 1) for cracks and security.
2. Inspect wires for deterioration and looseness.



1. Nut
2. Fore and aft proximity switch
3. Support assembly
4. Nut
5. Lateral proximity switch
6. Sensor plate

ICN-412-A-671200-B-97499-00016-A-001-01

Figure 1. Fore and Aft and Lateral Proximity Switches (Sheet 1 of 1)

Lateral Proximity Switch - Repair

Procedure

1. Tighten loose connections on wires.
2. Proximity switches are nonrepairable and shall be replaced if unserviceable.

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Lateral Proximity Switch - Installation

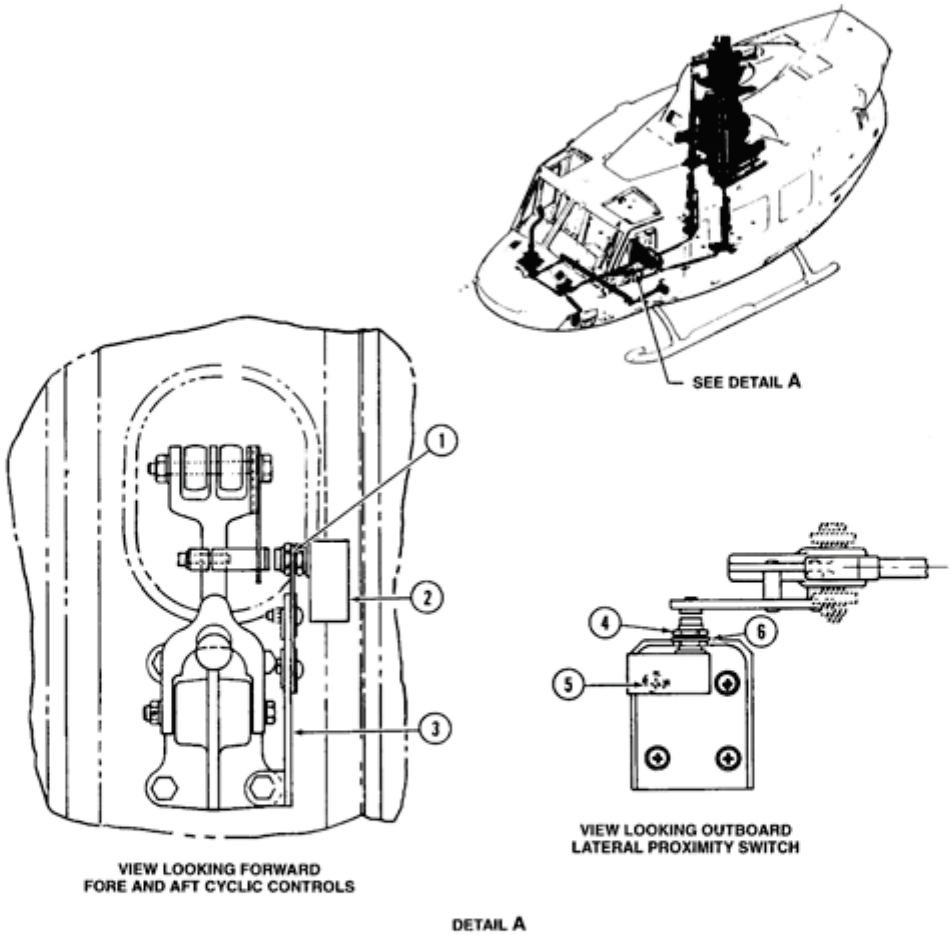
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Place lateral proximity switch (5, Figure 1) in sensor plate (6), and install nut (4). Do not secure nut at this time.
2. Connect wires at splice to switch.
3. Adjust switch as required (412-A-67-01-00-05A-276A-A / 01326). Tighten nut (4) and secure with lockwire (C-405).
4. Install access panel in center of cabin floor (412-A-52-40-00-02A-720A-A / 00840).



1. Nut
2. Fore and aft proximity switch
3. Support assembly
4. Nut
5. Lateral proximity switch
6. Sensor plate

ICN-412-A-671200-B-97499-00016-A-001-01

Figure 1. Fore and Aft and Lateral Proximity Switches (Sheet 1 of 1)

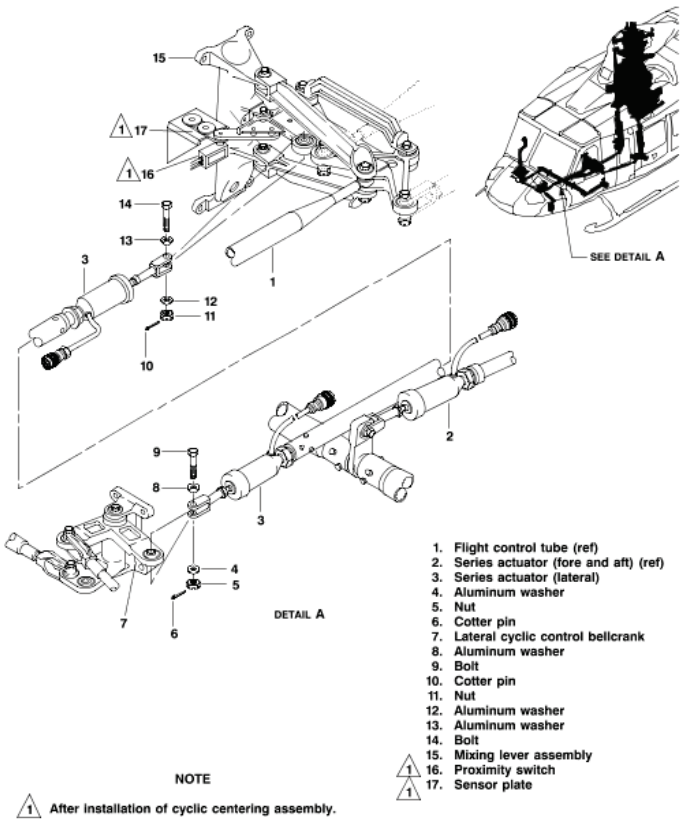
Lateral Series Actuator - Removal

Procedure



DISASSEMBLY OF SERIES ACTUATOR (3, Figure 1) IS NOT PERMITTED.

1. Remove access panel in center of aft cabin floor. (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect electrical connector.
3. Remove panel in cabin floor to provide access to lateral cyclic control bellcrank (7, Figure 1) (412-A-52-40-00-02A-520A-A / 00837).
4. Remove series actuator (3) from mixing lever assembly (15) by removing cotter pin (10), nut (11), aluminum washers (12 and 13) and bolt (14).
5. Remove series actuator (3) from lateral cyclic control bellcrank (7) by removing cotter pin (6), nut (5), aluminum washers (4 and 8) and bolt (9). Disconnect electrical connector. Slide series actuator forward and remove from helicopter.



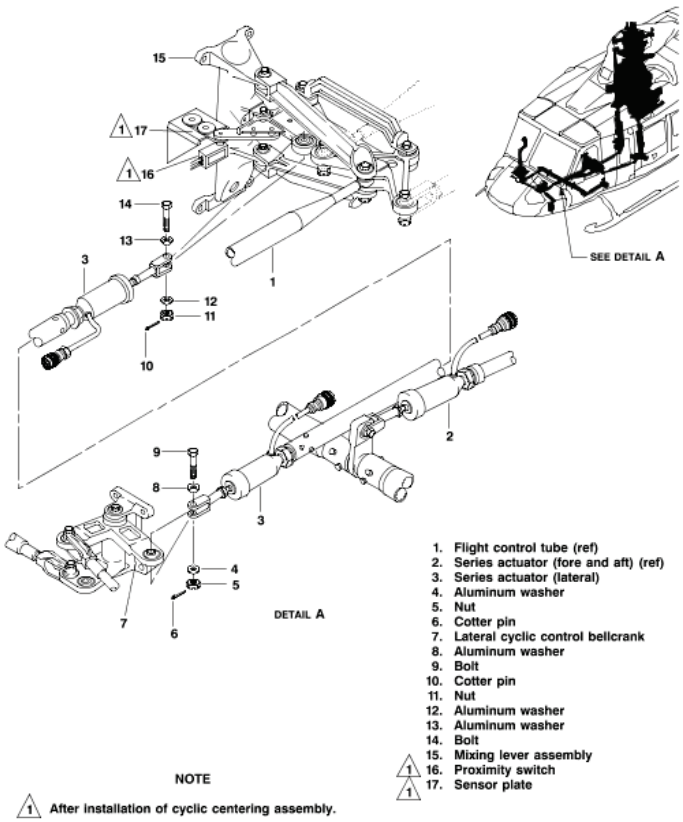
ICN-412-A-671200-B-97499-00017-A-001-01

Figure 1. Lateral Series Actuator and Proximity Switch (Sheet 1 of 1)

Lateral Series Actuator - Inspection

Procedure

1. Inspect series actuator (3, Figure 1) for damage, distortion, corrosion, and security of clevis (BHT-412-CR&O).



ICN-412-A-671200-B-97499-00017-A-001-01

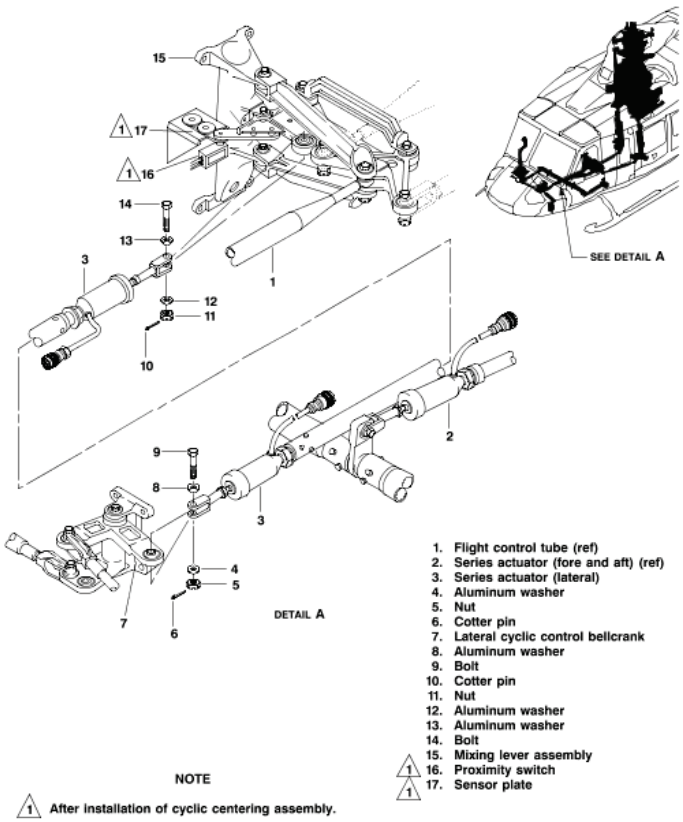
Figure 1. Lateral Series Actuator and Proximity Switch (Sheet 1 of 1)

Lateral Series Actuator - Repair Procedure



DISASSEMBLY OF SERIES ACTUATOR (3, Figure 1) IS NOT PERMITTED.

1. Series actuators are not field repairable and shall be replaced if unserviceable.



ICN-412-A-671200-B-97499-00017-A-001-01

Figure 1. Lateral Series Actuator and Proximity Switch (Sheet 1 of 1)

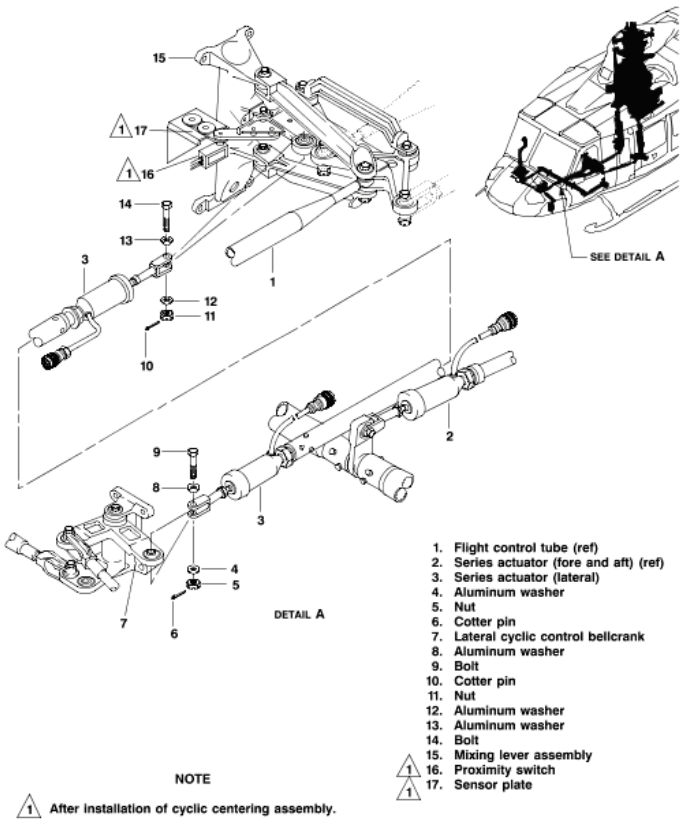
Lateral Series Actuator - Installation

Procedure



DISASSEMBLY OF SERIES ACTUATOR (3, Figure 1) IS NOT PERMITTED.

1. Place series actuator (3, Figure 1) into position in helicopter. Place clevis of actuator on lateral cyclic control bellcrank (7). Install bolt (9), aluminum washer (8), aluminum washer (4), nut (5), and cotter pin (6).
2. Place aft clevis of series actuator onto mixing lever assembly (15). Install bolt (14), aluminum washer (13), aluminum washer (12), nut (11), and cotter pin (10).
3. Connect electrical connector.
4. Install cabin floor panels (412-A-52-00-00-00A-009A-A / 00774).



ICN-412-A-671200-B-97499-00017-A-001-01

Figure 1. Lateral Series Actuator and Proximity Switch (Sheet 1 of 1)

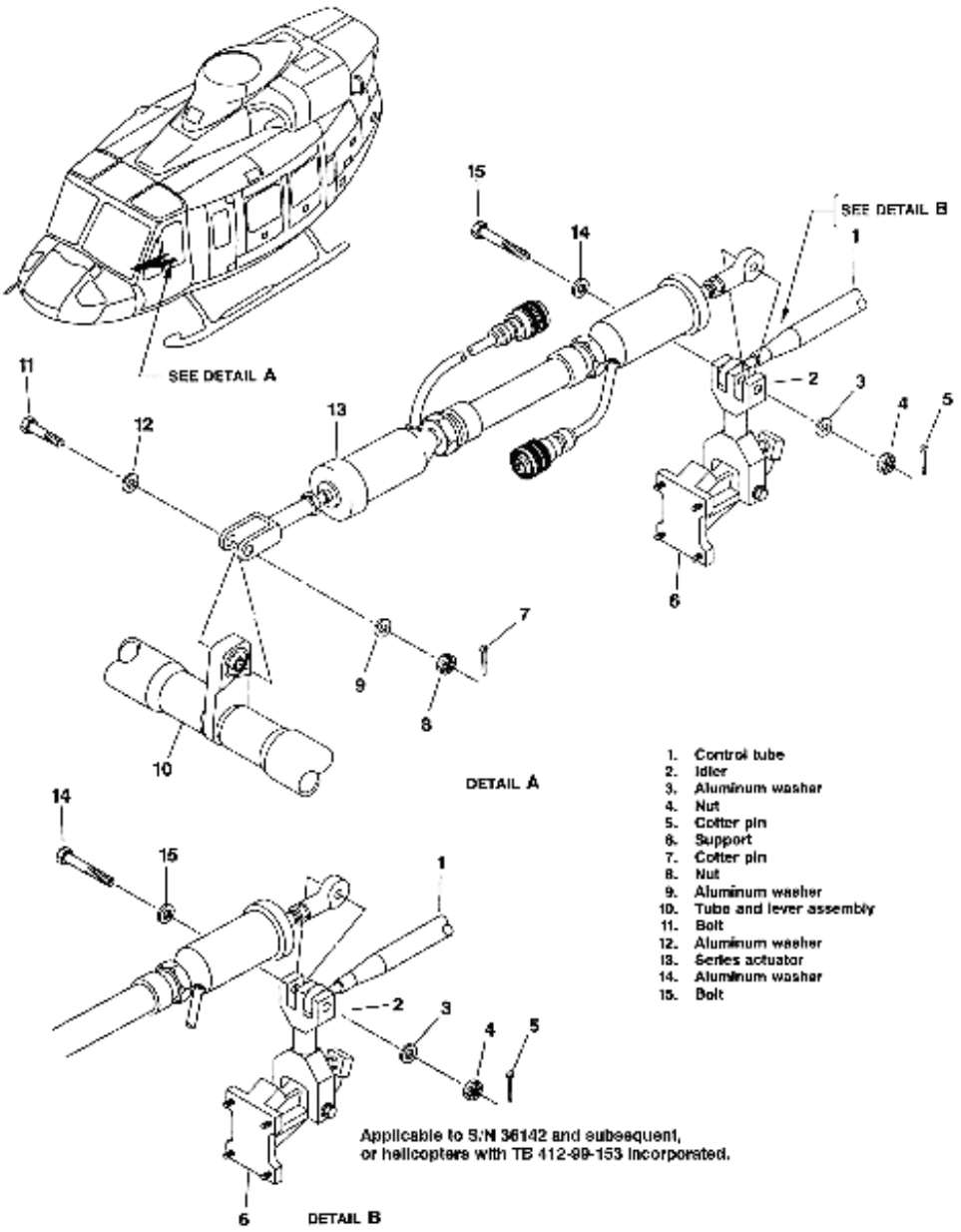
Longitudinal Series Actuator - Removal

Procedure



DISASSEMBLY OF SERIES ACTUATOR (13, Figure 1) IS NOT PERMITTED.

1. Remove access panel in center of aft cabin floor (412-A-52-40-00-02A-520A-A / 00837).
2. Remove panel from left side of console (412-A-52-40-00-02A-520A-A / 00837).
3. Remove cotter pin (5, Figure 1), nut (4), aluminum washers (3 and 14), and bolt (15).
4. Remove cotter pin (7), nut (8), aluminum washers (9 and 12), and bolt (11).
5. Disconnect electrical connectors. Slide series actuator (13) aft and remove from helicopter.



ICN-412-A-671200-B-97499-00018-A-001-01

Figure 1. Longitudinal Series Actuator (Sheet 1 of 1)

Longitudinal Series Actuator - Inspection

Procedure

1. Inspect series actuator in accordance with the BHT-412-CR&O.
2. Inspect rod end bearings on series actuator for wear (BHT-ALL-SPM, Standard Practices Manual).

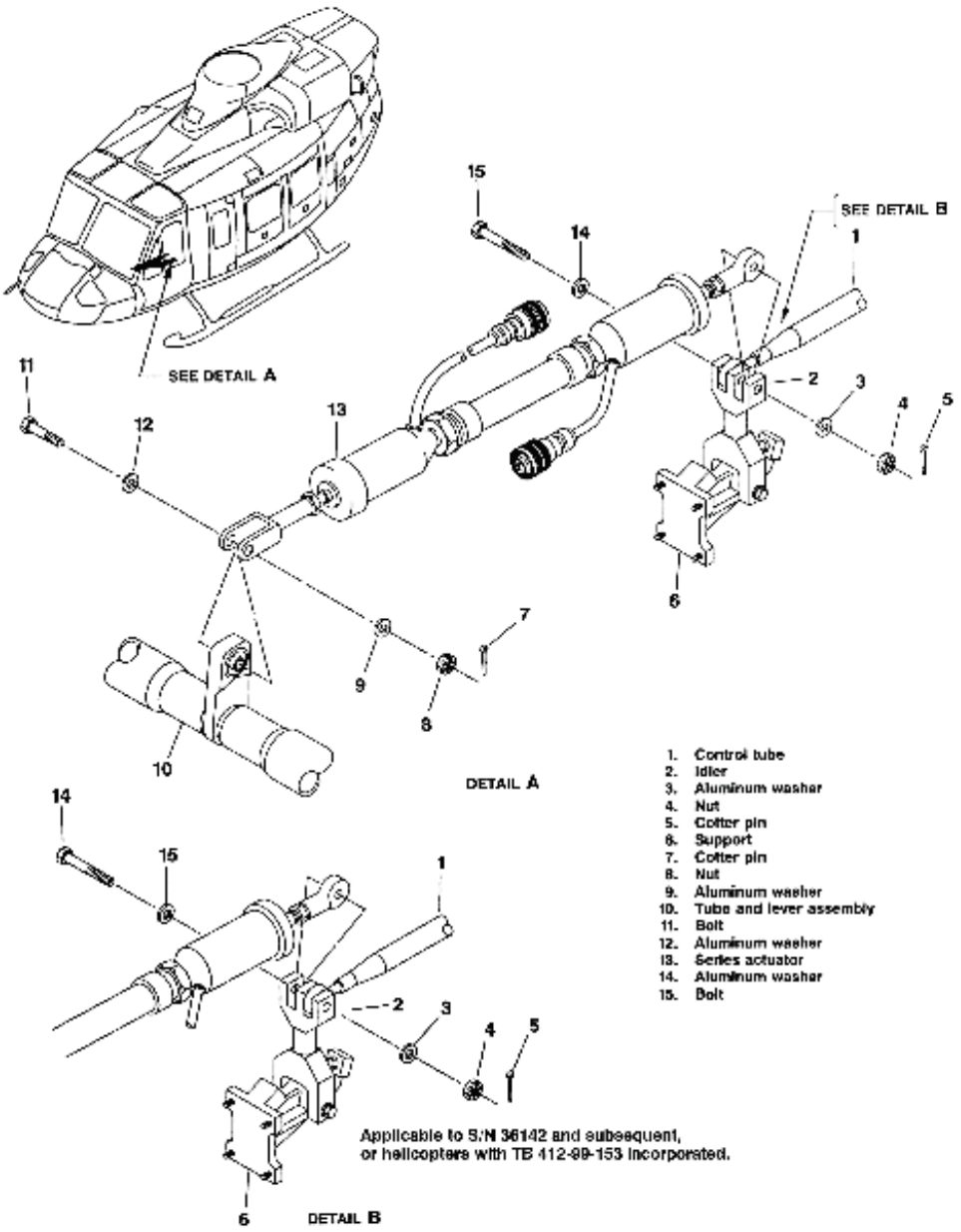
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Longitudinal Series Actuator - Repair Procedure



DISASSEMBLY OF SERIES ACTUATOR (13, Figure 1) IS NOT PERMITTED.

1. Series actuators are not field repairable and shall be replaced if unserviceable.



ICN-412-A-671200-B-97499-00018-A-001-01

Figure 1. Longitudinal Series Actuator (Sheet 1 of 1)

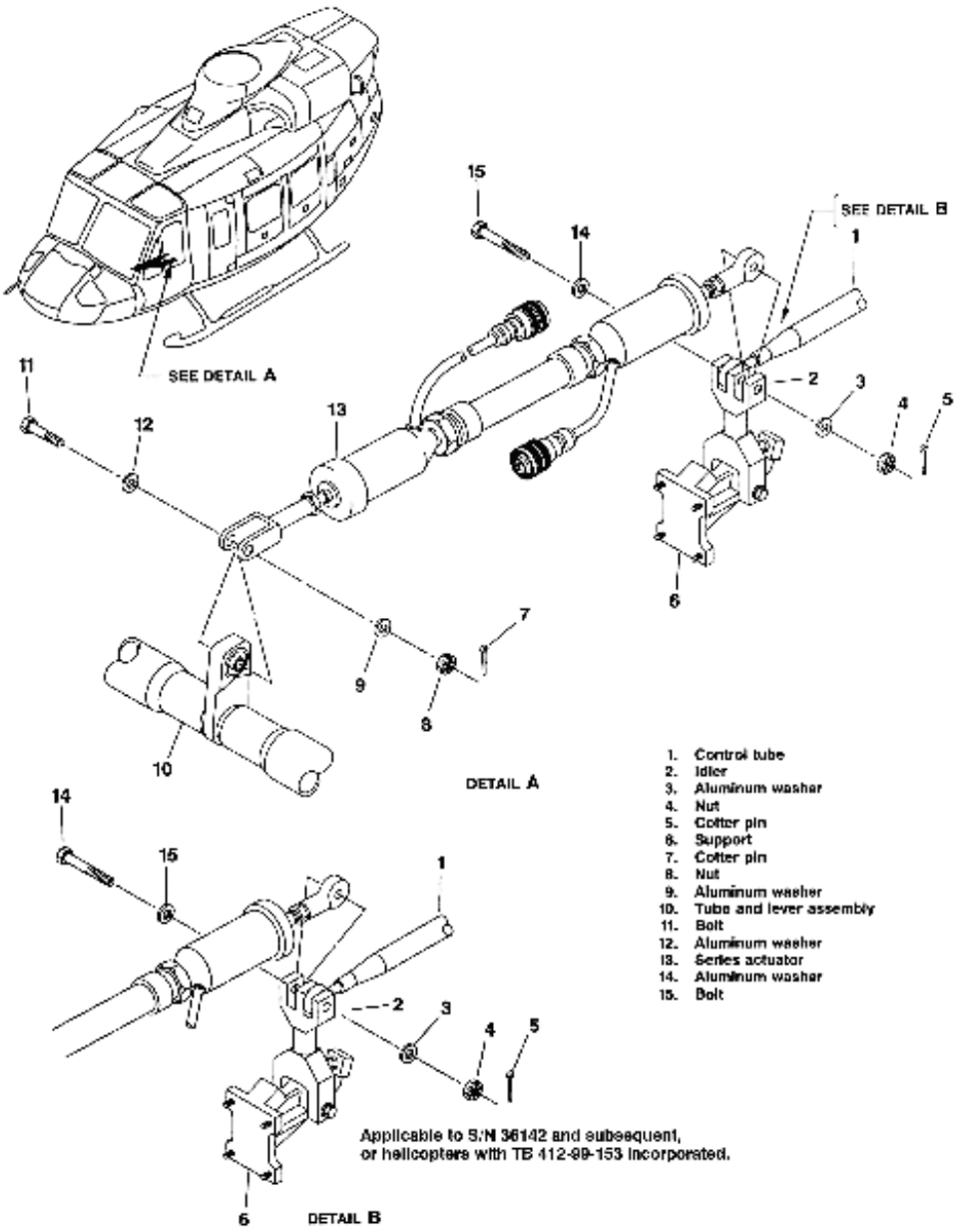
Longitudinal Series Actuator - Installation

Procedure



DISASSEMBLY OF SERIES ACTUATOR (13, Figure 1) IS NOT PERMITTED.

1. Place series actuator (13, Figure 1) into position in helicopter. Place forward end onto arm of tube and lever assembly (10) and install bolt (11), aluminum washer (12), aluminum washer (9), nut (8), and cotter pin (7).
2. Place aft end into idler (2), install bolt (15), aluminum washer (14), aluminum washer (3), nut (4), and cotter pin (5). Connect electrical connectors.
3. Verify cyclic control rigging within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).
4. Move controls through full travel and check for binding and obstructions.
5. Install access panel in center of cabin floor. (412-A-52-40-00-02A-720A-A / 00840).
6. Install panel on left side of console. (412-A-52-40-00-02A-720A-A / 00840).



ICN-412-A-671200-B-97499-00018-A-001-01

Figure 1. Longitudinal Series Actuator (Sheet 1 of 1)

Rotary Actuator (Lateral Cyclic) - Removal

Procedure

1. Remove access panel from bottom of helicopter just aft and inboard of left lower nose window (412-A-52-40-00-02A-520A-A / 00837).

2. Remove cotter pin (4, Figure 1) and nut (3).

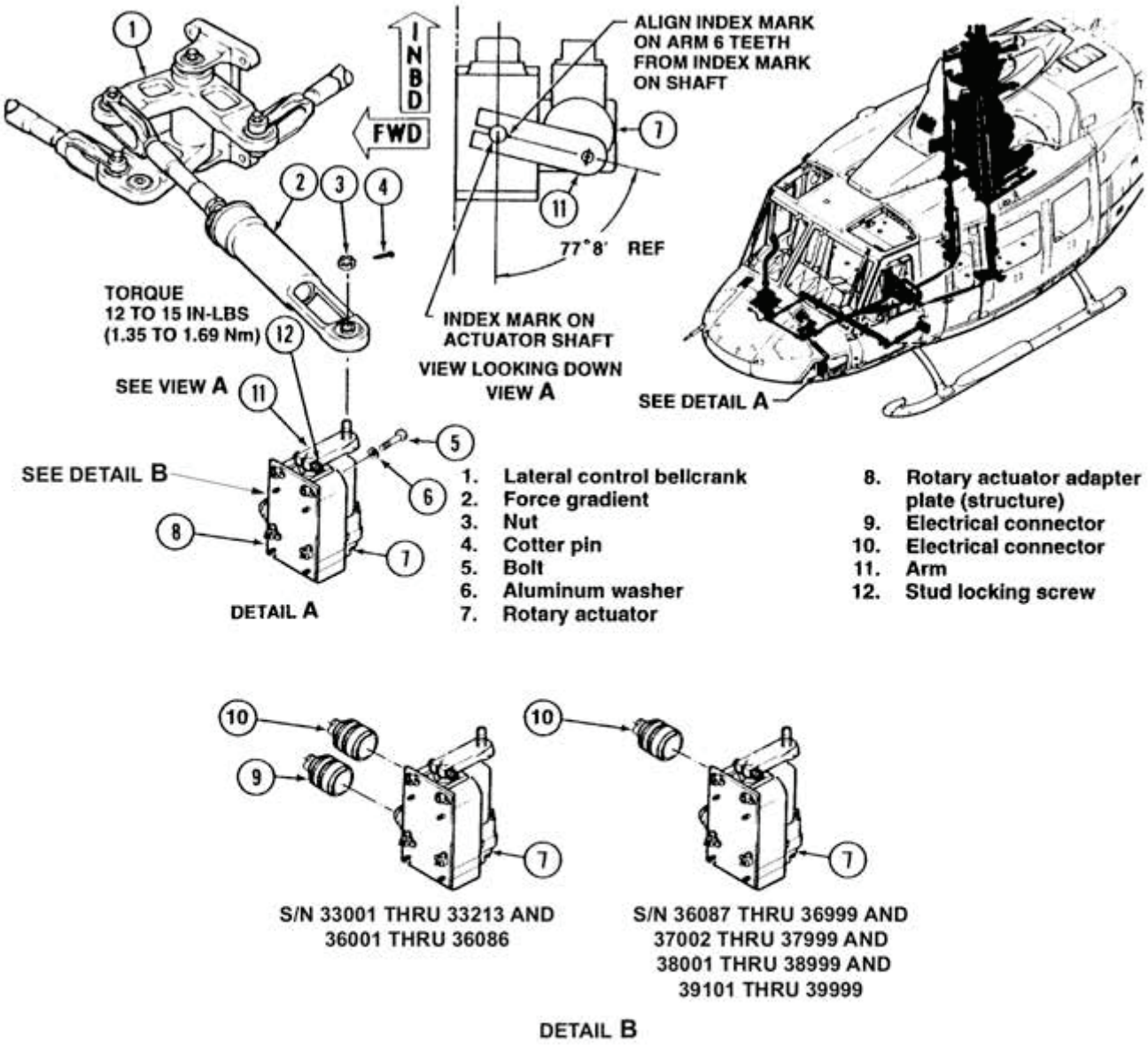
Applicable: (S/N 33001 THRU 33213, 36001 THRU 36086) :

3. *Disconnect electrical connector (9 and 10) from rotary actuator (7).*

Applicable: (S/N 36087 THRU 36999, 37002 THRU 37999, 38001 THRU 38999, 39101 THRU 39999) :

4. *Disconnect electrical connector (10) from rotary actuator (7).*

5. Remove four bolts (5) and four aluminum washers (6). Remove rotary actuator (7) and rotary actuator adapter plate (8).



ICN-412-A-671200-B-97499-00019-A-002-01

Figure 1. Lateral Cyclic Rotary Actuator (Sheet 1 of 1)

Rotary Actuator (Lateral Cyclic) - Inspection

Procedure

1. Inspect rotary actuator for mechanical and corrosion damage (BHT-412-CR&O).

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Rotary Actuator (Lateral Cyclic) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Inspect rotary actuator (7, Figure 1) to ensure arm (11) is installed with index mark on arm positioned six teeth from index mark on shaft (View A). Loosen stud locking screw (12) and reposition arm if index marks are not properly located.

2. Torque rotary actuator arm stud locking screw (12) 12 to 15 inch-pounds (1.35 to 1.69 Nm). Secure with lockwire (C-405) as required.

3. Position rotary actuator (7) on rotary actuator adapter plate (8). Install bolts (5) and aluminum washers (6).

4. Adjust length of force gradient (2) (412-A-67-01-00-03A-276A-A / 01324) for rigging instructions. After force gradient length is set, position force gradient and tube and lever assembly (1) on arm (11). Install nut (3) and cotter pin (4).

Applicable: (S/N 33001 THRU 33213, 36001 THRU 36086) :

5. *Connect electrical connector (9 and 10) to rotary actuator (7).*

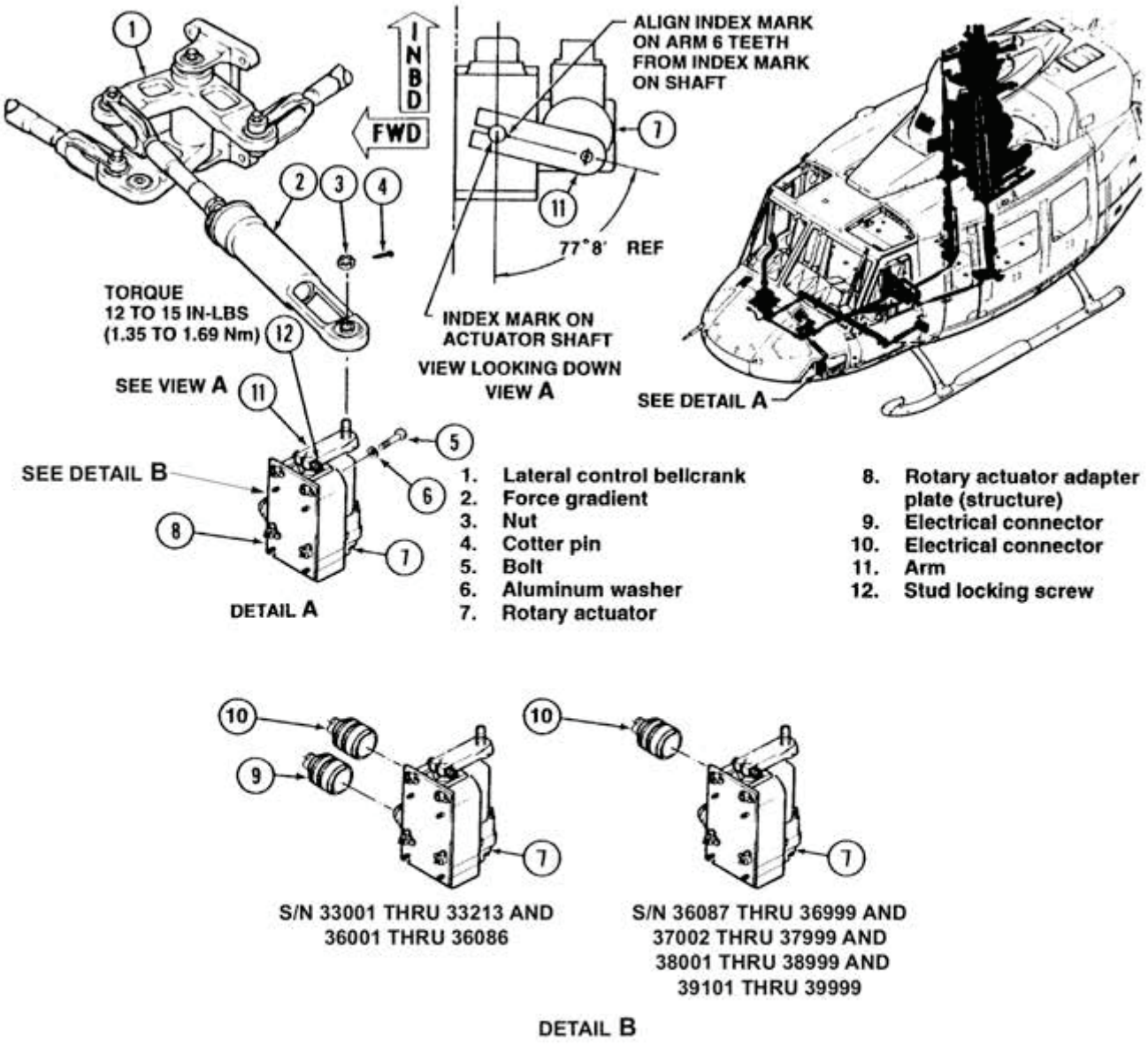
Applicable: (S/N 36087 THRU 36999, 37002 THRU 37999, 38001 THRU 38999, 39101 THRU 39999) :

6. *Connect electrical connector (10) to rotary actuator (7).*

7. Move cyclic controls through full travel and check for binding and/or obstructions.

8. Verify cyclic control rigging within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).

9. Install access panel (412-A-52-40-00-02A-720A-A / 00840).



ICN-412-A-671200-B-97499-00019-A-002-01

Figure 1. Lateral Cyclic Rotary Actuator (Sheet 1 of 1)

Rotary Actuator (Fore and Aft Cyclic) - Removal

Procedure

1. Remove access panel from cabin floor under left crew seat. Remove access panel from bottom of left side of helicopter just forward of forward landing gear crosstube (412-A-52-40-00-02A-520A-A / 00837).

2. Disconnect hook (16, Figure 1), spring (19) from bolt (13).

3. Remove cotter pin (3) and nut (4) and separate force gradient (2) from arm (6).

Applicable: (S/N 33001 THRU 33213, 36001 THRU 36086) :

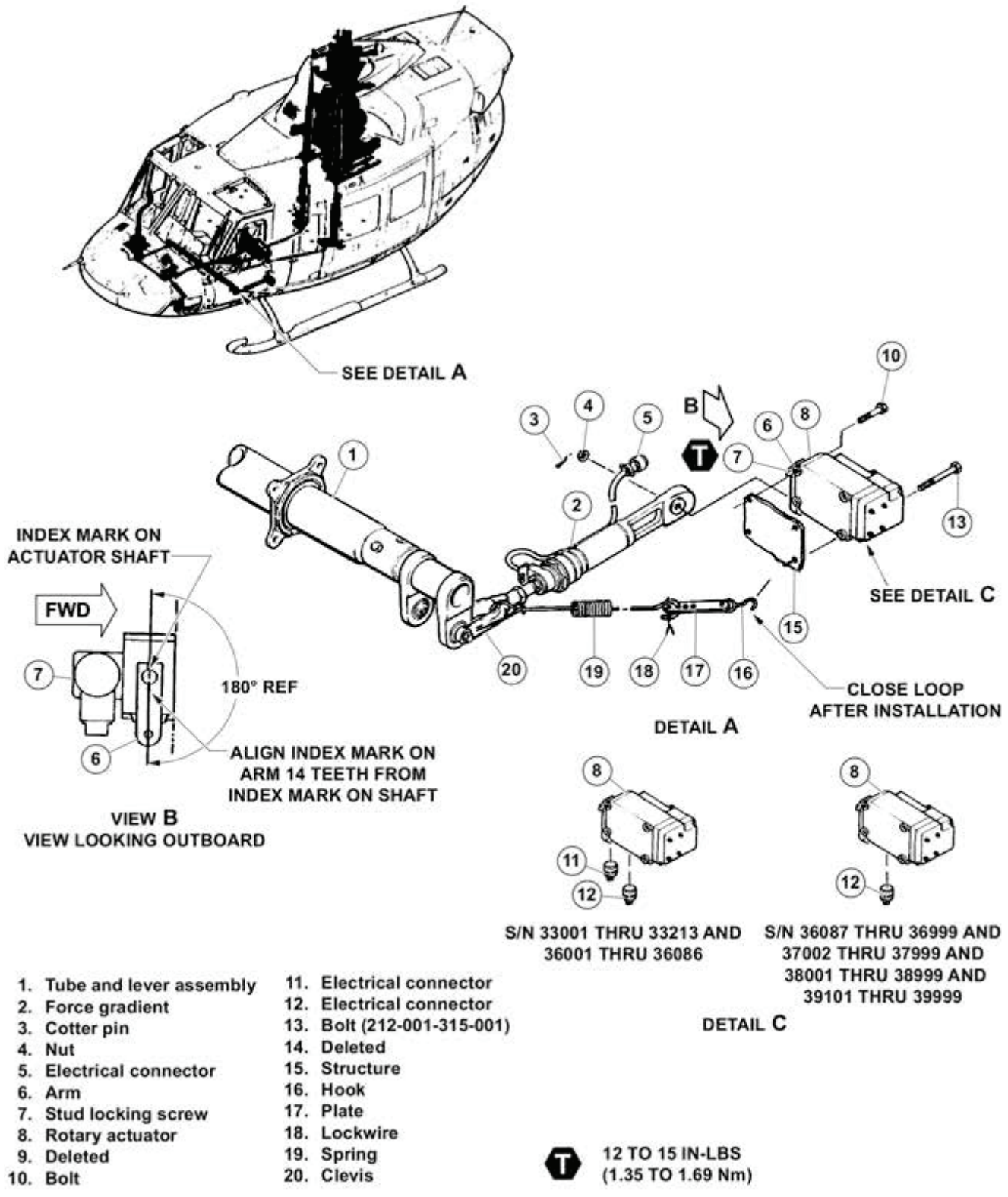
4. *Disconnect electrical connectors (5, 11 and 12).*

Applicable: (S/N 36087 THRU 36999, 37002 THRU 37999, 38001 THRU 38999, 39101 THRU 39999) :

5. *Disconnect electrical connectors (5 and 11).*

6. Remove bolts (10 and 13).

7. Remove rotary actuator (8) from structure (15).



1. Tube and lever assembly
2. Force gradient
3. Cotter pin
4. Nut
5. Electrical connector
6. Arm
7. Stud locking screw
8. Rotary actuator
9. Deleted
10. Bolt
11. Electrical connector
12. Electrical connector
13. Bolt (212-001-315-001)
14. Deleted
15. Structure
16. Hook
17. Plate
18. Lockwire
19. Spring
20. Clevis

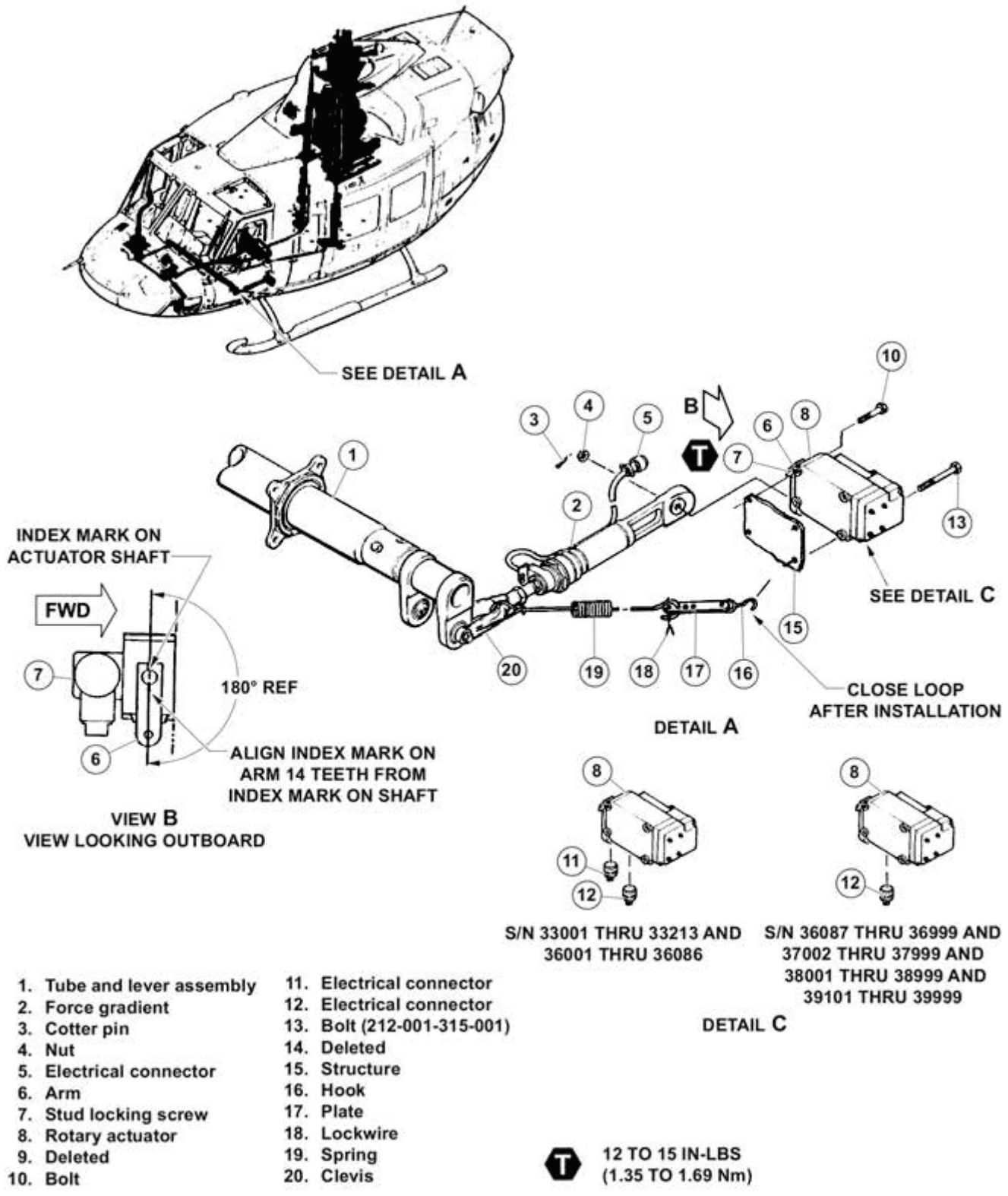
ICN-412-A-671200-B-97499-00020-A-002-01

Figure 1. Fore and Aft Cyclic Rotary Actuator (Sheet 1 of 1)

Rotary Actuator (Fore and Aft Cyclic) - Inspection

Procedure

1. Inspect rotary actuator (8, Figure 1) for mechanical and corrosion damage (BHT-412-CR&O).



ICN-412-A-671200-B-97499-00020-A-002-01

Figure 1. Fore and Aft Cyclic Rotary Actuator (Sheet 1 of 1)

Rotary Actuator (Fore and Aft Cyclic) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Inspect rotary actuator (8, Figure 1) to ensure arm (6) is installed with index mark on arm positioned 14 teeth from index mark on shaft (View A). Loosen stud locking screw (7) and reposition arm if index marks are not properly located.

2. Torque rotary actuator arm stud locking screw (7), 12 to 15 inch-pounds (1.35 to 1.69 Nm). Secure with lockwire (C-405) as required.

3. Position rotary actuator (8) on structure (15).

4. Install special bolt (13) in lower outboard hole.

5. Install bolts (10) in remaining holes.

6. Adjust length of force gradient (2) (412-A-67-01-00-03A-276A-A / 01324). After force gradient length is set, position force gradient and tube and lever assembly (1) on arm (6).

7. Install nut (4) and cotter pin (3).

8. Install spring (19), plate (17), and hook (16) on bolt (13). Refer to 412-A-67-12-00-17A-720A-A / 01462 for procedure to position spring (19) in correct hole in plate.

NOTE

For initial installation, install spring in fourth hole of plate (17). With hydraulic boost on or during ground run, adjust spring tension to position pilot cyclic (longitudinal only) perpendicular to deck within 2°.

9. Close loop on hook (16) on bolt (13). Secure spring (19) to plate (17) with lockwire (C-405).

Applicable: (S/N 33001 THRU 33213, 36001 THRU 36086) :

10. *Connect the electrical connectors (5, 11 and 12).*

Applicable: (S/N 36087 THRU 36999, 37002 THRU 37999, 38001 THRU 38999, 39101 THRU 39999) :

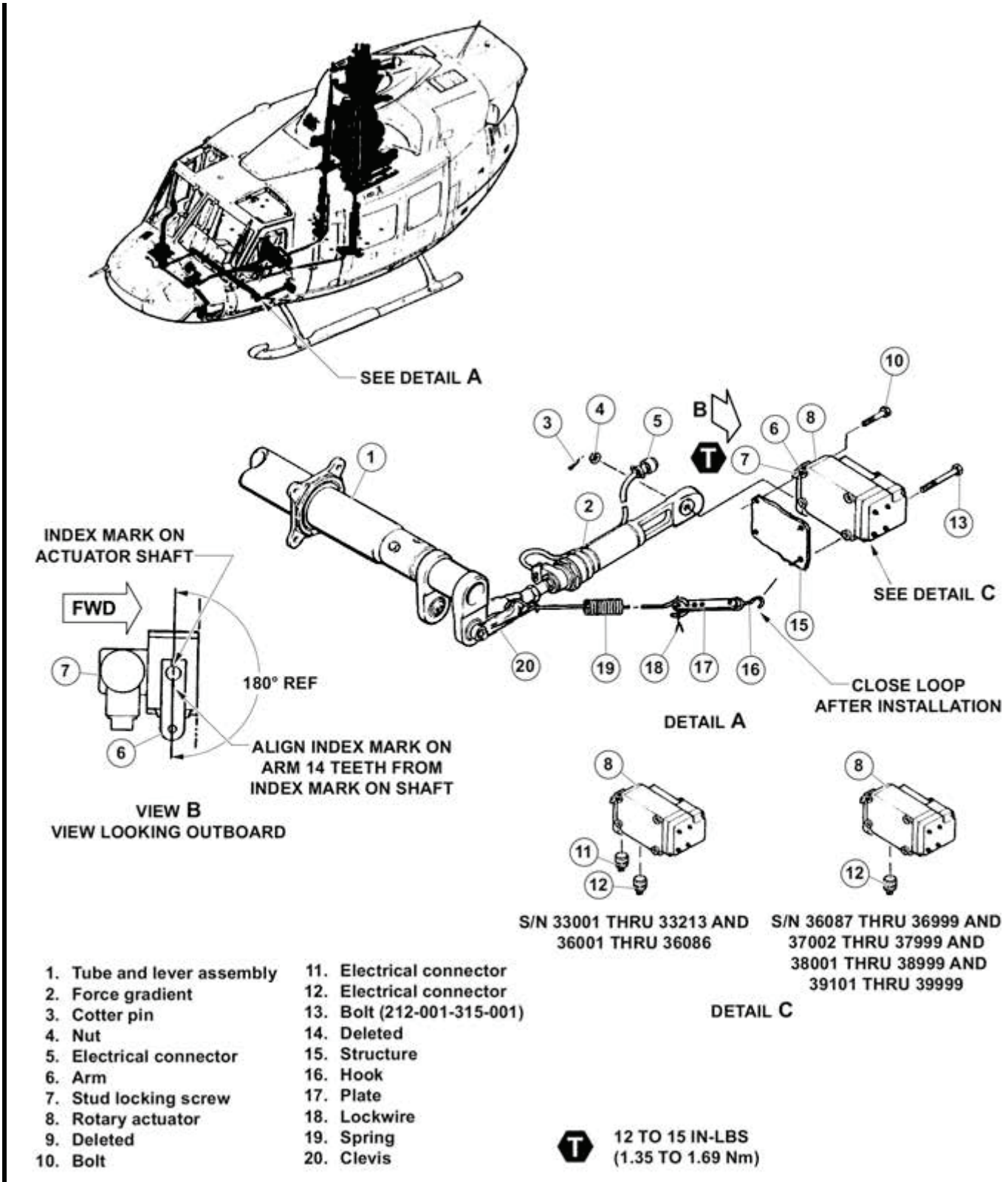
11. *Connect the electrical connectors (5 and 11).*

12. Move cyclic controls through full travel and check for binding and/or obstructions.

13. Verify cyclic rigging within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).

14. Install access panel on cabin floor under left crew seat (412-A-52-40-00-02A-720A-A / 00840).

15. Install access panel on bottom of helicopter (412-A-52-40-00-02A-720A-A / 00840).



1. Tube and lever assembly
2. Force gradient
3. Cotter pin
4. Nut
5. Electrical connector
6. Arm
7. Stud locking screw
8. Rotary actuator
9. Deleted
10. Bolt
11. Electrical connector
12. Electrical connector
13. Bolt (212-001-315-001)
14. Deleted
15. Structure
16. Hook
17. Plate
18. Lockwire
19. Spring
20. Clevis

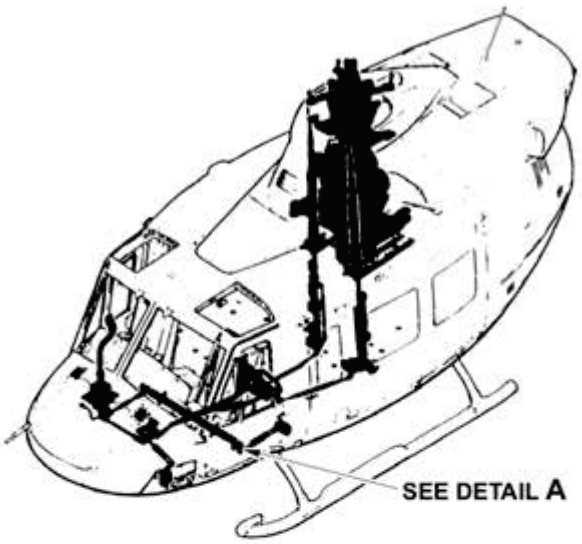
ICN-412-A-671200-B-97499-00020-A-002-01

Figure 1. Fore and Aft Cyclic Rotary Actuator (Sheet 1 of 1)

Force Gradient (Fore and Aft Cyclic) - Removal

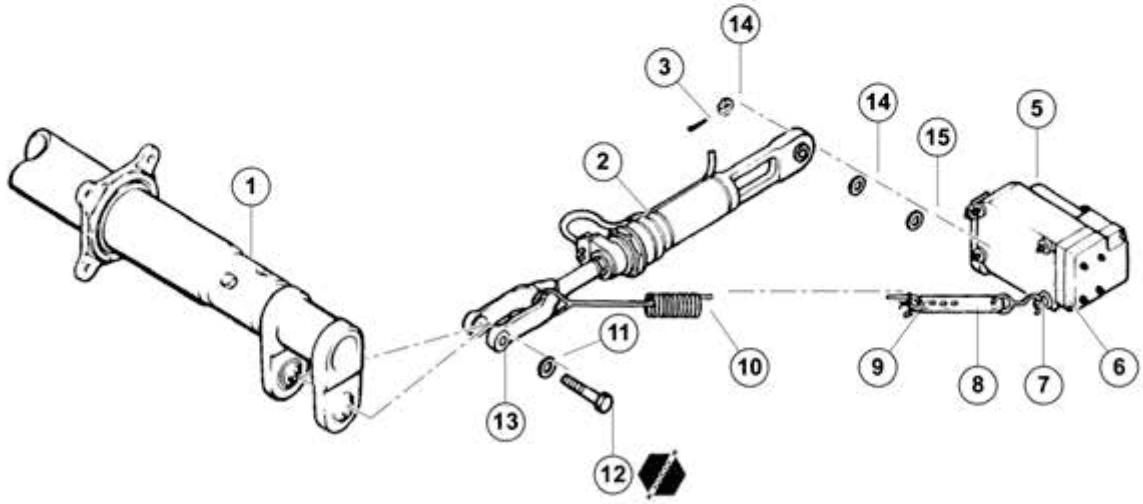
Procedure

1. *Remove access panel from cabin floor below the left crew seat (*412-A-52-40-00-02A-520A-A / 00837*).*
2. Remove access panel from bottom of helicopter on left side forward of forward landing gear crosstube. (412-A-52-40-00-02A-520A-A / 00837).
3. Remove hook (7, Figure 1) from bolt (6).
4. Disconnect electrical connector.
5. Remove cotter pin (3), nut (4), and washers (14 and 15) from rotary actuator (5).
6. Remove aluminum washer (11) and bolt (12).
7. Remove fore and aft cyclic controls force gradient (2) from tube and lever assembly (1).



1. Tube and lever assembly
2. Fore and aft cyclic controls force gradient
3. Cotter pin
4. Nut
5. Rotary actuator
6. Bolt (212-001-315-1)
7. Hook
8. Plate
9. Lockwire
10. Spring
11. Aluminum washer
12. Bolt
13. Clevis
14. Washer
15. Washer

LOCKWIRE (C-405)



DETAIL A

ICN-412-A-671200-B-97499-00021-A-001-01

Figure 1. Fore and Aft Cyclic Force Gradient - Removal (Sheet 1 of 1)

Force Gradient (Fore and Aft Cyclic) - Inspection

Procedure

1. Inspect fore and aft cyclic controls force gradient (2, Figure 1) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion is not acceptable and force gradient shall be replaced.

2. Inspect clevis (13) for damaged threaded helicoil insert and for elongated hole in bushing opposite the threaded helicoil insert.

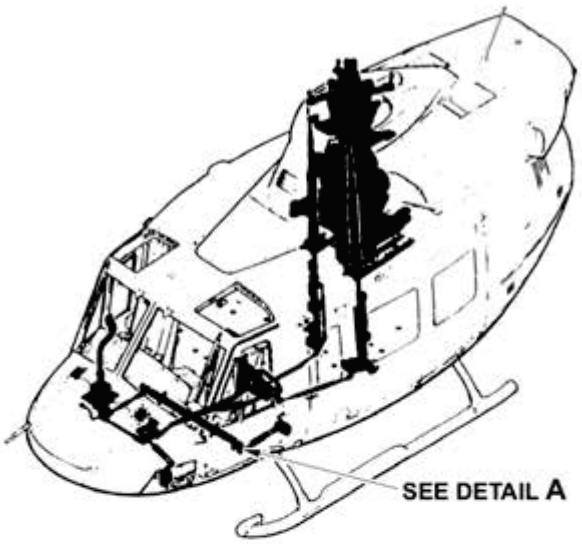
3. Inspect hook (7), plate (8), and spring (10) for obvious mechanical damage and for loss of tension in spring. Spring tension at 10.36 inches (263.1 mm) is 6.4 ±0.5 pounds (28.4 ±2.2 N).

4. Inspect bearing for wear and for secure installation in cylinder (5, Figure 2) (BHT-ALL-SPM, Standard Practices Manual).

5. Inspect two nuts (6 and 8) for secure installation. If adjustment of nuts (2 and 3, spring preload) is suspected of being incorrect, adjust spring preloads (412-A-67-12-00-17A-720A-A / 01462).

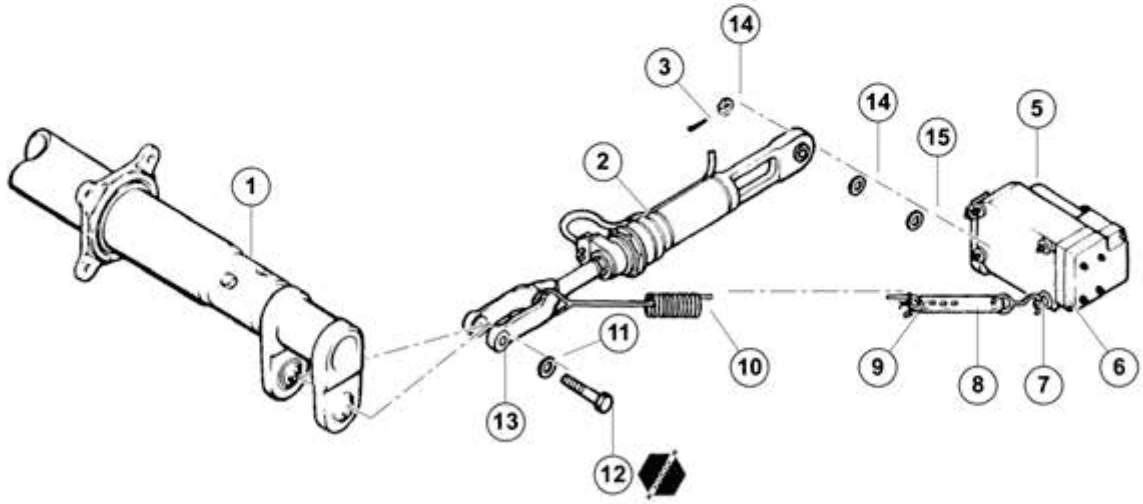
6. Inspect jam nut (1) and cylinder (5) for lockwire installation.

7. Inspect switch and electrical cable assembly (18) for frayed wires and broken pins in connector. Check continuity from connector pins through switch.



1. Tube and lever assembly
2. Fore and aft cyclic controls force gradient
3. Cotter pin
4. Nut
5. Rotary actuator
6. Bolt (212-001-315-1)
7. Hook
8. Plate
9. Lockwire
10. Spring
11. Aluminum washer
12. Bolt
13. Clevis
14. Washer
15. Washer

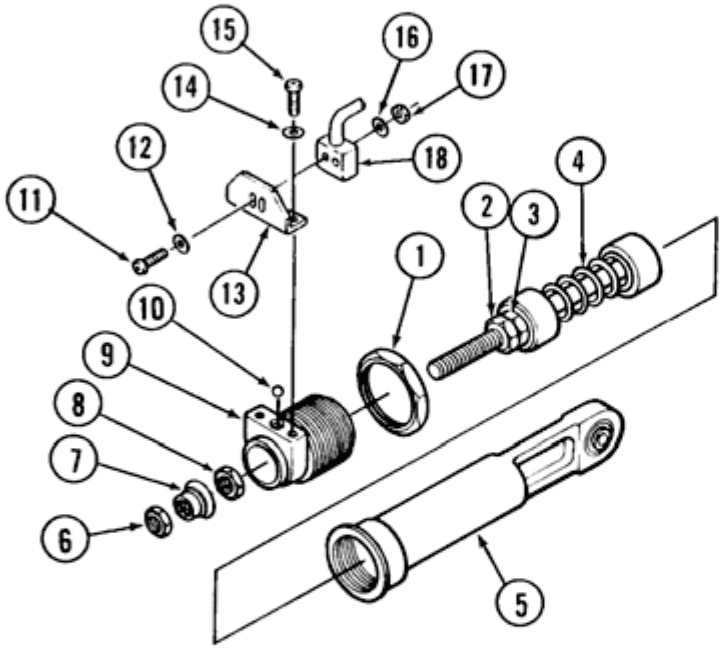
LOCKWIRE (C-405)



DETAIL A

ICN-412-A-671200-B-97499-00021-A-001-01

Figure 1. Fore and Aft Cyclic Force Gradient (Sheet 1 of 1)



1. Jam nut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Screw
12. Washer
13. Retainer
14. Washer
15. Screw
16. Washer
17. Nut
18. Cable assembly (switch)

ICN-412-A-671200-B-97499-00022-A-001-01

Figure 2. Fore and Aft Cyclic Force Gradient Assembly (Sheet 1 of 1)

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Force Gradient (Fore and Aft Cyclic) - Repair

Procedure

1. Repair of force gradient consists of replacement of unserviceable components and adjustment of spring preload (412-A-67-12-00-17A-720A-A / 01462).

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Force Gradient (Fore and Aft Cyclic) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

Fore and aft cyclic controls force gradient (2, Figure 1) is similar in appearance to force gradient in lateral cyclic controls, but the force gradients have different part numbers. Make sure correct part number force gradient is installed at lateral or fore and aft positions (BHT-412-IPB).

1. If not previously accomplished, adjust spring preload of force gradient as follows:

1.1. Cut and remove lockwire, loosen jam nut (1, Figure 2) and remove cap (9).

1.2. Remove spring assembly (4) from cylinder (5).

1.3. Adjust inboard nut (3) to provide 7.5 to 8.5 pounds (33.4 to 37.8 N) preload against spring assembly (4). Tighten outboard nut (2) against inboard nut.

1.4. Position spring assembly (4) in cylinder (5) and install cap (9) with jam nut (1) until all end play is eliminated. Tighten jam nut and secure to cylinder with lockwire (C-405).

2. If not previously accomplished, adjust actuator (7) on force gradient as follows:

2.1. Loosen nuts (6 and 8) on shaft of spring assembly (4).

2.2. Make sure retainer (13) is secured by screws (15) and washers (14).

2.3. Align actuator (7) on shaft of spring assembly (4) with ball (10) and plunger of cable assembly switch (18).

2.4. Loosen screws (11), washers (12 and 16) and nut (17) cable assembly switch (18) to retainer (13).

2.5. Push and hold actuator (7) down (180°) against cap (9) at point opposite location of cable assembly switch (18). Hold this position to initially adjust cable assembly switch then adjust switch to actuate within ±0.050 inch (1.27 mm) linear travel of spring assembly (4). Secure cable assembly switch to retainer (13) with screws (11), washers (12 and 16) and nut (17). Secure actuator with nut (8).

2.6. Install clevis (13, Figure 1) on shaft of spring assembly (4, Figure 2). Secure with nut (6).

3. After assembly of force gradient, prior to installation, verify spring assembly (4) preload of 7.5 to 8.5 pounds (33.36 to 37.81 N).

4. *Put the spring (10, Figure 1) in its position on clevis (13) as illustrated.*

5. *Put the fore and aft cyclic controls force gradient (2) in its position on tube and lever assembly (1). Clevis (13) has a threaded helicoil insert in one side. Position clevis with threaded insert inboard. Install bolt (12) and aluminum washer (11). Secure bolt head to hole in clevis (13) with* lockwire (C-405).



MAKE SURE FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO INSTALLATION OF FORCE GRADIENT ON ROTARY ACTUATOR.

6. If not previously accomplished, adjust length of force gradient as outlined in rigging instructions (412-A-67-01-00-03A-276A-A / 01324).

7. *Put the fore and aft cyclic controls force gradient (2) in its position on rotary actuator (5). Install the washers (14 and 15) and nut (4). Tighten the nut. Safety the nut with a new cotter pin (3).*

8. Install plate (8) on spring (10) with spring through fourth hole in plate. Do not lockwire spring to plate at this time.

9. Attach plate (8) with hook (7) to bolt (6). Do not close loop on hook at this time.

10. Connect electrical connector.

11. Move fore and aft cyclic controls through full travel and check for binding and obstructions.

12. Apply external hydraulic power to helicopter (412-A-29-00-00-00A-028A-A / 00611).

13. Loosen cyclic friction and observe cyclic stick position. If cyclic stick is not perpendicular (longitudinally) within 2°, move spring (10) to different hole in plate (8) to cause cyclic stick to be vertical (longitudinally).

14. When tension on spring (10) is correctly adjusted, secure spring to plate (8) with lockwire (C-405) (9) and close loop on hook (7) to prevent accidental disengagement of hook.

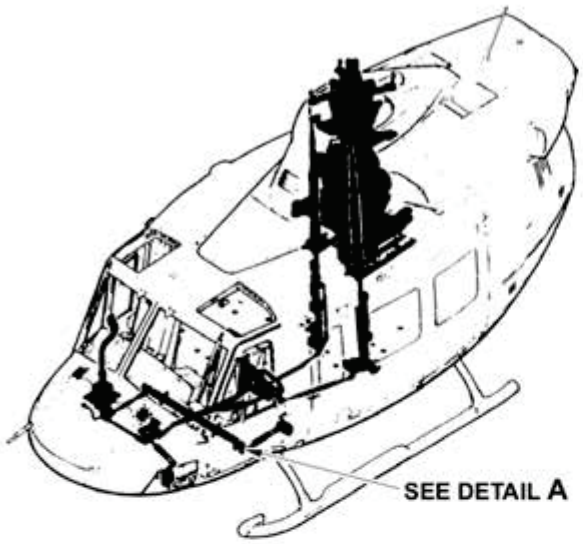
15. Move cyclic controls through full travel and check for binding and obstructions.

16. Verify cyclic control rigging is within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).

17. Remove external power from helicopter.

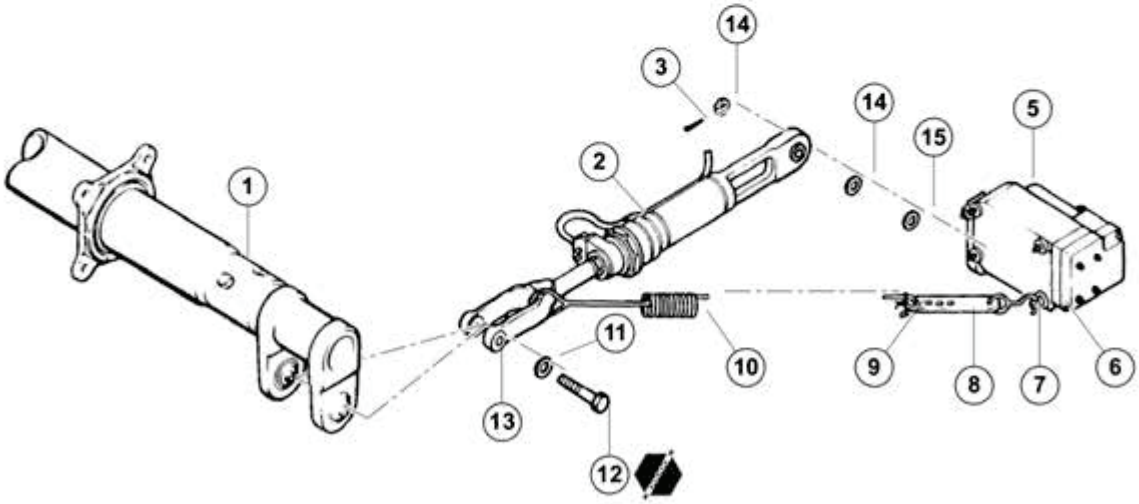
18. Install access panel on bottom of helicopter on left side forward of forward landing gear crosstube. (412-A-52-40-00-02A-720A-A / 00840).

19. Install access panel on cabin floor under left crew seat (412-A-52-40-00-02A-720A-A / 00840).



1. Tube and lever assembly
2. Fore and aft cyclic controls force gradient
3. Cotter pin
4. Nut
5. Rotary actuator
6. Bolt (212-001-315-1)
7. Hook
8. Plate
9. Lockwire
10. Spring
11. Aluminum washer
12. Bolt
13. Clevis
14. Washer
15. Washer

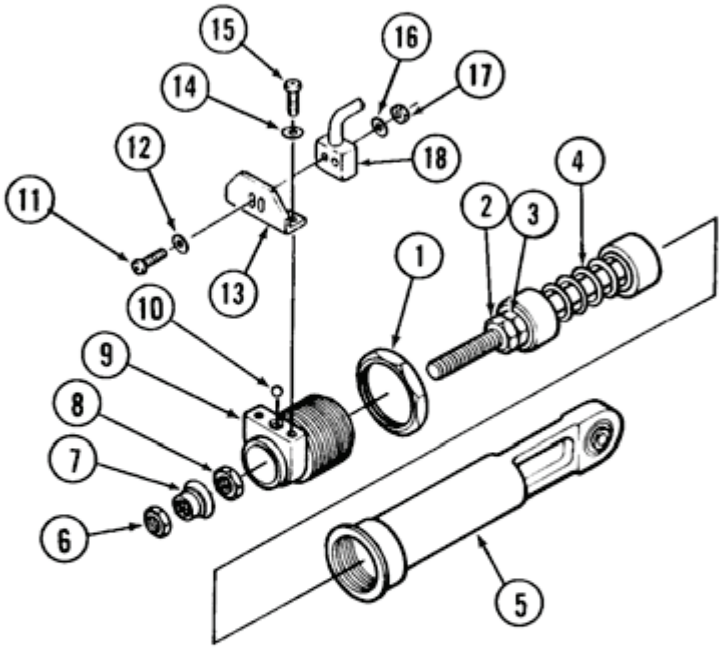
LOCKWIRE (C-405)



DETAIL A

ICN-412-A-671200-B-97499-00021-A-001-01

Figure 1. Fore and Aft Cyclic Force Gradient - Installation (Sheet 1 of 1)



1. Jam nut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Screw
12. Washer
13. Retainer
14. Washer
15. Screw
16. Washer
17. Nut
18. Cable assembly (switch)

ICN-412-A-671200-B-97499-00022-A-001-01

Figure 2. Fore and Aft Cyclic Force Gradient Assembly (Sheet 1 of 1)

Force Gradient (Lateral Cyclic) - Removal

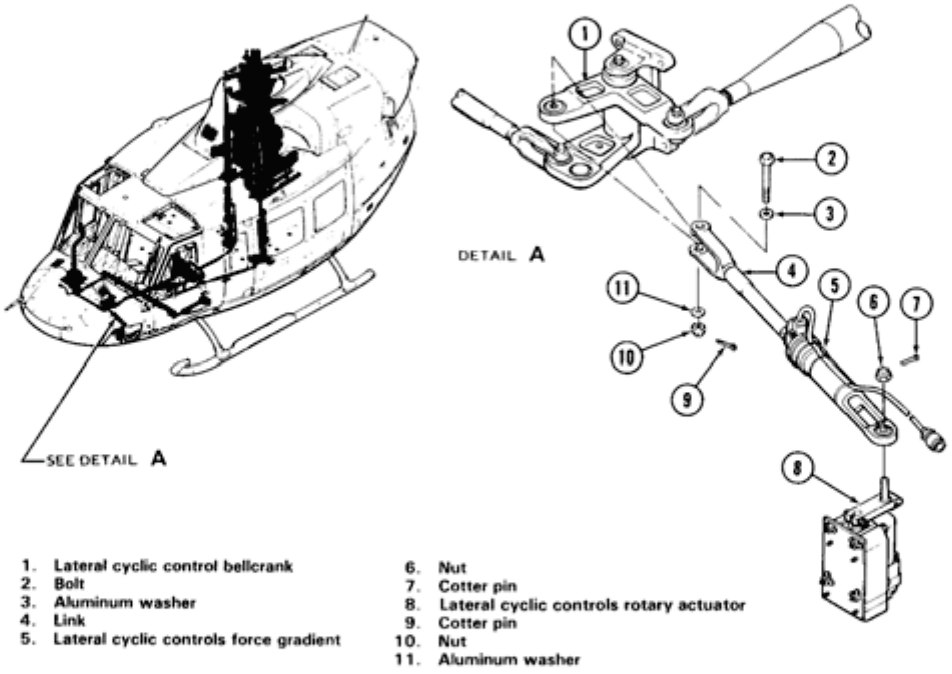
Procedure

1. Remove panel from bottom of helicopter below lateral cyclic control force gradient (5, Figure 1), (412-A-52-40-00-02A-520A-A / 00837).

2. Disconnect electrical connector.

3. Remove cotter pin (7) and nut (6). Separate lateral cyclic control force gradient (5) and lateral cyclic controls rotary actuator (8).

4. Remove cotter pin (9), nut (10), aluminum washers (3 and 11), and bolt (2). Remove lateral cyclic control force gradient (5).



1. Lateral cyclic control bellcrank
2. Bolt
3. Aluminum washer
4. Link
5. Lateral cyclic controls force gradient
6. Nut
7. Cotter pin
8. Lateral cyclic controls rotary actuator
9. Cotter pin
10. Nut
11. Aluminum washer

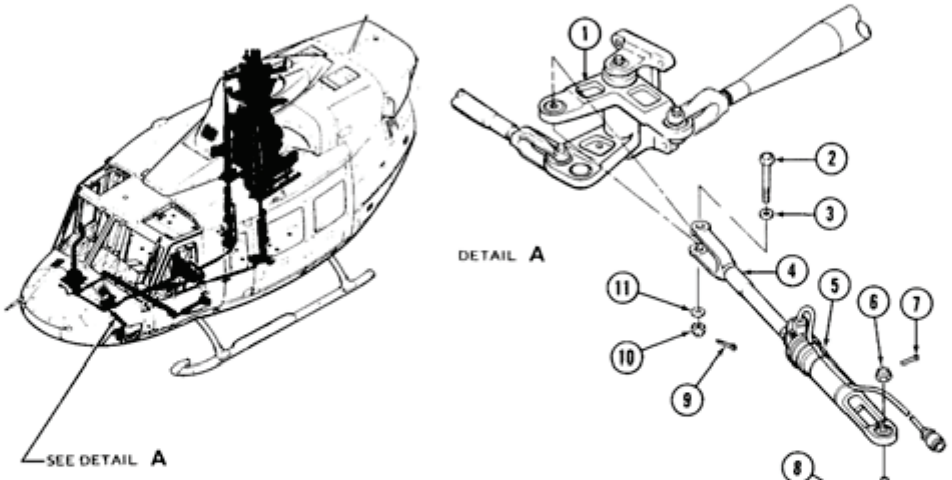
ICN-412-A-671200-B-97499-00023-A-001-01

Figure 1. Lateral Cyclic Force Gradient (Sheet 1 of 1)

Force Gradient (Lateral Cyclic) - Inspection

Procedure

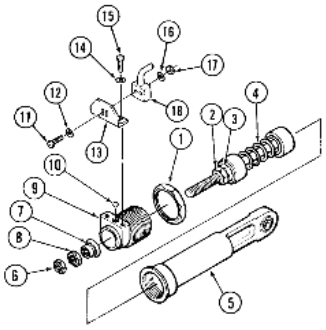
1. Inspect lateral cyclic control force gradient (5, Figure 1) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion is not acceptable.
2. Inspect clevis on link (4) for elongated holes.
3. Inspect bearing for wear and for secure installation in cylinder (5, Figure 2) (BHT-ALL-SPM, Standard Practices Manual).
4. Inspect two nuts (6 and 8) for secure installation. If adjustment of nuts (2 and 3) (spring preload) is suspected of being incorrect, adjust spring preload (412-A-67-12-00-18A-720A-A / 01466).
5. Inspect jam nut (1) and cylinder (5) for correct lockwire installation.
6. Inspect switch and electrical cable assembly (18) for frayed wires and broken pins in connector. Check continuity from connector pins through switch.



1. Lateral cyclic control bellcrank
2. Bolt
3. Aluminum washer
4. Link
5. Lateral cyclic controls force gradient
6. Nut
7. Cotter pin
8. Lateral cyclic controls rotary actuator
9. Cotter pin
10. Nut
11. Aluminum washer

ICN-412-A-671200-B-97499-00023-A-001-01

Figure 1. Lateral Cyclic Force Gradient (Sheet 1 of 1)



1. Jam nut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Screw
12. Washer
13. Retainer
14. Washer
15. Screw
16. Washer
17. Nut
18. Cable assembly (switch)

ICN-412-A-671200-B-97499-00024-A-001-01

Figure 2. Lateral Cyclic Force Gradient Assembly (Sheet 1 of 1)

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Force Gradient (Lateral Cyclic) - Repair

Procedure

1. Repair of lateral cyclic control force gradient is restricted to replacement of faulty parts and adjustment of spring preload.

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Force Gradient (Lateral Cyclic) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

Lateral cyclic controls force gradient (5, Figure 1) in lateral cyclic controls is similar in appearance to force gradient in fore and aft cyclic controls, but the force gradients have different part numbers. Make sure the force gradient with the correct part number is installed in the lateral cyclic control system position (BHT-412-IPB).

1. The instructions in Step 1.1.. are for lateral cyclic controls force gradient (5, Figure 1).

1.1. If not previously accomplished, adjust spring preload of lateral cyclic control force gradient as follows:

1.1.1. Cut and remove lockwire, loosen jam nut (1, Figure 2) and remove cap (9).

1.1.2. Remove spring assembly (4) from cylinder (5).

1.1.3. Adjust inboard nut (3) to provide 3.5 to 4.5 lbs. (15.6 to 20.0 N) preload against spring assembly (4). Tighten outboard nut (2) against inboard nut (3).

1.1.4. Position spring assembly (4) in cylinder (5) and install cap (9) with jam nut (1) until all end play is eliminated. Install jam nut (1) and secure with lockwire (C-405) to cylinder (5).

1.2. If not previously accomplished, adjust actuator (7) on force gradient as follows:

1.2.1. Loosen nuts (6 and 8) on shaft of spring assembly (4).

1.2.2. Ensure retainer (13) is secured by screws (15) and washer (14).

1.2.3. Align actuator (7) on shaft of spring assembly (4) with ball (10) and plunger of cable assembly (18) (switch).

1.2.4. Loosen screws (11), washers (12 and 16) and nut (17) mounting switch (18) to retainer (13).

1.2.5. Push and hold actuator (7) down (180°) against cap (9) at point opposite location of cable assembly (18). Hold this position to initially adjust switch (18) then adjust switch to actuate within ± 0.050 in. (1.27 mm) linear travel of spring assembly (4). Secure switch (18) to retainer (13) with screws (11), washers (12 and 16) and nut (17). Secure actuator (7) with nut (8).

1.2.6. Install clevis (4, Figure 1) on shaft of spring assembly (4, Figure 2), secure with nut (6).

1.3. After assembly of force gradient, prior to installation, verify spring assembly (4) preload of 3.5 to 4.5 pounds (15.57 to 20.02 N).

1.4. Position the lateral cyclic control force gradient (5, Figure 1) on the lateral cyclic control bellcrank (1). Install the bolt (2), washer (3), washer (11), nut (10), and cotter pin (9).



MAKE SURE FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO FINAL INSTALLATION OF LATERAL CYCLIC CONTROL FORCE GRADIENT ON ROTARY ACTUATOR.

1.5. If not previously accomplished, adjust the length of the lateral cyclic control force gradient as outlined in the rigging instructions (412-A-67-01-00-03A-276A-A / 01324).

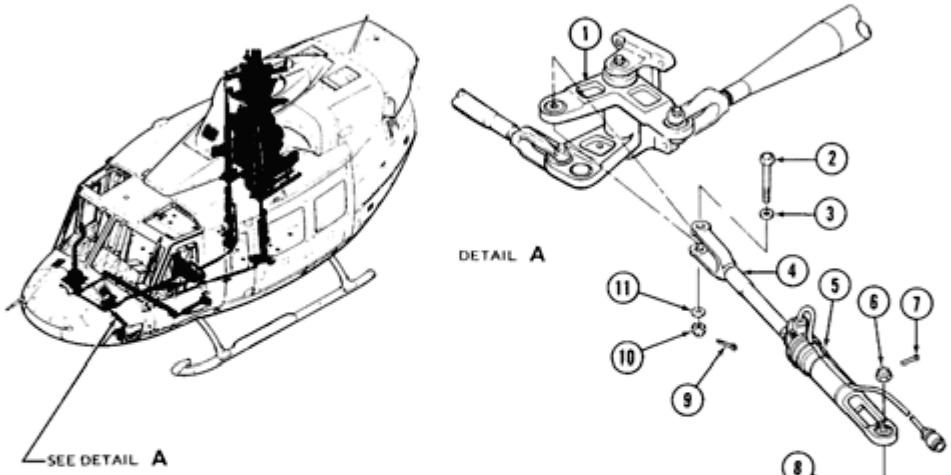
1.6. Position the lateral cyclic control force gradient (5) on the lateral cyclic control rotary actuator (8). Install the nut (6) and the cotter pin (7).

1.7. Connect the electrical connector.

1.8. Move the lateral cyclic controls through full travel and check for binding and/or obstructions.

1.9. Verify the cyclic control rigging is within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).

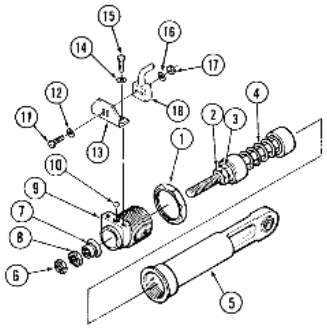
1.10. Install the access panel on the bottom of the helicopter below the force gradient (412-A-52-40-00-02A-720A-A / 00840).



1. Lateral cyclic control bellcrank
2. Bolt
3. Aluminum washer
4. Link
5. Lateral cyclic controls force gradient
6. Nut
7. Cotter pin
8. Lateral cyclic controls rotary actuator
9. Cotter pin
10. Nut
11. Aluminum washer

ICN-412-A-671200-B-97499-00023-A-001-01

Figure 1. Lateral Cyclic Force Gradient (Sheet 1 of 1)



1. Jam nut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Screw
12. Washer
13. Retainer
14. Washer
15. Screw
16. Washer
17. Nut
18. Cable assembly (switch)

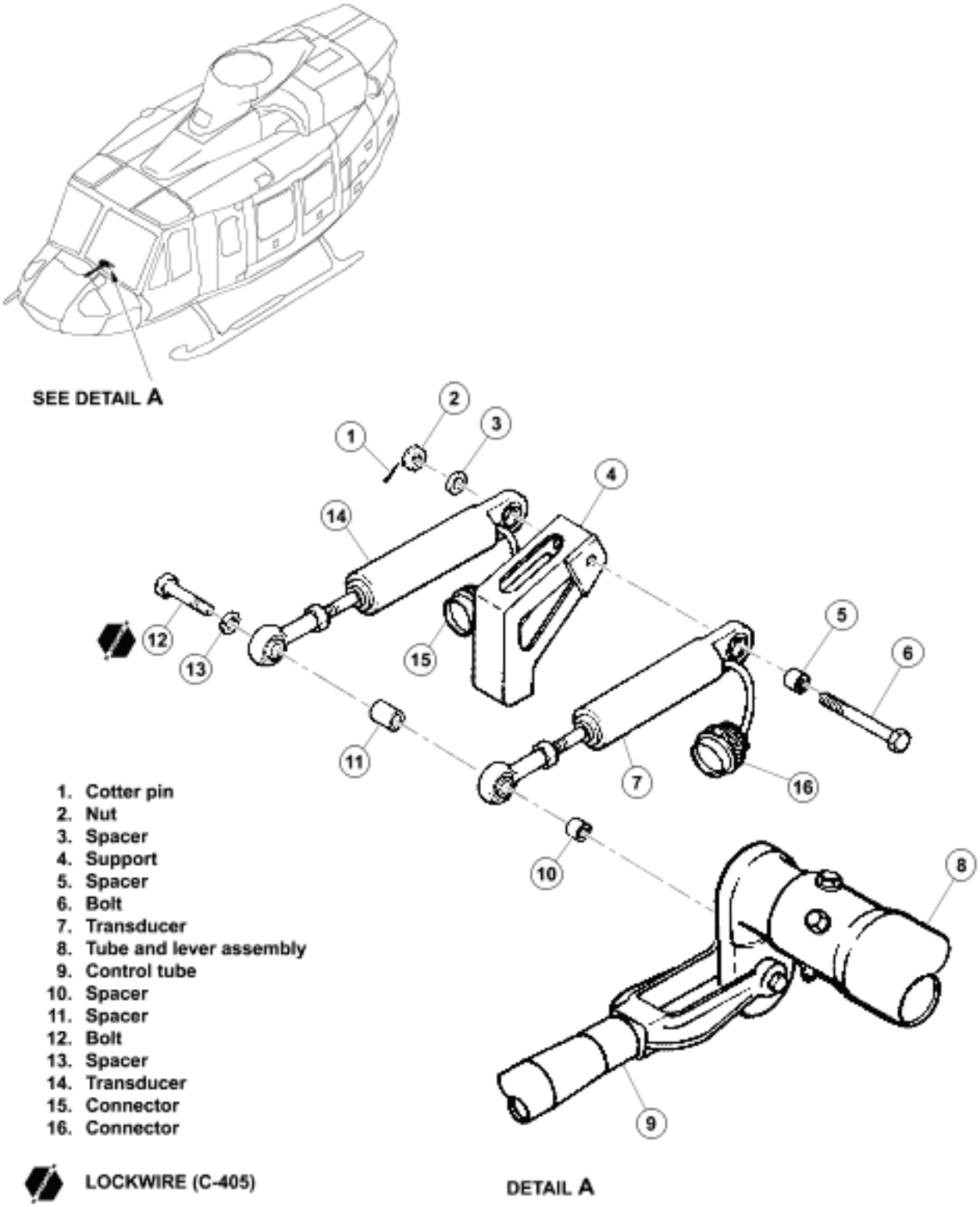
ICN-412-A-671200-B-97499-00024-A-001-01

Figure 2. Lateral Cyclic Force Gradient Assembly (Sheet 1 of 1)

Transducers (Fore and Aft Cyclic) - Removal

Procedure

1. Remove the access panel from the cabin floor under the pilot seat (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect the electrical connectors (15 and 16, Figure 1).
3. Remove the cotter pin (1), nut (2), spacers (3 and 5), and bolt (6) from the support (4) and transducers (7 and 14).
4. Cut and remove the lockwire.
5. Remove the bolt (12) and spacers (10, 11, and 13). Remove the transducer (14) and transducer (7) from the tube and ever assembly (8) and the control tube (9).



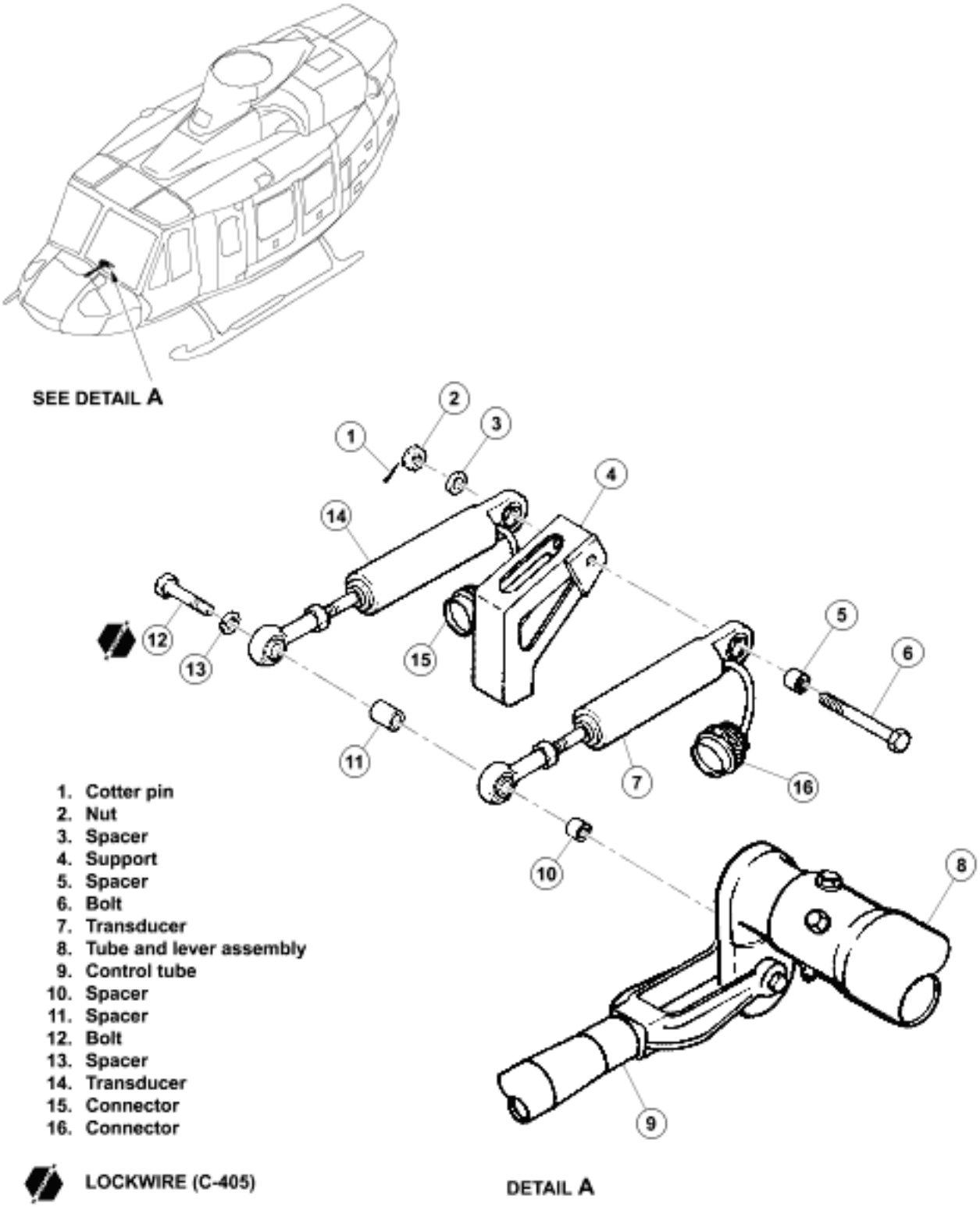
ICN-412-A-671200-B-97499-00025-A-001-01

Figure 1. Fore and Aft Cyclic Transducers (Sheet 1 of 1)

Transducers (Fore and Aft Cyclic) - Inspection

Procedure

1. Inspect the transducers (7 and 14, Figure 1) for damage.
2. Inspect the electrical cable and connectors (15 and 16) for damage.
3. Inspect the bearings for looseness (BHT-ALL-SPM, Standard Practices Manual).



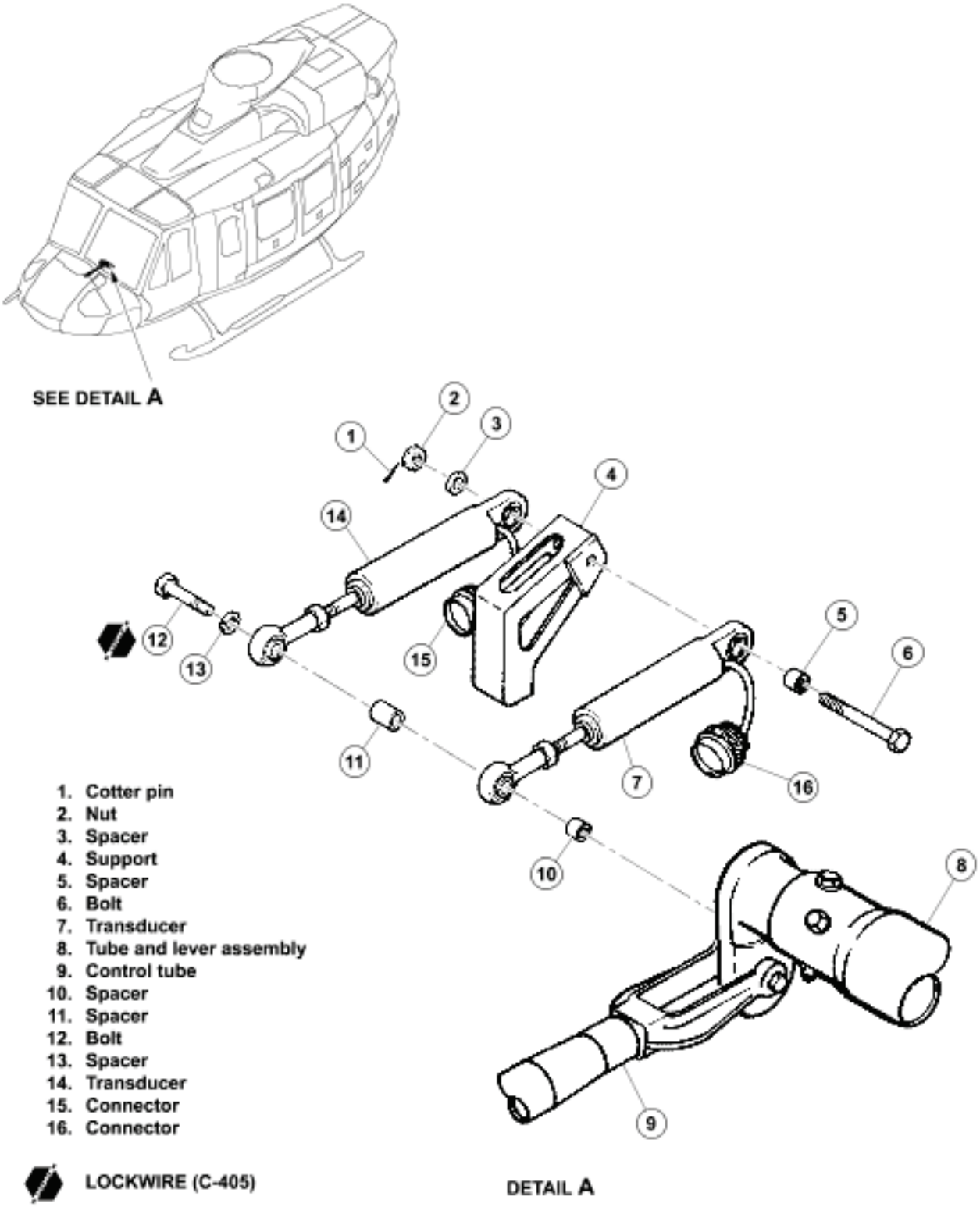
ICN-412-A-671200-B-97499-00025-A-001-01

Figure 1. Fore and Aft Cyclic Transducers (Sheet 1 of 1)

Transducers (Fore and Aft Cyclic) - Repair

Procedure

1. The fore and aft cyclic transducers (7 and 14, Figure 1) are nonrepairable and shall be replaced when defective.



ICN-412-A-671200-B-97499-00025-A-001-01

Figure 1. Fore and Aft Cyclic Transducers (Sheet 1 of 1)

Transducers (Fore and Aft Cyclic) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Position the transducers (7 and 14, Figure 1) on the support (4) and install the bolt (6), spacers (5 and 3), nut (2). Tighten the nut.

2. Safety the nut (2) with a new cotter pin (1).

3. Install bolt (12), spacers (10, 11, and 13), and transducers (7 and 14) on the tube and lever assembly (8). Tighten the bolt.

4. Secure the bolt (12) with lockwire (C-405).

NOTE

SN: 33001-33213 or 36001-36036:

If not previously accomplished, perform transducer adjustment. For helicopters with transducer P/N 412-074-101-101 installed, refer to 412-A-67-01-00-03A-276A-A / 01324.

NOTE

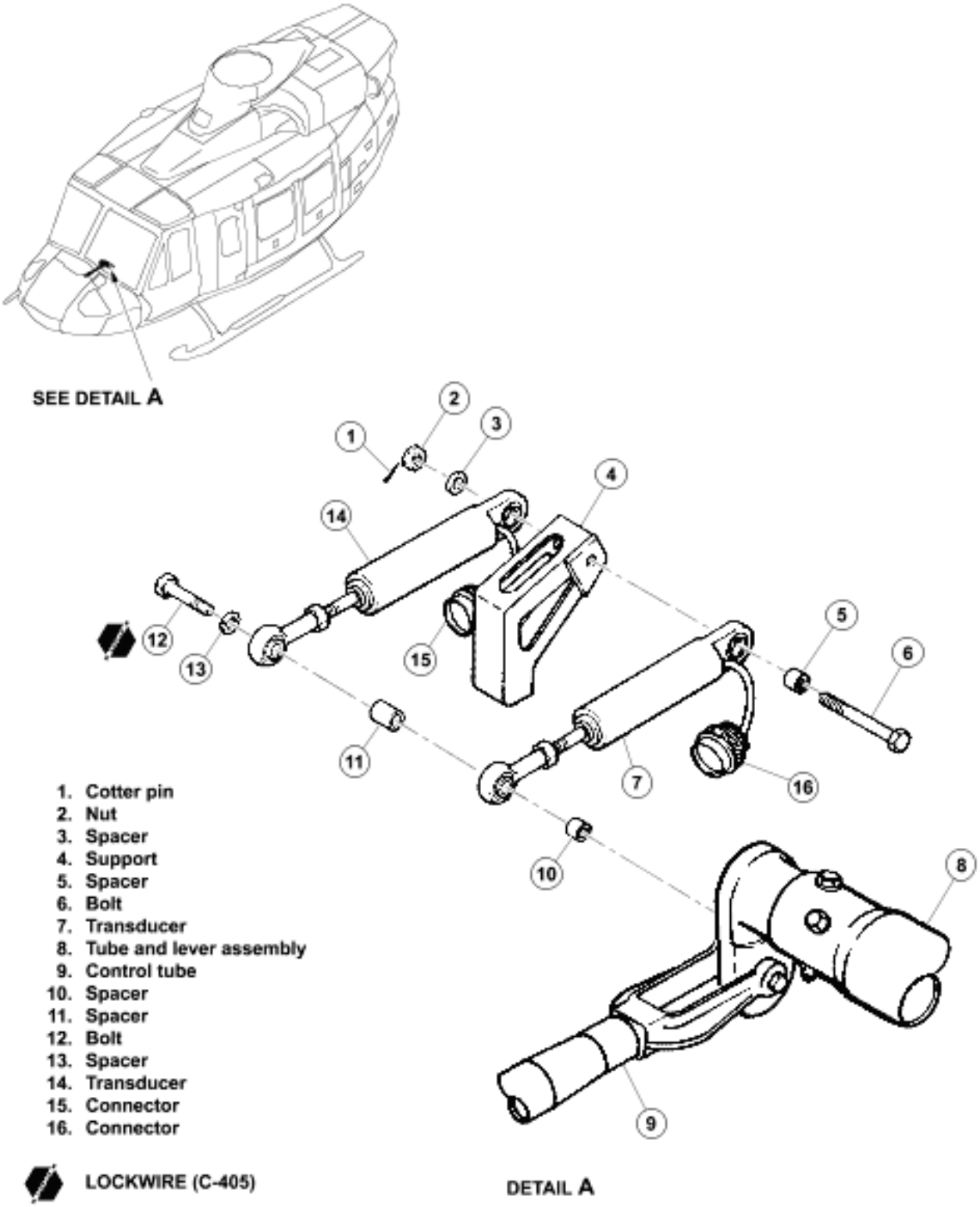
SN: 36037-Subsequent:

For helicopters with transducer P/N 412-074-101-103 installed, refer to 412-A-67-01-00-10A-276A-A / 01550.

5. Connect the electrical connectors (15 and 16).

6. Move the controls through full travel and check for binding and obstructions.

7. Install the access panel on the cabin floor under the pilot seat (412-A-52-40-00-02A-720A-A / 00840).



ICN-412-A-671200-B-97499-00025-A-001-01

Figure 1. Fore and Aft Cyclic Transducers (Sheet 1 of 1)

Jackshaft (Fore and Aft) Tube and Lever Assembly (Right Side) - Removal

Procedure

1. Remove access panel from cabin floor under pilot seat. Remove access panel from cabin floor at aft end of console. Remove access panel from left side of console. Remove oval access panel from right side of helicopter below pilot door (412-A-52-40-00-02A-520A-A / 00837).

2. For removal of control motion transducers refer to 412-A-67-12-00-19A-520A-A / 01467, 412-A-67-12-00-19A-280A-A / 01468, 412-A-67-12-00-19A-685A-A / 01469 and 412-A-67-12-00-19A-720A-A / 01470.

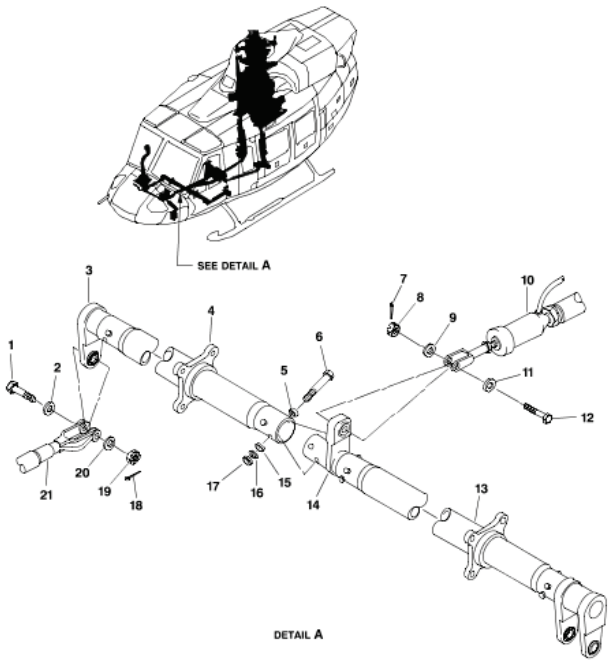
3. Remove cotter pin (18, Figure 1), nut (19), washers (20 and 2) and bolt (1) from tube and lever assembly (right) (3) and flight control tube (21).

4. Remove cotter pin (7), nut (8), washers (9 and 11) and bolt (12) from series actuator (10) and arm assembly (14).

5. Remove two nuts (17), two washers (16) two bolts (6), bushings (5 and 15). Index bushings for reinstallation in original location.

6. Tap on tube and lever assembly (3) with a plastic mallet to loosen inboard joint and to slide tube and lever assembly (3) outboard.

7. Carefully work tube and lever assembly (3) outboard to disengage arm assembly (14) and housing (4). Remove tube and lever assembly (3) from helicopter.



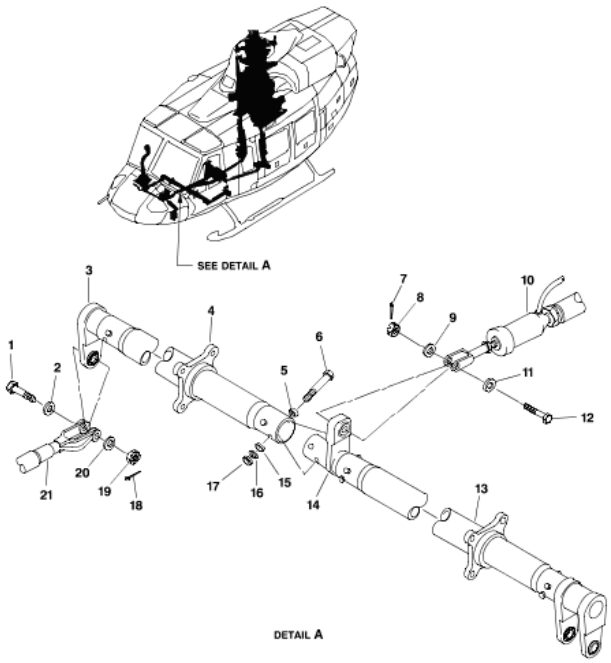
1. Bolt
2. Aluminum washer
3. Tube and lever assembly (right)
4. Housing
5. Tapered bushing
6. Bolt
7. Cotter pin
8. Nut
9. Aluminum washer
10. Series actuator
11. Aluminum washer
12. Bolt
13. Tube and lever assembly (left)
14. Arm assembly
15. Tapered bushing
16. Steel washer
17. Nut
18. Cotter pin
19. Nut
20. Aluminum washer
21. Flight control tube

ICN-412-A-671200-B-97499-00026-A-001-01

Figure 1. Fore and Aft Cyclic Jackshaft Tube and Lever Assembly (Right Side) (Sheet 1 of 1)

Jackshaft (Fore and Aft) Tube and Lever Assembly (Right Side) - Inspection Procedure

1. Inspect tube and lever assembly (3, Figure 1) for mechanical and corrosion damage.
2. Inspect bolt holes for elongation.
3. Inspect bearing for secure installation in tube and for wear (BHT-ALL-SPM, Standard Practices Manual).



1. Bolt
2. Aluminum washer
3. Tube and lever assembly (right)
4. Housing
5. Tapered bushing
6. Bolt
7. Cotter pin
8. Nut
9. Aluminum washer
10. Series actuator
11. Aluminum washer
12. Bolt
13. Tube and lever assembly (left)
14. Arm assembly
15. Tapered bushing
16. Steel washer
17. Nut
18. Cotter pin
19. Nut
20. Aluminum washer
21. Flight control tube

ICN-412-A-671200-B-97499-00026-A-001-01

Figure 1. Fore and Aft Cyclic Jackshaft Tube and Lever Assembly (Right Side) (Sheet 1 of 1)

Jackshaft (Fore and Aft) Tube and Lever Assembly (Right Side) - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Reamer	T27872-2	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Epoxy polyamide primer	C-204	AR	

Procedure

NOTE

Inspect tube and lever (3, Figure 1) to determine whether holes for bolts (6) have been drilled. If holes have been drilled and reamed, proceed to Step 14.. If tube and lever (3) has not been drilled and reamed, accomplish Step 1.. through Step 13..

1. Measure and record gap between arm assembly (1, Figure 2) and tube and lever assembly (3).

2. Remove arm assembly (1) from helicopter to use as an aid in drilling new tube and lever assembly (412-A-67-12-00-23A-520A-A / 01480).

3. Fit arm assembly (1) into tube and lever assembly (4) and establish dimension obtained in Step 1.. Set angle between arm assembly (1) and tube and lever assembly (3).

4. Holding lateral and angular dimensions, mark and center punch top holes in tube and lever assembly (4).

5. Drill through top surface of tube and lever assembly (4) using a No. 20 drill (0.161 in.) as a pilot. Insert drill through top of tube and lever assembly (4) and drill bottom hole using a No. 20 drill (0.161 in.).

6. Turn tube and lever assembly (4) and arm assembly over, keeping everything aligned and redrill bottom hole with an F drill (0.257 in.).

7. Inspect hole in top of tube and lever assembly (4). Verify hole is centered in existing hole in arm assembly (1). If hole is not centered, use a small grinder or a router bit in a drill motor to center hole. Redrill top hole with an F drill (0.257 in.).

8. Insert reamer (T27872-2) through hole so pilot projects through far side and ream hole to a smooth finish and until tapered bushings extend 0.030 to 0.060 in. (0.762 to 1.524 mm) (Figure 2).

9. Turn arm assembly (1) and tube and lever assembly (4) over. Place one tapered bushing in hole already reamed and insert reamer (T27872-2) so pilot projects into bushing. Ream hole to a smooth finish. Remove all burrs.

10. Repeat Step 4.. through Step 11.. to drill and ream holes for remaining bolt.

11. Temporarily install bolts (6, Figure 1), tapered bushings (5 and 15), washers (16), and nuts (17). Ensure bushings protrude 0.030 to 0.060 in. (0.762 to 1.524 mm) out of tube and lever assembly (3). Remove and retain bolts (6), washers (16), nuts (17), and bushings (5 and 15).

12. Place arm assembly (14) in position and mount to tube and lever assembly (13), being sure washer bonded to arm assembly (14) is on left side.

13. Apply epoxy polyamide primer (C-204) to end of arm assembly (14) and tube and lever assembly (3) being sure to coat all holes and bare surfaces.

14. Install tube and lever assembly (3) in helicopter, guiding it through housing (4) and aligning holes in tube and lever assembly (3) and arm assembly (14).

NOTE

After final installation of hardware, bushings (5 and 15) must protrude from jackshaft tube within the 0.030 to 0.060 in. (0762 to 1.524 mm) limits shown in Figure 2 view A-A.

15. Install bolts (6), bushings (5 and 15), washers (16), and nuts (17).

16. Place tube (21, Figure 1) onto tube and lever assembly (3) and install bolt (1), washers (2 and 20), nut (19), and cotter pin (18).

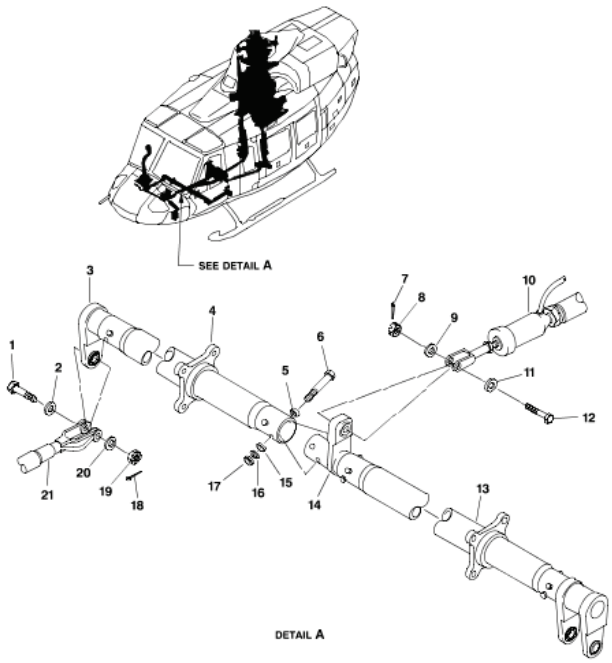
17. Place series actuator (10) on arm assembly (14) and install bolt (12), washers (11 and 9), nut (8), and cotter pin (7).

18. Move cyclic controls through full travel and check for interference and binding.

19. For installation of control motion transducers refer to 412-A-67-12-00-19A-520A-A / 01467, 412-A-67-12-00-19A-280A-A / 01468, 412-A-67-12-00-19A-685A-A / 01469 and 412-A-67-12-00-19A-720A-A / 01470.

20. Verify cyclic rigging within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).

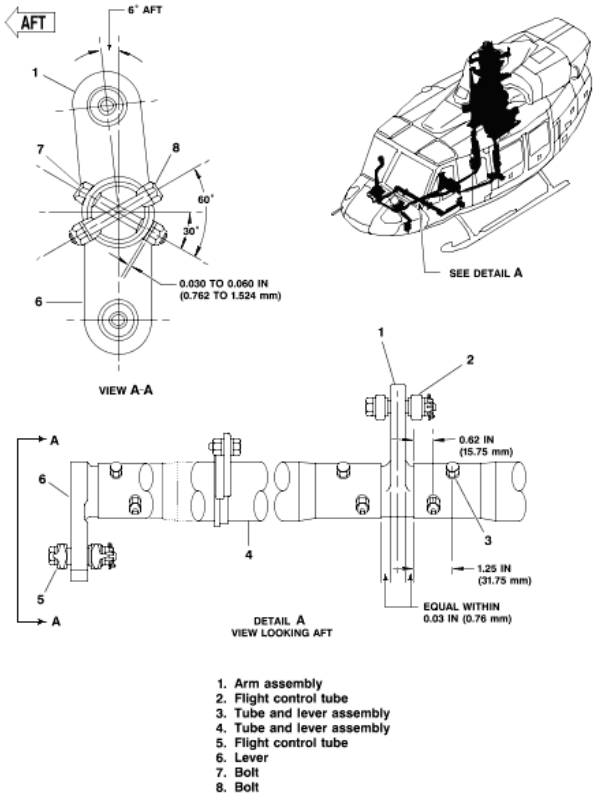
21. Install access panel on cabin floor under pilot seat. Install access panel on cabin floor aft of console. Install access panel on left side of console. Install oval access panel on right side of helicopter below pilot door (412-A-52-40-00-02A-720A-A / 00840).



1. Bolt
2. Aluminum washer
3. Tube and lever assembly (right)
4. Housing
5. Tapered bushing
6. Bolt
7. Cotter pin
8. Nut
9. Aluminum washer
10. Series actuator
11. Aluminum washer
12. Bolt
13. Tube and lever assembly (left)
14. Arm assembly
15. Tapered bushing
16. Steel washer
17. Nut
18. Cotter pin
19. Nut
20. Aluminum washer
21. Flight control tube

ICN-412-A-671200-B-97499-00026-A-001-01

Figure 1. Fore and Aft Cyclic Jackshaft Tube and Lever Assembly (Right Side) (Sheet 1 of 1)



1. Arm assembly
2. Flight control tube
3. Tube and lever assembly
4. Tube and lever assembly
5. Flight control tube
6. Lever
7. Bolt
8. Bolt

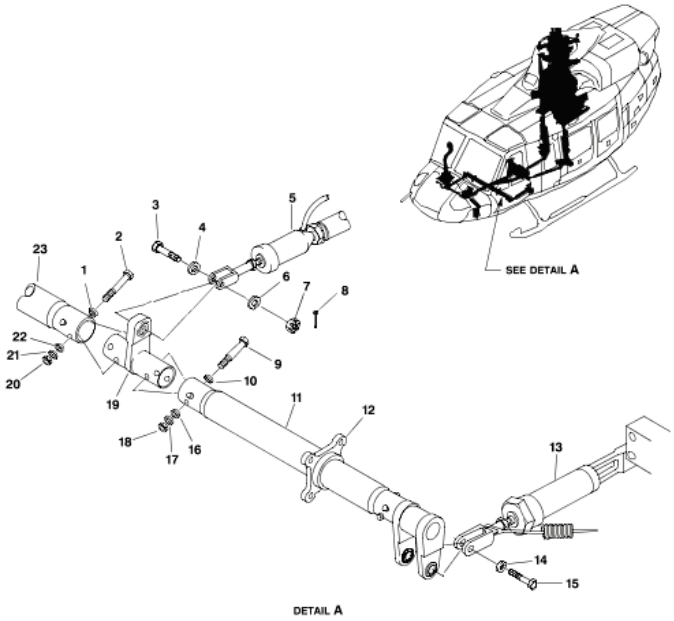
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Figure 2. Fore and Aft Cyclic Jackshaft Tube and Lever Assembly (Right Side) Alignment (Sheet 1 of 1)

Jackshaft (Fore and Aft) Tube and Lever Assembly (Left Side) - Removal

Procedure

1. Remove access panel from cabin floor under copilots seat, access panel from under aft end of console and oval access panel from lower left side of helicopter under left crew door (412-A-52-40-00-02A-520A-A / 00837).
2. Remove lockwire, bolt (15, Figure 1) and washers (14) from force gradient (13).
3. Remove cotter pin (8), nut (7), washers (4 and 6) and bolt (3) from series actuator (5) and arm (19).
4. Remove bolts (9), tapered bushings (10 and 16), washers (17), and nuts (18). Index bushings (10 and 16) for reinstallation in same location.
5. If copilot cyclic controls are installed, disconnect control tube (not illustrated) from tube and lever assembly (11).
6. Tap tube and lever assembly (11) with a plastic mallet to loosen it from arm assembly (19). Carefully work tube and lever assembly outboard and out of helicopter.



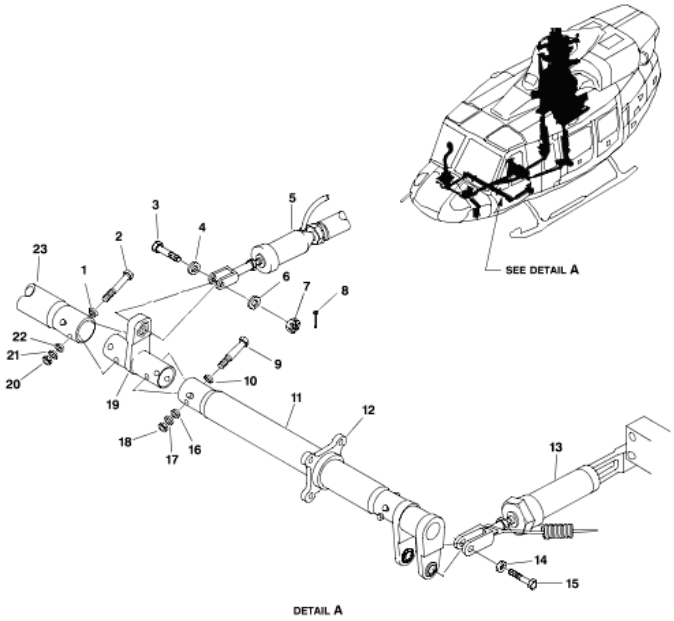
1. Tapered bushing
2. Bolt
3. Bolt
4. Aluminum washer
5. Series actuator
6. Aluminum washer
7. Nut
8. Cotter pin
9. Bolt
10. Tapered bushing
11. Tube and lever assembly (left)
12. Housing
13. Force gradient
14. Aluminum washer
15. Bolt
16. Tapered bushing
17. Washer
18. Nut
19. Arm
20. Nut
21. Washer
22. Tapered bushing
23. Tube and lever assembly (right)

ICN-412-A-671200-B-97499-00028-A-001-01

Figure 1. Fore and Aft Cyclic Jackshaft Tube and Lever Assembly (Left Side) (Sheet 1 of 1)

Jackshaft (Fore and Aft) Tube and Lever Assembly (Left Side) - Inspection Procedure

1. Inspect tube and lever assembly (11, Figure 1) for mechanical and corrosion damage.
2. Inspect bearings for freedom of movement and excessive wear.



1. Tapered bushing
2. Bolt
3. Bolt
4. Aluminum washer
5. Series actuator
6. Aluminum washer
7. Nut
8. Cotter pin
9. Bolt
10. Tapered bushing
11. Tube and lever assembly (left)
12. Housing
13. Force gradient
14. Aluminum washer
15. Bolt
16. Tapered bushing
17. Washer
18. Nut
19. Arm
20. Nut
21. Washer
22. Tapered bushing
23. Tube and lever assembly (right)

ICN-412-A-671200-B-97499-00028-A-001-01

Figure 1. Fore and Aft Cyclic Jackshaft Tube and Lever Assembly (Left Side) (Sheet 1 of 1)

Jackshaft (Fore and Aft) Tube and Lever Assembly (Left Side) - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Reamer	T27872-2	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Epoxy polyamide primer	C-204	AR	
Lockwire	C-405	AR	

Procedure

NOTE

Inspect tube and lever assembly (11, Figure 1) to determine whether holes for two bolts (9) have been drilled. If holes have been drilled and reamed, proceed to Step 14.. If tube and lever assembly (11) has not been drilled and reamed, accomplish Step 1.. through Step 13..

1. Measure distance between inboard end of tube and lever assembly (8, Figure 2) and arm assembly (1).

2. Remove arm assembly (1) from helicopter (412-A-67-12-00-23A-520A-A / 01480).

3. Fit arm assembly (1) into end of new tube and lever assembly (8). Set distance between inboard end of tube and lever assembly (8) and arm assembly (1) to correspond with dimension obtained in Step 1..

4. Set arm assembly (1) at angle shown. Holding angular and lateral dimension, measure holes for bolts (9 and 10) to pick up existing holes in arm assembly. Center punch top holes for bolts (9 and 10).

5. Drill a pilot hole through top surface of tube and lever assembly (8) with a No. 20 drill (0.161 in.) for bolts (9 and 10). Drill holes in bottom of tube and lever assembly (8) using a No. 20 drill (0.161 in.).

6. Turn over tube and lever assembly (8) and arm assembly (1) and redrill the holes in bottom of tube and lever assembly (8) using an F drill (0.257 in.).

7. Verify holes in top of tube and lever assembly (8) are centered with existing holes in arm assembly (1). If not centered, use a small hand grinder or a router bit to center holes. Redrill holes with an F drill (0.257 in.).

8. Insert reamer (T27872-2) so end of pilot projects through opposite side of tube and lever assembly (8).

9. Ream one side to a smooth finish. Remove all burrs. Install a tapered bushing and insert reamer (T27872-2) so end of pilot projects into tapered bushing.

10. Ream opposite side to a smooth finish. Remove all burrs.

11. Repeat Step 4. through Step 10. to drill and ream holes for remaining bolts.

12. Temporarily install bolt (9, Figure 1), tapered bushings (10 and 16), washer (17) and nut (18). Check tapered bushings (10 and 16) protrude 0.030 to 0.060 in. (0.762 to 1.524 mm). Remove bolts (9), washers (17), nuts (18), and bushings (10 and 16).

13. Coat inboard end of tube and lever assembly (23) and arm assembly (19) with epoxy polyamide primer (C-204). Insert arm assembly (19) into tube and lever assembly (23) with washer on left side of arm. Install bolts (2), bushings (1 and 22), washers (21), and nuts (20).

14. Coat inside of inboard end of tube and lever assembly with epoxy polyamide primer (C-204). Carefully guide tube and lever assembly (11) into helicopter through housing (12) and onto end of arm assembly (19).

NOTE

After final installation of hardware, bushings (10 and 16) must protrude from jackshaft tube within the 0.030 to 0.060 in. (0.762 to 1.524 mm) limits shown in Figure 3 view A-A.

15. Install bolts (9, Figure 1), bushings (10 and 16), washers (17), and nuts (18).

16. Install series actuator (5) on arm assembly (19) and secure with bolt (3), washers (4 and 6), nut (7), and cotter pin (8).

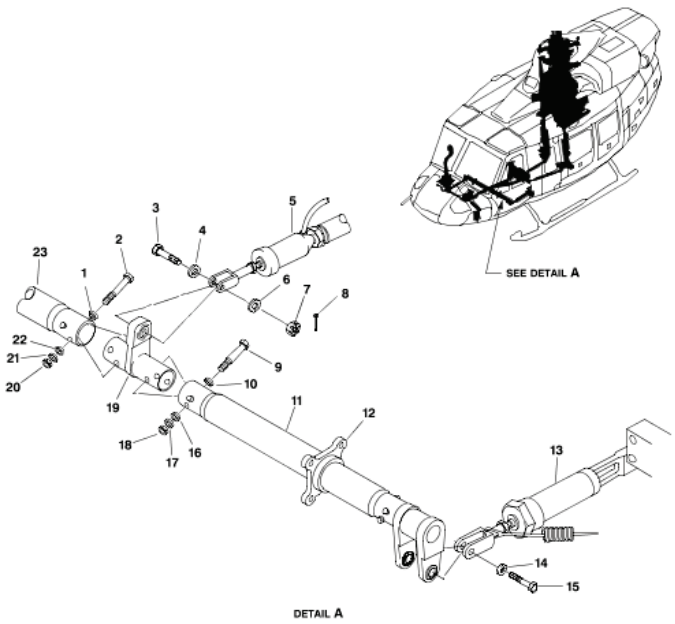
17. Place clevis of force gradient (13) on tube and lever assembly (11), install bolt (15) and washer (14). Secure bolt (15) with lockwire (C-405).

18. Attach control tube (not illustrated) from copilot cyclic controls to tube and lever assembly (11) arm adjacent to force gradient (13).

19. Move cyclic controls through full travel and check for interference and binding.

20. Verify cyclic rigging within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).

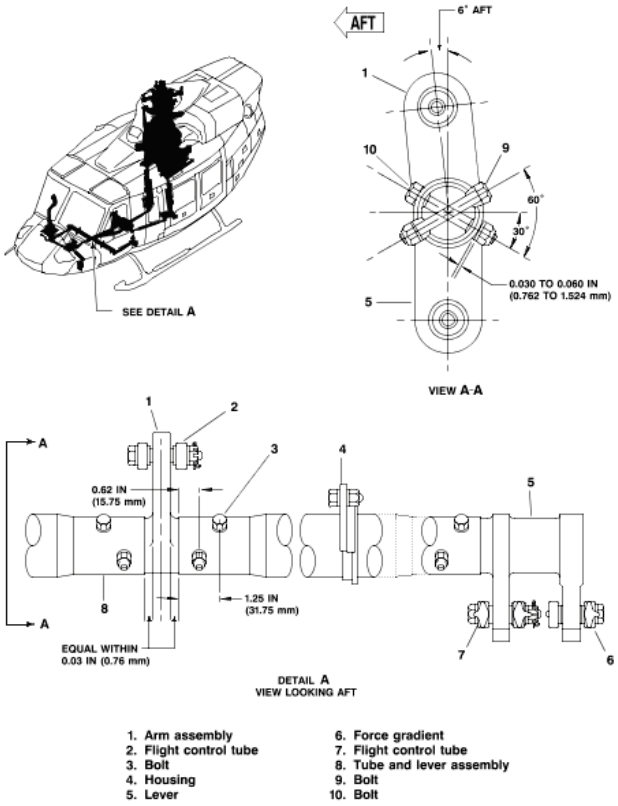
21. Install access panel from cabin floor under copilots seat, access panel from under aft end of console and oval access panel from lower left side of helicopter under left crew door (412-A-52-40-00-02A-720A-A / 00840).



1. Tapered bushing
2. Bolt
3. Bolt
4. Aluminum washer
5. Series actuator
6. Aluminum washer
7. Nut
8. Cotter pin
9. Bolt
10. Tapered bushing
11. Tube and lever assembly (left)
12. Housing
13. Force gradient
14. Aluminum washer
15. Bolt
16. Tapered bushing
17. Washer
18. Nut
19. Arm
20. Nut
21. Washer
22. Tapered bushing
23. Tube and lever assembly (right)

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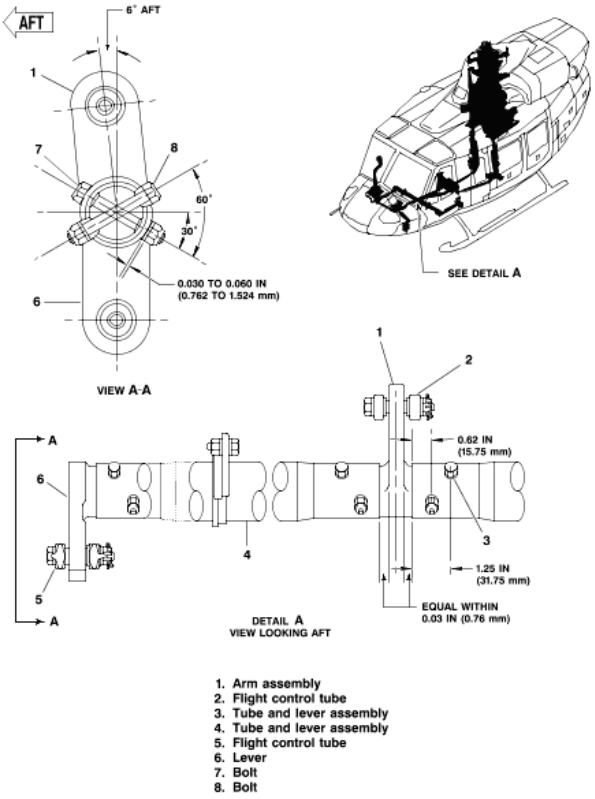
Figure 1. Fore and Aft Cyclic Jackshaft Tube and Lever Assembly (Left Side) (Sheet 1 of 1)



1. Arm assembly
2. Flight control tube
3. Bolt
4. Housing
5. Lever
6. Force gradient
7. Flight control tube
8. Tube and lever assembly
9. Bolt
10. Bolt

ICN-412-A-671200-B-97499-00029-A-001-01

Figure 2. Fore and Aft Cyclic Jackshaft Tube and Lever Assembly (Left Side) Alignment (Sheet 1 of 1)



1. Arm assembly
2. Flight control tube
3. Tube and lever assembly
4. Tube and lever assembly
5. Flight control tube
6. Lever
7. Bolt
8. Bolt

ICN-412-A-671200-B-97499-00027-A-001-01

Figure 3. Fore and Aft Cyclic Jackshaft Tube and Lever Assembly (Right Side) Alignment (Sheet 1 of 1)

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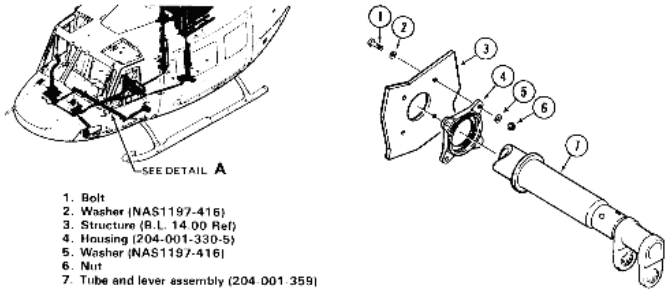
Jackshaft Housing - Removal

Procedure

1. Remove access panel from cabin floor under pilot seat. Remove access panel from cabin floor at aft end of console. Remove access panel from left side of console. Remove oval access panel from right side of helicopter below pilot door. Remove access panel from cabin floor under copilots seat, access panel from under aft end of console and oval access panel from lower left side of helicopter under left crew door (412-A-52-40-00-02A-520A-A / 00837).

2. Remove left tube and lever assembly (7, Figure 1) (412-A-67-12-00-21A-520A-A / 01474).

3. Remove nuts (6), washers (2 and 5) and bolts (1). Remove housing (4) from structure (3). Remove housing (4) from tube and lever assembly (7).



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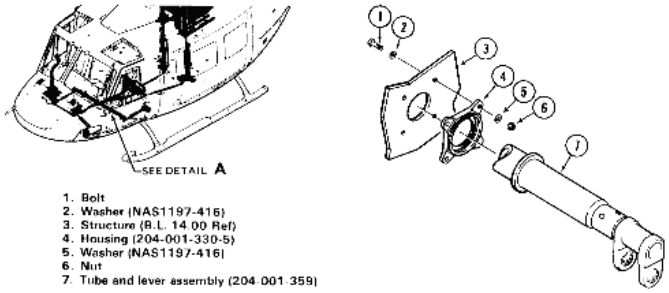
Figure 1. Cyclic Jackshaft Housing (Sheet 1 of 1)

Jackshaft Housing - Inspection

Procedure

1. Inspect housing (4, Figure 1) for mechanical and corrosion damage.

2. Inspect bearing for secure installation in housing and for wear (BHT-ALL-SPM, Standard Practices Manual).



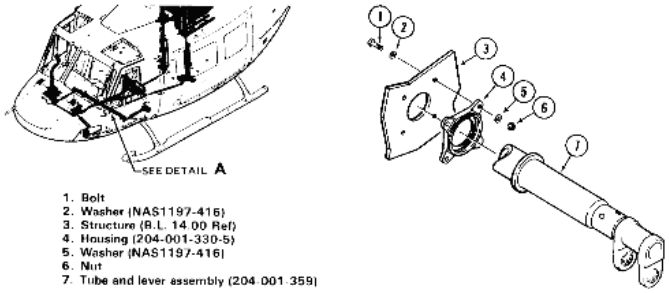
ICN-412-A-671200-B-97499-00030-A-001-01

Figure 1. Cyclic Jackshaft Housing (Sheet 1 of 1)

Jackshaft Housing - Installation

Procedure

1. Place housing (4, Figure 1) on structure (3) and install bolts (1), washers (2 and 5), and nuts (6).
2. Install tube and lever assembly (7) (412-A-67-12-00-21A-720A-A / 01476).
3. Move cyclic controls through full travel and check for interference and binding.
4. Verify cyclic rigging within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).
5. Install access panel from cabin floor under pilot seat. Install access panel from cabin floor at aft end of console. Install access panel from left side of console. Install oval access panel from right side of helicopter below pilot door. Install access panel from cabin floor under copilots seat, access panel from under aft end of console and oval access panel from lower left side of helicopter under left crew door (412-A-52-40-00-02A-720A-A / 00840).



ICN-412-A-671200-B-97499-00030-A-001-01

Figure 1. Cyclic Jackshaft Housing (Sheet 1 of 1)

Arm Assembly - Removal

Procedure

1. Remove access panel from cabin floor under pilot seat. Remove access panel from left side of pedestal (412-A-52-40-00-02A-520A-A / 00837).

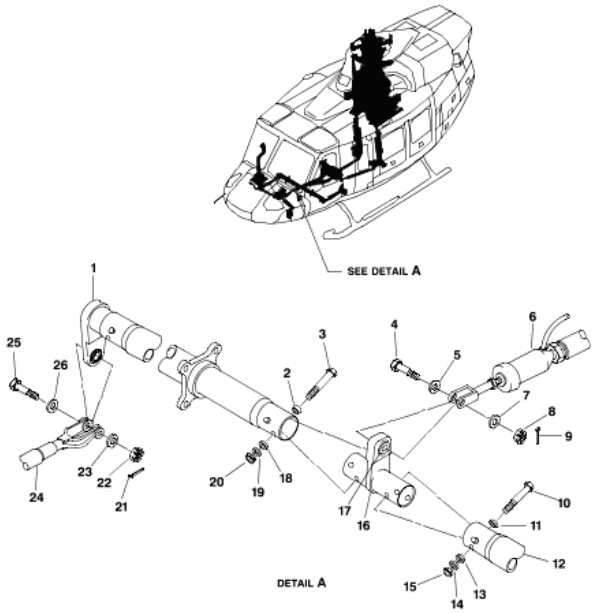
2. Remove cotter pin (21, Figure 1), nut (22), washers (23 and 26) and bolt (25). Remove flight control tube (24) from tube and lever assembly (1).

3. Remove cotter pin (9), nut (8), washers (5 and 7) and bolt (4). Remove series actuator (6) from arm assembly (16).

4. Remove bolts (10), tapered bushings (11 and 13), washers (14) and nuts (15). Index tapered bushings (11 and 13) for reinstallation in same location.

5. Remove nuts (20), tapered bushings (2 and 18), washers (19) and bolts (3). Index tapered bushings (2 and 18) for reinstallation in same location.

6. Tap tube and lever assembly (1) with a plastic mallet to slide out of arm assembly (16). Tap arm assembly (16) with a plastic mallet to disengage it from tube and lever assembly (12). Remove arm assembly.



1. Tube and lever assembly
2. Tapered bushing
3. Bolt
4. Bolt
5. Aluminum washer
6. Series actuator
7. Aluminum washer
8. Nut
9. Cotter pin
10. Bolt
11. Tapered bushing
12. Tube and lever assembly
13. Tapered bushing
14. Washer
15. Nut
16. Arm assembly
17. Washer
18. Tapered bushing
19. Washer
20. Nut
21. Cotter pin
22. Nut
23. Aluminum washer
24. Flight control tube
25. Bolt
26. Aluminum washer

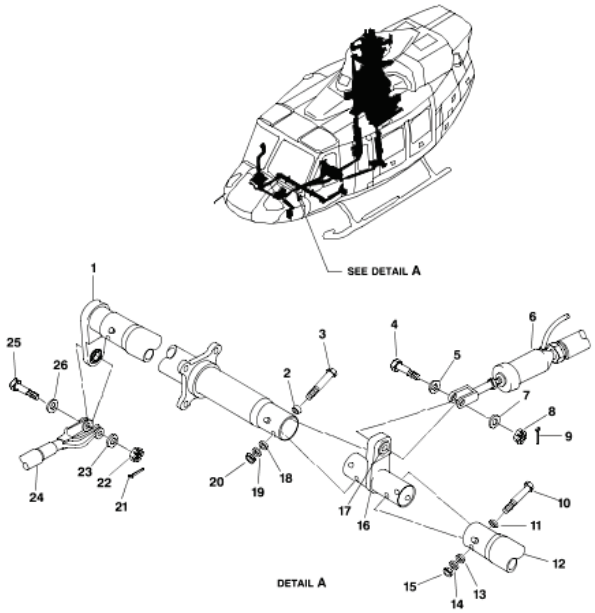
ICN-412-A-671200-B-97499-00031-A-001-01

Figure 1. Cyclic Controls Arm (Sheet 1 of 1)

Arm Assembly - Inspection

Procedure

1. Inspect arm assembly (16, Figure 1) for mechanical and corrosion damage (BHT-412-CR&O).
2. Inspect washer (17) for secure installation on arm (16).
3. Inspect bearing in arm assembly (16) for secure installation in arm and for wear (BHT-ALL-SPM, Standard Practices Manual).



1. Tube and lever assembly
2. Tapered bushing
3. Bolt
4. Bolt
5. Aluminum washer
6. Series actuator
7. Aluminum washer
8. Nut
9. Cotter pin
10. Bolt
11. Tapered bushing
12. Tube and lever assembly
13. Tapered bushing
14. Washer
15. Nut
16. Arm assembly
17. Washer
18. Tapered bushing
19. Washer
20. Nut
21. Cotter pin
22. Nut
23. Aluminum washer
24. Flight control tube
25. Bolt
26. Aluminum washer

ICN-412-A-671200-B-97499-00031-A-001-01

Figure 1. Cyclic Controls Arm (Sheet 1 of 1)

Arm Assembly - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Reamer	T27872-2	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Epoxy polyamide primer	C-204	AR	

Procedure

NOTE

If arm assembly is being replaced, tube and lever assemblies must be used as a jig to drill and ream holes for tapered bushings.

1. Remove tube and lever assembly (1, Figure 1) (412-A-67-12-00-20A-520A-A / 01471).

2. Remove tube and lever assembly (12) (412-A-67-12-00-21A-520A-A / 01474).

3. Fit new arm assembly (1, Figure 2) into tube and lever assemblies (5 and 8) with washer (2) on left side.

4. With levers (3 and 7) vertical and arm assembly (1) 6° aft, establish lateral dimensions as shown. Clamp all components in place.

5. Center punch top bolt holes in center of existing holes in tube and lever assemblies (5 and 8).

6. Drill a pilot hole through top surface of tube and lever assembly (3) with a No. 20 drill (0.161 in.) for bolts (9 and 10). Drill holes in bottom of tube and lever assembly (3) using a No. 20 drill (0.161 in.).

7. Turn over tube and lever assembly (3) and arm assembly (1) and redrill the holes in bottom of tube and lever assembly (3) using an F drill (0.257 in.).

8. Verify holes in top of tube and lever assembly (3) are centered with existing holes in arm assembly (1). If not centered, use a small hand grinder or a router bit to center holes. Redrill holes with an F drill (0.257 in.).

9. Insert reamer (T27872-2) so end of pilot projects through opposite side of tube and lever assembly (5).

10. Ream four top bolt holes so tapered bushings extend 0.030 to 0.060 in. (0.762 to 1.524 mm) as illustrated.

11. Insert a tapered bushing into top holes and insert reamer (T27872-2) so pilot projects into bushing. Ream the four bottom holes to same dimension described in preceding step.

12. Fit tube and lever assembly (1) and tube and lever assembly (12) into arm assembly (16). Install bolts (10), bushings (11), bushings (13), washers (14), and nuts (15).

13. Fit flight control tube (24) on tube and lever assembly (1). Install bolt (25), washers (23 and 26), nut (22), and cotter pin (21).

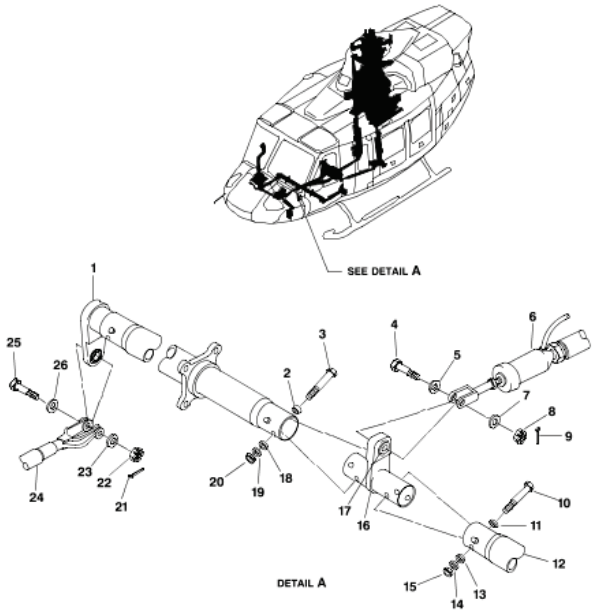
14. Install bolts (3), bushings (2), bushings (18), washers (19), and nuts (20).

15. Install series actuator (6) on arm and washer assembly (16) using bolt (4), washers (5 and 7), nut (8), and cotter pin (9).

16. Move cyclic controls through full travel and check for obstructions and binding.

17. Verify cyclic rigging within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).

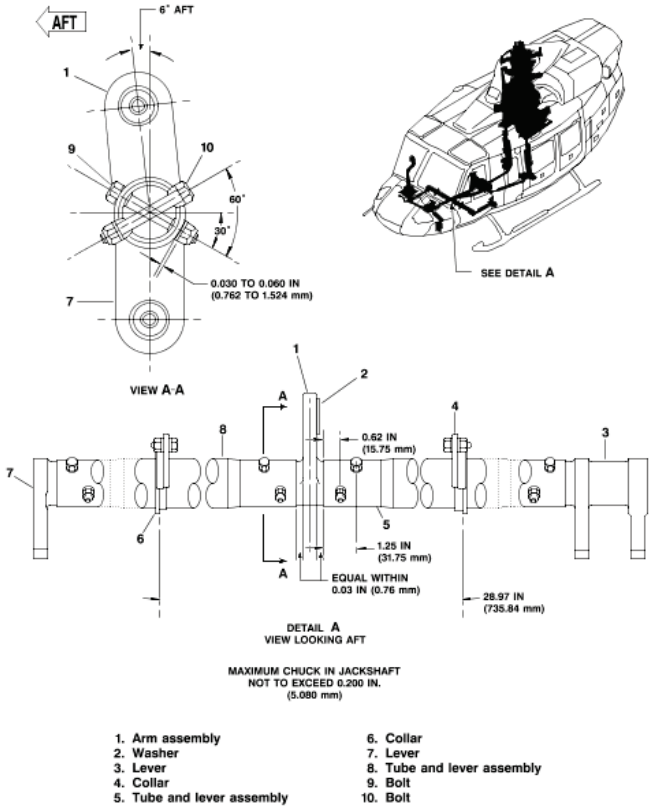
18. Install access panel on cabin floor under pilot seat (412-A-52-40-00-02A-720A-A / 00840).



1. Tube and lever assembly
2. Tapered bushing
3. Bolt
4. Bolt
5. Aluminum washer
6. Series actuator
7. Aluminum washer
8. Nut
9. Cotter pin
10. Bolt
11. Tapered bushing
12. Tube and lever assembly
13. Tapered bushing
14. Washer
15. Nut
16. Arm assembly
17. Washer
18. Tapered bushing
19. Washer
20. Nut
21. Cotter pin
22. Nut
23. Aluminum washer
24. Flight control tube
25. Bolt
26. Aluminum washer

ICN-412-A-671200-B-97499-00031-A-001-01

Figure 1. Cyclic Controls Arm (Sheet 1 of 1)



1. Arm assembly
2. Washer
3. Lever
4. Collar
5. Tube and lever assembly
6. Collar
7. Lever
8. Tube and lever assembly
9. Bolt
10. Bolt

ICN-412-A-671200-B-97499-00032-A-001-01

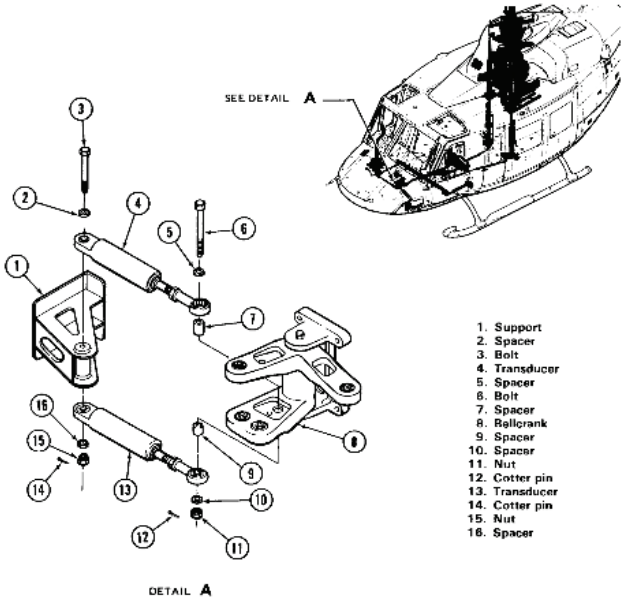
Figure 2. Cyclic Controls Arm and Washer Positioning (Sheet 1 of 1)

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Lateral Transducers (Cyclic Controls) - Removal

Procedure

1. Remove access panel from bottom of helicopter immediately aft of lower nose window line and right of helicopter centerline (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect electrical connector.
3. Remove bolt (3, Figure 1) by removing cotter pin (14), nut (15). Remove spacer (2) and spacer (16).
4. Remove bolt (6) by removing cotter pin (12), and nut (11). Remove spacers (5 and 10).
5. Remove transducer (4) with spacer (7) and transducer (13) with spacer (9).



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Figure 1. Lateral Transducer (Sheet 1 of 1)

Lateral Transducers (Cyclic Controls) - Inspection

Procedure

1. Inspect transducer for damage.
2. Inspect electrical cable and connectors for damage.
3. Inspect bearings for looseness (BHT-ALL-SPM, Standard Practices Manual).

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Lateral Transducers (Cyclic Controls) - Repair

Procedure

1. Lateral cyclic transducers are nonrepairable and shall be replaced when defective.

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Lateral Transducers (Cyclic Controls) - Installation

Procedure

1. Position transducer (4, Figure 1) and transducer (13) on support (1) and install bolt (3), spacer (2), spacer (16), nut (15), and cotter pin (14).

2. Position spacer (7) and spacer (9) on bellcrank (8). Install transducer (4) and transducer (13) onto bellcrank (8) with bolt (6), spacer (5), spacer (10), nut (11), and cotter pin (12).

NOTE

SN: 33001-33213 or 36001-36036:

If not previously accomplished, perform transducer adjustment. For helicopters with transducer P/N 412-074-101-101 installed, refer to 412-A-67-01-00-03A-276A-A / 01324.

NOTE

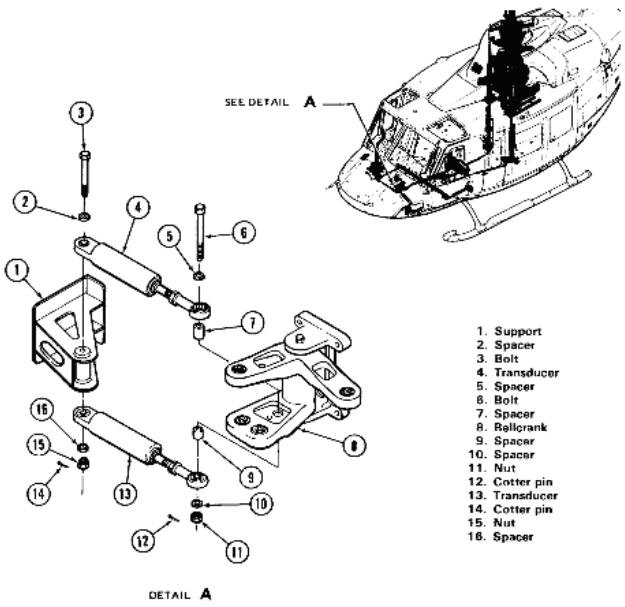
SN: 36037-Subsequent:

For helicopters with transducer P/N 412-074-101-103 installed, refer to 412-A-67-01-00-08A-276A-A / 01548.

3. Connect electrical connectors.

4. Move controls through full travel and check for binding and obstructions.

5. Install access panel on bottom of helicopter aft of lower nose window line and to right of helicopter centerline (412-A-52-40-00-02A-720A-A / 00840).



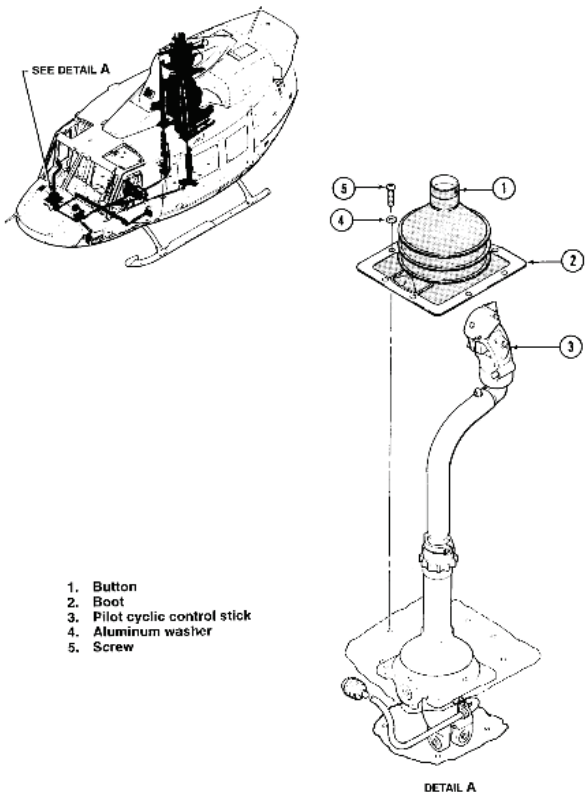
ICN-412-A-671200-B-97499-00033-A-001-01

Figure 1. Lateral Transducer (Sheet 1 of 1)

Cyclic Stick Boot - Removal

Procedure

1. Unsnap buttons (1, Figure 1).
2. Remove screws (5) and washers (4).
3. Remove boot (2) from pilot cyclic control stick (3).



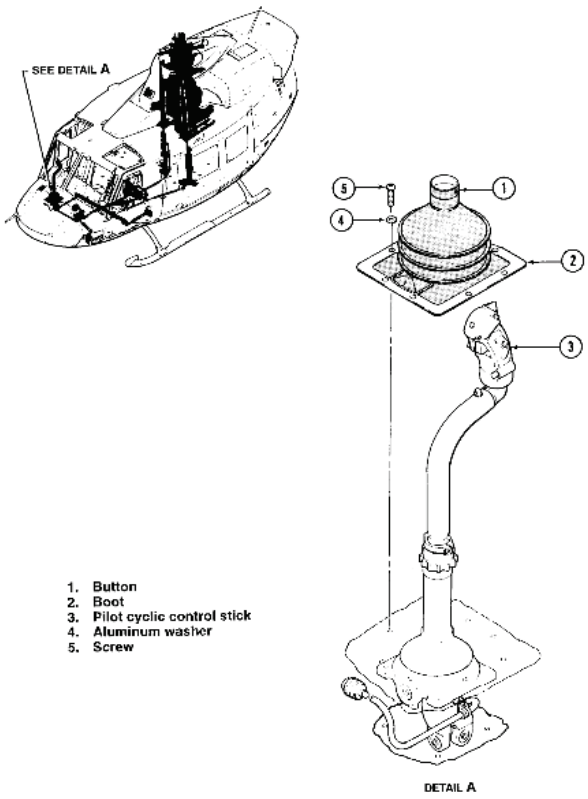
ICN-412-A-671200-B-97499-00034-A-001-01

Figure 1. Cyclic Stick Boot (Sheet 1 of 1)

Cyclic Stick Boot - Inspection

Procedure

1. Inspect boot (2, Figure 1) for tears, deterioration, and defective buttons.
2. Replace boot if unserviceable.



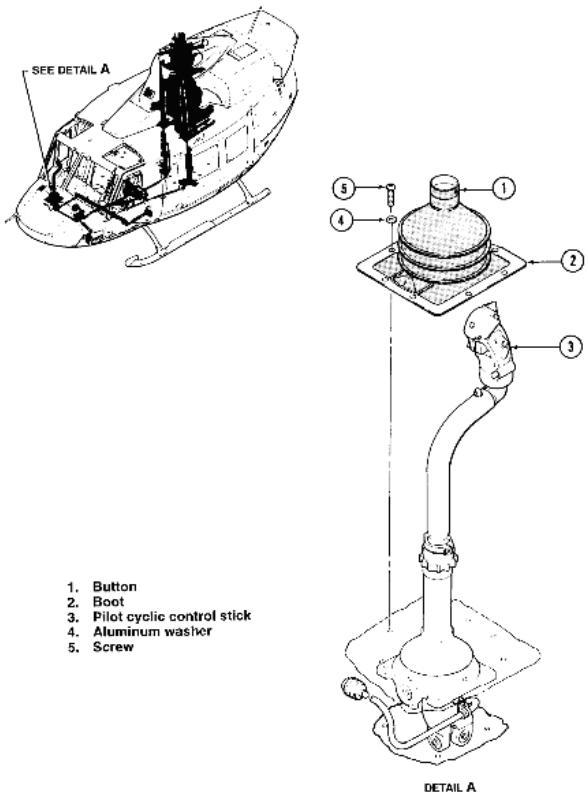
ICN-412-A-671200-B-97499-00034-A-001-01

Figure 1. Cyclic Stick Boot (Sheet 1 of 1)

Cyclic Stick Boot - Installation

Procedure

1. Slide boot (2, Figure 1) down over cyclic stick (3).
2. Align holes in boot (2) and in floor and install screws (5) and washers (4).
3. Snap buttons (1) around stick (3).



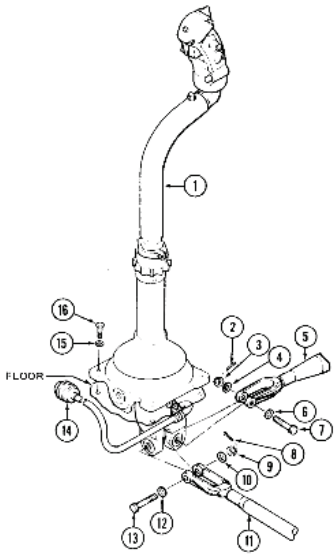
ICN-412-A-671200-B-97499-00034-A-001-01

Figure 1. Cyclic Stick Boot (Sheet 1 of 1)

Pilot Cyclic Control Stick Assembly - Removal

Procedure

1. Remove access panel from bottom of helicopter under pilot cyclic stick (412-A-52-40-00-02A-520A-A / 00837).
2. Remove boot (412-A-67-12-00-25A-520A-A / 01487).
3. Remove bolt (7, Figure 1) by removing cotter pin (2) and nut (3). Remove washers (4 and 6). Remove flight control tube (5) clevis from control stick (1).
4. Remove cotter pin (8), nut (9) and washer (10). Remove bolt (13) and washer (12). Remove flight control tube (11) clevis from control stick (1).
5. Disconnect electrical connector (14).
6. Remove bolts (16) and washers (15). Remove pilot cyclic control stick (1).



1. Pilot cyclic control stick
2. Cotter pin
3. Nut
4. Aluminum washer
5. Flight control tube
6. Aluminum washer
7. Bolt
8. Cotter pin
9. Nut
10. Aluminum washer
11. Flight control tube
12. Aluminum washer
13. Bolt
14. Electrical connector
15. Aluminum washer
16. Bolt

ICN-412-A-671200-B-97499-00035-A-001-01

Figure 1. Pilot Cyclic Stick (Sheet 1 of 1)

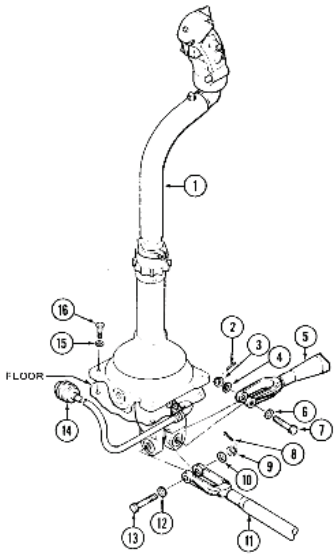
Pilot Cyclic Control Stick Assembly - Inspection

Procedure

1. Inspect stick assembly (1, Figure 1) for mechanical and corrosion damage. Superficial mechanical and corrosion damage is acceptable if repaired in accordance with BHT-412-CR&O.

2. Inspect bearings for secure installation in stick and for wear (BHT-ALL-SPM, Standard Practices Manual).

3. Inspect electrical connector (14) and wiring for security.



1. Pilot cyclic control stick
2. Cotter pin
3. Nut
4. Aluminum washer
5. Flight control tube
6. Aluminum washer
7. Bolt
8. Cotter pin
9. Nut
10. Aluminum washer
11. Flight control tube
12. Aluminum washer
13. Bolt
14. Electrical connector
15. Aluminum washer
16. Bolt

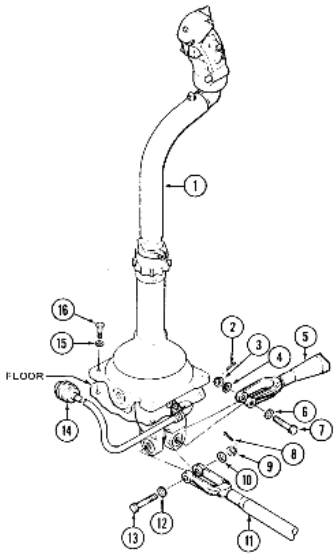
ICN-412-A-671200-B-97499-00035-A-001-01

Figure 1. Pilot Cyclic Stick (Sheet 1 of 1)

Pilot Cyclic Control Stick Assembly - Installation

Procedure

1. Place stick (1, Figure 1) in position and install bolts (16) and washers (15).
2. Place tube (5) on arm on stick (1) and install bolt (7), washers (6 and 4), nut (3) and cotter pin (2).
3. Place tube (11) on arm of stick (1) and install bolt (13), washers (12 and 10), nut (9) and cotter pin (8).
4. Connect electrical connector (14) to airframe mounted receptacle.
5. Verify cyclic rigging within limits (412-A-67-01-00-03A-276A-A / 01324 or 412-A-67-01-00-04A-276A-A / 01325).
6. Install boot (412-A-67-12-00-25A-720A-A / 01489).
7. Install access panel on bottom of helicopter under pilot control stick (412-A-52-40-00-02A-720A-A / 00840).



1. Pilot cyclic control stick
2. Cotter pin
3. Nut
4. Aluminum washer
5. Flight control tube
6. Aluminum washer
7. Bolt
8. Cotter pin
9. Nut
10. Aluminum washer
11. Flight control tube
12. Aluminum washer
13. Bolt
14. Electrical connector
15. Aluminum washer
16. Bolt

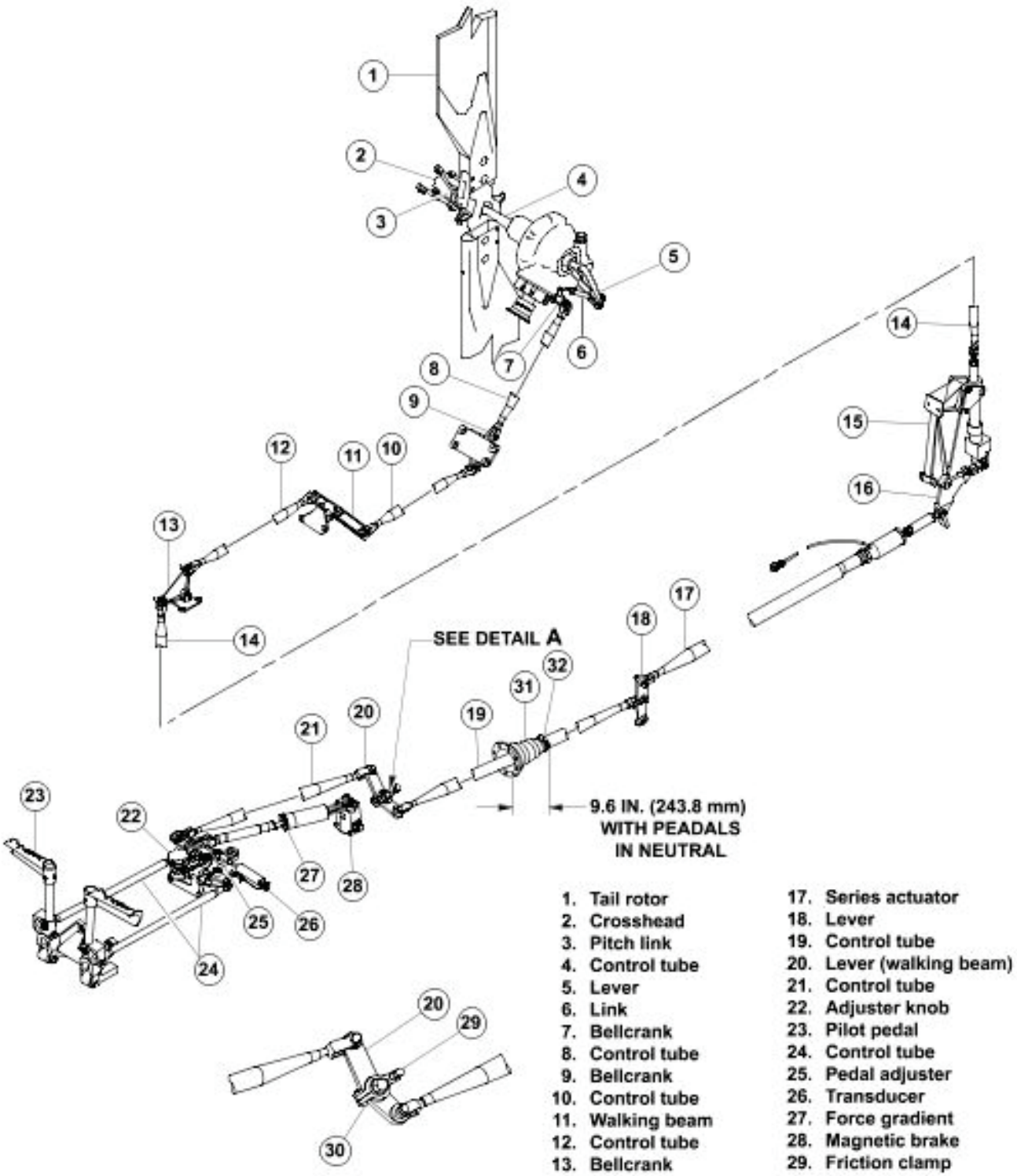
ICN-412-A-671200-B-97499-00035-A-001-01

Figure 1. Pilot Cyclic Stick (Sheet 1 of 1)

Anti-Torque Controls - Description

The anti-torque control system (Figure 1) consists of control linkage and components from pilot pedals (23) aft to tail rotor assembly. The AFCS provides a single electrical series actuator and a transducer. Series actuator is located forward of cylinder and support (15). A transducer is connected to airframe structure and bellcrank located aft of pedals.

Movement of pedals causes a hydraulic power assisted pitch change of tail rotor blades to offset main rotor torque and to control directional heading of helicopter.



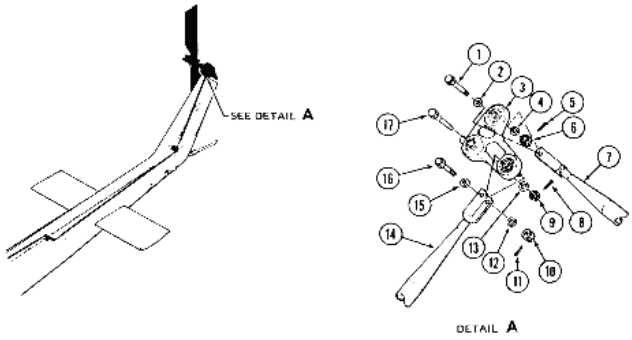
ICN-412-A-672000-B-97499-00026-A-001-01

Figure 1. Anti-Torque Controls (Sheet 1 of 1)

Pitch Change Control Bellcrank - Removal

Procedure

1. Remove cotter pin (11, Figure 1), nut (10), and washers (12 and 15), and bolt (16). Remove flight control tube (14).
2. Remove cotter pin (5), nut (6), washers (2 and 4), and bolt (1). Remove link (7).
3. Remove cotter pin (8), nut (9), washer (13), and bolt (17). Remove anti-torque control bellcrank (3).



1. Bolt
2. Steel washer
3. Anti-torque controls bellcrank
4. Steel washer
5. Cotter pin
6. Nut
7. Link
8. Cotter pin
9. Nut
10. Nut
11. Cotter pin
12. Washer (NAS1197-516)
13. Washer (AN960C416)
14. Flight control tube
15. Washer (AN960C416)
16. Bolt
17. Bolt

ICN-412-A-672000-B-97499-00027-A-001-01

Figure 1. Pitch Change Control Bellcrank (Sheet 1 of 1)

Pitch Change Control Bellcrank - Inspection

Procedure

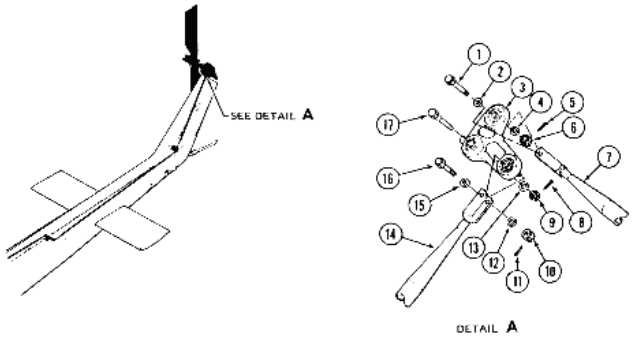
1. Inspect anti-torque control bellcrank (3) for corrosion and mechanical damage (BHT-412-CR&O).
2. Inspect bearings in bellcrank (BHT-ALL-SPM, Standard Practices Manual).

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Pitch Change Control Bellcrank - Repair

Procedure

1. Repair anti-torque control bellcrank (3, Figure 1) in accordance with BHT-412-CR&O.



1. Bolt
2. Steel washer
3. Anti-torque controls bellcrank
4. Steel washer
5. Cotter pin
6. Nut
7. Link
8. Cotter pin
9. Nut
10. Nut
11. Cotter pin
12. Washer (NAS1197-516)
13. Washer (AN960C416)
14. Flight control tube
15. Washer (AN960C416)
16. Bolt
17. Bolt

ICN-412-A-672000-B-97499-00027-A-001-01

Figure 1. Pitch Change Control Bellcrank (Sheet 1 of 1)

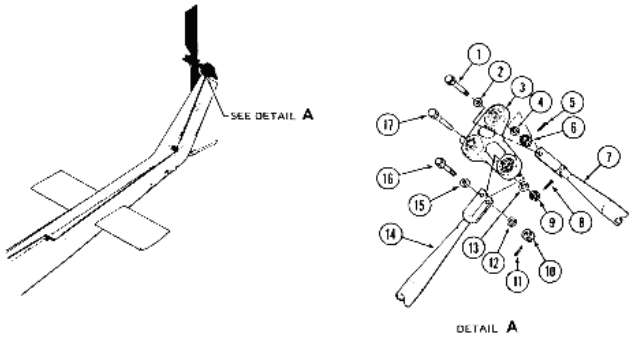
Pitch Change Control Bellcrank - Installation

Procedure

1. Place anti-torque control bellcrank (3, Figure 1) into position on fin. Install bolt (17), washer (13), and nut (9). Torque to 100 to 140 inch-pounds (11.3 to 15.8 Nm), and secure with cotter pin (8).

2. Place tube (14) onto bellcrank (3). Install bolt (16), washer (15), washer (12), nut (10), and cotter pin (11).

3. Place link (7) onto anti-torque control bellcrank (3). Install bolt (1), washer (2), and washer (4). Torque nut (6) and secure with cotter pin (5).



1. Bolt
2. Steel washer
3. Anti-torque controls bellcrank
4. Steel washer
5. Cotter pin
6. Nut
7. Link
8. Cotter pin
9. Nut
10. Nut
11. Cotter pin
12. Washer (NAS1197-516)
13. Washer (AN960C416)
14. Flight control tube
15. Washer (AN960C416)
16. Bolt
17. Bolt

ICN-412-A-672000-B-97499-00027-A-001-01

Figure 1. Pitch Change Control Bellcrank (Sheet 1 of 1)

Anti-Torque Control Tube Guides - Removal

Procedure

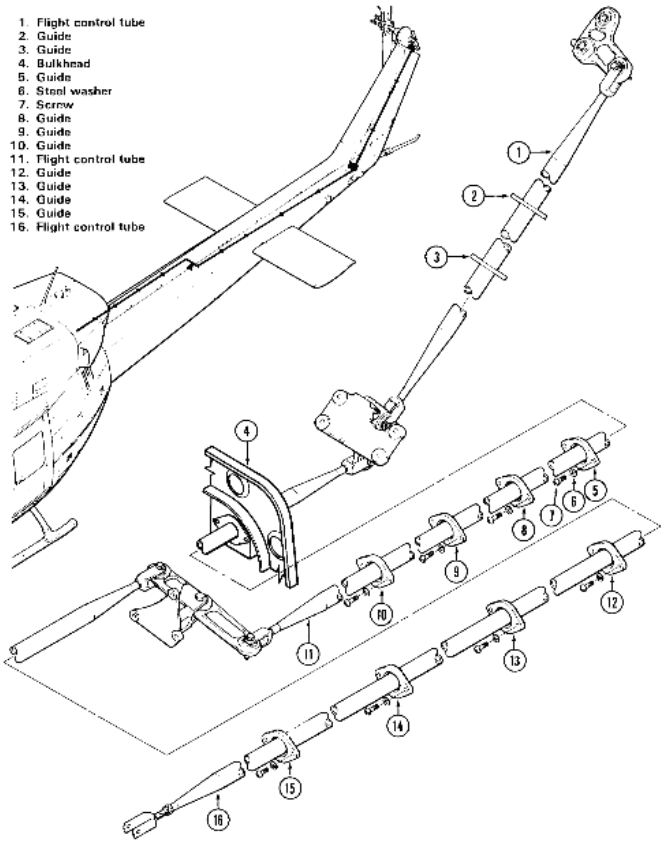
NOTE

Maintenance procedures for guide (5, Figure 1) is given. Maintenance for remaining guides is similar with the exception that appropriate flight control tube(s) (1, 11, or 16) must be removed.

1. Remove access door from lower side of tailboom (412-A-52-40-00-02A-520A-A / 00837).

2. Remove flight control tube (11) (412-A-67-20-00-03A-520A-A / 01502).

3. Remove two screws (7) and two washers (6). Remove guide (5).



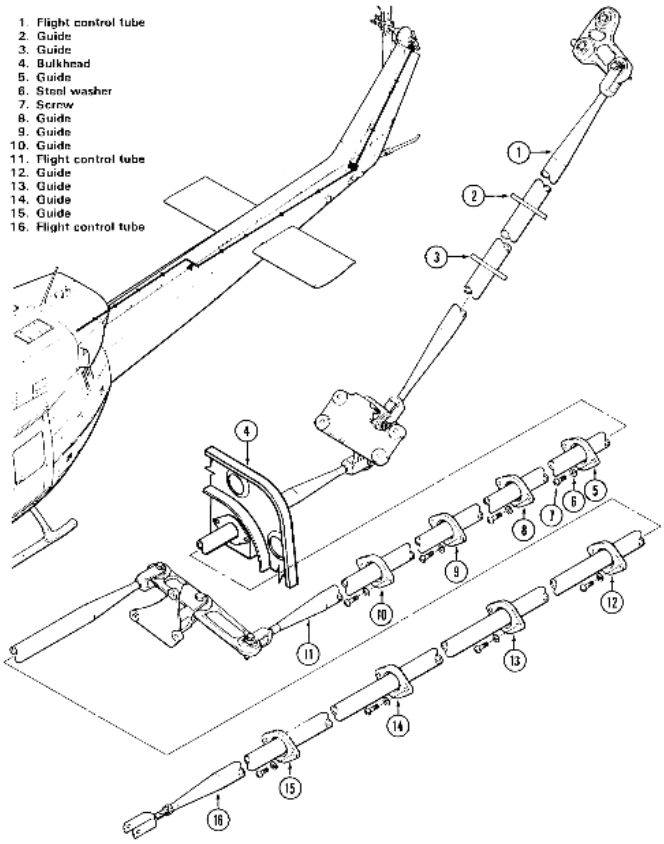
ICN-412-A-672000-B-97499-00028-A-001-01

Figure 1. Control Tube Guides (Sheet 1 of 1)

Anti-Torque Control Tube Guides - Inspection

Procedure

1. Inspect guide (Figure 1) for wear, distortion and cracks. Damage that can be detected visually is cause to replace guide. Guides are fabricated from nylatron material and are not repairable.



ICN-412-A-672000-B-97499-00028-A-001-01

Figure 1. Control Tube Guides (Sheet 1 of 1)

Anti-Torque Control Tube Guides - Installation

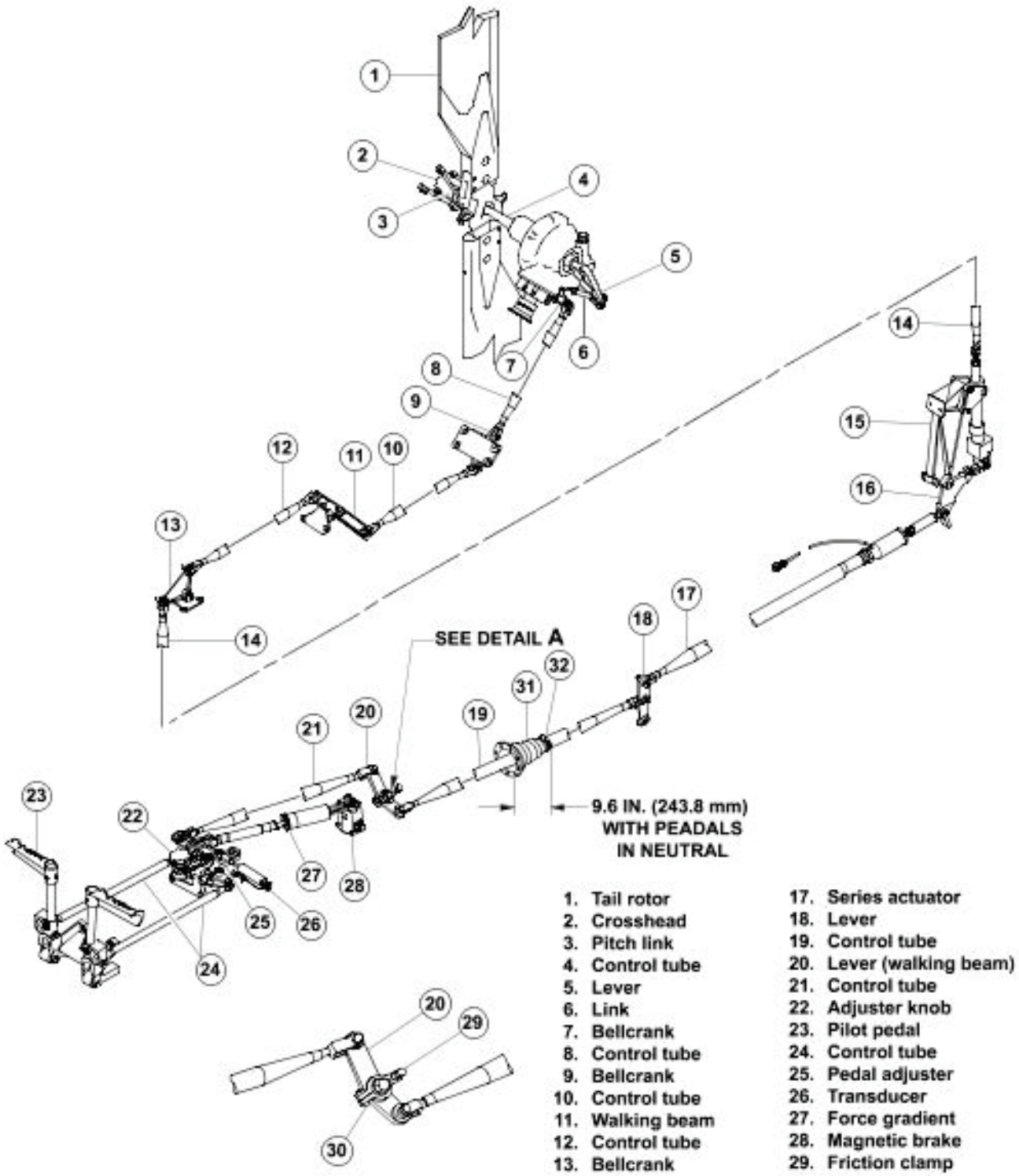
Procedure

1. Position guide (5) on bulkhead (4). Install two screws (7) and two washers (6).
2. Install flight control tube (11) (412-A-67-20-00-03A-720A-A / 01504).
3. Verify anti-torque rigging within limits (412-A-67-01-00-06A-276A-A / 01327).
4. Install access door on lower side of tailboom (412-A-52-40-00-02A-720A-A / 00840).

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Flight Control Tubes and Bellcranks - Description

Anti-torque control tubes and bellcranks transmit pilot motion at the tail rotor pedals to pitch change mechanism at tail rotor gearbox (Figure 1).



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Figure 1. Anti-Torque Controls (Sheet 1 of 1)

Flight Control Tubes and Bellcranks - Removal

Procedure

1. Remove appropriate access panel(s) and door(s) to gain access to desired control tube or bellcrank (412-A-52-40-00-02A-520A-A / 00837).

2. Remove cotter pins, nuts, washers, and bolts and remove desired component.

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Flight Control Tubes and Bellcranks - Inspection

Procedure

1. Inspect components for corrosion and mechanical damage (BHT-412-CR&O).
2. Inspect abrasion strip(s) for wear and for secure installation on flight control tube(s).
3. Inspect clevises for secure installation on tubes.

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Flight Control Tubes and Bellcranks - Installation

Procedure

1. Position control tube (or bellcrank) in fuselage or tailboom as appropriate.
2. Retain component in position with bolts, washers, and nuts. Secure with cotter pins.
3. Install components, panels, or doors removed for access (412-A-52-40-00-02A-720A-A / 00840).
4. Move anti-torque controls through full travel and check for interference and binding.
5. Verify anti-torque control rigging within limits (412-A-67-01-00-06A-276A-A / 01327).

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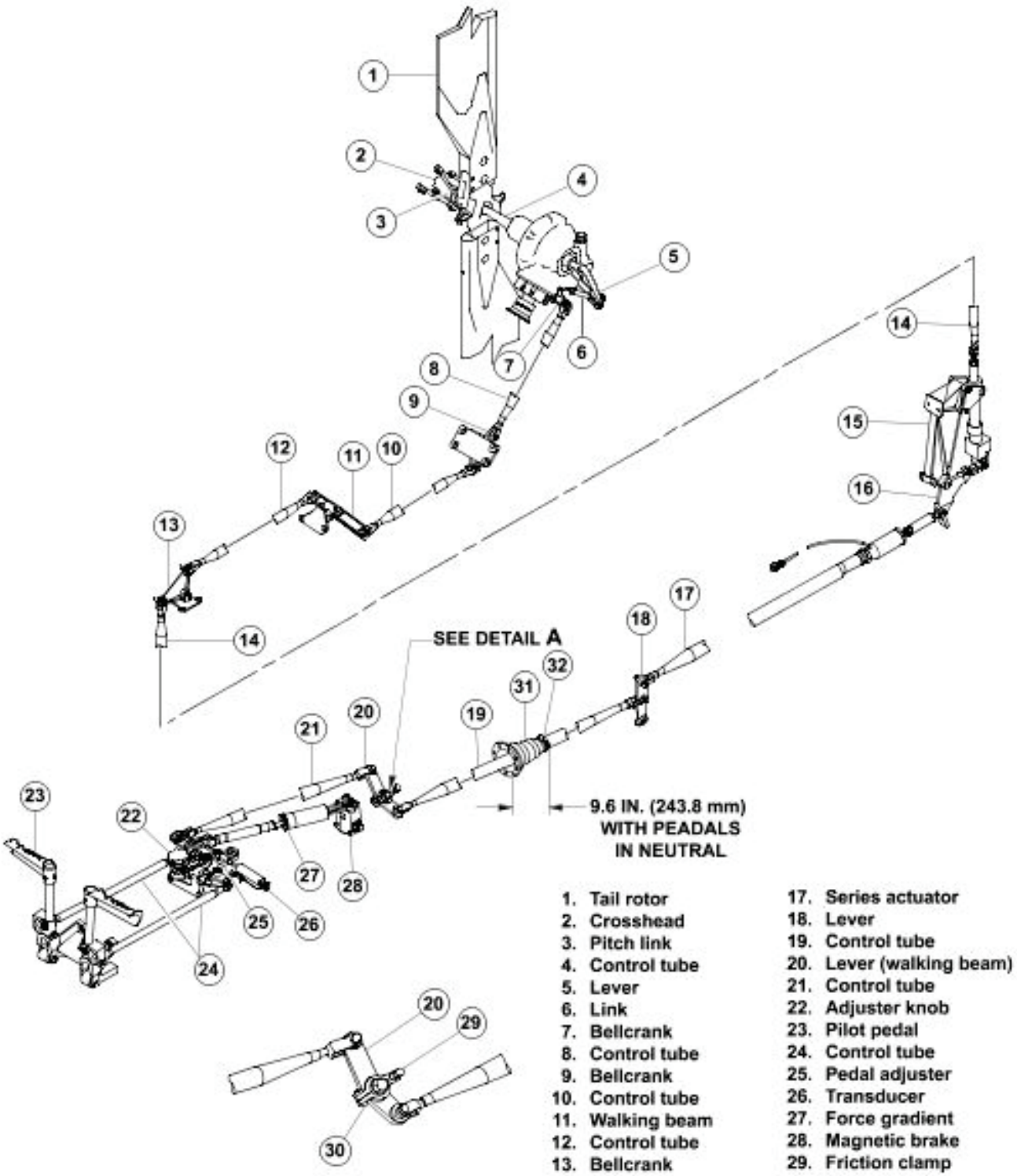
Applicable to: ALL
ECCN EAR99

Walking Beam Support Assembly - Removal

Procedure

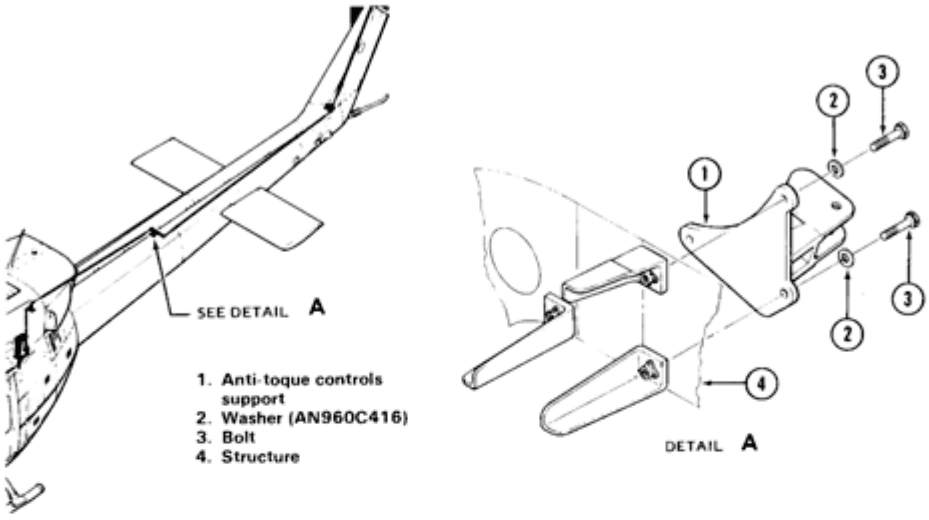
1. Remove the control tubes (10 and 12, Figure 1) and walking beam (11) (412-A-67-20-00-07A-520A-A / 01516).

2. Remove bolts (3, Figure 2) and washers (2). Remove anti-torque controls support (1).



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Figure 1. Anti-Torque Controls (Sheet 1 of 1)



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Figure 2. Anti-Torque Control Support (Sheet 1 of 1)

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Walking Beam Support Assembly - Inspection and Repair

Preliminary requirements

Consumables, materials and expendables

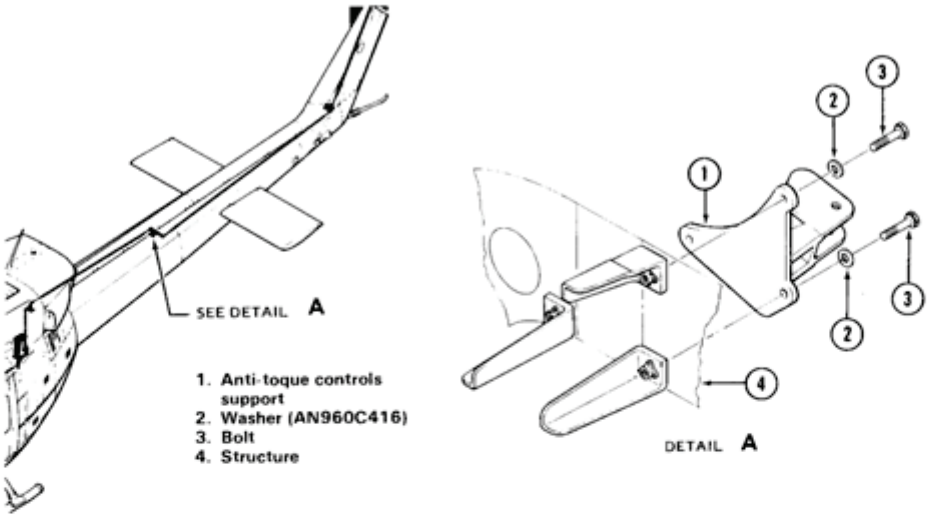
Nomenclature	Identification No.	Qty	Remarks
Barrier tape	C-456	AR	

Procedure

1. Inspect and repair support (1, Figure 1) mechanical and corrosion damage (BHT-412-CR&O).

2. Inspect bolt holes for elongation (BHT-412-CR&O). Damage which can be detected visually is not acceptable and is cause for replacement of support.

3. Inspect barrier tape (C-456) for adhesion to mating surface of support (1).



ICN-412-A-672000-B-97499-00029-A-001-01

Figure 1. Anti-Torque Control Support (Sheet 1 of 1)

Walking Beam Support Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Barrier tape	C-456	AR	

Procedure

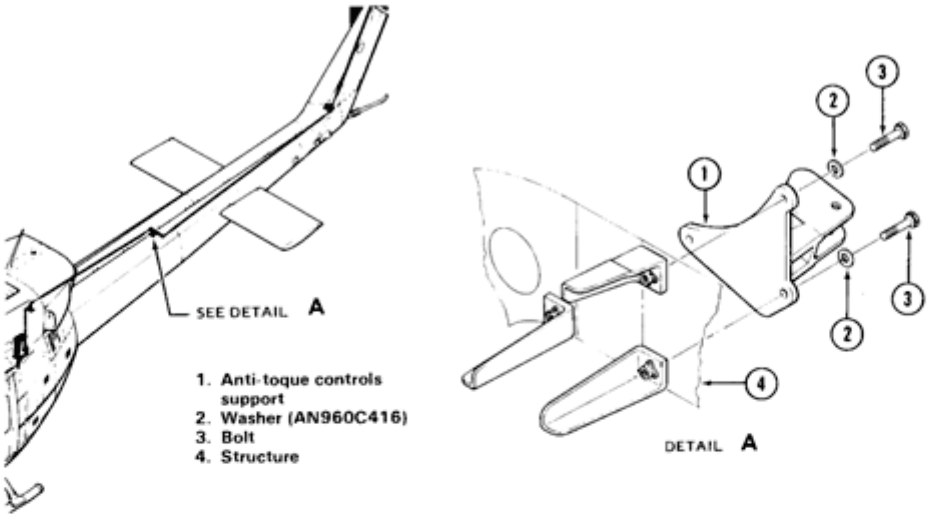
1. Apply barrier tape (C-456) to mating surface of support (1, Figure 1).

2. Place support (1) in position on structure (4). Install bolts (3) and washers (2).

3. Install the control tubes (10 and 12, Figure 2) walking beam (412-A-67-20-00-07A-720A-A / 01519).

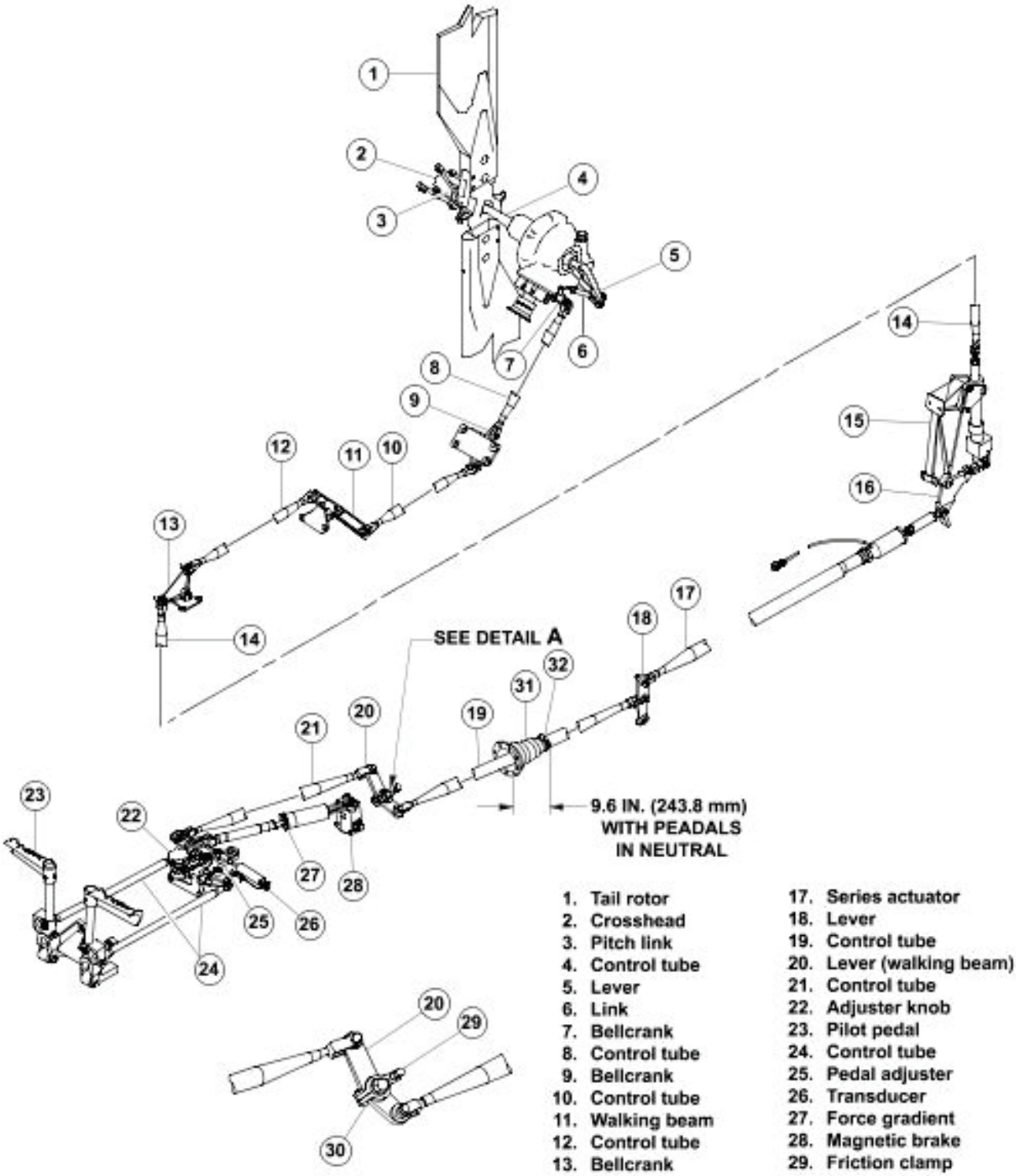
4. Move anti-torque pedals through full range of travel and check for interference and binding.

5. Verify anti-torque rigging within limits (412-A-67-01-00-06A-276A-A / 01327).



ICN-412-A-672000-B-97499-00029-A-001-01

Figure 1. Anti-Torque Control Support (Sheet 1 of 1)



ICN-412-A-672000-B-97499-00026-A-001-01

Figure 2. Anti-Torque Controls (Sheet 1 of 1)

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Control Cylinder - Removal

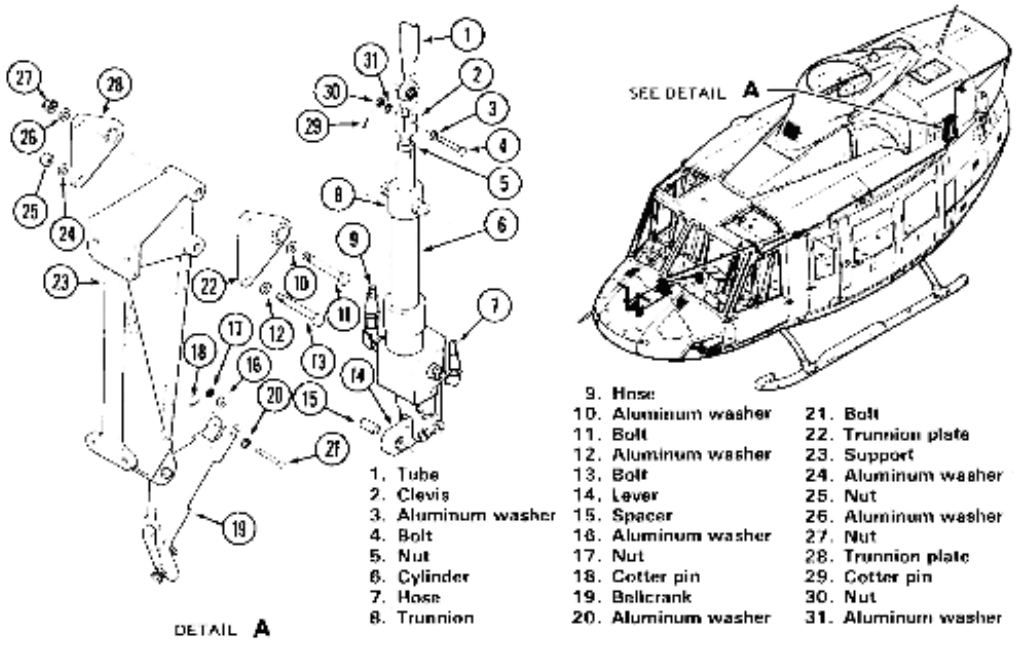
Procedure

1. Open door on aft right side of fuselage.

NOTE

Refer to 412-A-29-00-00-00A-042A-A / 00612 for general maintenance practices when in hydraulic areas.

2. Disconnect hoses (7 and 9, Figure 1). Plug or cap open ports to prevent contamination.
3. Remove bolt (4) by removing cotter pin (29), nut (30), and washers (3 and 31).
4. Remove cotter pin (18), nut (17), washer (16), and remove bolt (21) and washer (20), including spacer (15).
5. While supporting cylinder (6), remove nut (27), bolt (11) with washers (10 and 26). Remove nut (25), bolt (13) with washers (12 and 24).
6. Disengage trunnion plates (22 and 28) from trunnion (8). Remove cylinder (6).
7. If cylinder (6) is being replaced, loosen nut (5) and remove clevis (2).



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Figure 1. Anti-Torque Hydraulic Cylinder (Sheet 1 of 1)

Control Cylinder - Inspection

Procedure

1. Inspect cylinder (6, Figure 1) for damage and signs of leakage.

2. Inspect trunnion (8) and plates (22 and 28) for wear. Fit plates onto trunnion and feel for looseness and smoothness of movement.

3. Inspect bearing in lever (14) for freedom of movement.

4. Perform the following inspections only when high frequency vibrations due to anti-torque controls are suspected and/or anti-torque controls series actuator motors. Refer to troubleshooting chart in 412-A-29-00-00-00A-422A-A / 00613.

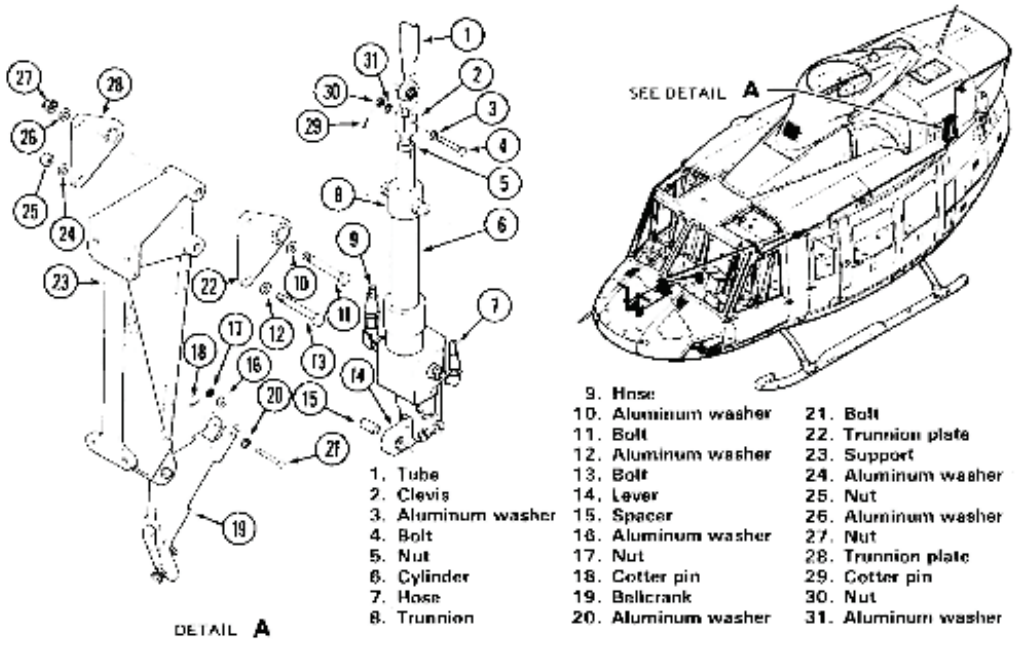
4.1. Grasp cylinder (6, Figure 1) just below plates (22 and 28) and move cylinder laterally. Lateral movement of cylinder should be 0.003 to 0.050 in. (0.076 to 1.27 mm). If binding is noted, refer to 412-A-67-20-00-05A-685A-A / 01510. There should not be any evidence of binding.

4.2. Inspect bearing in lever (14) for freedom of movement. Maximum acceptable force to rotate bearing is eight oz. (2.22 N).

4.2.1. Remove cotter pin (18), nut (17), aluminum washer (16), and bolt (21) with aluminum washer (20). Separate bellcrank (19) from lever (14).

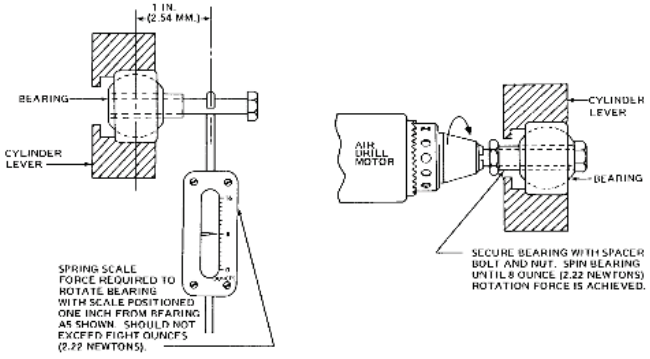
4.2.2. Install bolt (21) in bearing in lever (14).

4.2.3. Attach a spring scale to bolt (21) (Figure 2). Measure force required to rotate bearing. If force required to rotate bearing is in excess of value shown on illustration, refer to 412-A-67-20-00-05A-685A-A / 01510.



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Figure 1. Anti-Torque Hydraulic Cylinder (Sheet 1 of 1)



ICN-412-A-672000-B-97499-00031-A-001-01

Figure 2. Anti-Torque Hydraulic Cylinder Bearing Inspection Tool (Sheet 1 of 1)

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Control Cylinder - Repair

Preliminary requirements

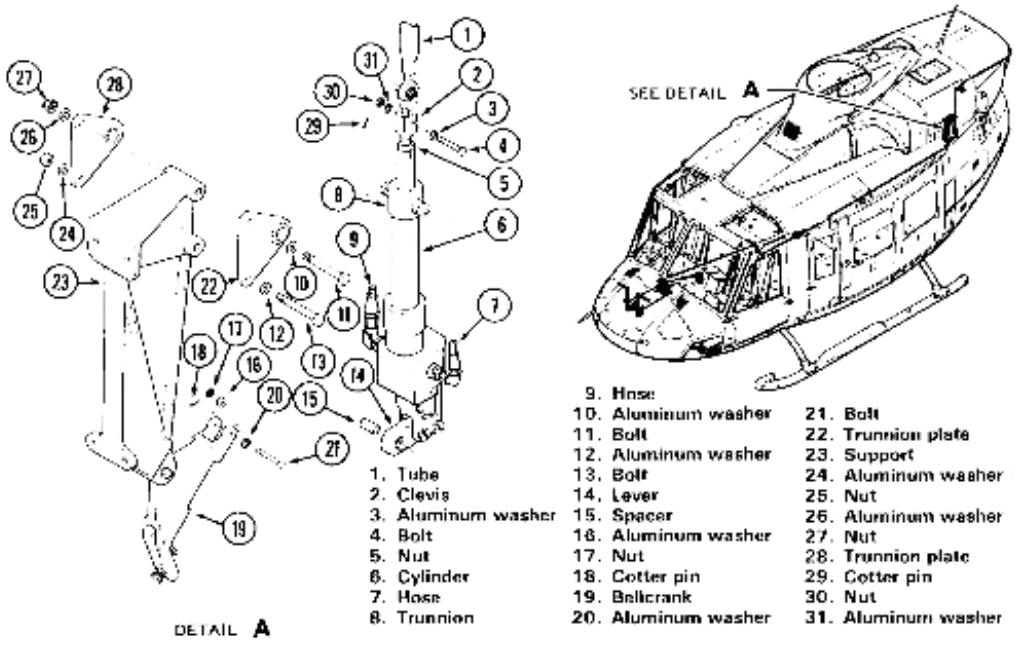
Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Abrasive pad	C-407	AR	

Procedure

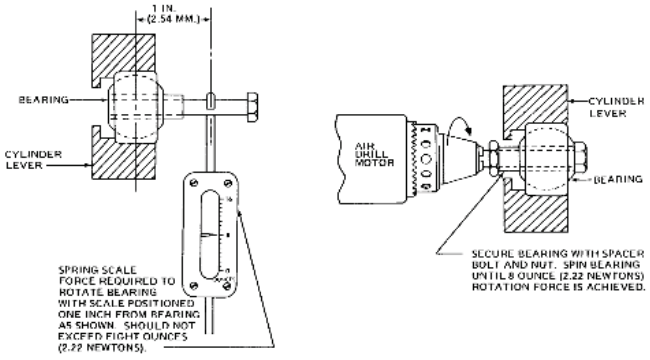
1. If binding was noted during accomplishment of step 4 of 412-A-67-20-00-05A-280A-A / 01509, clean trunnion plates (22 and 28, Figure 1) with abrasive pad (C-407).

2. If force to rotate bearing during accomplishment of step 4 exceeds value shown on illustration, attach an air drill motor to bolt as shown on Figure 2. Spin bearing with drill to loosen bearing until it rotates within the limits.



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Figure 1. Anti-Torque Hydraulic Cylinder (Sheet 1 of 1)



ICN-412-A-672000-B-97499-00031-A-001-01

Figure 2. Anti-Torque Hydraulic Cylinder Bearing Inspection Tool (Sheet 1 of 1)

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Control Cylinder - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Clean cloth	C-516	AR	

Procedure

Applicable: SN: 33001-33213 or 36001-36086:

1. If cylinder (6, Figure 1) is a replacement, install clevis (2) in top of piston and adjust to obtain 1.22 inches (30.99 mm) from bolt hole centerline to top of piston. Secure with nut (5).

Applicable: SN: 36087-Subsequent:

2. Obtain 1.29 inches (32.76 mm) from bolt hole centerline to top of piston. Secure with nut (5).

3. Thoroughly clean and dry trunnion (8) and trunnion plates (22 and 28) with aliphatic naphtha (C-305) and clean cloth (C-516).

4. Place trunnion plates (22 and 28) over trunnion (8) and position plates onto support (23). Install bolts (11 and 13), aluminum washers (10 and 12), aluminum washers (24 and 26), and nuts (25 and 27).

5. Position bellcrank (19) onto lever (14). Install spacer (15), bolt (21), washer (20), washer (16), nut (17), and cotter pin (18).

6. Position tube (1) into clevis (2). Install bolt (4), aluminum washer (3), aluminum washer (31), nut (30), and cotter pin (29).

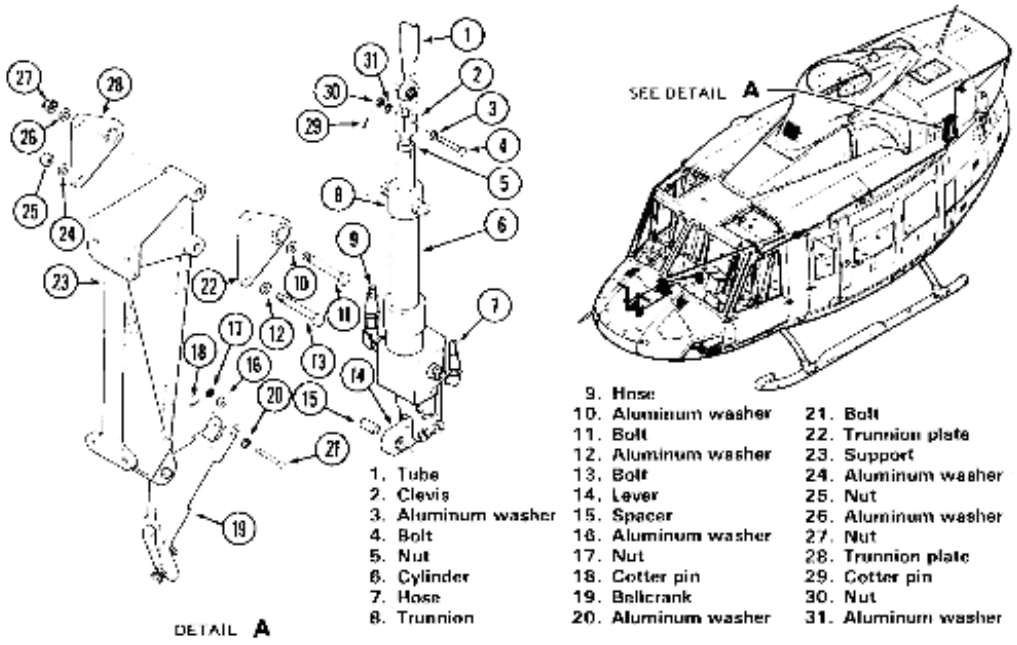
7. Remove plugs or caps and connect hoses (7 and 9).

8. Service hydraulic system reservoir.

9. Pressurize hydraulic system No. 1 and operate anti-torque controls through 10 complete cycles (412-A-29-00-00-00A-320B-A / 00617).

10. Verify anti-torque rigging within limits (412-A-67-01-00-06A-276A-A / 01327). Bleed hydraulic system (412-A-29-00-00-00A-320B-A / 00617).

11. Close door on right aft side of fuselage (412-A-52-40-00-02A-720A-A / 00840).



ICN-412-A-672000-B-97499-00030-A-001-01

Figure 1. Anti-Torque Hydraulic Cylinder (Sheet 1 of 1)

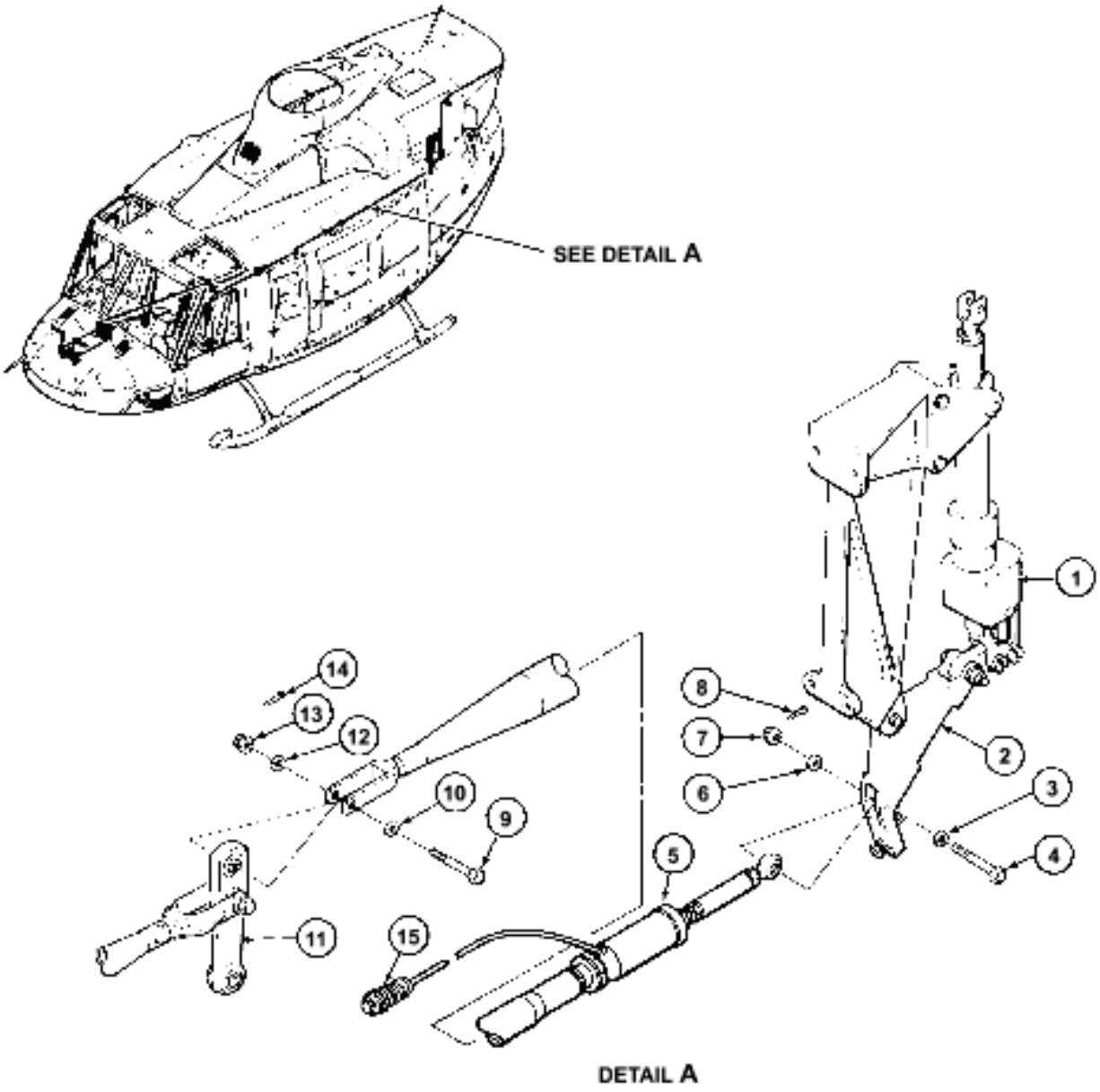
Anti-torque Series Actuator - Maintenance

Procedure



DISASSEMBLY OF SERVO ACTUATOR (5, Figure 1) IS NOT PERMITTED.

1. Series actuators are not field repairable and shall be replaced if unserviceable.



1. Directional hydraulic flight control cylinder
2. Bellcrank
3. Aluminum washer
4. Bolt
5. Servo actuator
6. Aluminum washer
7. Nut
8. Cotter pin
9. Bolt
10. Aluminum washer
11. Anti-torque control lever
12. Aluminum washer
13. Nut
14. Cotter pin
15. Connector

ICN-412-A-672000-B-97499-00032-A-001-01

Figure 1. Directional Series Actuator (Sheet 1 of 1)

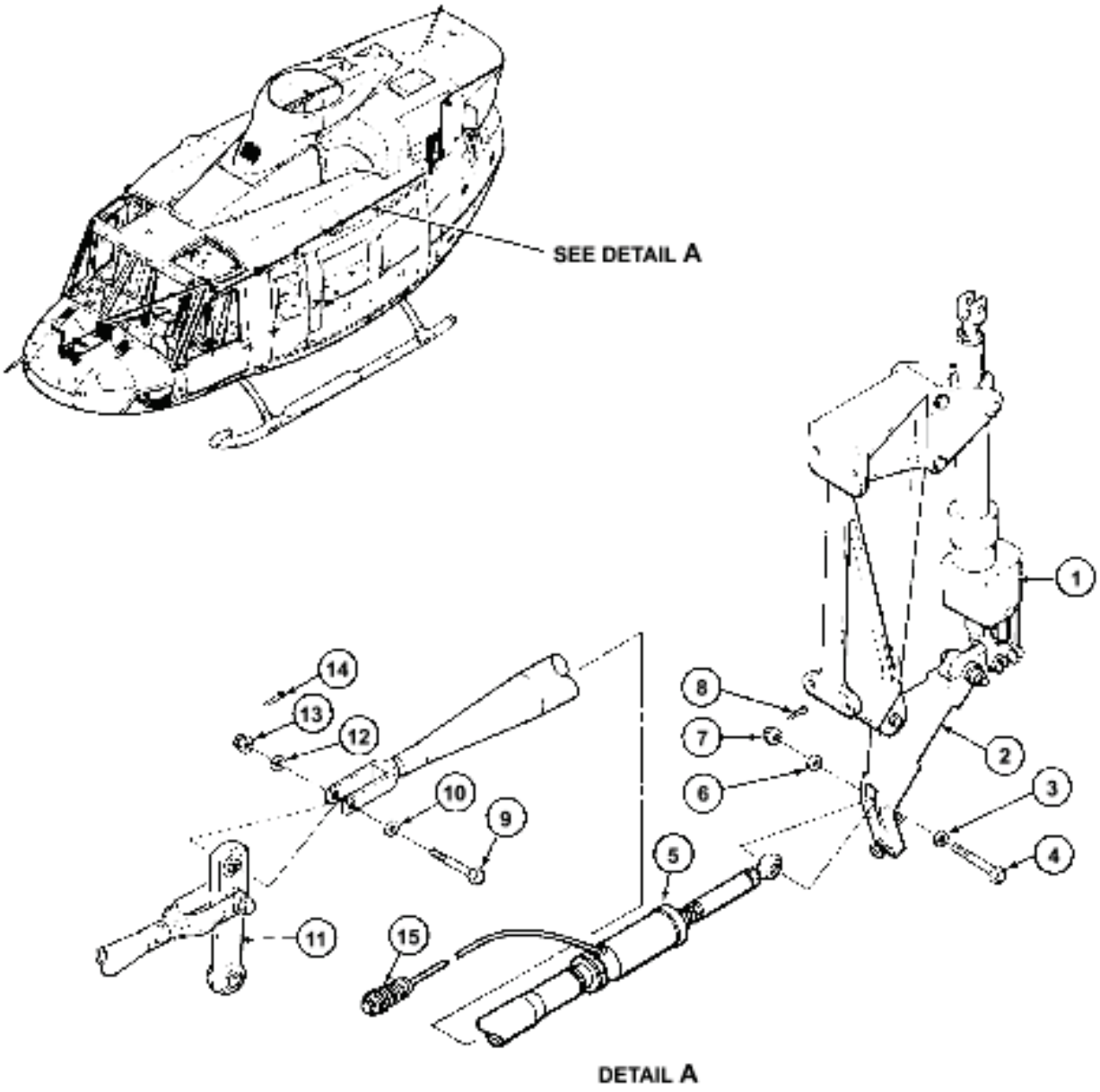
Anti-torque Series Actuator - Removal

Procedure



DISASSEMBLY OF SERVO ACTUATOR (5, Figure 1) IS NOT PERMITTED.

1. Remove large round access panel from bottom of aft fuselage. Open access door on right lower side of aft fuselage (412-A-52-40-00-02A-520A-A / 00837).
2. Remove tailboom assembly (412-A-53-50-00-00A-520A-A / 00912).
3. Disconnect electrical connector (15, Figure 1).
4. Remove cotter pin (14), nut (13), aluminum washer (12), and remove bolt (9) and aluminum washer (10).
5. Remove cotter pin (8), nut (7), aluminum washer (6), and bolt (4) with aluminum washer (3).
6. Remove servo actuator (5).



1. Directional hydraulic flight control cylinder
2. Bellcrank
3. Aluminum washer
4. Bolt
5. Servo actuator
6. Aluminum washer
7. Nut
8. Cotter pin
9. Bolt
10. Aluminum washer
11. Anti-torque control lever
12. Aluminum washer
13. Nut
14. Cotter pin
15. Connector

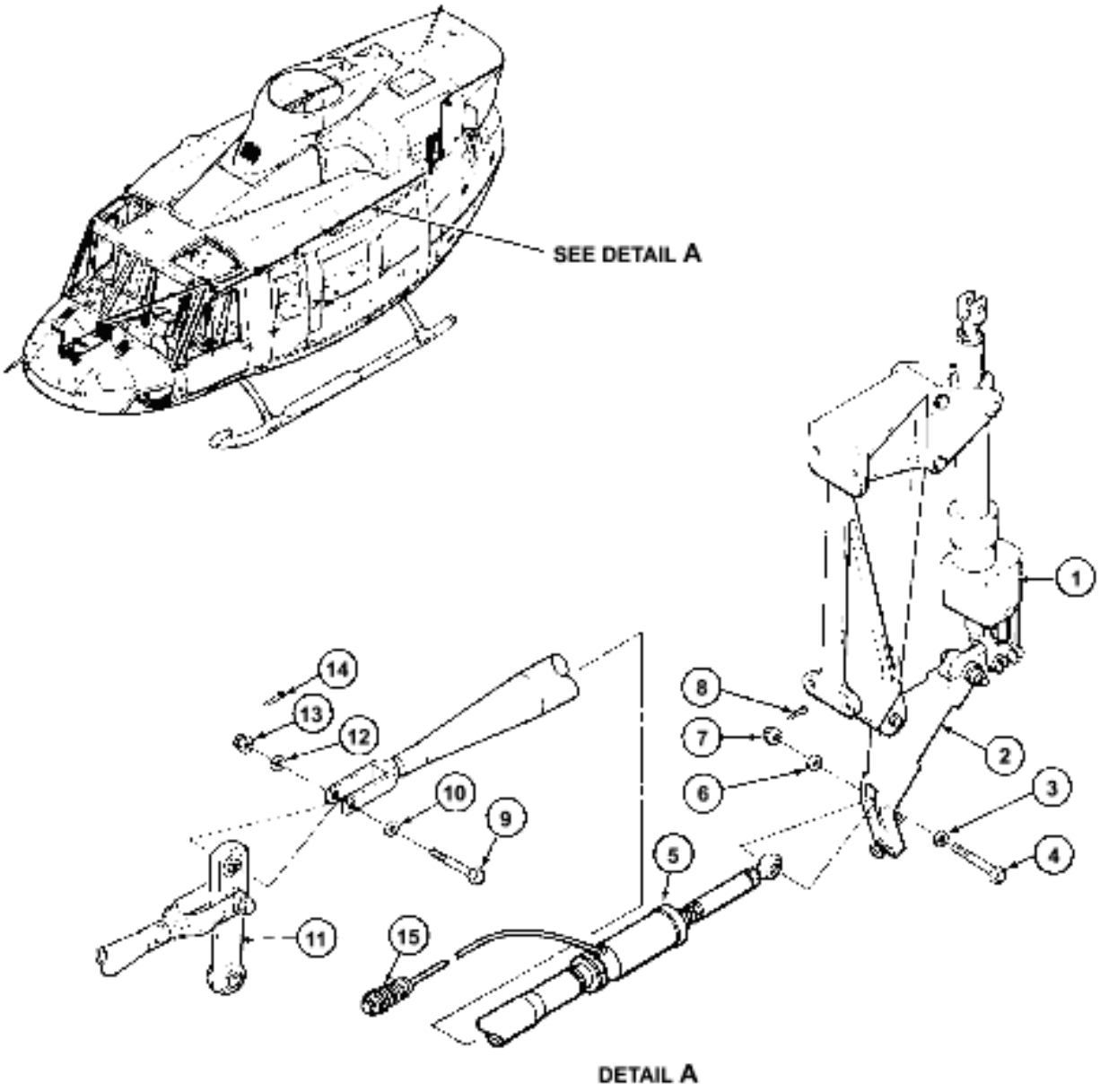
ICN-412-A-672000-B-97499-00032-A-001-01

Figure 1. Directional Series Actuator (Sheet 1 of 1)

Anti-torque Series Actuator - Inspection

Procedure

1. Inspect servo actuator (5, Figure 1) (BHT-412-CR&O).



1. Directional hydraulic flight control cylinder
2. Bellcrank
3. Aluminum washer
4. Bolt
5. Servo actuator
6. Aluminum washer
7. Nut
8. Cotter pin
9. Bolt
10. Aluminum washer
11. Anti-torque control lever
12. Aluminum washer
13. Nut
14. Cotter pin
15. Connector

ICN-412-A-672000-B-97499-00032-A-001-01

Figure 1. Directional Series Actuator (Sheet 1 of 1)

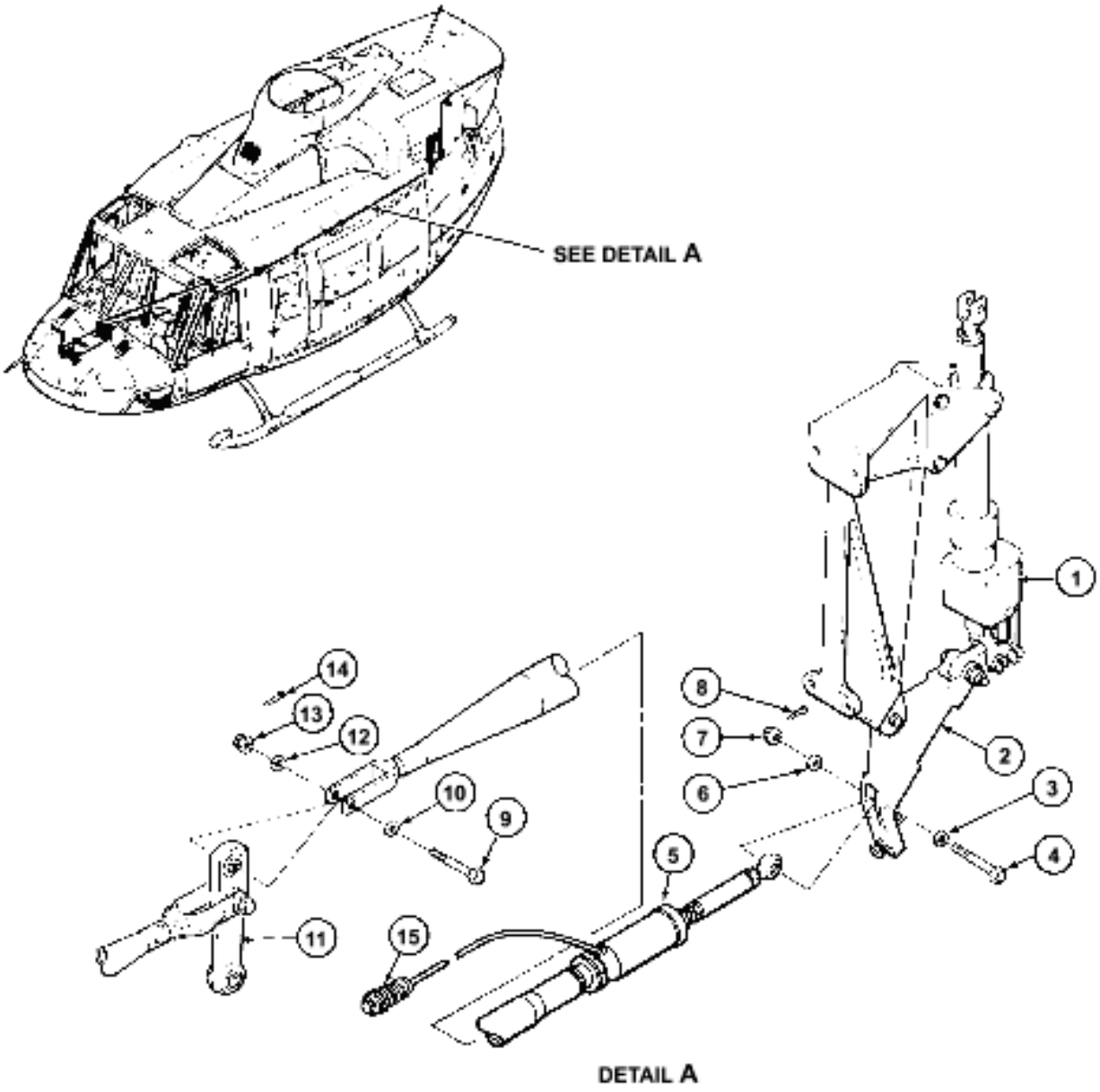
Anti-torque Series Actuator - Installation

Procedure



DISASSEMBLY OF SERIES ACTUATOR (5, Figure 1) IS NOT PERMITTED.

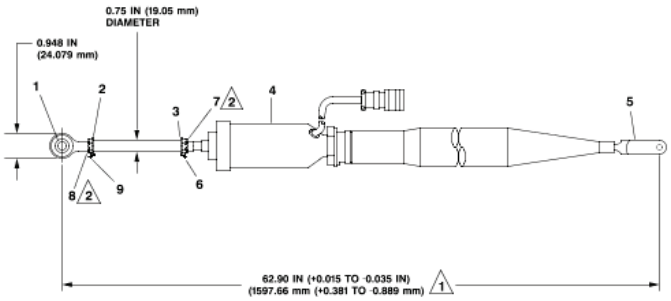
1. Establish overall dimensions shown in Figure 2.
2. Place servo actuator (5, Figure 1) into position with rod end bearing in bellcrank (2).
3. Install bolt (4), aluminum washer (3), aluminum washer (6), nut (7) and cotter pin (8).
4. Place forward end of servo actuator (5) on anti-torque controls lever (11).
5. Install bolt (9), aluminum washer (10), aluminum washer (12), nut (13), and cotter pin (14).
6. Move anti-torque controls through full travel and check for obstructions and binding.
7. Connect electrical connector (15).
8. Install tailboom assembly (412-A-53-50-00-00A-720A-A / 00915).
9. Verify anti-torque rigging within limits (412-A-67-01-00-06A-276A-A / 01327).
10. Install large round access panel on bottom of aft fuselage (412-A-52-40-00-02A-720A-A / 00840).
11. Close access door on lower right side of aft fuselage (412-A-52-40-00-02A-720A-A / 00840).



1. Directional hydraulic flight control cylinder
2. Bellcrank
3. Aluminum washer
4. Bolt
5. Servo actuator
6. Aluminum washer
7. Nut
8. Cotter pin
9. Bolt
10. Aluminum washer
11. Anti-torque control lever
12. Aluminum washer
13. Nut
14. Cotter pin
15. Connector

ICN-412-A-672000-B-97499-00032-A-001-01

Figure 1. Directional Series Actuator (Sheet 1 of 1)



1. Bearing
2. Washer
3. Washer
4. Directional series actuator
5. Clevis
6. Lockwire
7. Nut
8. Nut
9. Lockwire

NOTES

1 Length shown is with actuator at mid-position.

2 Nut to have engagement of threads.

3. Actuator length is 63.062 to 63.112 inches (1601.775 to 1603.045 mm) fully extended. Minimum length is 62.668 to 62.718 inches (1591.767 to 1593.073 mm) fully retracted.

ICN-412-A-672000-B-97499-00033-A-001-01

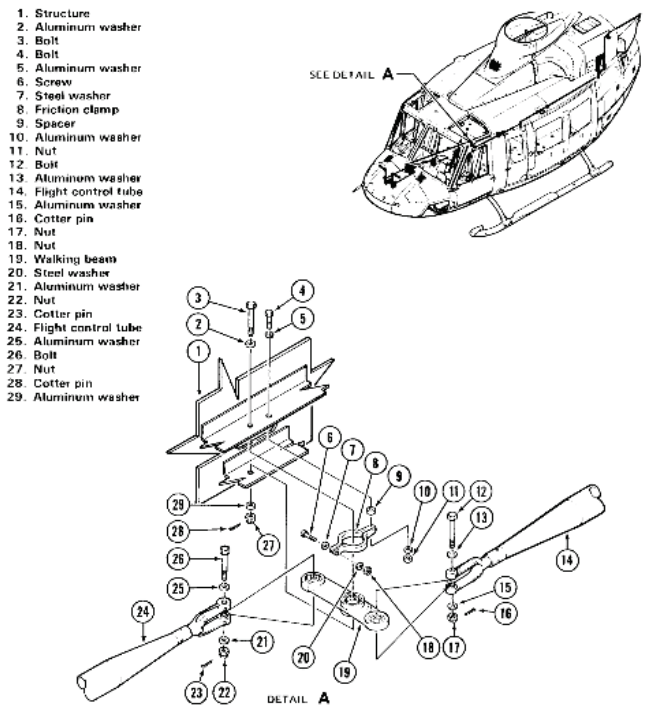
Figure 2. Directional Series Actuator Assembly (Sheet 1 of 1)

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Walking Beam and Friction Clamp - Removal

Procedure

1. Remove access panel from center of aft cabin floor. Remove access panel from bottom of helicopter under walking beam (19, Figure 1) (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect flight control tube (24) by removing cotter pin (23), nut (22), aluminum washer (21). Remove bolt (26) and aluminum washer (25).
3. Disconnect flight control tube (14) by removing cotter pin (16), nut (17), and aluminum washer (15). Remove bolt (12) and aluminum washer (13).
4. Remove nut (11) and aluminum washer (10). Remove bolt (4) and aluminum washer (5).
5. Remove cotter pin (28), nut (27), and washer (29). Remove bolt (3) and washer (2).
6. Remove walking beam (19) and friction clamp (8) with spacer (9).
7. Loosen screw (6) and remove friction clamp (8) from walking beam (19).



ICN-412-A-672000-B-97499-00034-A-001-01

Figure 1. Walking Beam and Friction Clamp Assembly (Sheet 1 of 1)

Walking Beam and Friction Clamp - Inspection

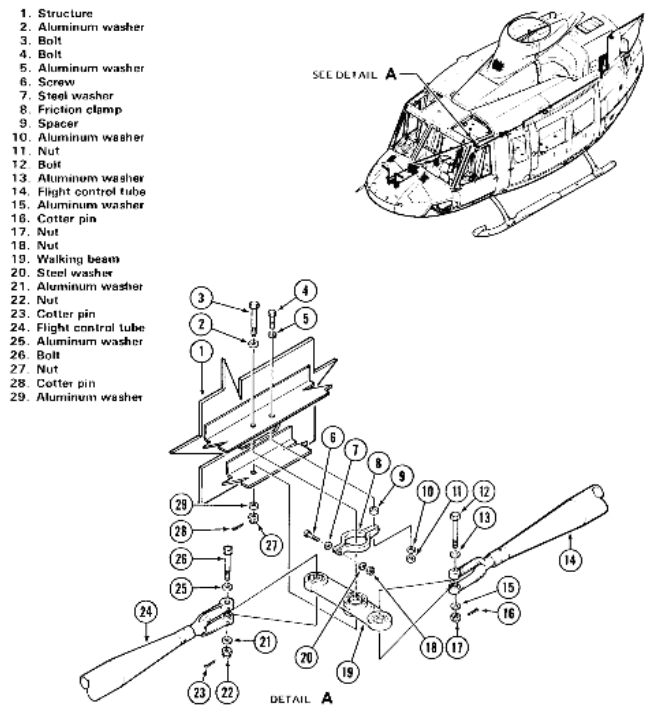
Procedure

1. Inspect walking beam (19, Figure 1) for mechanical and corrosion damage. Superficial mechanical and corrosion damage is acceptable if repaired (BHT-412-CR&O).

2. Elongated bolt hole damage that can be detected visually is cause to replace walking beam.

3. Inspect bearings for secure installation in beam and for wear (BHT-ALL-SPM, Standard Practices Manual).

4. Inspect friction clamp (8) for mechanical and corrosion damage and for elongated bolt holes.



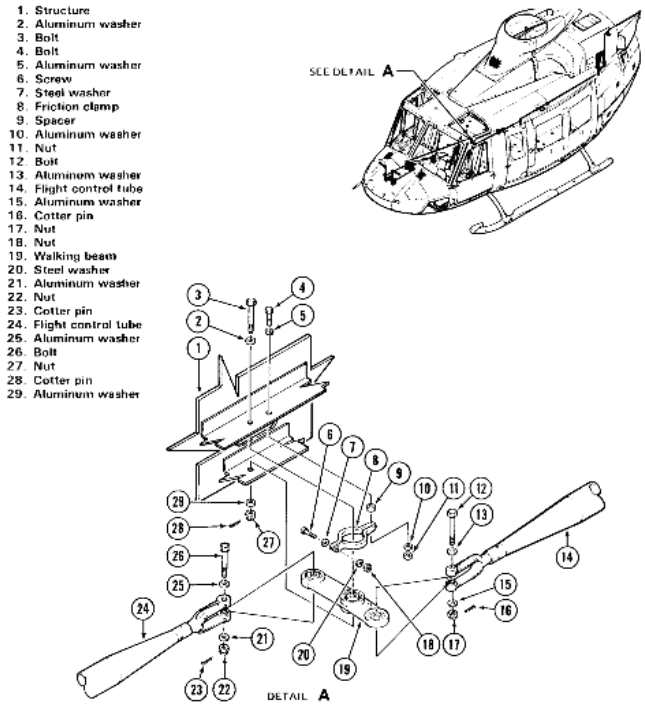
ICN-412-A-672000-B-97499-00034-A-001-01

Figure 1. Walking Beam and Friction Clamp Assembly (Sheet 1 of 1)

Walking Beam and Friction Clamp - Repair

Procedure

1. Repair walking beam (19, Figure 1) and clamp (8) in accordance with BHT-412-CR&O.



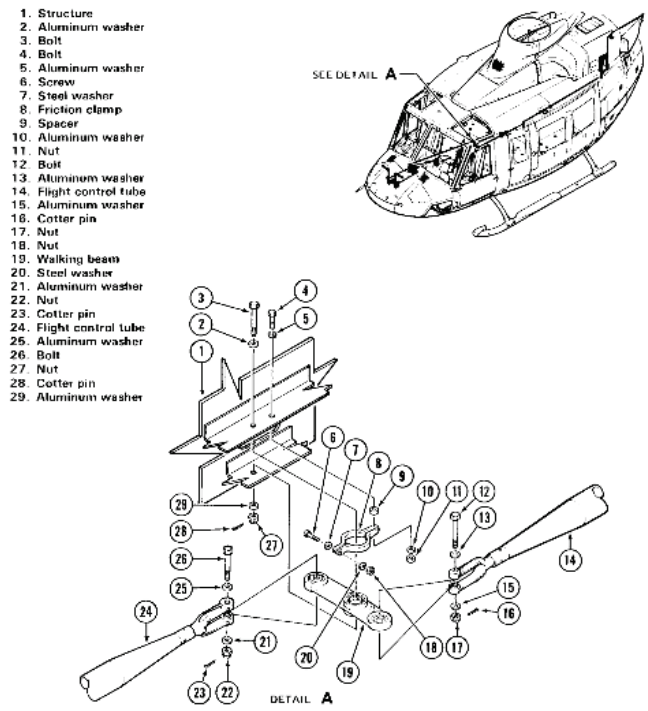
ICN-412-A-672000-B-97499-00034-A-001-01

Figure 1. Walking Beam and Friction Clamp Assembly (Sheet 1 of 1)

Walking Beam and Friction Clamp - Installation

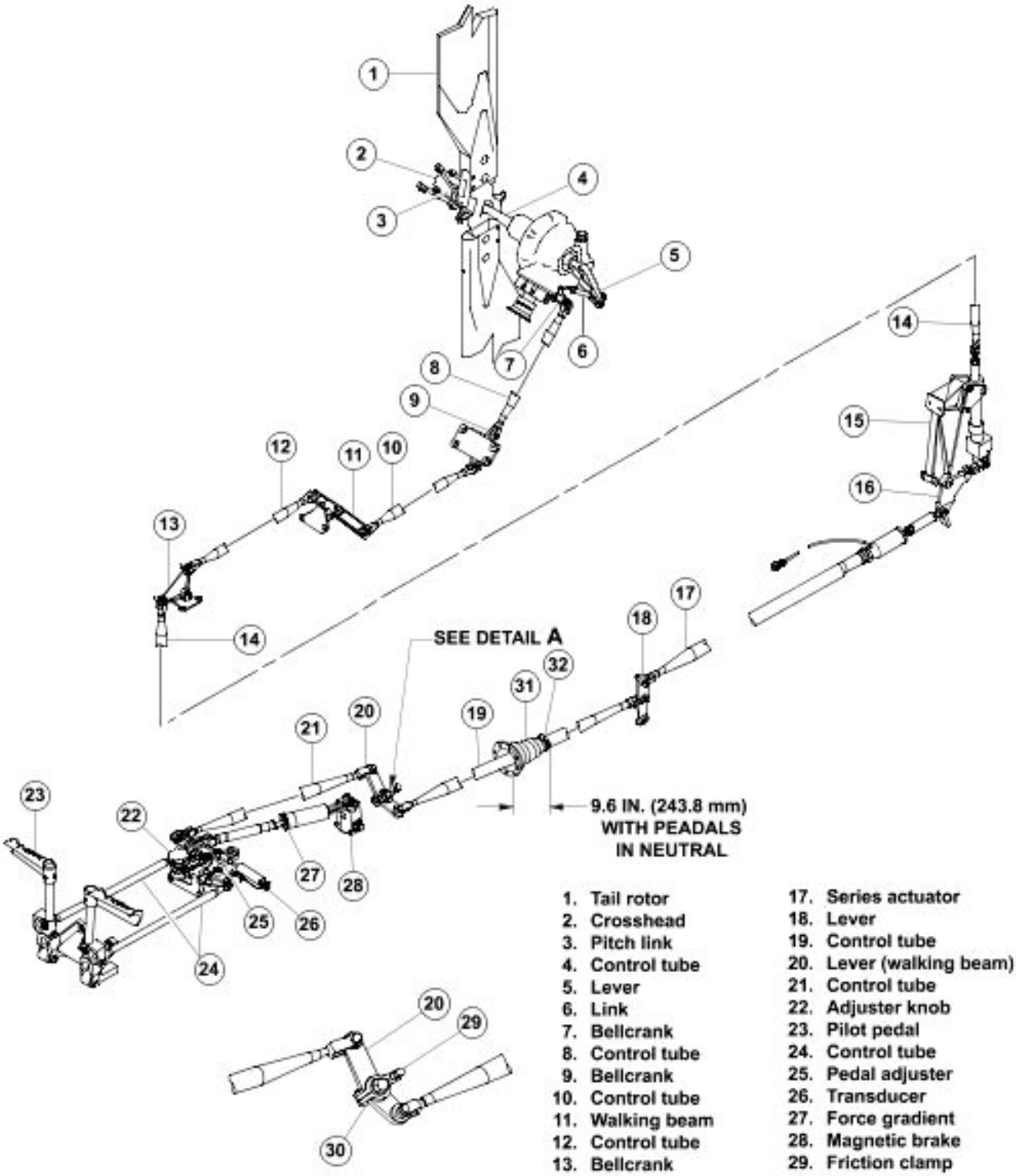
Procedure

1. Place friction clamp (8, Figure 1) onto walking beam (19). Place walking beam and clamp into structure (1). Install bolt (3) and aluminum washer (2) to hold walking beam in position.
2. Install bolt (4), aluminum washer (5), spacer (9), aluminum washer (10), and nut (11). Install aluminum washer (29), nut (27), and cotter pin (28) on bolt (3).
3. Place tube (24) on walking beam (19). Install bolt (26), aluminum washer (25), washer (21), nut (22), and cotter pin (23).
4. Disconnect force gradient (412-A-67-20-00-10A-520A-A / 01528).
5. Adjust friction screw (6) in friction clamp (8) until force of 4.25 to 4.75 pounds (18.90 to 21.13 N) is measured at pedals when moved through neutral position.
6. Install force gradient (412-A-67-20-00-10A-720A-A / 01531).
7. Place flight control tube (14) onto walking beam (19). Install bolt (12), aluminum washer (13), aluminum washer (15), nut (17), and cotter pin (16).
8. With anti-torque pedals in central position, locate aft end of boot (31, Figure 2) on control tube (19) 9.6 inches (243.3 mm) aft of bulkhead attachment for the boot. Tighten the clamp (32).
9. Move anti-torque controls through full travel and check for interference and binding.
10. Verify anti-torque rigging within limits (412-A-67-01-00-06A-276A-A / 01327).
11. Install access panel on bottom of helicopter under walking beam (19) (412-A-52-40-00-02A-720A-A / 00840).
12. Install access panel in center of aft cabin floor (412-A-52-40-00-02A-720A-A / 00840).



ICN-412-A-672000-B-97499-00034-A-001-01

Figure 1. Walking Beam and Friction Clamp Assembly (Sheet 1 of 1)



ICN-412-A-672000-B-97499-00026-A-001-01

Figure 2. Anti-Torque Controls (Sheet 1 of 1)

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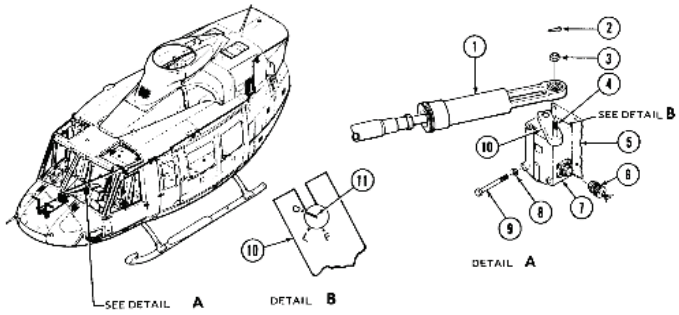
Magnetic Brake - Removal

Procedure

NOTE

The following procedure is applicable to magnetic brakes used in the anti-torque system. For anti-torque rotary actuators used when the optional 4-axis AFCS/Flight Director kit is installed, refer to 412-A-67-12-02-03A-520A-A / 01577.

1. Remove access panel from cabin floor under pilot seat (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect electrical connector (6, Figure 1).
3. Remove cotter pin (2) and nut (3).
4. Lift force gradient (1) off arm (10).
5. Remove four bolts (9) with aluminum washers (8).
6. Remove magnetic brake (7) from structure (5).



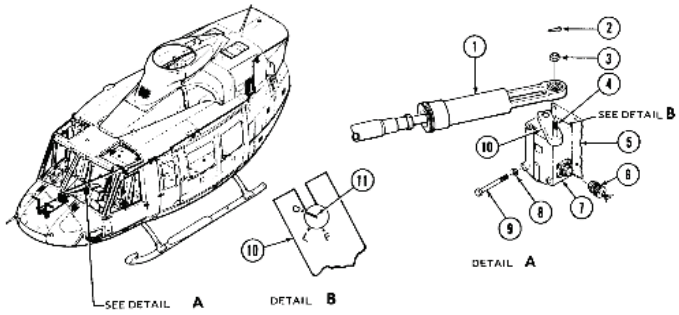
ICN-412-A-672000-B-97499-00035-A-001-01

Figure 1. Magnetic Brake (Sheet 1 of 1)

Magnetic Brake - Inspection

Procedure

1. Inspect magnetic brake (7, Figure 1) for mechanical and corrosion damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is not acceptable.
2. Inspect stud (4) for damaged threads. No damage that can be visually detected is acceptable.
3. Inspect electrical receptacle for bent pins and damaged threads. Superficial damage is acceptable without repair.



ICN-412-A-672000-B-97499-00035-A-001-01

Figure 1. Magnetic Brake (Sheet 1 of 1)

Magnetic Brake - Repair

Procedure

1. Refer to vendor manual for magnetic brake repair instructions.

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Magnetic Brake - Installation

Procedure

NOTE

The following procedure is applicable to magnetic brakes used in the anti-torque system. For anti-torque rotary actuators used when the optional 4-axis AFCS/Flight Director kit is installed, refer to 412-A-67-12-02-03A-720A-A / 01580.

1. Inspect magnetic brake (7, Figure 1) to ensure arm (10) is installed on magnetic brake with letter D positioned in relation to index mark (11) (Detail B). Reposition arm on magnetic brake if index mark is not properly aligned.

2. Position magnetic brake (7) on structure (5). Install four bolts (9) and aluminum washers (8) as follows:

2.1. Install aluminum washer (8) with each bolt (9).

NOTE

Three different length bolts are required. Install bolts of correct length to obtain proper thread engagement.

2.2. Install bolts (9) with drilled heads in two inboard holes in magnetic brake (7).

3. Adjust length of force gradient (1) (412-A-67-01-00-06A-276A-A / 01327). After force gradient length is set, position force gradient on arm (10). Install nut (3) and cotter pin (2).

4. Install electrical connector (6).

5. Move anti-torque controls through full travel and check for binding and obstructions.

6. Verify anti-torque rigging is within limits:

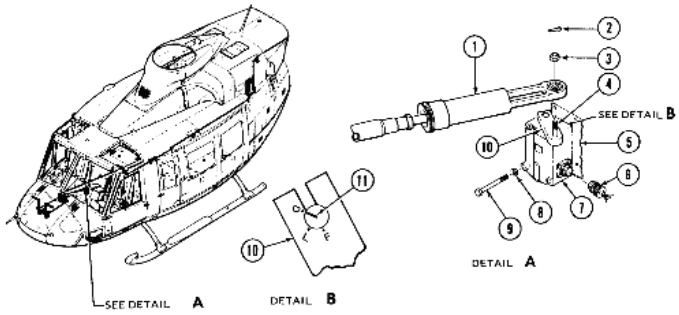
Applicable: SN: 36001-36086:

6.1. Refer to 412-A-67-01-00-06A-276A-A / 01327.

Applicable: SN: 36087-Subsequent:

6.2. Refer to 412-A-67-01-00-11A-276A-A / 01551.

7. Install access panel on cabin floor under pilot seat (412-A-52-40-00-02A-720A-A / 00840).



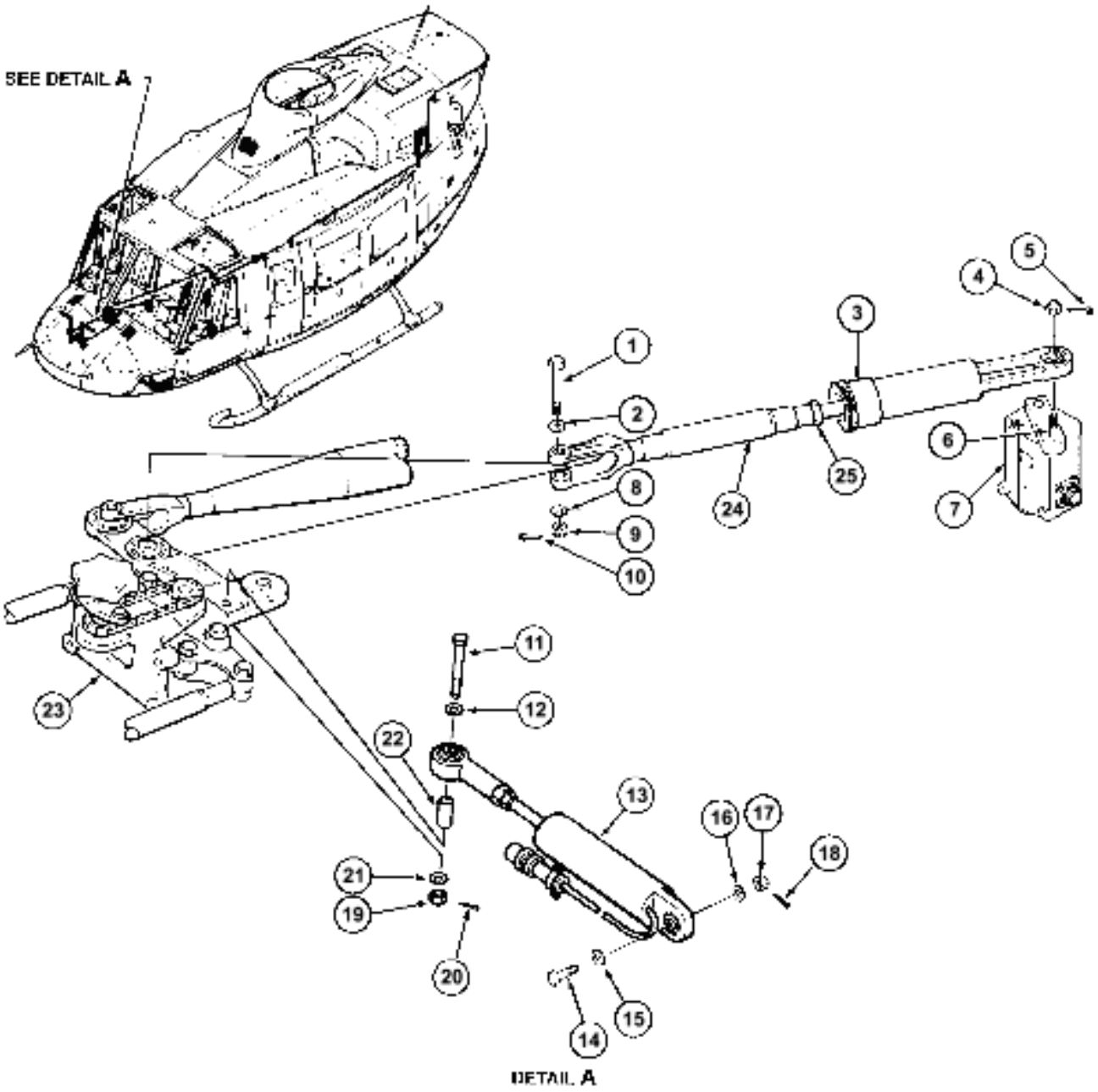
ICN-412-A-672000-B-97499-00035-A-001-01

Figure 1. Magnetic Brake (Sheet 1 of 1)

Control Transducer - Removal

Procedure

1. Remove access panel from cabin floor under pilot seat (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect electrical connector.
3. Remove cotter pin (20, Figure 1), nut (19) with washer (21) and remove bolt (11) and spacer (12).
4. Remove bolt (14) and aluminum washer (15) by removing cotter pin (18), nut (17), and spacer (16). Remove transducer (13).



1. Bolt
2. Aluminum washer
3. Force gradient
4. Nut
5. Cotter pin
6. Stud
7. Magnetic brake
8. Aluminum washer
9. Nut
10. Cotter pin
11. Bolt
12. Spacer
13. Transducer
14. Bolt
15. Aluminum washer
16. Washer
17. Nut
18. Cotter pin
19. Nut
20. Cotter pin
21. Washer
22. Spacer
23. Pedal adjuster assembly
24. Clevis
25. Nut

ICN-412-A-672000-B-97499-00036-A-001-01

Figure 1. Control Transducer and Force Gradient (Sheet 1 of 1)

Control Transducer - Inspection

Procedure

1. Inspect transducer for damage.
2. Inspect electrical cable and connectors for damage.
3. Inspect bearings for looseness (BHT-ALL-SPM, Standard Practices Manual).

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Control Transducer - Repair

Procedure

1. Transducer is nonrepairable and shall be replaced when unserviceable.

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Control Transducer - Installation

Procedure

1. Place transducer (13, Figure 1) into helicopter onto airframe structure. Install bolt (14) with aluminum washer (15), washer (16), nut (17), and secure with cotter pin (18).

2. Place spacer (22) and rod end bearing of transducer on pedal adjuster assembly (23). Install bolt (11), spacer (12), washer (21), nut (19) and cotter pin (20).

NOTE

SN: 33001-33213 or 36001-36036:

If not previously accomplished, perform transducer adjustment. refer to 412-A-67-01-00-06A-276A-A / 01327.

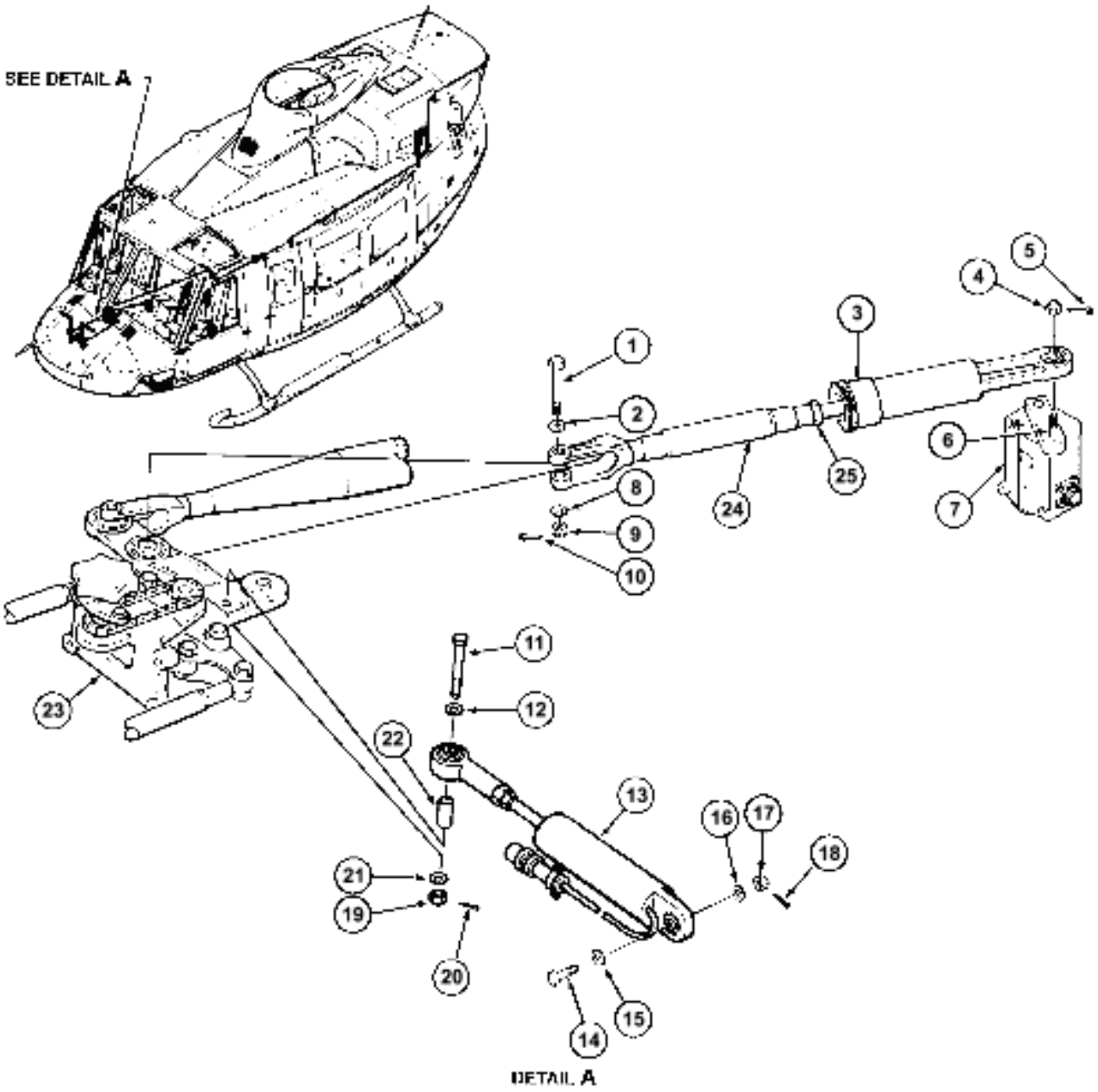
NOTE

SN: 36037-Subsequent:

Refer to 412-A-67-01-00-11A-276A-A / 01551.

3. Connect electrical connector.

4. Install access panel on floor under pilot seat (412-A-52-40-00-02A-720A-A / 00840).



1. Bolt
2. Aluminum washer
3. Force gradient
4. Nut
5. Cotter pin
6. Stud
7. Magnetic brake
8. Aluminum washer
9. Nut
10. Cotter pin
11. Bolt
12. Spacer
13. Transducer
14. Bolt
15. Aluminum washer
16. Washer
17. Nut
18. Cotter pin
19. Nut
20. Cotter pin
21. Washer
22. Spacer
23. Pedal adjuster assembly
24. Clevis
25. Nut

ICN-412-A-672000-B-97499-00036-A-001-01

Figure 1. Control Transducer and Force Gradient (Sheet 1 of 1)

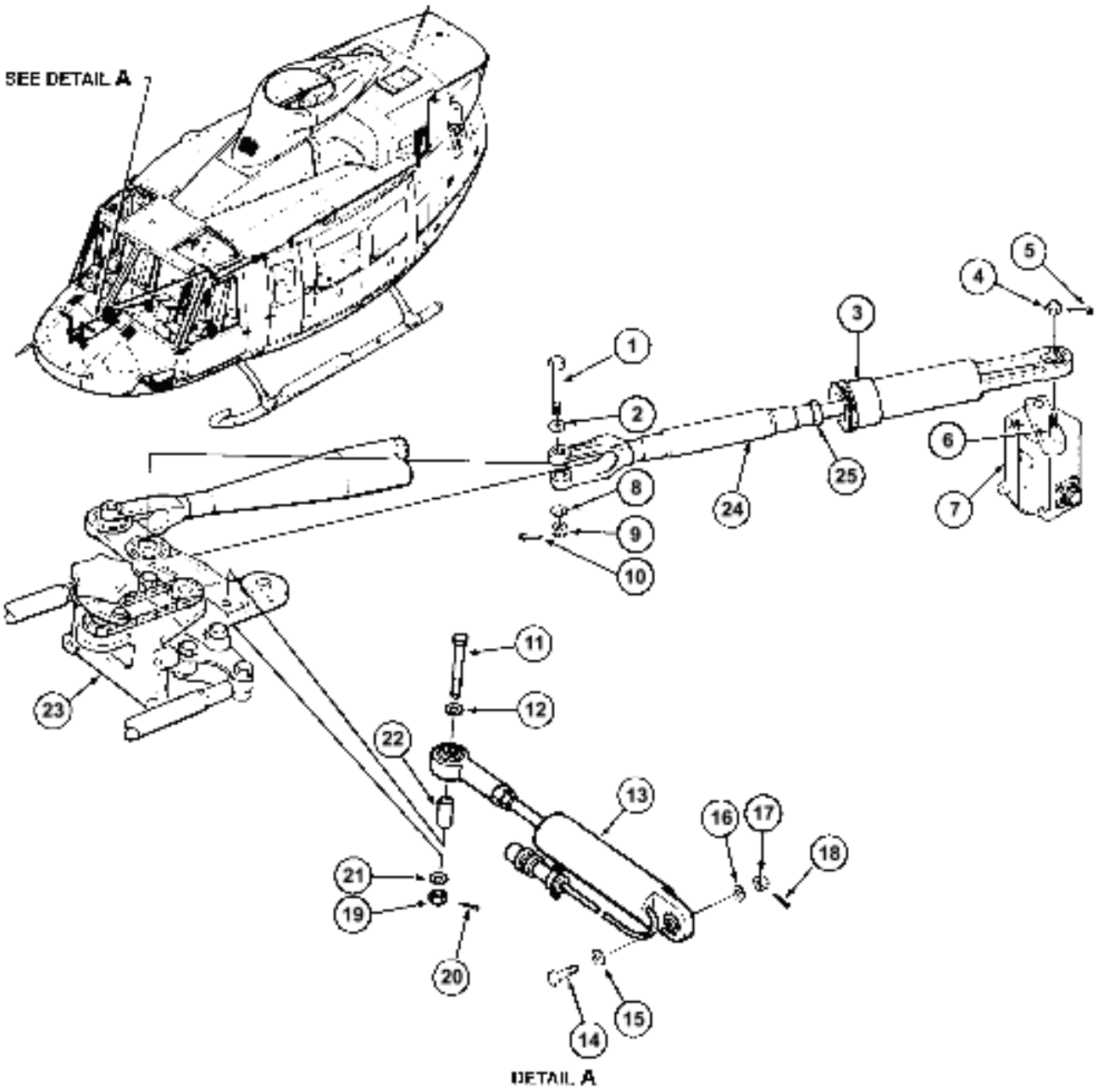
Force Gradient - Removal

Procedure

NOTE

The following procedure is applicable to anti-torque force gradients not incorporating a detent switch for anti-torque force gradients (incorporating a detent switch) used when the optional 4-axis AFCS/Flight Director kit is installed, refer to 412-A-67-12-02-02A-520A-A / 01573.

1. Remove access panel from cabin floor under pilot seat (412-A-52-40-00-02A-520A-A / 00837).
2. Remove cotter pin (10, Figure 1), nut (9), washer (8), and bolt (1) with washer (2) from pedal adjuster (23).
3. Remove cotter pin (5) and nut (4) from stud (6).
4. Remove force gradient (3) from magnetic brake (7).
5. Record length of force gradient assembly.
6. Loosen nut (25) and remove clevis (24).



1. Bolt
2. Aluminum washer
3. Force gradient
4. Nut
5. Cotter pin
6. Stud
7. Magnetic brake
8. Aluminum washer
9. Nut
10. Cotter pin
11. Bolt
12. Spacer
13. Transducer
14. Bolt
15. Aluminum washer
16. Washer
17. Nut
18. Cotter pin
19. Nut
20. Cotter pin
21. Washer
22. Spacer
23. Pedal adjuster assembly
24. Clevis
25. Nut

ICN-412-A-672000-B-97499-00036-A-001-01

Figure 1. Control Transducer and Force Gradient (Sheet 1 of 1)

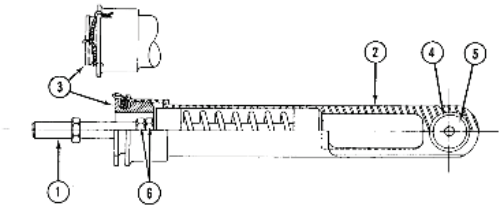
Force Gradient - Inspection

Procedure

1. Inspect force gradient assembly (Figure 1) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is not acceptable and shall be repaired or unit replaced.

2. Inspect bearing (5, Figure 1) and sleeve (4) for secure installation in housing (2) and for wear (BHT-ALL-SPM, Standard Practices Manual).

3. Inspect cap (3) for correct two way lockwire installation.



ICN-412-A-672000-B-97499-00037-A-001-01

Figure 1. Force Gradient Assembly (Sheet 1 of 1)

Force Gradient - Repair

Procedure

1. Repair of force gradient consists of replacement of unserviceable components and adjustment of spring preload in accordance with 412-A-67-20-00-10A-720A-A / 01531.

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Force Gradient - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

The following procedure is applicable to anti-torque force gradients not incorporating a detent switch. Anti-torque force gradients, which incorporate a detent switch, are used when the optional 4-axis AFCS/Flight Director kit is installed, refer to 412-A-67-12-02-02A-720A-A / 01576.

NOTE

Force gradient (3, Figure 1) is similar in appearance to force gradients used in cyclic controls but has a different part number. Ensure force gradient with correct part number is installed for anti-torque control system (BHT-412-IPB).

1. If not previously accomplished, adjust spring preload on force gradient as follows:

1.1. Cut and remove lockwire from cap (3, Figure 2).

1.2. Remove cap (3) and spring assembly (1) from housing (2).

1.3. Adjust inboard nut (6) to provide 2.5 to 3.0 pounds (11.12 to 13.34 N) preload against spring assembly (1). Tighten outboard nut (6) against inboard nut (6).

1.4. Position spring assembly (1) in housing (2). Install cap (3). Tighten cap until end play of spring is eliminated. Secure cap with lockwire (C-405) to prevent turning in either direction (Figure 2).



MAKE SURE FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO INSTALLATION ON MAGNETIC BRAKE.

2. Install clevis (24, Figure 1) on shaft of force gradient (3), adjust to previously recorded length, and secure with nut (25).

3. Position force gradient (3) on pedal adjuster assembly (23).

4. Install bolt (1), aluminum washer (2), aluminum washer (8), nut (9), and cotter pin (10).

5. If not previously accomplished, adjust length of force gradient as outlined in rigging instructions (412-A-67-01-00-06A-276A-A / 01327).

6. Position force gradient (3) on stud (6) of magnetic brake (7). Install nut (4) and cotter pin (5).

7. Move anti-torque controls through full travel and check for obstructions and binding.

8. Verify anti-torque rigging is within limits.

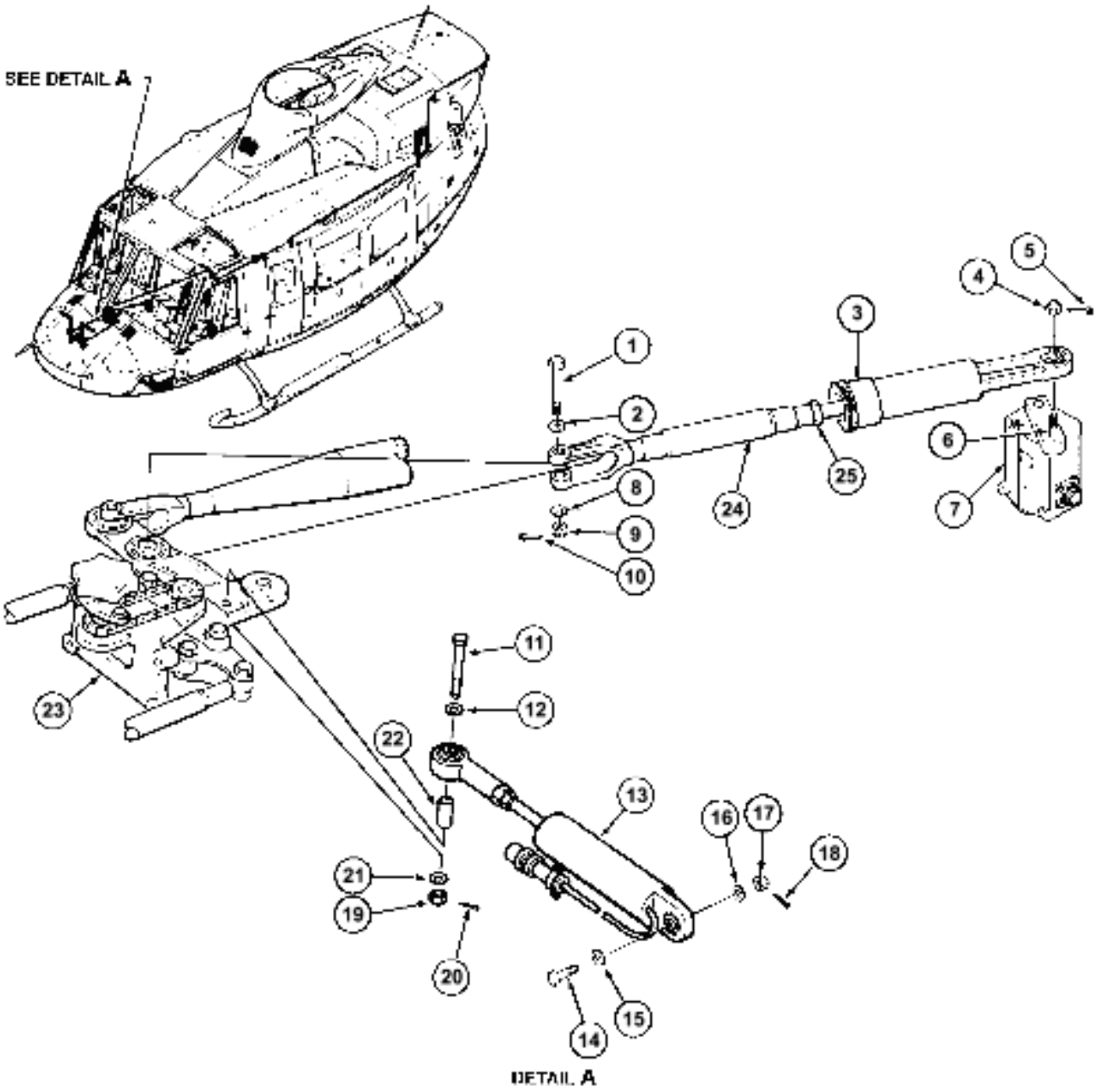
Applicable: SN: 36001-36086:

8.1. Refer to 412-A-67-01-00-06A-276A-A / 01327.

Applicable: SN: 36087-Subsequent:

8.2. Refer to 412-A-67-01-00-11A-276A-A / 01551.

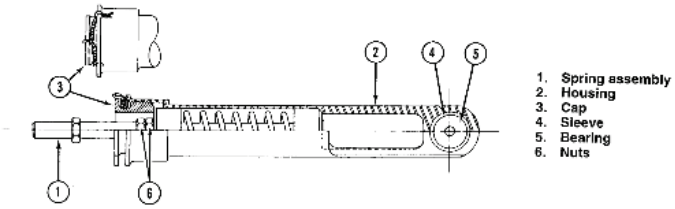
9. Install access panel on cabin floor under pilot seat (412-A-52-40-00-02A-720A-A / 00840).



1. Bolt
2. Aluminum washer
3. Force gradient
4. Nut
5. Cotter pin
6. Stud
7. Magnetic brake
8. Aluminum washer
9. Nut
10. Cotter pin
11. Bolt
12. Spacer
13. Transducer
14. Bolt
15. Aluminum washer
16. Washer
17. Nut
18. Cotter pin
19. Nut
20. Cotter pin
21. Washer
22. Spacer
23. Pedal adjuster assembly
24. Clevis
25. Nut

ICN-412-A-672000-B-97499-00036-A-001-01

Figure 1. Control Transducer and Force Gradient (Sheet 1 of 1)



ICN-412-A-672000-B-97499-00037-A-001-01

Figure 2. Force Gradient Assembly (Sheet 1 of 1)

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Applicable to: ALL
ECCN EAR99

Pedal Adjuster Assembly - Removal

Procedure

1. Remove access panel from cabin floor under pilot seat (412-A-52-40-00-02A-520A-A / 00837).

2. Remove access panel from bottom of helicopter under pilot cyclic control stick (412-A-52-40-00-02A-520A-A / 00837).

3. Remove pilot cyclic stick boot (412-A-67-12-00-19A-520A-A / 01467).

4. Disconnect flight control tube (3, Figure 1) from anti-torque control pedal adjuster assembly (41), as follows:

4.1. Remove cotter pin (6), nut (5), and aluminum washer (4).

4.2. Remove bolt (1) with aluminum washer (2) and disconnect flight control tube (3).

5. Disconnect force gradient tube (9), as follows:

5.1. Remove cotter pin (11), nut (12), and washer (10).

5.2. Remove bolt (7) with aluminum washer (8) and disconnect force gradient tube (9).

6. Disconnect transducer (15) from adjuster, as follows:

6.1. Remove cotter pin (31), nut (29), and washer (28).

6.2. Remove bolt (13), spacer (14), and spacer (21). Disconnect transducer (15).

7. Disconnect flight control tube (32), as follows:

7.1. Remove cotter pin (30), nut (27), and washer (26).

7.2. Remove bolt (22) with washer (23) and disconnect flight control tube (32).

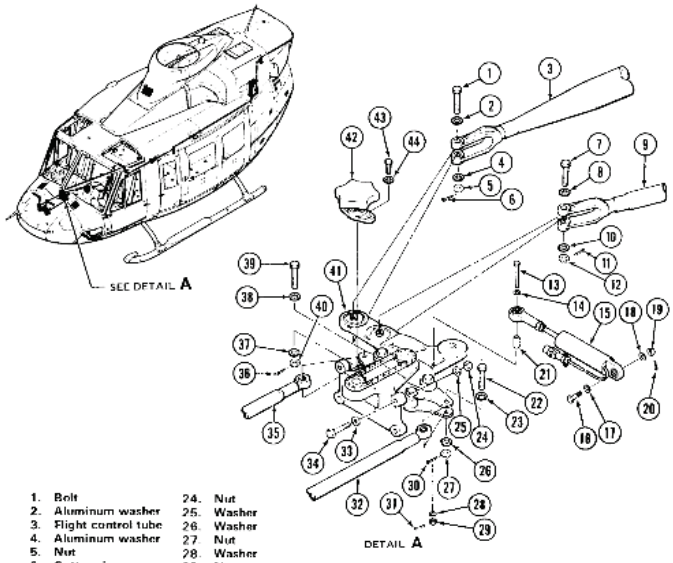
8. Disconnect flight control tube (35), as follows:

8.1. Remove cotter pin (36), nut (40), and aluminum washer (37).

8.2. Remove bolt (39) with aluminum washer (38) and disconnect flight control tube (35).

9. Remove nuts (24), washers (25 and 33), and bolts (34).

10. Remove anti-torque control pedal adjuster assembly (41) from helicopter.



1. Bolt
2. Aluminum washer
3. Flight control tube
4. Aluminum washer
5. Nut
6. Cotter pin
7. Bolt
8. Aluminum washer
9. Force gradient
10. Washer
11. Cotter pin
12. Nut
13. Bolt
14. Spacer
15. Transducer
16. Bolt
17. Aluminum washer
18. Spacer
19. Nut
20. Cotter pin
21. Spacer
22. Bolt
23. Washer
24. Nut
25. Washer
26. Washer
27. Nut
28. Washer
29. Nut
30. Cotter pin
31. Cotter pin
32. Flight control tube
33. Washer
34. Bolt
35. Flight control tube
36. Cotter pin
37. Aluminum washer
38. Aluminum washer
39. Bolt
40. Nut
41. Anti-torque control pedal adjuster assembly
42. Adjuster knob
43. Bolt
44. Washer

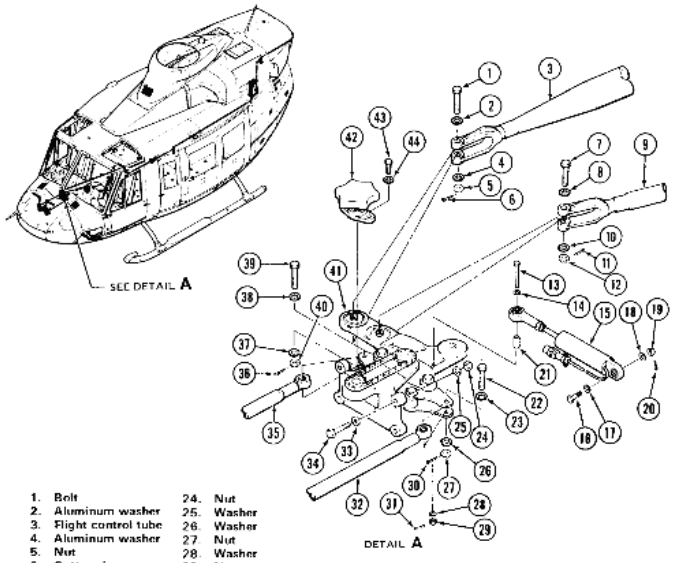
ICN-412-A-672000-B-97499-00038-A-001-01

Figure 1. Pedal Adjuster (Sheet 1 of 1)

Pedal Adjuster Assembly - Inspection

Procedure

1. Inspect components of anti-torque control pedal adjuster assembly (41, Figure 1) for mechanical and corrosion damage (BHT-412-CR&O).
2. Inspect chain for wear. Wear that can be detected visually is not acceptable.
3. Inspect gears for wear. Wear that can be detected visually is not acceptable.
4. Inspect bearings for secure installation and for wear (BHT-ALL-SPM, Standard Practices Manual).
5. Inspect bolt holes for elongation. Elongation that can be detected visually is not acceptable.



1. Bolt
2. Aluminum washer
3. Flight control tube
4. Aluminum washer
5. Nut
6. Cotter pin
7. Bolt
8. Aluminum washer
9. Force gradient
10. Washer
11. Cotter pin
12. Nut
13. Bolt
14. Spacer
15. Transducer
16. Bolt
17. Aluminum washer
18. Spacer
19. Nut
20. Cotter pin
21. Spacer
22. Bolt
23. Washer
24. Nut
25. Washer
26. Washer
27. Nut
28. Washer
29. Nut
30. Cotter pin
31. Cotter pin
32. Flight control tube
33. Washer
34. Bolt
35. Flight control tube
36. Cotter pin
37. Aluminum washer
38. Aluminum washer
39. Bolt
40. Nut
41. Anti-torque control pedal adjuster assembly
42. Adjuster knob
43. Bolt
44. Washer

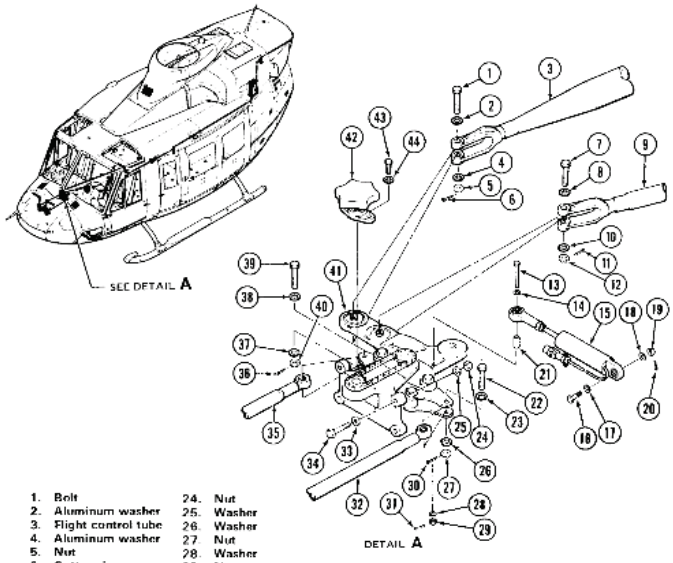
ICN-412-A-672000-B-97499-00038-A-001-01

Figure 1. Pedal Adjuster (Sheet 1 of 1)

Pedal Adjuster Assembly - Installation

Procedure

1. Place adjuster assembly (41, Figure 1) in position in helicopter. Install four bolts (34), washers (33 and 25), and nuts (24).
2. Place tubes (35 and 32) in pedal adjuster assembly (41). Install bolts (39 and 22), washers (38 and 23), washers (37 and 26), nuts (40 and 27), and cotter pins (36 and 30).
3. Place force gradient (9) on adjuster assembly (41). Install bolt (7), washers (8 and 10), nut (12), and cotter pin (11).
4. Place control tube (3) on pedal adjuster assembly (41). Install bolt (1), washers (2 and 4), nut (5), and cotter pin (6).
5. Place spacer (21) and rod end bearing of transducer (15) on pedal adjuster assembly (41). Install bolt (13), spacer (14), washer (28), nut (29) and cotter pin (31).
6. Install pilot cyclic stick boot (412-A-67-12-00-25A-720A-A / 01489).
7. Move anti-torque controls through full travel and check for interference and binding.
8. Verify anti-torque rigging within limits (412-A-67-01-00-06A-276A-A / 01327).
9. Install access panel on bottom of helicopter below pilot seat (412-A-52-40-00-02A-720A-A / 00840).
10. Install access panel on cabin floor under plot seat (412-A-52-40-00-02A-720A-A / 00840).



1. Bolt
2. Aluminum washer
3. Flight control tube
4. Aluminum washer
5. Nut
6. Cotter pin
7. Bolt
8. Aluminum washer
9. Force gradient
10. Washer
11. Cotter pin
12. Nut
13. Bolt
14. Spacer
15. Transducer
16. Bolt
17. Aluminum washer
18. Spacer
19. Nut
20. Cotter pin
21. Spacer
22. Bolt
23. Washer
24. Nut
25. Washer
26. Washer
27. Nut
28. Washer
29. Nut
30. Cotter pin
31. Cotter pin
32. Flight control tube
33. Washer
34. Bolt
35. Flight control tube
36. Cotter pin
37. Aluminum washer
38. Aluminum washer
39. Bolt
40. Nut
41. Anti-torque control pedal adjuster assembly
42. Adjuster knob
43. Bolt
44. Washer

ICN-412-A-672000-B-97499-00038-A-001-01

Figure 1. Pedal Adjuster (Sheet 1 of 1)

Pedals - Description

NOTE

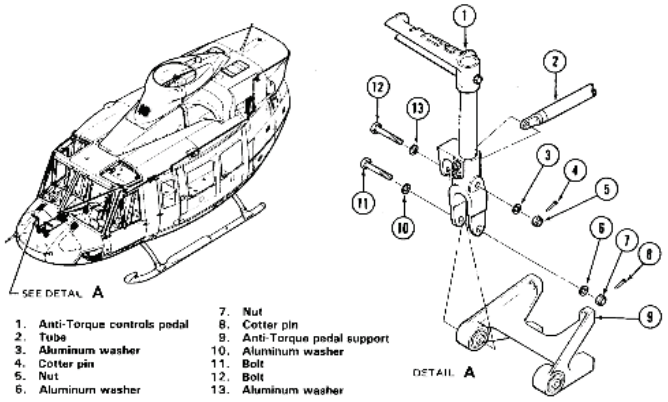
Maintenance procedure for anti-torque control right pedal is given. The procedure for left pedal is similar.

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Pedals - Removal

Procedure

1. Remove cotter pin (4, Figure 1), nut (5), washer (3), and bolt (12) with washer (13).
2. Remove cotter pin (8), nut (7), washer (6) and bolt (11) with washer (10). Remove anti-torque control pedal (1).



1. Anti-Torque controls pedal
2. Tube
3. Aluminum washer
4. Cotter pin
5. Nut
6. Aluminum washer
7. Nut
8. Cotter pin
9. Anti-Torque pedal support
10. Aluminum washer
11. Bolt
12. Bolt
13. Aluminum washer

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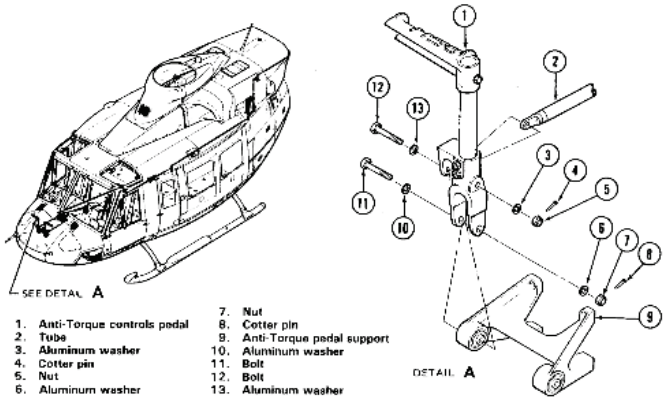
Figure 1. Anti-Torque Pedals (Sheet 1 of 1)

Pedals - Inspection

Procedure

1. Inspect pedal (1, Figure 1) for mechanical and corrosion damage. Superficial mechanical damage is acceptable without repair. Superficial corrosion damage is acceptable if repaired (BHT-412-CR&O).

2. Inspect bushings for elongation and security. Damage that can be detected visually is not acceptable.



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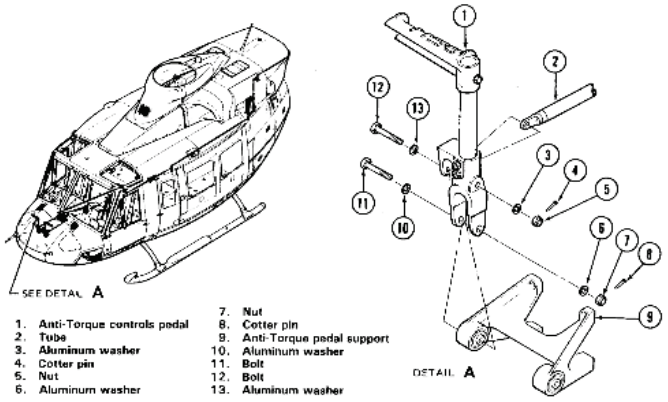
Figure 1. Anti-Torque Pedals (Sheet 1 of 1)

Pedals - Installation

Procedure

1. Place pedal (1, Figure 1) on anti-torque pedal support (9). Install bolt (11), washers (10 and 6), nut (7), and cotter pin (8).

2. Place tube (2) into pedal (1). Install bolt (12), washers (13 and 3), nut (5), and cotter pin (4).

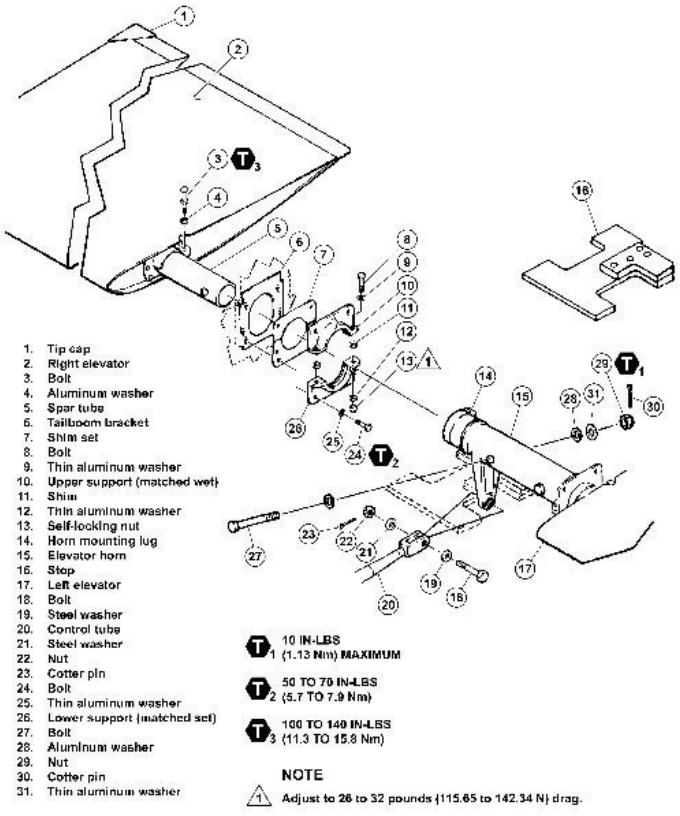


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Figure 1. Anti-Torque Pedals (Sheet 1 of 1)

Aerodynamically Actuated Elevator - Description

The aerodynamically actuated elevator installation (Figure 1) consists of left and right elevator assemblies (2 and 17), a horn assembly (15), two horn support sets (10 and 26), and spring-loaded control tube (20). The horn is mounted in the support set which secures horn to tailboom structure and permits horn to move to the extent allowed by the spring-loaded control tube. Each elevator is built up on a spar tube which is inserted into end of horn and is secured by two bolts (27). Each of the two bolts is installed into holes drilled through both the elevator spar and the horn assembly to provide a fail safe condition for the elevator.



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Figure 1. Aerodynamically Actuated Elevator and Horn (Sheet 1 of 1)

Aerodynamically Actuated Elevator - Removal

Procedure

NOTE

Perform inspections described in steps 1. through 4. of 412-A-67-02-00-01A-280A-A / 01541 prior to removing elevators.

1. Remove access panels from tailboom in area below elevators (412-A-52-40-00-02A-520A-A / 00837).

2. Remove bolt (3, Figure 1) and washer (4).

3. Remove cotter pin (30), nut (29), washers (28), bolt (27) and washer (31) from horn (15).

4. Pull elevator (2) straight outboard until spar tube (5) disengages from elevator horn (15).

5. Remove elevator (17) in same manner outlined in Step 2., Step 3., and Step 4..

6. Remove cotter pin (23), nut (22), washer (21), and bolt (18) with washer (19). Disconnect tube (20).



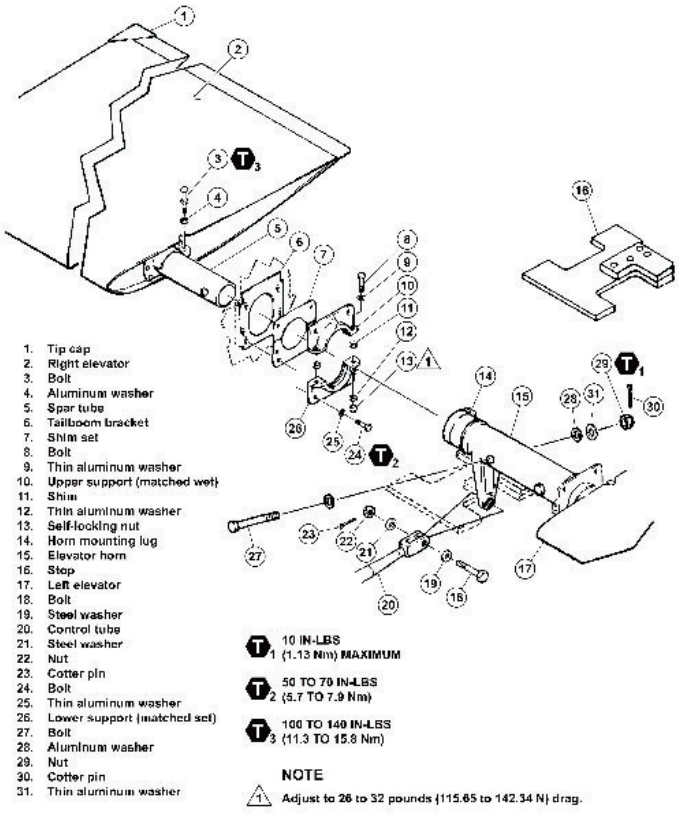
HANDLE SUPPORTS (10 AND 26) WITH CARE AFTER REMOVAL TO AVOID DAMAGE TO BUSHINGS THAT ARE BONDED TO SUPPORTS.

7. Remove self-locking nuts (13), washers (12) and bolts (8) and washers (9). Remove bolts (24) and washers (25).

8. Remove supports (10 and 26) and shims (7 and 11). Index shims (7 and 11) and supports (10 and 26) for installation as set in same location.

9. Remove supports on opposite end of horn (15). Index shims and supports as a set in same manner outlined in Step 8..

10. Remove horn (15) from helicopter.



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Figure 1. Aerodynamically Actuated Elevator and Horn (Sheet 1 of 1)

Aerodynamically Actuated Elevator - Inspection

Procedure

NOTE

Prior to removal of elevator, perform inspection outlined in Step 1.. through Step 4..

1. Inspect for loose rivets and/or cracks in skin at attachment points for bracket (6, Figure 1) and similar bracket on opposite side. Press down on each elevator (2 and 17) trailing edge to cause elevator to defect maximum of 0.040 in. (1.02 mm) and at same time inspect for loose rivets. It is recommended that a dial indicator be used to measure deflection. If loose rivets or cracks are found, refer to 412-A-67-02-00-01A-685A-A / 01542 for repair procedure.

2. Measure axial play of elevator horn (15) as follows:

2.1. Mount dial indicator on tailboom so stylus of indicator touches inboard edge of elevator near bolt (3).

2.2. Apply light force to move elevator inboard and outboard spanwise, as shown in Figure 2. Record dial indicator reading. Acceptable axial play is 0.005 to 0.030 inch (0.127 to 0.762 mm). If axial play is not within limits, adjust shims (7, Figure 1), as outlined in 412-A-67-02-00-01A-720A-A / 01543.

2.3. Remove dial indicator.

3. Measure radial play at elevator horn (15) as follows:

3.1. Mount dial indicator on tailboom, so stylus of indicator touches upper surface of right elevator (2) near bolt (3).

NOTE

Do not use heavy force to move elevator up and down when accomplishing the following step or false dial indicator readings will result.

3.2. Apply light force to move elevator up and down to measure radial play, as shown in Figure 2. Record dial indicator reading. Maximum acceptable radial play is 0.010 inch (0.254 mm).

3.3. Measure radial play on opposite elevator in same manner outlined in Step 3.1. and Step 3.2.. Remove dial indicator.

3.4. If radial play on either elevator exceeds limits, adjust shims (11, Figure 1) as outlined in 412-A-67-02-00-01A-720A-A / 01543.

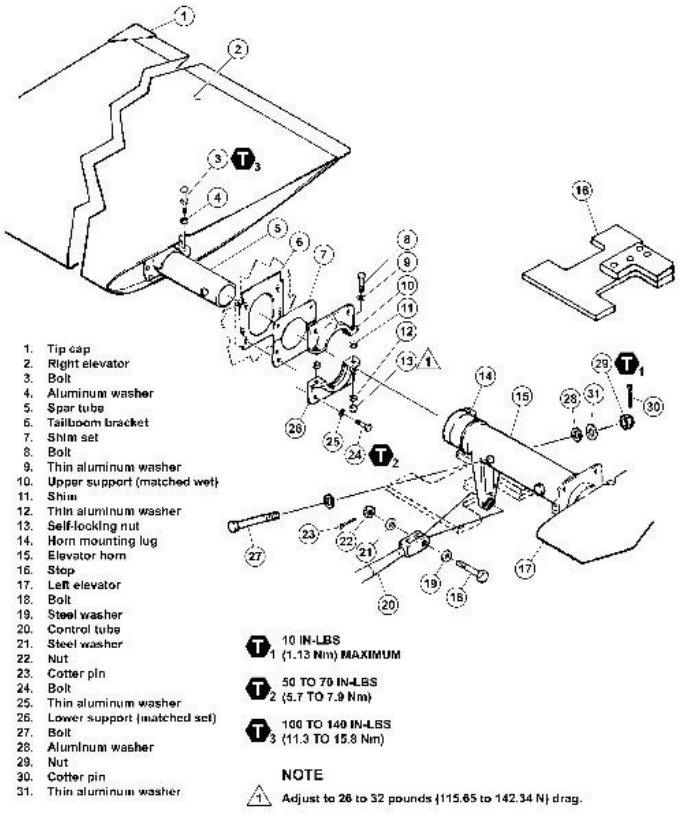
4. Inspect spring action of control tube assembly (20). Refer to 412-A-67-02-00-02A-280A-A / 01545 for inspection procedure.

5. Inspect elevators (2 and 17) as follows:

5.1. Inspect for loose and/or missing rivets, cracks, and holes in skin. Minor damage that does not result in distortion of elevator surface is acceptable if repaired as outlined in 412-A-67-02-00-01A-685A-A / 01542.

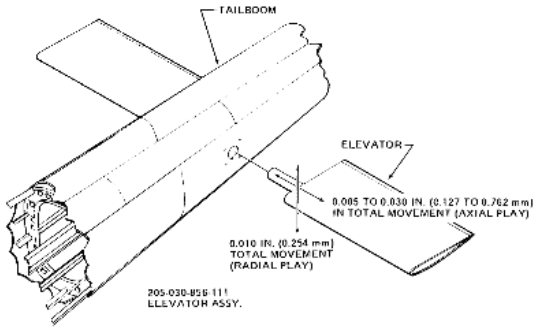
5.2. Inspect tip cap (1) and similar tip cap on left elevator for security.

6. For detailed inspection criteria, refer to the BHT-412-CR&O.



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Figure 1. Aerodynamically Actuated Elevator and Horn (Sheet 1 of 1)



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Figure 2. Elevator Inspection (Installed) (Sheet 1 of 1)

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Aerodynamically Actuated Elevator - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Aliphatic naphtha	C-305	AR	
Toluene	C-306	AR	
Adhesive	C-310	AR	

Procedure

1. Repair minor damage to skins and structure. Refer to the BHT-412-CR&O for materials.

2. Repair loose or missing elevator tip cap (1, Figure 1) and similar tip cap on left elevator (17) as follows:

2.1. Clean bonding area on elevator tip using aliphatic naphtha (C-305).

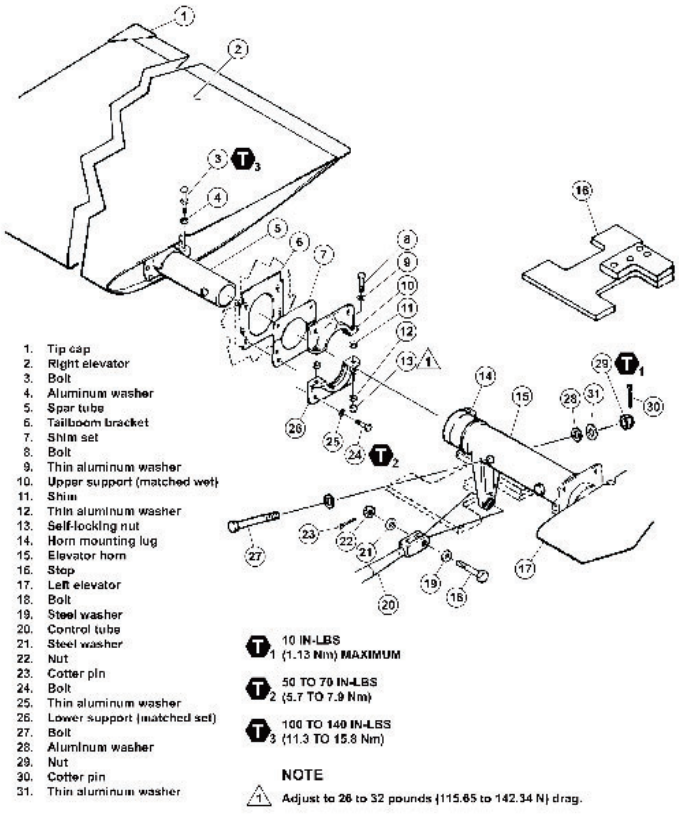
2.2. Clean inside of tip cap (1) using toluene (C-306).

2.3. Brush thin coat of thermosetting adhesive (C-310) about 0.010 inch (0.254 mm) thick inside tip cap (1).

2.4. Allow about 1 hour drying time for adhesive to get tacky, then install tip cap (1).

2.5. Hold tip cap (1) in place at least 15 seconds.

2.6. Allow to dry at least 24 hours prior to flight or operation.



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Figure 1. Aerodynamically Actuated Elevator and Horn (Sheet 1 of 1)

Aerodynamically Actuated Elevator - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Push-Pull Dial Gauge (Spring Scale)	DPPH200	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-104	AR	
Epoxy primer	C-202	AR	
Epoxy polyamide primer	C-204	AR	
Drycleaning solvent	C-304	AR	
MEK	C-309	AR	
Zinc chromate primer	C-201	AR	

Procedure

NOTE

If horn (15, Figure 1) and control tube (20) were previously installed, proceed to Step 12..

1. Place horn (15) in tailboom with horn arm on right side and pointing down, as illustrated.



HANDLE SUPPORTS (10 AND 26) CAREFULLY TO AVOID DAMAGE TO BUSHINGS, THAT ARE BONDED TO SUPPORTS.

2. Install supports (10 and 26), and shims (11) on horn (15) with two bolts (8), washers (9 and 12), and nuts (13). Install nuts snugly but do not torque at this time.

3. Install supports and shims on opposite end of horn in same manner outlined in Step 2..

4. Position shim set (7) and supports (10 and 26) on bracket (6) and install bolts (24) and washers (25). Install bolts snugly but do not torque at this time.

5. Install supports and shims on opposite end of horn (15) in same manner outlined in Step 4..

6. Measure axial play of horn (15) as outlined in 412-A-67-02-00-01A-280A-A / 01541, step 2. If play is not within acceptable limits, adjust thickness of shim set (7). Snug bolts (24) but do not torque at this time.

7. Install two nuts (13). Loosen two similar nuts in opposite end of horn (15).

8. Attach spring scale to hole in arm of horn (15) where control tube (20) is normally installed. Pull scale at right angle to arm and record force required to rotate horn. Acceptable force is 13 to 16 lbs. (57.82 to 71.17 N). If force is not within limits, adjust thickness of shims (11) (add shims if force is too great, subtract shims if force is less than acceptable).

9. After shims (11) are adjusted within acceptable limits, tighten two nuts corresponding to nut (13), but on left end of horn (15).

10. Repeat force check with spring scale as outlined in Step 8., but now acceptable force to rotate horn with both supports tight is 26 to 32 lbs. (115.65 to 142.34 N). If force is not within limits, adjust thickness of shims on left end of horn (15) that are opposite shims (11) (add shims if force is too great, subtract shims if force is less than acceptable).

11. After shims (11) are adjusted within acceptable limits as specified in Step 10., torque bolts (24) 50 to 70 in.-lbs. (5.7 to 7.9 N-m).

NOTE

If forward end of control tube (20) has not been previously installed on tube support fitting on bulkhead, install tube (20) on fitting (412-A-67-02-00-02A-720A-A / 01546).

12. Position control tube (20) on elevator horn (15). Install bolt (18), washers (19 and 21), and nut (22). Install cotter pin (23).

NOTE

The only authorized organic finishes for elevator spar tubes (5) are epoxy primer (C-202) and epoxy polyamide primer (C-204).

13. Prior to installation, clean both elevator spar tubes (5) and horn (15) as follows:

13.1. If either tube (5) has been erroneously painted or primed with zinc chromate primer (C-201) or any material other than epoxy primer (C-202) or epoxy polyamide primer (C-204), remove all traces of existing paint or primer with clean cloth moistened with MEK (C-309). Also, clean portion of horn (15) that is contacted by either tube (5). If bare metal is exposed, touch up with epoxy primer (C-202) or epoxy polyamide primer (C-204). Allow primer to thoroughly and completely dry before installing elevator into horn.

13.2. If tubes (5) are coated only with epoxy primer (C-202), clean tubes (5) and portion of horn (15) that is contacted by either tube (5) with drycleaning solvent (C-304).

14. Apply thin coat of corrosion preventive compound (C-104) to spar tube (5) on elevators and to internal surfaces of horn (15).

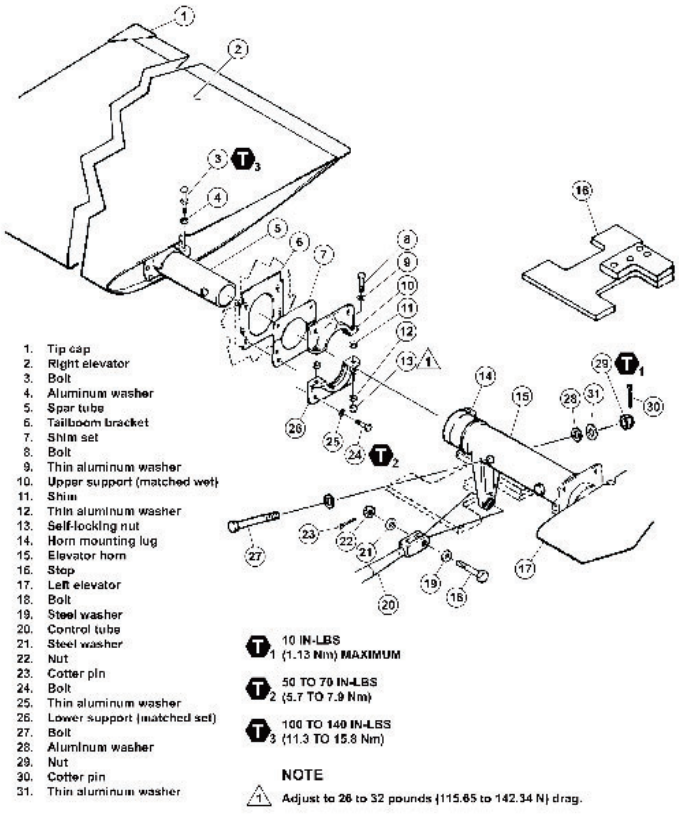
15. Insert spar tube (5) into horn and align horn mounting lug (14) for bolt (3). Install bolt (3) and washer (4). Torque bolt 100 to 140 in.-lbs. (11.3 to 15.8 N-m).

16. Install bolt (27) with washer from horn (15), washer (28), and nut (29). Torque nut (29) to 10.0 in.-lbs. (1.13 N-m) maximum. Install cotter pin (30).

17. Install elevator (17) in same manner outlined in Step 13.. through Step 16..

18. Inspect spring action of control tube (20) (412-A-67-02-00-02A-280A-A / 01545).

19. Install access doors on bottom of tailboom below elevators (412-A-52-40-00-02A-720A-A / 00840).



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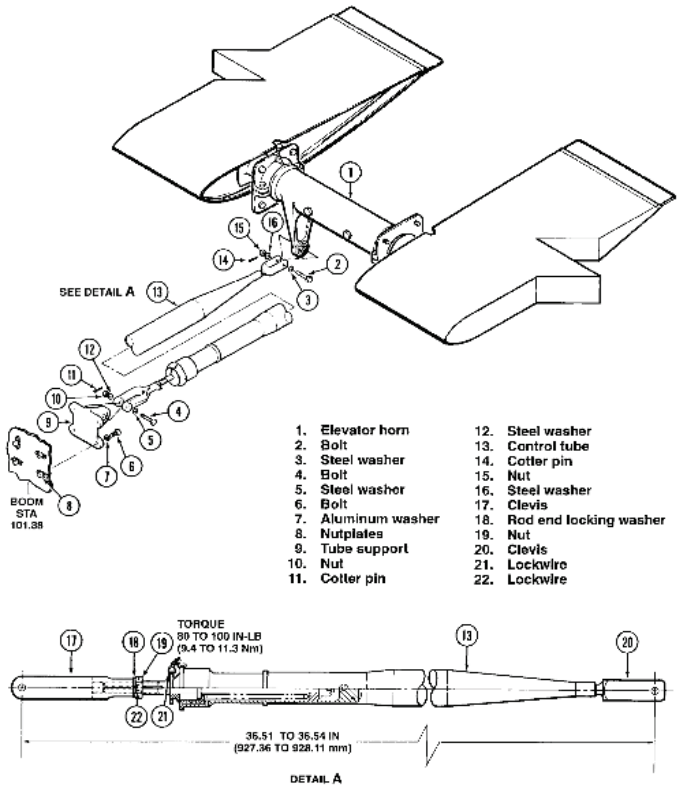
Figure 1. Aerodynamically Actuated Elevator and Horn (Sheet 1 of 1)

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Elevator Control Tube - Removal

Procedure

1. Remove access door from tailboom in area below elevators (412-A-52-40-00-02A-520A-A / 00837).
2. Remove cotter pin (14, Figure 1), nut (15), washer (16) and bolt (2) and washer (3).
3. Remove cotter pin (11), nut (10), washer (12), bolt (4), and washer (5) and remove control tube (13) from helicopter.



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Figure 1. Elevator Control Tube (Sheet 1 of 1)

Elevator Control Tube - Inspection

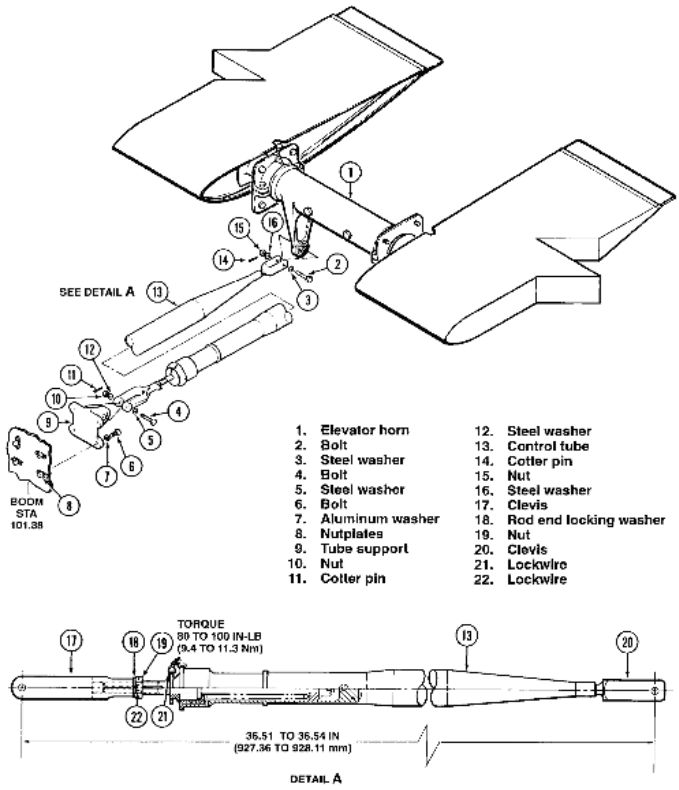
Procedure

NOTE

Perform inspections described in step 1. of 412-A-67-02-00-02A-520A-A / 01544 prior to removing elevators.

1. Inspect operation of tube (13, Figure 1). Move elevator to approximately 3° nose down position and slowly release. Spring action of tube (13) should return elevator to original position. Failure to return to original position, binding, and/or lock-up are not acceptable. If any of these conditions exist, adjust length of tube as shown in detail A. If this does not correct discrepancy, replace tube (13).

2. Inspect support (9, Figure 1) (BHT-412-CR&O).



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Figure 1. Elevator Control Tube (Sheet 1 of 1)

Elevator Control Tube - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-104	AR	

Procedure

1. If control tube (13, Figure 1) was not previously installed in same helicopter, adjust length of control tube (13) to dimensions shown on detail A. Align clevis (17) and clevis (20) so bolt holes are parallel. Tighten nut (19) but do not install lockwire (22) at this time.

2. Apply corrosion preventive compound (C-104) to nut (19) and adjacent threads.

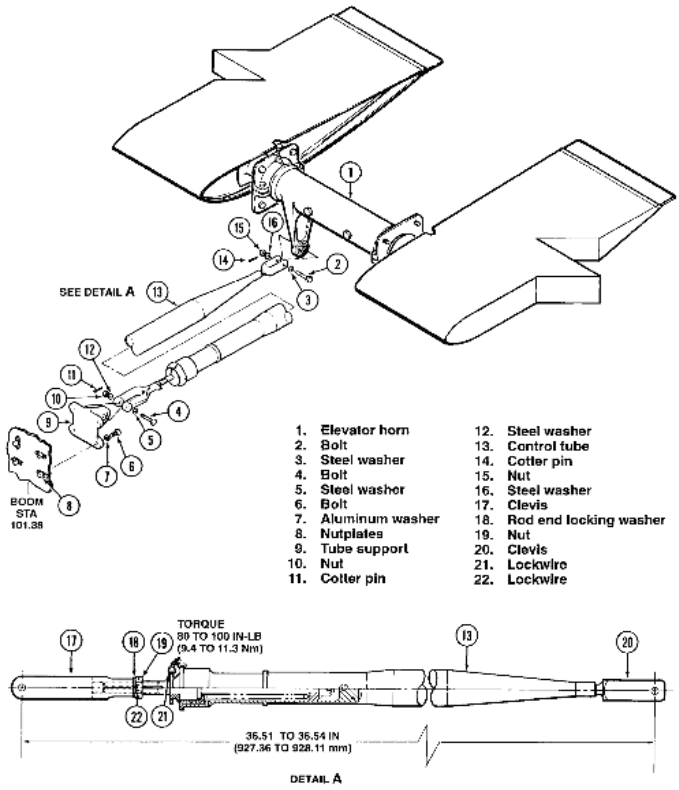
3. Position tube (13) in tailboom with non-adjustable clevis (20) on elevator horn (1). Install bolt (2), washers (3 and 16) and nut (15). Install cotter pin (14).

4. Align adjustable clevis (17) on tube support (9) and install bolt (4), washers (5 and 12), and nut (10). Do not install cotter pin (11) at this time.

5. Rig elevator controls (412-A-67-01-00-07A-276A-A / 01328).

6. After rigging is complete, ensure cotter pins (11 and 14) and lockwires (21 and 22) are installed.

7. Install access doors (412-A-52-40-00-02A-720A-A / 00840).



ICN-412-A-670200-B-97499-00003-A-001-01

Figure 1. Elevator Control Tube (Sheet 1 of 1)

Digital AFCS Flight Controls - Description

The following paragraphs contain information and instructions for maintenance of the Digital Automatic Flight Control System (DAFCS) which includes the standard flight controls with added components in the cyclic, directional and collective control systems.

Cyclic control system incorporates four electrical servo actuators, four transducers, a force gradient assembly and an electromechanical rotary actuator.

Anti-toque directional system incorporates two electrical servo actuators, two transducers, a force gradient assembly and a magnetic brake.

There are two transducers in the collective control system.

Forward electrical servo actuators in the cyclic and directional systems are controlled by Autopilot Flight Control Computer (FCC) number 1 (AP1). Aft actuators are controlled by AP2.

If the optional 3-axis Flight Director Kit is installed there are no changes to the flight control system.

If the optional 4-axis Flight Director kit is installed: It includes a collective trim system consisting of an electromechanical rotary actuator and a force gradient with a detent switch connected in parallel to the primary collective controls. The rotary actuator repositions the collective controls in response to a signal from either the pilot or the AFCS system. Actuation of the force gradient detent switch either by the rotor actuator driving to its internal stop or by a pilot control input will disengage the rotary actuator. In the directional control system the standard force gradient is replaced by a force gradient incorporating a detent switch and the magnetic brake is replaced by an electromechanical rotary actuator.

Refer to 412-A-96-10-00-00A-028A-A / 02172 for further description of the Digital Automatic Flight Control System and 412-A-98-00-00-13B-051A-A / 02573 for DAFCS system wiring diagrams.

For additional information on the Digital Automatic Flight Control System refer to Honeywell SPZ-7600 Integrated Flight Control System Bell 412 System Maintenance Manual.

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Flight Controls - Rigging

Procedure

1. Rigging procedures for 412EP incorporating the Digital Automatic Flight Control System (DAFCS) are the same as AFCS rigging except where noted. Refer to 412-A-67-01-00-00A-042A-A / 01322.

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Collective Controls - Rigging

Procedure

1. Basic rigging procedures for the collective control system in helicopters are the same as rigging instructions in 412-A-67-01-00-01A-276A-A / 01323. Adjustment of the dual element control motion transducers is accomplished in accordance with 412-A-67-11-00-18A-720C-A / 01394.

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Cyclic Controls - Rigging

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Calibration Tool (Pot Calibrator)	412-274-001-101	1	

Procedure

NOTE

Basic rigging procedures for the cyclic Digital Automatic Flight Control System (DAFCS) are the same as AFCS rigging except where noted. Refer to 412-A-67-01-00-03A-276A-A / 01324.

NOTE

For maintenance procedures of cyclic dual controls, refer to the BHT-412-SI-13 and to the BHT-412-IPB for illustrated parts list.

1. Perform fore and aft cyclic control motion transducer (2, Figure 1, Detail A) adjustment as follows:

1.1. Position pilot cyclic stick (1) perpendicular to the deck within degree longitudinally. Loosen jam nut (5) on fore and aft cyclic transducer (2) so the transducer shaft (7) can rotate freely.

NOTE

Dual element control motion transducers (412-074-101-103) are aligned and positioned using calibration tool (pot calibrator) (412-274-001-101). This calibration tool is used to electrically align the control motion transducers installed on the helicopter to be used as control position sensors for the flight data recorder. This tool will ensure the control motion transducers are aligned within ±0.5% of the prescribed position. Fore and aft position is set at 50%. Lateral position is set at 50%.

1.2. Using calibration tool (9) calibration tool (pot calibrator) (412-274-001-101) turn calibration power switch (10) to the off position. Connect calibration box connector (8) to one of the fore and aft cyclic transducers (2) electrical connector (6). Turn selector knob (11) to fore and aft (F/A) position.

1.3. Turn calibration power switch (10) to on position.

NOTE

Any time the LO BAT lamp comes on, replace batteries (use 9 VDC alkaline batteries). Low battery is set to illuminate at 12 ±1 volt.

1.4. If a RED lamp comes on, the transducer shaft (7) shall be adjusted in the direction indicated. An extend indication means extend the transducer shaft (7) by screwing it into rod end bearing (4) (in clockwise direction when looking at rod end). A retract indication means retract the transducer shaft by unscrewing it from the rod end bearing (in counterclockwise direction when looking at the rod end). Adjust transducer shaft until GREEN IN RANGE lamp comes on.

1.5. Tighten jam nut (5) on transducer shaft (7) and verify that GREEN IN RANGE lamp is on and RED lamps are off.

1.6. Move calibration power switch (10) to off position. Disconnect calibration box connector (8) from transducer electrical connector (6).

1.7. Connect transducer electrical connector (6) to electrical connector on the bulkhead.

1.8. Perform calibration adjustment on remaining fore and aft cyclic transducer (2) using Step 1.1. through Step 1.7..

2. Alternate method for fore and aft cyclic transducer (2) adjustment is as follows:

2.1. Connect hydraulic power cart and turn on.

2.2. Install fore and aft cyclic transducers (2) on structure only. Place pilot cyclic stick (1) against extreme forward stick stop, then to extreme aft stick stop and check to see if transducers would bottom out if they were connected to jackshaft. If bottoming would occur, adjust rod end bearings (4) on transducers until bottoming is eliminated, then connect transducers to the jackshaft. Overtravel of transducers should be approximately equal.

2.3. Position pilot cyclic stick perpendicular to the deck within 0.5° longitudinally and laterally. Loosen jam nut on transducer so shaft can rotate freely.

2.4. Measure and record resistance (T) between pins D and F of the transducer electrical connector. Resistance should be 5000 ± 10% ohms.

2.5. Multiply resistance (T) x 50% = __. Record value of T.

2.6. Measure and record resistance (R) between pins E and F of the transducer electrical connector. Adjust the transducer shaft until resistance (R) equals the value of Step 2.5. ± 25 ohms.

2.7. Tighten jam nut on the transducer shaft and verify that the resistance is still within limits.

2.8. Connect the transducer electrical connector to the bulkhead.

2.9. Turn off hydraulic power cart and disconnect from helicopter.

3. Perform lateral cyclic transducer (3) adjustment as follows:

3.1. Position pilot cyclic stick (1) perpendicular to deck within 0.5° longitudinally and laterally. Loosen jam nut (5) on lateral cyclic transducer (3) so transducer shaft (7) can rotate freely.

3.2. Using calibration tool (pot calibrator) (412-274-001-101) turn calibration power switch (10) to off position. Connect calibration box connector (8) to one of the lateral cyclic transducer (3) electrical connectors (6). Turn selector knob (11) to LAT position.

3.3. Perform calibration adjustment on lateral cyclic transducer (3) by repeating Step 1.3. through Step 1.7..

3.4. Perform calibration adjustment on remaining lateral cyclic transducer (3) by repeating Step 1.3. through Step 1.7..

4. Alternate method for lateral cyclic transducer (3) adjustment is as follows:

4.1. Connect hydraulic power cart and turn on.

4.2. Install lateral cyclic transducers (3) on structure only. Place pilot cyclic stick against extreme left lateral stick stop and check to see if lateral cyclic transducers would bottom out if they were connected to bellcrank. If bottoming would occur, adjust rod end bearings on lateral cyclic transducers until bottoming is eliminated, then connect lateral cyclic transducers to the bellcrank. Overtravel of lateral cyclic transducers should be approximately equal.

4.3. Position pilot cyclic stick perpendicular to the deck within 0.5° longitudinally and laterally. Loosen jam nut on lateral cyclic transducer (3) so shaft can rotate freely.

4.4. Measure and record resistance (T) between pins D and F of the lateral cyclic transducer (3) electrical connector (6). Resistance should be 5000 ± 10% ohms.

4.5. Multiply resistance (T) x 49.7% = __. Record value of T.

4.6. Measure and record resistance (R) between pins E and F of the lateral cyclic transducer (3) electrical connector (6). Adjust the transducer shaft until resistance (R) equals the value of Step 4.5. ± 25 ohms.

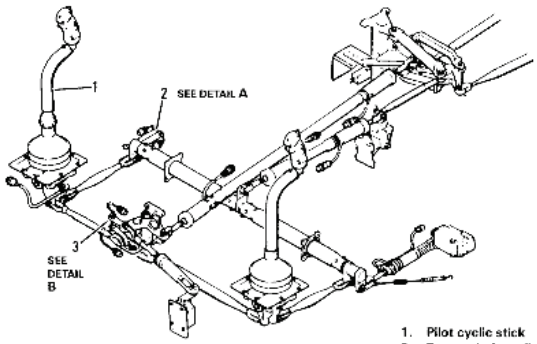
4.7. Tighten jam nut on the lateral cyclic transducer (3) shaft and verify that the resistance is still within limits.

4.8. Connect the lateral cyclic transducer (3) electrical connector (6) to bulkhead.

4.9. Turn off hydraulic power cart and disconnect from helicopter.

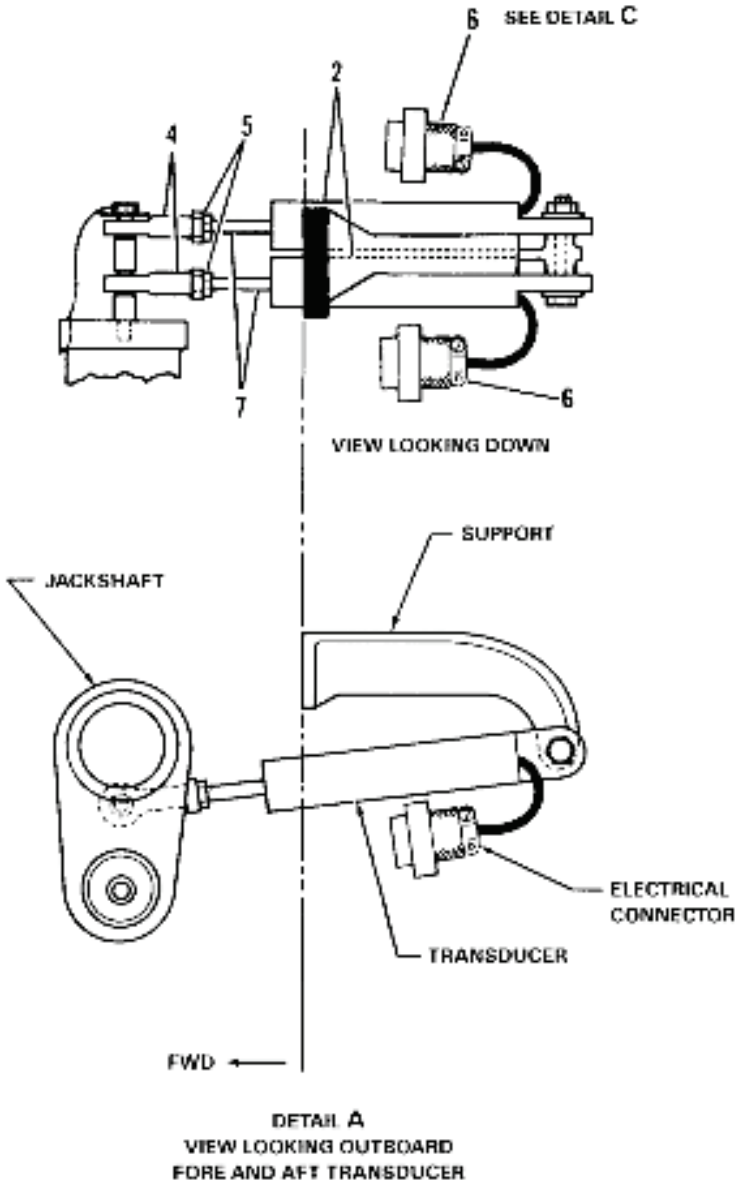
5. Install cyclic fore and aft force gradient and adjust spring preload (412-A-67-11-00-02A-520A-A / 01333, 412-A-67-11-00-02A-280A-A / 01334 and 412-A-67-11-00-02A-720A-A / 01335).

6. Install cyclic lateral force gradient and adjust spring preload (412-A-67-11-00-03A-520A-A / 01336).



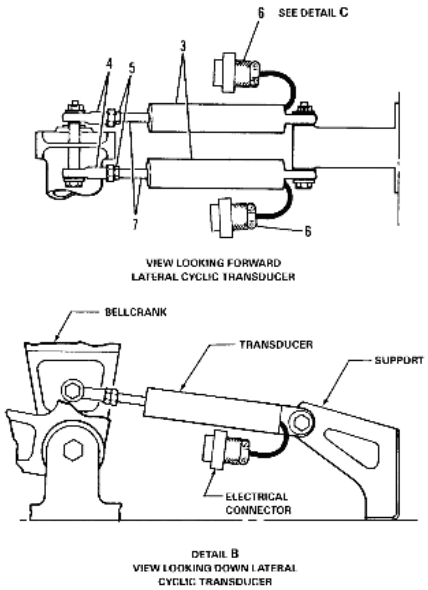
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Figure 1. Cyclic Controls Rigging (36087 and Subsequent) (Sheet 1 of 4)



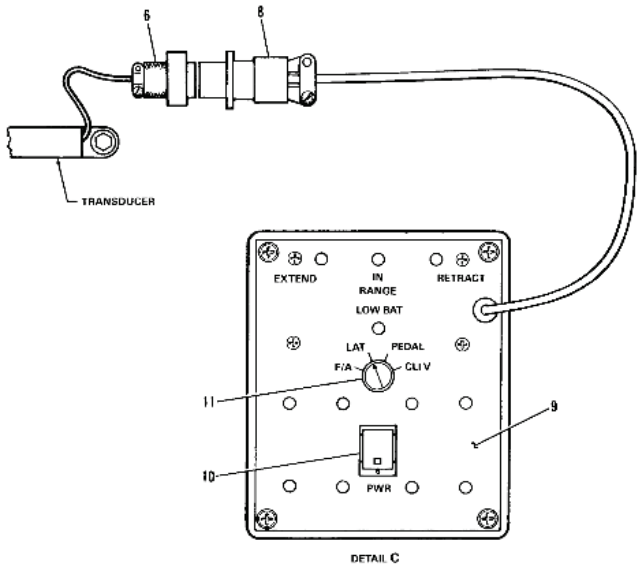
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Figure 1. Cyclic Controls Rigging (36087 and Subsequent) (Sheet 2 of 4)



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Figure 1. Cyclic Controls Rigging (36087 and Subsequent) (Sheet 3 of 4)



4. Rod end bearing
5. Jam nut
6. Electrical connector
7. Transducer shaft
8. Calibration box connector
9. Calibration tool (412-274-001-101)
10. Calibration power switch
11. Selector knob

ICN-412-A-670100-B-97499-00034-A-001-01

Figure 1. Cyclic Controls Rigging (36087 and Subsequent) (Sheet 4 of 4)

Anti-Torque Controls - Rigging

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Tool Assembly	412-201-700-101	1	
Tool Assembly	412-201-701-101	1	
Tail Rotor Rigging Tool	T103206-101	1	
Spring Scale	Commercial	1	Range to measure 4 to 5 pounds (17.80 to 22.24 N)

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Corrosion preventive compound	C-104	AR	
Sealant	C-308	AR	
Lockwire	C-405	AR	

Procedure

NOTE

Rigging procedures contained herein, pertain to helicopters with DAFCS and dual controls installed in basic controls systems.

Refer to 412-A-52-40-00-02A-520A-A / 00837 for removal of required access panel.

1. Comply with the following general procedures when rigging anti-torque controls:

1.1. When a rigging procedure requires a specific position to accomplish the following steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

1.2. Tolerance for rigging dimensions is ±0.03 inch (0.76 mm).

1.3. All adjustable control tubes must have a maximum of 1.0 inch (2.54 mm) of exposed threads after adjustment. If control tubes have inspection holes, threads shall be engaged sufficiently to cover holes.

1.4. Apply corrosion preventive compound (C-104) to threads of all adjustable rod bearings and clevises after adjustment.

1.5. Torque control jam nuts to 80 to 100 inch-pounds (9.04 to 11.30 Nm) after adjustment unless other torque is specified.

1.6. Control tubes must be free to rotate several degrees about longitudinal axis. Loosen jam nuts, adjust end fittings, and torque jam nuts if necessary.

2. Disconnect the following control tubes and force gradient.

2.1. Disconnect anti-torque force gradient (14, Figure 1) (412-A-67-20-00-10A-520A-A / 01528) and anti-torque transducers (20) (412-A-67-20-00-09A-520A-A / 01524) from pilot pedal adjuster (21).

2.2. Disconnect control tube (24) from pilot pedal adjuster (21) (412-A-67-20-00-11A-720A-A / 01534).

2.3. Disconnect control tube (26) from bellcrank (25) (412-A-67-01-00-06A-276A-A / 01327).

2.4. Disconnect link (4) from lever (3) (412-A-67-20-00-01A-520A-A / 01494).

3. Verify length of centered series actuator (10), 62.865 to 62.915 inches (1596.77 to 1598.04 mm).

4. Connect hydraulic power cart to helicopter. Do not turn on the hydraulic power cart until directed.

5. Position copilot and pilot left directional pedals (18 and 22) against extreme forward stop. Adjust control tube (16) to fit, torque jam nut, and install control tube.

6. Connect control tube (24) to pilot pedal adjuster (21) (412-A-67-20-00-11A-720A-A / 01534).

7. Retract two stop bolts on bellcrank (9, Detail A).

8. Install tool assembly (412-201-701-101) (27, Detail B) on pilot pedals (22 and 23).

9. Remove bolt (28, Detail A) that secures series actuator (10) to bellcrank (9).

10. Remove bolt (31) which secures bellcrank (9) to support (8).

11. Retract both anti-rotation screws (32) in tool assembly (412-201-700-101) (29).

12. Position tool assembly (412-201-700-101) (29) against the inboard surface of support (8). Install bolt (NAS6604-30D) through tool assembly, support, and bellcrank (9), as shown in Detail A, but do not completely tighten at this time. Rotate tool assembly body until the indexing pin (30) engages the center slot of the tool assembly.

13. Adjust both tool assembly (29) anti-rotation screws (32) to make contact with the support (8) as shown in Detail A. As the anti-rotation screws are tightened, ensure indexing pin (30) is free to disengage and re-engage the center slot in the tool assembly body, then tighten bolt (NAS6604-30D).

14. Remove tool assembly (412-201-701-101) (27) from the pilots left and right pedals (22 and 23, Detail C).

15. Retract indexing pin (30) until it clears tool assembly (412-201-700-101) (29) body.

16. Slowly move pilots left pedal (22) forward until indexing pin (30) engages the forward slot in tool assembly (29) body. Adjust lower forward stop bolt in bellcrank (9) to the left pedal stop within 0.000 to 0.002 inch (0.000 to 0.051 mm) as shown in Detail A.

17. Retract indexing pin (30) until it clears tool assembly (29) body.

18. Slowly move pilots right pedal (23) forward until indexing pin (30) engages the aft slot in tool assembly (29) body. Adjust upper aft stop in bellcrank (9) to the right pedal stop within 0.000 to 0.002 inch (0.000 to 0.051 mm) as shown in Detail A.

19. Remove tool assembly (412-201-700-101) (29) and install bolt (28) and bolt (31) that were removed in Step 10. and Step 11..

20. Position and hold pilots left pedal (22) forward against stop.

NOTE

TB 412-01-173: Post:

Adjustment of pitch change link length not required for fixed length pitch link.

21. Check two pitch links (2, Detail E) for 6.180 ±0.010 inches length (156.97 ±0.254 mm) between bolt hole centers.

22. Remove bolt (34) retaining pitch link to crosshead (1).

23. Install tail rotor rigging tool (T103206-101) (35) to crosshead (1) and retain with bolt (34). Position tail rotor rigging tool (T103206-101) as shown in Figure 1 against tail rotor hub trunnion (37). Hold in this position by pushing inboard on tail rotor crosshead.

NOTE

If tail rotor rigging tool (T103206-101) (35) is not available, position crosshead (1) to trunnion (37) to 4.334 ±0.010 inches (110.08 ±0.254 mm).

24. Adjust link (4, Detail F) to obtain 0.58 inch (14.73 mm) clearance with bellcrank (6). Torque jam nut on link, and install link on bellcrank and lever (3). Turn hydraulic power cart on.

25. With pilot left pedal (22) positioned (Step 21.), and link (4) adjusted (Step 25.), adjust control tube (26) to fit to bellcrank (25). Torque jam nut and install control tube. Turn hydraulic power cart off and disconnect from helicopter.

NOTE

Alternate method of rigging directional controls without pressurizing the hydraulic system: With pilot left pedal (22) positioned (Step 21.), the hydraulic series actuator valve bottomed by pushing down on actuator piston rod, and link (4) adjusted (Step 25.), adjust tube (26) to fit to bellcrank (25), then shorten 1/2 turn of clevis. Torque jam nut and install control tube.

26. Remove tail rotor rigging tool (35) and install pitch link (2) to crosshead (1).

27. Flap tail rotor assembly to one extreme and check for clearance between blade and tail rotor yoke using a 0.010 inch (0.254 mm) feeler gauge (Figure 2). Perform clearance check on the same blade with tail rotor flapped to opposite extreme. Accomplish clearance check on opposite blade in the same manner.

28. If clearance is grater than 0.010 inch (0.254 mm), rigging has been performed correctly.

29. Rig anti-torque trim system as follows:

29.1. Install tool assembly (412-201-701-101) (27) on pilot left and right pedals (22 and 23, Detail B).

29.2. Position output arm (33) on magnetic brake or rotary actuator (15), as shown in Detail C.

29.3. Place output arm (33) of magnetic brake or rotary actuator(15) at center of travel, adjust force gradient (14) to fit the pilot pedal adjuster (21). Torque jam nut and install force gradient on pedal adjuster.

29.4. Check clearance between force gradient (14) and structure at extreme position of the output arm (33). Re-index output arm relative to shaft by one serration, if required for clearance, then repeat Step 29.2. through Step 29.4..

29.5. Remove tool assembly (27) from pilot left and right pedals (22 and 23, Detail B).

30. Perform anti-torque control motion transducer (20, Detail D) adjustment as follows:

30.1. Install transducers (20) to structure only. Place pilot pedal against one stop, then the other and check to see if transducers would bottom out in either retracted or extended position if transducers were connected to bellcrank (48). If bottoming would occur, adjust rod end bearing (46) on transducers until bottoming is eliminated. Then connect transducers to bellcrank. Overtravel of transducers should be approximately equal.

NOTE

Dual element control motion transducers 412-074-101-103 are aligned and positioned using calibration tool 412-274-001-101.

30.2. Position pilot left pedal forward against left pedal stop.

NOTE

This calibration tool 412-274-001-101 is used to electrically align the control motion transducers installed on the helicopter. The transducers are used as control position sensors for the flight data recorder. This tool will insure that the control motion transducers are aligned within ±0.5% of the prescribed position. Pedal position is set at 12%.

30.3. Using calibration tool (9, Figure 3) 412-274-001-101 (pot calibrator) turn calibrator power switch (10) to the off position. Connect calibrator box connector (8) to one of the transducers (20, Figure 1) electrical connector (42). Turn selector knob (11, Figure 3) to pedal position.

30.4. Turn calibrator power switch (10) to on position.

NOTE

Any time the LO BAT lamp comes on, replace batteries (use 9 VDC alkaline batteries). Low battery is set to illuminate at 12 ±1 volt.

30.5. If a RED lamp comes on, the transducer shaft shall be adjusted in the direction indicated. An extend indication means extend the transducer shaft by screwing it into rod end bearing (46, Figure 1) (clockwise looking at rod end). A retract indication means retract the transducer shaft by unscrewing it from the rod end bearing. Adjust transducer shaft until GREEN, IN RANGE lamp comes on.

30.6. Tighten jam nut (41) on transducer shaft (7, Figure 3) and verify that GREEN, IN RANGE lamp is on and RED lamps are off.

30.7. Move calibrator power switch (10) to off position. Disconnect calibrator box connector (8) from transducer electrical connector (42, Figure 1).

30.8. Connect transducer electrical connector (42) to electrical connector on the bulkhead.

30.9. Perform calibration adjustment on remaining transducer (20) using the previous Step 30.1. through Step 30.8..

31. Make sure that all anti-torque controls bolts are installed and cotter pins, sealant (C-308), lockwire (C-405) is installed where applicable.

32. Alternate method anti-torque control motion transducer (20, Detail D) adjustment as follows:

32.1. Install transducers (20) to structure only. Place pilots pedal against one stop, then the other and check to see if transducers would bottom out in either retracted or extended position if transducers were connected to bellcrank (48).

If bottoming would occur, adjust rod end bearing (46) on transducers until bottoming is eliminated. Then connect transducers to bellcrank. Overtravel of transducers should be approximately equal.

32.2. Position pilot left pedal forward against left pedal stop.

32.3. Measure and record resistance (T) between pins D and F of the transducer electrical connector. Resistance should be 5000 ±10% ohms.

32.4. Multiply resistance (T) x 12.0% = Record value of T.

32.5. Measure and record resistance (R) between pins E and F of the transducer electrical connector. Adjust the transducer shaft until resistance (R) equals the value of Step 32.4. ±25 ohms.

32.6. Tighten jam nut on the transducer shaft and verify that the resistance is still within limits.

32.7. Connect the transducer electrical connector to bulkhead.

33. Adjust anti-torque minimum friction as follows:

33.1. Disconnect force gradient (14) from pedal adjuster (21).

33.2. Disconnect control tube (12) from walking beam (13).

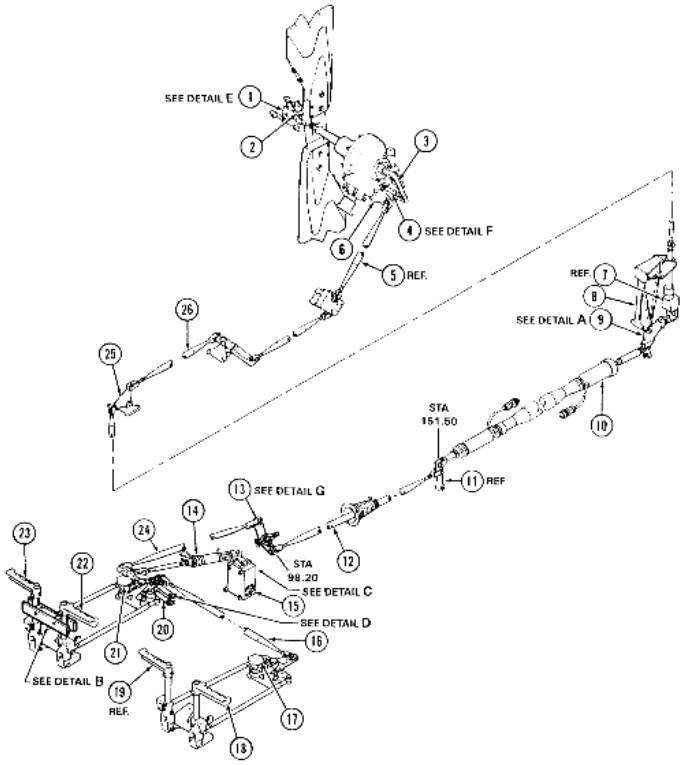
33.3. Attach spring scale to one pilot anti-torque pedal. Measure force required to move pedal through neutral. Acceptable force range is 4.25 to 4.75 pounds (18.91 to 21.13 N). Adjust screw (39, Detail G) on clamp (38) to obtain friction within acceptable range.

33.4. Install control tube (12) on walking beam (13).

33.5. Install force gradient (14) on pedal adjuster (21).

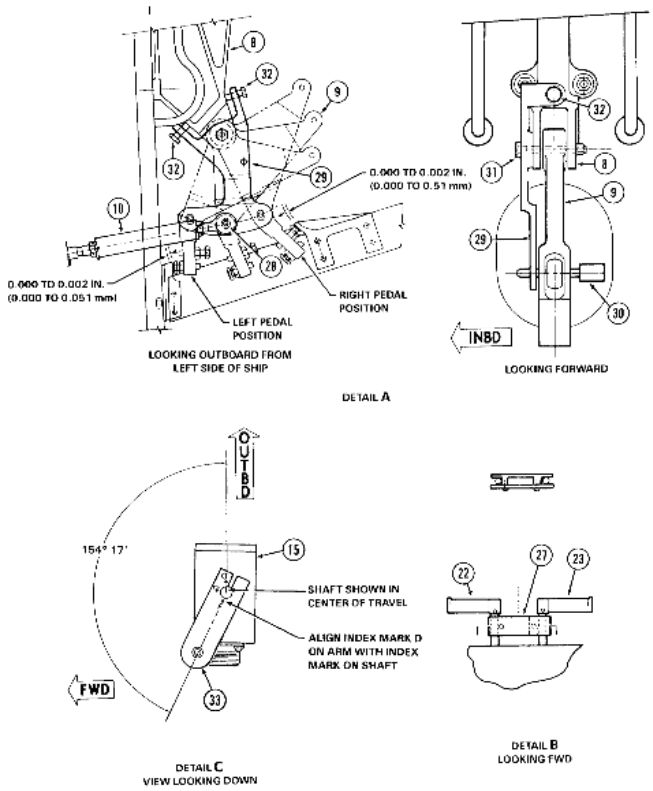
34. Move anti-torque controls through full travel and check for binding and/or obstruction of control movement.

35. Make sure that all anti-torque controls bolts are installed and cotter pins, sealant (C-308), lockwire (C-405) is installed where applicable.



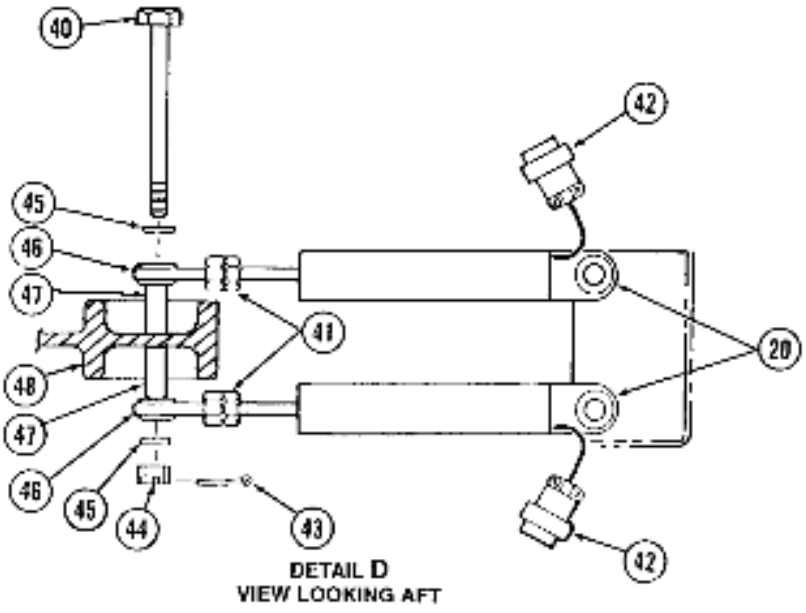
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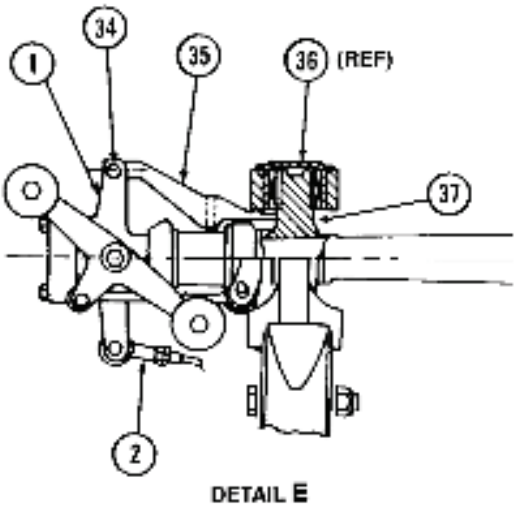
Figure 1. Anti-Torque Controls Rigging (36087 and Subsequent) (Sheet 1 of 4)

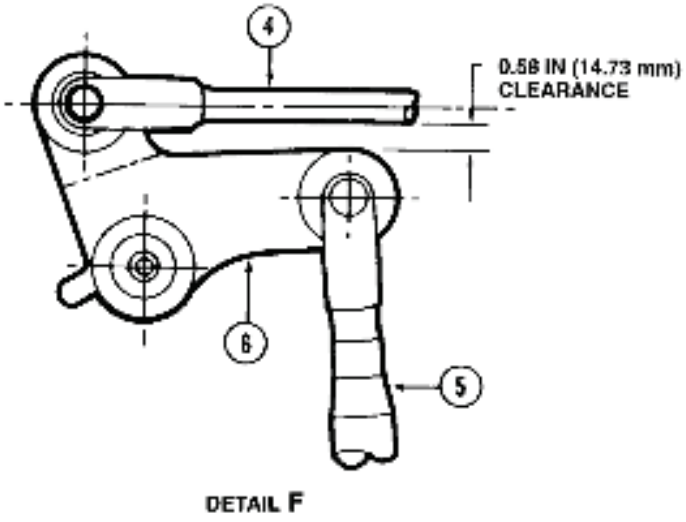


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Figure 1. Anti-Torque Controls Rigging (36087 and Subsequent) (Sheet 2 of 4)

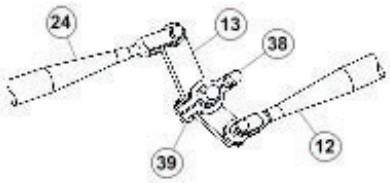






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Figure 1. Anti-Torque Controls Rigging (36087 and Subsequent) (Sheet 3 of 4)



DETAIL G

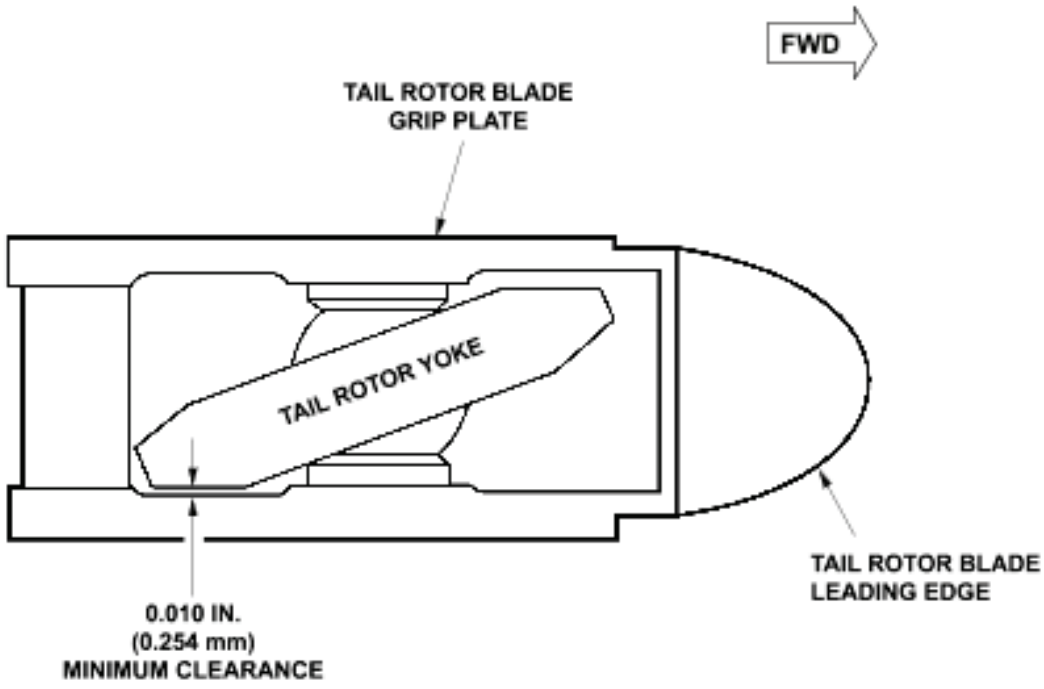
1. Crosshead
2. Pitch links
3. Lever
4. Link
5. Control tube
6. Bellcrank
7. Hydraulic cylinder
8. Support
9. Bellcrank
10. Series actuator
11. Lever
12. Control tube
13. Walking beam
14. Force gradient
15. Magnetic brake/rotary actuator 1
16. Control tube
17. Copilot pedal adjuster
18. Copilot left pedal
19. Copilot right pedal
20. Transducers
21. Pilot pedal adjuster
22. Pilot left pedal
23. Pilot right pedal
24. Control tube
25. Bellcrank
26. Control tube
27. Tool assembly (412-201-701-101)
28. Bolt (20-057-4-18D)
29. Tool assembly (412-201-700-101)
30. Indexing pin
31. Bolt (20 057 4 23D)
32. Anti-rotation screws
33. Output arm
34. Bolt
35. Tail rotor rigging tool (T103206-101)
36. Tail rotor
37. Tail rotor hub trunnion
38. Clamp
39. Screw
40. Bolt
41. Jam nuts
42. Electrical connectors
43. Cotter pin
44. Nut
45. Spacers
46. Rod end bearings
47. Spacers
48. Bellcrank

NOTE

1 Rotary actuator used when optinal 4-axis AFCS/flight director kit installed.

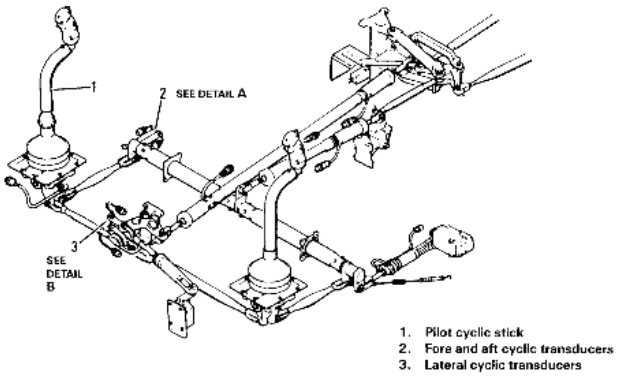
ICN-412-A-670100-B-97499-00038-A-001-01

Figure 1. Anti-Torque Controls Rigging (36087 and Subsequent) (Sheet 4 of 4)



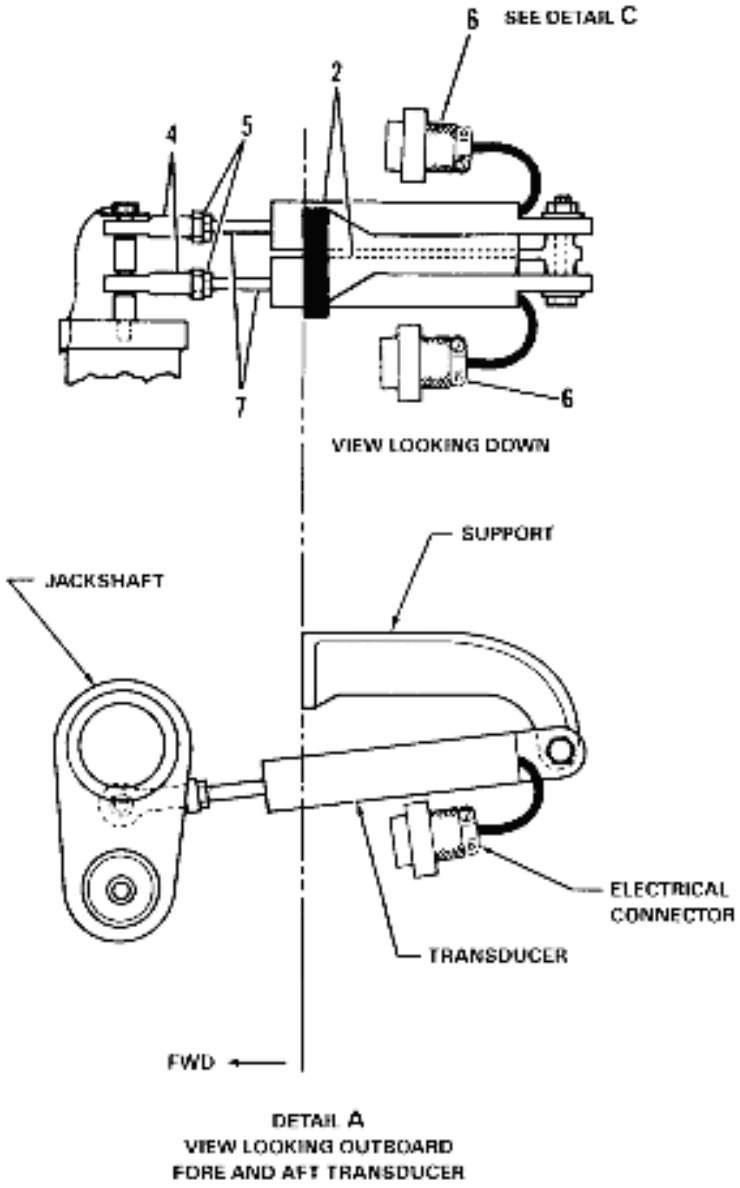
ICN-412-A-670100-B-97499-00027-A-001-01

Figure 2. Tail Rotor Blade to Yoke Clearance (Sheet 1 of 1)



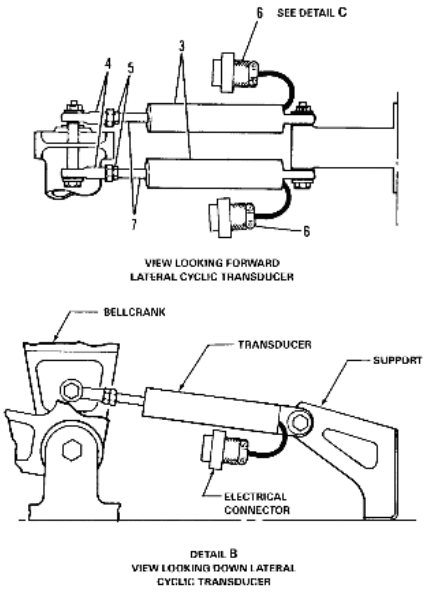
ICN-412-A-670100-B-97499-00031-A-001-01

Figure 3. Cyclic Controls Rigging (36087 and Subsequent) (Sheet 1 of 4)



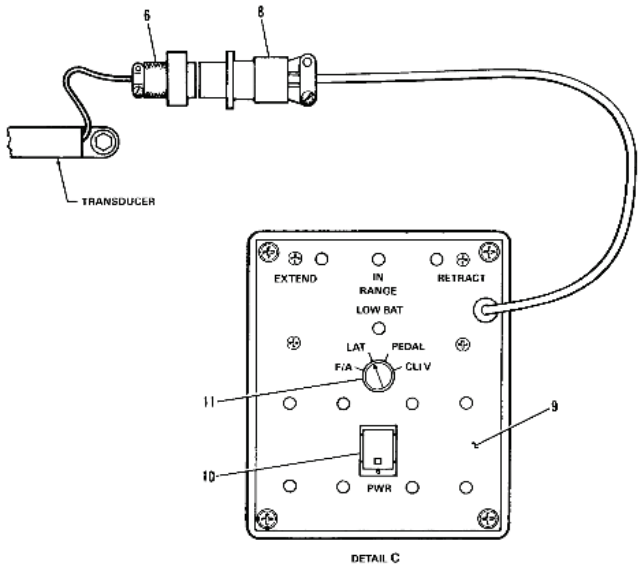
ICN-412-A-670100-B-97499-00032-A-001-01

Figure 3. Cyclic Controls Rigging (36087 and Subsequent) (Sheet 2 of 4)



ICN-412-A-670100-B-97499-00033-A-001-01

Figure 3. Cyclic Controls Rigging (36087 and Subsequent) (Sheet 3 of 4)



4. Rod end bearing
5. Jam nut
6. Electrical connector
7. Transducer shaft
8. Calibration box connector
9. Calibration tool (412-274-001-101)
10. Calibration power switch
11. Selector knob

ICN-412-A-670100-B-97499-00034-A-001-01

Figure 3. Cyclic Controls Rigging (36087 and Subsequent) (Sheet 4 of 4)

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Cyclic Controls - Description

The cyclic controls installation (Figure 1) consist of the controls and components from the pilot and copilot cyclic stick (15 and 7) up to but not including the swashplate (1).

The cyclic control system incorporates two cyclic series actuators (6, fore and aft and 17, lateral), one connected in roll system and one connected in pitch system. A control tube and idler (19) are attached to series actuators (6) in pitch system.

Four transducers, two lateral cyclic transducers (13) in roll system and two fore and aft cyclic transducers (16) in pitch system, are used to convert mechanical movement to electrical signals for injecting pilot and copilot control commands to the DAFCS.

Two fore and aft cyclic transducers (16) are attached to right end of cyclic jackshaft (8) providing a fore and aft cyclic signal. Two lateral cyclic transducers (13) are attached to a forward cyclic bellcrank (14) providing a lateral cyclic signal.

A force gradient assembly connected to an electromechanical rotary actuator is also installed in each cyclic system. Cyclic roll system force gradient attaches to a bellcrank forward of jackshaft. The cyclic pitch system force gradient attaches directly to the left end of cyclic jackshaft.

A support assembly and idler (19) are used to connect the fore and aft cyclic series actuators (6) and pitch control tube which is attached to the lever assembly (21).

Two proximity switches (18, fore and aft and 20, lateral), and two cyclic centering (CYC CTR) caution lights are installed for aid in centering cyclic sticks. The lights are located in the pilots and copilots instrument panel. The two proximity switches (18 and 20) are located underneath the floor near the mixing lever assembly (21).

NOTE

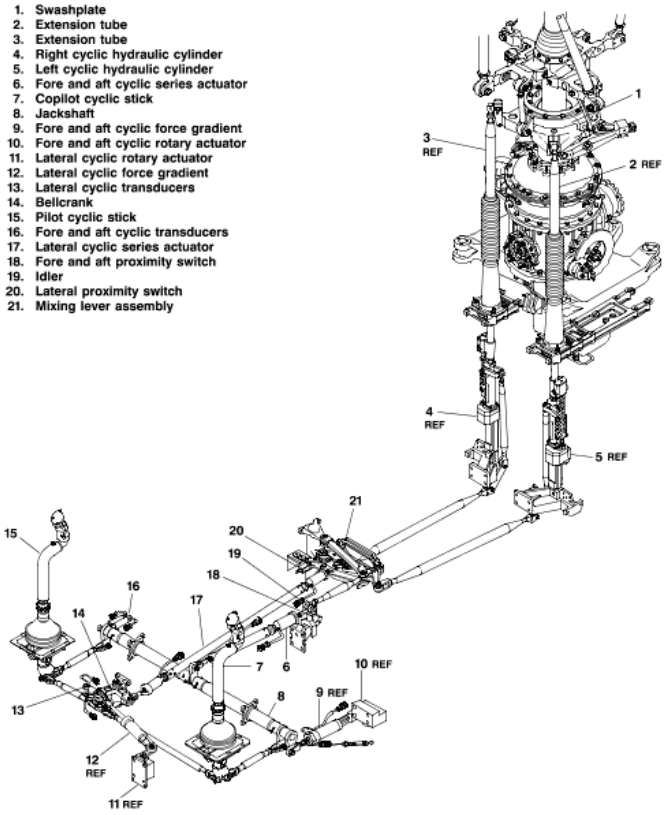
Cyclic control system components forward of mixing lever assembly (21) are identified as lateral cyclic or fore and aft cyclic as applicable, i.e., lateral cyclic force gradient.

Cyclic control system components aft of mixing lever assembly (21) are identified as left cyclic or right cyclic as applicable, i.e., left cyclic flight control cylinder.

The fore and aft control linkage are separate from the cyclic stick to the mixing lever assembly (21). From the mixing lever assembly (21) to the swashplate (1) the effects of the control linkage are combined and cannot be considered separately as to effect.

Actuation of the pilot and copilot cyclic sticks (15 and 7) causes hydraulic power assisted cyclic pitch change of the main rotor blades.

Dual controls are installed for copilot control of the helicopter. Operation of dual controls system is basically the same as operation of single system. Refer to Flight Manual for operating procedures.



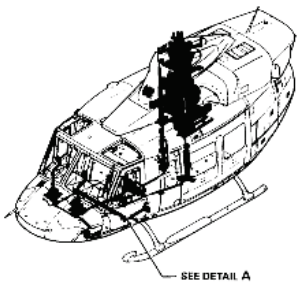
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Figure 1. Cyclic Controls (36087 and Subsequent) (Sheet 1 of 1)

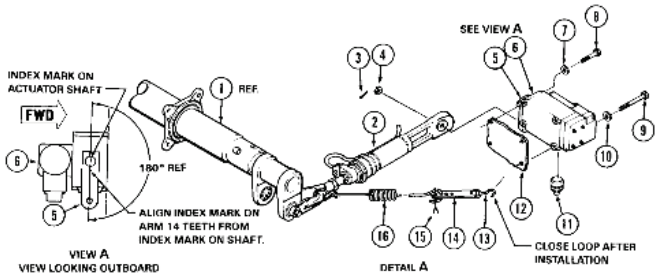
Rotary Actuator (Fore and Aft Cyclic) - Removal

Procedure

1. Remove access panel from cabin floor under left crew seat. Remove access panel from bottom left side of helicopter just forward of front landing gear crosstube (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect electrical connector (11, Figure 1).
3. Remove cotter pin (3), nut (4) and separate force gradient (2) from arm (5).
4. Disconnect hook (13) from bolt (9).
5. Remove three bolts (8) and washers (7).
6. Remove one bolt (9) and washer (10).
7. Remove rotary actuator (6).



1. Jackshaft
2. Force gradient
3. Cotter pin
4. Nut
5. Arm
6. Rotary actuator
7. Aluminum washer
8. Bolt
9. Bolt (212-001-315-1)
10. Aluminum washer
11. Electrical connector
12. Structure
13. Hook
14. Plate
15. Lockwire
16. Spring



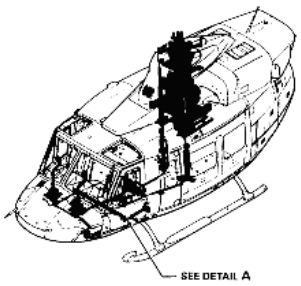
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Figure 1. Fore and Aft Cyclic Rotary Actuator Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

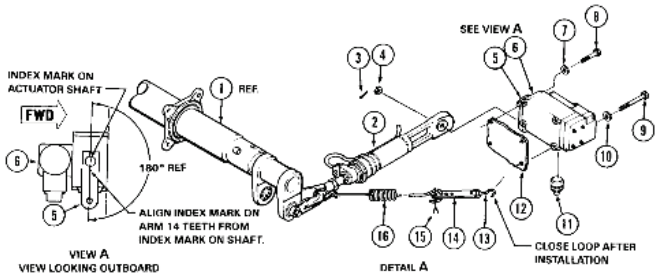
Rotary Actuator (Fore and Aft Cyclic) - Inspection

Procedure

1. Inspect rotary actuator (6, Figure 1) for mechanical and corrosion damage and wear (BHT-ALL-SPM, Standard Practices Manual).
2. Inspect rotary actuator arm (5) stud for damaged threads. No damage that can be detected visually is acceptable.
3. Inspect electrical receptacle for bent pins and damaged threads. Superficial damage is acceptable without repair.



1. Jackshaft
2. Force gradient
3. Cotter pin
4. Nut
5. Arm
6. Rotary actuator
7. Aluminum washer
8. Bolt
9. Bolt (212-001-315-1)
10. Aluminum washer
11. Electrical connector
12. Structure
13. Hook
14. Plate
15. Lockwire
16. Spring



ICN-412-A-671201-B-97499-00002-A-001-01

Figure 1. Fore and Aft Cyclic Rotary Actuator Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Rotary Actuator (Fore and Aft Cyclic) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Inspect rotary actuator (6, Figure 1) to ensure arm (5) is installed on rotary actuator with index mark on arm position 14 teeth from index mark on rotary actuator shaft shown on view A. Reposition arm (5) on rotary actuator if index marks are not properly located.

2. Position rotary actuator (6) on structure (12). Install special bolt (9) and aluminum washer (10) in lower out board hole. Install three bolts (8) and aluminum washers (7) in remaining holes.

3. Position force gradient (2) on arm (5) stud. Install nut (4) and cotter pin (3).

4. Install spring (16), plate (14), and hook (13) on bolt (9). Refer to 412-A-67-12-00-16A-720A-A / 01458 for procedure to position spring (16) in correct hole in plate (14).

NOTE

For initial installation, install spring in fourth hole of plate (14). With hydraulic boost on, or during ground run, adjust spring tension to position pilot control stick (longitudinal cyclic only) perpendicular to deck within 2 degrees.

5. When tension on spring (16) is correctly adjusted, secure spring to plate (8) with lockwire (C-405) (15).

6. Close loop on hook (13) through bolt (9).

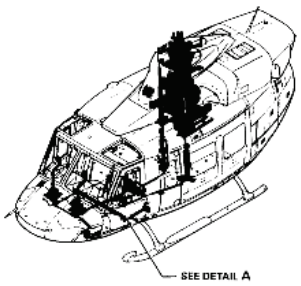
7. Install electrical connector (11).

8. Move cyclic controls through full travel and check for binding and/or obstructions.

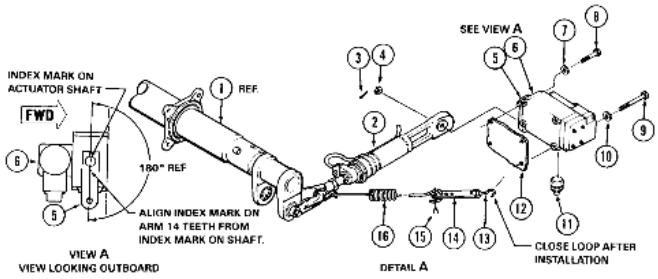
9. Verify that cyclic control rigging is within limits (412-A-67-01-00-08A-276A-A / 01548).

10. Install access panel on cabin floor under left crew seat (412-A-52-40-00-02A-720A-A / 00840).

11. Install access panel bottom left side of helicopter just forward of forward landing gear crosstube (412-A-52-40-00-02A-720A-A / 00840).



1. Jackshaft
2. Force gradient
3. Cotter pin
4. Nut
5. Arm
6. Rotary actuator
7. Aluminum washer
8. Bolt
9. Bolt (212-001-315-1)
10. Aluminum washer
11. Electrical connector
12. Structure
13. Hook
14. Plate
15. Lockwire
16. Spring



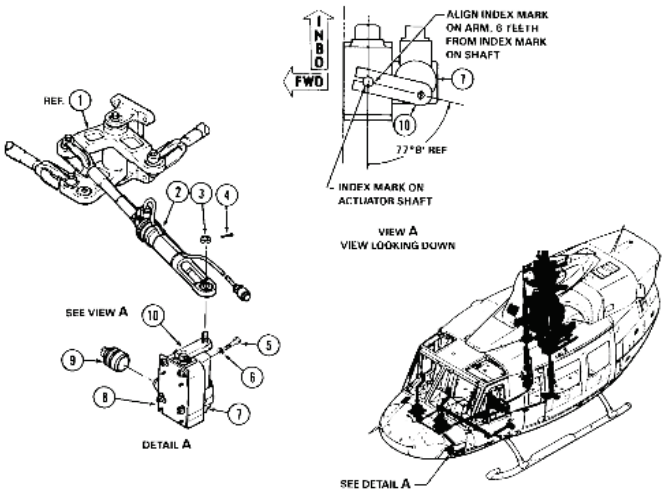
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Figure 1. Fore and Aft Cyclic Rotary Actuator Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Rotary Actuator (Lateral Cyclic) - Removal

Procedure

1. Remove access panel from bottom of helicopter just aft and inboard of left lower nose window (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect electrical connector (9, Figure 1) from rotary actuator (7).
3. Remove cotter pin (4) and nut (3) and separate force gradient (2) from arm (10).
4. Remove four bolts (5) and washers (6).
5. Separate rotary actuator (7) from rotary actuator adapter plate (8).



1. Lateral control bellcrank
2. Force gradient
3. Nut
4. Cotter pin
5. Bolt
6. Aluminum washer
7. Rotary actuator
8. Rotary actuator adapter plate (structure)
9. Electrical connector
10. Arm

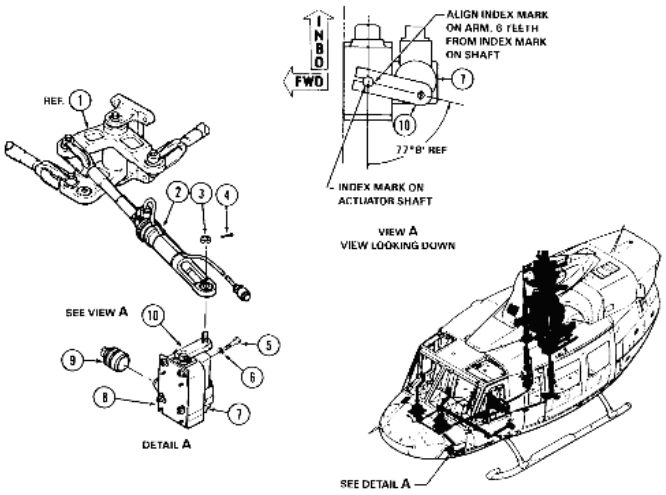
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Figure 1. Lateral Cyclic Rotary Actuator Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Rotary Actuator (Lateral Cyclic) - Inspection

Procedure

1. Inspect rotary actuator (7, Figure 1) for mechanical and corrosion damage and wear (BHT-ALL-SPM, Standard Practices Manual).
2. Inspect rotary actuator arm (10) stud for damaged threads. No damage that can be detected visually is acceptable.
3. Inspect electrical connector (9) and receptacle for bent pins and damaged threads. Superficial damage is acceptable without repair.



1. Lateral control bellcrank
2. Force gradient
3. Nut
4. Cotter pin
5. Bolt
6. Aluminum washer
7. Rotary actuator
8. Rotary actuator adapter plate (structure)
9. Electrical connector
10. Arm

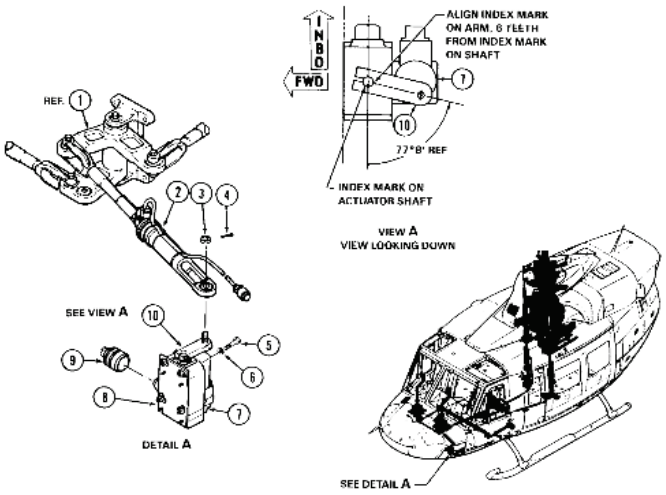
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Figure 1. Lateral Cyclic Rotary Actuator Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Rotary Actuator (Lateral Cyclic) - Installation

Procedure

1. Inspect rotary actuator (7, Figure 1) to ensure that arm (10) is installed on rotary actuator with index mark on arm positioned six teeth from index mark on rotary actuator shaft as shown in view A. Reposition arm (10) on rotary actuator if index marks are not properly located.
2. Position rotary actuator (7) on adapter plate (8). Install four bolts (5) and four aluminum washers (6).
3. Position force gradient (2) on arm (10). Install nut (3) and cotter pin (4).
4. Install electrical connector (9).
5. Move cyclic controls through full travel and check for binding and/or obstructions.
6. Verify that cyclic control rigging is within limits (412-A-67-01-00-08A-276A-A / 01548).
7. Install access panel on bottom of helicopter just aft and inboard of left lower nose window (412-A-52-40-00-02A-720A-A / 00840).



1. Lateral control bellcrank
2. Force gradient
3. Nut
4. Cotter pin
5. Bolt
6. Aluminum washer
7. Rotary actuator
8. Rotary actuator adapter plate (structure)
9. Electrical connector
10. Arm

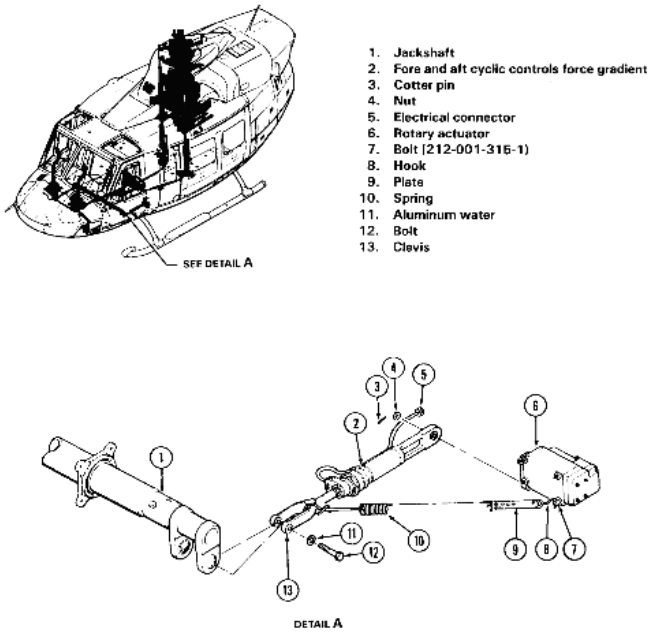
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Figure 1. Lateral Cyclic Rotary Actuator Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Force Gradient (Fore and Aft Cyclic) - Removal

Procedure

1. Remove access panel from cabin floor under left crew seat (412-A-52-40-00-02A-520A-A / 00837).
2. Remove access panel from bottom of helicopter on left side forward of forward landing gear crosstube (412-A-52-40-00-02A-520A-A / 00837).
3. Disconnect electrical connector (5, Figure 1).
4. Remove hook (8) from bolt (7).
5. Remove cotter pin (3) and nut (4) from rotary actuator (6).
6. Cut and discard lockwire. Remove bolt (12) and washer (11) from clevis (13).
7. Remove fore and aft cyclic force gradient (2), spring assembly (10) and plate (9) from jackshaft (1). Measure and record overall length of force gradient.



ICN-412-A-671201-B-97499-00004-A-001-01

Figure 1. Fore and Aft Cyclic Force Gradient Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Force Gradient (Fore and Aft Cyclic) - Inspection

Procedure

1. Inspect force gradient (2, Figure 1) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion is not acceptable.

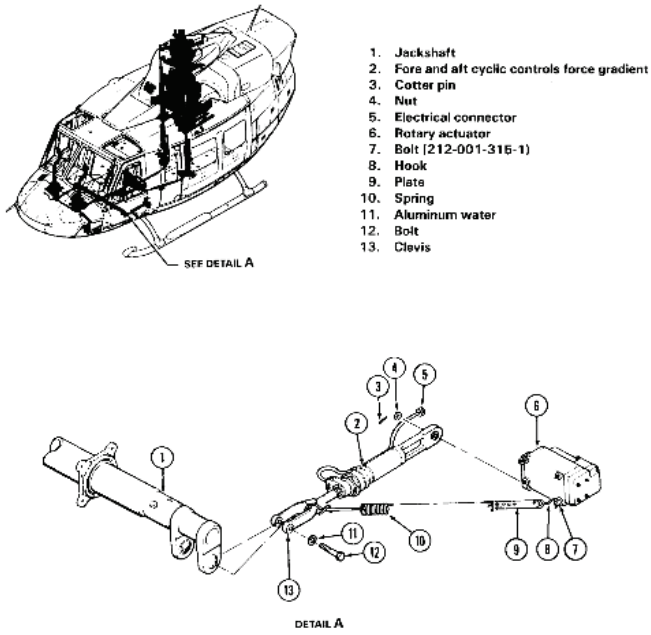
2. Inspect for thread damage in threaded hole and for elongation of hole in opposite side of clevis (13).

3. Inspect hook (8), plate (9), and spring (10) for obvious mechanical damage and for loss of tension in spring. Spring tension at 10.36 in. (263.14 mm) is 6.4 ± 0.5 lbs. (28.47 ± 2.22 kg).

4. Inspect bearings for wear and secure installation in cylinder (5). Refer to Bearing Wear Limit Table in BHT-ALL-SPM, Standard Practices Manual for acceptable limits.

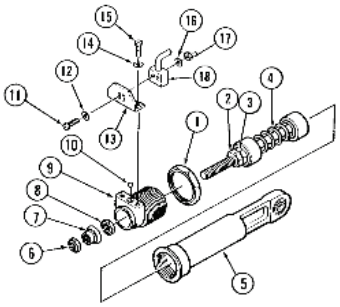
5. Inspect jam nut (1) and cylinder (5) for secure installation.

6. Inspect switch and electrical cable assembly (18, Figure 2) for frayed wires and broken pins in connector. Check continuity from connector pins through switch.



ICN-412-A-671201-B-97499-00004-A-001-01

Figure 1. Fore and Aft Cyclic Force Gradient Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)



1. Jam nut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Screw
12. Washer
13. Retainer
14. Washer
15. Screw
16. Washer
17. Nut
18. Cable assembly (switch)

ICN-412-A-671201-B-97499-00005-A-001-01

Figure 2. Fore and Aft Cyclic Force Gradient (36087 and Subsequent) (Sheet 1 of 1)

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Force Gradient (Fore and Aft Cyclic) - Repair

Procedure

1. Repair of force gradient consists of replacement of unserviceable components and adjustment of spring preload in accordance with 412-A-67-12-01-03A-720A-A / 01562.

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Force Gradient (Fore and Aft Cyclic) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

Force gradient (2, Figure 1) in the fore and aft cyclic controls is similar in appearance to the force gradient in lateral cyclic controls, but has different part numbers. Make sure that correct part number force gradient is installed in the fore and aft position (BHT-412-IPB).

1. If not previously accomplished, adjust spring preload of force gradient as follows:

1.1. Cut and remove lockwire, loosen jam nut (1, Figure 2) and remove cap (9).

1.2. Remove spring assembly (4) from cylinder (5).

1.3. Adjust inboard nut (3) to provide 8.0 to 11.0 pounds (35.59 to 48.93 N) preload against spring assembly (4). Tighten outboard nut (2) against inboard nut (3).

1.4. Position spring assembly (4) in cylinder (5) and install cap (9) with jam nut (1) until all end play is eliminated. Tighten jam nut and secure with lockwire (C-405) to cylinder.

2. If not previously accomplished, adjust actuator (7) on force gradient as follows:

2.1. Loosen nuts (6 and 8) on shaft of spring assembly (4).

2.2. Make sure retainer (13) is secured by screws (15) and washers (14).

2.3. Align actuator (7) on shaft of spring assembly (4) with ball (10) and plunger of cable assembly (switch) (18).

2.4. Loosen screws (11), washers (12 and 16), nut (17), and cable assembly (switch) (18) from the retainer (13).

2.5. Push and hold actuator (7) down (180°) against cap (9) at point opposite location of cable assembly (switch) (18). Hold this position to initially adjust switch (18) then adjust switch to actuate within ±0.050 inch (1.27 mm) linear travel of spring assembly (4). Secure switch to retainer (13) with screws (11), washers (12 and 16), and nut (17). Secure actuator with nut (8).

2.6. Install clevis (13, Figure 1) on shaft of spring assembly (4, Figure 2), secure with nut (6).

3. After assembly of force gradient, prior to installation, verify spring assembly (4) preload of 8.0 to 11.0 pounds (35.59 to 48.93 N).

4. Position spring (10, Figure 1) on clevis (13).

5. Position fore and aft cyclic controls force gradient (2) on jackshaft (1). Clevis (13) has a threaded hole in one side. Position the clevis with the threaded hole inboard. Install bolt (12) and aluminum washer (11). Secure bolt head to hole in clevis with lockwire (C-405).



MAKE SURE FORCE GRADIENT LENGTH IS ADJUSTED TO RECORDED LENGTH PRIOR TO INSTALLATION OF FORCE GRADIENT ON ROTARY ACTUATOR.

6. Position fore and aft cyclic controls force gradient (2) on rotary actuator (6). Install nut (4) and cotter pin (3).

7. Install plate (9) on spring (10) with spring through fourth hole in plate. Do not lockwire spring to plate at this time.

8. Attach plate (9) with hook (8) to bolt (7). Do not close loop on hook at this time.

9. Connect electrical connector (5).

10. Move fore and aft cyclic controls through full travel and check for binding and obstructions.

11. Apply external hydraulic power to helicopter.

12. Loosen cyclic friction and observe cyclic stick position. If cyclic stick is not perpendicular (longitudinally) within 2°, move spring (10) to different hole in plate (9) to cause cyclic stick to be vertical (longitudinally).

13. When tension on spring (10) is correctly adjusted, secure spring to plate (9) with lockwire (C-405). Close loop on hook (8) to prevent accidental disengagement of hook.

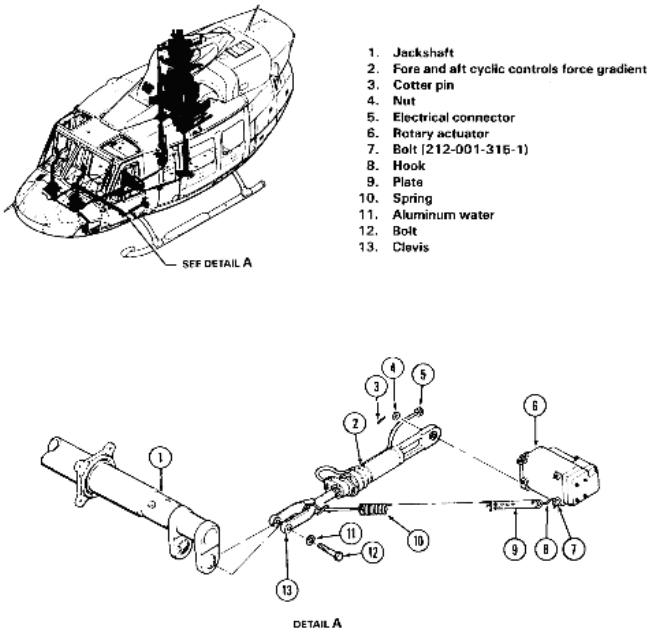
14. Move fore and aft cyclic controls through full travel and check for binding and obstructions.

15. Confirm that cyclic control rigging is within limits (412-A-67-01-00-08A-276A-A / 01548).

16. Remove external power from helicopter.

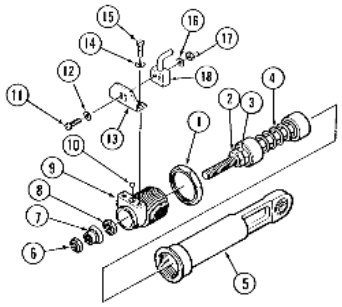
17. Install access panel on bottom of helicopter on left side forward of forward landing gear crosstube (412-A-52-40-00-02A-720A-A / 00840).

18. Install access panel on cabin floor under left crew seat (412-A-52-40-00-02A-720A-A / 00840).



ICN-412-A-671201-B-97499-00004-A-001-01

Figure 1. Fore and Aft Cyclic Force Gradient Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)



1. Jam nut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Screw
12. Washer
13. Retainer
14. Washer
15. Screw
16. Washer
17. Nut
18. Cable assembly (switch)

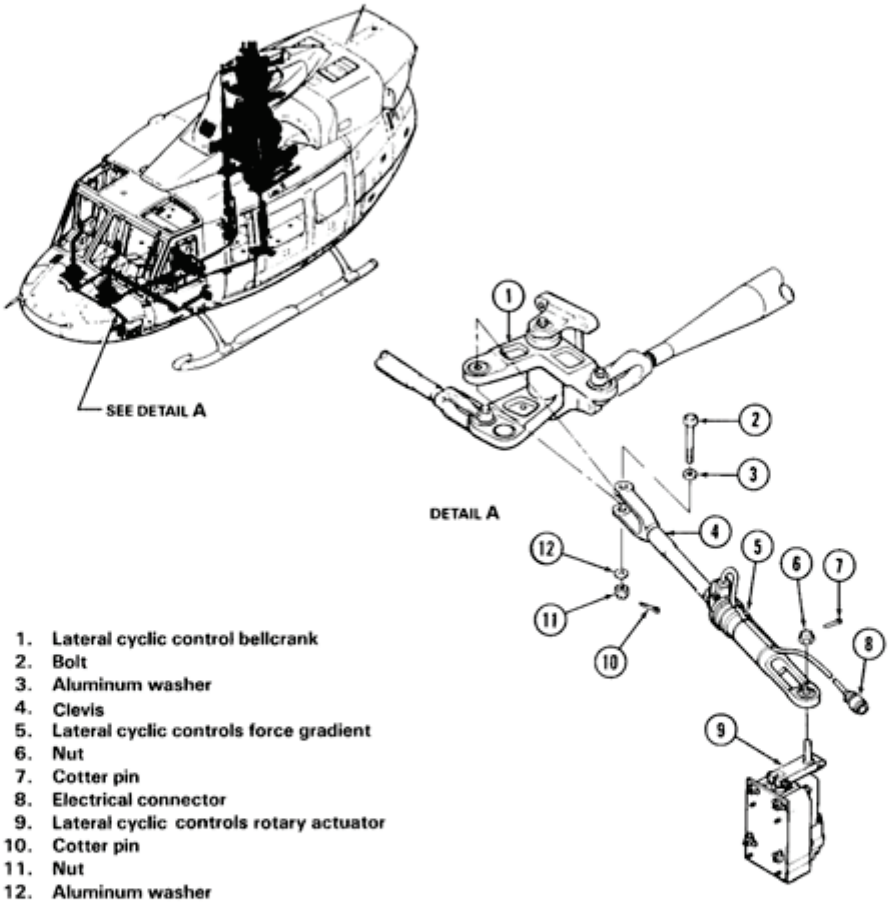
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Figure 2. Fore and Aft Cyclic Force Gradient (36087 and Subsequent) (Sheet 1 of 1)

Force Gradient (Lateral Cyclic) - Removal

Procedure

1. Remove access panel from bottom of helicopter below lateral cyclic controls force gradient (5, Figure 1) (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect electrical connector (8).
3. Remove cotter pin (7) and nut (6).
4. Remove cotter pin (10), nut (11), and aluminum washer (12).
5. Remove bolt (2) with aluminum washer (3) from lateral cyclic control bellcrank (1).
6. Remove lateral cyclic controls force gradient (5).
7. Record over all length of force gradient.



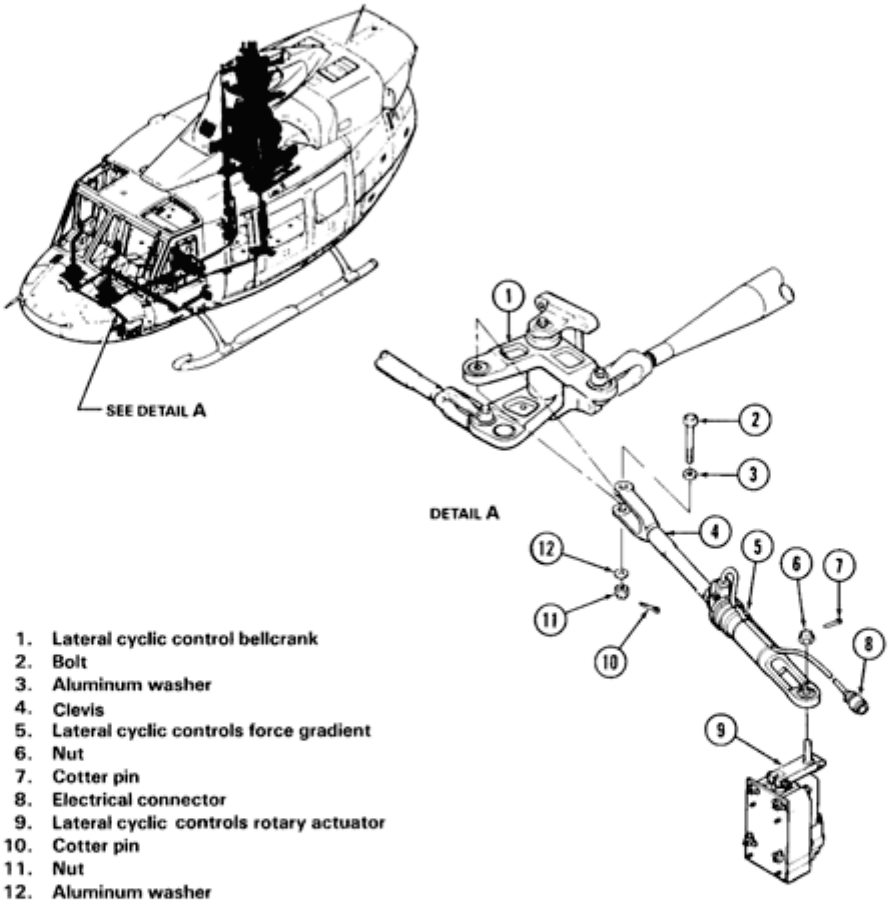
ICN-412-A-671201-B-97499-00006-A-001-01

Figure 1. Lateral Cyclic Force Gradient Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Force Gradient (Lateral Cyclic) - Inspection

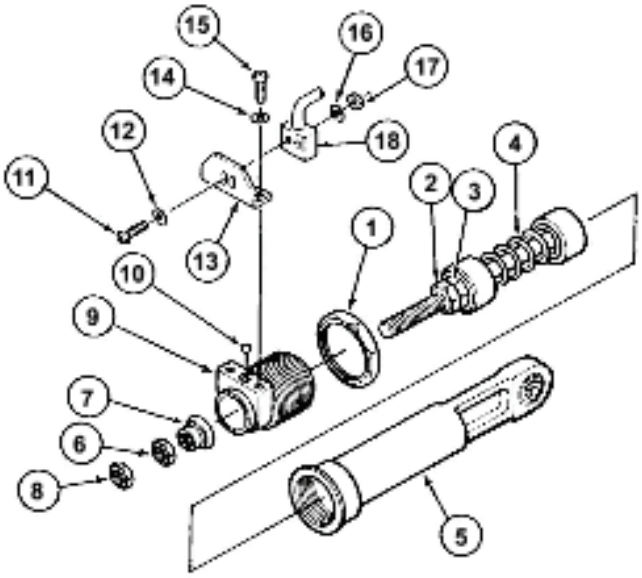
Procedure

1. Inspect force gradient (5, Figure 1) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion is not acceptable.
2. Inspect clevis (4) for damage.
3. Inspect bearings for wear and secure installation in cylinder (5, Figure 2). Refer to Bearing Wear Limit Table in BHT-ALL-SPM, Standard Practices Manual for acceptable limits.
4. Inspect jam nut (1) and cylinder (5) for secure installation.
5. Inspect switch and electrical cable assembly (18, Figure 2) for frayed wires and broken pins in connector. Check continuity from connector pins through switch.



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Figure 1. Lateral Cyclic Force Gradient Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)



1. Jamnut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Screw
12. Washer
13. Retainer
14. Washer
15. Screw
16. Washer
17. Nut
18. Cable assembly (switch)

ICN-412-A-671201-B-97499-00007-A-001-01

Figure 2. Lateral Cyclic Force Gradient (36087 and Subsequent) (Sheet 1 of 1)

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Force Gradient (Lateral Cyclic) - Repair

Procedure

1. Repair of force gradient consists of replacement of unserviceable components and adjustment of spring preload in accordance with 412-A-67-12-01-04A-720A-A / 01566.

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Force Gradient (Lateral Cyclic) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

Force gradient (5, Figure 1) in the lateral cyclic controls is similar in appearance to the force gradient in the fore and aft cyclic controls, but the force gradient has different part numbers. Ensure that correct part number force gradient is installed in lateral position. Refer BHT-412 IPB.

1. If not previously accomplished, adjust spring preload on force gradient as follows:

1.1. Cut and remove lockwire, remove cap (9, Figure 2).

1.2. Remove spring assembly (4) from cylinder (5).

1.3. Adjust inboard nut (3) to provide 6.0 to 7.0 pounds (26.7 to 31.1 N) preload against spring assembly (4). Tighten outboard nut (2) against inboard nut (3).

1.4. Position spring assembly (4) in cylinder (5) and install cap (9) with jam nut (1) until all end play is eliminated. Tighten jam nut (1) and secure with lockwire (C-405) to cylinder (5).

2. If not previously accomplished, adjust actuator (7) on force gradient as follows:

2.1. Loosen nuts (6 and 8) on shaft of spring assembly (4).

2.2. Ensure retainer (13) is secured by screws (15) and washers (14).

2.3. Align actuator (7) on shaft of spring assembly (4) with ball (10) and plunge of cable assembly (switch) (18).

2.4. Loosen screws (11), washers (12 and 16), and nut (17) cable assembly (switch) (18) to retainer (13).

2.5. Push and hold actuator (7) down (180°) against cap (9) at point opposite location of cable assembly (18). Hold this position to initially adjust switch (18) then adjust switch to actuate within ±0.050 inch (1.27 mm) linear travel of spring assembly (4). Secure switch (18) to retainer (13) with screws (11), washers (12 and 16), and nut (17). Secure actuator (7) with nuts (6 and 8).

2.6. Install clevis (4, Figure 1) on shaft of spring assembly (4, Figure 2), adjust to length recorded during removal, and secure with nut (8).

3. After assembly of force gradient, prior to installation, verify spring assembly (4) preload of 6.0 to 7.0 pounds (26.69 to 31.14 N).

4. Position force gradient (5, Figure 1) on lateral cyclic control bellcrank (1). Install bolt (2), washers (3 and 12), nut (11) and cotter pin (10).



ENSURE THAT FORCE GRADIENT LENGTH IS ADJUSTED TO RECORDED LENGTH PRIOR TO INSTALLATION OF FORCE GRADIENT ON ROTARY ACTUATOR.

5. Position force gradient (5) on rotary actuator (9). Install nut (6) and cotter pin (7).

6. Connect electrical connector (8).

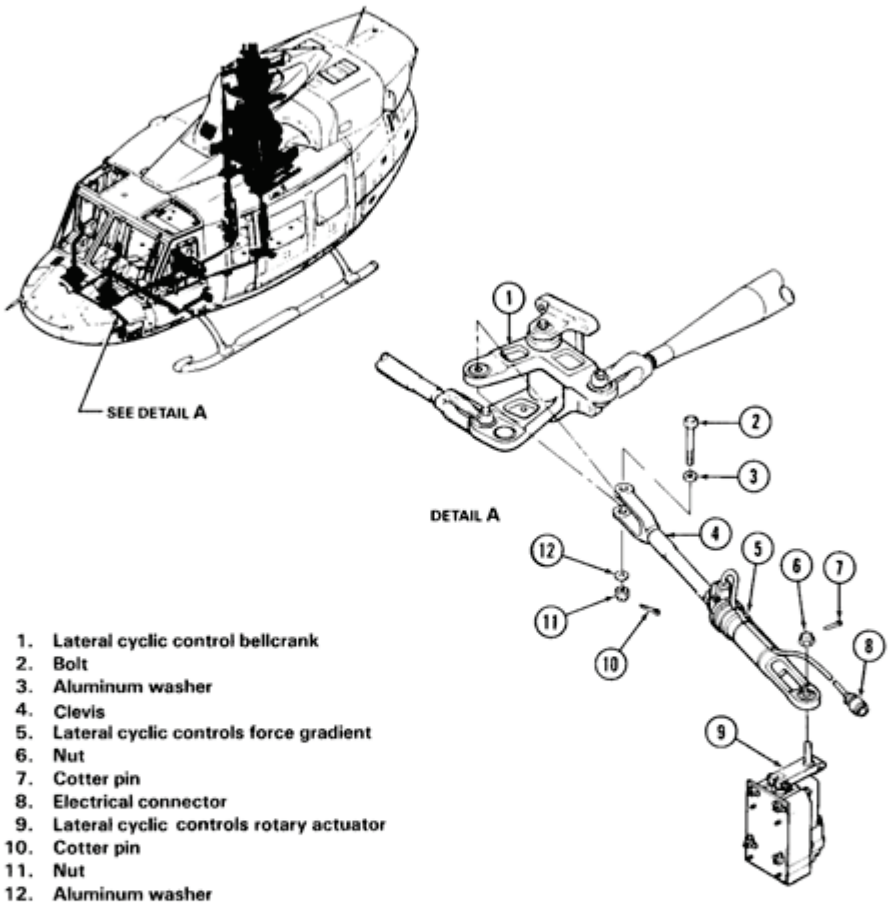
7. Apply external hydraulic power to helicopter.

8. Move cyclic controls through full travel and check for binding and/or obstructions.

9. Verify that cyclic control rigging is within limits (412-A-67-01-00-08A-276A-A / 01548).

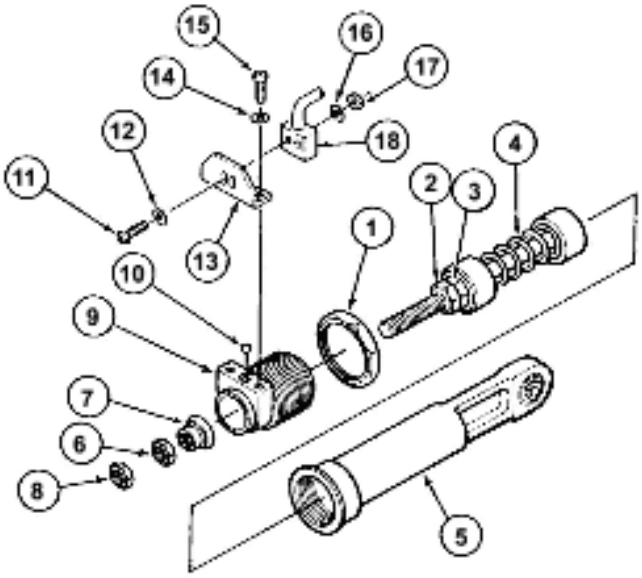
10. Remove external hydraulic power from helicopter.

11. Install access panel on bottom of helicopter below force gradient (412-A-52-40-00-02A-720A-A / 00840).



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Figure 1. Lateral Cyclic Force Gradient Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)



1. Jamnut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Screw
12. Washer
13. Retainer
14. Washer
15. Screw
16. Washer
17. Nut
18. Cable assembly (switch)

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Figure 2. Lateral Cyclic Force Gradient (36087 and Subsequent) (Sheet 1 of 1)

Anti-Torque Controls - Description

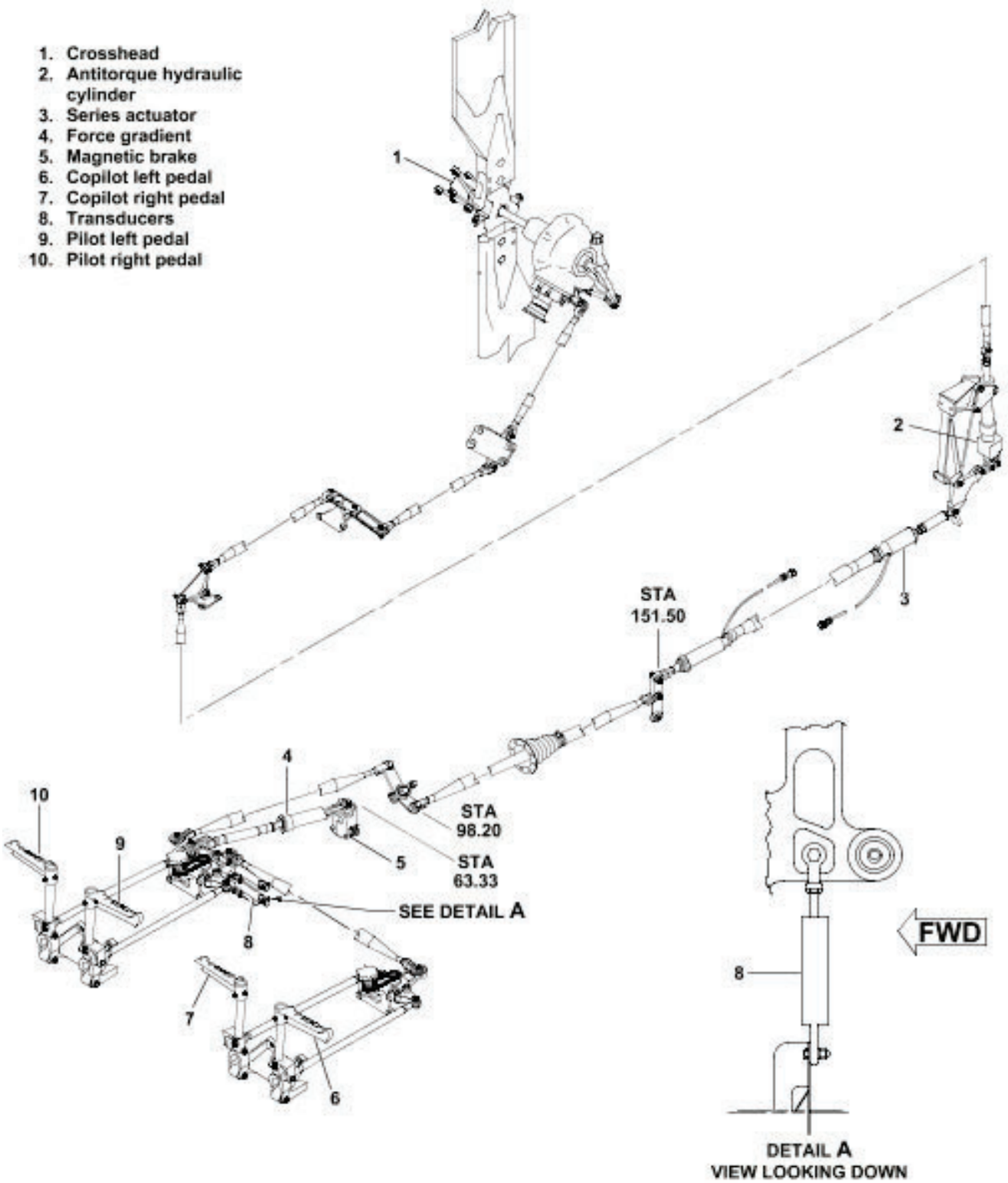
The anti-torque control system consists of control linkage and components from pilot and copilot pedals (9, 10, 6, and 7, Figure 1) aft to crosshead (1) on tail rotor. The DAFCS incorporates electromechanical series actuators (3), magnetic brake (5), force gradient (4), and two transducers (8). The series actuators (3) is located forward of anti-torque hydraulic cylinder (2) and support. The magnetic brake (5) is located on main bulkhead under access panel below pilot seat. The force gradient (4) is connected to magnetic brake (5) and directional control bellcrank. The two transducers (8) are connected to directional control bellcrank located aft of pilot left and right pedals (9 and 10) and to main beam of airframe structure.

When the optional 4-axis AFCS/Flight Director kit is installed, the force gradient (3, Figure 2) incorporates a detent switch and the magnetic brake is replaced by an electromechanical rotary actuator (8).

Actuation of the pilot or copilot pedals causes a hydraulic power assisted pitch change of the tail rotor blades to offset main rotor torque and to control the directional heading of the helicopter.

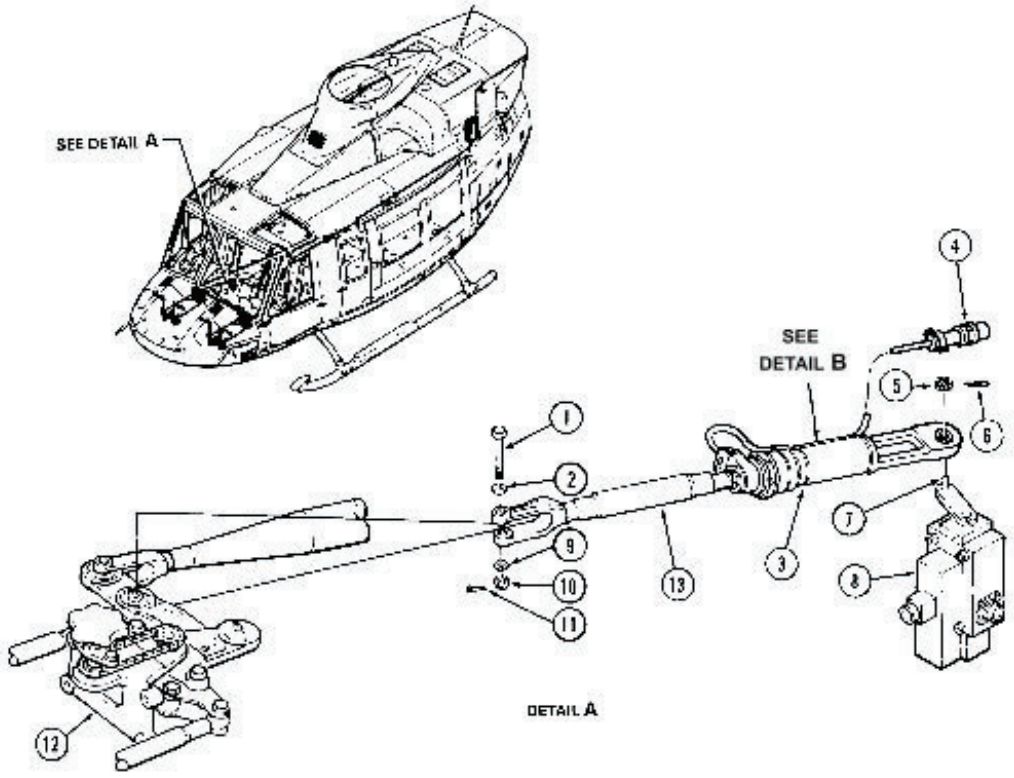
Dual controls are installed for copilot control of helicopter.

Operation of anti-torque dual controls system is basically the same as operation of single system. Refer to Flight Manual for operating procedures.



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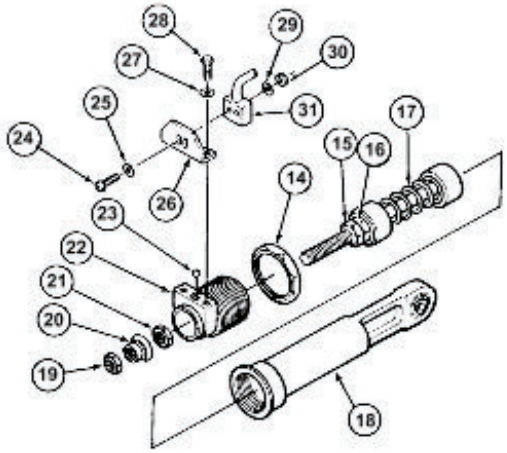
Figure 1. Anti-Torque Controls (36087 and Subsequent) (Sheet 1 of 1)



1. Bolt
2. Aluminum washer
3. Force gradient
4. Electrical connector
5. Nut
6. Cotter pin
7. Stud
8. Rotary actuator
9. Aluminum washer
10. Nut
11. Cotter pin
12. Pedal adjuster
13. Clevis

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Figure 2. Anti-Torque Force Gradient (4-Axis Configuration) (Sheet 1 of 2)



DETAIL B

14. Jam nut
15. Outboard nut
16. Inboard nut
17. Spring assembly
18. Cylinder
19. Nut
20. Actuator
21. Nut
22. Cap
23. Ball
24. Screw
25. Washer
26. Retainer
27. Washer
28. Screw
29. Washer
30. Nut
31. Cable assembly (switch)

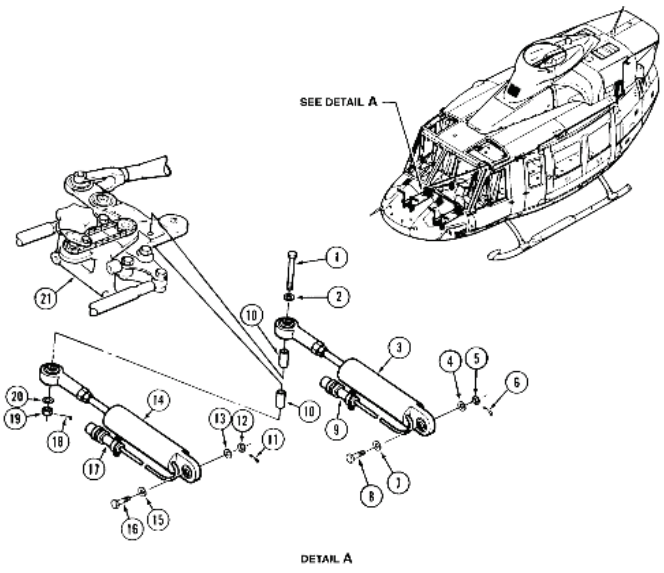
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Figure 2. Anti-Torque Force Gradient (4-Axis Configuration) (Sheet 2 of 2)

Anti-Torque Transducers - Removal

Procedure

1. Remove access panel from cabin floor under pilot seat (412-A-52-40-00-02A-520A-A / 00837).
2. Disconnect electrical connectors (9 and 17, Figure 1).
3. Remove cotter pin (18), nut (19), washer (20), spacers (10) and bolt (1) with washer (2).
4. Remove cotter pin (6), nut (5), spacer (4) washer (7) and bolt (8).
5. Remove cotter pin (11), nut (12), washers (13 and 15) and bolt (16).
6. Remove transducers (3 and 14).



1. Bolt
2. Aluminum washer
3. Transducer
4. Spacer
5. Nut
6. Cotter pin
7. Aluminum washer
8. Bolt
9. Electrical connector
10. Spacer
11. Cotter pin
12. Nut
13. Washer
14. Transducer
15. Aluminum washer
16. Bolt
17. Electrical connector
18. Cotter pin
19. Nut
20. Aluminum washer
21. Pedal adjuster

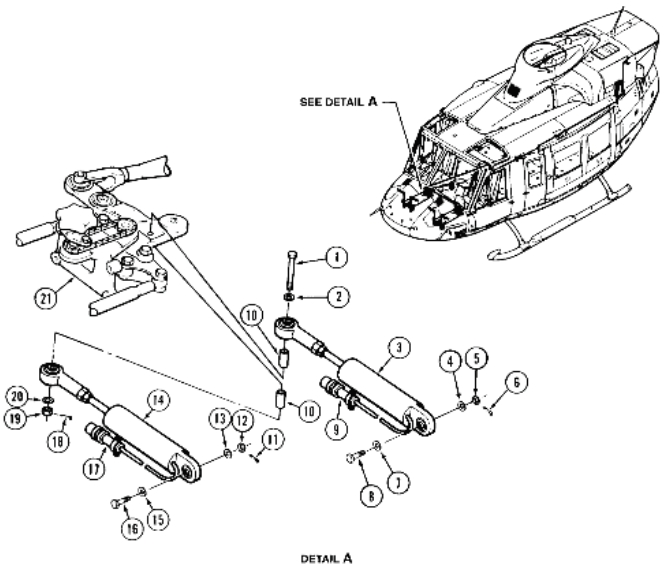
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Figure 1. Anti-Torque Transducer Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Anti-Torque Transducers - Inspection

Procedure

1. Inspect transducers (3 and 14, Figure 1) for damage and secured electrical plugs.
2. Inspect bearings for looseness (BHT-ALL-SPM, Standard Practices Manual).



1. Bolt
2. Aluminum washer
3. Transducer
4. Spacer
5. Nut
6. Cotter pin
7. Aluminum washer
8. Bolt
9. Electrical connector
10. Spacer
11. Cotter pin
12. Nut
13. Washer
14. Transducer
15. Aluminum washer
16. Bolt
17. Electrical connector
18. Cotter pin
19. Nut
20. Aluminum washer
21. Pedal adjuster

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Figure 1. Anti-Torque Transducer Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Anti-Torque Transducers - Repair

Procedure

1. Anti-torque control transducers are nonrepairable and must be replaced when defective.

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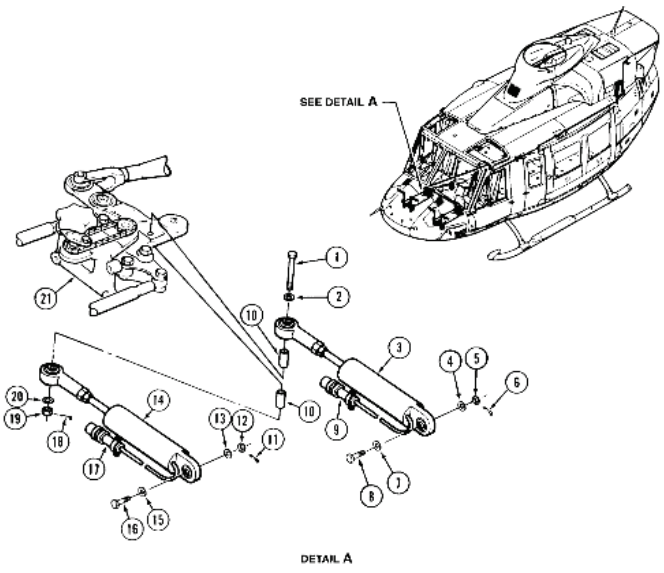
Anti-Torque Transducers - Installation

Procedure

NOTE

If not previously accomplished perform transducer adjustment, refer to 412-A-67-01-00-11A-276A-A / 01551.

1. Position transducers (3 and 14, Figure 1) on bracket of main beam of airframe structure.
2. Install bolt (8), washer (7), spacer (4), nut (5), and cotter pin (6).
3. Install bolt (16), aluminum washers (13 and 15), nut (12), and cotter pin (11).
4. Position transducers (3 and 14) on pedal adjuster (21) bellcrank.
5. Install bolt (1), washer (2), spacers (10), washer (20), nut (19), and cotter pin (18).
6. Connect electrical connectors (9 and 17).
7. Install access panel on floor under pilot seat (412-A-52-40-00-02A-720A-A / 00840).



1. Bolt
2. Aluminum washer
3. Transducer
4. Spacer
5. Nut
6. Cotter pin
7. Aluminum washer
8. Bolt
9. Electrical connector
10. Spacer
11. Cotter pin
12. Nut
13. Washer
14. Transducer
15. Aluminum washer
16. Bolt
17. Electrical connector
18. Cotter pin
19. Nut
20. Aluminum washer
21. Pedal adjuster

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Figure 1. Anti-Torque Transducer Removal/Installation (36087 and Subsequent) (Sheet 1 of 1)

Anti-Torque Force Gradient - Description

NOTE

The maintenance procedure is applicable to anti-torque force gradients (incorporating a detent switch) used when the optional 4-axis AFCS/Flight Director kit is installed. For anti-torque force gradients not incorporating a detent switch, refer to 412-A-67-20-00-10A-520A-A / 01528.

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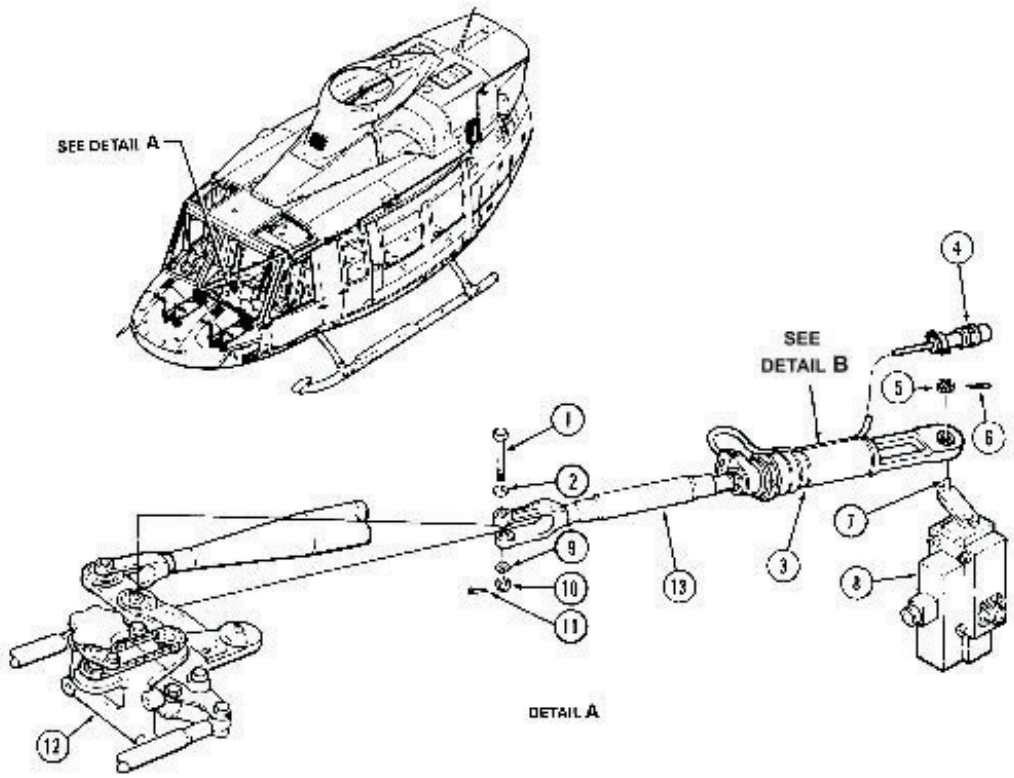
Anti-Torque Force Gradient - Removal

Procedure

NOTE

The following removal procedures are applicable to anti-torque force gradients (incorporating a detent switch) used when the optional 4-axis AFCS/Flight Director kit is installed. For anti-torque force gradients not incorporating a detent switch, refer to 412-A-67-20-00-10A-520A-A / 01528.

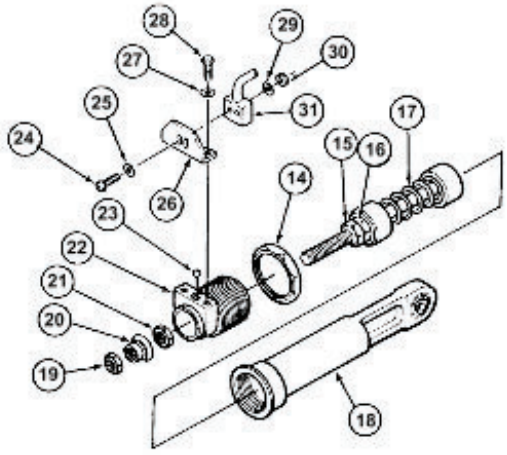
1. Remove access panel from cabin floor under pilot seat (412-A-52-40-00-02A-520A-A / 00837).
2. Remove cotter pin (11, Figure 1), nut (10), washer (9) and bolt (1) with washer (2) from pedal adjuster (12).
3. Remove cotter pin (6) and nut (5) from stud (7).
4. Remove force gradient (3) from magnetic brake (8).
5. Record length of force gradient assembly.
6. Loosen nut (19) and remove clevis (13) from spring assembly (17).



1. Bolt
2. Aluminum washer
3. Force gradient
4. Electrical connector
5. Nut
6. Cotter pin
7. Stud
8. Rotary actuator
9. Aluminum washer
10. Nut
11. Cotter pin
12. Pedal adjuster
13. Clevis

ICN-412-A-671202-B-97499-00003-A-001-01

Figure 1. Anti-Torque Force Gradient (4-Axis Configuration) (Sheet 1 of 2)



DETAIL B

14. Jam nut
15. Outboard nut
16. Inboard nut
17. Spring assembly
18. Cylinder
19. Nut
20. Actuator
21. Nut
22. Cap
23. Ball
24. Screw
25. Washer
26. Retainer
27. Washer
28. Screw
29. Washer
30. Nut
31. Cable assembly (switch)

ICN-412-A-671202-B-97499-00004-A-001-01

Figure 1. Anti-Torque Force Gradient (4-Axis Configuration) (Sheet 2 of 2)

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